Phraseology

An interdisciplinary perspective

Edited by Sylviane Granger
and Fanny Meunier

John Benjamins Publishing Company

Phraseology

Phraseology

An interdisciplinary perspective

Edited by

Sylviane Granger
Université catholique de Louvain

Fanny Meunier
Université catholique de Louvain

John Benjamins Publishing Company
Amsterdam / Philadelphia

The paper used in this publication meets the minimum requirements of American National Standard for Information Sciences — Permanence of Paper for Printed Library Materials, ANSI Z39.48-1984.

Library of Congress Cataloging-in-Publication Data

Phraseology : an interdisciplinary perspective / edited by Sylviane Granger, Fanny Meunier.
p. cm.
Includes bibliographical references and index.
1. Phraseology. I. Granger, Sylviane, 1951- II. Meunier, Fanny.
P326.5.P45P475 2008
415--dc22 2008005167
ISBN 978 90 272 3246 5 (Hb; alk. paper)

© 2008 – John Benjamins B.V.
No part of this book may be reproduced in any form, by print, photoprint, microfilm, or any other means, without written permission from the publisher.

John Benjamins Publishing Co. · P.O. Box 36224 · 1020 ME Amsterdam · The Netherlands
John Benjamins North America · P.O. Box 27519 · Philadelphia PA 19118-0519 · USA

We wish to dedicate this volume to the memory of John Sinclair, who has played a key role in giving centre stage to phraseology and inspired much of the work covered in the volume.

Table of contents

List of contributors xi

Acknowledgements xiii

Preface xv
John Sinclair

Introduction: The many faces of phraseology xix
Sylviane Granger and Fanny Meunier

Section I
Phraseology: Theory, typology and terminology

1. Phraseology and linguistic theory: A brief survey 3
Stefan Th. Gries
2. Disentangling the phraseological web 27
Sylviane Granger and Magali Paquot
3. A unified approach to semantic frames and collocational patterns 51
Willy Martin
4. Processing of idioms and idiom modifications: A view from cognitive linguistics 67
Marija Omazić
5. A very complex criterion of fixedness: Non-compositionality 81
Maria Helena Svensson
6. Reassessing the canon: 'Fixed' phrases in general reference corpora 95
Gill Philip

Section II
Corpus-based analysis of phraseological units

7. Adjective + Noun sequences in attributive or NP-final positions: Observations on lexicalization 111
Pierre J. L. Arnaud, Emmanuel Ferragne, Diana M. Lewis, and François Maniez
8. Phrasal similes in the BNC 127
Kay Wikberg

9. *Foot* and *mouth*: The phrasal patterns of two frequent nouns 143
Hans Lindquist and Magnus Levin
10. The Good Lord and his works: A corpus-driven study of collocational resonance 159
Geoffrey C. Williams
11. Fixed expressions, extenders and metonymy in the speech of people with Alzheimer's disease 175
Margaret Maclagan, Boyd Davis, and Ron Lunsford

Section III
Phraseology across languages and cultures

12. Cross-linguistic phraseological studies: An overview 191
Jean-Pierre Colson
13. Figurative phraseology and culture 207
Elisabeth Piirainen
14. Critical observations on the culture-boundness of phraseology 229
Annette Sabban
15. Phraseology in a European framework: A cross-linguistic and cross-cultural research project on widespread idioms 243
Elisabeth Piirainen
16. Free and bound prepositions in a contrastive perspective: The case of *with* and *avec* 259
Christelle Cosme and Gaëtanelle Gilquin
17. Contrastive idiom analysis: The case of Japanese and English idioms of anger 275
Priscilla Ishida
18. Automatic extraction of translation equivalents of phrasal and light verbs in English and Russian 293
Olga Mudraya, Scott S. L. Piao, Paul Rayson, Serge Sharoff, Bogdan Babych, and Laura Löfberg

Section IV
Phraseology in lexicography and natural language processing

19. Dictionaries and collocation 313
Rosamund Moon
20. Computational phraseology: An overview 337
Ulrich Heid
21. A computational lexicography approach to phraseologisms 361
Cornelia Tschichold
22. Extracting specialized collocations using lexical functions 377
Brigitte Orliac

23. Combined statistical and grammatical criteria for the retrieval of phraseological units in an electronic corpus 391
José-Manuel Pazos Bretaña and Antonio Pamies Bertrán

Envoi
The phrase, the whole phrase, and nothing but the phrase 407
John Sinclair

Author index 411
Subject index 417

List of contributors

Pierre J. L. Arnaud
Université Lumière, Lyon, France

Bogdan Babych
University of Leeds, United Kingdom

Jean-Pierre Colson
Institut libre Marie Haps, Brussels /
Université catholique de Louvain, Belgium

Christelle Cosme
Université catholique de Louvain, Belgium

Boyd Davis
University of North Carolina at Charlotte,
United States of America

Emmanuel Ferragne
Université Lumière, Lyon, France

Gaëtanelle Gilquin
Université catholique de Louvain, Belgium

Sylviane Granger
Université catholique de Louvain, Belgium

Stefan Th. Gries
University of California, Santa Barbara,
United States of America

Ulrich Heid
Universität Stuttgart, Germany

Priscilla Ishida
University of Tsukuba, Japan

Magnus Levin
Växjö University, Sweden

Diana M. Lewis
Université Lumière, Lyon, France

Hans Lindquist
Växjö University, Sweden

Laura Löfberg
University of Tampere, Finland

Ron Lunsford
University of North Carolina at Charlotte,
United States of America

Margaret Maclagan
University of Canterbury, New Zealand

François Maniez
Université Lumière, Lyon, France

Willy Martin
Vrije Universiteit Amsterdam,
The Netherlands

Fanny Meunier
Université catholique de Louvain, Belgium

Rosamund Moon
University of Birmingham, Great Britain

Olga Mudraya
Lancaster University, United Kingdom

Marija Omazić
Josip Juraj Strossmayer University of Osijek,
Croatia

Brigitte Orliac
Rochester School, Vermont,
United States of America

Magali Paquot
Université catholique de Louvain, Belgium

Antonio Pamies Bertrán
Universidad de Granada, Spain

José-Manuel Pazos Bretaña
Universidad de Granada, Spain

Gill Philip
University of Bologna, Italy

Scott S. L. Piao
University of Manchester, United Kingdom

Elisabeth Piirainen
Steinfurt, Germany

Paul Rayson
Lancaster University, United Kingdom

Annette Sabban
University of Hildesheim, Germany

Serge Sharoff
University of Leeds, United Kingdom

John Sinclair
Tuscan Word Centre, Italy

Maria Helena Svensson
Umeå University, Sweden

Cornelia Tschichold
Swansea University, United Kingdom

Kay Wikberg
University of Oslo, Norway

Geoffrey C. Williams
Université de Bretagne Sud, France

Acknowledgements

We are indebted to the Communauté française de Belgique for funding the concerted action research project on 'Foreign Language Learning: Phraseology and Discourse' (N° 03/08-301) within which this volume and the conference that gave rise to it have been produced. We would also like to thank all the contributors to this volume for their diligence in keeping to deadlines and patience in complying with our editorial demands. We are also indebted to the many reviewers who provided insightful comments on preliminary versions of the chapters and thereby helped increase the quality of the volume. Special thanks are due to Claire Hugon for her meticulous examination of the typescript and help in formatting the chapters. Finally, we would like to express our gratitude to Kees Vaes at Benjamins for his trust, support and patience.

Preface

John Sinclair

> "But, Count," I said, "You know and speak English thoroughly!" He bowed gravely.
> "I thank you, my friend, for your all too-flattering estimate, but yet I fear that I am but a little way on the road I would travel. True, I know the grammar and the words, but yet I know not how to speak them."

How many learners have felt the same! The Count is here not talking about his pronunciation, but about the deployment of his hard-learned resources. He can make grammatical utterances using a wide vocabulary, but he realises that this is but Stage One. Now he must learn how to deploy this skill to make effective and naturally-phrased utterances. Until recently, this was uncharted territory.[1]

This volume focuses on the general issues raised by the new and growing interest in Phraseology.

Phraseology has been a neglected area of language description, at least in Western traditions of scholarship, for a long time. Among several possible reasons, I would like to mention two. One is that phraseology does not make a sharp division between grammar and lexis/semantics. It takes a holistic view, starting with phrases taken directly from text; since there is not a strong tradition of description of this kind, it allows, even encourages, a wide range of approaches, as the papers which follow testify. But there is a penalty for adopting a holistic strategy; there is no place for phraseology in the traditional apparatus of language analysis, so it is often just ignored.

Secondly, phraseology, in sharp contrast to most grammars, prioritises syntagmatic patterns over paradigmatic ones. Most grammars are paradigmatic, in that they see textual items as the realisation of choices; all the alternatives are rejected, but because they are set out as alternatives in the grammar, their absence defines the one chosen precisely. 'Positive' means 'not negative', and 'negative' means 'not positive', and that is that.

---

1. "The Count" is Count Dracula, quoted by Bram Stoker.

Syntagmatic patterns do not depend on possible alternatives; they are, to begin with, phrasal combinations on the surface of the text. In analysis they do not easily group themselves into neat small systems but form sets of related phrases, often large and sprawling. They make meaning by their combinations, by inclusion rather than exclusion.

Perhaps the main reason for the rapid increase in interest in phraseology is the present-day use of text corpora as the principal data-source for language analysis. Phraseology is the ideal point of contact between a corpus and a description, because it accepts surface phenomena, and this, initially, is what a corpus provides; no pre-processing is required, no abstractions, no information such as parts of speech added. This is made possible because corpora, especially large corpora, open up new lines of argument, based heavily on the perception of stretches of text as repeated events, and being able to compute frequencies and use statistical measures.

Early signs of the importance of syntagmatic patterns came with collocation, which came to the notice of a number of English teachers in Japan around 1930.[2] Twenty years later collocation was incorporated in Firth's 'spectrum of meaning', in his famous phrase "the company words keep" (Firth 1951).[3] This gave the impetus to early lexical work on corpora (Krishnamurthy et al. 2004), and to some interesting collections which were assembled in Eastern Europe.[4]

All these studies and collections shared the feature that they were organised grammatically; they were the contents of grammatical frames like 'adjective-noun collocations'. The lexical choices were not permitted to spill over into the cotext, although it is obvious that they do. There was no alternative model, and the dominance of grammar is clearly seen here, as it is still in most dictionaries. The tradition continues, as some of the following papers illustrate, while treatments based on lexical models are developing (e.g. Sinclair 1998).

These observations are from a fairly technical linguistic perspective, but this book spreads the net much wider, noting that phraseology is by no means the exclusive province of linguists; it attracts scholars in several disciplines, and the papers herein should cross-fertilise, pass on useful ideas and clarify differences.

In the first section, there are theoretical positions outlined from more than one perspective, and a number of tricky issues are tackled and clarifications offered. The matter of terminology is currently a problem because in such a recent area of development, with contributions from a range of disciplines, there is overlap and lack of fit,

---

2. Cowie (2004) gives a brief account of this work, and some references to longer treatments.

3. It is not clear whether or not Firth was aware of the earlier work, which did not come to full-scale publication.

4. e.g. Dzierżanowska and Kozłowska (1982), Perebeinos et al. (1986). The latter is remarkable, because if memory serves me it was assembled by hand by a group of language teachers who went through a great deal of text, noting frequencies, which are precisely recorded; the first collection in UK was Benson et al. also 1986.

as we saw a moment ago with collocation. It will be some time before this area settles down. 'Fixedness' is a notion that seems simple enough, but keeps cropping up because it has serious implications for studies such as idiom definition, the specification of lexical items, formulaic language and the 'lexical approach' in language teaching (Lewis 1993).

Fixedness is further dealt with in the second section, which concerns the units of analysis in phraseological studies, but opens up some wider issues as well. Some chapters are descriptive, using familiar units, but others invoke related linguistic concepts and disciplines.

The third section opens up language comparison from a phraseological point of view. At the word level we all know the 'false friends' between pairs of languages, traps for the unwary. We must expect similar quirks at the phrase level, and since that level is compounded by variation, there is a lot of tricky description ahead. As well as just misleading translation equivalence, there are all sorts of differences in scope, range, connotation and usage conventions, and the influence of the wider culture to take into account. The section contains two overview chapters: one on cross-linguistic studies and the other on cross-cultural issues. Then follow a series of papers dealing with language pairs which are fairly close culturally (e.g. English and French), or more exotic with respect to each other (e.g. English and Japanese). The remaining papers in Section 3 address issues of wider cultural scope, contrasting phrases that operate within cultural constraints to those that are more widespread.

The fourth section returns to more linguistic topics, lexicography and natural language processing (NLP), two areas which have had a head start in the retrieval of phrasal evidence from corpora. The section opens with an overview chapter for each topic. Although traditional lexicography was not very sophisticated in handling multi-word expressions, restricting the headwords as far as possible to single words, there is gradual movement towards a loosening up, despite the considerable problems that go with the development. Both monolingual and bilingual lexicography come under scrutiny. In natural language processing, automatic processes are developed for retrieval of recurrent patterns from corpora, usually large corpora. Statistical measurement is prominent in this work and the section includes a paper on the application of such measures to phrases, rather than single words – a process which incurs similar problems to those faced by the lexicographers. A number of other NLP approaches are illustrated in the other papers in this section.

It is clear from these remarks that many of the papers and all of the sections in this volume are springboards for stimulating further study, with guidelines, descriptive frameworks, and exemplars provided to support the exploration both of the wider issues raised by phraseology and the details of a range of relevant and popular methodologies.

One of the great strengths of a phraseological approach is the preservation of the integrity of text for much longer than alternative approaches to description, and in turn this entails the preservation of meaning. Eventually the analytic processes of abstraction, generalisation and formalisation will replace the text with an intralinguistic

representation, but the specific details of the text will be able to exert maximum effect on the description. The energy and novelty of the phraseological perspective on language is unmistakeable in this fine collection.

John Sinclair
The Tuscan Word Centre
March 2007

References

Benson, M., E. Benson & R. Ilson (1986). *Dictionary of English Word Combinations*. Amsterdam: John Benjamins.

Cowie, A. (2004). Phraseology – the Hornby legacy. In Williams, G. & S. Vessier (eds.) *Proceedings of the Eleveth EURALEX International Congress*, Vol. 1, 37–52. Université de Bretagne-Sud.

Dzierżanowska, H. & C. Kozłowska (1982). *Selected English Collocations*. Warsaw: Państwowe Wydawnictwo Naukowe.

Firth, J. (1951). Modes of Meaning. In *Essays and Studies, The English Association*. Reprinted in Firth, J. *Papers in Linguistics 1934–1951*, 190–215. Oxford: Oxford University Press.

Krishnamurthy, R, J. Sinclair, S. Jones & R. Daley (eds.) (2004 [1970]). *English Collocational Studies*. London: Continuum.

Lewis, M. (1993) *The Lexical Approach*, Hove: LTP.

Perebeinos, V., S. Hidekel, L. Weiner, H. Holkova, I. Keselman & I. Sloninskaya (1986). *A Deskbook of Most Frequent English Collocations*. Moscow: Prosveshenie.

Sinclair, J. (1998). The lexical item. In Weigand, E. (ed.) *Contrastive Lexical Semantics* [Current Issues in Linguistic Theory 17], 1–24. Amsterdam: John Benjamins.

Introduction

The many faces of phraseology

Sylviane Granger and Fanny Meunier

Phraseology is pervasive in all language fields and yet despite this fact – or perhaps precisely because of it – it has only relatively recently become established as a discipline in its own right. It is usually presented as a subfield of lexicology dealing with the study of word combinations rather than single words. These multi-word units (MWUs) are classified into a range of subtypes according to their degree of semantic non-compositionality, syntactic fixedness, lexical restrictions and institutionalization. Long regarded as a peripheral issue, phraseology is now taking centre stage in a wide range of fields, from natural language processing to foreign language teaching and now, 25 years after the publication of Pawley & Syder's (1983) seminal article, it is gradually acquiring the place it deserves in linguistic theory.

The impetus for this volume came from an **interdisciplinary** conference on phraseology entitled *Phraseology 2005. The Many Faces of Phraseology* organized in Louvain-la-Neuve in October 2005. The 170 participants, gathered from a wide variety of countries and specialist research areas, were there to scrutinize the field of phraseology from a wide range of perspectives. Three volumes emanated from the conference: a volume in French entitled *La phraséologie dans tous ses états* edited by Catherine Bolly, Jean René Klein and Béatrice Lamiroy (Cahiers de l'Institut de Linguistique de Louvain, Peeters, 2005), and two volumes in English, one entitled *Phraseology in Foreign Language Learning and Teaching* (Meunier, F. & S. Granger eds. 2008) entirely devoted to the role played by phraseology in L2 learning and teaching, and the current volume that purports to stress the multi-faceted nature of phraseology.

Our aim in organizing the conference was to take stock of the fast growing role of phraseology in a wide range of linguistic disciplines, a development that undoubtedly has a great deal to do with **corpus linguistics** research, which has both demonstrated the key role of phraseological expressions in language and also provided researchers with the automated methods of extraction and analysis they need to study them successfully, a development which has led to a major expansion of the field. Whereas previously phraseology had encompassed the study of only the most fixed and opaque

multi-word units, it now covers a much wider range of lexical units, many of which display a high degree of syntactic variability and semantic compositionality.

The current phraseological boom is evidenced by the numerous specialist publications and conferences on the subject. There are many niche areas of research buzzing with activity. It would seem however, that there is very little contact between these different areas of activity. Natural language processing researchers are often unfamiliar with work related to the typology of phraseological expressions. Researchers trying to draw up rigorous phraseological typologies are often equally unfamiliar with work being carried out in the automatic extraction of phraseological units. Similarly, there is very little contact between psycholinguistic researchers attempting to define the role of phraseology in language acquisition, comprehension and production and educational researchers aiming to give phraseology a higher profile in language teaching. In general terms, corpus linguistics studies describing phraseological expressions in large computer corpora are undeservedly little known. This lack of contact between different areas of phraseological research is problematic for two reasons: first, it means there is a very real chance of researchers 'reinventing the wheel'; second, and more importantly, it increases the likelihood of researchers coming up with erroneous data analyses. The aim of the conference was thus to enable researchers working in the field of phraseology to meet other researchers studying the same types of expressions from perhaps quite different perspectives. The current volume is meant to reflect this interdisciplinary dimension. Most of the chapters in the volume are based on presentations made at the conference. There are, however, a series of specially commissioned chapters that aim to give an overview of the different perspectives on phraseology.

The volume targets both would-be and experienced phraseologists and provides readers with a variety of ways into the field. For those who are intrigued by the general idea of phraseology without fully knowing what it encompasses, the volume provides a rich overview and introduces a wide range of methodological approaches. For those already experienced phraseologists, the volume is an invitation to look at the field with different eyes. Despite its wide scope, the volume does not claim exhaustivity, however. For a fully comprehensive view of phraseology, the reader is referred to other volumes (notably the other two volumes emanating from the conference and Burger et al.'s recent two-volume handbook of phraseology).

One particularly appealing aspect of the volume is the range of languages it covers: not only English, but also Dutch, French, German, Japanese, Russian and Spanish. This **plurilingual** perspective provides insights into the workings of phraseology in different languages and, at the same time, into a wide range of phraseological traditions: Anglo-Saxon, East European, French, German, but also Asian.

The volume opens with a preface to the volume by the late **John Sinclair**, who was a keynote speaker at the conference. John had too many other commitments to write a chapter but he kindly agreed to write the preface to the volume. Although he was already very unwell and would have had every reason not to write the preface, he sent us the first draft of his text on 7 March, exactly six days before he died. It must

have been one of the very last pieces he wrote and we have decided to publish it 'as is', despite the changes which inevitably took place during the editing phase.

The volume contains four major sections. Section 1, 'Phraseology: theory, typology and terminology' sets the scene: it delimits the field, introduces major categories of word combinations and the terms used to refer to them as well as highlighting the role played by phraseology in several linguistic theories. Section 2, 'Corpus-based analyses of phraseological units' turns theory into practice by means of a series of corpus-based case studies of different categories of word combinations. Section 3, 'Phraseology across languages and cultures', focuses on a crucial perspective in phraseology, viz. the contrastive perspective, cast in two different lights: purely linguistic and cultural. Section 4, as its title 'Phraseology in lexicography and natural language processing' suggests, brings together lexicographical and natural language processing perspectives, two perspectives which used to be separate but are now moving progressively closer. At the end of the volume, we have reproduced the extended abstract of John Sinclair's plenary presentation at the *Phraseology 2005* conference, whose title "The phrase, the whole phrase and nothing but the phrase" seems to sum up his legacy to linguistics quite perfectly.

The volume contains six overview chapters: Gries on phraseology and linguistic theory, Granger & Paquot on categorization and terminology, Colson on crosslinguistic approaches, Piirainen on cross-cultural issues, Moon on phraseology and lexicography and Heid on phraseology and natural language processing (NLP). These chapters provide an excellent starting point for researchers who are not particularly familiar with phraseological studies. The other chapters tackle more specific aspects of phraseology, particular theoretical approaches, methodologies, research frameworks or particular categories of word combinations. While each chapter has been classified into one of the four sections, it is worth noting that many are at the intersection of two or more sections. The volume is interdisciplinary, not only because it brings together studies from different disciplines but also, encouragingly, because several studies are intrinsically interdisciplinary.

The next section of this introduction briefly outlines each chapter and highlights some major trends emerging from the volume.

Section 1 starts with an overview chapter by **Stefan Gries**, which tackles phraseology from three major angles: definition, theory and identification. As regards the definition of phraseological units, Gries deplores the general lack of rigour and argues convincingly that the only way of ensuring the comparability of phraseological studies is to make the criteria used to define phraseological units maximally explicit. To this end, he proposes six parameters: nature and number of elements, frequency of occurrence, distance between elements, lexical and syntactic flexibility, semantic unity and non-compositionality. As regards theory, he demonstrates the growing role played by the notion of phraseologism, from TGG, where it has been on the fringe, to the more recent linguistic frameworks of cognitive linguistics and construction grammar and the methodological paradigm of corpus linguistics, where it is much more central. As to the identification of phraseologisms, the author is critical of the lack

of sophisticated statistical techniques and advocates a greater use of NLP techniques and more generally, increased dialogue between linguistic trends and disciplines. In the second chapter of the section, **Sylviane Granger & Magali Paquot** aim to disentangle the phraseological web. They first present the main traditions of phraseological studies and address the differences in the approaches adopted. The authors argue that the variations in scope are a result of the field's fuzzy borders with four neighbouring disciplines (semantics, morphology, syntax and discourse). They then introduce some influential typologies of phraseology used in English lexicology and give an overview of the categories of multi-word units used in data-driven phraseological studies. The final sections of the article offer suggestions for reconciling the different approaches and propose a most welcome clarification of the terminology. The following two chapters delve more deeply into two theoretical frameworks within which phraseological studies can be conducted. **Willy Martin** highlights the contribution of frame semantics to phraseology. He shows how a representation of the meaning of lexical items in terms of conceptual semantic frames can help distinguish between the different types of word combinations. Focusing more particularly on lexical collocations, he distinguishes between type-bound collocations like *koffie malen* (E. 'grind coffee'), token-bound collocations like *slappe koffie* (E. 'weak coffee') and in-between cases like *koffie drinken* (E. 'drink coffee'), which are essentially type-bound but intrude into the token-bound category because coffee is a prototypical drink. This qualitative frame-based approach is a useful way of interpreting the data extracted automatically from corpora and can therefore be viewed as a valuable complement to quantitative corpus-based approaches. **Marija Omazić** takes the standpoint of cognitive linguistics and establishes the role of two theories – metaphor theory and conceptual integration theory - in the processing of figurative phraseological units. The two theories are presented as complementary. The theory of metaphor accounts for the processing of a wide range of conventional phraseological units like *to burn with love* while the conceptual integration theory caters for modified units like *be born with a wooden* (instead of *silver*) *spoon in one's mouth.* A range of attested modified idioms are used to describe the factors involved in the interpretation of these idioms and the different stages of the 'unpacking process' that underlies the processing of idiom modifications. The final two chapters in the section focus on two major features of phraseological units: non-compositionality and fixedness. In Chapter 5 **Maria Helena Svensson** attempts to get to grips with the highly complex notion of non-compositionality, a recurrent criterion in phraseological studies, but nevertheless often ill-defined or simply taken for granted. To help clarify the notion, Svensson suggests breaking it down into four scaled dimensions: motivation/non motivation, transparency/opacity, analyzability/unanalyzability and literal/figurative meaning. These notions are closely related but not exact equivalents and therefore need to be clearly distinguished in phraseological studies. The interaction of a series of other notions such as prototypicality, salience and frequency is also discussed as well as related notions like encyclopedic non-compositionality. In conclusion, Svensson argues against using non-compositionality as an all-embracing term and suggests specifying which aspect of non-compositionality is intended. In

Chapter 6 **Gill Philip** tackles the notion of fixedness, a crucial notion which has long been considered as the defining feature of phraseological units together with non-compositionality. Recent corpus-based studies have shown, however, that beside their canonical forms, so-called 'fixed phrases' display a wide range of variants and that variation within phraseological units is the rule rather than the exception. The chapter focuses on multi-word units referred to as 'phraseological skeletons' which include collocational frameworks, lexico-grammatical frames and semi-prepackaged phrases. The challenge posed by these types of phrases is that they are very difficult to extract automatically as variants are largely unpredictable. In her chapter Philip states the case for phrase deconstruction and suggests ingenious search strategies for extracting variants of idiomatic phrases, thereby paving the way for further exploration of the role of creativity in phraseology.

Section 2 contains a number of case studies which differ in the types of word combinations they analyze but share the characteristic of being corpus-based. One of the interests of the section is that it illustrates a range of different corpus methodologies that can be used to identify and analyze phraseological units in corpora. The opening chapter by **Pierre J. L. Arnaud, Emmanuel Ferragne, Diana M. Lewis & François Maniez** analyzes the little explored field of incipient lexicalization of phraseological units on the basis of Adj + N sequences extracted from the British National Corpus (BNC). All Adj + N sequences containing a highly frequent central adjective are extracted and further categorized syntactically. Analysis of the structures points to varying degrees of linear fusion of certain sequences which manifest themselves as syntactic recategorization as Adj, N or Adv, loss of compositionality and loss of semantic transparency. Interestingly, no evidence of phonological coalescence was found, which suggests that syntactic and semantic shifts may be interdependent but phonological change may be independently motivated. In Chapter 8 **Kay Wikberg** makes use of the same corpus to throw light on the category of phrasal simile which figures in all typologies of multi-word units but had not previously featured in a large-scale corpus-based investigation. The concept of simile is defined and set apart from the neighbouring concepts of literal comparison and metaphor. Automatic extraction of four simile patterns from the BNC is followed by a time-consuming and complex procedure aimed at singling out figurative occurrences. The analysis of the selected similes highlights interesting differences between the patterns in terms of frequency, lexicalization and register. Figurativeness also lies at the heart of the following chapter by **Hans Lindquist & Magnus Levin** which offers a detailed analysis of the two frequent body part nouns *foot* and *mouth*. The analytic framework used is a combination of concepts from cognitive linguistics and methods from corpus linguistics. Here too, the corpus used is the BNC supplemented with British, American and Australian newspaper data. N-grams of different lengths are extracted and further scanned to single out the phrases that display linguistic integrity. The minute linguistic analysis to which the phrases are submitted highlights the rich phraseology displayed by the two nouns and draws attention in particular to the major role played by metonymy and metaphor in the creation and extension of new phrasal patterns. In Chapter 10 **Geoffrey C. Williams** introduces

the notion of collocational resonance, a notion that draws on his own work on collocational networks and Hoey's lexical priming. The result could be considered as a phraseological manifestation of the literary notion of intertextuality. Using a corpus-driven methodology based on the z-score co-occurrence statistical measure, Williams analyzes the effect of resonance of New Testament formulae with *God* as the central node and compares the collocational networks from those extracted from the works of Shakespeare and the BNC. The study highlights interesting changes from contextual to restricted collocations, a gradual shift to formulaic use and loss of religious resonance. The study opens new vistas in understanding the link between language and collective memory. In the last chapter of the section, **Margaret Maclagan, Boyd Davis & Ron Lunsford** offer an innovative perspective on a little researched field, that of the role of phraseology in pathological speech. They analyze a range of multi-word units in a corpus of speech of people with Alzheimer's disease on the basis of a pre-established list of multi-word units supplemented with phrases noted by the analysts. The study shows that multi-word units are pervasive in impaired speech and serve an important social-interactional function. Extenders like *things like that* or *all that sort of thing* help impaired speakers maintain the appearance of competence. As the corpus is longitudinal, the authors are able to trace the evolution in the use of these units, from perfect mastery to reduced pragmatic appropriateness and semantic range.

Section 3, devoted to cross-linguistic and cross-cultural approaches to phraseology, opens with two overview chapters. The first, by **Jean-Pierre Colson**, focuses on studies in cross-linguistic and contrastive phraseology. Colson criticizes the frequent lack of theoretical framework in many of the studies in the domain and argues that proper theoretical foundations could be put in place drawing from the fields of cognitive semantics and corpus linguistics. After a critical presentation of the strengths and weaknesses of those two approaches, the author calls for greater interaction between contrastive and translation studies, as the concept of phraseology is still notably absent from studies on translation theory and practice. Colson also stresses the need for more phraseologically-oriented cross-linguistic and contrastive research on non-European language families in order to assess the universality of phraseology. In the second overview chapter, **Elisabeth Piirainen** demonstrates the relevance of studying conventional figurative language to reveal its cultural content and analyses this connection between figurative language and culture from various angles. She presents a typology of the cultural elements underlying phrasemes (such as for instance textual dependence, pre-scientific conceptions of the world or aspects of material culture). She then highlights the fact that various types of phrasemes are unequally affected by aspects of culture and also illustrates some of the manifestations of culture in language starting either from source concepts (like taurine phraseology in Spanish) or from semantic fields. Finally, she tackles the disputed link between phraseology and the worldview of a language community, and also addresses issues in historical and etymological research. Five other chapters follow the overview chapters, two with a cultural focus and the other three with a linguistic contrastive one. **Annette Sabban**, in her chapter on the culture-bound nature of phraseology, starts with a discussion of terminological issues

related to the terms 'culture', 'culture-specific' and 'culturally-bound'. She then focuses on idioms as particularly relevant in examining the phenomenon and deals with some of the problems that may arise in the interpretation of phrasemes in terms of culture, including differences in cultural knowledge and in speakers' motivation of idiomatic meaning. Sabban also insists on the importance of distinguishing between concepts in language (i.e. results of modes of thinking which may no longer be relevant to the users of a language) and current concepts of thought (which may no longer coincide with concepts in language). In Chapter 15, **Elisabeth Piirainen**'s second contribution to the volume, the author examines phraseology in a European framework in the light of a cross-linguistic and cross-cultural research project on widespread idioms, i.e. idioms displaying similar lexical structure and figurative meaning in various languages, be they genetically and geographically close or distant. The chapter addresses one of the issues mentioned in her overview chapter, viz. the fact that classical phraseology too often associates a (national) language and its linguistic or cultural specificities with space. The author first criticizes the use of terms such as 'Europeanism' or even 'universalism' to refer to cross-linguistic similarities of idioms in only a few (often rather closely related) languages. She then presents the *Widespread Idioms* project which aims to systematically investigate the similarities among idioms in as many languages as possible, with a view to either corroborating or refuting the often claimed uniformity of European phraseologies based on a common European cultural heritage. Preliminary results are presented and possible explanations for widespread cross-linguistic similarities are put forward. As for the three contrastively-oriented chapters of Section 3, they all use corpus methodology, at least in part. In Chapter 16 **Christelle Cosme & Gaëtanelle Gilquin** present the results of a contrastive corpus analysis of the free and bound uses of the English preposition 'with' and its intuitive French equivalent 'avec'. Cosme and Gilquin stress the very low mutual translatability of the prepositions and partly attribute this lack of equivalence to polysemy and phraseology factors. The authors also propose concrete pedagogical applications of their findings, notably in bilingual lexicography and foreign language teaching methodology. In the next contrastive chapter, **Priscilla Ishida** analyses Japanese and English anger idioms. After an introduction on past approaches to the cross-linguistic analysis of idioms, she presents a 4-step method which focuses on both the L1 and L2 semantic networks and which is complemented by textual and discourse analysis. Ishida demonstrates that whilst full semantic correspondence is rare, many Japanese and English idioms overlap partially in meaning. The final part of the chapter also addresses semantic correspondence and the translation problems they pose. The last contrastive chapter, by **Olga Mudraya, Scott S. L. Piao, Paul Rayson, Serge Sharoff, Bogdan Babych & Laura Löfberg** deals with the automatic extraction of reliable lists of multilingual phraseological units. The authors focus on the translation equivalents of phrasal and light verbs in English and Russian and address a series of issues such as the different morpho-syntactic structures and the varying semantic properties. The use of the English Semantic Tagger developed at Lancaster University is illustrated and the results of several case studies are presented. Mudraya and her colleagues also stress the value of

cross-linguistic corpus-based studies for theoretical and applied studies in contrastive linguistics, multilingual lexicon extraction and language teaching.

Section 4 groups together lexicography and natural language processing (NLP). This combination seems rather natural firstly because the lexicon constitutes a key component of any NLP tool and, secondly, because the fast growing development of electronic dictionaries allows for the direct integration of NLP technology. As was the case in Section 3, the section opens with two overview chapters, the first by **Rosamund Moon** on dictionaries and collocation, and the second by **Ulrich Heid** on an overview of phraseology and NLP. **Moon**'s chapter starts with the analysis of the collocational behaviour of the three English words, *river, rivet* and *riven*, in large corpora. She then compares her results to the collocational representations of these words in monolingual dictionaries for natives and for learners, in bilingual dictionaries and in specialized dictionaries of collocations. Through her analysis, Moon provides the readers with a diachronic and synchronic perspective on the place of phraseology in dictionaries. She offers a balanced and critical approach to the challenges met by lexicographers and to the benefits and limits of corpus-based lexicography. **Heid**'s overview chapter on phraseology and NLP begins with a definition of 'computational phraseology' and a discussion of the value of the term multi-word expressions (MWEs), a term often preferred to phraseology in NLP circles. After a brief discussion of the role of phraseology in NLP applications such as machine translation or natural language generation, Heid addresses an impressive number of issues linked to multi-word expressions: issues of formal representation and annotation (lexical, morphosyntactic, syntactic and semantic idiosyncrasies and variation), aspects of lexicographical representation for human users, frequency and productivity issues, and finally, issues related to the (semi)-automatic identification and classification of MWEs. The problems of identification, classification and annotation of MWEs are further explored in **Cornelia Tschichold**'s contribution. She criticizes the lack of good computational tools for the study of the variability of MWEs in large corpora. After a section on data collection issues, she focuses on lexicographic work per se and on the creation of a lexicographic database. She describes the *WordManager-PhraseManager,* a language independent system that can help lexicographers and linguists to carry out a tailor-made annotation of the internal structure of MWEs. The system can capture the variability of MWEs and hence favours maximum reusability of the phraseological database. **Brigitte Orliac** also deals with the automatic extraction of specialized collocations and presents a method for extracting verb plus noun collocations in a specialized corpus of computer science texts. After a discussion of various extraction methodologies (where statistical methods usually precede linguistic analysis or filtering), she proposes a method based on the use of the lexical functions of the Meaning-Text theory put forward by Mel'čuk. The collocation extractor, *Colex*, combines linguistic annotation and statistical methods, with linguistic annotation taking chronological precedence. The statistical measures used by *Colex* in later stages of the analysis make it possible to distinguish between bound and free combinations. In the last NLP-oriented chapter of the volume **José-Manuel Pazos Bretaña & Antonio Pamies Bertrán** also analyse combined statistical

and grammatical criteria for the retrieval of phraseological units in corpora. They first assess the success of purely statistical methods and measures for the automatic extraction of phraseological units and show that whilst these methods can be effective and provide the user with lists of n-grams in decreasing order of frequency, the number of unexpected non-phraseological combinations is too high. To remedy this lack of precision in the results, the authors propose a preliminary three-step linguistic annotation of the data (lemmatization, grammatical filtering and part-of-speech tagging) and demonstrate that a combined approach (preliminary linguistic treatment followed by statistical analysis) yields much better results than a purely statistical one.

Although this volume contains a highly diverse range of articles, some key ideas emerge. First, as regards the scope of the field, the volume brings support to the wide view of phraseology. It recognizes the importance of a whole range of largely compositional multi-word units over and above the more traditional types. This wider view is in keeping with the corpus approach to phraseology which is very much in evidence in the volume. The corpora used to investigate phraseological units in the different chapters of the volume are extremely varied: monolingual vs. bilingual, written vs. spoken, large representative corpora like the British National Corpus vs. small locally-collected corpora, native vs. learner corpora, impaired vs non-impaired speech, etc. However, as many types of phraseological units are highly infrequent (cf. Moon 1998), corpus data are often complemented by other types of data, such as newspaper databases or the web. Second, the volume shows that multi-word units can be extracted using a wide range of methods and that the number and types of units extracted varies greatly according to the method used. The starting-point can be word forms, lemmas, syntactic structures or variable patterns. Alternatively, using a more corpus-driven approach, multi-word units can be extracted via the application of statistical tests, such as mutual information or the t-score or a combination of statistical tests and linguistic criteria. A third major thread running through the volume is that there is a limit to the information that can be derived from corpora and that more controlled data types like elicitation techniques may prove very useful. Surprisingly perhaps in a volume where corpora are so much in evidence, the word intuition or introspection is often used, even in the most NLP-oriented chapters. As pointed out by Lindquist (cf. Chapter 9), "corpus-drivenness has its limits" and recourse to intuition and subjective judgement is needed. This shows that that the two approaches to phraseology, the traditional approach and the quantitative approach, should not be viewed as conflicting but as complementary. Rather than replace older approaches by newer ones, one should examine how they can be made to cross-fertilize, with the more traditional phraseologists learning about interesting new techniques and new types of units, and proponents of the distributional approach learning to dissect the results of their queries in linguistically interesting ways.

John Sinclair has been a tremendous source of inspiration for us like for so many researchers around the world. It therefore seems quite natural to give him the last word by quoting the concluding words of an article he wrote 40 years ago but which still has a thoroughly modern resonance today: "[t]he theory of lexis opens up exciting areas

for describing language more accurately and more usefully. The practical problems are immense, and no secret has been made of them here, but the results that they promise are, possibly because of their novelty, no less fascinating than those of any other branch of linguistics" (Sinclair 1966: 429).

References

Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) (2007). *Phraseologie/Phraseology. Ein internationales Handbuch zeitgenössischer Forschung / An International Handbook of Contemporary Research*. Vols 1 & 2. Berlin: Mouton de Gruyter.

Moon, R. (1998). *Fixed Expressions and Idioms in English. A Corpus-based Approach*. Oxford: Clarendon Press.

Pawley, A. & F. Syder (1983). Two puzzles for linguistic theory: Native-like selection and native-like fluency. In J. Richards & R. Schmidt (Eds.), *Language and Communication*, 191–226. London: Longman.

Sinclair, J. McH. (1966). Beginning the Study of Lexis. In C. E. Bazell, J. C. Catford, M. A. K. Halliday & P. H. Robins (eds.) *In Memory of J. R. Firth*, 410–430. London: Longman.

SECTION I

Phraseology: Theory, typology and terminology

Phraseology and linguistic theory

A brief survey

Stefan Th. Gries

This chapter has three objectives. First, it argues in favor of more rigorous definitions of the term 'phraseologism' on the basis of six dimensions and exemplifies these dimensions for several different kinds of phraseologism. Second, it reviews the ways in which phraseologisms as defined here have figured in three different linguistic approaches: generative linguistics, cognitive linguistics, and corpus linguistics. Finally, it discusses some shortcomings in the identification of phraseologisms and points to relevant work to overcome these shortcomings.

1. Introduction

Interest in phraseology has grown considerably over the last twenty years or so. While the general linguists' view of phraseology before that time can probably be caricatured as 'idiom researchers and lexicographers classifying and researching various kinds of fairly frozen idiomatic expressions', this view has thankfully changed. Nowadays, the issues of identifying and classifying phraseologisms as well as integrating them into theoretical research and practical application has a much more profound influence on researchers and their agendas in many different sub-disciplines of linguistics as well as in language learning, acquisition, and teaching, natural language processing, etc. However, this influence is often not fully recognized or acknowledged, or reflected terminologically. This is undesirable, not only because it is often not easy to recognize the domains where research on phraseology has left its marks, but also because it renders the overlap of assumptions, concepts, and findings less transparent than is desirable.

This chapter attempts to take a modest step in this direction. I will try to identify and make explicit six crucial dimensions, or defining parameters, of phraseologisms. I think these actually underlie most phraseological work – if only implicitly – but I would like phraseologists to always be maximally explicit about which parameter settings are adopted in order to (i) render their definitions maximally precise and (ii) allow researchers from other frameworks to more easily recognize potential areas of overlap, or indeed conflict. In Section 3 I will then use the suggested parameters to discuss the role phraseologisms have played in different linguistic frameworks, viz. transformational-generative grammar, cognitive linguistics and

Construction Grammar, as well as corpus linguistics. While most scholars do not view corpus linguistics as a linguistic theory but rather as a methodology, it has given rise to many theoretical assumptions that, I believe, warrant its inclusion here. In Section 4, I will briefly, but critically, evaluate the methods practitioners in these three approaches have used to identify phraseologisms. Section 5 will conclude.

2. The notion of phraseology

While the notion of *phraseology* is very widespread, just as with other linguistic concepts, different authors have defined it differently, sometimes not providing a clear-cut definition, or conflating several terms that many scholars prefer to distinguish.[1] However, a closer comparative look at the vast majority of studies that exist allows us to identify a set of parameters that are typically implicated in phraseological research. I believe a rigorous definition of co-occurrence phenomena in general, and phraseology in particular, needs to take a stand regarding at least the following six parameters (cf. Howarth 1998: 25 for a similar critique of the absence of defining criteria and an alternative proposal).

i. the *nature* of the elements involved in a phraseologism;
ii. the *number* of elements involved in a phraseologism;
iii. the *number of times* an expression must be observed before it counts as a phraseologism;
iv. the permissible *distance* between the elements involved in a phraseologism;
v. the degree of *lexical and syntactic flexibility* of the elements involved;
vi. the role that *semantic unity* and *semantic non-compositionality / non-predictability* play in the definition.[2]

---

1. A case in point is Stubbs (2001). According to the index, the term *phraseology* and the cross-referenced *extended lexical unit* are first mentioned on p. 59 and p. 31f. respectively. However, no explicit definition of *phraseology* is provided on these pages nor on the page where *phraseology* is first mentioned (p. 24). Another example is Hunston (2002: 137f.), who first discusses "some examples" she categorizes as "[c]ollocation", "[p]hrases and variation", "[t]he tendency of certain verbs to occur in the passive rather than the active, or in the negative rather than the positive" (i.e., what is usually referred to as colligation), and "[t]he occurrence of complementation patterns", but then merely states that "[t]hese and the other consequences of sequence preference together might be called 'phraseology'". As will become apparent below, I largely agree with Hunston's inclusion of these examples as phraseologisms, but the definition as such is not as explicitly delineated as it could be and leaves much to inference processes on the part of the reader.

2. Additional or alternative criteria one might wish to invoke are a possible separation of lexical flexibility and syntactic flexibility (or commutability/substitutability) of the elements involved in potential phraseologisms and/or the distinction between encoding and decoding idioms.

As to the first criterion, the definition of a phraseologism I will adopt is among the broadest conceivable. I consider a phraseologism to be the co-occurrence of a form or a lemma of a lexical item and any other kind of linguistic element, which can be, for example,

– another (form of a) lexical item (*kith and kin* is a very frequently cited example of a nearly deterministic co-occurrence of two lexical items, as is *strong tea*);
– a grammatical pattern (as opposed to, say, a grammatical relation), i.e. when a particular lexical item tends to occur in/co-occur with a particular grammatical construction (the fact that the verb *hem* is mostly used in the passive is a frequently cited case in point).

Note that this definition does not distinguish between lexical items and grammatical patterns that co-occur with a lexical item. Also, note that the definition does not commit to a particular level of granularity regarding the lexical elements involved: both can involve either all forms of a lemma or just particular morphological forms (cf. Rice & Newman 2005 and Gries, to appear, for conflicting points of view on this matter).

As to the second criterion, it is important to decide whether, for example, phraseologisms can consist of only two elements (such as word pairs) or can include a larger number of elements. I will assume that phraseologisms can contain more than two elements (as in, say, *to eke out a living*, which contains a verb slot filled with some form of *to eke*, a direct object slot filled with DET *living*, and a slot for the particle *out* complementing the transitive phrasal verb).

As to the third criterion, it is probably fair to say that there is little work which has defined phraseologisms solely on the basis of some quantitative criterion based on their frequency of occurrence (and/or additional frequency information). True, some scholars have used a threshold of absolute frequency of occurrence (usually defined arbitrarily or not at all; cf. Hunston & Francis 2000: 37, for example). Others, most notably British and Scandinavian scholars from the Sinclairian/Cobuild tradition, have argued that observed frequencies must exceed frequencies expected on the basis of chance (significantly or just at all), but most previous work has restricted itself to reporting frequencies or percentages of occurrence of phraseologisms. In order to avoid an inflation of what could be considered phraseological, I will consider an expression a phraseologism if its observed frequency of occurrence is larger than its expected one.

As to the fourth criterion, some work (especially *n*-gram-based studies in natural language processing) concerns itself only with immediately adjacent elements, but I will adopt the more widespread broader perspective which also recognizes discontinuous phraseologisms.

As to the fifth criterion, studies that are only concerned with completely inflexible patterns such as the standardly quoted example of *by and large* can be distinguished from studies that include relatively flexible patterns such as *kick the bucket* (which allows different tenses but no passivization), studies (also) involving partially lexically-filled patterns such as the *into*-causative ([$_{VP}$ V DO *into* V-*ing*]), and finally studies (also) including completely lexically unspecified and thus maximally flexible

expressions, such as the English ditransitive pattern [$_{VP}$ V OBJ_1 OBJ_2] (cf. Section 3.2 for references). My definition of phraseologisms excludes only the last of these because they do not involve at least one lexically specified element (as required by the first parameter).

As to the final, and for many researchers probably most important, criterion, the elements of a phraseologism – however they are distributed across a clause or sentence – are usually assumed to function as a semantic unit, i.e. to have a sense just like a single morpheme or word. However, one can distinguish between studies in which the sense of a phraseologism is by definition non-compositional (cf. Fraser's 1976: v definition of an idiom as "a single constituent or series of constituents, whose semantic interpretation is independent of the formatives which compose it") from studies where non-compositional semantics is not a necessary condition for phraseologisms (cf. Nunberg, Sag, & Wasow 1994: 499ff. as well as Wulff to appear and below for further discussion). For something to count as a phraseologism, I will require semantic unity, but not non-compositional semantics.

In sum, a phraseologism is defined as the co-occurrence of a form or a lemma of a lexical item and one or more additional linguistic elements of various kinds which functions as one semantic unit in a clause or sentence and whose frequency of co-occurrence is larger than expected on the basis of chance.

While this definition is maximally explicit with respect to the above-mentioned parameters, it also follows that, as in Hunston's (2002) approach, the range of phenomena regarded as phraseologisms is very large. An example from the inflexible end of the continuum of phraseologisms is the largely fixed expression *to run amok,* which can be analyzed with respect to the six above criteria as follows:

– nature of the elements: words;
– number of elements: two;
– frequency of occurrence: the two parts of the expression co-occur more often than expected by chance: in the British National Corpus World Edition (BNC WE), any form of *to run* and *amok* occur in 38,088 and 43 of all 6,051,206 sentence units (lines beginning with "<s n=") respectively; thus, one would expect 0.27 within-sentence unit co-occurrences, but one actually obtains 40;[3]

3. All corpus data have been retrieved with regular expression searches performed with R for Windows 2.4 (cf. R Development Core Team 2006). The data discussed here are based on a retrieval of all case-insensitive matches for ">amok\\W" and "<w (VV.|VV.-...)> (ran|run(s|ning)?)\\W" in all lines with sentence units from the BNC WE. This, like the additional corpus data reported below, is of course only an approximation because it is only most, but not all within-sentence-unit co-occurrences that instantiate the construction in question. The expected frequency has been computed as is customary in nearly all measures of collocational strength or chi-square tests by multiplying the observed marginal totals of the two individual items in question and dividing by the corpus size. One might suspect that the frequency of co-occurrence criterion could be problematic for cases where the potential phraseologism involves one or more high-frequency items such as *to break the ice*. However, even in

– distance of elements: the two parts of the phraseologism usually co-occur adjacently (in all but one case, where *dangerously* intervenes);
– flexibility of the elements: *to run* can occur in various morphological forms, but *amok* can apparently not be preposed (judging from the hits in the BNC WE, that is);
– semantics: *to run amok* functions as one semantic unit, meaning roughly 'to behave violently and uncontrollably'.

Another group of examples are transitive phrasal verbs such as *to pick up*, *to give up*, or the concrete example of *to eke out* (usually *a living* or *an existence*), etc.:

– nature of the elements: words and phrases in a transitive phrasal verb frame (the direct object can be an NP or a clause);
– number of elements: two lexical elements and one syntactic slot to be filled;
– frequency of occurrence: in the BNC WE, any form of the verb *to eke* ("<w (VV.|VV.-...)>ek(e[sd]?|ing)\\W") and *out* tagged as an adverbial particle ("<w AVP>out\\W") occur in 78 and 140,975 of all 6,051,206 sentence units respectively (with a case-insensitive search), which is why one would expect 1.8 co-occurrences, but one actually obtains 70;
– distance of elements: the verb and the particle can occur right next to each other or with intervening material (the maximum length of direct objects in verb-particle constructions in general reported by Gries 2003 is 21);
– flexibility of the elements: the verb, the direct object, and the particle allow for constituent order variation in that they need not be adjacent, allow passivization, ...;
– semantics: transitive phrasal verbs function as one semantic unit, which is evidenced by (i) the well-known fact that many have a one-word near synonym (*to pick up*: *to lift/elevate*; *to give back*: *to return*, *to put down*: *to deposit*) and (ii) by the fact that many have non-compositional readings (or even a compositional and a non-compositional reading such as *to hold up* or *to throw up)*.

A final group of examples are patterns which (i) are lexically partially filled, (ii) require the insertion of additional lexical material, and (iii) allow for syntactic variation, such

such cases the number of observed co-occurrences exceeds the expected value, as can be seen by retrieving all case-insensitive matches for "<w (VV.|VV.-...)>(break(s|ing)?)|(broken?)\\W" and "<w (N..|N..-...)>ice\\W" in all sentence unit lines of the BNC WE. These searches yielded 22,256 and 4,392 matches respectively, so 16.2 co-occurrences would be expected, but in fact 125 sentence units with both search strings were observed, approximately half of which instantiated the idiom in question.

as the comparative clause construction (i.e., [$_{XP}$ *the* Adj$_{comparative}$, *the* Adj$_{comparative}$]) or the *into*-causative (i.e., [$_{VP}$ V DO *into* V*ing*]).[4]

Given the broad definition proposed above, it might seem as if now everything in language is phraseological and *phraseologism* is a futile catch-all term devoid of empirical content and unworthy of empirical study. However, this is not the case. On the one hand, the definition does not include highly frequent co-occurring expressions such as *of the* or *in the*, because these do not function as a semantic unit. Also, the definition does not include completely lexically unspecified patterns such as those that Construction Grammarians refer to as argument structure constructions (cf. below Section 3.2; examples include the ditransitive construction [$_{VP}$ V OBJ$_1$ OBJ$_2$] or the caused-motion construction [$_{VP}$ V DO OBL]), which bridge the gap to the patterns posited in Pattern Grammar. As such and in other words, the definition of phraseologism proposed above serves as a convenient cover term for co-occurrence phenomena at the *syntax-lexis interface* (since at least one lexical element must be specified) as opposed to the *syntax-semantics interface*, to which lexically unspecified patterns from Pattern Grammar or argument structure constructions from Construction Grammar would be associated. On the other hand, the present definition does cover particular words' significant attraction to argument structure constructions (cf. Stefanowitsch & Gries's (2003) collostructions) and completely lexically filled and frozen expressions which, although diachronically derived from collocations, are synchronically single lexemes (e.g. *of course*, *at least*).

Not all researchers would of course subscribe to the parameter settings I have proposed and/or would prefer to exclude some of these and/or include additional ones (see Note 2). If, for example, non-compositionality were taken as a necessary condition for something to count as a phraseologism, many highly frequent but fully compositional elements would no longer count as phraseologisms anymore. Similarly, if the requirement for at least one specified lexical element were dropped, argument structure constructions would belong to the realm of phraseologisms.

However, phraseologisms as defined above are worthy of empirical study because the present definition does not single out any particular level of granularity at which co-occurrences, and thus phraseologisms, may be observed. This has two interesting consequences. First, it means that phraseologists must carefully define the linguistic level(s) at which they observe a potential phraseologism. In the case of the phrasal verb *to eke out a living*, for example, one could recognize at least the following co-occurrences as potential phraseologisms:

– *to* [$_{VP}$ *eke out a living*];
– *to* [$_{VP}$ *eke out* DO];
– *to* [$_{VP}$ V *out* DO];

4. Given particular lexical material and some syntactic structure, such phraseologisms may well develop into completely frozen units, as exemplified by the proverbial instance of the comparative clause construction *the more, the merrier*.

– *to* [$_{VP}$ V Particle DO]; or even
– *to* [$_{VP}$ V Particle DO_{NP}] (while the DO does not have to be an NP, it usually is and one may want to include this probabilistic information); ...[5]

The crucial question is to decide which level of resolution to focus on, an issue that will sometimes be decided on the basis of a particular researcher's interest but which can also be decided purely quantitatively by, say, measuring the level of granularity at which the attraction between the elements involved is highest.

As another example, if one retrieved from a corpus many instances of the ditransitive pattern [$_{VP}$ V OBJ_1 OBJ_2] and inspected the verbs occurring in them, one could draw many different probabilistic conclusions about co-occurrence preferences. One could concentrate on

– the strong positive correlation between the verb form *gave* and the ditransitive;
– the strong positive correlation between the verb lemma GIVE and the ditransitive (as in collexeme analysis; cf. Stefanowitsch & Gries 2003);
– the strong positive correlation between the semantic class of transfer verbs and the ditransitive;
– etc.

However, not all of these are theoretically revealing or relevant (cf. Gries 2006b & to appear for discussion and exemplification of differences between word-form specific and lemma-specific results as well as differences between speaking and writing). On the whole, I think it is fair to say that there is as yet little empirically rigorous work on this issue.

Phraseologists must also decide how many elements a phraseologism is supposed to comprise. The potential phraseologisms listed above, derived from the example of *to eke out a living*, all involved three elements, but on occasion this may not be the most revealing analysis. Similarly, if semantic unity were not required for something to count as a phraseologism, one could posit that *in spite* is a phraseologism: it involves two words (number and nature of elements) that co-occur more often than expected by chance,[6] are adjacent and inflexible. However, it is obvious that a more reasonable assumption would be that the 'real' phraseologism is *in spite of*, which is what statistically more sophisticated approaches would recognize (cf. Mason's work on lexical gravity and Kita et al.'s cost criterion mentioned below in Section 4.3).

---

5. Of course, not all these examples qualify as a phraseologism according to my definition (some are not a single semantic unit and the last two do not involve at least one specific lexical item). However, they may be phraseologisms according to other scholars' definitions.

6. This claim is based on retrievals of all case-insensitive matches for "<w PRP>in\\W", "<w PRP>in spite\\W", and "\\Wspite\\W" in all lines of corpus files from the BNC WE that begin with "^<s n=", yielding 1,361,163, 2,683, and 2,897 matches respectively. In other words, nearly all occurrences of *spite* are actually instances of *in spite*.

Thus, even though the above definition may seem overly powerful at first, it still delimits the possible space of phraseologisms effectively and leaves room for many issues to be discussed. The central point to be made here, however, is that phraseologists should formulate more definitions of this kind. By this I do not mean that phraseologists should necessarily adopt *my* definition – but that it is essential that we, who are interested in something as flexible as patterns of co-occurrence, always make our choice of parameter settings maximally explicit to facilitate both the understanding and communication of our work. Using the above six parameters, the following section will explore how phraseologisms figure – sometimes rather implicitly – in different linguistic approaches.

3. The role of phraseology in linguistic theory

The role phraseology has played in linguistic theory is quite varied. On the one hand, it is varied because theoretical frameworks or approaches in linguistics differ widely in terms of the importance attached to phraseologisms. On the other hand, the importance that phraseology can play in a framework also crucially depends, of course, on how phraseologisms are defined, which is why I devoted so much space to the question of definition in Section 2. Space does not allow for a comprehensive comparison of the role of phraseology in many different frameworks so I have to be selective. Section 3.1 looks at transformational-generative linguistics. Section 3.2 discusses phraseology in cognitive linguistics and Construction Grammar while Section 3.3 is concerned with phraseology in corpus linguistics.

3.1 Generative linguistics

It is probably fair to say that phraseology has generally played a rather limited role in the development of the various versions of generative grammar. Given a conception of the linguistic system which crucially involves only

- a grammar, i.e. a set of algorithmic rules that combines linguistic elements only with respect to their structural characteristics and irrespective of their meaning; and
- a lexicon, i.e. a repository of all non-compositional irregularities that must be rote-learned;

it comes as no surprise that, of the above six parameters, the only one which plays a role for generative linguistics is the last one, semantic unity and non-compositionality. In this conception, an expression such as *to bite the dust* is recognized as an idiom, a non-compositional semantic unit as defined by Fraser (1976: v), and is thus stored with its syntactic characteristics as a separate item in the lexicon. Note also that this conception of the linguistic system is somewhat at odds with my definition of phraseologisms which does not treat grammatical and lexical elements as different in kind.

This generative conception of phraseologisms comes with a few problems. On the one hand, it is much more difficult to draw a strict dividing line between what is idiomatic and what is not than one may initially assume; cf. Fraser (1966: 59, n. 3) for the difficulty of obtaining unanimous judgments as well as Cowie & Mackin (1993: ix) and Gibbs (1994: Ch. 5–6) for discussion. On the other hand, research has shown that phraseologisms/idioms vary considerably in terms of the syntactic operations they allow for (cf. the seminal paper by Nunberg et al 1994 for discussion), and since not all of these can be explained away by straightforward performance factors, one would have to postulate that the lexicon contains for each putative unit a list of what operations are licensed, an option that is particularly unattractive for an approach that otherwise eschews redundant representation.

It is only in more recent developments of the generative framework that the importance of phraseologisms has come to be recognized more openly. For example, Culicover (1999) insightfully discusses a variety of patterns that are usually classified as phraseologisms (examples include *had better*, *not*-topics, etc.) and points out that they pose serious challenges to a modular organization of language in terms of an algorithmic grammar and a lexicon because they appear to cut across this supposedly well-established boundary. A similar tack is taken in some recent work by Jackendoff. To name but one example, Jackendoff (1997) is concerned with a phraseological expression – the 'time' *away* construction exemplified by *We're twistin' the night away*, which, given its properties with respect to the above parameters, would certainly be recognized as a phraseologism by most phraseologists:

- nature of the elements: words and phrases in a transitive phrasal verb frame;
- number of elements: three: the V-slot must be filled with an intransitive verb; the DO slot must be filled with a time expression; the particle is *away*;
- frequency of occurrence: In the BNC WE, *night/nights* tagged as nouns ("<w NN[12]>night(s)?\\W") and *away* tagged as an adverbial particle ("<w AV0>away \\W") occur in 36,265, and 34,343 of all 6,051,206 sentence units respectively, which is why one would expect 206 co-occurrences, but one actually obtains 512;[7]
- distance of elements: the intransitive verb, the direct object, and *away* occur right next to each other;
- flexibility of the elements: just like regular transitive phrasal verbs, the intransitive verb, the direct object, and the particle can occur in the order [$_{VP}$ V DO Particle] or in the order [$_{VP}$ V Particle DO]; passivization and tough movement are possible, but rare;
- semantics: the pattern of transitive phrasal verbs with time expressions as direct object and *away* functions as a semantic unit, as is evidenced by the fact that this pattern forces a particular interpretation of the clause such that the referent of the

7. Again, the resulting frequencies need of course not all be instances of the 'time' *away* construction but only serve to make the point that the observed co-occurrence frequency of *night* and *away* most likely exceeds chance levels considerably.

> subject is understood to act volitionally; the verb must denote an activity, not a state, and the referent of the subject uses up the whole time denoted by the time expression (cf. Jackendoff 1997: 534–7).

I am not aware that this has been recognized or even openly acknowledged by transformational-generative grammarians, but it is interesting to note that the notion of phraseologism, which has been rather on the fringe of transformational-generative grammar in particular and most of theoretical linguistics in general, is so crucial to the revision of the most dominant linguistic paradigm of the 20th century and, thus, to the way the linguistic system proper is viewed. More specifically, it is, among other things of course, the recognition of phraseologisms as theoretically relevant entities in their own right that begins (i) to undermine the modular organization of the linguistic system into a grammar and a lexicon and (ii) to make linguists aware of the way in which the analysis of phraseologisms in performance data reveals many subtle interdependencies on different levels of linguistic analysis. While this interpretation may be controversial, I believe it is supported by the fact that the next two frameworks or approaches to be discussed – cognitive linguistics / Construction Grammar and corpus linguistics – also rely heavily on the notion of phraseologism as I have defined it above, even though the term *phraseologism* is not always used. These parallels will be outlined in more detail in the following two sections.

3.2 Cognitive linguistics and Construction Grammar

As mentioned above, the discussions by Culicover and Jackendoff of what we have been referring to as phraseologisms have not made use of this term. However, the way their analyses are phrased makes the connections not only to the notion of phraseologism, but also to other theoretically related concepts rather obvious. Two related theoretical frameworks whose practitioners are currently very much concerned with phraseologisms are cognitive linguistics and Construction Grammar.

Cognitive linguistics as such is not so much a single theory as a set of related approaches that share several fundamental assumptions which set it apart from other competing frameworks. The same is true of Construction Grammar, where one may distinguish at least between, say, the version of Construction Grammar by Goldberg (1995, 2006), that of the Berkeley school (cf. the references to the works by Fillmore and Kay below), Croft's (2001) Radical Construction Grammar, and maybe others. My discussion of cognitive linguistics and Construction Grammar cannot encompass all the different approaches. Instead, for cognitive linguistics, I will focus on what I consider the most thoroughly developed approach, namely Langacker's Cognitive Grammar as outlined in Langacker (1987, 1991); my discussion of Construction Grammar will focus on Goldberg's version. As will become more apparent below, these two theories' equivalents of the notion of *phraseologism* are very similar, but until recently differed with respect to one of the above defining parameters of phraseologisms, viz. non-compositionality.

Cognitive Grammar as a discipline does not really have a theoretical notion that is a precise equivalent of *phraseologism*. Rather, it has a more general term, of which phraseologisms constitute a subset. As I did above, Cognitive Grammar does away with a strict separation between lexicon and grammar. The only kinds of element the linguistic system is said to contain are symbolic units. A unit is defined as

> a structure that a speaker has mastered quite thoroughly, to the extent that he can employ it in largely automatic fashion, without having to focus his attention specifically on its individual parts for their arrangement [...] he has no need to reflect on how to put it together. (Langacker 1987: 57)

A symbolic unit in turn is a pairing of a form and a meaning/function, i.e. a conventionalized association of a phonological pole (i.e., a phonological structure) and a semantic/conceptual pole (i.e., a semantic/conceptual structure). The more often a speaker/hearer encounters a particular symbolic unit, the more entrenched this symbolic unit becomes in his or her linguistic system and the more automatically the unit is accessed. Thus, unit status correlates positively with a speaker/hearer not analyzing the internal structure of a unit. Crucially for our present purposes, the notion of symbolic unit is not restricted to morphemes or words, but comprises more abstract grammatical patterns such as transitive constructions, reference-point constructions (such as the *s*-genitive in English), idioms, etc. Using my defining parameters of a phraseologism, a symbolic unit can be defined as follows:

- nature of the elements: no restrictions as long as the forms in the expression are paired with some meaning;
- number of elements: no restrictions;
- frequency of occurrence: a symbolic unit must have occurred frequently enough for it to be entrenched in a speaker/hearer's linguistic system (I am, however, not aware of any rigorous operationalization of a sufficient frequency threshold);
- distance of elements: no restrictions as long as the speaker/hearer categorizes the parts as making up one symbolic unit;
- flexibility of the elements: no restrictions as long as the speaker/hearer can form one or more generalizations (a schema in Langacker's parlance) which sanction the concrete instances; for example, if a speaker recognizes that two expressions instantiate transitive constructions, it is unimportant that the two instances may contain different verbs in different tenses with different and/or differently long objects etc.;
- semantics: by definition, the symbolic unit must have a semantic pole or meaning/function, but non-compositionality is not required.

This definition is of course not only Langacker's; other scholars such as Bybee (1985) also subscribe to this kind of definition. This definition of a symbolic unit is nearly identical to that of a phraseologism given above: it is only somewhat broader, including as it does simple words/morphemes and also lexically unspecified patterns. However, given this definition, phraseologisms do not enjoy a special status within

Cognitive Grammar: they are just one kind of symbolic unit, requiring the same descriptive apparatus as the more specific categories of morphemes or words or the more general categories of argument structure constructions or clause patterns. In terms of what they consider the central units of analysis, Cognitive Grammar and phraseology research are, thus, nearly maximally compatible.

As will become obvious below, there is a similar degree of compatibility between Construction Grammar and phraseological research. Given the theoretical affinity of Cognitive Grammar and Construction Grammar and the parallel evolution of the two theories, this should not come as a big surprise, and the main difference between how Cognitive Grammar and Construction Grammar define their objects of study as compared to phraseological research is largely terminological. The central linguistic unit of Construction Grammar – the analogon to symbolic units in Cognitive Grammar – is the construction. A construction in the sense of Goldberg's (1995) Construction Grammar is defined as follows:

> C is a construction iff$_{def}$ C is a form-meaning pair <F_i, S_i> such that some aspect of F_i or some aspect of S_i is not strictly predictable from C's component parts or from other previously established constructions. (Goldberg 1995: 4)

The only major difference between this definition and those of symbolic units and phraseologisms discussed above is that a construction as defined here requires non-compositionality or, in Goldberg's terminology, non-predictability while this was not required of symbolic units or phraseologisms.[8] This difference certainly has implications for the nature of the linguistic system postulated but is not a major qualitative difference. Put differently, *symbolic unit* is a general notion, *construction* as defined above is slightly more specific by requiring one non-predictable aspect, and *phraseologism* as defined here is also more specific by not requiring non-predictability, but at least one lexically specified element. It remains obvious, though, that there is again a high degree of compatibility between phraseological research and construction grammarians. In a way, this is not even surprising given that it was prime examples of phraseologisms whose analysis 'gave rise' to Construction Grammar in the first place. As a particularly obvious example, some of the earlier publications that are now understood to have laid the foundations of at least one of the schools of Construction Grammar, viz. the Fillmore-Kay kind of Construction Grammar, took as their starting point elements that would uncontroversially be considered phraseologisms by most scholars; cf. Fillmore et al (1988) on *let alone* or Kay & Fillmore (1999) on the *What's X doing Y?* construction.

Finally, there is another aspect of both Cognitive Grammar and Construction Grammar that is worth pointing out here and will become more relevant below, viz.

---

8. In her most recent work, Goldberg has actually revised her approach such that now non-compositionality is not required anymore provided that the frequency of an expression is large enough for it to become entrenched and, thus, attain construction status (cf. Goldberg 2006: 5). This move renders constructions and symbolic units even more similar.

the importance both theories attach to actual frequencies of usage or occurrence. As mentioned above, Langacker's Cognitive Grammar is explicitly usage-based in the sense that (i) exposure to, and use of, symbolic units, i.e. performance, is assumed to shape the linguistic system of speakers and hearers and (ii) sufficient frequency of occurrence is a necessary condition for entrenchment and, in turn, unit status of a linguistic expression. In this respect, Goldberg's (2006) approach does not differ from Langacker's approach, and while non-compositionality was an additional necessary condition for constructionhood in Goldberg's (1995) Construction Grammar, sufficient frequency was of course also a necessary condition for construction status. Thus, many construction grammarians have made extensive use of the frequency distribution and behavior of constructions in authentic language data (i.e., natural language corpora) in the theoretical literature (cf., e.g., Brenier & Michaelis 2005) and in other domains such as first language acquisition (cf., e.g., Tomasello 2003), language change (cf., e.g., Israel 1996), etc.

By way of an interim summary, we can conclude that, unlike the transformational-generative paradigm, both Cognitive Grammar and Construction Grammar are highly compatible with phraseological research. True, terminologies differ and definitions are not completely identical, but it is easy to see that phraseologisms do not just have a marginal status in these two theories but are rather at the core of what they consider to be their fundamental entities. From this, it of course also follows in turn that phraseological research has a lot to offer to these theories in terms of descriptive work and exploration of the ontological status of phraseological elements. In the opposite direction, phraseological research can benefit from the elaborate theoretical apparatus and the cognitively plausible background provided by Cognitive Grammar and Construction Grammar.

The next section will be concerned with the approach that is probably most intimately connected to phraseological research, viz. corpus linguistics, and we shall see that there is again a high degree of both theoretical and practical overlap, testifying even more to the relevance of phraseological research.

3.3 Corpus linguistics

While the two previous sections have been concerned with different linguistic theories (from the opposite ends of virtually all conceivable dimensions), this section will look at the relation of phraseologisms to a methodological paradigm, that of corpus linguistics.

While much of 20th century linguistics was characterized by the strong methodological predominance of acceptability/grammaticality judgments, corpus linguistics as a method has constantly increased in importance in most fields of linguistics, and to my mind at least is currently the single most frequently used method employed in the study of phraseology. This predominance of corpus-linguistic methods within phraseological research is of course not accidental. Corpora as such can only provide

frequency information – frequencies of occurrence and frequencies of co-occurrence.[9] From this, it is a relatively small conceptual leap to the above definition of phraseologism as a co-occurrence phenomenon. As a matter of fact, some of the most central notions in corpus linguistics can be straightforwardly compared to phraseologisms on the basis of the six criteria discussed above. The terms *word clusters*/n-*grams* and *collocations*, for example, refer to frequent co-occurrences of this kind:

- nature of the elements: words;
- number of elements: *n* (usually, that means 'two or more');
- frequency of occurrence: sufficiently frequent to be recognized as a combined element;
- distance of elements: for clusters/*n*-grams, the distance is usually 0 (i.e., the elements are immediately adjacent); for collocations, the distance between the elements involved can vary, but usually exhibits one or a few preferred distances;
- flexibility of the elements: for clusters/*n*-grams, there is usually no flexibility; for collocations, one usually allows for some flexibility: the collocation of *strong* and *tea* would be instantiated both by *strong tea* and *the tea is strong;*
- semantics: *n*-grams are usually retrieved for natural language processing purposes where the issue of non-compositional semantics is only sometimes relevant; for collocations, researchers differ as to whether they require some non-predictable behavior (*strong tea* is acceptable but *powerful tea* is not) or not.

Similarly, the notion of *colligation* is not nowadays usually used as it was originally defined by Firth (1968: 182) – as the co-occurrence of grammatical patterns – but also as a particular kind of phraseologism, namely one in which one or more words habitually co-occur with a grammatical pattern (cf. the example of *to hem*'s preference for passives mentioned at the beginning of this chapter).[10] From these brief remarks about the nature and the number of elements involved, it is clear that much work in corpus linguistics cuts across the boundary of syntax and lexis upheld in formal approaches

9. By this I of course do not mean that corpus linguistics only provides frequency lists – quite the contrary. What I mean is that whatever information is gleaned from corpora is ultimately based on frequencies. Put differently, a corpus search does not output meaning, or pragmatic intention – it only yields frequency lists, concordances, or collocates, i.e., elements and their frequencies of occurrences or frequencies of co-occurrence of elements of various degrees of abstraction. All linguistic inferences in, say, the domains of morphology, syntax, semantics etc. are thus dependent on statistical information of some kind.

10. Recently, Hoey (2004) has extended the notion of colligation to also include the co-occurrence of words with, say, particular grammatical slots (e.g., subject, object, complement etc.) and positions within sentences or paragraphs. Such co-occurrences would not qualify as phraseologisms as defined here.

to language,[11] and that there is a considerable overlap between the assumptions made by cognitive linguists, phraseologists, and, as we now see, also corpus linguists.

Another central notion in contemporary corpus linguistics, the *pattern*, involves additional parameters of the above set, viz. the parameter of non-compositionality/non-predictability. This is the definition of a pattern according to Hunston & Francis (2000: 37):

> The patterns of a word can be defined as all the words and structures which are regularly associated with the word and contribute to its meaning. A pattern can be identified if a combination of words occurs relatively frequently, if it is dependent on a particular word choice, and if there is a clear meaning associated with it.

An expression that would therefore not count as a pattern according to this definition is the adjective *available* followed by spatial prepositions such as *at* or *from* (cf. Hunston & Francis 2000: 72f.) simply because (i) the information provided by the PP headed by these prepositions is straightforwardly and compositionally providing the place where something is available and (ii) the PPs are fairly freely movable within the clause. I would imagine that many, if not most, phraseologists would also not consider *CDs are available at the store* as an instance of a phraseologism. This definition of a pattern fulfils the above six defining parameters and is virtually the same as my definition of phraseologism as well as that of symbolic units in Cognitive Grammar and constructions in Construction Grammar. This testifies strongly to the fact that phraseology is one of the key concepts in both theoretical linguistics and in the method of corpus linguistics, although differing terminology may sometimes render this fact more opaque than is desirable.

In fact, the range of correspondences is even larger. For example, we have seen above that the notion of a (symbolic) unit in Cognitive Grammar involves a degree of automaticity in accessing a structure as well as the absence of the need to analyze the internal structure of a unit. Exactly these notions figure in the formulation of one of the most prominent principles in contemporary corpus linguistics, Sinclair's 'idiom principle'. This principle states that "a language user has available to him or her a large number of semi-preconstructed phrases that constitute single choices, even though they might appear to be analyzable into segments" (Sinclair 1991: 110) and contrasts with the open-choice principle, which states that "[a]t each point where a unit is completed (a word or a phrase or a clause), a large number of choices opens up and the only restraint is grammaticalness" (Sinclair 1991: 109). In other words, phraseological research quickly leads to evidence for a claim by Pawley & Syder (1983: 213–215ff.), which Langacker (1987: 29–42) has discussed extensively under the heading

11. In corpus linguistics, this assumption is often attributed to Sinclair (1991). Hunston & Francis (2000) are also often quoted in this connection. However, a wide range of earlier studies make the same assumption, as is obvious from Altenberg & Eeg-Olofsson's (1990: 1) introductory paragraph where several even earlier references building on the very same assumption are cited.

of "rule-list fallacy", that speakers' mental lexicons do contain much more than just lexical primitives, namely hundreds of thousands of prefabricated items that *could* be productively assembled but are, as a result of frequent encounter, redundantly stored and accessed. Thus, the analysis of phraseologisms does not only reveal patterns, and maybe peculiarities, of usage, but can also ultimately lead to more refined statements about matters of mental representation within the linguistic system.

A final parameter which is relevant in this connection and shared across disciplines is one that has just been mentioned, viz. the role that frequency plays for phraseologisms. As mentioned above, a phraseologism is characterized by a sufficiently high frequency of co-occurrence (even when a strict frequency threshold is not provided; cf. Hunston & Francis's definition of a pattern above). A corpus-based approach to phraseologisms and related phenomena such as collocations, colligations, and patterns provides just the frequency data that are needed. Similarly, in cognitive linguistics, the frequency with which a speaker/hearer encounters a particular symbolic unit is assumed to be positively correlated with the degree of cognitive entrenchment of that unit in the speaker's cognitive system, an assumption that has more recently been labeled the 'From-Corpus-to-Cognition Principle' (Schmid 2000: 39).

In sum, Sections 3.1 and 3.2 showed that phraseologisms are relevant not only because they have led some generative linguists to a critical re-evaluation of their framework, but also in the sense that phraseologisms constitute the core of theoretical frameworks that are currently becoming increasingly popular. Section 3.3 has now demonstrated that phraseology and cognitively-inspired linguistic theories overlap almost completely with much of the descriptive and theoretical apparatus of contemporary corpus linguistics. Section 4 will focus on this overlap from a methodological perspective, addressing the question of which strategies different frameworks employ to identify phraseologisms and how these may be improved or refined.

4. The identification of phraseologisms

As is obvious from the six parameters discussed in Section 2, there are many different ways of defining the notion of phraseologism. In addition, we have seen in Section 3 that the number of concepts in the semantic space around the words *phraseology* and *phraseologism* is huge. It, therefore, comes as no surprise that a considerable arsenal of empirical approaches has been used to identify phraseological units. It is useful to briefly review these, bearing in mind the role phraseology has played in different theoretical frameworks.

4.1 The identification of phraseologisms in transformational-generative linguistics

In transformational-generative linguistics, the identification of phraseologisms has been rather eclectic. Given (i) a linguistic system involving only perfectly productive rules and a lexicon as a grab bag of exceptions and (ii) the objective of developing a language-independent / universal grammar, there has never been a systematic identification of the inventory of phraseologisms in a language within transformational-generative grammar. Indeed, from this perspective, why should there be? Phraseologisms are by most accounts not productive, and thus can be relegated to the role of exceptions; moreover, they are by their very nature not universal and, thus, of little relevance to the core objective of the whole generative enterprise. The lack of a comprehensive identification procedure does therefore not come as a big surprise, and it is probably fair to say that the identification of phraseologisms has been largely based on recognizing that a particular semantic unit's behavior – be it a single- or multi-word unit – defies characterization in terms of the hard-and-fast rules of the grammar that are thought to be necessary on syntactic grounds alone.

4.2 The identification of phraseologisms in cognitive linguistics/Construction Grammar

In cognitive linguistics and Construction Grammar, the level of sophistication of the identification and analysis of phraseologism is certainly somewhat higher. First, since the architecture of these approaches does not simply allow their practitioners to simply relegate irregular elements to a peripheral component of the linguistic system, much more energy has been devoted to the identification of phraseological elements in the first place. Second, since both Cognitive Grammar and Construction Grammar are usage-based, their practitioners have relied on authentic corpus data much more than practitioners of many other linguistic theories. This has, of course, necessitated a higher degree of awareness and sophistication regarding the identification and analysis of potential phraseological expressions – as anybody who has ever compared the clarity of invented examples to the messiness of authentic data can confirm. However, it is also interesting to note that, in spite of this recent upsurge of usage-based approaches within cognitive linguistics, the notions of phraseologism, pattern, collocation, and colligation are not particularly frequent and the degree of methodological rigor may sometimes leave something to be desired. For example, Gries et al. (2005, to appear) exemplify this in their discussions of the so-called *as*-predicative – the phraseologism exemplified by *I never saw myself as a costume designer* or *He sees this as a problem* or *Politicians are regarded as being closer to actors* – and show that the most frequent kind of corpus-based quantitative data in Construction Grammar does not always correlate well with experimental results.

4.3 The identification of phraseologisms in corpus linguistics

The most comprehensive identification procedures for phraseologisms are certainly found in corpus linguistics. This is to be expected given that corpus linguistics is a methodology that is mostly concerned with lexical (co-)occurrences. Several levels of sophistication are discernible. As in cognitive linguistics and Construction Grammar, the most basic approaches are, it seems, also the most widely used ones.

First, much work in this area, such as that by Stubbs and his colleagues (e.g. Stubbs 2001, 2002), involves the generation of frequency lists of *n*-grams, i.e. uninterrupted sequences of word forms; the upper limit of *n* is usually five.

Secondly, most studies involve the generation of concordances, although the ways in which concordances are generated or processed differ a lot. Some studies just generate a concordance of one or several word forms and interpret the data on the basis of various sorting styles to arrive at generalizations governing the preferred uses of the word forms in question (see, for example, Sinclair 1991: 53ff., Hunston & Francis 2000). Other, slightly more complex, approaches generate concordances on the basis of uninterrupted sequences of part-of-speech tags (e.g., Justeson & Katz's (1995a, b) work involving the frames [Adj N] as well as [N N], [N P N], etc.).

Yet other work is based on a mixture of these two approaches. On the one hand, there is what Renouf & Sinclair (1991) call *collocational frameworks*, namely patterns matching [*a* N *of*] or [*be* Adj *to*]. On the other hand, there is work invoking the above notion of colligation. Like all methods employed so far, these approaches are also usually based on frequencies and percentages.

While these methods are no doubt the most widespread ones, they do have some methodological shortcomings. One of the most severe is their limited degree of quantitative sophistication. For example, Stubbs (2002) and Stubbs & Barth (2003) largely ignore the immensely interesting work that has been done on the automatic or semi-automatic identification of multi-word units (cf. below for a range of relevant references). Similarly, Hunston & Francis's (2000: 37) above formulation that a combination of words needs to be "relatively frequent" to qualify as a pattern is so vague as to be practically vacuous. Hunston (2002: 147) discusses the frequencies of *after a moment*, *after a few moments*, and *after a few moments of*, and then asks "[h]ow many examples of a three-, four, or five-word sequence are necessary for it to be considered a phrase?"

This is all the more regrettable because there is a very large body of research employing sophisticated tools for the identification of phraseologisms.[12] For example, there is a vast array of studies researching how and which collocational statistics improve on the predominant approach of just reporting observed frequencies. Church et al. (1991) is an example of an early study in this vein. More recent examples include Evert and colleagues (e.g. Evert & Krenn 2001) and Gries & Stefanowitsch's work on collostructional analysis, a family of methods concerned with measuring and inter-

12. See. Gries (2006a) for a brief overview of several related problems.

preting the statistical association of words to constructions/patterns (cf. Stefanowitsch & Gries 2003, to appear and Gries & Stefanowitsch 2004) as well as Gries et al. (2005, to appear) for experimental confirmation. In addition, Mason (1997, 1999) and Cantos & Sánchez (2001) discuss a variety of issues concerning the overall validity of collocational studies.

Most of these studies are based on a particular search span or presuppose a particular length for the collocation being investigated. However, the definition of phraseologisms in Section 2 requires decisions about both the length of phraseologisms *and* the different levels of granularity at which co-occurrences can be observed. In addition, a top-down, or *a priori*, approach may not always be the most useful strategy. Sometimes it may be more revealing to let the data – rather than the preconceptions of any particular researcher – decide what the potentially most revealing pattern is. There is a large body of immensely interesting work in this area: for example, Kita et al.'s (1994) cost criterion serves to identify *in a bottom-up manner* the size of interesting uninterrupted multi-word units, which are prime candidates for phraseologisms. Similarly, Mason's (1997, 1999) notion of lexical gravity (cf. Sinclair & Jones 1974 for an earlier though simpler approach) helps to identify the range of collocates – the span – of a word that exhibits interesting distributional patterns. This notion has unfortunately never received the recognition it deserves. Also, the methods proposed by Dias et al (1999), Nagao & Mori (1994), Ikehara et al. (1996), to name but a few, all contain interesting concepts and methodological tools concerning the (semi-)automatic identification of phraseologisms that most corpus-linguistic, let alone cognitive-linguistic, work has not even begun to recognize or utilize to their fullest potential. I hope that the ideas developed in these and similar studies find their way into phraseological research soon and that this chapter, as well as the one specifically addressing this area (cf. Heid this volume), will help promote these approaches.

5. Concluding remarks

This chapter has provided a brief and necessarily selective discussion of the notion of phraseologism. I have suggested six parameters that I consider to be the characteristics that every definition of phraseologisms should include. While these parameters lay no claim to originality, I hope that they both underscore the linguistic dimensions that are relevant to phraseology and provide a unified framework of reference for definitions. By choosing different settings of these parameters, it is possible to define a variety of interrelated concepts from different frameworks including, but not limited to, idioms, word-clusters, *n*-grams, collocations, colligations, collostructions, constructions, patterns, fixed expressions and phraseologisms.

While space has precluded a more exhaustive discussion of how all these notions relate to phraseology, I have briefly discussed the role that phraseology plays both in different theoretical schools of thought within linguistics and in the methodological paradigm of corpus linguistics. I have pointed out how the notion of phraseologism –

even if labeled differently – plays a role for different linguistic frameworks that can hardly be overestimated, either by providing motivation to critically revise the assumptions of a framework or by integrating seamlessly into, and enriching, the framework. The overlap in terms of the relevant theoretical assumptions and parameters of Cognitive Grammar and Construction Grammar on the one hand and phraseology research and Pattern Grammar on the other is actually so enormous that it is amazing that up to now phraseologists and cognitively inspired linguists have worked on similar issues largely separately. They may not even have recognized or topicalized the overlap, although Schönefeld (1999) and Mukherjee (2004) have made laudable attempts to bridge the gap.

My hope is that the large conceptual overlap of especially corpus-based phraseologists and usage-based cognitive linguists will henceforth be recognized more explicitly and will stimulate a larger amount of cross-disciplinary work. Cognitive linguists as well as construction grammarians have often been relatively lenient and shown little rigor in their handling of frequency data. They, thus, have much to gain from looking at how natural language processing researchers interested in phraseologisms use frequencies and other more elaborated statistics to identify recurrent patterns that are prime candidates for phraseologisms and symbolic units and constructions within cognitive linguistics. In addition, especially some of the early work in Construction Grammar has been concerned with relatively marginal constructions. Phraseologists are probably in an ideal position to provide less specific patterns against which cognitive linguists and construction grammarians can test their analyses. On the other hand, many phraseologists – often with an applied linguistics or lexicographic background – have focused on rather descriptive work on phraseologisms (or, more narrowly, idioms) and have often not been concerned with integrating their accounts of phraseologisms in particular and other patterns more generally into a larger theory of the linguistic system. Given the cognitive linguist's cognitive commitment "to make one's account of human language accord with what is generally known about the mind and the brain, from other disciplines as well as our own" (Lakoff 1990: 40), it seems to me as if this theory would be the one which usage-based phraseologists could work with best. Thus, if only a few of these gaps between these different fields are bridged, this chapter has served its purpose.

Acknowledgements

I am grateful to Sylviane Granger and especially Stefanie Wulff for feedback and discussion. Two anonymous reviewers provided a wealth of comments that made me clarify many issues considerably. I also owe much to Anatol Stefanowitsch for extensive cooperation and thus much input and discussion in the overall subject area over the past few years. The usual disclaimers apply.

References

Altenberg, B. & M. Eeg-Olofsson (1990). Phraseology in spoken English: Presentation of a project. In Aarts, J. & W. Meijs (eds.) *Theory and Practice in Corpus Linguistics,* 1–27. Amsterdam: Rodopi.

Brenier, J. M. & L. A. Michaelis (2005). Optimization via syntactic amalgam: Syntax-prosody mismatch and copula doubling. *Corpus Linguistics and Linguistic Theory* 1(1): 45–88.

Bybee, J. (1985). *Morphology*: *A Study into the Relation of Meaning and Form*. Amsterdam: John Benjamins.

Cantos, P. & A. Sánchez (2001). Lexical constellations: What collocates fail to tell. *International Journal of Corpus Linguistics* 6(2): 199–228.

Church, K. W., W. Gale, P. Hanks & D. Hindle (1991). Using statistics in lexical analysis. In Zernik, U. (ed.) *Lexical Acquisition: Exploiting On-Line Resources to Build a Lexicon,* 115–164. Hillsdale, NJ: Lawrence Erlbaum.

Cobuild on Compact Disc V1.2 (1995). HarperCollins.

Cowie, A. P. & R. Mackin (1993). *Oxford Dictionary of Phrasal Verbs*. Oxford: Oxford University Press.

Croft, W. (2001). *Radical Construction Grammar: Syntactic Theory in Typological Perspective*. Oxford: Oxford University Press.

Culicover, P. W. (1999). *Syntactic Nuts: Hard Cases, Syntactic Theory, and Language Acquisition*. Oxford: Oxford University Press.

Dias, G., J. G. Pereira Lopes & S. Guilloré (1999). Mutual expectation: A measure for multiword lexical unit extraction. In *VEXTA: Proceedings of the Conference, 22–24 November 1999, Università Ca' Foscari, San Servolo, VIU, Venezia, Italia*, 133–138. Padova: Unipress.

Evert, S. & B. Krenn (2001). Methods for the qualitative evaluation of lexical association measures. In *Proceedings of the 39th Annual Meeting of the Association for Computational Linguistics*, 188–195. Morristown, NJ: Association for Computational Linguistics.

Fillmore, C. J., P. Kay & M. C. O'Connor (1988). Regularity and idiomaticity in grammatical constructions: The case of *let alone*. *Language* 64(3): 501–538.

Firth, J. R. (1968). *Selected Papers of J. R. Firth 1952–59*. F. R. Palmer (ed.). London: Longman.

Fraser, B. (1966). Some remarks on the VPC in English. In Dinneen, F. P. (ed.) *Problems in Semantics, History of Linguistics, Linguistics and English*, 45–61. Washington, DC: Georgetown University Press.

Fraser, B. (1976). *The Verb-Particle Combination in English*. New York, NY: Academic Press.

Gibbs, R. W. Jr. (1994). *The Poetics of Mind: Figurative Thought, Language, and Understanding*. Cambridge: Cambridge University Press.

Goldberg, A. E. (1995). *Constructions*: *A Construction Grammar Approach to Argument Structure*. Chicago, IL: The University of Chicago Press.

Goldberg, A. E. (2006). *Constructions at Work: The Nature of Generalization in Language*. Oxford: Oxford University Press.

Gries, S. T. (2003). *Multifactorial Analysis in Corpus Linguistics*: *A Study of Particle Placement*. London: Continuum Press.

Gries, S. T. (2006a). Some proposals towards more rigorous corpus linguistics. *Zeitschrift für Anglistik und Amerikanistik* 54(2): 191–202.

Gries, S. T. (2006b). Exploring variability within and between corpora: Some methodological considerations. *Corpora* 1(2): 109–51.

Gries, S. T. (to appear). Corpus data in usage-based linguistics: What's the right degree of granularity for the analysis of argument structure constructions?

Gries, S. T., B. Hampe & D. Schönefeld (2005). Converging evidence: Bringing together experimental and corpus data on the association of verbs and constructions. *Cognitive Linguistics* 16(4): 635–676.

Gries, S. T., B. Hampe, & D. Schönefeld (to appear). Converging evidence II: More on the association of verbs and constructions. In Newman, J. & S. Rice (eds.) *Experimental and Empirical Methods in the Study of Conceptual Structure, Discourse, and Language*. Stanford, CA: CSLI.

Gries, S. T. & A. Stefanowitsch (2004). Extending collostructional analysis: A corpus-based perspectives on 'alternations'. *International Journal of Corpus Linguistics* 9(1): 97–129.

Hoey, M. (2004). Textual colligation: A special kind of lexical priming. *Language and Computers* 49(1): 71–94.

Howarth, P. (1998). Phraseology and second language proficiency. *Applied Linguistics* 19(1): 24–44.

Hunston, S. (2002). *Corpora in Applied Linguistics*. Cambridge: Cambridge University Press.

Hunston, S. & G. Francis (2000). *Pattern Grammar: A Corpus-Driven Approach to the Lexical Grammar of English*. Amsterdam: John Benjamins.

Ikehara, S., S. Shirai, & H. Uchino (1996). A statistical method for extracting uninterrupted and interrupted collocations from very large corpora. In *Proceedings of the 16th Conference on Computational Linguistics – Volume 1*, 574–579. Morristown, NJ: Association for Computational Linguistics.

Israel, M. (1996). The *way* constructions grow. In Goldberg, A. E. (ed.) *Conceptual Structure, Discourse, and Language*, 217–230. Stanford, CA: CSLI.

Jackendoff, R. S. (1997). Twistin' the night away. *Language* 73(3): 534–559.

Justeson, J. S. & S. M. Katz (1995a). Technical terminology: Some linguistic properties and an algorithm for identification in text. *Natural Language Engineering* 1: 9–27.

Justeson, J. S. & S. M. Katz (1995b). Principled disambiguation: Discriminating adjective senses with modified nouns. *Computational Linguistics* 21(1): 1–27.

Kay, P. & C. J. Fillmore (1999). Grammatical constructions and linguistic generalizations: The *What's X doing Y?* construction. *Language* 75(1): 1–33.

Kita, K., Y. Kato, T. Omoto & Y. Yano (1994). A comparative study of automatic extraction of collocations from corpora: Mutual information vs. cost criteria. *Journal of Natural Language Processing* 1(1): 21–33.

Lakoff, G. (1990). The invariance hypothesis: Is abstract reason based on image schemas? *Cognitive Linguistics* 1(1): 39–74.

Langacker, R. W. (1987). *Foundations of Cognitive Grammar: Theoretical Prerequisites*. Stanford, CA: Stanford University Press.

Langacker, R. W. (1991). *Foundations of Cognitive Grammar: Descriptive Application*. Stanford, CA: Stanford University Press.

Mason, O. (1997). The weight of words: An investigation of lexical gravity. In Lewandowska-Tomaszczyk, B. & P. J. Melia (eds.) *PALC'97: Practical Applications in Language Corpora*, 361–75. Lodz: Lodz University.

Mason, O. (1999). Parameters of collocation: The word in the centre of gravity. In Kirk, J. (ed.) *Corpora Galore: Analyses and Techniques in Describing English,* 267–280. Amsterdam: Rodopi.

Mukherjee, J. (2004). Corpus data in a usage-based cognitive grammar. In Aijmer, K. & B. Altenberg (eds.) *The Theory and Use of Corpora: Papers from the 23rd ICAME Conference,* 85–100. Amsterdam: Rodopi.

Nagao, M. & S. Mori (1994). A new method of *n*-gram statistics for large number of *n* and automatic extraction of words and phrases from large text data of Japanese. *COLING* 14 Vol.1, 611–615. San Mateo, CA: Morgan Kaufmann.

Nunberg, G., I. A. Sag, & T. Wasow (1994). Idioms. *Language* 70(3): 491–538.

Pawley, A. & F.H. Syder (1983). Two puzzles for linguistic theory: Nativelike selection and nativelike fluency. In Richards, J.C. & R. W. Schmidt (eds.) *Language and Communication,* 191–226. London: Longman.

R Development Core Team (2006). *R: A language and Environment for Statistical Computing*. Vienna: R Foundation for Statistical Computing.

Renouf, A. & J. M. Sinclair (1991). Collocational frameworks in English. In Aijmer, K. & B. Altenberg (eds.) *English Corpus Linguistics,* 128–143. London: Longman.

Rice, S. & J. Newman (2005). Inflectional islands. Paper presented at the International Cognitive Linguistics Conference, Seoul, South Korea.

Schmid, H.-J. (2000). *English Abstract Nouns as Conceptual Shells: From Corpus to Cognition*. Berlin: Mouton de Gruyter.

Schönefeld, D. (1999). Corpus linguistics and cognitivism. *International Journal of Corpus Linguistics* 4(1): 131–71.

Sinclair, J. M. (1991). *Corpus, Concordance, Collocation*. Oxford: Oxford University Press.

Sinclair, J. M. & S. Jones (1974). English lexical collocations: A study in computational linguistics. *Cahiers de Lexicologie* 24(2): 15–61.

Stefanowitsch, A. & S. T. Gries (2003). Collostructions: Investigating the interaction between words and constructions. *International Journal of Corpus Linguistics* 8(2): 209–43.

Stefanowitsch, A. & S.T. Gries (to appear). Corpora and grammar. In Lüdeling, A. & M. Kytö (eds.) *HSK Korpuslinguistik – Corpus Linguistics*. Berlin: Mouton de Gruyter.

Stubbs, M. (2001). *Words and Phrases: Corpus Studies of Lexical Semantics*. Oxford: Blackwell.

Stubbs, M. (2002). Two quantitative methods of studying phraseology in English. *International Journal of Corpus Linguistics* 7(2): 215–44.

Stubbs, M. & I. Barth (2003). Using recurrent phrases as text-type discriminators. *Functions of Corpus Language* 10(1): 61–104.

Tomasello, M. (2003). *Constructing a Language*: *A Usage-Based Theory of Language Acquisition*. Cambridge, MA: Harvard University Press.

Wulff, S. (to appear). *Rethinking Idiomaticity*: *A Usage-Based Approach*. London: Continuum Press.

Disentangling the phraseological web

Sylviane Granger and Magali Paquot

Although phraseology has recently begun to establish itself as a field in its own right, this process is being hindered by two main factors: the highly variable and wide-ranging scope of the field and the vast and confusing terminology associated with it. This chapter tackles these two issues successively in an attempt to disentangle the 'phraseological web'. We first draw up a clear distinction between two major approaches to the study of multi-word units, i.e. the phraseological approach and the distributional or frequency-based approach, which have set quite different boundaries to the field. We then argue that the variations in scope that characterize the field of phraseology are a direct result of its fuzzy borders with four neighbouring disciplines: semantics, morphology, syntax and discourse. We describe some of the most influential typologies of word combinations within the phraseological approach and present a categorization of multi-word units emerging from the distributional approach. We conclude with suggestions regarding both the scope of the field and the terminology used.

1. Introduction

Although the study of multi-word units has a long history, with Bally distinguishing between the fully fixed 'unités phraséologiques' and the looser 'séries phraséologiques' as early as 1909, phraseology has only recently begun to establish itself as a field in its own right. This process is being hindered by two main factors however: the highly variable and wide-ranging scope of the field on the one hand and on the other, the vast and confusing terminology associated with it.

Phraseology can be loosely defined as "the study of the structure, meaning and use of word combinations" (Cowie 1994: 3168). As word combinations come in many different shapes and forms, the scope of the field is a function of the criteria used by linguists to distinguish phraseological units from non-phraseological ones. While the East European tradition (see Section 2) has tended to favour fairly fixed combinations like idioms or proverbs, the more recent corpus-based approaches have adopted a much wider perspective and included many word combinations that would traditionally be considered to fall outside the scope of phraseology. Even neighbouring countries like Great-Britain and France have quite different traditions: the notion of fixedness ('figement') lies at the heart of the French tradition (Gross 1996), while the

Anglo-Saxon tradition has from the start attached great importance to the less fixed category of collocation (Palmer 1933 in Cowie 1998a: 210–212). This diversity is a source of richness but it also hinders communication between linguists and generally increases the impression of fuzziness in the field. This impression is amplified by the unwieldy terminology employed, with different terms covering the same units and the same terms used to denote quite different units. This situation is deplored by Cowie (1998a: 210) who refers to phraseology as "a field bedevilled by the proliferation of terms and by the conflicting uses of the same term". As pointed out by Wray & Perkins (2000: 3), one of the pernicious effects of the loose terminology is that it makes it "extremely difficult to be sure when like is being compared with like".

The purpose of this chapter is to give a brief overview of the field of phraseology, introduce some major typologies and make suggestions regarding both the scope of the field and the terminology used. In Section 2 we briefly present the two major approaches to phraseology. Section 3 argues that the variations in scope that characterize the field of phraseology are a direct result of its fuzzy borders with four neighbouring disciplines: semantics, morphology, syntax and discourse. In Section 4 some influential typologies are presented and in Section 5 we make suggestions as to how current (perceptions of) confusion in the field might be addressed. Section 6 offers some conclusions and suggestions for new ways forward.

2. Two major approaches to phraseology

The traditional approach to phraseology is greatly indebted to scholars from the former Soviet Union and other countries of Eastern Europe (cf. Cowie 1998b: 1). Russian scholars like Vinogradov and Amosova are at the foundations of a view of phraseology that restricts the scope of the field to a specific subset of linguistically defined multi-word units and sees phraseology as a continuum along which word combinations are situated, with the most opaque and fixed ones at one end and the most transparent and variable ones at the other. Cowie's (1981) continuum, which goes from free combinations to pure idioms through restricted collocations and figurative idioms, is a direct descendent of these early Russian schemes (see Section 4.1). One of the main preoccupations of linguists working within that tradition has been to find linguistic criteria for distinguishing one type of phraseological unit from another and especially for distinguishing the most variable and transparent multi-word units from free combinations, which have only syntactic and semantic restrictions and are therefore considered as falling outside the realm of phraseology (Cowie 1998b: 6). In this tradition, the most idiomatic units, whose meanings cannot be derived from the meanings of the constituents, are often presented as the most 'core'. This appears clearly from the following statement by Gläser (1998: 126): "Idioms form the majority and may be regarded as the prototype of the phraseological unit". This tradition deserves much of the credit for having established phraseology as a discipline in its own right, created a terminology for the field and provided linguists with a set of discrete criteria which can be

used to categorize and analyze phraseological units. To refer to this tradition, we adopt Nesselhauf's (2004) term 'phraseological approach'.

A more recent approach to phraseology, which originated with Sinclair's pioneering lexicographic work, has literally turned phraseology on its head. Instead of adopting a top-down approach which identifies phraseological units on the basis of linguistic criteria, it uses a bottom-up corpus-driven approach to identify lexical co-occurrences (Sinclair 1987). This inductive approach, which is referred to as the distributional (Evert 2004) or frequency-based (Nesselhauf 2004) approach, generates a wide range of word combinations, which do not all fit predefined linguistic categories. It has opened up a "huge area of syntagmatic prospection" (Sinclair 2004: 19) encompassing sequences like frames, collocational frameworks, colligations and largely compositional recurrent phrases (see Section 4.2). All these sequences illustrate Sinclair's (1991) idiom principle, a principle that views language as essentially made up of strings of co-selected words that constitute single choices. This new approach has "pushed the boundary that roughly demarcates the 'phraseological' more and more into the zone previously thought of as free" (Cowie 1998b: 20). Many of the units that were traditionally considered as peripheral or falling outside the limits of phraseology have now become central as they have revealed themselves to be pervasive in language, while many of the most restricted units (idioms, proverbs) have proved to be highly infrequent (Moon 1998). Unlike proponents of the classical approach to phraseology, Sinclair and his followers are much less preoccupied with distinguishing between different linguistic categories and subcategories of word combinations or more generally setting clear boundaries to phraseology. In Sinclair's model of language, phraseology is central: phraseological items, whatever their nature, take precedence over single words. This radical view has been criticized. Gaatone (1997: 168), for instance, welcomes the growing importance attached to multi-word units but warns against considering everything as phraseological.

3. The fuzzy borders of phraseology

The two approaches to phraseology set quite different boundaries to the field. In Figure 1 phraseology is represented as a field that has fuzzy borders – hence the dotted lines – with four other major fields: semantics, morphology, syntax and discourse.[1] The territory covered by the frequency-based approach (represented by the light grey circle) is much wider than that of the traditional view (in dark grey). Figure 1 highlights the inherently multidisciplinary nature of the field. As pointed out by Mel'čuk (1995: 227), phraseology has to deal with everything, which makes it "so difficult, but so appealing!"

1. There are other relevant fields, notably phonology/prosody, sociolinguistics and psycholinguistics (cf. Wray 2002).

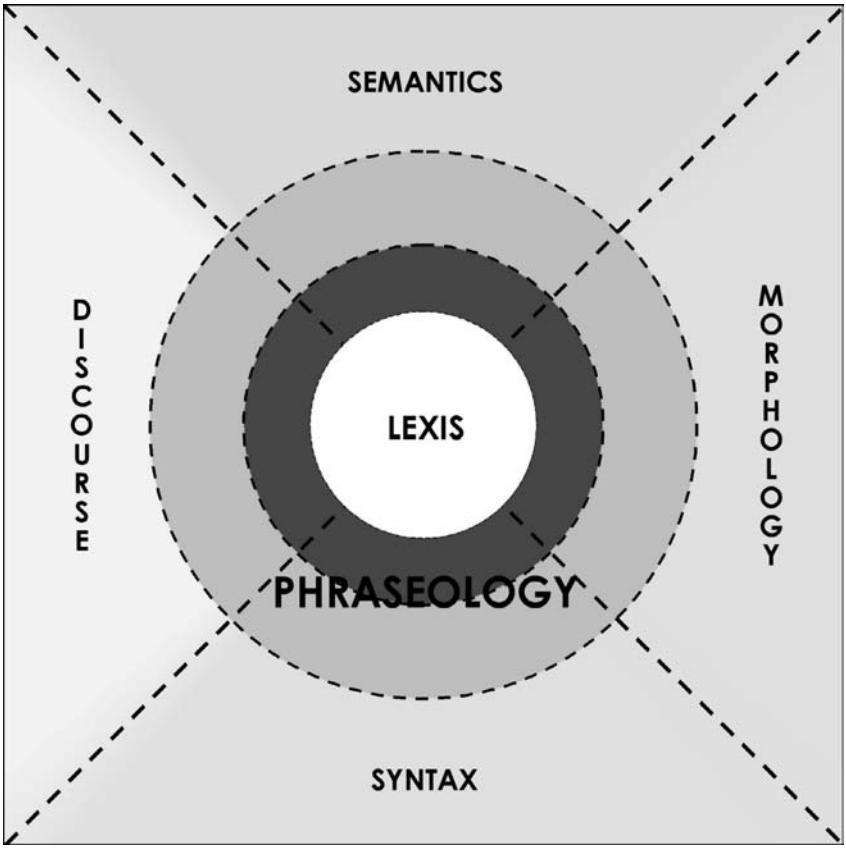


Figure 1. Phraseology wide and narrow

3.1 Phraseology and semantics

The field with which phraseology has arguably the strongest – and at the same time fuzziest – links is semantics. Recourse to semantics is essential to distinguish between different types of lexical affinity. Allerton (1984) sets out semantic co-occurrence restrictions, which can be logically predicted from the lexical meaning and semantic traits of a given lexeme (e.g. in its literal meaning the adjective *pregnant* can only be used to describe female beings), from locutional co-occurrence restrictions, which cannot be generalized and should be described for every single lexeme. Only usage can explain why we say *strong coffee* and not **powerful coffee* or why we prefer to speak of *a chestnut horse* rather than of *a brown horse*. This distinction lies at the heart of the traditional approach. It makes a sharp distinction between free combinations like *spend a day/year* or *spend money/two pounds*, which are only governed by semantic co-occurrence restrictions and are thus considered as falling outside the realm of

phraseology, and other multi-word units whose co-occurrence cannot be accounted for by semantics and qualify as phraseological units or phrasemes.

Another semantic notion that lies at the heart of phraseology is non-compositionality. A lexical item is said to be non-compositional if its global meaning is different from the sum of its individual parts (for a thorough discussion, cf. Svensson this volume). Non-compositionality is considered by some linguists as the defining criterion of phraseological units while others view it as a secondary feature that characterizes some, but definitely not all units. Mel'čuk (1998: 24) clearly holds the former view: "the main substantive property of a phraseme is its non-compositionality". However, all linguists, including Mel'čuk, recognize that non-compositionality is a cline, ranging from fully compositional to fully non-compositional with several intermediate categories: "there is no clear dividing-line between idioms and non-idioms: they form the end-points of a continuum" (Cowie et al. 1983: xiv). This notwithstanding, full compositionality is sometimes used as a factor of exclusion from phraseology in the traditional approach. For example, Cowie (2005) includes *face pack* in phraseology but excludes *face flannel* on the grounds that in *face flannel* the meaning is entirely compositional.

By contrast, all types of word combinations are part and parcel of the distributional approach which does not use semantic criteria to identify multi-word units. This is not to say that meaning plays no part. Rather it is a different view of meaning that prevails, the Firthian contextual theory of meaning, according to which "the formalisation of contextual patterning of a given word or expression is assumed to be relevant to the identification of the meaning of that word or expression" (Tognini-Bonelli 2001: 4). In this framework meaning extends well beyond the limits of the word. The relationship between a lexical item and a lexical set of semantically related words is what Sinclair (1996, 1998) and Partington (2004) refer to as semantic preference. For example, Partington (2004: 148) observes that collocates of the maximizers *utterly, totally, completely* and *entirely* share the semantic preference of 'absence/change of state', e.g. *totally uneducated* and *completely lacking*. The "proximity of a consistent series of collocates" (Louw 2000: 57) may establish yet another form of meaning, i.e. semantic prosody, whose primary function is "the expression of the attitude of its speaker or writer towards some pragmatic situation" (ibid.) (see also Louw 1993). Partington (2004: 150–151) illustrates the interdependence of semantic preference and semantic prosody using the example of the verb *undergo*, which collocates with, and thus shows semantic preference for, items from the lexical sets of 'change' (e.g. *dramatic changes, a historic transformation*), 'medicine' (e.g. *treatment, brain surgery*), 'testing' (e.g. *examinations*) and 'involuntariness' (e.g. *must, forced to, required to*). All these semantic preferences imbue the item *undergo* with a very strong unfavourable semantic prosody. Other often cited examples of words with negative semantic prosody include *happen*, *set in* (cf. Sinclair 1991) and *cause* (cf. Stubbs 1995). The systematicity of these relationships between a word and its environment has led Sinclair and his colleagues to postulate the existence of an extended unit of meaning "where collocational and colligational patterning (lexical and grammatical choices respectively) are intertwined to

build up a multi-word unit with a specific semantic preference, associating the formal patterning with a semantic field, and an identifiable semantic prosody, performing an attitudinal and pragmatic function in the discourse" (Tognini-Bonelli 2002: 79).

3.2 Phraseology and morphology

The definition of phraseology as the study of word combinations entails that phraseological units are made up of at least two words. Polylexicality is generally described as one of the first necessary conditions for inclusion in the phraseological spectrum (cf. Gross 1996; Mejri 2005; Montoro del Arco 2006).[2] However, in view of the ambiguity surrounding the definition of the concept of word, this definition is not as helpful as may seem at first sight. The scope of phraseology varies according to whether 'word' is taken in the meaning of orthographic word (separated by blanks on either side) or in the meaning of "unit characterized by internal stability and uninterruptability" (Lyons 1968: 202). In the first meaning, *of course* or *letter box* are considered to be made up of two words and hence part of phraseology, in the second they are monolexemic and hence fall outside its scope. The situation is complicated by the fact that compounds can be written in three different ways (solid as in *bookstore*, hyphenated as in *father-in-law* and open as in *high school*) and regularly have more than one spelling form (*good will, good-will, goodwill*). Although crucial, this issue is rarely tackled explicitly in phraseological studies. One regularly has to scan through the examples given by the authors to find out whether or not (solid, hyphenated and/or open) compounds are included in the range of phraseological units covered. The traditional view either excludes compounds from phraseology altogether (Barkema 1996: 133) or only keeps units that meet some well-defined criteria (stress, meaning, etc). Others seem to exclude compounds written as one word, viz. solid compounds, but include open and hyphenated compounds (e.g. Mel'čuk 1995; Gläser 1998). In the distributional approach, all sequences made up of two or more graphic words are extracted if they meet some recurrence or co-occurrence threshold. As a result, a wide range of phraseological units are extracted, including open compounds (and possibly hyphenated ones) but excluding solid compounds.

It is important to note that compounds are not the only category to pose problems because of their uncertain status as single or multi-word units. Equally problematic and less often referred to are the categories of complex prepositions (*due to*), adverbs (*in fact*) and conjunctions (*even if*), which are generally either totally disregarded or

2. Note, however, that some linguists include single words in phraseology. For example, Zuluaga (1980) includes *Salud* (En. 'cheers') and *Adiós* (En. 'bye bye') in the phraseological spectrum on the basis that these words display pragmatic fixedness (Sp. 'fijación pragmática').

regarded as minor categories (see Section 4.1).[3] Linguists often make quite arbitrary decisions as to what they include and exclude. For instance, Moon (1998: 79) excludes "for practical reasons" compound nouns, adjectives, and verbs such as *civil servant, self-raising*, and *freeze-dry*, but includes units such as *at last* and *in fact,* which she calls 'grammatical collocations'. The reason she invokes to exclude compounds is that "[t]he interest in compound words seems to me to rest largely in morphology" (ibid.: 3). However, her study of fixed expressions and idioms turns out to include units such as *ivory tower*, *trump card* or *full stop*, which would normally be classified as compounds. The issue of compounding is thus undoubtedly a major factor in blurring the line between phraseology and morphology.

3.3 Phraseology and syntax

As phraseology and syntax both deal with syntagmatic relations, it is normal that a clear demarcation line between the two fields should be difficult to draw. The whole debate centres on the looser, less idiomatic phraseological units, in particular collocations. While collocations are usually defined as arbitrarily restricted combinations of lexical words like *strong tea* or *dispel fear,* some linguists, like Benson et al. (1986), subdivide collocations into lexical collocations, which contain two lexical words, and grammatical collocations, which are made up of a lexical word and a grammatical word (*aim at, afraid that*) or structure (*avoid* + -ing, *necessary* + infinitive).[4] This view is shared by several linguists (cf. Gries this volume; Hunston 2002) but criticized by others who, like Heid (2002), consider that Benson's grammatical collocations in fact pertain to syntax. The fuzzy area that is at stake here is that of word grammar, i.e. the syntactic constraints on the use of lexis or, to use Woolard's (2000: 45) term, the "grammatical signatures" of words. This area is very close to that of valency patterns, which describe words in terms of the obligatory and optional arguments they accept (cf. Herbst et al. 2004). The whole debate should also be put in relation to Hoey's (2005) notion of 'lexical priming', which posits that words are primed to favour particular collocates, grammatical roles and positions, semantic associations, etc.

Another fuzzy area between phraseology and syntax is that between compounds and syntactic phrases. In principle, the situation is clear: productive and regular phrases belong to syntax and compounds originate in the lexicon. However, Giegerich

3. Spanish and French phraseologists have tended to give more prominence to complex prepositions, complex conjunctions and complex adverbials and to classify them as phraseological units (cf. Gross 1996; Montoro del Arco 2006).

4. The term 'colligation' is often used as a near-synonym of grammatical collocation. For example, Stefanowitsch and Gries (2003: 210) define colligations as "linear co-occurrence preferences and restrictions holding between specific lexical items and the word-class of the items that precede or follow them". In their system, the word *involvement* is said to colligate with prepositions but to collocate with *in* and *with*.

(2004, 2005) demonstrates that some constructions, for example Adj + N constructions like *dental building, mental hospital* or *financial advisor*, straddle the lexicon-syntax divide. The general issue at play is that of syntactic flexibility, a feature regularly presented as a determinant of phraseological status and more particularly of idiom status. It involves determining the extent to which word combinations are allowed to undergo syntactic variation (e.g. passivisation, insertion, deletion, pronominalisation) without losing their phraseological status. Recent corpus-based studies, notably that of Moon (1998), have shown up "the fallacy of the notion of fixedness of form" (ibid.: 47) and degrees of inflexibility now tend to be considered as an indication rather than a criterion of phrasemes (Svensson 2002 and this volume).

Generally speaking, the traditional approach tends to adopt a stricter attitude, clearly distinguishing between phraseology and syntax, while the distributional approach gives more emphasis to the lexico-grammar interface. Stefanowitsch and Gries (2003), for example, have recently proposed applying collocational analysis within a constructional view of language, viz. collostructional analysis (a blend of 'construction' and 'collocational analysis'), with the aim of providing "an objective approach to identifying the meaning of a grammatical construction and of determining the degree to which particular slots in a grammatical structure prefer, or are restricted to, a particular set or semantic class of lexical items" (ibid.: 211). Collostructional analysis can be performed on single words in specific constructions (e.g. *cause* in transitive constructions) and what the authors call 'variable idioms' (e.g. the [X *think nothing of* V_{gerund}] construction), partially filled and unfilled argument structure constructions (e.g. the *into*-causative, the ditransitive), and tense, aspect and mood (e.g. lexemes attracted by the progressive form, the imperative or past tense).

3.4 Phraseology and discourse

Phraseology has close links with discourse, a field which centres on the organisation of language above the sentence or above the clause, and therefore studies larger linguistic units, such as conversational exchanges or written texts (Stubbs 1983: 1).

The traditional approach to phraseology has tended to favour units that reflect discourse as interaction. Cowie (1988) distinguishes a category of 'formulae', viz. pragmatically specialized units like *good morning* or *how do you do,* whose meanings "are largely a reflection of the way they function in discourse (as greetings, enquiries, invitations, etc.)" (ibid: 132). Another category of formulae is that of units such as *are you with me* or *would you mind repeating that,* which are used in "organizing turn-taking, indicating a speaker's attitude to other participants, and generally ensuring the smooth conduct of interaction" (ibid: 133). The same emphasis on interactional phrasemes is found in Mel'čuk (1998). These studies typically focus on the most fixed units typical of speech. It is revealing in this respect that Gramley & Pätzold's (1992: 58–61) section on what they call "pragmatic idioms" deals almost exclusively with spoken interaction.

With the distributional approach the focus moves from pragmatics to stylistics or rhetoric. While recognizing the importance of routine formulae, corpus analysts

attribute equal – and in many cases even greater – importance to text structuring multi-word units. Using automatic methods Biber et al. (1999: 990ff.) extract a wide range of prefabricated sequences, called 'lexical bundles', which they define as "simple sequences of word forms that commonly go together in natural discourse". These sequences, which mostly display syntactic and semantic regularity, fulfil a range of major discourse functions, such as hedging, organizing, etc. They are typically verbal and clausal units in speech (e.g. *I don't know, I think I might, what's the matter with*) and extended nominal phrases and prepositional phrases in writing (e.g. *the effect of, in the case of, the extent to which*). A similar study by Altenberg (1998) uncovers "a large stock of recurrent word-combinations that are seldom completely fixed but can be described as 'preferred' ways of saying things – more or less conventionalized building blocks that are used as convenient routines in language production. These building blocks come in all forms and sizes, from complete utterances to short snatches of words, and they display varying degrees of flexibility" (see also De Cock 2003 & 2004 for an analysis of recurrent word combinations in native and learner writing and speech). All these studies have highlighted the role of 'preferred ways of saying things' as key register markers and have contributed significantly to widening the scope of phraseology.

4. Categories of word combinations

As word combinations are highly heterogeneous, linguists have quite naturally felt the need to subcategorize them. Typologies abound in the literature: some are designed for lexicological or lexicographic purposes (Gläser 1986; Cowie 1988; Moon 1998), others are pedagogically-oriented (Nattinger & DeCarrico 1992; Lewis 1993) or take a psycholinguistic perspective (Wray & Perkins 2000; Wray 2002). Several 'ad hoc' descriptions have also been proposed within the field of natural language processing (cf. Sag et al. 2002; Tschichold 2000).

Differences between the typologies largely correspond to differences in the selection of the features used to categorize multi-word units and the prioritization of selected features. Most classifications give prominence to one or more of five features of phrasemes: (1) internal structure (e.g. verb + noun or verb + preposition); (2) extent: phrase- vs. sentence-level; (3) degree of semantic (non-)compositionality; (4) degree of syntactic flexibility and collocability; (5) discourse function. Among the terms used to refer to subcategories of multi-word units, some appear to have acquired a relatively stable core meaning (e.g. idioms are usually defined as being non-compositional), others are much more confusing (e.g. collocations are used in a large number of different meanings).

The following sections are not intended as a comprehensive survey of the many typologies of phraseological units that have been proposed in different fields. Instead, Section 4.1 focuses on some influential traditional typologies which are deeply rooted in lexicology and lexicography. In Section 4.2, we propose a categorization of multi-word units emerging from the distributional approach.

4.1 Some influential typologies

One of the most influential typologies in English lexicology and lexicography is that of Cowie (e.g. 1988, 1994), which makes a primary distinction between composites, which function syntactically at or below the level of the sentence, and formulae, which function pragmatically as autonomous utterances.

As shown in Figure 2, composites are further subdivided into restricted collocations, figurative idioms and pure idioms, three categories which form a phraseological continuum, with the most transparent and variable at one hand and the most opaque and fixed at the other, as illustrated in Figure 3. The category of restricted collocations, often referred to simply as 'collocations', includes combinations such as *perform a task* or *heavy rain,* which are characterized by restricted collocability and figurative or specialized meaning of one of the elements. It includes verb-noun combinations with a delexical verb (e.g. *make a comment*). Figurative idioms have a figurative meaning but also preserve a literal interpretation (e.g. *do a U-turn*). They resist substitution of their components. Pure idioms such as *spill the beans* or *blow the gaff* are semantically non-compositional. The category of formulae includes 'sentence-like' units, "which function pragmatically as sayings, catchphrases, and conversational formulae" (Cowie 1998b: 4). Cowie (2001) later subdivides the category of formulae into routine formulae, like *good morning,* or *see you soon*, which perform speech-act functions, and speech formulae, which are used to organize messages and indicate speakers' or writers' attitudes (*you know what I mean, are you with me?)*.

Another influential model is that proposed by Mel'čuk (1995, 1998) within the meaning-text theory. Although it uses a different terminology, it is very similar to Cowie's, notably in the primary distinction made between semantic phrasemes, which

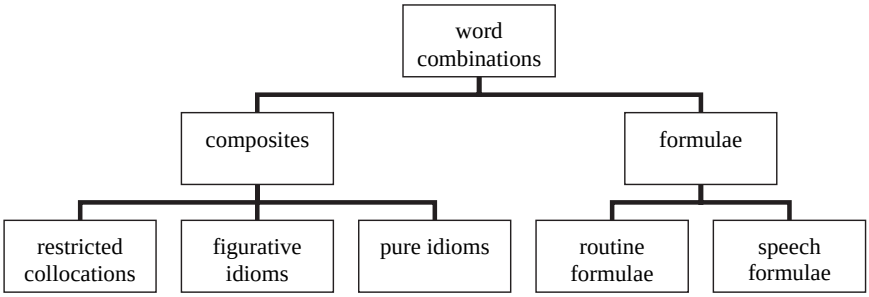


Figure 2. Cowie's (1988, 2001) classification of word combinations

free combination	‖	restricted collocation	>>	figurative idiom	>>	pure idiom
blow a trumpet		*blow a fuse*		*blow your own trumpet*		*blow the gaff*

Figure 3. Cowie's (1981) phraseological continuum

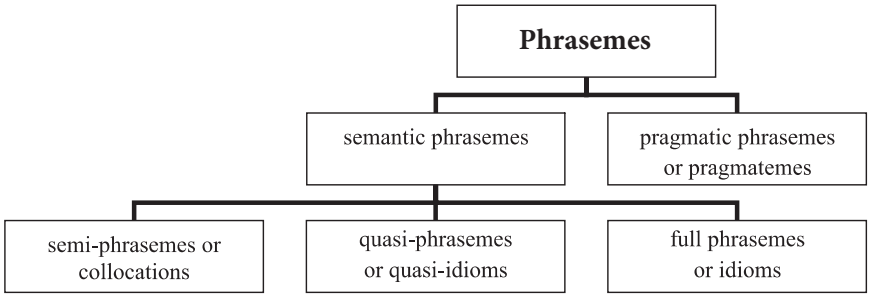


Figure 4. Mel'čuk's (1998) typology

roughly correspond to composites, and pragmatic phrasemes or pragmatemes, which are very close to Cowie's formulae.

One highly influential aspect of Mel'čuk's work is his treatment of collocations by means of lexical functions. When a native speaker of English wants to express the fact that somebody smokes a lot, he usually says that this person is a *heavy smoker* rather than a *big smoker*. By contrast, he will most probably speak of a *big eater* rather than a *heavy eater*. Mel'čuk (1995, 1998) attempts to describe these lexical preferences with lexical functions. A lexical function is "a very general and abstract meaning that can be expressed in a large variety of ways depending on the lexical unit to which this meaning applies" (Mel'čuk 1995: 186). Examples of lexical functions are:

– **Magn** which expresses the meaning of 'intense(ly)' or 'very' and functions as an intensifier, e.g. **Magn**(shave$_N$) = *close, clean*; **Magn**(easy) = *as pie, as 1-2-3*; **Magn**(to condemn) = *strongly*
– **Oper** which expresses the meaning of 'do/perform', e.g. **Oper$_1$**(cry) = *to let out* [ART∼]
– **Real** which conveys the meaning of 'fulfil the requirement of X' or 'do with X what you are supposed to do with X', e.g. **Real$_1$**(car) = *to drive* [ART∼]; **Real$_1$**(accusation) = *to prove* [ART∼]

Unlike Cowie's and Mel'čuk's typologies, Burger's (1998) typology is primarily based on the function of phraseological units in discourse. As shown in Figure 5, the top subdivision distinguishes between the following three functional categories: referential units, communicative units and structural units. Referential phraseological units are divided into two sub-categories according to a syntactico-semantic criterion. First, nominative phraseological units are constituents of the sentence and refer to objects, phenomena or facts of life (e.g. *Schwarzes Brett* 'billboard' or *jemanden übers Ohr hauen* 'to rip somebody off'). This category broadly corresponds to Cowie's 'composites' and Gläser's (1998) 'nominations'. Following the Russian tradition and phraseologists such as Cowie and Mel'čuk, nominative phraseological units are sub-divided into idioms, partial idioms and collocations. Second, propositional phraseological units generally function at sentence level but a few propositional phraseological units

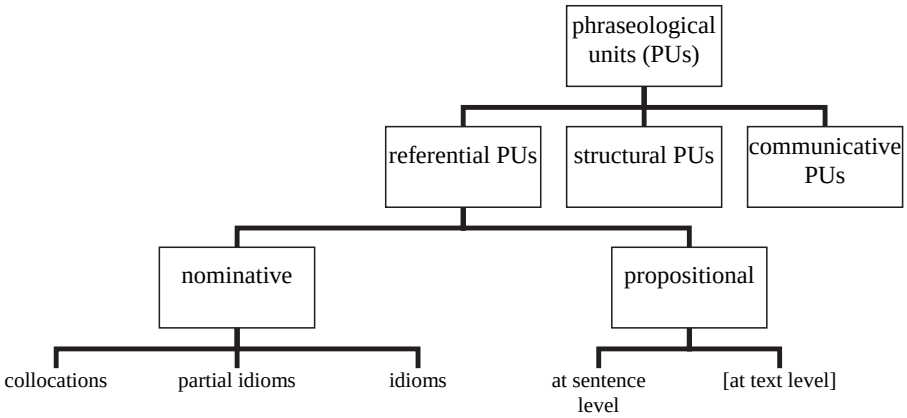


Figure 5. Burger's (1998) typology

function at text level; they refer to a statement or an utterance about these objects or phenomena (*Morgenstund hat Gold im Mund* 'the early bird catches the worm'). Propositional units include proverbs and idiomatic sentences, two broad categories that are classified as 'formulae' or 'pragmatic phrasemes' in models such as those put forward by Cowie and Mel'čuk which use both the criteria of function in discourse and function in the sentence. Communicative phraseological units or routine formulae fulfil an interactional function: they are typically used as text controllers to initiate, maintain and close a conversation or to signal the attitude of the addressor. Examples are *Guten Morgen* ('Good morning') and *Ich meine ...* ('Well, I mean...'). Unlike Cowie and Mel'čuk, Burger creates a third category of structural phraseological units which includes word combinations that establish grammatical relations, e.g. *in Bezug auf* ('concerning') and *sowohl ... als auch* ('as well ... as ...'). However, he regards structural phraseological units as the smallest and least interesting[5] category and does not go into any further detail.

4.2 Distributional categories

No categorization of phraseological units has emerged from studies rooted in the distributional approach to phraseology. It is, however, possible to draw up a typology of the types of units obtained by the different extraction procedures. As shown in Figure 6, a main subdivision can be made between two main extraction methods: n-gram analysis and co-occurrence analysis (cf. Stubbs 2002).

N-gram analysis is a method which allows for the extraction of recurrent continuous sequences of two or more words, viz. "recurrent expressions, regardless of

5. "Von den drei Gruppen ist [strukturelle Phraseologismen] die kleinste und am wenigsten interessante." (Burger 1998: 37)

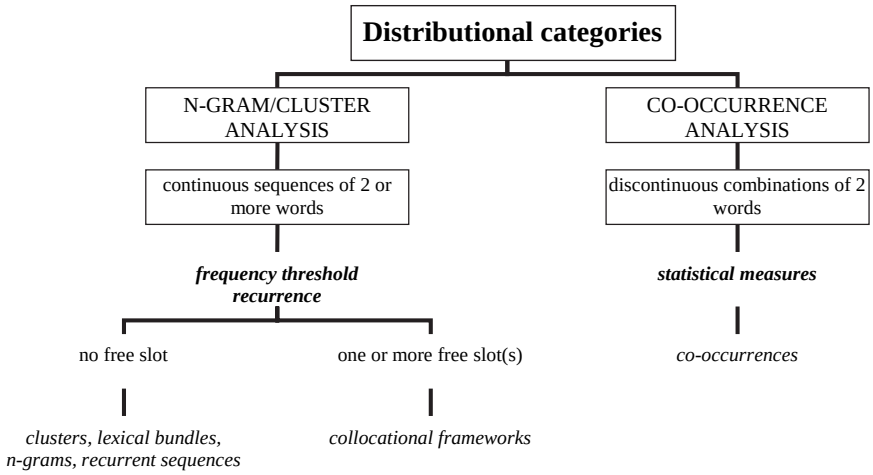


Figure 6. Distributional categories

their idiomaticity, and regardless of their structural status" (Biber et al. 1999: 990). It has been used by a wide range of authors for a variety of purposes: terminology extraction, variation study, interlanguage study, information retrieval, etc. The extracted sequences are called n-grams (or the more specific terms bigrams or trigrams) (cf. Stubbs 2007a, 2007b), lexical bundles (Biber & Conrad 1999; Biber et al. 2003; Biber 2004), clusters (Scott and Tribble 2006), chains (Stubbs 2002; Stubbs & Barth 2003), recurrent sequences (De Cock 2003), recurrent word combinations (Altenberg 1998), etc. Examples of retrieved sequences are *I don't know what, I thought that was, can I have a, in the case of, on the other hand, the use of, the fact that.* A special category of recurrent sequences is that of collocational frameworks, which consist of sequences containing one or more free slots (Renouf & Sinclair 1991: 128).[6] Examples include '*a* + ? + of', '*an* + ? + *of*', '*be* + ? + *to*', and '*too* + ? + *to*'. Stubbs (2007a, 2007b) has recently referred to these multi-word sequences as 'phrase-frames'.

Co-occurrence analysis can be roughly defined as the statistical uncovering of significant word co-occurrences. To refer to the retrieved units, the term 'collocation' or 'collocate' are used (cf. Manning & Schütze 1999; Stubbs 2002). This type of analysis lay at the core of the COBUILD dictionary project, which relied on the following definition of collocates: "[t]he definition of regular or significant collocates was 'lexical items occurring within five words either way of the headword with a greater frequency than the law of averages would lead you to expect'. (...) Collocation was established

6. Although a collocational framework is defined by Renouf & Sinclair as "a discontinuous sequence of two words, positioned at one word remove from each other" (1991: 128), we have classified them as recurrent sequences because they are not usually extracted by co-occurrence analysis but rather by means of n-gram analysis software tools (e.g. Fletcher's website: Phrases in English <http://pie.usna.edu/>).

only on the basis of corpus evidence" (Krishnamurthy 1987:70). While collocation and collocates are the most widely used terms, some linguists (cf. Schmid 2003; Evert 2004) prefer to use the term 'co-occurrence' or 'co-occurrent' and, for reasons that will be made clear in Section 5, it is these terms which we have decided to include in our typology (cf. Figure 6).

These quantitative methods constitute fantastic heuristic devices that show their full potential in a program like the *Sketch Engine*, which provides lexicographers with "corpus-based summaries of a word's grammatical and collocational behaviour"

Table 1. A sample of the word sketch for the noun *evidence*

object of	**5522**	**2.7**	**subject of**	**1982**	**1.9**	**adj. modifier**	**6173**	**2.4**
adduce	64	46.31	suggest	412	51.95	circumstantial	83	54.25
provide	622	40.61	support	117	33.69	conclusive	94	51.82
give	941	39.04	indicate	82	32.04	empirical	163	50.61
obtain	130	29.55	point	59	29.77	anecdotal	67	50.26
gather	68	28.89	show	146	28.6	ample	91	45.51
produce	187	28.44	exist	43	26.81	archaeological	75	41.35
find	334	27.62	emerge	40	26.76	forensic	57	40.86
present	120	27.49	accumulate	20	26.33	further	283	40.76
hear	144	26.96	implicate	16	25.85	sufficient	148	39.12
collect	62	24.57	relate	52	24.82	supporting	67	38.98
n. modifier	**820**	**0.4**	**pp_of-p**	**3614**	**3.3**	**pp_on-p**	**282**	**1.4**
documentary	115	59.59	senses	24	23.66	oath	9	24.54
hearsay	30	47.97	efficacy	13	20.82	behalf	9	22.15
expert	62	36.45	infection	25	19.99	issue	10	13.54
affidavit	21	35.9	abuse	26	19.89	matter	7	11.52
dating	19	32.79	damage	31	18.96	subject	7	11.46
research	72	30.47	ischaemium	6	18.7	point	9	11.27
fossil	20	29.48	witness	20	18.6	ground	5	10.15
confession	14	26.24	nephropathy	5	18.08	nature	5	9.32
parole	5	25.92	competence	15	17.86	effect	6	9.1
video	21	22.34	disease	34	17.08	side	5	7.31
pp_in-p	**393**	**0.8**	**pp_obj_to-p**	**187**	**0.7**	**pp_obj_by-p**	**248**	**1.6**
case	49	26.39	relate	15	21.81	support	66	38.83
court	41	25.69	point	12	21.49	unsupported	10	34.21
trial	21	24.61	regard	7	20.73	substantiate	5	20.74
proceedings	14	22.43	listen	9	20.12	contradict	5	18.55
prosecution	7	17.04	refer	8	16.25	convince	6	17.31
favour	6	15.93	reference	6	13.86	justify	5	13.31
chief	6	13.41	apply	6	12.33	prove	6	12.27
form	9	10.16	add	6	11.43	confirm	5	11.99
action	6	8.57	give	8	8.15	establish	5	9.83
area	7	7.05	make	5	4.11	suggest	5	9.42

(Kilgarriff et al. 2004: 105). Table 1 gives a sample of the word sketch for the noun *evidence* based on the British National Corpus (BNC).

A word of caution is needed, however. In both types of analysis, the set of retrieved units depends on the settings adopted. N-gram analysis often relies on a relatively high frequency threshold. Biber et al. (2004: 376), for example, make use of a frequency cut-off of 40 times per million words to extract lexical bundles. A number of parameters may influence the outcome of a co-occurrence analysis. They include the size of the co-occurrence window or span used, the use of filters such as a minimum frequency threshold or a stopword list and, more importantly, the statistical measure used (e.g. mutual information, log-likelihood, t-test). For example, the association measure implemented in the *Sketch Engine* is the log-log. If other statistical measures such as the log-likelihood or mutual information (MI) were used, the word sketch for the noun *evidence* might look quite different (cf. McEnery et al. 2006: 208–226). The choice of an association measure clearly depends on the objectives of a co-occurrence analysis. As McEnery et al. (2006: 217) have suggested, word pairs that are significant when MI is used are generally interesting for lexicographical purposes while they are of secondary importance for pedagogical purposes. By contrast, they argue that word pairs highlighted by MI3, a purely heuristic variant of MI, are probably "more useful for second language learners at beginning and intermediate levels." Other researchers have suggested that it is "difficult, if not impossible, to select one measure which provides the best assessment of the collocates" and that it is "probably better to use as much information as possible in exploring collocation, and to take advantage of the different perspectives provided by the use of more than one measure" (Barnbrook 1996: 101). Similarly, Bartsch (2004) uses three association measures to ensure identification of relevant co-occurrence data. She uses the MI score as the prime statistic for filtering what she calls 'collocation candidates' from the BNC word pairs and the t-test and chi-square scores for cross-checking purposes, as "these can support and sometimes supplement the data identified by MI" (ibid.: 112).

5. Reconciling the two approaches

The emergence of a new approach to phraseology is proving to be of immense value to the field. However, proponents of the two approaches are still too wide apart and both sides have a great deal to gain from a rapprochement. Many linguists working in the traditional framework seem to be largely unaware of the benefit they could derive from automatic corpus-based methods of extraction and analysis. Conversely, linguists working in the distributional framework seem not to appreciate how much they stand to benefit from the fine-grained linguistic analyses of the traditional approach. However, any rapprochement will only be fruitful if it is accompanied by some rigorous clarification of the terminology. We suggest making a clear distinction between two typologies: one for automated extraction and one for linguistic analysis.

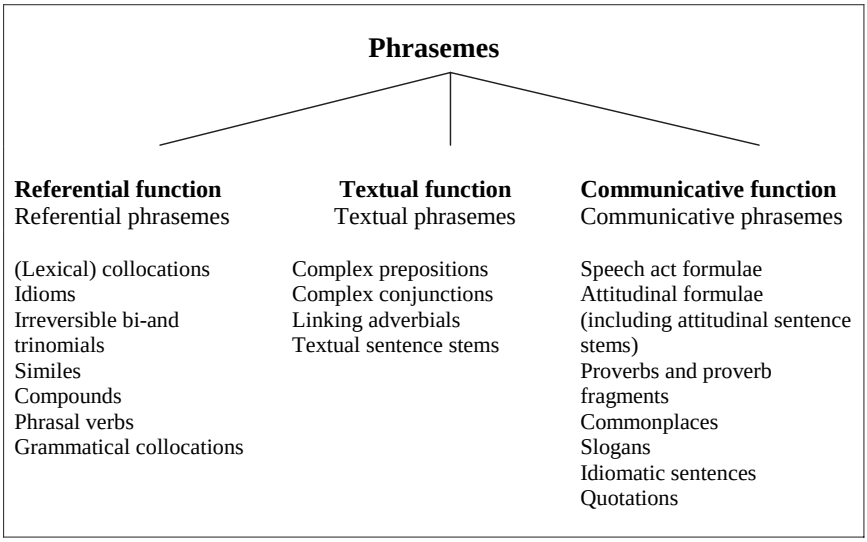


Figure 7. The phraseological spectrum

To refer to the results of automated extraction, we advocate the use of the terms in Figure 6. This means that in our view the term 'collocation' should not be used to refer to statistical word co-occurrences but instead kept in its traditional meaning of usage-based lexically restricted combination. We agree with Schmid (2003: 239) that "[i]t is not clear what is gained by calling co-occurrences of words 'collocations', when the term 'combination', or indeed 'co-occurrence' itself, covers the same range of phenomena."

As regards the linguistic classification, we think it is essential to integrate the new insights derived from the corpus-based approach. We propose an extended version of Burger's (1998) classification, as represented in Figure 7. Phraseological units are assigned to one of three major categories: referential phrasemes, textual phrasemes (an extension of Burger's category of 'structural phrasemes') and communicative phrasemes. Referential phrasemes are used to convey a content message: they refer to objects, phenomena or real-life facts. They include lexical and grammatical collocations, idioms, similes, irreversible bi- and trinomials, compounds and phrasal verbs. Textual phrasemes are typically used to structure and organize the content (i.e. referential information) of a text or any type of discourse; they include grammaticalized sequences such as complex prepositions and complex conjunctions, linking adverbials and textual sentence stems. Communicative phrasemes are used to express feelings or beliefs towards a propositional content or to explicitly address interlocutors, either to focus their attention, include them as discourse participants or influence them. They include speech act formulae, attitudinal formulae, commonplaces, proverbs and slogans.

Table 2. Categories of referential phrasemes

Category	Definition and illustration
(Lexical) collocations	(Lexical) collocations are usage-determined or preferred syntagmatic relations between two lexemes in a specific syntactic pattern. Both lexemes make an isolable semantic contribution to the word combination but they do not have the same status. Semantically autonomous, the 'base' of a collocation is selected first by a language user for its independent meaning. The second element, i.e. the 'collocate' or 'collocator', is selected by and semantically dependent on the 'base'. Examples: *heavy rain, closely linked, apologize profusely.*
Idioms	The category of idioms is restricted to phrasemes that are constructed around a verbal nucleus. Idioms are characterized by their semantic non-compositionality, which can be the result of a metaphorical process. Lack of flexibility and marked syntax are further indications of their idiomatic status. Examples: *to spill the beans, to let the cat out of the bag, to bark up the wrong tree*
Irreversible bi- and trinomials	Irreversible bi- and trinomials are fixed sequences of two or three word forms that belong to the same part-of-speech category and are linked by the conjunction 'and' or 'or'. Examples: *bed and breakfast, kith and kin, left, right and centre.*
Similes	Similes are sequences of words that function as stereotyped comparisons. They typically consist of sequences following the frames '*as* ADJ *as* (DET) NOUN' and 'VERB like a NOUN'. Examples: *as old as the hills, to swear like a trooper.*
Compounds	Compounds are morphologically made up of two elements which have independent status outside these word combinations. They can be written separately, with a hyphen or as one orthographic word. They resemble single words in that they carry meaning as a whole and are characterized by high degrees of inflexibility, viz. set order and non-interruptibility of their parts. Examples: *black hole*, *goldfish, blow-dry.*
Grammatical collocations	Grammatical collocations are restricted combinations of a lexical and a grammatical word, typically verb/noun/adjective + preposition, e.g. *depend on, cope with, a contribution to, afraid of, angry at, interested in*. The term 'grammatical collocation' is borrowed from Benson et al. (1986) but our definition is slightly more restricted as these authors also use the term to refer to other valency patterns, e.g. avoid + *-ing* form, which we do not consider to be part of the phraseological spectrum.
Phrasal verbs	Phrasal verbs are combinations of verbs and adverbial particles. Examples: *blow up, make out, crop up*.

Table 3. Categories of textual phrasemes

Category	Definition and illustration
Complex prepositions	Complex prepositions are grammaticalized combinations of two simple prepositions with an intervening noun, adverb or adjective. Examples: *with respect to, in addition to, apart from, irrespective of*
Complex conjunctions	Complex conjunctions are grammaticalized sequences that function as conjunctions. Examples: *so that, as if, even though, as soon as, given that.*
Linking adverbials	Linking adverbials include various types of phrasemes such as grammaticalized prepositional phrases, adjectival phrases, adverbial phrases, finite and non-finite clauses that play a conjunctive role in the text. Examples: *in other words, last but not least, more accurately, what is more, to conclude.*
Textual sentence stems	Textual sentence stems are routinized fragments of sentences that are used to serve specific textual or organizational functions. They consist of sequences of two or more clause constituents, and typically involve a subject and a verb. Examples: *the final point is . . . ; another thing is . . . ; it will be shown that . . . .; I will discuss . . . .*

Table 4. Categories of communicative phrasemes

Category	Definition and illustration
Speech act formulae	Speech act formulae (or routine formulae) are relatively inflexible phrasemes which are recognized by the members of a language community as preferred ways of performing certain functions such as greetings, compliments, invitations, etc. They display different degrees of compositionality. Examples: g*ood morning!, take care!, happy birthday!, you're welcome, how do you do*?
Attitudinal formulae	Attitudinal formulae are phrasemes used to signal speakers' attitudes towards their utterances and interlocutors. Examples: *in fact*, *to be honest, it is clear that*, *I think that*.
Commonplaces	Commonplaces are non-metaphorical complete sentences that express tautologies, truisms and sayings based on everyday experience. Examples: *Enough is enough, We only live once*, *It's a small world*.
Proverbs	Proverbs express general ideas by means of non-literal meaning (metaphors, metonymies, etc.). They are equivalent to complete sentences but are often abbreviated. Examples: *A bird in the hand is worth two in the bush, When in Rome*.
Slogans	Short directive phrases made popular by their repeated use in politics or advertising Example: *Make love, not war*.

Categories of multi-word units have received a wide range of definitions in the literature and a detailed survey would not be possible within the scope of this chapter. Tables 2–4 contain a set of working definitions, which draw heavily on the work of major phraseologists, notably Cowie, Mel'čuk and Burger. For in-depth discussion of each category, we refer the readers to publications by these authors and others in Cowie (1998), Allerton et al. (2004), Burger et al. (2007a) and several articles in this volume.

6. Conclusion

The major and rapid expansion of the field of phraseology in the last 25 years has resulted in the co-existence of two approaches – one linguistically-based, the other data-driven. While this development has undeniably further blurred the boundaries of a field whose inherent fuzziness has long been recognized, the resulting cross-fertilization should be viewed as a unique opportunity to lead the field of phraseology into pastures new.

The unwieldy terminology used to refer to the different types of multi-word units is a direct reflection of the wide range of theoretical frameworks and fields in which phraseological studies are conducted and can be seen as a sign of the vitality of the field. To some extent, however, it impedes the process of cross-fertilization and hinders the smooth integration of phraseological insights into other fields, notably a field like language teaching where phraseology is taking on an increasingly important role. Some terminological order is clearly needed. However, we agree with Burger et al. (2007b: 18) that "an international uniformity of terminology and classification is only possible and desirable to a certain degree." More than a unified terminology, what the field needs is some clear indication from researchers of the criteria used to identify multi-word units, as Gries (this volume) argues convincingly.

As regards the scope of the field, there can be little doubt that the new types of units uncovered by corpus-based approaches need to be fully incorporated into the mainstream of phraseology. Overemphasis on fixedness and semantic non-compositionality has tended to obscure the role played by a wide range of recurrent and co-occurrent units which are fully regular, both syntactically and semantically, and yet clearly belong to the field of phraseology. The crucial role played by these units in language is beginning to be recognized, as evidenced by a range of new publications (cf. Siepmann 2006; Gilquin et al. 2007; Pecman 2008). This remodelling of the field should go hand in hand with a better appropriation of the highly-developed analytical instruments provided by the traditional approach. Combining the best of the two worlds is the surest way of giving phraseology the place it deserves in linguistic theory and practice.

Acknowledgements

We gratefully acknowledge the support of the Communauté française de Belgique, which funded this research within the framework of the 'Action de recherche concertée' project entitled 'Foreign Language Learning: Phraseology and Discourse' (No. 03/08-301).

References

Allerton, D. J. (1984). Three (or four) levels of word co-occurrence restriction. *Lingua* 63: 17–40.

Allerton, D. J., N. Nesselhauf & P. Skandera (eds.) (2004). *Phraseological Units: Basic Concepts and their Application.* Basel: Schwabe.

Altenberg, B. (1998). On the phraseology of spoken English: The evidence of recurrent word-combinations. In Cowie, A. P. (ed.) *Phraseology. Theory, Analysis, and Applications*, 101–122. Oxford: Oxford University Press.

Bally, C. (1909). *Traité de stylistique française*. Paris: Klincksieck.

Barkema, H. (1996). Idiomaticity and terminology: A multi-dimensional descriptive model. *Studia Linguistica* 50(2): 125–160.

Barnbrook, G. (1996). *Language and Computers*. Edinburgh: Edinburgh University Press.

Bartsch, S. (2004). *Structural and Functional Properties of Collocations in English*. Tübingen: Gunter Narr Verlag.

Benson, M., E. Benson & R. Ilson (1986). *The Lexicographic Description of English*. Amsterdam: John Benjamins.

Biber, D. (2004). Lexical bundles in academic speech and writing. In Lewandowska-Tomaszczyk, B. (ed.) *Practical Applications in Language and Computers (PALC 2003)*, 165–178. Frankfurt: Peter Lang.

Biber, D. & S. Conrad (1999). Lexical bundles in conversation and academic prose. In Hasselgård, H. & S. Oksefjell (eds.) *Out of Corpora: Studies in Honour of Stig Johansson*, 181–190. Amsterdam: Rodopi.

Biber, D., S. Conrad & V. Cortes (2003). Lexical bundles in speech and writing: An initial taxonomy. In Wilson, A., P. Rayson & T. McEnery (eds.) *Corpus Linguistics by the Lune: A Festschrift for Geoffrey* Leech, 71–92. Frankfurt: Peter Lang.

Biber, D., S. Conrad & V. Cortes (2004). *If you look at . . ..*: Lexical bundles in university teaching and textbooks. *Applied Linguistics* 25(3): 371–405.

Biber, D., S. Johansson, G. Leech, S. Conrad & E. Finegan (1999). *Longman Grammar of Spoken and Written English.* Longman: Harlow.

Burger, H. (1998). *Phraseologie. Eine Einführung am Beispiel des Deutschen*. Berlin: Erich Schmidt.

Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) (2007a). *Phraseology: An International Handbook of Contemporary Research*. Berlin: Mouton de Gruyter.

Burger H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (2007b). Phraseology: Subject area, terminology and research topics. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseology: An International Handbook of Contemporary Research*, 10–19. Berlin: Mouton de Gruyter.

Cowie, A. P. (1981). The treatment of collocations and idioms in learners' dictionaries. *Applied Linguistics* 2(3): 223–235.

Cowie, A. P. (1988). Stable and creative aspects of vocabulary use. In Carter, R. & M. J. McCarthy (eds.) *Vocabulary and Language Teaching*, 126–137. London: Longman.

Cowie, A. P. (1994). Phraseology. In Asher, R. E. (ed.). *The Encyclopedia of Language and Linguistics*, 3168–3171. Oxford: Oxford University Press.

Cowie, A. P. (ed.) (1998). *Phraseology: Theory, Analysis and Applications*. Oxford: Oxford University Press.

Cowie, A. P. (1998a). Phraseological dictionaries: Some East-West comparisons. In Cowie, A. P. (ed.) *Phraseology: Theory, Analysis and Applications*, 209–228. Oxford: Oxford University Press.

Cowie, A. P. (1998b). Introduction. In Cowie A. P. (ed.). *Phraseology: Theory, Analysis and Applications,* 1–20. Oxford: Oxford University Press.

Cowie, A. P. (2001). Speech formulae in English: problems of analysis and dictionary treatment. In van der Meer, G. & A. G. B. ter Meulen (eds.) *Making Senses: From Lexeme to Discourse. In Honor of Werner Abraham*, 1–12 [Groninger Arbeiten zur germanistishen Linguistik 44]. Groningen: Center for language and Cognition.

Cowie, A. P. (2005). Review of S. Nuccorini (ed.) *Phrases and Phraseology – Data and Descriptions*. *International Journal of Lexicography* 18(1): 103–106.

Cowie, A., R. Mackin & I. R. McCaig (1983). *Oxford Dictionary of Current Idiomatic English.* Oxford: Oxford University Press.

De Cock, S. (2003). Recurrent Sequences of Words in Native Speaker and Advanced Learner Spoken and Written English. PhD dissertation, Université catholique de Louvain.

De Cock, S. (2004). Preferred sequences of words in NS and NNS speech. *Belgian Journal of English Language and Literatures (BELL)*, New Series 2: 225–246.

Evert, S. (2004). The Statistics of Word Cooccurrences: Word Pairs and Collocations. PhD dissertation, University of Stuttgart.

Gaatone, D. (1997). La locution: Analyse interne et analyse globale. In Martins-Baltar, M. (ed.) *La locution entre langue et usages*, 165–177. Fontenay-Saint Cloud: ENS éditions.

Giegerich, H. Z. (2004). Compound or phrase? English noun-plus-noun constructions and the stress criterion. *English Language and Linguistics* 8(1): 1–24.

Giegerich, H. Z. (2005). Associative adjectives and the lexicon-syntax interface. *Journal of Linguistics* 41: 571–591.

Gilquin, G., S. Granger & M. Paquot (2007). Learner corpora: The missing link in EAP pedagogy. In Thompson, P. (ed.) *Corpus-based EAP Pedagogy*. Special issue of *Journal of English for Academic Purposes* 6(4): 319–335.

Gläser, R. (1986). *Phraseologie der englischen Sprache*. Tübingen: Max Niemeyer.

Gläser, R. (1998). The stylistic potential of phraseological units in the light of genre analysis. In Cowie A. P. (ed.) *Phraseology. Theory, Analysis, and Applications*, 125–143. Oxford University Press: Oxford.

Gramley, S. & K.-M. Pätzold (1992). *A Survey of Modern English.* London: Routledge.

Gross, G. (1996). *Les expressions figées en français: Noms composés et autres locutions*. Paris: Ophrys.

Heid, U. (2002). Collocations in lexicography. Presentation given at Colloc02, workshop on computational approaches to collocations, 23 August 2002. Austria: Vienna. Retrieved October 2007 from http://www.ofai.at/~brigitte.krenn/colloc02/workshop_prog.html

Herbst, T., D. Heath, I. Roe & D. Götz (2004). *A Valency Dictionary of English: A Corpus-Based Anaysis of the Complementation Patterns of English Verbs, Nouns and Adjectives.* Berlin: Mouton de Gruyter.

Hoey, M. (2005). *Lexical Priming: A New Theory of Words and Language*. London: Routledge.

Hunston, S. (2002). *Corpora in Applied Linguistics*. Cambridge: Cambridge University Press.

Kilgarriff, A., P. Rychly, P. Smrz & D. Tugwell (2004). The Sketch Engine. In Williams, G. & S. Vessier (eds.) *Proceedings of the Eleventh EURALEX International Congress*, 105–116. Lorient: Université de Bretagne-Sud.

Krishnamurthy, R. (1987). The process of compilation. In Sinclair, J. (ed.). *Looking Up. An Account of the COBUILD Project in Lexical Computing*, 62–85. London: Collins ELT.

Lewis, M. (1993). *The Lexical Approach: The State of ELT and a Way Forward.* Hove: Language Teaching Publications.

Louw, B. (1993). Irony in the text or insincerity in the writer? In Baker, M., G. Francis & E. Tognini-Bonelli (eds.) *Text and Technology: In Honour of John Sinclair*, 157–176. Amsterdam: John Benjamins.

Louw, B. (2000). Contextual prosodic theory: Bringing semantic prosodies to life. In Heffer, C., H. Sauntson & G. Fox (eds.) *Words in Context: A Tribute to John Sinclair on his Retirement*. Birmingham: University of Birmingham.

Lyons, J. (1968). *Introduction to Theoretical Linguistics*. Cambridge: Cambridge University Press.

Manning, C. & H. Schütze (1999). *Foundations of Statistical Natural Language Processing*. Cambridge, MA: The MIT press.

McEnery, A., R. Xiao & Y. Tono (2006). *Corpus-based Language Studies: An Advanced Resource Book*. London: Routledge.

Mejri, S. (2005). Introduction: polysémie et polylexicalité. In Mejri, S. (ed.) *Polysémie et Polylexicalité. Syntaxe et Sémantique* 5: 13–30.

Mel'čuk, I. (1995). Phrasemes in language and phraseology in linguistics. In Everaert, M., E. J. Van der Linden & A. Schenk (eds.) *Idioms: Structural and Psychological Perspectives*, 167–232. Hillsdale, NJ: Lawrence Erlbaum Associates.

Mel'čuk, I. (1998). Collocations and lexical functions. In Cowie, A. P. (ed.) *Phraseology. Theory, Analysis, and Applications*, 23–53. Oxford: Oxford University Press.

Montoro del Arco, E. T. (2006). *Teoría fraseológica de las locuciones particulares: Las locuciones prepositivas, conjuntivas y marcadoras en español*. Frankfurt: Peter Lang.

Moon, R. (1998). Frequencies and forms of phrasal lexemes in English. In Cowie, A. P. (ed.) *Phraseology. Theory, Analysis, and Applications*, 79–100. Oxford: Oxford University Press.

Nattinger, J. R. & J. S. DeCarrico (1992). *Lexical Phrases and Language Teaching.* Oxford: Oxford University Press.

Nesselhauf, N. (2004). What are collocations? In Allerton, D. J., N. Nesselhauf & P. Skandera (eds.) *Phraseological Units: Basic Concepts and their Application,* 1–21. Basel: Schwabe.

Palmer, H. E. (1933). *Second Interim Report on English Collocations*. Tokyo: Kaitakusha.

Partington, A. (2004). "Utterly content in each other's company": Semantic prosody and semantic preference. *International Journal of Corpus Linguistics* 9(1): 131–156.

Pecman, M. (2008). Compilation, formalisation and presentation of bilingual phraseology: Problems and possible solutions. In Meunier, F. & S. Granger (eds.) *Phraseology in Foreign Language Learning and Teaching*. Amsterdam: John Benjamins.

Renouf, A. & J. Sinclair (1991). Collocational frameworks in English. In Aijmer, K. & B. Altenberg (eds.) *English Corpus Linguistics: Studies in Honour of Jan Svartvik,* 128–143. London: Longman.

Sag, I., T. Baldwin, F. Bond, A. Copestake & D. Flickinger (2002). Multi-word expressions: A pain in the neck for NLP. In *Proceedings of the Third International Conference on Intelligent Text Processing and Computational Linguistics* (CICLING 2002), Mexico City, 1–15.

Schmid, H.-J. (2003). Collocation: Hard to pin down, but bloody useful. *ZAA* 51(3): 235–258.

Scott, M. & C. Tribble (2006). *Textual Patterns: Key Words and Corpus Analysis in Language Education*. Amsterdam: John Benjamins.

Siepmann, D. (2006). Collocation, colligation and encoding dictionaries. Part II: lexicographical aspects. *International Journal of Lexicography* 19(1): 1–39.

Sinclair, J. (ed.) (1987). *Looking Up. An Account of the COBUILD Project in Lexical Computing.* London: Collins Cobuild.

Sinclair, J. (1991). *Corpus, Concordance, Collocation*. Oxford: Oxford University Press.

Sinclair, J. (1996) The search for units of meaning. *TEXTUS* IX: 75–106.

Sinclair, J. (1998). The lexical item. In Weigand, E. (ed.) *Contrastive Lexical Semantics*, 1–24. Amsterdam: John Benjamins.

Sinclair, J. (2004). Trust the text. In Sinclair, J. & R. Carter (eds.) *Trust the Text – Language, Corpus and Discourse*, 9–23. London: Routledge.

Stefanowitsch, A. & S. Gries (2003). Collostructions: Investigating the interaction between words and constructions. *International Journal of Corpus Linguistics* 8(2): 209–243.

Stubbs, M. (1983). *Discourse Analysis: The Sociolinguistic Analysis of Natural Language.* Oxford: Basil Blackwell.

Stubbs, M. (1995). Collocations and semantic profiles: On the cause of the trouble with quantitative studies. *Functions of Language* 2(1): 23–55.

Stubbs, M. (2002). Two quantitative methods of studying phraseology in English. *International Journal of Corpus Linguistics* 7(2): 215–244.

Stubbs, M. (2007a). Quantitative data on multi-word sequences in English: The case of the word 'world'. In Hoey, M., M. Malhberg, M. Stubbs & W. Teubert (eds.) *Text, Discourse and Corpora: Theory and Analysis*. London: Continuum.

Stubbs, M. (2007b). An example of frequent English phraseology: Distribution, structures and functions. In Facchinetti, R. (ed.) *Corpus Linguistics 25 Years on*, 89–105. Amsterdam: Rodopi.

Stubbs, M. & I. Barth (2003). Using recurrent phrases as text-type discriminators: A quantitative method and some findings. *Functions of Language* 10(1): 61–104.

Svensson, M. H. (2002). Critères de figement et conditions nécessaires et suffisantes. *Romansk Forum* 16(2): 777–783.

Tognini-Bonelli, E. (2001). *Corpus Linguistics at Work*. Amsterdam: John Benjamins.

Tognini-Bonelli, E. (2002). Functionally complete units of meaning across English and Italian: Towards a corpus-driven approach. In Altenberg, B. & S. Granger (eds.) *Lexis in Contrast: Corpus-based Approaches*, 73–95. Amsterdam: John Benjamins.

Tschichold, C. (2000). *Multi-word Units in a Lexicon for Natural Language Processing*. Olms: Hildesheim.

Woolard, G. (2000). Collocation – encouraging learner independence. In Lewis M. (ed.). *Teaching Collocation: Further Developments in the Lexical Approach*, 28–46. Hove: Language Teaching Publications.

Wray, A. (2002). *Formulaic Language and the Lexicon*. Cambridge: Cambridge University Press.

Wray, A. & M. Perkins (2000). The functions of formulaic language: An integrated model. *Language and Communication* 20: 1–28.

Zuluaga, A. (1980). *Introducción al estudio de las expresiones fijas*. Frankfurt: Peter Lang.

A unified approach to semantic frames and collocational patterns

Willy Martin

The aim of this chapter is to show that frames cannot only offer advantages in explaining the meaning of words (as is usually claimed), but also in clarifying their combinatorial behaviour. In order to do so, different kinds of frames are presented, ranging from the more language-oriented ones à la Fillmore to the more knowledge-in-general oriented ones à la Minsky to finally end up with what I call 'conceptual semantic frames'. The latter do not only serve as a background for definitions but for word combinations and more particularly collocations as well. In the frame-based approach presented here I try to show that the more the collocator is, conceptually speaking, type-bound and the more it is, lexically speaking, token-bound, the more we are dealing with a collocation that forms a conceptual and lexical unit and therefore qualifies as a lexical collocation.

1. Introduction

In a paper entitled 'Double-decker definitions: The role of frames in meaning explanations', Fillmore (2003: 263) maintains that "certain kinds of structured background information (or 'frames') should be treated as essential components or accomplishments of word definitions". Mutatis mutandis in what follows, I would like to show that frames can offer advantages not only in explaining the meaning of words but also in clarifying their combinatorial behaviour.

2. Types of frames

There exist at least two 'schools' in frame linguistics, the one more language (i.e. syntax) oriented (Fillmore and colleagues), the other more 'knowledge' (i.e. cognition) oriented (e.g. Minsky).[1] The Fillmorian school eventually led to the FrameNet project

1. Frames as organisational devices are used not only in linguistics and cognitive studies but also in political and economic studies. For the latter see, for example, *Choices, Values and Frames*, ed. by D. Kahneman and A. Tversky, Cambridge/New York: Cambridge University Press, 2000.

Table 1. Example of a 'Fillmorian' frame (based on Fillmore, Johnson & Petruck 2003)

SELL	
ISA	commercial transaction
SELLER	who sells?
GOODS	what is sold?
BUYER	to whom is sold?
CONDITION	for what is sold?

in which semantic frames of the *sell* type, for instance, were structured (see Table 1) and defined as: "schematic representations of situations involving various participants, props and other conceptual roles, each of which is a frame element. The semantic arguments of a predicating word correspond to the frame elements of the frame or frames, associated with that word" (from the FrameNet website[2]). Linked to the frame elements are their syntactic realizations (eg. PrepNP, NP etc. for BUYER). In the words of the FrameNet makers: "FrameNet identifies and describes semantic frames, and analyzes the meanings of words by directly appealing to the frames that underlie their meanings and studying the syntactic properties of words by asking how their semantic properties are given syntactic form" (Fillmore et al. 2003: 235).

Contrasting with Fillmore's syntactico-semantic frame is the knowledge frame that Minsky introduced in the mid-seventies in the domains of computational linguistics and artificial intelligence. As the name suggests, Minsky's aim was to develop a model "that would enable to represent in a formalised way knowledge aspects linked to human perception and the understanding of texts. This knowledge then would become available for computers as well" (Wermuth 2005: 159–160; my translation from Dutch into English). Although Fillmore subsequently moved from small-scale frames (such as those used in Case Grammar) towards more large-scale cognitive ones as introduced by Minsky, differences in orientation and depth between the two approaches still remain (see Fillmore 2003: 288).

In my own search for an adequate lexical model I have tried to reconcile the more general organisational Minskyan approach to knowledge with the more specific lexicological aim of Fillmore. First of all I have taken over from Minsky the form, viz. the slot-filler format where slots represent general conceptual relations and fillers the specifications of these general categories. As to the contents, Minsky sees frames as a kind of expectation pattern that is evoked or adapted in a new situation. In Minsky's own words: "A frame is a data structure for representing a stereotyped situation (...). It is a collection of questions to be asked about a hypothetical situation. It can be viewed as an organised matrix of slots for given states of affairs" (Minsky 1977: 355). These states of affairs can be quite diverse with Minsky. For instance, they can be entities such as 'living room' or events (scenes) such as 'birthday party'. What the two entities have in

2. FrameNet website: http://framenet.icsi.berkeley.edu/~framenet

Table 2. Example of a 'conceptual semantic' frame (based on Martin 2003: 14)

MUSICAL INSTRUMENT	
SLOTS	FILLERS
ISA	wind/percussion/stringed/plucked/keyboard . . . instrument
FUNCTION	special function other than musical
MATERIAL	
SIZE	as compared to other members of the (sub)type
FORM	
PARTS	
MATERIAL	
SIZE	
FORM	
NUMBER	
ACCESSORIES	
MATERIAL	
SIZE	
FORM	
NUMBER	
SOUND	typical sound produced
USER	typical user
MANNER OF PLAYING	
POSITION OF PLAYER	
POSITION OF INSTRUMENT	vis-à-vis player
MECHANISM	
ORIGIN	
TYPICAL GENRE	the instrument is used in
SIMILARITY	similar to

common is that they can only be fully understood in a frame, a context, a background of common knowledge based on cultural conventions.

Mainly inspired by Minsky, I have tried to represent the conceptual meaning of lexical items by means of what I call 'conceptual semantic frames'. To illustrate what I mean, see the frame for musical instruments in Table 2. This shows the most important features of these frames, viz.:

– Conceptual semantic frames are knowledge clusters that represent the conceptual meaning of words in a slot-filler format; they do not make a strict distinction between linguistic and encyclopaedic knowledge, but reflect the subjective, stereotypical knowledge of language users as evidenced in their usage (both passive and active) of language.
– Conceptual semantic frames are definition models: they define the meaning of words (as far as possible) by means of relations (slots) with other words, i.e. lexicalised concepts.

Table 3. Semantic frame for *violin* using *Collins Cobuild English Dictionary for Advanced Learners, 3rd edition* (2001) as a knowledge source

VIOLIN	
ISA	musical instrument
MATERIAL	wood
PARTS	
	string
	NUMBER: four
ACCESSORIES	bow
MANNER-OF-PLAYING	hold violin under chin and move bow across strings

– Conceptual semantic frames are type- or category-bound. In this respect the frame in Table 2 is an empty, underspecified frame, the fillers only containing clarifications or pick lists. A frame can only get its specific fillers when dealing with a concrete token or member of the category/type. So, for instance, the frame for the token *violin* can be specified as shown in Table 3.

Using conceptual semantic frames transforms the lexicon into one large relational network, frame or web with each word having its own vertical (paradigmatic) and horizontal (syntagmatic) links. In the next section I will deal with these horizontal, i.e. combinatorial, links in particular.

3. Frames and the derivation of collocations

3.1 Frames and collocational patterns

The advantages of a frame-based approach to the lexicon lie not only at the level of representation (cf. Section 2), but also at the level of production. In particular, frames can be extremely useful for producing collocations. As is well-known, constructing a collocational dictionary or database is not an easy task. It does not suffice to take a corpus and select the most frequent/statistically significant combinations from it. Apparently there is no simple one-to-one relationship between frequency/statistics and relevancy. Frames can be of help here in that they can evoke a collocational pattern. When dealing with the frame for musical instruments, it becomes obvious that, as with other artefacts, the function-slot takes a prominent place in defining the concept. Corresponding to this definitional slot there is a collocational counterpart that can be evoked. The following examples taken from German illustrate this. We start from the generally accepted assumption that collocations consist of two parts: a base (the part that is specified) and a collocator (the part that specifies):

Base	**Collocator**
[MI = musical instrument]	[F = typical functioning]
↓	↓
the instantiation of a particular MI	the lexicalization of a typical functioning of the particular MI
↓	↓
der Gong	ertönt/schlägt
(E. gong)	(E. sounds)

From the slots given for musical instruments in Table 2, the following 'collocations' can now be derived:

– from the slot FUNCTION:

→ [MI] [typically functions]
e.g. der Gong ertönt/schlägt (the gong sounds)
→ [MI] does not function
e.g. der Gong schweigt (the gong is silent)
→ X [causes to function better] [MI]
e.g. eine Geige stimmen (to tune a violin)
→ [MI] [functions well]
e.g. die Geige hat einen guten Klang (the violin sounds pure)
→ [MI] [typically malfunctions]
e.g. die Geige klingt falsch (the violin sounds out of tune)

– from the slot SOUND:

→ [MI] [makes typical sound]
e.g. die Trompete schmettert (the trumpet blares)

– from the slot USER:

→ X [typically makes use of] [MI]
e.g. (auf der) Trompete blasen (to blow the trumpet)

As may be clear from the preceding discussion, the knowledge stored in a (type-bound) frame not only helps us to understand utterances better, it also helps us to produce utterances, in particular collocations. Of course, not all the combinations mentioned above are considered to be collocations to the same degree, but here too frames can be of help in differentiating between them.

3.2 Frames and collocations

To understand which combinations are a subset of collocational patterns and therefore qualify as collocations, we need to define what collocations are. Although Mel'čuk rightly points out that there is "no universally accepted formal definition of

collocations nor a proposal for their uniform and systematic treatment in dictionaries" (Mel'čuk 1998: 23), most scholars do agree on two points. First, it is generally accepted that collocations consist of two parts: a collocator and a base (see, inter alia, Hausmann 2004 (quoting Schneider) using the term binemes as contrasted with monemes). Second, it is usually agreed that these two elements show a degree of binding/fixation or restriction to each other, thus forming a unit that fits somewhere in between idioms and free combinations. In this respect, some authors use the terms 'semi-phrasemes' (Mel'čuk 1998: 30) or 'encoding idioms' (Croft & Cruse 2004: 250), implying that collocations, like idioms and phrasemes, behave as units.

As it is this latter feature in particular – the degree of binding/fixation/unification – that remains vague and problematic, I will focus on it in some detail, starting from the following working definition.

A collocation is

- a word group consisting of two conceptual elements: a collocator (determinans) and a base (determinatum)
- of a certain syntactic type (N+N, V+N, A+N, Adv+V, Adv+A)
- showing a semantic, type-based, relationship between the two elements
- the more the collocator is, conceptually speaking, type-bound (bound to a conceptual type or category) and the more it is, lexically speaking, token-bound (bound to a lexical token/item), the more we are dealing with a collocation that conceptually and lexically forms a unit, i.e. one that is a lexical collocation.

In the following I will comment on each part of the definition separately.

- A collocation is a word group consisting of two conceptual elements: a collocator (determinans) and a base (determinatum)

By this I mean that a collocation is the combination of two concepts or frames which, as a rule, are in a dependency relation, one (the collocator or modifier) modifying the other (the base or head). So, for instance, in *commit suicide* we will consider *suicide* to be the base or topic triggering the frame. This means that there is a dominant frame, evoked by *suicide* and a dependent one (*commit*) zooming in on a specific aspect of the dominant frame.[3] Defining collocations as the combination of two concepts or frames distinguishes them from idioms, which, although showing a compositional structure, contain only one concept or frame.[4]

- of a certain syntactic type (N+N, V+N, A+N, Adv+V, Adv+A)

3. Some authors, such as Poulsen 2005, prefer to speak of an 'interdependent' relationship, even if there is a bias towards/dominance of one of the elements.

4. Notice, however, that collocations can be recursively defined so that one collocation can be embedded in another and thus lead to more than just two elements, as is the case in (take ((strong) (measures))).

This part of the definition implies that:

a. We call the combinations we are dealing with lexical in the sense that they combine lexical items; combinations of a lexical item and a function word (preposition, conjunction etc.) are not taken into account here and will be regarded as grammatical collocations.
b. The order in which the elements are given reflects a semantic dependency relationship, not a syntactical one: the left member is dependent on the right one. In other words, the right member is considered to be the base (being the independent or dominant frame), the left member the collocator (the dependent frame).
c. The dominant item here is the one that most strongly evokes the frame. Which item is independent (dominant) and which is not is often a matter of dispute (see e.g. Poulsen 2005: 271). In this discussion I take a functional point of view: in a combination in which a noun and a verb occur, it is the noun, as a rule, that triggers the frame functioning as the topic, while the verb fulfils the role of comment.[5]

– showing a semantic, type-based, relationship between the two elements

In a frame-based approach the combinatorial possibilities of lexical units (LUs) to form constructions such as compounds and collocations are defined by the conceptual structure of the component LUs. In other words, given two LUs, X and Y, each with their own conceptual frame, X and Y will only combine and unify if the modifying dependent LU fits the slot of the modified dominant LU, thus specifying, among other things, the meaning of the latter. In this respect there is no fundamental difference in combinatorial behaviour between collocations and (a substantial subset of) compounds, although their function may differ, collocations having a more characterising and compounds a more categorising function.[6]

– the more the collocator is, conceptually speaking, type-bound (bound to a conceptual type or category) and the more it is, lexically speaking, token-bound (bound to a lexical token/item), the more we are dealing with a collocation that conceptually and lexically forms a unit, i.e. a lexical collocation.

Although most definitions of collocations mention a degree of binding as a characteristic feature, they usually remain rather vague in their elaboration of this phenomenon. In fact, the more collocational a collocation is, the more its collocator is, conceptually

5. In this respect it should be noted that the more concrete the meaning of the verb, the easier it is to consider it the topic of the discourse. Compare for instance *take* in *take measures* (where *take* is an abstract support verb) with *take the car* (*take* = make use of) and *take a book* (*take* = grasp). In the last case (*take a book*) one can argue that *take* triggers the frame and *book* is one of the many fillers of the slot/class of 'graspable objects'.

6. See, for example, Feilke 2004 who states, referring to Barz 1996, that 'die Kollokation eher charakterisiert, das Kompositium eher kategorisiert (z.B. *dicke Milch* vs. *Dickmilch*).' (Feilke 2004: 54) [collocations rather characterise while compounds rather categorise].

speaking, type-bound and, lexically speaking, token-bound. This can only be understood fully against the background of frames with their slot-filler format. In a concrete combination such as *drink coffee*, for instance, *drink* acts as a filler for the slot 'WAY OF CONSUMING', a slot that is typical for the type DRINK. In other words, for the meaning of the word *coffee*, it is very relevant, just as it is for other drinks, that we *drink* it. On the other hand, the word *drink* is not bound to the word *coffee* in an exclusive way. As a matter of fact, *drink* is a default value for all drinks and so it is more type- than token-bound. As a consequence, although there is a binding between the elements of *drink coffee*, this binding is less strong than that between *weak* and *coffee*. This could suggest that lexical combinatorics are subject to (type-bound) rules and exceptions. However, linguistic reality is less straightforward. Besides rules and exceptions, the productivity/generality of rules also plays a role, as do preferences and prototypicality. I discuss these phenomena below, illustrating them with Dutch examples (also see Martin 2006 and Martin 2007 for a more detailed discussion).

3.2.1 *Type-bound regularity*

Type-bound regularity in this context means that the lexicalisation of the collocator is not an isolated phenomenon only applying to the base in question, but also characterises the other members of the type the base belongs to. So, for instance, one may suppose that the type 'drink that has to be prepared' has a 'rule' in Dutch that stipulates that the slot 'PREPARE' is realized through the collocator (*klaar*) *maken*. As expected, in Dutch we find words such as *cocktail*, *soep* (E. *soup*), *bouillon* (E. *broth*), *gin-tonic, grog* etc. to which the rule applies, realizing the slot 'PREPARE' by means of the filler *maken*.[7] However, if we take *koffie* (E. *coffee*) or *thee* (E. *tea*) as possible tokens of the type mentioned, we see that Dutch uses *zetten* here instead of *maken*. To explain this, we can either take the stance that *koffie* and *thee* belong to a much more restricted type than the one presented above, viz. something like 'a drink that has to be prepared according to a certain very specific procedure, i.e. by pouring hot water over leaves (tea) or ground beans (coffee)' and that therefore another rule applies, according to which 'PREPARE' is realized as *zetten.* Or we can adopt the point of view that rules must have a sufficient degree of productivity/generality to be efficient/useful/acceptable. As this is not true for *koffie/thee zetten*, we will regard the collocator as not being predictable by a (more general) type-bound rule, but as an exception to such a rule. In other words, *zetten* in the case of *koffie/thee* will be taken to be token- rather than type-bound.

3.2.2 *Preferences*

In the preceding discussion I suggested that the less productive the rules are that govern the lexicalization of collocators, the more these lexicalisations can be regarded as token- rather than type-bound. There exists moreover another kind of token-

7. In Dutch, as well as *maken* (E. *make*), we can also use *mixen* (E. *mix*) for *cocktail* and *trekken* (E. *make*) for *soep* (E. *soup*) and *bouillon* (E. *stock*).

boundness. Consider the following case: in Dutch we have many words which belong to the type 'social allowance'. For instance: *salaris* (E. *salary*), *uitkering* (E. *allowance*), *pensioen* (E. *pension*), *kinderbijslag* (E. *child benefit*), *vakantietoeslag* (E. *holiday allowance*) etc. All these tokens have the slot 'SIZE' (the amount of the allowance) and select a filler/collocator from a set of intensifiers such as *groot, goed, hoog* (for 'high') and *klein, laag* (for 'low').[8] However, if we take a closer look at these examples, we see that there are differences in the preferred combinatorial behaviour of the members. *Salaris*, for instance, prefers the company of *goed, hoog* and *riant* (for 'high'), and *klein, laag*, not *slecht* (for 'low'); *pensioen* goes with *goed* and *groot*, less often with *hoog* and *riant* (for 'high'), and is only used with *klein* (for 'low'), etc. What is clear here is that, when different collocators can occur, their distribution over the various tokens can lead to different predilections/preferences. In as far as this occurs, the combinatorial behaviour is only partially (if at all) type-bound and so expresses another form of token-boundness.

3.2.3 *Prototypicality*

From this discussion a certain lexical categorisation in which rules, constraints and preferences play a role, emerges. Another organising principle that we have to take into account when dealing with collocations is that of prototypicality. Consider the following concrete illustration: given a type 'liquid to be consumed', one expects a slot '(TYPICAL) WAY OF CONSUMING'. In Dutch this slot is filled by *drinken* (E. *drink*). *Drinken* is thus a typical type-bound filler/collocator in this respect. Yet, depending on the socio-cultural context, combinations as *wijn* (E. *wine*) *drinken*, *wodka* (E. *vodka*) *drinken*, *bier* (E. *beer*) *drinken*, *koffie* (E. *coffee*) *drinken*, *thee* (E. *tea*) *drinken* etc. can all, to a greater or lesser extent, also be considered to be token-bound. This is so because the rule component of the lexicon need not just be taken to consist only of type-bound abstract rules, but can also include concrete cases, and good prototypical/illustrative token-bound examples for the rule in question. As one of my PhD students, Jeroen Redel, has put it (personal communication): "this leads to the collocational paradox: importing type-bound combinations into the more token-bound category via prototypes". In this sense *koffie drinken*, which is a typical type-bound collocation, can also be considered token-bound, in as far as it acts as a prototypical example. In the same vein, prototypical examples of free combinations can 'intrude' into semantic frames and thus become type-bound. All in all, this leads to a typology and model such as that presented and illustrated in Section 3.3 below.

---

8. The choice of the collocator is, of course, also grammatically defined. Uncountables such as *kinderbijslag* (E. *child benefit*) only take quantifiers such as *veel* (E. *much*) and *weinig* (E. *little*).

3.3 Typology, model and illustration

3.3.1 *Introduction*

The typology for collocational patterns presented in Figure 1 emerges from the preceding discussion. Although this typology reflects the well-known idea of collocations as 'in-betweens' situated on a continuum, it does so in a dynamic way: type-bound collocations 'intruding' into the token-bound class via prototypes (e.g. *koffie drinken* (E. *drink coffee*)) and free (non-bound) combinations doing the same with the type-bound class (e.g. *koffie inschenken* (E. *pour out coffee*)), as shown in Figure 2.

In the next section I will take up the Dutch word *koffie* (E. *coffee*) to illustrate the model. Of course, if the aim is to scale things up, a whole system of types, subtypes, subsubtypes etc. is needed together with their frames (slots plus (default) fillers). Therefore, I give below some heuristic clues which may serve as short cuts for rapidly selecting token-bound lexical collocations from the set of potential ones. The following clues can be used:

– The vaguer the collocators, the more difficult it is to predict (from the type) their occurrence. Therefore, as a rule, light verbs (such as *do* a favour, *take* a look, *have* breakfast etc.) will be token-bound collocators.

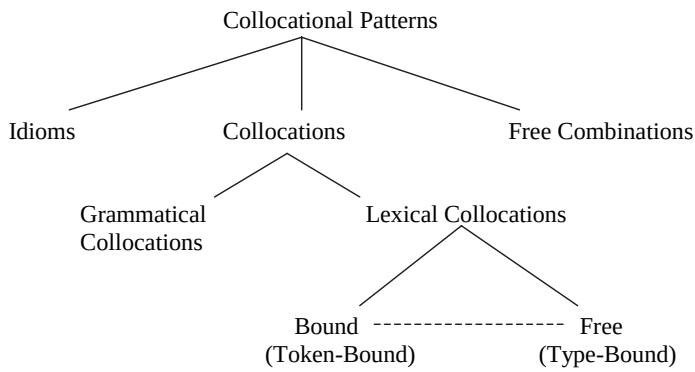


Figure 1. Types of collocational patterns

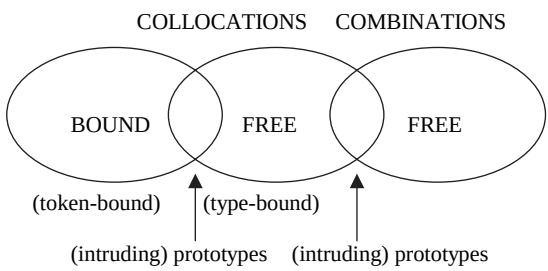


Figure 2. Dynamic model for lexical collocations

– The more fillers there are for a certain slot (of a certain type), the more difficult it is to predict their distribution and the preferences that play a role. As a consequence, there is a high probability that, for example, intensifiers will be token-bound.
– The lexical functions as found in the work of Mel'čuk and his collaborators (see, for instance, Mel'čuk 1998 and Mel'čuk & Zholkovsky 1988) and, in particular, the syntagmatic standard lexical functions, have both a wide range of application (they are general) and a strong impact on the meaning of the base-word. This means that they typically yield type-bound fillers. However, the latter may result in token-bound ones when they are strongly restricted and/or preferred.

In the following section I illustrate the model by taking all combinations of the word *koffie* (E. *coffee*) as they occur in a well-known database for Dutch, the Referentie Bestand Nederlands (RBN) (see Martin & Maks 2005), characterising them according to the categories mentioned in my typology/model. Figure 3 presents the combinations in diagram format.

3.3.2 *Examples and characterisation*

Idioms and grammatical collocations

One of the data categories to be found in the RBN is called 'combinatorics'. In fact, this section covers more than just lexical collocations. It also includes idioms, pragmatic formulae, proverbs, clichés, multi-word terms and the like, thus bringing this section closer to phraseologisms in the broad sense, for which Schmitt and Carter (2004) use the term 'formulaic sequences'. I will start by examining two examples that occur in the combinatorics section but, stricto sensu, fall outside my central object of investigation.

– *dat is geen zuivere koffie* (E. *this looks suspicious;* lit. this is not pure coffee).

What we have here is not a collocation, but an idiom operating as a single frame. It could be argued that this is a figurative expression in which two literal concepts/frames (that of *coffee* and that of *flavour*) are still visible. Figurative expressions thus bridge the gap between idioms in the strict sense and collocations.

– *op de koffie komen (E. to be invited for coffee*).

Koffie/coffee is a polysemous word showing such meaning aspects as: the drink, the product (beans/seeds) the drink is made of, the time when the drink is usually drunk, the container the drink is served in, etc. In its temporal meaning *koffie* is typically used with temporal prepositions such as *na* (E. *after*) and *op* (E. *on*). These prepositions function as fillers for slots that are highly relevant for this particular meaning of *koffie*. As they form combinations of function words plus lexical words they are considered to be grammatical, not lexical, collocations. A highly relevant slot (from a conceptual point of view) in combination with a filler that is difficult to predict (Du. *op*), here yields a bound, albeit grammatical, collocation.

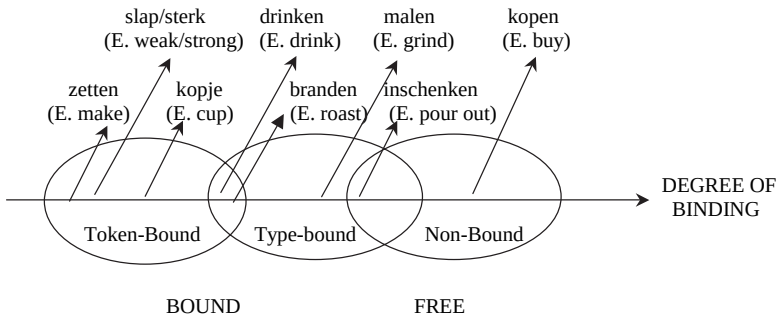


Figure 3. Lexical collocations and combinations with *koffie* (E. *coffee*) as overlapping categories

Token- and type-bound lexical collocations

- *koffie zetten* (E. *make coffee*)
- *slappe/sterke koffie* (E. *weak/strong coffee*)
- *koffie drinken* (E. *drink coffee*)
- *(een) kopje koffie* (E. *(a) cup of coffee*)
- *koffie branden* (E. *roast coffee*)

All these cases are – at the very least – type-bound in a prominent way. The reason for this is that in the knowledge frame, *koffie* (E. *coffee*) as a drink is most prominently specified by the following slots: PREPARE (coffee is a drink that is prepared in a certain way), FLAVOUR (one expects coffee to have a certain prototypical flavour), WAY OF USING/CONSUMING (coffee is consumed in a certain way), CONTAINER (coffee is drunk out of a certain specific container). From the above series *zetten*, *slappe/sterke*, and *kopje* are the most token-bound. *Zetten* because it is idiosyncratic and highly restricted (see Section 3.2.1), *slappe/sterke* because its distribution is irregular,[9] and *kopje* because drinks typically have their own containers in the referential world that is reflected in language usage. In English, we drink a *glass of milk*, a *cup of tea or coffee*, *a bottle of coke* (in Dutch we use the diminutive here: *flesje – little bottle*) etc. This heterogeneity makes it difficult to come up with a general, type-bound, prediction. The more so, as we also drink *coffee* out of *mugs* and *plastic beakers* but still say that we are drinking a *cup* of coffee.

As to *koffie drinken* (E. *drink coffee*) this is typically a type-bound combination (see the discussion in Section 3.2.3). However, as I have argued there, it can 'intrude' into the token-bound class because of the fact that *koffie* (E. *coffee*) is one of our prototypical drinks (see the discussion on prototypicality). Thus, *koffie drinken* (E. *drink coffee*) becomes an in-between or transit case and is situated in the overlapping zone between token- and type-bound collocations in Figure 3.

9. Less common alternatives for *slap* are, for instance, *flauw* and *flets* (E. *weak*, *watery*).

The last example, *koffie branden* (E. *roast coffee*) differs from the preceding ones in as far as it concerns another meaning of *koffie*, viz. the beans or seeds that have to be prepared before they can become 'proper' ingredients for the drink. In this respect there is another frame at play that triggers both *koffie branden* (E. *roast coffee*) and *koffie malen* (E. *grind coffee*), *branden* (E. *roast*) and *malen* (E. *grind*) being typical fillers for the slot PROCESS. The more predictable these fillers are – which is certainly the case for *malen* (E. *grind*) and to a lesser degree also for *branden* (E. *roast*)[10] – the more we are dealing here with collocators that are more conceptually rather than lexically bound, and can therefore be considered to be (more) type-bound.

Free lexical collocations

– *koffie inschenken* (E. *pour out coffee*).

As *inschenken* only has an indirect conceptual link/relationship with *koffie*, viz. via the 'CONTAINER' slot which is presupposed when one pours out a drink, it cannot be regarded as (strongly) type-bound. Therefore its meaning impact is minor and I will consider it to be free, although its lexical filler is not fully predictable.[11]

Free combinations

– *koffie kopen* (E. *buy coffee*)[12]

Although *koffie* is a well known product that is sold and bought much in the Low Countries, it is not a lexical collocation at all. Indeed, as *kopen* no longer can be predicted by a particular slot in the frame of *koffie*, no filler can be formulated/expected here. Instead *koffie kopen* (E. *buy coffee*) is an example of a free combination between *kopen* (E. *buy*) and *koffie* (E. *coffee*), the latter functioning as a non-prototypical filler in the 'GOODS' slot of *kopen*.

4. Concluding remarks and further prospects

In this chapter I have tried to make clear why not only word meanings but also word combinations, in particular, collocations can profit from one and the same frame-based approach. In doing so I have stressed both the representational aspects

10. In Dutch we use *roosteren* (E. *roast*) with *pinda's* (E. *peanuts*) and *branden* (E. *roast*) with *amandelen* (E. *almonds*) for comparable processes.

11. Instead of ***in**schenken*, ***uit**schenken* is also possible (cf. E. *pour* ***out***) but there is a difference in perspective. Yet, as often is the case with 'alternatives', there might also be a semantic difference as *uitschenken* often has a resultative aspect: 'pour out until there is no more left/ until the container is empty'. This might have an impact on the choice. Anyway, in Dutch we prefer to say *inschenken*, at least with *koffie*.

12. This 'free' example does not occur in the RBN under the entry *koffie*.

of collocations by introducing a new model, and the acquisitional aspects by using frames both as an analytical instrument and as a heuristic device.

The new model integrates rules, constraints, preferences and prototypes in its descriptive apparatus and thus enhances the purely quantitative, statistical, corpus-based approach with a more qualitative, cognitive, frame-based one. This approach does not exclude the quantitative approach but complements it. So, for instance, frames as analytical instruments allow us to interpret/structure/analyse collocational candidates as found in corpora or gathered by intuition. Frames, in other words, present a framework for reading the data (see the *koffie* examples). The fact that they do so in a systematic way, using 'degree of binding' with slots and fillers as a guiding principle, leads not only to greater consistency but also to greater explanatory power for the data under study (motivation of the combinatorial data).

As a heuristic device, frames offer the possibility of superseding the data as found in corpora, by intuition or otherwise. In other words, collocational patterns as generated by frames act as expectation patterns, not only for drawing the attention to what (frequently) occurs, but also to what does *not* occur. In this respect the 'generation' of these patterns is at least as important as the selection of lexical collocations as such, as the former is a pre-requisite to the latter.

However, it will have become clear that, in order to make the frame-based approach fully operational, three other things are needed:

1. First, more empirical research is needed in order to build a frame-based lexicon with types, subtypes, frames, slots and fillers. Failing this, a more heuristics-based approach will have to be followed for most languages for the time being.
2. Second, in order to test and validate the relevance of the collocational model/hypothesis presented, psycholinguistic research is needed.
3. When it comes to applications of the approach, be they in language didactics or in language technology, an adapted user model will be needed in order to find out what is redundant/informative given the knowledge level of the intended users (native speakers, machines, second language users, etc.).

Acknowledgements

I would like to thank Dr Hennie van der Vliet and Drs Jeroen Redel for our productive discussions on this subject. I am, of course, solely responsible for all possible errors or misinterpretations.

References

Croft, W. & D. A. Cruse (2004). *Cognitive Linguistics*. Cambridge: Cambridge University Press.

Feilke, H. (2004). Kontext – Zeichen – Kompetenz. Wortverbindungen unter sprachtheoretischem Aspekt. In Steyer, K. (ed.) *Wortverbindungen – mehr oder weniger fest*, 41–64. Berlin: De Gruyter.

Fillmore, C. (1977). Scenes-and-frames semantics. In Zampolli, A. (ed.) *Linguistic Structures Processing*, 55–81. Amsterdam: North Holland.

Fillmore, C. (2003). Double-decker definitions: The role of frames in meaning explanations. *Sign Language Studies* 3(3): 263–295.

Fillmore, C., C. Johnson & M. Petruck (2003). Background to FrameNet. *International Journal of Lexicography* 16(3): 235–250.

Hausmann, F. J. (2004). Was sind eigentlich Kollokationen? In Steyer, K. (ed.) *Wortverbindungen – mehr oder weniger fest*, 309–334. Berlin: De Gruyter.

Kahneman, D. & A. Tversky (eds.) (2000). *Choices, Values and Frames*. Cambridge: Cambridge University Press.

Martin, W. (2003). Definitions and collocations in dictionaries. *Lexicographica*. Series Maior 113: 3–23.

Martin, W. (2006). Een kwestie van frames. Valedictory Lecture, Vrije Universiteit Amsterdam.

Martin, W. (2007). The lexicon is a (kind of) frame. In Miyares, L.R. & al. (eds.) *Actas X simposio internacional communicación social*, 410–418. Santiago de Cuba: Centro de Linguistica Aplicada.

Martin, W. & I. Maks (2005). *Referentie Bestand Nederlands. Documentatie*. Amsterdam: Vrije Universiteit, Research Group Lexicology/Terminology.

Mel'čuk, I. (1998). Collocations and lexical functions. In Cowie, A. (ed.) *Phraseology. Theory, Analysis and Applications*, 23–53. Oxford: Clarendon Press.

Mel'čuk, I. & A. Zholkovsky (1988). The explanatory combinatorial dictionary. In Evens, M. (ed.) *Relational Models of the Lexicon*, 41–74. Cambridge: Cambridge University Press.

Minsky, M. (1977). Frame-system theory. In Johnson-Laird, P. N. & P. C. Watson (eds.) *Thinking: Readings in Cognitive Science*, 355–376. Cambridge: Cambridge University Press.

Poulsen, S. (2005). Collocation as a Language Resource. A Functional and Cognitive Study in English Phraseology. PhD dissertation, University of Southern Denmark.

Schmitt, N. & N. Carter (2004). Formulaic sequences in action: An introduction. In Schmitt, N. (ed.) *Formulaic Sequences*, 1–22. Amsterdam: John Benjamins.

Steyer, K. (ed.) (2004). *Wortverbindungen – mehr oder weniger fest.* Berlin: De Gruyter.

Wermuth, C. (2005). Een framegebaseerde benadering van classificatiesrubrieken: cardiovasculaire rubrieken als case study. PhD dissertation, Vrije Universiteit Amsterdam.

Processing of idioms and idiom modifications

A view from cognitive linguistics

Marija Omazić

This chapter attempts to establish the role of two cognitive linguistic theories, metaphor theory and conceptual integration theory, in the processing of both conventional and modified phraseological units. It investigates the range of applications and interpretations of the existing cognitive models, furnishing them with attested phraseological material to test their efficiency and applicability as processing templates. It is argued that the two theories can be viewed as complementary: whereas the theory of metaphor is productive in accounting for the processing of a wide range of conventional phraseological units, the theory of conceptual integration can be applied to the analysis of modified phraseological units.

1. Introduction

Research on figurative language processing has always been challenging, and at times even controversial. Some of the existing models of idiom interpretation are the standard pragmatic model, idiom list hypothesis, lexical representation hypothesis, direct access view or figurative thought model (Gibbs 1994), phrase-induced polysemy model of idiom comprehension (Glucksberg 1993: 11), graded salience hypothesis (Giora 1997), and the idiom decomposition approach (Gibbs 1995: 97–116).

Two of the more recent developments stemming from the psychological and cognitive traditions are the constraint satisfaction model (Katz & Feretti 2001) and the space structuring model (Coulson & Matlock 2001). In the constraint satisfaction model the best interpretation of a figurative expression is the one that offers the most coherent account of what a speaker or writer is saying, taking into account both linguistic and non-linguistic information. The space structuring model, on the other hand, is more in line with the cognitive conceptual integration theory or blending theory (Fauconnier & Turner 1998), involving more than just simple mappings between source and target domains.

The aim of this chapter is to extend the application of the conceptual integration theory to idiom modification processing in particular. Section 2 provides a summary of relevant aspects of the theory of metaphor and metonymy and the conceptual

integration theory. In Section 3 we discuss the processing of idioms, followed by a discussion of idiom modification processing in Section 4, with concluding remarks in Section 5.

2. Theory of metaphor and metonymy and conceptual integration theory

The theory of conceptual metaphor and metonymy (Lakoff & Johnson 1980) is one of the central theories of cognitive linguistics, which managed to explain and account for the ubiquity of the two mechanisms in language use. Lakoff & Johnson propose that the production and processing of figurative expressions are mediated by metaphorical and metonymic correspondences that are part of the human conceptual system, i.e. that figurative expressions are interpreted as instantiations of deep conceptual metaphors or metonymies.

A more recent development of cognitive linguistics is the conceptual integration theory, or blending theory (Fauconnier & Turner 1998, 2002). This is currently one of the most promising theoretical frameworks, which provides insights into how we think, create, and understand the world around us, aiming to account for both linguistic and non-linguistic blends. A conceptual integration network is an array of mental spaces, which usually includes two or more input spaces structured by information from different cognitive domains, a generic space and a blended space. The structure of the generic space is common to all input spaces, and the structure of the blended

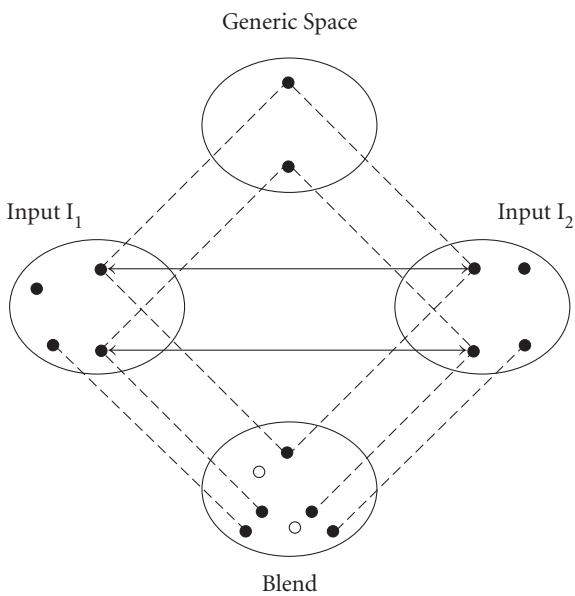


Figure 1. The conceptual integration network (Fauconnier & Turner 2002)

space inherits elements from all inputs, developing a novel, emergent structure (see Figure 1).

The theory offers a model of meaning construction which operates in analogy, metaphor, metonymy, counterfactuals, and other phenomena (Coulson & Oakley 2005; Fauconnier & Turner 1998). Fauconnier & Turner (2002) designed blending theory as a set of principles for combining cognitive models in a network of mental spaces containing partial representations of entities and their relationships in a scenario, as they may be perceived, imagined, remembered, or understood by speakers. The theory can provide an explanation of how a participant in an exchange of information might encode this information on a referential level by dividing it into concepts relevant to different aspects of the scenario. The central component of the theory is the mechanism of cross-space mappings, which makes it possible to understand how an element in one mental space corresponds to an element in another mental space. Constitutive principles at work within the conceptual integration network include matching and counterpart connections in cross-space mapping, selective projection from inputs, composition, completion and elaboration, finally integrated into an emergent structure.

3. Processing of idioms

Most research focusing on the mechanisms of idiom processing investigates only one factor in the processing: context, individual word meanings, cognitive mechanisms, etc. I will here argue for an interplay of factors that jointly lead to idiom interpretation. Additional relevant factors that are often neglected are individual differences between language users (age, sex, education, knowledge of the language(s)), processing of native vs. foreign language idioms, processing of familiar vs. unfamiliar expressions and real time vs. retrospective processing. It cannot be assumed that the processing of idioms will be the same for all users and for all kinds of expressions, in all the languages they use. For processing idioms in a foreign language, it is expected that the native language, or the command of other foreign languages, will play a role in the processing. The first time understanding of an idiom in a foreign language may involve a longer process and incorporate actions such as consulting monolingual and bilingual dictionaries and native speakers, which are not necessary the second time the idiom is encountered, depending, of course, on memory and other factors influencing idiom acquisition. In this chapter I will, however, only consider idiom processing by adult users in their native language.

Challenging the notion of pure idioms and their apparent semantic opacity (*to burn the candle at both ends* or *to hear something straight from the horse's mouth* are often used as examples of this type), I argue here that the meanings of their constituents, although insufficient for their final interpretation, do play a role in their processing. Evidently, the meanings *to hear* and *straight from* are retained in the interpretation of *to hear something straight from the horse's mouth*, whereas *the horse's mouth* should then

have little or nothing to do with a real horse but rather refer to a reliable source. This is evidently not the case, as the *horse's mouth*, a piece of racing slang, alludes to the fact that a horse's age could only be discovered by inspecting its teeth. Most users do not make this link, but it is still true that, based on the meanings of *hear straight from,* some reliable assumptions can be made about the meaning of the whole phrase.

Sometimes the etymological links are rather crooked, as in *to eat humble pie*, another seemingly opaque idiom (Hendrickson 1998:226). The word *humble* in this expression has nothing to do etymologically with the modern English word *humble*. Umbles or numbles were the innards of deer and were often used in pies made for servants. Therefore, anyone who ate umble pie was considered to be in an inferior position. As a result of the play upon words and phonetic similarity that led to substituting *humble* for *umbles*, we today speak of *eating humble pie*, meaning suffering humiliation, apologising, or abasing oneself. Knowledge of the world and language, in this case of history and etymology, can therefore be used as an analytical tool for idiom interpretation. Of course, the origin of most opaque idioms is unknown even to native speakers and appears to be nonsensical, but they still evoke images and are based on the metaphorical or metonymic concepts that we subconsciously use to interpret them. The etymology of opaque idioms, however, is not always helpful or accessible in real time processing for most users, but may be used only in retrospect. Research suggests that etymology as a tool for idiom interpretation is more likely to be used by non-native speakers as it can prove useful in helping learners learn and remember idioms in a foreign language (Boers 2001; Boers et al. 2004).

Contextual clues are also relevant and helpful in the process of interpretation. If the idiom *to burn the candle at both ends* is encountered in a context such as *If you keep burning the candle at both ends you will end up all stressed out and exhausted,* the context provides extra references to the meaning of the idiom. Clues can even be found beyond the limits of the particular sentence in broader discourse.

From the traditional perspective, an idiom such as *to burn a candle at both ends* should not have anything to do whatsoever with the actual burning of candles. And yet, there appears to be a certain logic behind the image of literally trying to burn candles at both ends (of the day) and the current meaning of this idiom. The links here are not clearly etymological, but it is evident that there is so much more at stake here: word meanings are loaded with powerful symbols, relationships, and images that we evoke when we attempt to interpret an idiom in which a word occurs. The interpretation thus also relies on our general knowledge of the world embodied in images. The verb *to burn* itself here rests on a powerful metaphor that can be found in a number of expressions that denote zealous activity and selfless dedication to work, even across languages.

In contrast to the traditional view of idiom opacity, cognitive linguistics treats the phenomenon of understanding or processing idioms differently. From the cognitive perspective, even seemingly opaque idioms can be interpreted based on the cognitive concepts and mechanisms speakers intuitively use when they analyse the meanings of utterances. Many authors have stressed and recorded the systematic clustering of figu-

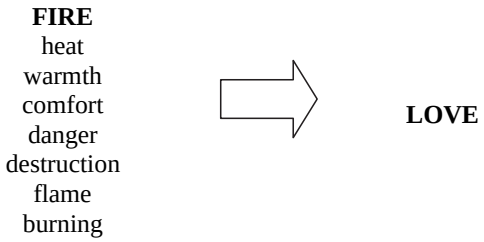


Figure 2. Cross-domain metaphorical mapping from the domain of FIRE to the domain of LOVE

rative expressions around conceptual metaphors and metonymies (Lakoff & Johnson 1980; Kövecses 1986; Gibbs 1995). This implies that many of these expressions have a common underlying mechanism which is activated automatically and subconsciously in real time processing. In this view, conceptual metaphors and metonymies are both available and accessible in any context and can serve as a basis for understanding figurative language.

Figure 2 illustrates how the more concrete domain of FIRE, with several different salient aspects of fire, is mapped onto the more abstract domain of LOVE. There are a number of linguistic expressions in which the underlying concept is that of the conceptual metaphor LOVE IS FIRE: *to carry a torch for someone, the fire goes out between two people, to burn with love*, *the flames are gone, be someone's old flame, to have the hots for someone, be on fire*. As a result of the relative universality of this conceptual metaphor across languages, it is possible, even for non-native speakers, to make appropriate connections on the conceptual level and interpret some of these expressions. This is the case even when the expressions are encountered for the first time, and even when they do not have a full lexical and structural equivalent in their own language, or indeed any equivalent at all. This view, however, focuses on only one processing input, that of conceptual mechanisms, leaving out a number of factors I discussed above (the meanings of the individual lexemes that compose the idiom, contextual clues, etymology).

Dobrovol'skij & Piirainen (2005) challenge the postulates of the cognitive theory of metaphor (CTM) as proposed by Lakoff & Johnson (1980), arguing that despite claims that many conceptual metaphors are universal (e.g. ANGER IS THE HEAT OF A FLUID IN A CONTAINER as the cognitive basis for *someone's gall/bile flows over*, which is not recognised by many native speakers of English, but is common in many European languages), many others are based on historical knowledge that, although perhaps no longer current, still contributes to the interpretation. The authors embrace the CTM as a valid and useful tool for analysing conventional figurative language, but argue that the knowledge of underlying conceptual metaphors is insufficient and/or not linguistically relevant in all cases. The main points of criticism of the CTM are that it cannot account, or can only partly account, for the wealth and breadth of phraseological data across languages, and that it does not incorporate enough of the relevant cultural inputs that help shape the conceptual systems of different languages and cultures.

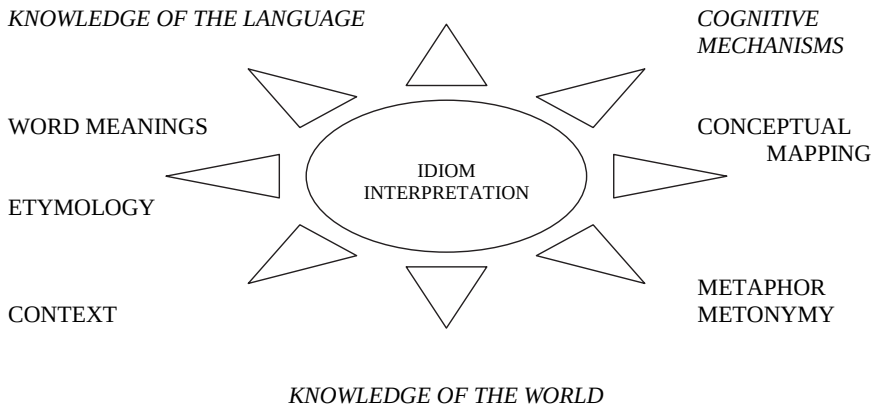


Figure 3. Factors that influence idiom interpretation

Metaphorically speaking, in order to determine the meaning of an idiom we need a complex functional relationship, or factor analysis, rather than a straightforward operation of addition. In sum, idiom interpretation relies on some or all of the following interrelated operations or factors (see Figure 3):

– cognitive mechanisms – conceptual metaphor and metonymy, conceptual mapping between and within domains;
– knowledge of the language – semantics, syntax, etymology, discourse analysis (contextual clues);
– knowledge of the world – cultural and historical background, imagery, symbolism.

4. Processing of idiom modification

None of the models proposed for idiom processing fully accounts for the specificities of processing idiom modification. All of the factors listed above as relevant to idiom processing also play a role in the processing of modifications, but they fail to account for the wealth and breadth of modified expressions. In order to process a modification a listener or reader must be able to decipher a reason for the modification, i.e. to be aware of its communicative purpose. Consider a practical example of how we manage to understand an expression like *to be born with a wooden spoon*. Clearly, the meaning of this modification is derived from the canonical form *to be born with a silver spoon in one's mouth* by substitution. As the modified meaning cannot be recalled from memory, the meanings of constituent words and the relation between substitutes, among other things, must play a role in determining the meaning of this modification. The 'unpacking process' thus involves a degree of comparison and inference. The meaning of the modification is accessed with respect to and via the meaning of the original. Langacker (1987) in his imagistic theory of grammar uses the term 'summary images'

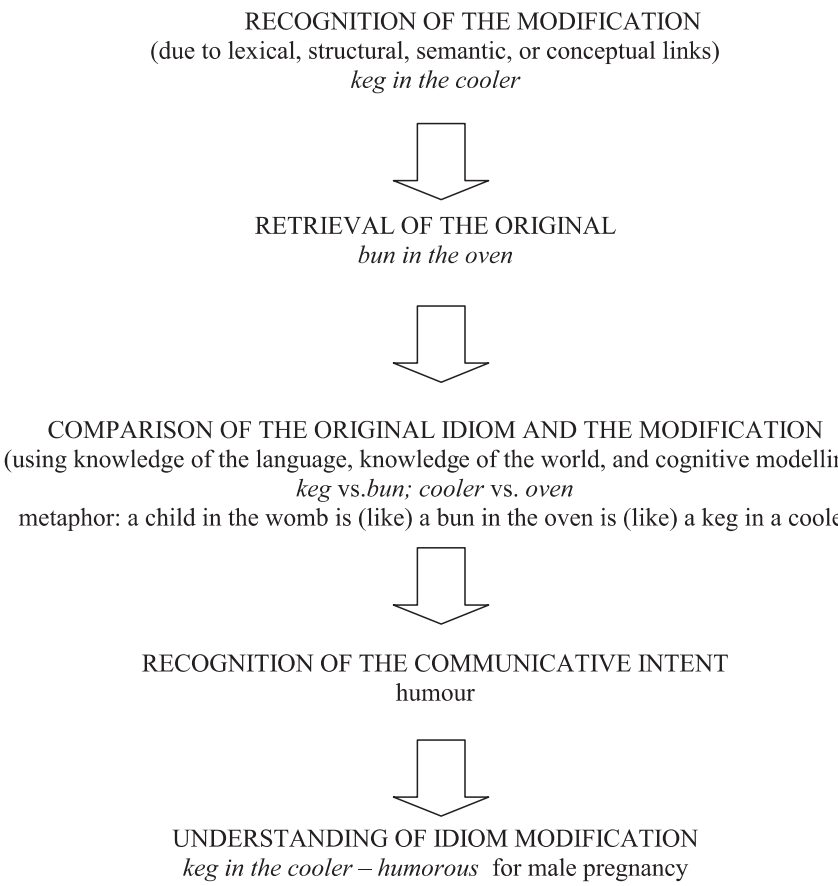


Figure 4. Idiom modification processing steps

to refer to the human ability to retain past mental events so that the new information can later on be inscribed in the same location, much like a multiple exposure photograph. This ability is exploited in our interpretation of modifications, when the novel, 'one-shot image' is 'exposed' on top of the old and already familiar one, creating a new picture, a new image, and a new meaning. These processing steps refer to the understanding of lexical substitution only, and should be tailored to suit each particular type of modification, with complete lexical and structural modification, which relies on the constraints of the conceptual nature of a given idiom, being the most taxing and the most complex of all. The entire process of modification processing is possible thanks to the same factors as play a role in the interpretation of idioms, but reinforced by the process of comparison, as illustrated in Figure 4.

After recognising, by its lexical, structural, semantic, or conceptual correspondence with the original, that an expression is a modification, the user retrieves the original idiom, and then compares the differences and similarities using the same

factors at play in idiom comprehension, i.e. knowledge of the language, knowledge of the world, and cognitive modelling. In order to fully appreciate the modification, the user also needs to recognise the communicative intent. Idioms are modified to achieve certain effects, ranging from irony, humour, surprise and novelty, to fill empty contextual slots, or for pure enjoyment of language play. If the reason behind the modification is fuzzy or hidden, the recipient may take it to be no more than a mistake.

There is no absolute freedom in applying different modification mechanisms to idioms. The regulating principle seems to be to keep enough elements of the canonical form (lexis, structure, semantics, or most importantly, underlying cognitive concepts) to ensure the recognizability of the original in the modification. This means that there are structural, semantic and cognitive limits within which the modification has to be kept to guarantee the recipients' understanding and appreciation of the idiom. In seeking answers to the question of how we process idiom modifications, conceptual integration theory presents itself as both useful and relevant.

Conceptual blending can be seen as a powerful on-line meaning construction process. As such, it is perfectly suited to accounting for on-line linguistic creativity. It may yield conventional blends that may serve as blueprints or templates for new blends. Blending theory is intended to account for cases of novel inferences, but they need not be entirely novel or built from scratch, which makes it suitable for accounting for instances of idiom modification. For example, a conventional idiomatic expression like *the skeleton in someone's closet* can provide the frame onto which other input spaces can project other elements of knowledge. Consider, for example, the example *This article describes the real 'Blue Dress' in Bush's closet*, for which a cross-space mapping is provided in Figures 5 and 6 (also discussed in Omazić 2005 and Omazić & Delibegović, forthcoming).

As illustrated in Figure 6, the emergent structure in the blended space inherits the idiom structure from input space 3, and lexical projections from input spaces 1 and 2: blue dress is projected from input space 1, Bush from input space 2. The blend inherits its structure and meaning from input space 3 and different meaning elements (knowledge of world affairs and culture) from inputs 1 and 2. Clear alignment exists between the topologies in input spaces 1, 2 and 3: the blue dress, the weapons scandal and skeletons are linked by analogy, as all three are/were to remain hidden. Presidential roles change from Clinton to Bush from inputs 1 to 2, the type of scandal has different values, but similarity and analogy are still preserved. The resulting blend is a compression of all three inputs: input 3 provides the organising frame, whereas selective salient projections come from the other two inputs. Within input 1 there is the part-for-whole inner-space vital relation, where the dress stands for the Clinton-Lewinsky scandal. This inner-space relation is compressed in the blend where the blue dress stands for the scandal, but is set in a new frame. What the blend implies is that the Bush weapons scandal is as shameful as the Clinton sex scandal, that there were attempts to cover it up, that it still surfaced and that it may cause damage to his reputation and career. Looking at the blend itself we see that it is grammatically well-formed, and semantically transparent, albeit highly complex.

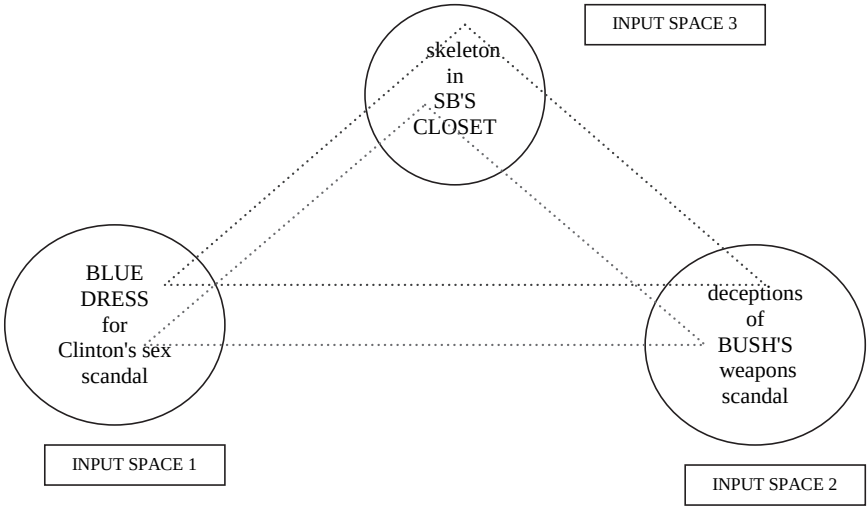


Figure 5. Cross-space mappings for *the real 'Blue Dress' in Bush's closet*

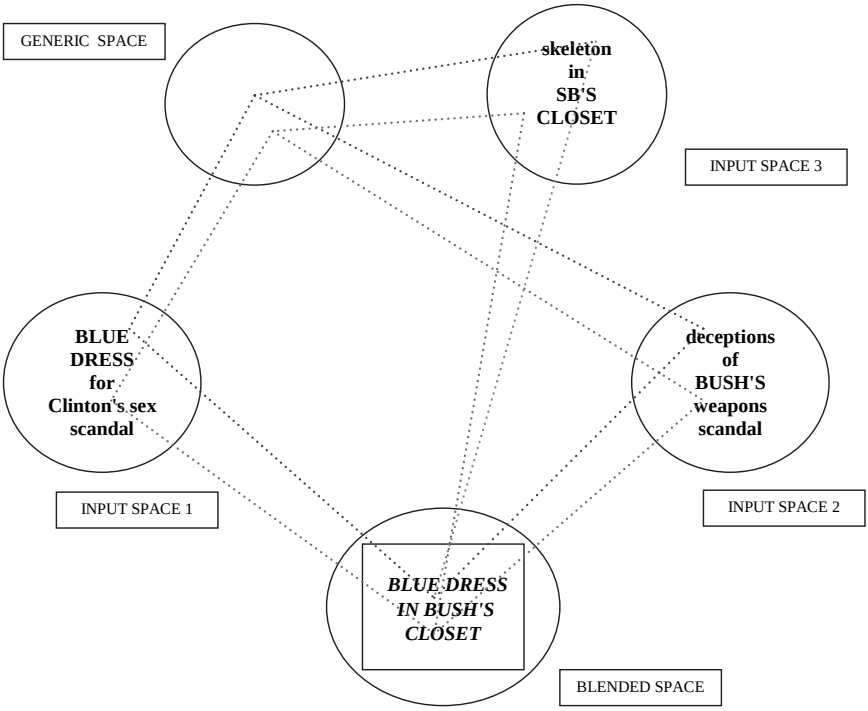


Figure 6. The conceptual integration network for *blue dress in Bush's closet*

Further evidence for the claim that blending is not only a meaning construction process but also an 'unpacking mechanism' for interpreting idiom modifications was found in a larger study of the modifications of phraseological units, which looked at mechanisms that underlie idiom modifications and their consequences for the processing of modifications. The analysis was performed for a number of idiom modifications analysed in detail as part of a large study of the modifications of phraseological units (Omazić & Delibegović, forthcoming). The theory of conceptual integration appears to account well for the mechanisms at work in the modification of phraseological units, as well as their unpacking, i.e. processing. Examples like that above, where the idiom frame is recycled and furnished with new elements, are among the most complex, with other single modification mechanisms (substitution, deletion, insertion) having less complex networks. Very often a proverb or an idiom is modified in such a way that only one word is substituted by another, as in *to be born with a wooden spoon.* This is a clear instance of blending in which the idiom *to be born with a silver spoon* provides the frame, and there are two other input spaces, viz. that of our knowledge of silver and its value contrasted to that of our knowledge of wood and its value. The inconsistency of mapping between these two inputs provides the desired effect in the blend.

The theory of conceptual integration not only shows us how we arrive at modifications and how we unpack them, but its tools and principles (vital relations, governing principles, and constitutive principles) may also be used as constraints that keep the scope of modification under control (Omazić & Delibegović, forthcoming).

5. Conclusion

Having looked at a number of existing theories of idiom processing, we can draw the following conclusions. Idiom processing involves three interrelated factors:

– cognitive mechanisms – conceptual metaphor and metonymy, conceptual mapping between and within domains;
– knowledge of the language;
– knowledge of the world.

The processing as described here is not viewed as a single activity but as a multitude of processes that occur in real time. The same factors also play a role in the processing of idiom modifications, but are insufficient to fully account for it. The processing of idiom modifications can be presented as the following step sequence:

– recognition of the modification (using lexical, structural, semantic, or conceptual links);
– retrieval of the original;
– comparison of the original idiom and the modification (using knowledge of the language, knowledge of the world, and cognitive modelling);

– recognition of the communicative intent;
– understanding of idiom modification.

Looking at my selected examples of idiom modifications, it can tentatively be suggested that the principles and constraints to idiom modification designed within the blending theory can also account for instances of idiom modification. As for constitutive principles, there is one or more canonical idiom forms that serve as input(s) or organizing frame(s), plus one or more inputs that provide additional contextual or cultural domains of knowledge. These then project selective salient elements into the organising frame. Projected elements are combined and elaborated, and form a manageable blend with a new emergent meaning.

Obviously, this begs many questions, such as criteria for mobilising different inputs into the network. Are all inputs (all idioms and all domains of knowledge) equally available for blending in any network? The answer to this and similar questions comes in the form of vital relations and governing principles. Not everything goes. Indeed only the things that can establish relevant web links and inner-space relations go. Only the blends that successfully prompt for the entire network go. There is also the question of which slots in the frame-providing idioms are open, and to what extent they are open. The answer to this may be that the open slots are those that allow the efficient compression of vital relations (analogy, disanalogy, cause-effect, part for whole, time, role, etc.).

Most accounts of idiom modification revolve around the discussion of their syntactic resilience or flexibility, as well as the issue of the range of lexical items that are allowed or forbidden in a particular slot. All these discussions are actually instantiations of only two aspects of blending, viz. the requirement that the blend must have relevance and that it must be well-integrated, i.e. manageable syntactically as a single unit. These two governing principles subsume the semantic, grammatical and lexical constraints and set limits to how far we can go in modifying an expression. It should not be modified beyond recognition as this would violate the relevance principle. Whatever new element is found in an idiom modification it has to be there for a reason (relevance), and the modification must be in line with the rules of grammar, i.e. syntax (a coherent unit). Grave violations of these principles may cause the failure of a blend – if an idiom is modified beyond recognition, violating the syntax and showing no evidence of the reason for modification, it may be perceived as a mistake or not appreciated at all.

As opposed to earlier idiom modification accounts, the blending theory not only regulates the operability of the modified form, but also sheds light on the cognitive operations leading to modification. Unlike earlier accounts that deal, for example, with lexical substitution, addition, or formal structural blending, it clearly shows us what unfolds behind these surface operations.

The modified idioms in my case studies can be presented using the conceptual integration theory apparatus as follows:

- a modified idiom is a compressed version of the canonical form(s) and a new context(s);
- a modified idiom is a complete, well-integrated, manageable language unit;
- a modified idiom preserves and intensifies one or more of the vital relations;
- new elements appearing in a modified idiom have to be relevant;
- a modified idiom must prompt for its own unpacking, i.e. processing.

I hope to have shown that both cognitive linguistic theories – metaphor theory and conceptual integration theory – can be used to analyse phraseological material, conventional and modified, so as to provide insights into how it is processed. Metaphor theory gives insights into the mechanisms of conventional figurative language creation as well as its processing. Conceptual integration theory provides insights into the meaning creation of modified figurative expressions, and as it easily incorporates all the processing factors and processing steps, it suggests itself as a template for processing the rich idiomatic meanings of idiom modifications.

References

Boers, F. (2001). Remembering figurative idioms by hypothesising about their origins. *Prospect* 16(3): 35–43.

Boers, F., M. Demecheleer & J. Eyckmans (2004). Etymological elaboration as a strategy for learning idioms. In Bogaards, P. & B. Laufer (eds.) *Vocabulary in a Second Language: Selection, Acquisition and Testing,* 53–78. Amsterdam: John Benjamins.

Cacciari, C. & P. Tabossi (eds.) (1993). *Idioms: Processing, Structure and Interpretation*. Hillsdale, NJ: Lawrence Erlbaum.

Coulson, S. & T. Matlock (2001). Metaphor and the space structuring model. *Metaphor and Symbol* 16(3&4): 295–316.

Coulson, S. & T. Oakley (2005). Blending and coded meaning: Literal and figurative meaning in cognitive semantics. *Journal of Pragmatics* 37(10): 1510–1536.

Dobrovol'skij, D. O. & E. Piirainen (2005). *Figurative Language: Cross-Cultural and Cross-Linguistic Perspective* [Current Research in the Semantics/Pragmatics Interface Vol. 13]. Amsterdam: Elsevier.

Fauconnier, G. & M. Turner (1998). Conceptual integration networks. *Cognitive Science* 22(2): 133–187.

Fauconnier, G. & M. Turner (2002). *The Way We Think: Conceptual Blending and the Mind's Hidden Complexities*. New York, NY: Basic Books.

Gibbs, R. W. Jr. (1994). *The Poetics of Mind: Figurative Thought, Language, and Understanding.* Cambridge: Cambridge University Press.

Gibbs, R. W. Jr. (1995). Idiomaticity and human cognition. In Everaert, M., E. J. van der Linden, A. Schenk & R. Schreuder (eds.) *Idioms: Structural and Psychological Perspectives,* 97–116. Hillsdale, NJ: Lawrence Erlbaum.

Glucksberg, S. (1993). Idiom meanings and allusional content. In Cacciari, C. & P. Tabossi (eds.) *Idioms: Processing, Structure and Interpretation*, 3–26. Hillsdale, NJ: Lawrence Erlbaum.

Giora, R. (1997). Understanding figurative and literal language: The graded salience hypothesis. *Cognitive Linguistics* 8(3): 183–206.

Grady, J. E. (2005). Primary metaphors as inputs to conceptual integration. *Journal of Pragmatics* 37(10): 1595–1614.

Hendrickson, R. (ed.) (1998). *Encyclopedia of Word and Phrase Origins*. New York, NY: Quality Paperback Book Club.

Katz, A. N. & T. R. Feretti (2001). Moment by moment reading of proverbs in literal and non-literal contexts. *Metaphor and Symbol* 16(3&4): 193–221.

Kövecses, Z. (1986). *Metaphors of Anger, Pride, and Love. A Lexical Approach to the Structure of Concepts.* Amsterdam: John Benjamins.

Lakoff, G. & M. Johnson (1980). *Metaphors We Live By.* Chicago, IL: University of Chicago Press.

Lakoff, G. & M. Turner (1989). *More than Cool Reason: A Field Guide to Poetic Metaphor*. Chicago, IL: University of Chicago Press.

Langacker, R. W. (1987). *Foundations of Cognitive Grammar: Theoretical Prerequisites.* Stanford, CA: Stanford University Press.

Omazić, M. & N. Delibegović (forthcoming). Constraints to mechanisms of idiom modification. In Földes, C. (ed.) *Disciplinary and Interdisciplinary Phraseology.* Tübingen: Günter Narr.

A very complex criterion of fixedness: Non-compositionality

Maria Helena Svensson

The terminology used in definitions of different types of fixed expressions (such as idioms, locutions, proverbs etc.) is complex and sometimes ambiguous. Within the field of phraseology, it is therefore subject to discussion. This chapter deals with the complexity of the notion of non-compositionality. Non-compositionality is a criterion that often appears in definitions of fixed expressions. It will be shown that there are four dichotomies of particular interest within the very notion of non-compositionality. These dichotomies are: motivation/non-motivation, transparency/opacity, analysability/unanalysability and literal/figurative meaning. Other notions of relevance to non-compositionality, such as common meaning, prototypicality, salience and frequency as well as encyclopaedic compositionality will also be described and problematised. Finally, the notions of partial compositionality, holistic or atomistic compositionality, and lexical content or grammatical form, all describing different aspects of compositionality, will be evoked, to further demonstrate the complexity of this criterion of fixedness.

1. Introduction

Problems of terminology are frequently encountered within the field of phraseology. When trying to define the notion of fixed expression or when searching for specific fixed expressions of any kind, researchers very often run into those problems. The purpose of this chapter is to show the complexity of the notion of non-compositionality, a recurrent criterion in studies of fixed expressions. By comparing the criteria proposed by linguists specialising in phraseology and by scrutinising certain fixed expressions, it is possible to show that the complexity of the notion of non-compositionality is due to the fact that it covers several aspects that do not necessarily function in the same way. Sometimes only one of these aspects is considered in definitions, sometimes more than one. In some cases, it is clear which aspect(s) are taken into account, but often this is not stated explicitly. This makes it hard to define non-compositionality. Revealing the different aspects involved and specifying which aspect of non-compositionality is intended in a given problem, definition or discussion, may lead to ways of avoiding confusion.

2. The criterion of non-compositionality

Compositionality can be regarded as part of a dichotomy, in which the counterpart is non- compositionality. It is, of course, the latter notion that is relevant in the context of phraseology. When describing the notion of compositionality, Lyons (1995: 204) writes "the meaning of a composite expression is a function of the meanings of its component expressions". Freely generated language is viewed as compositional, whereas fixed expressions are said to be non-compositional (this, however, is sometimes disputed; see, for instance, Gibbs 1994; Mejri 1997 and Svensson 2004).

In definitions of fixed expressions (Hudson 1998; Moon 1998) or prefabs (Erman & Warren 2000) of different types (idioms, locutions, proverbs etc.), the criterion of non-compositionality is often mentioned (Gross 1996; Hudson 1998; Moon 1998; Gonzáles Rey 2002; Svensson 2004). In some cases, it is the unmarked term 'compositionality' that receives the most attention, but of course facts about compositionality often yield relevant knowledge about non-compositionality as well.

The notion of non-compositionality is a complex one, which can be described by as many as four dichotomies, and also by other features, which will be described in this chapter. What is particularly problematic is that very often it is not clear that more than one dichotomy is involved. It will be shown that it is possible to give each of the four dichotomies definitions of their own. These definitions may overlap and be related to one another, but they are not exact equivalents.

The notions of common meaning, prototypicality, salience and frequency, and the notion of encyclopaedic compositionality are less confusing, but will also be discussed, as well as other notions of importance, such as partial compositionality, lexical content or grammatical form, and atomistic and holistic perspectives.

3. Defining features of (non-)compositionality

3.1 Four dichotomies associated with non-compositionality

The four dichotomies that will be discussed here are: motivation/non-motivation, transparency/opacity, analysability/unanalysability and literal/figurative meaning. For the sake of simplification, the pairs of notions examined are presented as dichotomies, although two of them should rather be described as continuums (see Section 4.1 below).

The notions of non-motivation, opacity, unanalysability and figurative meaning are often present in discussions on fixed expressions and their definitions (see Langacker 1987; Nunberg et al. 1994; Mejri 1997; Hudson 1998; Moon 1998; Le Pesant 2003; Petit 2003). They are often associated with non-compositionality and appear in definitions where non-compositionality is a defining feature (Gross 1996: 10–11; Martin 1997: 293; Gonzáles Rey 2002: 56–57). For instance, the notion of opacity is often associated with figurative meaning. The fact that these notions appear under

the same word in indexes or glossaries is also of importance here. Hudson (1998: 176) says, for instance, in her index: "compositionality [...] *see also* analysability". In Moon's (1998: 334) index, we read: "compositionality [...] *see also* motivation, transparency". The glossary of Gross (1996: 154) relates the terms compositionality and opacity: "Compositionnalité [...] (voir Opacité)".

3.1.1 *Motivation vs. non-motivation*

The definition of compositionality by Nunberg et al. (1994: 498) can be used to establish a relation between compositionality and motivation: "[...] compositionality – that is, the degree to which the phrasal meaning, *once known* [my italics], can be analysed in terms of the contribution of the idiom parts". This quote points to a relation between the criterion of compositionality and the possibility of accounting for the contribution of each word to the whole, once the meaning of the expression in question (in this case an idiom) is known.

More explicitly, their definition can be used in the following way: first the meaning of an expression or prefab is learned. If, after this, it is possible to motivate the meaning of the expression by making sense of the meaning of each word included in it, the expression/prefab is motivated. The term 'motivatable' (see for instance Farø 2005: 123) is even more appropriate, since 'motivating an expression' is often about finding a possible explanation for the appearance of the words in the expression. The explanation found is not necessarily the one behind the original use of the expression, but can serve as a motivation, correct or not. The expression is then 'motivatable' rather than 'motivated'. This implies that the motivatability can vary from one language-user to another – some are more inclined to find possible explanations than others. As a consequence, non-motivatability does not function well in definitions of fixed expressions.

Colour expressions in which a colour term is used as an attribute of an abstract noun can illustrate motivatability rather efficiently. Of course there is no actual, physical colour present since there is no concrete object to describe. In these expressions the meaning of the colour term cannot be concrete and literal. Still, expressions of this type can be used in at least two ways. One use is when the colour term can be motivated, even in a synchronic perspective. The expression, or more specifically, the use of the colour term in question, is motivatable if there is a relation between the colour term and the meaning of the expression. There is often a conventional use of colour expressions where the colour is associated with a particular meaning. For instance, the colour term *white* is often associated with innocence or purity, *black* with pessimism or illegality, and *green* with plants or youth. Therefore, expressions like *white wedding*, *black market* and *green policy* are motivated (or motivatable), yet figurative (see Section 3.1.4 below; see also Steinvall 2002, Chapter 7, for an analysis of the figurative use of colour terms).

In other types of colour expressions, the choice of colour seems completely arbitrary. Even if there was once a logical reason for the colour term to appear in the

expression, the origin has been forgotten. This might be the case in expressions such as *purple prose* or the French *nuit blanche* ('white night' – a night when one doesn't sleep).

3.1.2 *Transparency vs. opacity*

When talking about the notion of 'locution', Martin (1997) writes that even if each word in the expression is easy to understand, this is not sufficient to understand the meaning of the whole French expression *tirer le diable par la queue* ('pull the devil's tail' – have difficulty living within small resources). In this context, he states that a prototypical locution is characterised by its non-compositionality:

> a prototypical locution is characterized [...] by its non-compositionality. Even if one understands all of the words that make up the expression *tirer le diable par la queue* ['pull the devil's tail'], this is not enough to make the locution comprehensible. [my translation] (Martin 1997: 293)[1]

This implies that non-compositionality is related to the difficulty with which an expression can be understood. Non-compositionality is explained in terms of comprehension, which in turn can be associated with the dichotomy transparency/opacity. If, when presented with an expression, a language user understands it without any problems, without any other previous knowledge than understanding the separate words that make up the expression, then it is transparent. Conversely, as Gross (1996: 155) puts it: "a given sequence is said to be opaque when, from the meaning of the composing elements, one cannot reconstruct the meaning as a whole" [my translation]. As Moon (1998: 22–23) points out, not all speakers of a language understand all the expressions containing certain metaphors. Her examples of some semi-transparent metaphors are *grasp the nettle, on an even keel, the pecking order, throw in the towel* and *under one's belt*.

One group of expressions that can be transparent despite not always being motivatable are similes. Often, one characteristic is expressed and then reinforced by something that is supposedly a good example of the characteristic (although this is sometimes more of a habit than an actual fact). Examples are *as good as gold, as easy as pie, dry as a bone* or *white as snow*. Comparable French examples are *fort comme un Turc,* ('strong as a Turk') *soûl comme un Polonais* ('drunk as a Pole') and *bête comme ses pieds* ('stupid like one's feet'). There are similar expressions which reinforce a verb: *swear like a trooper* and *sleep/cry like a baby.* Some French examples of this kind are: *pleurer comme une madeleine* ('cry one's heart/eyes out') and *jurer comme un charretier* ('swear like a carter').

While this kind of expression is often easy to understand for a learner of a new language (if the learner recognises a pattern, where for instance *as* ADJECTIVE *as* NOUN – *as good as gold* equals *very* ADJECTIVE - *very good*), it is not obvious which simile is estab-

1. "Une locution prototypique est caractérisée [...] par sa non-compositionnalité. On a beau comprendre tous les mots qui entrent dans *tirer le diable par la queue*, cela ne suffit pas pour comprendre ce que cette locution veut dire."

lished and conventionalised in the language in question. One possible explanation for the conventionalisation of some of the examples given above is that of alliteration (see Svensson (2004: 95) on phonological structure and Boers & Lindstromberg (2005), who mention alliteration as a cognitively salient feature within another criterion of fixedness, that of memorability).

3.1.3 *Analysability vs.unanalysability*

Compositionality can also be associated with the analysability of an expression (and hence, non-compositionality can be associated with unanalysability). Here, we are concerned with the contribution of each word in the expression to the meaning of the whole. However, it is possible to separate these two notions, as for instance Langacker does: "Compositionality [...] is to be distinguished from analysability, which pertains instead to the extent to which speakers are cognizant [...] of the contribution that *individual component structures* [my italics] make to the composite whole" (Langacker 1987: 457). According to Langacker's definition, each individual component must then contribute to the meaning of the expression as a whole. Gibbs (1994: 278) uses the term 'decomposition', but his definition of 'decomposable' is comparable to Langacker's definition of 'analysable': "Idioms like *pop the question*, *spill the beans*, and *lay down the law* are 'decomposable', because each component obviously contributes to the overall figurative interpretation".

These examples and their analysability can be further examined. According to Gibbs et al. (1989: 578), there is no foolproof method of deciding the exact analysability of an expression. There are, however, certain ways of examining expressions such as idioms. One is to use semantic fields. The more the separate parts of an idiom share the same semantic field as their idiomatic references, the easier it is to analyse the idiom. In this perspective, both *pop the question* and *spill the beans* are decomposable (or analysable) (Gibbs 1994: 278). It is easy to figure out which part of the idiom stands for the verb and which part contains the noun, which makes it easy to figure out which part contributes which meaning. In the case of *pop the question*, *pop* can be taken to mean 'ask' and *question* means 'wedding proposal'. *Spill the beans* can be decomposed in the same way: *spill* means 'reveal' and *beans* means 'secret'. But the semantics of the words making up these idioms and their idiomatic references are quite different from each other. In the first example, it is fairly easy to establish a semantic relationship between *pop* and 'ask' as well as between *question* and 'marriage proposal', because a marriage proposal is actually a question. In the case of *spill the beans*, the semantic relations are not as obvious. There is a semantic relation between *spill* and 'reveal', this being possibly an action done involuntarily. There is, however, no natural semantic relationship between *beans* and 'secret'. In the latter example it is therefore only the structure of the phrase that makes it decomposable. One way of making this more explicit is to say that the syntactic structure only defines the notion of 'analysability'. In my view, this is the most efficient way of dealing with this aspect of (non)-compositionality.

The example *lay down the law* is equally interesting. Since Gibbs (1994: 278) states that each word contributes independently to the figurative interpretation of the expression as a whole, each of the words *lay*, *down, the* and *law* should contribute to the meaning of the idiom. At a closer look, it can be difficult to see how each part contributes independently to the meaning of the whole. The last part of this expression is less problematic. *The law* refers to laws, rules or principles that must be applied in a given situation. But how does *lay* contribute to the meaning? And *down*? *Lay* can mean 'put' or 'place', but how the word *down* contributes independently to the meaning is not clear. *Lay* and *down* together can also mean 'place' or 'put', so this part of the expression can be said to mean 'impose' or something similar. But there is no obvious relation between the *down* part of the expression and the figurative interpretation. Another possible meaning of *lay down* is 'put aside', which would imply the opposite of 'impose'. A comparison with the Swedish expression *lägga ner* (literally 'lay down') is interesting here. These two words can be parts of expressions like *lägga ner fabriken* (literally 'lay down the factory' – close the factory) and *lägga ner sin röst* (literally 'lay down one's voice' – refrain from voting). In other words, it is not obvious that the words in the expression *lay down the law* each contribute individually to the meaning of the expression.

More obvious examples of unanalysable expressions are *bite the dust* and *saw logs*. They are both unanalysable, since it is the expression as a whole that makes up the sense of the expression. The words contribute to the meaning, but not individually, since it is not possible to say which part means what.

3.1.4 *Literal meaning vs. figurative meaning*

The last dichotomy examined here is that between literal and figurative meaning. The problem of defining literal language is well known, and some researchers even reject the view that a phrase can have a literal meaning independent of its context (see for instance Searle 1978: 207 and Recanati 1995: 209–210). Defining a figurative or (often) metaphorical meaning is not necessarily less problematic, but there are ways of dealing with it. Beardsley (1967: 285) suggests that one way of recognising a figurative or metaphorical use of language, is to identify something in an utterance that would be a logical contradiction or an absurdity if interpreted literally. According to Black (1993: 34), it is not just an obvious contradiction of the intended meaning that is decisive for an interpretation to become figurative, but also that it would become meaningless if interpreted literally. Yet another indication of a figurative meaning is when an utterance deviates from its context (see Ariel 2002: 362–364).

There is, supposedly, a relation between literal and compositional as well as between figurative and non-compositional. Gross illustrates this with the following explanation, where he states that word units that cannot be interpreted literally (from the separate words that make up the unit) do not have a compositional meaning:

> [...] in any language, there are many sequences that a foreigner cannot interpret literally, even if he or she knows the common meaning of all the words that make them up. This is the case in the following sentence: *La moutarde lui monte au nez* ['The mustard goes up his nose']. The 'ordinary' meaning of the words in this sentence does not allow one to conclude that the whole sentence means that a person is getting angry. We will say that this sentence does not have a compositional meaning. [my translation] (Gross 1996: 11)[2]

The feature 'not having a literal interpretation' must be equivalent to 'having a figurative interpretation', which implies that the word unit in question has a figurative meaning.

3.1.5 *Combinations of the dichotomies*

One effect of separating the notions in the dichotomies above is that several of them can be activated at the same time. What is interesting here is not that they can be defining features of the same expression, but the fact that in one of the dichotomies the part normally describing freely generated language can be used to describe an expression that clearly falls under the description of a fixed expression according to another dichotomy. One and the same expression can be motivatable, yet have a figurative meaning. The expression *a shot in the arm* can be used as an example. The word *shot* is not to be taken here as something that comes out of a firearm. The *shot* in question is supposed to be an injection, meant to give energy. This makes the expression motivatable. Still, the expression is at the same time opaque to some language users. In the same way, many expressions including colour terms can be figurative yet motivatable. They can be transparent or not, depending on the knowledge of the speaker or listener. As we will see (see Section 4.2 below), the example *saw logs* can be said to be motivatable or not, depending on the perspective with which we look upon the expression (atomistic or holistic). Taken separately, the words do not contribute to the meaning 'snore', so it is unanalysable. Taken together, as a unit, the expression can be motivatable. The meaning is clearly figurative.

When combining several properties from the dichotomies, it is possible to find examples that fit the description of most of the combinations. Below are some combinations of the terms in the dichotomies and examples of English and French fixed expressions that fit under their description. This list is simplified and it is necessary to bear in mind that an expression can be partly motivatable (see Section 4.1 below):

Figurative, motivatable, transparent and analysable:
green policy

2. "Dans les langues, il existe un grand nombre de suites qu'un étranger ne peut pas interpréter littéralement, même s'il connaît le sens habituel de tous les mots qui les composent. Il en est ainsi de la phrase suivante: *La moutarde lui monte au nez*. Le sens 'ordinaire' des mots de cette phrase ne permet pas de conclure que la phrase dans son ensemble signifie que l'on parle d'une personne qui se fâche. Nous dirons que cette phrase n'a pas de lecture compositionnelle."

Figurative, motivatable, opaque and analysable:
marcher sur des œufs ['to walk on eggs' – see Section 4.1, below]
Figurative, non-motivatable, transparent and analysable:
soûl comme un Polonais ['drunk as a Pole']
Figurative, non-motivatable, opaque and analysable:
put the kibosh on something
Figurative, motivatable, transparent and unanalysable:
bite the dust
Figurative, motivatable, opaque and unanalysable:
spill the beans, lay down the law
Figurative, non-motivatable, transparent and unanalysable:
? (Fixed expressions of this combination are particularly hard to find)
Figurative, non-motivatable, opaque and unanalysable:
blue movie, purple patch

Expressions with a literal meaning are not included in this list. They are not necessarily transparent, but they are motivatable and analysable. It is hard to imagine an expression with a literal meaning that would be unanalysable and non-motivatable. Finding a figurative meaning that is analysable is, however, possible, as indicated in the list above. Another example is *spill the beans*, which, in my view is analysable (if only the syntactic structure is taken into account), yet figurative.

As indicated in the list, it is difficult to find examples of the combination figurative, non-motivatable, transparent and unanalysable. A transparent expression is likely to be motivatable. One exception to this might be some similes. They are easy to understand, but their origin might be based on old prejudices and stereotypes, making them non-motivatable from a modern perspective. But finding transparent expressions that are both non-motivatable and unanalysable is harder. This is a challenge for anyone reading the list above.

3.1.6 *The (in)efficiency of the dichotomies*

The dichotomies examined obviously have properties rather different from one another. While the criteria may appear in contexts where fixed expressions are discussed, and hence are relevant in a sense, they are not necessarily efficient. Furthermore, it is often possible to be more precise about which of these aspects of (non-)compositionality is the relevant one for a given example.

The dichotomies analysability/unanalysability and literal/figurative meaning meaning are inherent properties of an expression (either it is analysable or not, either it is figurative or not, though it can be partially so) that are quite objective and easy to use. The transparency or opacity is bound to vary from one receiver to another, although it might be possible to label certain expressions as opaque for the majority of language users who have not learned the meaning yet. The same can be said about motivatability. As we have seen, the motivatability of an expression depends on each speaker's inventiveness and creativity. The dichotomies motivation/non-motivation

and transparency/opacity are therefore more subjective and difficult to use as definite criteria. They may be of use in other contexts, but it is not certain that definitions of fixed expressions benefit from the use of the terms opaque or non-motivatable. I have stated elsewhere (see Svensson 2004) that none of the features non-motivatable, opacity, unanalysability and figurative meaning are both necessary and sufficient conditions for defining fixed expressions.

3.2 Common meaning, prototypicality, salience, and frequency

The notion of (non)-compositionality can be further problematised. Other terms that appear in definitions are (not having) common meaning, (not having) prototypical meaning or use and (not having) salient meaning or use. We have also considered Gross's (1996: 11) example *La moutarde lui monte au nez*. Gross states that the words do not have their 'common' meaning ('sens ordinaire') in this expression. The problem with the word 'common' is that it is not very well defined. It could refer to the original meaning, or the most frequent meaning or something else. Since the meaning of the phrase is clearly figurative, common meaning could refer to its literal meaning. Usually, concrete meaning is considered literal whereas abstract meaning is considered to be figurative. Hanks (2005: 182) says, for instance, "[i]f a term has both a concrete meaning and an abstract meaning, the concrete one generally has preferential status as literal".

To make things more interesting, it is not always the case that the literal meaning is the prototypical meaning. Hudson (1998: 66) points out that it is not always clear which meaning is prototypical. According to her, salience is related to prototypicality and frequency. One way of defining salient meaning is to let this term refer to the most frequent meaning. Consider Hudson's (1998: 66) example of *way*: "[way] can be used in a concrete sense ('path', 'road') or in an extended metaphorical sense ('fashion', 'means'). Corpus investigation shows [...] that the latter, metaphorical sense is much more frequently used in present-day English, and I therefore call this the salient sense". This means that if compositionality is linked with high frequency or prototypicality (and non-compositionality with low frequency and non-prototypical use of words and/or meanings), this has to be clearly stated in each case. If this is taken into account, the question of whether an expression like *in this way* is non-compositional or not will depend on the view adopted.

3.3 Encyclopaedic (non-)compositionality

There is yet another feature associated with the notion of (non-)compositionality. Some expressions have the property of being motivatable, transparent, literal and analysable but are still considered non-compositional. How is this possible? This occurs when the meanings of the separate words each contribute to the meaning of the expression but the expression also has an additional meaning. Langacker (1987: 455) states that "most composite expressions have a conventional meaning more specific

than their compositional value". His example with the (made up) expression "patriotic pole-climber" (Langacker 1987: 454) is rather specific. There are actual, existing expressions which are less spectacular. Consider for instance the compound *greenhouse*. It is fairly transparent, easily motivated, partially literal (the house in itself is not necessarily green, but its appearance might be because of its content) and both *green* and *house* contribute to the meaning. But it is, of course, not just a 'green house'. An even more obvious example is perhaps *the White House*, which can refer to something that is exactly that: a house that is white. In most contexts, there is also an important meaning which includes the American government or politics. This meaning, however, is not possible to derive just from the meaning of 'white' and the meaning of 'house'. Our encyclopaedic knowledge tells us about the additional meaning of this compound. Barkema (1996: 139) refers to this kind of expression as "pseudo-compositional", while Mel'čuk (1998: 30) calls them "quasi-phrasemes" or "quasi-idioms". Martin (1997: 297) gives some examples in French, for instance *attendre un bébé* which means 'expect a baby' (literally 'wait for a baby'), but also much more. *Attendre un bébé* is not just about expecting the arrival of (any) baby, but also implies (in the case of a mother-to-be) carrying a foetus in the body while it is developing. What one awaits is for the mother to give birth to the baby. In this context, Warren (2001: 110) talks about "unexpressed constraints" and "conventional meaning". For instance, the conventional meaning of *from coast to coast*, includes an unexpressed constraint, implying that such a journey should take place over land, and not over water, even though the latter could also describe travelling from coast to coast.

4. Other important notions

4.1 Partial compositionality

In the descriptions given above, the definitions are simplified. As Langacker (1987: 449) points out, the compositionality of an expression is often a question of partial compositionality rather than a clear-cut compositional/non-compositional dichotomy: "[l]inguistic phenomena lend themselves more easily to a claim of *partial* rather than full compositionality" (see also Barkema 1996: 140). The same is true for some of the components examined in the dichotomies. At least two of the dichotomies are more likely to be scalar than either/or. The motivation as well as the literal meaning of an expression can be partial. It might be more problematic to define a 'partial transparency'. Either you understand the expression or you do not. The same goes for analysability. Either you can tell which part contributes which meaning or you cannot.

The scalarity of motivation and literal meaning can be illustrated with the French expression *marcher sur des œufs* ('walk on eggs'), which can have several meanings, depending on context. It is, of course, possible to walk on eggs in the most literal and concrete sense of the words, although this probably does not occur very often. If this meaning is intended, the expression is literal and motivated. Another possible

meaning of this expression is 'to walk on the ground with caution'. This meaning is probably intended more often than the literal one. There are usually no eggs present in this context, but the reference to eggs can be said to be motivated (motivatable). Eggs are fragile and can therefore be associated with caution. The verb still has its literal meaning. Consequently, the expression is partially literal (or partly figurative), but still motivatable. The expression has one more possible meaning: 'to act with caution'. Here, the meaning of the verb is less salient. The action is not specified and *walk* must now be taken as a figurative meaning for 'act' in some way. The motivation for the verb is no longer obvious. The word *eggs* is still motivatable, since the meaning 'with caution' is still present. The third possible meaning of the expression is therefore figurative, and partly motivatable.

4.2 Holistic or atomistic compositionality

As we have seen, the expressions *saw logs* and *bite the dust* are unanalysable, which means that the words do not contribute individually to the meaning of the expressions in question, although the expressions taken as whole units might be motivatable and/or transparent. Two other terms that are sometimes evoked in this context are holistic and atomistic. Gosselin (1996:161) uses the term "atomistic compositional approach" to describe cases where the global meaning ('signification globale') is a direct result of the analyses of the individual meanings of the elements that make up an utterance. The terms atomistic and analysable seem to be related as do the terms holistic and unanalysable. An unanalysable expression such as *saw logs* (where neither *saw* nor *logs* make up the meaning 'snore') is nevertheless motivatable if looked upon in a holistic perspective. Taken as a whole, the expression has a logical interpretation via an association of the sound that a snoring person makes.

4.3 Lexical content or grammatical form

In order to describe yet another aspect of compositionality, let us look at two ways of deciding on the motivatability of an expression. Here, the French expression *les carottes sont cuites* (literally 'the carrots are cooked', meaning 'it's over', 'it's no use', 'it's too late') is a useful example. When looking at the meaning (that is, the semantic or lexical content) of each of the words in the expression, it is easy to draw the conclusion that it is non-motivatable. Nothing in the semantics of *carrots* or *cooking carrots* yields 'it's over' as a logical conclusion. However, the grammatical form of this expression is actually motivatable. The verb form in the expression implies that the action in the expression has already taken place and is indeed concluded. It would be less logical to use a progressive form or another choice of tense (such as the future or the present) to express the rather fatalistic state that is evoked by the expression. Likewise, using an adjective instead of the verb would not be a natural way to express the irreversibility of the event. This aspect should therefore be taken into account when the motivatability or transparency of an expression is being decided.

5. Conclusion

This chapter shows that there are many notions that appear in descriptions of the notion of (non)-compositionality, some of which are lexical content and grammatical form, partial compositionality, atomistic and holistic compositionality, prototypicality, salience and common meaning. Some of these notions, such as atomistic compositionality and analysability, are equivalents or near equivalents.

The properties most commonly associated with (non-)compositionality are probably the dichotomies motivation/non-motivation, transparency/opacity, analysability/unanalysability and literal/figurative meaning. In the definitions examined here, there is clearly an association between each of the four notions of non-motivation, opacity, unanalysability and figurative meaning and the notion of non-compositionality. Even so, these concepts are not equivalents. More than one of them can appear in the same definition. The proof that they are not synonyms is that the same expression can be classified on different sides of the dichotomies. It can thus be shown that the dichotomies are relevant notions on their own and that they exist independently.

There is no doubt that the notion of (non-)compositionality is very complex. While it is interesting and in many respects relevant in discussions about fixed expressions, the complexity makes it difficult to apply an all-embracing notion of non-compositionality as a criterion for fixed expressions and prefabs. Unless it is possible to be very specific about which aspect is intended, other criteria are probably better suited for defining fixed expressions (see Svensson 2004: 109 for further discussion of alternatives).

References

Ariel, M. (2002). The demise of a unique concept of literal meaning. *Journal of Pragmatics* 34(4): 361–402.

Barkema, H. (1996). Idiomaticity and terminology: A multi-dimensional descriptive model. *Studia Linguistica* 50(2): 125–160.

Beardsley, M. C. (1967). Metaphor. In Edwards, P. (ed.) *Encyclopedia of Philosophy,* Vol. 5. New York, NY: Macmillan.

Black, M. (1993 [1979]). More about metaphor. In Ortony, A. (ed.) *Metaphor and Thought,* 19–41. Cambridge: Cambridge University Press.

Boers, F. & S. Lindstromberg (2005). Finding ways to make phrase-learning feasible: The mnemonic effect of alliteration. *System* 33(2): 225–238.

Erman, B. & B. Warren (2000). The idiom principle and the open choice principle. *Text* 20(1): 29–62.

Farø, K. (2005). On motivation/iconicity and arbitrariness of idioms. In Cosme, C., C. Gouverneur, F. Meunier & M. Paquot (eds.) *Phraseology 2005. The Many Faces of Phraseology. An Interdisciplinary Conference*, 121–124. Louvain-la-Neuve, 13–15 October 2005.

Gibbs, R. W. Jr. (1994). *The Poetics of Mind: Figurative Thought, Language, and Understanding*. Cambridge: Cambridge University Press.

Gibbs, R. W. Jr., N. P. Nayak & C. Cutting (1989). How to kick the bucket and not decompose: Analyzability and idiom processing. *Journal of Memory and Language* 28(5): 576–593.

Gonzáles Rey, I. (2002). *La phraséologie du français.* Toulouse: Presses Universitaires du Mirail.

Gosselin, L. (1996). *Sémantique de la temporalité en français. Un modèle calculatoire et cognitif du temps et de l'aspect*. Louvain-la-Neuve: Duculot.

Gross, G. (1996). *Les expressions figées en français; Noms composés et autres locutions*. Paris: Éditions Ophrys.

Hanks, P. (2005). Resonance and the phraseology of metaphors. In Cosme, C., C. Gouverneur, F. Meunier & M. Paquot (eds.) *Phraseology 2005. The Many Faces of Phraseology. An Interdisciplinary Conference,* 181–184. Louvain-la-Neuve, 13–15 October 2005.

Hudson, J. (1998). *Perspectives on Fixedness: Applied and Theoretical* [Lund Studies in English 94]. Lund: Lund University Press.

Langacker, R. W. (1987). *Foundations of Cognitive Grammar: Theoretical Prerequisites*. Stanford, CA: Stanford University Press.

Le Pesant, D. (2003). La polysémie des phrases figées métaphoriques. In Mejri, S. (ed.) *Syntaxe & sémantique* 5. *Polysémie et polylexicalité,* 115–129. Caen: Presses Universitaires de Caen.

Lyons, J. (1995). *Linguistic Semantics.* Cambridge: Cambridge University Press.

Martin, R. (1997). Sur les facteurs du figement lexical. In Martins-Baltar, M. (ed.) *La locution entre langue et usages*, 291–305. Fontenay Saint-Cloud: ENS Éditions.

Mejri, S. (1997). *Le figement lexical. Descriptions linguistiques et structuration sémantique*. Tunis: Publications de la Faculté des Lettres de la Manouba.

Mel'čuk, I. (1998). Collocations and lexical functions. In Cowie, A. P. (ed.) *Phraseology: Theory, Analysis and Applications,* 23–53. Oxford: Clarendon Press.

Moon, R. (1998). *Fixed Expressions and Idioms in English. A Corpus-Based Approach*. Oxford: Clarendon Press.

Nunberg, G. D., I. Sag & T. Wasow (1994). Idioms. *Language* 70(3): 491–528.

Petit, G. (2003). La polysémie des séquences polylexicales. In Mejri, S. (ed.) *Syntaxe & Sémantique* 5. *Polysémie et polylexicalité*, 91–114. Caen: Presses Universitaires de Caen.

Récanati, F. (1995). The alleged priority of literal interpretation. *Cognitive Science* 19(1): 207–232.

Searle, J. R. (1978). Literal meaning. *Erkenntnis* 13(2): 207–224.

Steinvall, A. (2002). English Color Terms in Context. PhD dissertation, Umeå University.

Svensson, M. H. (2004). Critères de figement. L'identification des expressions figées en français contemporain. PhD dissertation, Umeå University.

Warren, B. (2001). Accounting for compositionality. In Aijmer, K. (ed.) *A Wealth of English: Studies in Honour of Göran Kjellmer,* 103–114. Göteborg: Acta Universitatis Gothoburgensis.

Reassessing the canon

'Fixed' phrases in general reference corpora

Gill Philip

This chapter sets forth the argument for revisiting fixed phrases in the light of the knowledge that their fixedness is not necessarily something to be taken for granted. It focuses on the location and analysis of variant forms in general reference corpora. Existing phraseological structures, including collocational frameworks, idiom schemas and semi-prepackaged phrases, are introduced by way of background before a procedure for retrieving non-canonical forms of fixed expressions in general reference corpora is presented. Some implications relating to the study of variant forms are presented, along with suggestions for future research directions.

1. Introduction

Although phraseology focuses primarily on phrase building, from word to collocate and beyond, this chapter sets out the case for phrase deconstruction. At first glance, this might appear to be the antithesis of what phraseology is about. Why dismantle multi-word expressions when so much effort has gone into identifying their most typical realisation – the canonical form?[1]

One reason is that corpus linguists have consistently noticed that canonical forms are not particularly common in language corpora and, crucially, they tend to be outnumbered by non-canonical variants and exploitations. These non-standard forms generally receive little consideration from linguists unless they happen to appear in a particularly eye-catching guise, for example in journalistic and advertising word-play.

Variations of, and deviations from the canonical form are more than simply 'alternative renderings' of the 'same' basic expression. Their existence fills a semantic and pragmatic need – that of personalising and making one's language use relevant to the

1. Throughout this chapter, the 'dictionary citation form' of a fixed expression is referred to as the 'canonical form' (though it should be remembered that established alternative forms can co-exist; see Moon 1998: 122–124). Variants are defined as 'marked' if the changes to the canonical form affect the semantic and/or pragmatic meaning conveyed (e.g. puns and word-play), and 'unmarked' when the changes cause little or no real change in meaning.

situation in which it is uttered. This chapter presents some of the ways in which variant forms of fixed expressions have been classified in existing literature (Section 2), and how they can be retrieved from general reference corpora using off-the-shelf corpus query applications (Section 3). The linguistic value of variant forms is discussed in Section 4, where particular attention is paid to the evaluation of word-play relative to variation in general (Section 4.1), and the insights that the observation of variant forms can provide in the study of semantic productivity (Section 4.2).

2. Recurrent and non-recurrent forms in language corpora

With the growth and increased availability of large, general reference corpora, phraseology scholars have been able to move away from the manual, serendipitous, collection of citations. The information that can be gleaned from studying multiple representations of an expression on a page of concordances is not only more detailed than that derived from smaller amounts of data supplemented by intuition; for many years, corpus data has also been providing evidence that "so-called 'fixed phrases'[2] are not in fact fixed" (Sinclair 1996: 83). Yet although linguists are aware of this fact, very little attention seems to have been given to the variant and anomalous forms except when they create stylistic effects such as puns, irony and humour (for example, Partington 1996: 121–143).

So why has less marked variation been overlooked or cast aside? The fact of the matter is that unmarked variation is seen to be of limited interest because it concerns non-recurrent forms, and as such it is generally believed to contribute little to the description of the canonical form. Canonical forms are linguistically important because of their regularity and consistency. Variants, on the other hand, are embellishments, additions or reductions whose meaning and function are ultimately dependent on, and reducible to, the canonical form from which they are derived. Insofar as corpus studies are concerned, there is a degree of incompatibility between a methodology which favours the identification of recurrent patterns, and the search for non-recurring variants. One of the most commonly-cited drawbacks of corpus-related research into phraseological variation is that "you find what you look for: search tools will only match the pattern sought. An over-restricted search for *a wolf in sheep's clothing* will not find *a sheep in wolf's clothing*" (Moon 1996: 252). Commercial corpus query software is designed for the extraction of lexically- and syntactically-related forms making it difficult to imagine how to retrieve variants based on semantics rather than gram-

2. The term 'fixed phrases' will be used here to include all types of conventional, phraseological chunks such as idioms, metaphors and similes, proverbs, sayings and clichés, bound collocations and binominals. See Moon (1998: 19–25) for a detailed definition of these types.

matical structure.[3] And so the location, or rather the discovery, of non-canonical forms in corpora is still generally considered to be a matter of good fortune (Moon 1998: 51), arrived at more by accident than by design. After all, how can one search for something without knowing what that something is?

2.1 Phraseological skeletons

The change from canonical form to variant, rather than being clear-cut, operates along a continuum. Canonical forms often include grammatical elements which inflect in text, and they may also have alternative lexical realizations. There are several existing studies investigating different types of phraseological frameworks and schemas which incorporate such 'regular irregularities', including Renouf & Sinclair (1991), Francis (1993), and Moon (1998). Each of these scholars focuses on a different type of collocational phenomenon, which can be referred to with the generic term *phraseological skeletons*. The core components of a phraseological unit constitute its skeleton, which is fleshed out by elements with a less central role in creating and maintaining the structure of a phraseological form. The different types of skeleton discussed in this subsection are all illustrated with examples for ease of reference.

Renouf & Sinclair (1991) introduced the concept of *collocational frameworks*. These are defined as "a discontinuous sequence of two words, positioned at one word remove from each other; they are therefore not grammatically self-standing; their well-formedness is dependent on what intervenes" (Renouf & Sinclair 1991: 128). Collocational frameworks are essentially grammatical collocations with a variable lexical 'slot' intervening, and can be seen as an early stage in the subsequent development of the definition of the *idiom principle* (Sinclair 1991), and the *extended unit of meaning* (Sinclair 1996). The example in Figure 1 (after Renouf & Sinclair 1991: 142) illustrates the way in which the slot-filler, *accident*, serves as an "idiomatic platform" (ibid.) for a series of semantically-related words to the immediate right of the framework, *an . . . of*; each of these choices in turn would presumably favour a restricted number of collocations, which again limit the successive options, and so on until the outer bounds of the unit of meaning are reached.

Lexicogrammatical frames (Moon 1998) are a particular type of phraseological collocation that are akin to collocational frameworks in that they are composed of a fixed part and a variable one. The difference between these models is that the variable elements in lexicogrammatical frames must be related, because the resulting clusters

3. Although there are several NLP applications which incorporate semantic tagging as an aid to identifying possible contenders for multi-word expressions, they do not resolve entirely the problem of locating variant forms: if statistically-based, they "are not accurate for dealing with MWEs of very low frequencies, particularly those occurring only once or twice" (Piao et al. 2005: 379); and if they are dependent on human judgements of lexical and semantic use, they are subject to the same shortcomings that befall manually-entered, trial-and-error corpus query searches – namely that one is unlikely to expect (and look for and find) the unexpected.

an accident of birth
an accident of history
an accident of history or birth
an accident of fate
an accident of post-war politics
an accident of war

Figure 1. Collocational framework

beyond belief
beyond description
beyond doubt
beyond question
beyond recognition
beyond repair

Figure 2. Lexicogrammatical frame

of phrases are related not only syntactically, but also semantically. As Moon explains (1998: 145–146), "[t]here is a common structure which contains a variable slot; the variable element is lexical, rather than grammatical, and the variations found tend to belong to the same semantic set. Because of the similarities in grammatical structure and lexical content, the meanings of the phrases can be said to be roughly synonymous". A lexicogrammatical frame with the fixed preposition, *beyond*, and variable nouns (after Moon 1998: 39) can be seen in Figure 2.

The type of collocation that Francis (1993) calls *semi-prepackaged phrases* takes us a further step away from the notion of collocation as word co-occurrence. Instead, she presents collocating semantic sets. Semi-prepackaged phrases are phrases which are understood to be variants of one another, but in which "there is no single lexical item which is essential" (ibid: 144). It is important to stress that this is not the same as semantic preference (Sinclair 1996): in semi-prepackaged phrases, the collocation itself is defined by semantic set, whereas semantic preference is the tendency for a collocation (in the received sense of the term) to co-occur with a restricted range of semantically-related words in the extended co-text: this preference for particular semantic sets contributes towards the definition of the extended unit of meaning, but does not affect the composition of the initial collocation.

The example in Figure 3 (after Francis 1993: 144) illustrates forms of the semi-prepackaged phrase, *the faintest idea*, and its variant wordings. Despite first impressions, which suggest that the phrase is quite fixed, it can be noted that "the only essential elements are the 'superlative' marking of the adjective and the definite article *the* which accompanies it." (ibid.). The method used for the extraction of these variants, in which there is no fixed element, will be examined in Section 3.

What the three phraseological skeletons discussed so far have in common is an increasing tendency towards idiomaticity: even the apparently innocuous grammatical

the faintest idea
the least idea
the slightest idea
the foggiest idea
the remotest idea
the slightest conception
the slightest notion
the foggiest notion
the remotest notion
the foggiest

Figure 3. Semi-prepackaged phrase

one sandwich short of a pic-nic
several cards short of a full deck
a few gallons shy of a full tank
two beanshoots short of a spring roll
a bishop short of a chess set
several hatstands short of a cloakroom
one number short of a logarithm

Figure 4a. Idiom schemas (i)

shake in one's shoes
quake in one's shoes
shake in one's boots
quake in one's boots
shiver in one's boots
quake in one's Doc Marten's
quake in one's size 11s

Figure 4b. Idiom schemas (ii)

framework *a... of*, once its lexical slot is filled, finds its meaning possibilities restricted. Lexicogrammatical frames are defined by a freer type of collocational regularity, which involves a fixed element which collocates with a range of words belonging to the same semantic set, resulting in the formation of a series of loosely synonymous expressions. This is further extended in semi-prepackaged phrases, where the collocation involves no fixed element, both slots being filled by members of a restricted semantic set, and a cluster of related, roughly synonymous expressions is generated.

This brings us to *idiom schemas* (Moon 1998), which again are defined by semantic similarity rather than word-form co-occurrence.

Idiom schemas "share an underlying metaphorical conceit and their lexicalizations are drawn from sets of co-hyponyms." (Moon 1996: 252). The schemas that Moon illustrates appear to be a sub-type of lexicogrammatical frame, in that there tends to be a base structure which supports the variant forms. The examples provided in Figures 4a and 4b demonstrate the schemas for *one [component] short/shy of a [whole]* (ibid: 252) and *[tremble] in one's [footwear]* (Moon 1998: 161) respectively, where the words enclosed in square brackets represent the variable slots.

2.2 Capturing the variety of non-canonical forms

While the kinds of schematic representation outlined in Section 2.1 are adequate for classifying variant forms from a lexicographic point of view, they leave little room for the inclusion of variants which, though exploiting the underlying conceit, do not

like a red rag to a bull
as a red rag to a bull
a red rag to the Unionist bull
the latest red rag from a bullish Beijing
like a red rag to the Euro-sceptics
like waving a red flag in the face of a bull
like waving a red rag at the bull
like putting a "red rag to a bull"
like bulls to a red rag
red flag to a dragon

Figure 5. Idiom theme and variations

adhere to the typical phraseological patterning (e.g. *the pan calling the kettle blackbottom*, related to *the pot calling the kettle black*: see Appendix; see also Moon 1998: 170–177). Conceit-based exploitations are recognisably related to the canonical form in much the same way as variant forms of semi-prepackaged phrases are appreciably 'the same but different', so it may not always be helpful to treat exploitations separately from schemas.

One way to include exploitations alongside schematic representations of idiomatic and figurative phraseology is to focus on the key components of the idiomatic 'theme' (Philip 2000): these are typically the most salient elements, and they may be syntactic or lexical. Sometimes the recognition of the canonical phrase is triggered by a core collocation (e.g. *red rag*), at other times a combination of salient words and grammatical elements have the same effect (e.g. *like a* [[colour] [fabric]] *to a* [NP]).[4] In contrast with the phraseological models discussed above, the reduction of phrases to key components is not designed for lexicographical description. It aims instead simply to extract as many phraseological permutations as possible in order to paint a comprehensive picture of everyday variation which can then be used as a benchmark for assessing stylistic effects in journalism, literature and translation.

Figure 5 shows variation to the idiom *like a red rag to a bull* (after Philip 2000: 231–232); here it can be observed that there is only one invariable element, *red* (no variations to this colour were attested in the corpus, though marked variation may well home in on this component; see Philip 2003: 201–242); the comparative appears to be optional, as indeed is *bull*, which is substituted in a semantically intriguing way: rather than being replaced by co-hyponyms as happens in idiom schemas, the semantic set is *attributive* rather than *taxonomic* (Glucksberg & Keysar 1993: 408–9; see also 4.2). The implications of this are considerable, especially given the current vogue for automated

4. The analysis of variants makes it possible to sketch out the trends that seem to be followed for individual phrases, but it is much more difficult to predict what these trends might be in the absence of relevant data. To complicate matters further, it is common for several types of variation, whether grammatical or semantic, or involving e.g. rhyme, inversion or truncation, to interact in a single example (Philip 2000: 223).

extraction of data from corpora. Attributive semantic sets are not fixed lists and do not conform to traditional notions of semantic relatedness as adopted in thesauri, but instead are determined and interpreted contextually. The variation found in idiomatic and figurative phrases is a combination of grammatical and lexical components, and with the added complication of attributive semantics, it is easy to see why "from a lexicographical viewpoint, they are simply nightmares" (Moon 1996: 252).

3. Searching general reference corpora for variant forms

Given the difficulties to be encountered in defining types of variation, it becomes clear why corpus searches generally produce only a limited range of non-canonical citations. The degree of semantic productivity to be encountered in non-canonical forms can be surprisingly complex and unpredictable, and the retrieval of variants becomes well-nigh impossible if no fixed element can be defined. However, if the search criteria insist on there being certain fixed elements present, any examples which do not contain these word forms (but which contain other key components, or use phonologically or graphically similar forms) will remain hidden. Tagging – both grammatical and semantic – can aid the process, but does not resolve the problem entirely. While both probabilistic algorithms and expert linguists are able to predict likely variants, the reality of variation as revealed in corpus data (as can be seen in the Appendix) often results in the formation of *ad hoc* semantic classes which evade prediction. So the corpus user has to define a search strategy which will maximise the retrieval of a useful data set.

For most corpus linguists, this means carrying out a single, wide-reaching and general search which is subsequently refined. What few attempt is to combine the results of successive related searches before embarking on the refining and selecting procedure. One such method was used by Francis (1993) for retrieving semi-prepackaged phrases.

> I concordanced *idea* and found this meaning with the adjectives *faintest*, *least*, *slightest*, *foggiest* and *remotest*. I then concordanced all these adjectives in order to find head-nouns combining with them to form the same meaning, and found *conception* and *notion*. (ibid.: 156)

The resulting data was then combined into a single file, from which a 'stepped concordance' – a concordance with no single, invariable node – was produced (ibid: 144).

Cignoni & Coffey (1998, 2000) adopted a different approach to extract idiom and proverb variants from the untagged Italian Reference Corpus (Bindi et al. 1991), which involved "making searches for one or more key words for each idiom and subsequently editing out irrelevant material with a word-processor" (Cignoni & Coffey 1998: 292). Manual editing is quite feasible with a corpus of this size (15 million words), but the larger the corpus, the more laborious the selecting becomes.

1a	red+1,5bull	[*red* followed by *bull*; five word window]
1b	bull+1,5red	[*bull* followed by *red*; five word window]
2a	rag+1,5bull	[*rag* followed by *bull*; five word window]
2b	bull+1,5rag	[*bull* followed by *rag*; five word window]
3a	flag+1,5bull	[*flag* followed by *bull*; five word window]
3b	bull+1,5flag	[*bull* followed by *flag*; five word window]
4	to+a+bull	[*to a bull*; no words intervening]
5	red+rag	[*red rag*; no words intervening]
6	red+flag	[*red flag*; no words intervening]
7	like+a+1,5to+a	[*like a* followed by *to a*; five word window]

Figure 6. Search procedures[5] for *like a red rag/flag to a bull*

Philip (2000, 2003) developed a procedure for retrieving variants of idiomatic phrases in the 450 million-word Bank of English. Given the size of the corpus, simple key-word searches would have been impossible to edit manually, so it was essential to formulate a series of searches that would be inclusive of all potentially relevant data, yet restrictive enough to exclude as much 'noise' as possible. The results of the searches were collated, then edited with a PC concordance package to eliminate duplicates and irrelevant concordances (Philip 2003: 127–129).

The queries shown in Figure 6 incorporate various lexical and phrasal elements of the canonical form, and exploit the maximum number of unspecified words (the 'five-word window') permitted between search terms in order to include as much data as possible.

Some justification needs to be made regarding the decidedly low-tech aspects of the search procedures illustrated in Figure 6. In the first place, they were devised to be used not only with the Bank of English, but also with corpora which had no tagging, making more sophisticated queries impossible to carry out. The advantage of this is however that they can be used on any corpus or text collection, using even the most rudimentary of concordance packages. They can also be run on Internet search engines, making it possible to use the web to verify findings derived from corpus data, and to provide supplementary data when the corpus is unable to provide sufficient examples for the study of longer phraseological units.

In the course of carrying out multiple searches of this type, it was discovered that some relevant examples featured none of the apparently essential key-words, and others lacked the expected syntactic patternings. This confirms the rationale behind the use of multiple queries to compile a reliable set of phraseological data. The many variants which do not conform to predictable patterns and standard synonyms can be found, but only though a combination of keywords, syntactic frameworks and wildcards over repeated searches.

5. The search routines are defined in the Look Up query language used with the Bank of English; an explanation of the formulae is provided in square brackets.

4. The linguistic value of phraseological variation

If finding the data has in itself been something of a barrier to studying variation, so too is the fairly low status attributed to non-standard forms. Perhaps surprisingly, canonical forms of idioms and other figurative phrases are actually quite uncommon in language corpora and are, as a general rule, outnumbered by their corresponding non-canonical forms (Moon 1998; Cignoni & Coffey 2000; Philip 2003). Yet they are considered to be exceptions to the norm because they are non-recurrent and ultimately reducible to the canonical form.

While there are very sound lexicographical and pedagogical reasons for concentrating on repeated patterns, variant forms reveal a great deal about human linguistic behaviour. Not all variants are deliberate, and not all are marked or ambiguous. Variant forms occupy a very substantial grey area lying between the extremes of the canonical form and the eye-catching puns that induce in us the "smugness effect" (Partington 1996: 140). Yet unmarked variation tends to be overlooked entirely, with marked variation typically being compared and contrasted with the canonical form alone. It is important to incorporate unmarked variation into phraseological description, because it allows marked forms to be judged with respect to other variants as well as with the canonical form from which it stems, as the discussion in Section 4.1 shows.

4.1 Variation and the open-choice principle

To illustrate the difference that an awareness of typical variation can contribute to the evaluation of marked forms, let us consider some examples of *the pot calling the kettle black:*

(1) The words kettle, black and pot suddenly spring to mind.

(2) the pan calling the kettle blackbottom

(3) Talk about Mr Pot and Mr Kettle

(4) POST CALLING THE KESTLE BLACK?

These examples give some idea of the range of variant forms that are encountered in the Bank of English (see Appendix). Considered individually, and with sole reference to the canonical phrase, each of these examples would be understood to be marked: example (1) exploits the keywords and the underlying conceit, but not the standard structure; example (2) replaces *pot* with the semantically-related *pan*, and elaborates the conceit by replacing *black* with *blackbottom*; example (3) personifies *pot* and *kettle*, and exploits the expression without specifying any other components of the phrase; example (4) maintains the overall phraseological patterning, replacing *pot* and *kettle* with the semantically unconnected but phonologically similar alternatives, *post* and *Kestle*.

How marked are these variants? If compared to the full cline of variation (Appendix) it becomes apparent that, with the exception of example (4), they are little more than instantiations of the variation tendencies associated with this particular

The case of Pot versus Black Kettle
Hello pot, my name's kettle.
pot, meet kettle.
Pat Pot meets Mariah Kettle.
Talk about Mr Pot and Mr Kettle?
dear pot, yours kettle.

Figure 7. Personification of *pot* and *kettle*

idiomatic phrase. Each can be read within the paradigm of its variation type as well as in relation to the canonical form. Taking example (3) as a case in point, the personification of *pot* and *kettle* (Figure 7) can be seen to be one of the tendencies that variations to this idiom follow. Read in this context, then, *Mr Pot and Mr Kettle* is not nearly as marked as it seems to be when evaluated against the canonical form alone.

Whereas examples (1)–(3) are contextualised but not wholly context-dependent, examples (4)–(7) are true puns as they incorporate allusive and connotative meanings into the interpretation of the variant form. Example (4) substitutes one pair of litigants, *pot* and *kettle* with another pair, the Post Office and a member of the public, as revealed in the subsequent context. And although they fall into the same variation type – colour-term substitution – examples (5)–(7) are also marked, as the substituted terms evoke connotative meanings which are central to the textual meaning of the variant (*grey* denoting dullness; *schwarz* and *noir* German and French cultural connotations respectively).

(5) The pot calling the kettle grey?

(6) Surely a case of the pot calling the kettle schwarz.

(7) It is time the pot stopped calling the kettle 'noir'

The creation of a pun may be considered by some as the reaffirmation of the *open-choice principle* (Sinclair 1991) within phraseological chunks. Certainly, the choice of substituted term appears to be very free indeed. But open choice implies far more freedom than is actually available in this sort of phraseological manipulation, because whatever element is substituted, its meaning is always read in relation to the canonical phrase. The new element forces the reader to analyse the phrase both compositionally and non-compositionally, and the overall meaning is a combination of the old phrase and the new, and not a new phrase in its own right (Philip 2003). This type of variation can be described as a *palimpsest effect*. Just as vellum was re-used in medieval times by over-writing the pages of old books – hiding but not erasing the original text – puns constitute a linguistic palimpsest in which the new meaning is written over the old one, but fails to cancel it out completely.

The study of canonical forms alongside non-canonical forms in all their guises highlights the relationship of phraseological items and their cotextual environments. The analysis of corpus data demonstrates that the core of an extended unit of meaning, typically taken to be a single word, can just as readily take the form of an entire phrase,

but not necessarily in its canonical form. In fact, various internal parameters are at work in ensuring that a variant retains enough of the canonical form to be recognised as relating to it, but a further factor is external: cotext. Just like single words, fixed phrases too attract colligational features, semantic preferences and semantic prosodies (Philip 2003: 239–40). If the cotext of an innovative use features the norms typically associated with the canonical form, then these norms offset the effects of the internal variation, inducing the reader to relate the variant to its canonical form. In example (8), the cotext provides enough of the expected patterning associated with *green with envy* to ensure that this is the expression that is interpreted alongside the colours of the Irish flag. Interestingly, word-play can be created by locating a canonical form in an atypical cotext, whereby contextually-relevant interpretations merge with the meaning of the (unchanged) phrase. Consider the effect in example (9), where the meaning of *in the pink* ('happy and healthy') undergoes forced reinterpretation due to its association with homosexuality.

(8) Stunning Miss Ireland Emir Holohan-Doyle wraps our national flag around her – hoping to make her Miss World rivals green, white and orange with envy!

(9) Peter Tachell is the author of Europe in the Pink – lesbian and gay equality in the new Europe.

It is extremely rare to encounter a non-canonical form in an atypical cotext, almost certainly because the proportion of open-choice to idiom principle would be too unbalanced in favour of open-choice, making real-time interpretation very difficult. When this occurs in corpus data, it tends to be found when two similar structures are fused during on-line processing, typically in transcribed spontaneous speech, and represents a 'crack' in the phraseological priming (Hoey 2005: 11).

4.2 Variation and the emergence of ad hoc semantic classes

The variation that occurs in phraseological skeletons often follows unpredictable patterns (see 2.2, 3), and one of the most interesting and potentially important features to emerge from the analysis of variant forms is the phenomenon of *ad hoc* semantic classes.

The *Class Inclusion Hypothesis* (Glucksberg & Keysar 1993) notes that semantic classes are often created attributively, especially in the case of metaphorical and figurative language. Attributive categories differ from taxonomic categories in that the metaphor schema or conceit that is in operation "is used to attribute an organized set of properties to the metaphor topic by projecting onto a target domain, such as *crime*, all of the relevant properties of a source domain, such as *disease*" (Glucksberg & Keysar 1993: 48). If the relationship between the substituted term and the canonical one is based on common attributes, rather than relations of co-hyponomy, then this accounts for much of the non-standard semantics that can be observed in corpus data.

black sheep of the family
black sheep of the Mitchell family
black sheep in the Compositae family
black sheep of Britain's financial services
the black sheep of the EU
black sheep of the sporting world.

Figure 8. Attributive semantics in *black sheep of the family*

It also helps to explain why most language users find little difficulty in interpreting and producing variants such as those in Figure 8, whereas the generation of such sets and the prediction of tendencies in variation, whether computationally or manually, continues to challenge.

The semantic productivity occurring along the paradigmatic axis, where terms are substituted not only by members of the same semantic set but also by apparently unrelated terms, is an area of study that is waiting to be explored, and to which corpus analysis can contribute enormously. *Ad hoc* semantic sets remain on the whole something of an unknown quantity. What do they tell us about the ways in which we classify the world around us? Can attributive semantic sets be predicted at all, and if so, how can such knowledge be incorporated into Artificial Intelligence and Natural Language Processing? Up until now, most research in the field has been based on opportunistically-collected and invented examples, and has not been detailed and exhaustive enough to tackle such questions. Using corpus data to study variants means that more examples are available, with the additional advantage that these belong to the same, homogeneous data set.

5. Concluding remarks

This chapter has shown that variant forms of fixed expressions can be found in corpora by following quite simple procedures, effectively debunking the myth that their retrieval is governed by happenstance. Non-canonical forms are indeed unpredictable, but they seem to follow tendencies in their variability, suggesting that their apparent randomness is in fact fairly systematic. The types of variation that emerge merit further study as they provide data that is otherwise difficult to access regarding how language users manipulate words and meanings.

The fact that variations tend to follow trends provides tantalising evidence of the idiom principle in operation. As demonstrated in Section 4.1, changes to the canonical form are necessarily restricted if the meaning value is to be preserved, and non-canonical forms are inclined to occur within a 'canonical' cotext, where the most typical features associated with the canonical form and its extended unit of meaning are all present. This suggests that the phraseology external to the fixed expression shares the role of transmitting meaning, exerting most influence when the internal phraseology is weakened due to variation.

Regularities found in corpus data serve as a benchmark in language description, where they illustrate normal language use. Where then does variation fit? While it is true that the canonical form is more important to document and learn, textual occurrences of fixed phrases are likely to involve variation. This opens up a need for pedagogical and lexicographical descriptions to address fixed phrases from a more inclusive viewpoint, where creativity is considered an integral feature of phraseology. The prospects are enticing.

References

Bindi, R., M. Monachini & P. Orsini (1991). *Italian Reference Corpus: General Information and Key for Consultation*. Pisa: CNR, Istituto di Linguistica Computazionale.

Cignoni, L. & S. Coffey (2000). A corpus study of Italian proverbs: Implications for lexicographical description. In *Euralex 2000 Proceedings*, 549–555. Stuttgart: Universität Stuttgart.

Cignoni, L. & S. Coffey (1998). A corpus-based study of Italian idiomatic phrases: From citation forms to 'real-life' occurrences. In *Euralex '98 proceedings*, 291–300. Liège: Université de Liège.

Francis, G. (1993). A corpus-driven approach to grammar – principles, methods and examples. In Baker, M., G. Francis & E. Tognini-Bonelli (eds.) *Text and Technology: In Honour of John Sinclair,* 137–154. Amsterdam: John Benjamins.

Glucksberg, S. & B. Keysar (1993). How metaphors work. In Ortony, A. (ed.) *Metaphor and Thought,* 401–424. Cambridge: Cambridge University Press.

Hoey, M. (2005). *Lexical Priming: A New Theory of Words and Language.* London: Routledge.

Hunston, S. & G. Francis (1999). *Pattern Grammar: A Corpus-Driven Approach to the Lexical Grammar of English*. Amsterdam: John Benjamins.

Moon, R. E. (1998). *Fixed Expressions and Idioms in English: A Corpus-Based Approach.* Oxford: Clarendon Press.

Moon, R. E. (1996). Data, description, and idioms in corpus lexicography. In *Euralex '96 Proceedings*, 245–256. Gothenburg: Göteborg University.

Partington, A. (1996). *Patterns and Meanings: Using Corpora for English Language Research and Teaching*. Amsterdam: John Benjamins.

Philip, G. (2003). Connotation and Collocation: A Corpus-Based Investigation of Colour Words in English and Italian. PhD dissertation, The University of Birmingham. <http://amsacta.cib.unibo.it/archive/00002266/01/Thesis.pdf>.

Philip, G. (2000). An idiomatic theme and variations. In Heffer, C. & H. Sauntson (eds.) *Words in Context: A Tribute to John Sinclair on His Retirement* [ELR Monograph 18], 221–233. Birmingham: The University of Birmingham.

Piao, S. S., P. Rayson, D. Archer & T. McEnery (2005). Comparing and combining a semantic tagger and a statistical tool for MWE extraction. *Computer Speech and Language* 19: 378–397.

Renouf, A. & J. M. Sinclair (1991). Collocational frameworks in English. In Aijmer, K. & B. Altenberg (eds.) *English Corpus Linguistics: Studies in Honour of Jan Svartvik*, 128–143. London: Longman.

Sinclair, J. M. (1996). The search for units of meaning. *TEXTUS* 9(1):75–106.

Sinclair, J. M. (1991). *Corpus, Concordance, Collocation*. Oxford: Oxford University Press.

Appendix

the pot calling the kettle black

```
   certainly a case of the pot calling the kettle black. Keatingspeak is now
       it is a case of the pot calling the kettle black \# mdash; P. Hudson,
   it is not a case of the pot calling the kettle black. <p> McEnroe genuinely
   is rather a case of the pot calling the kettle black? The RSPCA used to run
      s this a case of the pot calling the kettle black? Could holidaymakers
     a classic case of the pot calling the kettle black, MEPs were obliged to
    In a prime case of the pot calling the kettle black, 48-year-old Iglesias,
   which is a bit like the pot calling the kettle black. <p> As if to answer
    one. <p> It's like the pot calling the kettle black,'' said parish council
   her, because it was the pot calling the kettle black. <p> That Scotland
         Grove. January 12 Pot calling the kettle black? IT is not often that
      </dt> TALK about the pot calling the kettle black \# Linfield actually
            would be ''the pot calling the kettle black.'' It's hard to see how
   MPs, this really is the pot calling the kettle black.'' He believes that the
       WVW: Isn't that the pot calling the kettle black? After her time in
        <ZGY> Rather <ZG0> pot calling the kettle black <ZGY> <M0X>
   mind, that would be the pot calling the kettle black with a vengeance. And
          </date> Sir: The pot calling the kettle black! Press tells doctors
   look like a case of the pot calling the kettle. . . Hannahs make-over has
      Surely a case of the pot calling the kettle schwarz. <p> Germans have
     out of steam. <p> The pot calling the kettle grey? Labour will be a
  ''It is time the pot stopped calling the kettle 'noir \# The scandal is
     happy. I've heard of pots calling the kettle black, but this is more in
       in the kitchen and pots calling the kettle black. I wondered if it was
         because that would be calling the kettle black, but I don't like the
    been for years as well. So we've got a kettle calling the pot black round
         would be a classic example of the kettle calling the pot black. And
        that a case of the Doc calling the kettle black? <h> John, Neil or
    low she says to me the pan calling the kettle blackbottom and I had to
   is this a case of the pot-i calling the kettle black \# <p> Mahoney laughed
     know, really. This is the pot and the kettle getting together and
    new boss is a useless jerk - a pot and kettle case if ever there was to
    superiority. There is a bit of pot-and-kettle about its outrage. Growth
          North and south are like pot and kettle and neither out-shines the
      out demons and evil spirits. Pot and kettle, or what? KEITH PORTEOUS
    so clever and witty, the words pot and kettle do spring to mind about his
        spent on petrol the phrase, ''Pot, kettle and black,'' springs to mind.
          anything's gone wrong. The words kettle, black and pot suddenly
   awful afternoon, which called to mind ''kettle'' and ''pot'', and culminated
wonder, ever heard the words ''pot'' and ''kettle \# Take Bruce Anderson, the
    mind you. The case of Pot versus Black Kettle (1927). A BAND in Texas have
   no-smoking area?'' Hello pot, my name's kettle. I have a phobia of dirty
   the speaker ''sound stupid'' (pot, meet kettle). She was sitting next to
     of humility. Talk about Mr Pot and Mr Kettle? Finally I must chide you
       there was a case of dear pot, yours kettle. Mandy didn't bother too
   it,'' he declared. Pat Pot meets Mariah Kettle. liamfay@clubi.ie <xr> 9108
   o move along. </p> <h> POST CALLING THE KESTLE BLACK? </h> <p> SARAH Kestle
```

SECTION II

Corpus-based analysis of phraseological units

Adjective + Noun sequences in attributive or NP-final positions

Observations on lexicalization

Pierre J. L. Arnaud, Emmanuel Ferragne,
Diana M. Lewis, and François Maniez

Frequent Adjective + Noun sequences in the British National Corpus based on the most frequent 'central' adjectives were examined in their use as noun premodifiers and in other uses (predicative, adverbial, or with N being the head of a noun phrase). Hyphenated and non-hyphenated occurrences were taken into account. An investigation of the presence of these sequences in dictionaries shows a relatively strong correlation with frequency of occurrence, as well as with hyphenated spelling. Six Adj + N sequences were more closely examined in the Periodicals and Spoken sections of the BNC, and evidence of reanalysis was found in changes of scope, predicative uses of an adjectival nature, adverbial uses, and occurrences of Adj + N as subjects or objects with restricted sense. A small, but not negligible, proportion of unambiguously lexicalized occurrences was found. Finally, the accentual behaviour of Adj + N sequences was investigated in a laboratory experiment. The prevalence of the /12/ accentual pattern across the frequency of occurrence range and in attributive as well as predicative uses showed that degree of lexicalization had no influence on the stress pattern.

1. Introduction

There has been a recent emphasis in phraseology on the effects of frequency. The identification and study of frequently-occurring sequences of words, also known as 'chunks' (e.g. Bybee 2002) or 'prefabs' (e.g. Erman & Warren 2000) or 'multi-word items', has been enormously facilitated by the creation and exploitation of large corpora. It is well known from observations of fossilized expressions that high frequency of occurrence protects many sequences from the changes affecting analogous expressions. It has also been shown that very frequently occurring sequences can crystallize into new constructional categories (e.g. Krug 2001). Moreover, frequency of occurrence may well be directly correlated with type of storage – the "linear fusion hypothesis" in the words of Bybee (2002: 112). It has been suggested for some time that

many more sequences than traditionally thought are stored in memory as units and accessed as such, rather than stored as independent elements and assembled compositionally online (Pawley & Syder 1983). Among the possible evidence for the development of holistic storage are said to be phonological, syntactic and semantic shifts: (1) phonologically, the sequence becomes more tightly-knit (notably in stress pattern), a phenomenon often termed 'coalescence'; (2) syntactically, individual words of the sequence lose some of their positional freedom and this can lead to recategorisation; (3) semantically, the meaning of a sequence becomes more opaque and less easily computable from its component parts.

Much recent work on frequency has focused on grammaticalizing structures, but frequency clearly plays a very similar role in lexicalization (Brinton & Traugott 2005). It has been shown that the frequency of nouns as premodifiers of common nouns in newspaper language almost doubled in the twentieth century, resulting in dense information packaging (Biber & Clark 2002; Biber 2003; see also Quirk et al. 1985: 1330; Biber et al. 1999: 588). This compression leads to a loss of transparency insofar as the relationship between the premodifier noun and the modified noun is no longer explicit. Either the reader/hearer must infer the relationship from the context or it must be stored in memory (lexicalized) as a result of previous exposure to the sequence.

This chapter examines a slightly different case of increasing information compression. It examines some English Adjective + Noun sequences that are used as premodifiers of common nouns, to find out whether, and if so in which contexts, they appear to be lexicalizing.

Among the questions that are addressed are the semantic relationships between the Adj + N premodifiers and the Adj + N sequences from which they are seen to originate, the reanalysis that this implies, and whether lexicalized sequences exhibit a different stress pattern. Apparent lexicalization in the data is compared with relevant lexical entries in three monolingual dictionaries.

2. Corpus investigation

We selected the 31 adjectives exhibiting the behaviour of central members of the class (Pullum & Huddleston 2002: 528) among the 50 most frequent adjectives in the British National Corpus (BNC), viz. *last, new, good, old, great, high, small, large, long, young, big, right, early, able, full, far, late, little, low, bad, real, clear, sure, black, white, common, strong, free, true, open, short*. We searched for the occurrences in the BNC of the Adj + N sequences beginning with one of these adjectives (both hyphenated and non-hyphenated forms) and used as modifiers, and extracted concordance lines for them.

These concordance lines were then imported into a table for database processing in order to retrieve the following statistics:

– Frequency of use of each Adj + N sequence in which the sequence is used attributively; this revealed a relative variety in the use of adjectives in the most frequently

recorded sequences: *long-term, full-time, short-term, large-scale, full-scale, high-speed, small-scale, high-level, long-distance, low-cost* (forms with more than 200 occurrences). The complete results for the adjectives with 100 occurrences or more are to be found in Table 1.

– Frequency of use of each [Adj + N] + N sequence; this revealed very high values for a small set of Adj + N sequences in the most frequently recorded ones: among the 62 such sequences with at least 20 occurrences, 34 are based on *long-term*, 14 on *full-time* and only 5 adjectives are used more than once as the first component (*long, full, low, short, high*). The complete list of all such sequences with at least 25

Table 1. Frequency of hyphenated adjectives used attributively in the BNC

ADJ	N	Frequency
long	term	3108
full	time	1358
short	term	1270
large	scale	771
last	minute	325
full	scale	285
high	speed	280
small	scale	270
high	level	261
real	time	223
long	distance	209
low	cost	208
low	level	188
high	tech	187
high	quality	182
open	air	181
real	life	178
white	collar	174
late	night	155
long	range	151
full	length	149
high	end	140
long	stay	134
low	income	127
long	run	120
high	performance	120
high	rise	119
long	time	117
common	sense	114
high	risk	111
free	market	110
low	fat	100

occurrences is presented in Table 2. No attempt was made to group the singular and plural forms under the same lemma for modified nouns[1] (for instance, *long-term effect* and *long-term effects* are counted separately).

We also recorded the number of occurrences of the Adj + N sequence where it was not used as a modifier (i.e. predicatively, as an adverbial or as the modifier and head of a noun phrase). For instance, *last ditch* has 102 occurrences as a modifier (68 in its hyphenated form and 34 in its non-hyphenated form) while the sequence has 10 BNC occurrences in which it is not used as a modifier (such as in the expression *fight to the last ditch*). One phrase (*last ditch attempt*) accounts for one third (34) of all occurrences of *last ditch* as a modifier, and only three other phrases have more than one occurrence[2] (*last ditch effort, last ditch stand, last ditch defence)*.

The presence of those Adj + N sequences in three monolingual dictionaries was recorded.[3] Dictionaries attempt to reflect the inventory of lexical units of the language and, although they are not perfect and inevitably include a dose of arbitrariness, they constitute evidence for lexicality independently from our corpus data (none of the dictionaries we looked at was based on the BNC). Our aim here was therefore not to assess the dictionaries used but to obtain some data triangulation. Entries in these dictionaries show a relatively strong correlation with frequency in the BNC, since 66% (35/53) of the sequences with at least 50 occurrences in their hyphenated form were included in at least two of the dictionaries.

Non-hyphenated Adj + N sequences were not part of our original database, but we collected frequency data concerning their use as modifiers with the Phrases in English interface to the BNC developed by William Fletcher (see note 1). Columns 2 and 5 of Table 3 show the figures for each of the 53 forms with more than 50 occurrences

1. It is also possible to obtain a list of all such syntactic patterns using the Advanced Search module of the *Phrases in English* Web site (http://pie.usna.edu/explore.html). *Phrases in English* allows the user to obtain the frequency and view the context of any polylexical expression found in the British National Corpus by entering individual words and/or their part of speech (it was created by William Fletcher, who also designed the Web concordancer KwicFinder, downloadable at http://www.kwicfinder.com/KWiCFinder.html). The list is obtained by using the POS codes 'AJ0' and 'NN?' for the bigram and the string *-* for the adjective. Note that this also returns results that do not fit the Adj + N pattern for the compound adjective (*video-taped report, pre-tax profit*). For those adjectives that do fit the pattern, the numbers obtained using *Phrases in English* are usually lower than those quoted here, because we included both hyphenated and non-hyphenated forms in our counts.

2. We did not assess the relative concentration of use of the modifying adjective with a limited set of nouns as opposed to a large variety of Ns, but our data suggest that this might be a factor relevant to the perceived lexicality of [Adj-N] patterns.

3. Merriam Webster's 2000 (version 2.5), American Heritage Dictionary 1994 (version 3.6a), Collins COBUILD 2001 (version 3.1).

Table 2. Frequency of [ADJ + N] + N patterns in the BNC

[ADJ + N]	N	Frequency
full-time	employment	89
full-time	job	88
full-time	education	83
long-term	effects	80
long-term	future	68
short-term	memory	66
long-term	unemployment	63
full-time	staff	60
white-collar	workers	52
long-term	strategy	47
long-term	investment	47
full-time	study	46
full-time	students	43
long-term	debt	40
long-term	contracts	40
long-term	care	40
short-term	interest	40
full-time	work	39
full-time	courses	39
long-term	view	37
long-term	development	37
full-time	research	37
long-term	interests	36
long-term	interest	36
long-term	aim	35
long-term	plans	34
long-term	commitment	34
long-term	growth	33
full-time	basis	33
long-term	solution	32
long-term	memory	32
long-term	trend	30
real-time	unix	28
long-stay	hospitals	28
low-income	families	28
old-age	pensions	28
low-pass	filter	28
long-term	relationship	26
long-term	prospects	26
long-term	damage	26
long-term	capital	26
full-time	employees	26
high-speed	rail	26
long-term	planning	25
long-term	effect	25
full-time	jobs	25

as a hyphenated modifier.[4] The counts are mostly lower for the non-hyphenated use of the Adj + N sequence (*high-tech*, *high-end*, *long-run*, *high-rise*, *high-profile*, *low-key*, *low-end*, *short-run*, *old-style*, *high-class*, *open-loop*, *small-town*, *high-energy* and *high-pressure* have very few or no occurrences in the non-hyphenated form). Even though the hyphenation of such forms is notoriously variable in British English, most are present in at least two of the dictionaries of our sample. A strong prevalence of the hyphenated spelling is thus correlated with the presence of Adj + N in those dictionaries, suggesting that hyphen use reflects writers' perception of a sequence as lexical.

Column 4 of Table 3 indicates the ratio of the total number of occurrences of the Adj + N form as a modifier (*long-time admirer*, *long-time supporter*) to the total number of non-hyphenated forms, whether used as a modifier or not (*long time companion*, *for a long time*). Again, the expressions that are hyphenated relatively frequently are more likely to have dictionary entries. Thus, the average ratio for the adjectives that are included in at least two dictionaries out of three is 2.70 and the average HM (use as a hyphenated modifier) count in the BNC for that category is 311.74. The corresponding figures for the complementary set (the adjectives that are included in one dictionary or none at all) are 0.49 and 107.39 respectively.[5] It thus seems that the spelling of the Adj + N sequence in its hyphenated form is related to the presence of a given adjective as an entry in the dictionaries of our sample.

We also investigated the relative frequencies of predicative vs. attributive uses of those Adj + N sequences in the BNC. As the search for such a sequence as '*was* [Adj + N]' returned very low numbers, we recorded the number of occurrences of the sequence '*was* [Adj + N] *and*' on the World-Wide Web for each of those forms,[6] using the results provided by search engines (manual examination of the first 100 hits for this query showed that about two thirds of the results actually correspond to predicative uses[7]). While some of the sequences are frequently used predicatively, others

4. The counts provided by *Phrases in English* are slightly inaccurate for our purpose in that they include all sequences regardless of constituent structure (for instance *long time companion* and *long time period*).

5. The HM/NHF ratio also varies according to the overall frequency of the hyphenated ADJ-N form in the BNC (the ratio is 1.22 for the 27 most frequent adjectives in Table 4 and 0.57 for the next 26 – i.e. the complementary set, excluding the 56 value obtained for the adjective *open-loop*).

6. We had originally considered using the query '*the* * *was* [Adj-N]', but the Google search engine has recently changed the use of the wild card character (*) in an expression enclosed by quotation marks from one intervening word to several words. Thus, a query like 'the * was real time' will return hits containing sentences like *we know for certain that the demo was real time* as well as *He proved the tech demo was real time*. The query we used returned mostly attributive uses even though some of the ADJ-N patterns (*real life*, *late night*, *long time*) generated a sizable amount of noise.

7. The purpose of using Google counts here is mostly to obtain a general idea of relative frequencies of the forms under study. For more information on the reliability of Google

Table 3. Frequency of ADJ-N modifiers in the BNC and presence of the compound adjective in three monolingual dictionaries

[Adj-N]	HM (Hyphenated modifiers)	NHF (Non hyphenated forms)	HM /NHF ratio	Non hyphenated modifiers	Dictionaries		
					MW	AHD	CC
long-term	3108	1730	1,8	630	X	X	X
full-time	1358	592	2,29	175	X	X	X
short-term	1270	826	1,54	114	X	X	X
large-scale	771	500	1,54	46		X	X
last-minute	325	623	0,52	31	X	X	
full-scale	285	89	3,2	15	X	X	X
high-speed	280	415	0,67	75	X	X	
small-scale	270	241	1,12	12	X	X	X
high-level	261	962	0,27	25	X	X	
real-time	223	200	1,12	14	X	X	X
long-distance	209	246	0,85	55	X	X	X
low-cost	208	278	0,75	31			
low-level	188	482	0,39	56	X	X	
high-tech	187	104	1,8	9	X	X	
high-quality	182	864	0,21	158			
open-air	181	307	0,59	40	X	X	X
real-life	178	567	0,31	29	X		
white-collar	174	127	1,37	51	X		X
late-night	155	175	0,89	15			X
long-range	151	111	1,36	11	X	X	X
full-length	149	244	0,61	24	X	X	X
high-end	140	36	3,89	3	X	X	X
long-stay	134	83	1,61	27			
low-income	127	201	0,63	38		X	
long-run	120	618	0,19	0			
high-performance	120	270	0,44	85			X
high-rise	119	40	2,98	11	X	X	X
long-time	117	4242	0,03	23		X	X
common-sense	114	969	0,12	14	X	X	
high-risk	111	328	0,34	88		X	
free-market	110	417	0,26	121	X		
low-fat	100	145	0,69	69			
high-profile	96	222	0,43	0		X	X
low-key	88	58	1,52	0	X	X	X
low-end	80	34	2,35	0	X	X	
high-technology	80	244	0,33	71	X	X	
old-age	78	1261	0,06	170			
short-run	76	164	0,46	5	X		
whole-school	74	150	0,49	17			
old-style	73	88	0,83	0	X	X	X
common-law	72	1141	0,06	308	X	X	X
last-ditch	68	44	1,55	20	X	X	X
high-class	65	72	0,9	3	X	X	X
free-trade	62	569	0,11	259	X	X	
early-morning	61	518	0,12	125			
real-world	60	666	0,09	4	X		
full-size	60	80	0,75	11	X	X	X
full-year	57	230	0,25	18			
open-loop	56	1	56	0	X	X	
high-fibre	56	39	1,44	16			
small-town	53	293	0,18	0			X
high-energy	53	107	0,5	0	X	X	
high-pressure	51	141	0,36	0	X	X	

MW = Merriam Webster's 2000; AHD = American Heritage Dictionary 1994; CC = Collins COBUILD 2001

(*white-collar, long-stay, long-run, high-rise, long-time, free-market, old-age, last-ditch, free-trade, full-year, high-fibre*) seem to have no or very few such uses. Here again, the numbers suggest that a high number of predicative uses correlates with the Adj + N sequence being present as an entry in the dictionaries used in our study (the average Google count for the '*was* [Adj + N] *and*' sequence is 374 for the forms that were found in at least two dictionaries as opposed to 217 for those that were found in one or none at all. The results are shown in Table 4.

3. Some examples of Adj + N

3.1 Frequency and distribution of six Adj + N sequences

A few of the more frequent sequences were selected for a more detailed analysis. The distribution of the sequences across constructions was noted, and evidence sought that lexicalization is taking place. Six frequent Adj + N sequences containing the adjectives *high*, *low*, *long* and *short* were chosen. Occurrences were taken from the 30-m-word Periodicals section and the 10-m-word Spoken section of the BNC. The normalised frequencies are shown in Table 5.

All the sequences are more frequent in written discourse. And in each case, the unmarked adjective of the antonymous pair forms a more frequent sequence than the marked one.[8]

The distribution of the sequences across construction types is shown in Table 6. (In order to mitigate the effect of the 'burstiness' of the data, multiple identical sequences from the same text were treated as a single occurrence.) Here, there was little difference in proportions between the written and the spoken data. Overall, the construction [Adj +N] + N (Column 1 of the table) accounts for between half and three quarters of occurrences. This suggests that hearers experience the sequences most frequently as modifiers.

In the case of *high quality*, the constructions *a high quality NP* and *an NP of high quality* are clearly very close in meaning. In the cases of *high level* and *long term*, however (and *low level* and *short term* follow suit), there appears to be a functional split between the [Adj + N] + N construction and a second most frequent construction in each case. This type of distribution is likely to be conducive to reanalysis.

---

counts, it is worth consulting Jean Véronis's blog at http://aixtal.blogspot.com/2005/03/web-google-adjusts-its-counts.html. Other interesting comments are to be found at http://aixtal.blogspot.com/2005/02/web-le-mystre-des-pages-manquantes-de.html (sic).

8. 'Unmarked' in broadly the sense of Lyons (1977: 305–311), that is, the term that tends to be used to refer neutrally to the range, as in *How high is it?*

Table 4. Frequency of predicative use of ADJ-N patterns in the BNC and on the World Wide Web vs. presence of the compound adjective in three monolingual dictionaries

[ADJ-N]	predicative uses in the BNC ("*was* ADJ + N")	"*was* ADJ + N *and*" on Google	Dictionaries		
			MW	AHD	CC
long-term	2	1,200	X	X	X
full-time	3	1,020	X	X	X
short-term	1	906	X	X	X
large-scale	0	378		X	X
last-minute	1	553	X	X	
full-scale	0	73	X	X	X
high-speed	0	573	X	X	
small-scale	0	797	X	X	X
high-level	0	301	X	X	
real-time	0	432	X	X	X
long-distance	0	839	X	X	X
low-cost	1	537			
low-level	0	411	X	X	
high-tech	0	411	X	X	
high-quality	1	752			
open-air	0	851	X	X	X
real-life	0	589	X		
white-collar	0	82	X		X
late-night	0	826			X
long-range	0	277	X	X	X
full-length	1	538	X	X	X
high-end	0	526	X	X	X
long-stay	0	1			
low-income	0	289		X	
long-run	0	51			
high-performance	0	223			X
high-rise	0	13	X	X	X
long-time	0	572		X	X
common-sense	1	2,050	X	X	
high-risk	0	845		X	
free-market	0	89	X		

(MW = Merriam Webster's 2000; AHD = American Heritage Dictionary 1994; CC = Collins COBUILD 2001)

3.2 Reanalysis

That a reanalysis has occurred, of course, can only be inferred once the expression is found in contexts which preclude the use of the complex sequence. The data on the six sequences was therefore examined for contextual clues to the reanalysis of these sequences as single lexemes. Five main contextual clues were found.

Table 5. Frequencies (per million words) of some Adj(-)N sequences in the Periodicals section and the Spoken section of the BNC

	Periodicals	Spoken
high(-)quality	16	5
low(-)quality	1	< 1
high(-)level	16	4
low(-)level	10	2
long(-)term	76	29
short(-)term	28	14

Table 6. Distribution of Adj N sequences by construction

	Adj N N	NP P Adj N	Adj N P NP	Adverb constr.	other constr. or unclear
high quality n=456	353 (77%)	64 (16%)			39 (7%)
low quality n=27	15 (just over half)	4			8
high level n=503	222 (44%)		210 (42%)	47 (9%)	24 (5%)
low level n=259	151 (58%)		46 (18%)	42 (16%)	20 (8%)
long term n=1243	968 (78%)			207 (17%)	68 (5%)
short term n=875	625 (71%)			190 (22%)	60 (7%)

Scope

In a few cases, a premodifying adjective seemed to have scope over the whole Adj + N sequence, suggesting that the sequence was reanalysed as a simple noun (Example (1)):

(1) we provide a consistent high level of service (Web)

It looks as though the analysis of (1) is [[consistent] [high level]] [of [service]] and might be paraphrased as 'the high level of our service is consistent'. And occasionally, the [Adj + N] sequence acts as an adverbial modifier of an adjective, as in (2):

(2) disabled and elderly and long term ill people and single parent families (BNCPer)

Here, *long term* has scope over the adjective *ill* and thus acts as an adverb modifying *ill*. These examples suggest that the [Adj + N] sequences are perceived by the writers as single processing units.

Predicative use as an adjective (Adj + N > Adj)
In several cases, a sequence seemed to be analysed holistically as a gradable adjective that could be used predicatively, and preceded by *how* or *too* or *however* (Example (3)):

(3) a. some abuse was very much more long term (BNC Sp)
b. he argues that the benefits were 'too small and too short-term ...' (BNC Per)

In (3a) *long term* seems to be treated as a simple adjective forming its comparative with *more* (cf. *longer term*). In (3b) *too short term* clearly has the same structure for the writer as *too small*, so that *short term* is treated as a simple adjective.

Unexpected juxtapositions implying meaning shift
Occasionally Adj + N sequences applied as modifiers were found in unexpected contexts (Example (4)):

(4) a. The new system is designed to high quality standards (BNC)
b. in immunocompetent children, [...] viral infections are of short term duration (BNC Per)

In Example (4a) *high quality standards* seems anomalous if interpreted literally. Rather, *high quality* seems to mean 'very good'. Again, (4b) only makes sense if *short term* is interpreted as a single modifier describing the type of duration. The fact that the phrase is not unacceptably tautologous suggests that the sense of *term* as 'period of time' has weakened as *short term* has fused.

Adj + N as adverb
Long / short term are sometimes found as adverbs, modifying an Adj, as in (4a), a VP or a sentence as in (5b):

(5) a. I'm looking for proof that I am long-term lovable (BNC Per)
b. Short-term the outcome depends on the market (BNC Per)

The adverbial use for adjective modification (5a) was already noted in example (2); (5b) suggests that the adverbial phrase *in the short term* is sometimes replaced by the univerbation *short-term*.

Adj + N as NP subject or object with restricted sense
Even when the Adj + N sequence functions as a standard NP argument, a more specific sense is often evident. In (6a) and (6b), *the long term* and *the short term* refer not to periods of time, but to sets of events or activities perceived as contained in the period types.

(6) a. The 'ecology of leadership' as Helgerson labels it, promoted the long-term and sees daily events within a larger .. context (BNC Per)

b. ..instead of focusing on the short term and the next election (BNC Per) [cf. A company often thinks that it can let space for a short term (BNC Per)]

The constructions exemplified in (1-6) above suggest that the Adj + N sequences have not only fused in certain contexts into univerbal noun modifiers, but also and at the same time into other syntactic categories, notably adverb and noun, perhaps driven by the very high frequency of the collocation and the emergence of one or more new, non-compositional senses.[9]

There is, in fact, evidence of this kind of semantic shift and of incipient polysemy. For example, *high quality* and *high level* as premodifiers seem to indicate that the entity modified belongs to a particular category, rather than locate the entity on a scale. *High level* and *low level* show a marked polysemy between quantity (a lot/a little) and quality (degree of importance or prestige) in addition to the literal spatial sense. These polysemies correspond to the structural split noted above. *Long term* and *short term* as premodifiers have a more specialized and abstract sense than the adverbial phrases *in the long / short term*.

Overall, the proportion of unambiguously lexicalized uses is small but not negligible. In information structure terms, the premodifier use represents a decrease in informational salience; in fact there is loss of information regarding the relationship between the premodifier Adj + N and the following N (cf. the findings of Biber 2003 for NN sequences). The antonymy of *high-low* and *long-short* seems to be carried over to the lexicalizing items and maintained across the various constructions and uses. Finally, specialist or technical uses are evident, often carrying over into non-specialist contexts. For instance, *long term* / *short term* appears to have a particular use in medicine and to be more lexicalized in medical contexts than elsewhere; likewise *low-level* in aviation.

4. The phonology of Adj + N sequences

Compounding is known to cause phonological changes in the resulting units, in particular on the stress pattern of N + N and Adj + N sequences which becomes /12/, i.e. the first element has primary stress, as opposed to the /21/ pattern of occasional sequences of the same structure. Since high-frequency Adj + N sequences may be lexical or on their way to lexicalization, it is of interest to examine whether their stress patterns differ from those of low frequency ones. The hypothesis is that they would exhibit a /12/ pattern, while the low-frequency sequences would exhibit the /21/ pattern of occasional sequences. It is also interesting to determine whether this pattern holds for the predicative uses, as this would be a sign of frozenness.

9. Analysis of diachronic corpus data tracing the evolution of frequencies would be required to confirm or refute this hypothesis.

A laboratory experiment was undertaken in order to test these hypotheses. The study of stress is complicated by the fact that it results from a combination of variables. Fry (1955) demonstrated that stress in English involves duration, frequency of the fundamental, and intensity (see also Ferragne 2003). Comparisons of duration can only be made 'all other things being equal', i.e. in identical environments, which was precluded by the variety of the sequences under investigation and their contexts. Fundamental frequency changes due to stress are difficult to distinguish from those due to intonation, and, in addition, often require some subjective interpretation by the expert. For these reasons, it was decided to investigate intensity only as the one parameter of lexical stress that is relatively easy to measure in laboratory conditions, keeping in mind that some differences might be missed depending on the correlation with the two parameters that were not taken into account.

Three categories of targets were selected: [Adj + N] + N sequences with a frequent Adj + N modifier, ex. : *short term relief*, [Adj + N] + N sequences with infrequent Adj + N, ex. *short line store*, and the same frequent Adj + N sequences in predicative position. An Adj + N was considered as frequent if it had at least 100 occurrences as the premodifier of a noun in the BNC. A set of short texts containing the three categories of target items (n = respectively 16, 16 and 15) were extracted from the BNC (or, failing this, the Web), slightly shortened when necessary, controlled for focus and contrast, and together with distractor texts grouped in a thematic order so subjects would not guess the aim of the enquiry.[10] Six subjects, all speakers of British English, took part in the experiment. The subjects were equipped with head-mounted microphones and asked to read aloud the texts which appeared in succession on a computer screen as they pressed a key. None of the subjects reported having inferred what the experiment was about.

Recordings with hesitations during the pronunciation of the target sequence were rejected; in addition, a difference of 3 dB between the stressed syllables of a sequence was used as a reliability threshold as this is generally considered to be the minimal difference perceptible by the human ear in the case of speech sounds. In the case of attributive Adj + N sequences, two comparisons were made: a) between the intensities of Adj and N, b) between Adj + N as a unit and the following modified noun. Obviously, only comparison a) was possible in predicative cases. The results are presented in Table 7.

As stated above, a number of recordings had to be rejected and we need to keep in mind the possibility that the other two parameters of stress take on greater weight when intensity is not significant. The data, however, clearly fail to confirm the hypotheses. The fact that the /12/ intensity pattern typical of lexicalized compounds dominates in the case of infrequent Adj + N sequences as well as frequent ones suggests that degree of lexicalization is irrelevant here. In addition, the fact that this stress pattern is

10. For example: *Technical advances and the increasing demand for long distance communication meant but one thing: lower prices.*

Table 7. Stress patterns

	a) Adj ~ N			b) [Adj + N] ~ N		
	/ 1 2 /	/ 2 1 /	χ^2	/ 1 2 /	/ 2 1 /	χ^2
frequent attributive	32	4	20.25**	42	1	35.38**
infrequent attributive	25	3	15.75**	42	1	35.38**
predicative	25	9	6.61*			

* – p < .025 ; ** – p < .01

also found in predicative positions shows that it is the adjectival Adj + N pattern itself which causes it to be present.

The prevalence of /12/ when one compares Adj + N and the final (head) noun in [Adj + N] + N sequences is more difficult to account for. It apparently contradicts Liberman & Sproat (1992) who, in their brief mention of [Adj + N] + N sequences like *fresh fish shop* or *civil rights bill*, state that the main stress is on the middle word, but this may be due to the fact that their examples of Adj + N are not semantically adjectival, in contrast to ours. We suggest that the strength of the /12/ pattern which is associated with adjectival Adj + N sequences causes the reduction of the stress of the head-noun.

5. Conclusion

The corpus investigations reported in this chapter have shown relatively high frequencies for a small set of Adj + N sequences used as premodifiers of nouns, as well as a prevalence of this attributive pattern over other uses; in addition, Adj + N sequences are found in larger numbers in written discourse. The most frequent ones, especially those that occur in hyphenated form, tend to be included in dictionaries.

In a sample of frequent Adj + N sequences, some phenomena were observed that suggest a degree of lexicalization for the sequences: in particular, modifications of scope, use as adjective, use as adverb, and standard uses as NPs but with a restricted sense. These data provide evidence that the Adj + N sequences in question have two or more syntactic analyses and two or more distinct senses, at least for some speakers. They suggest a recategorization of Adj + N (as Adj, as N or as Adv), a loss of compositionality, and a loss of semantic transparency. In short, the data point to the 'linear fusion' of certain Adj + N sequences. However, there is no evidence of concomitant phonological change, as no difference in stress pattern was found between frequent sequences and infrequent ones. This suggests that while syntactic reanalysis and semantic shift may be interdependent, phonological change may be independently motivated.

References

Biber, D. (2003). Compressed noun-phrase structures in newspaper discourse: The competing demands of popularization vs. economy. In Aitchison, J. & D. M. Lewis (eds.) *New Media Language,* 169–181. London: Routledge.

Biber, D. & V. Clark (2002). Historical shifts in modification patterns with complex noun phrase structures: How long can you go without a verb? In Fanego, T., M.-J. López-Couso & J. Pérez-Guerra (eds.) *English Historical Syntax and Morphology. Selected Papers from 11 ICEHL*, 43–66. Amsterdam: John Benjamins.

Biber, D., S. Johansson, G. Leech, S. Conrad & E. Finegan (1999). *Longman Grammar of Spoken and Written English*. New York, NY: Longman.

Brinton, L. J. & E. C. Traugott (2005). *Lexicalization and Language Change*. Cambridge: Cambridge University Press.

Bybee, J. (2002). Sequentiality as the basis of constituent structure. In Givón, T. & B. F. Malle (eds.) *The Evolution of Language out of Pre-Language,* 109–134. Amsterdam: John Benjamins.

Erman, B. & B. Warren (2000). The idiom principle and the open-choice principle. *Text* 20: 29–62.

Ferragne, E. (2003). Les corrélats phonétiques de l'accent de mot en anglais contemporain. DEA dissertation, Université Lumière-Lyon 2.

Fry, D. B. (1955). Duration and intensity as physical correlates of linguistic stress. *Journal of the Acoustic Society of America* 27: 765–768.

Krug, M. (2001). Frequency, iconicity, categorization: Evidence from emerging modals. In Bybee, J. & P. Hopper (eds.) *Frequency and the Emergence of Linguistic Structure,* 309–335. Amsterdam: John Benjamins.

Liberman, M. & R. Sproat (1992). The stress and structure of modified noun phrases in English. In Sag, I.A. & A. Szabolcsi (eds.) *Lexical Matters,* 131–181. Stanford, CA: CSLI.

Lyons, J. (1977). *Semantics*. Cambridge: Cambridge University Press.

Pawley, A. & F. H. Syder (1983). Two puzzles for linguistic theory: Nativelike selection and nativelike fluency. In Richards, J.C. & R. W. Schmidt (eds.) *Language and Communication,* 191–226. London: Longman.

Pullum, G. K. & R. Huddleston (2002). Adjectives and adverbs. In Huddleston, R. & G. K. Pullum (eds.) *The Cambridge Grammar of the English Language,* 525–562. Cambridge: Cambridge University Press.

Quirk, R., S. Greenbaum, G. Leech & J. Svartvik (1985). *A Comprehensive Grammar of the English Language*. London: Longman.

Phrasal similes in the BNC

Kay Wikberg

This study focuses on similes, a category of phraseological units that has been amply studied in the phraseology literature but suffers from a lack of corpus-based description. The concept of simile is defined and the differences between similes and comparisons, on the one hand, and similes and metaphors, on the other, are described in some detail with reference to the literature. Then four simile patterns are extracted from the British National Corpus: *as* Adj/Adv *as*, *is like a(n), is like* V-*ing a(n)* and V *like a(n).* The selection procedure consists in sifting out literal comparisons using the reversibility test and the degree of figurativeness. This process highlights interesting differences between the patterns in terms of frequency, lexicalization and register. The *as* Adj/Adv *as*-pattern is the most frequent but the vast majority of the instances represent literal comparisons. The few per cent similes tend to be lexicalized and there is little innovation. By contrast, the *is like a(n)*-sequences are much less common but regularly figurative and innovative. The V *like a(n)*-pattern also proves to contain a high proportion of similes, both lexicalized and innovative.

1. Introduction

Similes are well-known figures of speech and are easy to recognize because of their explicit markers and the kind of comparisons they make. We shall begin by looking at some fairly recent definitions. Miller (1993:373) defines a simile as "a comparison statement involving two unlike things." Although he otherwise gives a detailed and illuminating semantic analysis of similes, his definition is inadequate. By comparison Wales (2001:358), writing for a dictionary, states that a "simile is a figure of speech whereby two concepts are imaginatively and descriptively compared: e.g. *My love is like a red, red rose; as white as a sheet,* etc." Both definitions refer to two aspects of a simile which are basic to its identification, i.e. comparison and, implicitly, the notion of comparability. The two things or concepts referred to, the topic (*my love*) and the vehicle (*a red, red rose*), are unlike but they must be similar enough for us to understand the grounds for the comparison. The figurative element refers to the fact that neither of Wales' examples really makes you think of colour. The red rose conveys BEAUTY or FRESHNESS, the white sheet PALENESS, if said about a person.

However, we need a definition which can distinguish a simile from a metaphor. Both are figurative expressions but the simile represents an explicit comparison as it must contain a marker. Thus we end up with this working definition:

> A simile can be defined as a figurative expression used to make an explicit comparison of two unlike things by means of the prepositions *like, (as) … as* or the conjunctions *as, as if, as though.*

It is the prepositions that mark a simile as phrasal whereas the conjunctions open clausal similes, which we can ignore in this context. We shall come back to the communicative impact of the marker in Section 3.

Another distinction that has to be made is that between a unit of description that can be part of phraseology and other freely formed sequences of words. To distinguish phrasal expressions from other kinds of strings, it is customary to use the following criteria: institutionalization, fixedness, and non-compositionality (Moon 1998). I shall illustrate the application of these criteria to this example: *Ruth* ***dropped like a stone*** *into the armchair.* When the lemma DROP is followed by *like a stone,* this constitutes a conventional or institutionalized expression. Evidence for this is found in the BNC, which contains 12 instances of *stone* co-occurring with DROP *like a* out of 25 hits. Fixedness applies in the sense that *stone* only occurs in the singular and only with the indefinite article. Moreover, *like* cannot be replaced with *as*. Finally, non-compositionality refers to cases when the meaning of the whole is not equal to the sum of the meanings of the individual elements. This is a slightly more controversial notion as it has to do with units of meaning (cf. Moon 1998). Our data shows that the meaning of the current string depends closely on the type of Subject. With Subjects referring to physical things (*Ruth*), DROP *like a stone* involves a sudden or quick fall or move down. However, when this is said about somebody's heart as in *Her heart dropped like a stone*, we get an instance of metonymy: the person whose heart is referred to suddenly got sad. Several abstract Subjects in the data result in different interpretations, such as *an economic crisis with the 'pound dropping like a stone'* ('decreasing in value', figurativeness marked in the text) and *Miss Jarman's words dropped like a stone in a pond, causing ripples of protest* ('had the same effect as'; simile and metaphor combined). Having said that, it is important to be aware that there are degrees of these properties and that one can be more dominant than the others.

The aim of this chapter is to show how a corpus study of some major simile structures in the BNC can be used to throw light on how similes function in discourse. My hypothesis is that by examining a relatively large number of authentic similes in relation to each other and to their co-text we will gain more specific knowledge than we have had so far of different simile structures, their frequency and communicative functions. The structures studied are those represented by these sentences:

Predicative uses

(1) He is **like a wolf**. (ADP [arts])[1]

(2) Sonny was tall and broad and **as strong as an ox**. (B1X [imag.])

(3) it's **like pushing a ten-ton truck with its brakes on**. (FEE [imag.])

Adverbial uses

(4) I have known horses I could read **like a book**. (G06 [imag.])

(5) Even in wellingtons, he could move **as lightly as a squirrel.** (AC4 [imag.])

That three of the above examples compare human beings to animals is not a coincidence. Animals represent a much wider variety of creatures and therefore make up a rewarding source of comparison when describing people and their behaviour. But whereas metaphors structurally vary from single words to long chunks of text, similes consist of a limited number of fixed comparative frames. It is the open slots in these frames that can be filled with material from the four major word classes and which result in a great deal of variation in content and images. The main focus will be on instances like (1) and (4). Example (3) contains the preposition *like* followed by a non-finite clause as Complement. The structure has been included although it is somewhat peripheral to phraseology. On the other hand, *as-* and *as if (though)*-clauses have nothing to do with phraseology and have therefore been omitted.

A typical limitation of linguistic simile research so far has been the use of simplified examples removed from their context. Another limitation is that there has been relatively little focus on the range of different simile structures such as the question of lexicalization versus innovation. It is expected that a corpus-based study will enrich our understanding of similes in these respects. Finally, as regards patterning in different genres, previous studies have shown that similes are most frequent in fiction (cf. Wikberg 2004) but the BNC can shed more light on this as well (cf. Section 5).

2. Literal comparison versus simile

What looks like a simile on the surface is often just a literal comparison, which is reversible:

(6) a. Olive oil is like a fine wine. →
b. A fine wine is like olive oil.

By contrast, a simile is irreversible:

(7) a. Kim is like a ray of sunshine. →
b. *A ray of sunshine is like Kim.

1. Each file in the BNC World Edition has a 3-character name. This is here followed by a domain reference as described in *The BNC Handbook* (Aston & Burnard 1998).

What makes the literal comparison possible is of course the fact that olive oil and wine are much more similar categories than Kim and a ray of sunshine. Olive oil and wine are related entities, a person and (a ray of) sunshine are not. An interesting fact about (6a) as compared with (7a) is that the former makes us expect an elaboration. Nobody drinks olive oil so how can it resemble 'fine wine'? The elaboration is actually given in the text:

(8) Olive oil is like a fine wine – you can buy 'plonk' for everyday cooking or an estate bottled vintage for the connoisseur. (ABB [Leisure])

This is not necessary with (7a), which is complete as it stands, a lexicalized simile used for positive evaluation. As regards this example and its metaphorical equivalent *Kim is a ray of sunshine*, the difference between the simile and the metaphor in this special case does not seem to be very significant.

The preposition *like* is used in a number of different senses, which could be summarized in the following way with examples from the BNC and *The Cambridge Advanced Learner's Dictionary (CALD):*

a. 'similar to': *Keeping a secret in the Manse is* ***like*** *hiding a fire under straw.* (GWF [imag.])
b. 'such as': *As children, we all got up to tricks* ***like*** *pinching a few apples or knocking on someone's door and running away...* (CF9 [leisure])
c. 'typical of': *That's just* ***like*** *Maisie to turn up half an hour late to her own party!* (CALD)
d. 'as if': *It looks* ***like*** *I'm going to be in the office until late tonight.* (CALD)
e. pragmatic marker: *... Cos what I need is* ***like*** *a block of wood.* (KNY [spoken demographic])

Only (a) is a valid reading in this context. Quirk et al. (1985: 698) point out that "[with] transitive and intransitive verbs, *like* can have the meaning of 'in a manner resembling'; with copular verbs, its meaning is purely that of 'resemblance'..." Sentences like (e) were dropped since *like* is not a preposition.

It is also worth mentioning that *like* can be used for morphological creativity with the same meaning as *like* in (a) above:

(9) Enamels of permanent **gem-like quality** can be made by adding the appropriate metals to powdered glass fused to a metal base. (FBA [world affairs])

The sentence is immediately preceded by a reference to 'precious stones', so 'gem' can be regarded as a given entity.

3. Similes versus metaphors

Traditionally similes have attracted a great deal of attention in philosophy and literature. It is to Aristotle's Rhetoric (337 BC) that we owe the view that metaphors are

elliptical similes, sometimes referred to as the Comparison Theory (Searle 1981: 264). However, as we shall see below, a number of scholars have shown that this view is not correct (e.g. Black 1981; Searle 1981 and Glucksberg 2001). In the choice between metaphor and simile in literature, Lodge (1977: 113) argues that a writer may "incline towards simile rather than metaphor proper when drawing attention to similarity between things dissimilar." Nowattny (1962: 66) expresses this in a way which is close to very recent findings:

> It may (...) well be true that suggestion is usually better done by simile. Simile (when simple) does not indicate the respect in which one thing is like another thing. It says the things are alike; it is up to us to see why; the things may be alike in a large number of ways. Thus simile in turn has its own advantages. It may be a considerable advantage to the poet to claim that likeness exists without indicating where it lies.

It has been argued that you can compare anything to anything else. In practice what happens in communication is that 'things dissimilar' will still have to share some properties for us to interpret them as somehow similar without making undue cognitive effort. Linguists working in the generative tradition in the early 1970's explained figurative usage as semantic deviance, whereas pragmaticians have considered metaphor a violation of the conversational maxim of Quality (Grice 1975) or 'loose use' of language (Carston 2002). Leech (1969: 156–157) lists some differences between metaphor and simile, such as the greater explicitness of similes, the specification of the ground in simile, and the inexplicitness of metaphor "with regard to both the ground of the comparison, and the things compared". He also mentions the occurrence of blends in literary language, simile and metaphor operating in the same sentence (as in sentences (14)–(15) below). But even in books on conceptual metaphor like Lakoff & Johnson (1980) and Kövecses (2002) it is hard to find examples of similes and if they occur, they are not referred to as a special figure of speech. In this respect the papers by Ortony (1993) and Miller (1993) are exceptions since both address the issue in some depth. Ortony considers similes 'nonliteral comparisons' whereas Miller argues that a metaphor is a comparison statement with parts left out. He also accounts for the formal semantic analysis of similes in terms of predicate logic. Goatly (1997) discusses the linguistic properties of both metaphor and simile with authentic examples from literary texts. Since then, what we have learnt about similes comes mainly from cognitive psychologists (Glucksberg 2001; Chiappe & Kennedy 2001).

Most linguistic treatments of similes concern the relationship between simile and metaphor such as the condensed simile hypothesis and the question whether a simile expresses something else than the corresponding metaphor, if there is one. There are several reasons for not believing in the condensed simile hypothesis. For one thing, metaphors range structurally from single lexical items to clauses and sentences. Although it is true that many metaphors of the *A is B*-type (e.g. *He is a wolf*) can be turned into similes, thinking in terms of simile in all cases would be difficult,

or impossible. Thus it would be hard to find a simile that would correspond to the orientational metaphor in this sentence:

(10) The number of crimes keeps going up.

A more interesting topic is no doubt the semantic and pragmatic aspects of a simile. There is recent research by Glucksberg (2001) and Chiappe & Kennedy (2001) which shows that a simile does not express the same thing as a metaphor. For instance, we have learnt that a metaphor of the *A is B*-type categorizes; by contrast, the *A is like B*-structure is used when A is similar to B with respect to some properties only. It is assumed that similes are understood in terms of property sharing between topic and vehicle. Thus, in *He is like a wolf* we are clearly not interested in the fact that a wolf has four legs, lives in the wilderness, or is predatory. *The New Oxford Dictionary of English* defines the figurative use of *wolf* as "used in similes and metaphors to refer to a rapacious, ferocious, or voracious person or thing." The simile can vaguely refer to any of these or other salient attributes connected with the wolf whereas the metaphor *He is a wolf* assigns wolf-like properties to the topic in a more general sense. Only the context could help us to understand what these are.

As regards their description, similes can be handled in the same way as metaphors using the traditional elements 'topic', 'vehicle' and 'grounds', or within conceptual metaphor theory, i.e. as a mapping between two semantic domains, the only formal differences being the constant presence of a marker and the occurrence of explicit grounds or correspondences, as in *as* Adj *as*-structures (see (2) and (5) above). However, as Steen (1999) has pointed out, a metaphor need not always have an explicit topic or vehicle. A simile needs both. A semantic difference between similes and metaphors has to do with the fact that unlike similes, some metaphors can express correlations rather than similarity, as in (10). Grady (1999) gives examples like MORE IS UP and DEATH IS A THIEF, which do not rely on similarity. A more important communicative difference is that similes can convey what is not possible to say by using the equivalent metaphor:

(11) **A rose is like a dog**: you do not and cannot get back more than you put in and, in the same way that a dog is more rewarding than most other animals, you get more back from a rose than other flowers. (CMM [leisure])

Obviously **A rose is a dog* would not make sense since roses and dogs are entirely different categories. But what (11) illustrates is the fact that the grounds for the simile are given after the actual comparison. The grounds are indeed necessary for this simile to work properly.

Similes and metaphors are both strongly context-dependent, which is sometimes forgotten. The language user draws on elements and aspects of conceptual domains that are currently activated or potentially accessible in a given context. Accessibility is dependent on common ground and a number of other factors such as cultural background, education, age, and personal experience.

4. Corpus-based analysis of four simile patterns

Finding the similes in the BNC is a selection procedure which starts out from comparative structures with the formal properties underlying examples (1)–(5), here repeated for convenience:

Predicative uses

1. He is **like a wolf**. (ADP [arts])
2. Sonny was tall and broad and **as strong as an ox**. (B1X [imag.])
3. It's **like pushing a ten-ton truck with its brakes on**. (FEE [imag.])

Adverbial uses

4. I have known horses I could read **like a book**. (G06 [imag.])
5. Even in wellingtons, he could move **as lightly as a squirrel**. (AC4 [imag.])

The BNC contains tens of thousands of such expressions and the researcher's most difficult and time-consuming task is therefore to single out the figurative occurrences. This is probably the chief difference between similes and most other fixed phrases. In spite of the reversibility criterion used for identification, identifying similes in discourse cannot be a completely objective task because of the enormous range of possible comparisons. Owing to the huge numbers, several formal constraints were imposed to reduce the data slightly. One was definiteness, another was tense. In the initial examples (1)–(5), the NP's following the markers are indefinite. In none of the examples is there reference to a specific wolf, ox, book or squirrel since no comparison is being made with a given exemplar of any of these entities. It is the indeterminate property of the vehicle that is at issue. Still, a look at the concordance lines shows that preference for indefiniteness varies slightly depending on the comparative structure. Thus it is hard to find a simile containing the sequence *as* Adj *as the*, which is preferred for literal comparison, whereas there are 143 instances of the string *is like the*, as in

(12) A long marriage **is like the** bath foam covering the water – until, in the end, it's all foam and no water. (AEO [imag.])

In consequence, in order to restrict the number of instances, only indefinite examples were selected, which also applies to the V *like a*-structure. As regards tense, only present tense instances were included of BE whereas in the V *like a*-structure, any form of an intransitive lexical verb other than BE was allowed.

Table 1 contains information on the number of comparisons, stages in the selection procedure and the resulting number or proportion of similes in the various patterns.

Table 1. Frequencies and selection procedure

Pattern	Number of comparisons	Comments on the selection procedure and numbers (proportion) of similes
as Adj *as*	10,341	Only the 200 most frequent adjectives selected. 5,257 of these occurring with BE. Thinning based on 1,000 instances gave 64 similes (6.4%).
as Adv *as*	12,901	Based on the 100 most frequent adverbs. After deletion of the most common literal sequences manual selection of a sample of adverbs gave an approximate figure of one per cent similes.
is like a	444	n=414 (93%)
is like V-*ing a*	69	n=58
V *like a*	6,608	3,429 (52%) are perception verbs. Thinning based on 1,000 instances of other main verbs gave 392 (39,2%) instances.

4.1 *as* Adj/Adv *as* (n=34,224)

The correlative *as* . . . *as* serves as the frame for both adjectives and adverbs, which make up the grounds for the comparison, as in *as **strong** as – as **strongly** as.* The main difference between the two subpatterns is obviously that one functions as a Complement, the other as an Adverbial. As the search for the 'as Adj/Adv as' pattern in the BNC yielded as many as 34,224 occurrences, the number of hits had to be reduced. How that was done for each category will be explained below. The total number for *as* Adj/Adv *as* combined (34,224) is higher than the sum of the figures in the table since a selection of adjectives/adverbs has already been made, as explained in column 3 of Table 1. A lot could also be said about differences in figurativeness and frequencies between the adjective and the derived adverb and how each category patterns with other words to form more or less fixed expressions. Some examples of this will be given in Sections 4.1.1 and 4.1.2.

4.1.1 *as* Adj *as* (n=10,341)

If we disregard degrees of figurativeness, what is characteristic of this comparison structure is first of all its high frequency. This is possibly due to the fixed frame formed by the correlative *as*. . . *as* combined with high-frequency adjectives. It turned out that the most frequent adjectives in these *as-as* sequences form fixed combinations which occupy a large proportion of the total number of instances. As appears from Table 2, the proportion of fixed combinations can be quite high: 60 % of the occurrences of the pattern *as simple as* occur in the fixed combination *as simple as that*. There are numerous other combinations, which in part demonstrates the aptness of these adjectives for comparison.

Because of the large number of instances only a sample of *as* Adj *as*-sequences was scanned, including the 200 most frequent adjectives, i.e. those occurring 11 times

Table 2. Recurrent *as....as* combinations

Recurrent combination	Total number of recurrent combinations	Total number of *as...as* patterns	% recurrent use
as simple as that	225	374	60.1%
not/almost/nearly as bad as	112	427	26.2%
as low as possible	47	261	18%
just as important as	53	344	15.4%

or more, which gave 10,341 instances. When forms of the lemma BE are included to the left of the *as-as* string, there remain 5,257 comparative instances. Even this number causes problems since the maximum number of items that the SARA server can handle simultaneously is 1,000. Thinning was therefore applied to the strings *be as* Adj *as* (n=1,373) and *is as* Adj *as* (n=1,025). A random sample of 500 instances from each gave altogether 64 similes (6.4 %). That the proportion of similes was so low was surprising.

If we now go back to example (2), the adjective in the simile *as **strong** as an ox* is salient and obviously has to be construed in relation to the previous topic (the man being referred to) and the following vehicle (*ox*). If the author had just written *tall and broad and strong*, the intensifying effect would have been lost. Other strong animals human beings are compared to in our data include lions, horses and bulls. What then are the typical properties of adjectives which promote the lexicalization of this pattern? There is a preference for monosyllabic and polysemous adjectives. One of the most common adjectives was *thick*, which occurred in idioms like *as thick as thieves* (x4), *as thick as shit* (x1), *as thick as two short planks* (x4). Typically it is extreme degrees of the salient property that are enhanced through such figures, often hyperbolical. The similes in examples (2) and (5) also belong to this category though (5) combines with the adverb.

Owing to the relatively low frequency of the *as* Adj *as*-simile pattern, it is not surprising that the data shows little creativity. When similes occur, they tend to be lexicalized (cf. Moon 1998, 2005). However, *white as snow* appears once as *white as newly fallen snow*, which might be a more realistic version today. Another example of a dynamic simile is

(13) even though he found his host about **as intelligent as a mule** and **as human as a boa-constrictor** (ASC [Biogr.])

This example illustrates two things, first that similes can be combined, second that they can be used ironically.

4.1.2 *as* Adv *as* (n=12,901)

The total number of *as* Adv *as*-sequences includes all comparative forms occurring 5 times or more, altogether 100 different adverbs. Since there were relatively few figurative occurrences, thinning was not applied in this case; instead a manual search of a selection of adverbs was made. This process demonstrated that adverbs have an even

lower proportion of similes than the corresponding adjectival category, about 1 per cent. The reason for this is that the vast majority of the instances are ready-made literal combinations (*as far as, as soon as, as well as,*...), which leave no space for figurative usage.[2] *As far as* (n=5,107) and *as soon as* (n=1,238) are both grammatical and together account for 56 per cent of all occurrences. There were numerous other adverbs which did not form any similes at all such as *completely, firmly, slowly, swiftly, evenly, frequently, accurately* and *readily*. The most frequent comparative strings include *possible* (n=3,036), as in *as quickly as possible* (392/636), or *as* Adv *as* + personal pronoun + *can/could* (e.g. *as best as he could*), which are literal expressions.

Among the dynamic occurrences some were used as 'precision similes' (Goatly 1997: 185–186) in the semantic domain of communicating (WORDS ARE THINGS):

(14) Blanche let the question drop **as softly as** a feather (G1W [imag.])

(15) Every word fell **as heavily as** slabs of marble (HA3 [imag.])

In each example a metaphor precedes the simile, which serves to give additional precision to the grounds. *Light as a feather* is a lexicalized simile (9 instances in the BNC) and it is this LIGHTNESS that allows the feather to drop *softly.* Other examples of occasional lexicalized adverbial similes include *as silently as a cat* and *as regularly as clockwork.* To conclude this section: the *as* Adv *as*-category is no doubt interesting in the way it forms high-frequency fixed literal expressions but hardly in terms of figurative creativity.

4.2 *is like a* (n=444)

This, then, is the favourite among the theoriticians, a sort of prototype of a simile. There are several reasons for this. One is the fact that the majority of the sentences that contain this string are figurative (over 90 per cent). Another is that it invites comparison with the corresponding metaphor. A third factor is the rich semantic environment provided by *like* ('similar to') and the following vehicle. This structure is far less transparent than *as* Adj *as*-sequences and can therefore express a higher degree of figurativeness although lower than the equivalent metaphor. The grounds are sometimes implicit, as in these more or less lexicalized examples:

(16) The place **is like a pigsty**. (A4W [world affairs])

(17) That room **is like a morgue**. (HWP [imag.])

(18) T'**is like a bread oven** in here. (HHI [imag.])

(19) Christ, this place **is like an icebox**. (BMW [imag.])

2. *As far as* is literal in *some of the escapees got as far as North Wales* and metaphorical in *as far as I can gather.* However, I have not included *as far as* at all since it is completely 'dead' as a figurative fixed expression.

What these similes share is vehicles representing unattractive places to be. Phrasal similes are widely used to express evaluation. Thus, (16) describes the Ford factory in Dagenham. Attributes mentioned in the immediate co-text are 'cramped', 'unpleasant' and 'inefficient'. *Like a pigsty* is a fixed combination, which has direct equivalents in other languages (N *som en svinesti*; Fi. *kuin sikala*; Fr. *comme une porcherie*).[3] (17) is used humorously according to *The Longman Dictionary of Contemporary English.* (18) is a longer version of the fixed expression *like an oven* ('very hot'). There is an even stronger version of this one in the data: *our cell . . . is like a sauna and remains so all night despite open windows* (AHC). Finally, (19) is an exclamation by a man in a bedroom; the meaning of the simile as the opposite of that in (18) does not require much inferencing.

However, the striking thing is that in the vast majority of cases the vehicle contains modifiers, usually postmodifiers, which are suggestive as to what sort of similarity is being referred to. If such information is not given in a postmodifier, it is provided in the next few sentences. An interesting aspect of the *is like a(n)*-pattern is that, in contrast to the structures described so far, most instances are innovative. The innovation must then hinge on the type and character of things compared and particularly on the noun or noun phrase that follows the verb. It is hard to generalize about the nouns in vehicle position. The new information they provide is often due to the vehicle being more specific or imaginative.

Example (20) shows what happens when a novel simile is used; the grounds are given in the immediate co-text, certainly necessary here:

(20) The **lunch hour is like a time warp** – give or take a few price increases. The walls are covered with original cream and emerald tiles. Without the restaurant's 10 grey and white marble tables, wooden pew-like seats and two whirring overhead fans, the interior would have the atmosphere of a beautiful – and meticulously clean – Victorian public convenience. "The hardest thing is keeping it looking the same," says Mr Poole, "mainly because of modern health and safety regulations." (A3C [leisure])

This is a description of a restaurant. A *time warp* is a semantically rich noun which as such evokes images of movement in time and space. The initial simile is like a trigger and makes us expect information about the place. In this respect it works like examples (11) and (12). A simile with a similar meaning in the data is *like a leap into the past.*

A spin-off effect of searching for the ***is like a***-pattern was a list of generic proverbial sentences. Some examples:

(21) a. An Emmental farm without woodland **is like** a bed without a blanket. (FTU [leisure])
b. The Parthenon without the Elgin Marbles **is like a** smile with a missing tooth. (AHN [world affairs])

3. On cultural differences between similes, see Piirainen (this volume, Chapter 13).

c. An apple pie without cheese, **is like a** kiss without a squeeze. (BPK [social science)
d. An army without secret agents **is like** a man without eyes or ears. (G04 [imag.])

In all of these analogies the language user makes us match two situations, each of which lacks some crucial part or element. Several of these are referred to explicitly as 'sayings' in their respective contexts. (21a) occurs at the end of a paragraph which has described the importance of forestry and timber trade in the Emmental in Switzerland. (21b) is taken from an interview with Kinnock, the former Labour leader, who was asked: "*Will Mr Kinnock, if he becomes Prime Minister, return the Elgin Marbles to Athens?*" and who uttered (21b) as part of his answer. (21c) is another saying, which merits a full quote:

> There is a saying in Yorkshire that "An apple pie without cheese, is like a kiss without a squeeze," and a slice of Wensleydale cheese is often offered in place of the usual cream.

Finally, (21d) is also a paragraph-final utterance. One function of generic sentences is to express an evaluation, which would be natural at the end of a passage.

4.2.1 *is like* V-*ing a* (n=58)

This minor category, often preceded by a form of BE, has an *ing*-construction after the preposition, resulting in analogies:

(22) Learning to relax is **like pulling a plug at the bottom of your stress glass,** gradually the tension and stress drain away. (EB1 [applied science])

In this example metaphor and simile merge into one, representing the conceptual metaphor STRESS IS LIQUID. The propositions underlying such analogies evoke various images. A few with pronominal topics are lexicalized:

(23) a. she feels it's a bit **like looking a gift horse in the mouth**. (FNC [imag.])
b. It's **like walking a tightrope in size-15 wellies**. (CH1 [leisure])

(23a) is said about a woman who was told she could not have children but who is now pregnant and even so not sure whether to have the child or not. (23b) refers to John Major, the former British Prime Minister, caught in a political dilemma.

4.3 V *like a* (n=6,608)

There are two subcategories depending on the verb. The **predicative** category consists of the perception verbs (LOOK, SOUND, FEEL, SMELL) and SEEM, which together make up 52 % of all the instances (n=3,429). The perception verbs are the only verbs to occur with a frequency of more than 1 in 1 million words in this pattern. Perception verbs are based on sensory-perceptual experiences, often hard or impossible to express by other means, which may in part account for their high frequency. In this respect

Cacciari (1998: 187–188) makes an important point, which relates to many instances of perception similes:

> referential-literary language is fundamentally inadequate to describe the 'qualities' and the nuances of many of our perceptions and body experiences (visual, haptic, kinesthetic, and so on)... Metaphor [including simile] ... provides a possible contribution to filling this gap between the complexity of the perceptual world and the limitations of our repertoire to describe it.

Biber et al. (1999) show that both LOOK and FEEL are most common in Fiction, least common in Academic discourse. In the BNC, a major proportion of LOOK and FEEL *like a/an*-sequences occur in Imaginative, Leisure and Arts.

The **adverbial** subcategory occurs with a very wide spread of main verbs which take the *like*-adverbial of manner, over 1,500 verb forms in all. A thinning procedure was adopted, which yielded 392 similes out of 1,000 hits (39.2%) when literal comparisons had been eliminated. The vast majority of the verb forms occur only once or twice in the BNC. They tend to express events, states and very specific types of behaviour, the following *like*-phrase adding further precision. The most common verbs are the general verbs BEHAVE and ACT, which require adverbs or prepositional phrases, including *like*-expressions, to construe precise but often literal meanings. Figurative meanings with BEHAVE contain references to *angel, bear, bull, child, monster, pig, shithead* and many others. The less frequent verb forms (1,012 occur only once) almost regularly form part of figurative expressions, as in:

(24) The smell of juniper, burned earlier as incense, still **lingered like** the warm breath of the forest in summer. (BNU [world affairs])

(25) Desire **licked like** a flame at her senses, sending dangerous signals to her brain. (JXW [imag.])

(26) All this was to seem academic by the Seventies when the boom finally stopped and the dream future of abundance and leisure **popped like** a rosy soap bubble. (ACS [leisure])

(27) Beneath me the Thames lassooed and **pulsed like** a human brain, sending signals, slipping veil after veil as if a heavier liquid had been sent to slide across its face of water, leaving no doubt that rivers are living things. (H0M [imag.])

(28) It **points like** a finger to heaven and **like** a magnet it **attracts** strangers passing by. (ECR [world affairs])

In most of these there is a blend of metaphor and simile. Goatly (1997) would refer to (24) and (26) as 'metaphorical frameworks', but (25)–(27) all use their verbs metaphorically. (28) describes the pinnacle of a crumbling abbey.

There are quite a few fixed expressions (*collapse like a pack of cards, grin like a Cheshire cat, sweat like a pig, shake/tremble like a leaf, smoke like a chimney, sleep like a baby*) in which the adverbial in colligation with the verb brings out behaviour typical of the topics.

Table 3. Frequencies per million words in two major simile patterns

Domains (genres)	*is like a(n)*	**V** *like a(n)*
Imaginative	5.61	13.3
Arts	8.74	4.45
Belief and thought	12.64	2.99
Leisure	6.32	4.27
Natural and pure science	8.19	1.59
Commerce and finance	2.07	0.69
Applied science	2.07	1.83
Social science	2.66	1.58
World affairs	2.45	1.34
Total	**4.62**	**4.32**

5. Register effects

Similes are figurative and it is no surprise that they are presented as most frequent in literary discourse. Corpus-based analyses make it possible to highlight the stylistic preferences of lexical items, phrases, and structures. Thus the BNC category distribution helps the researcher to find the 'text domains' from which the hits are drawn. In this chapter domains are given within square brackets after each citation. The domains or genres[4] used in the written texts in the BNC are listed in Table 3, which gives an overview of what I consider the two major simile patterns and their frequencies. They were selected because, as we have seen, the proportion of similes is very high in both. They are also easily comparable since the proportion of similes in each pattern is much the same. Still, the distributions differ. The main verb must be the decisive factor. 'V' here stands for a noncopular intransitive main verb. The variety of verbs mentioned in Section 4.4 expressing actions and processes would be a feature typical of Imaginative discourse, which clearly makes this domain stick out in its category.

By comparison, in the column under *is like a(n),* it is the domain of 'Belief and thought' that stands out as the domain with the highest frequency. This represents abstract discourse and BE as a main verb would therefore be more natural than a variety of dynamic verbs. The rhetorical function of the simile would also tend to be different. While a simile in the domain of 'Belief and thought' would rather be suggestive, sometimes explanatory, a simile in imaginative prose is often used for more typically descriptive purposes such as to express intensity or evaluation. Much more could obviously still be said about the communicative functions of similes in the various domains.

4. On 'domain' and 'genre' in the BNC, see Lee (2001).

6. Conclusion

We have seen that similes fall into a relatively limited range of syntactic patterns, a subset of comparative structures. These patterns should be paid more attention to when discussing the interpretation of similes and their status in relation to metaphors. This chapter shows that there is no close relation between the number of comparative instances of a given pattern and the number of figurative instances. Thus the *as* Adj/Adv *as*-sequences occur with a very high frequency in the BNC but only a few per cent of the instances are similes. By contrast, the *is like a(n)*-pattern is much less common but when it is used, the proportion of similes is very high, i.e. in the range of about 90 per cent. This is evidence that supports the frequent reference to this structure in the literature. However, what the data also shows is that when the co-text is examined, the world of comparisons is more complex than the impression one gets from stereotypical equations.

Although similes have strict syntactic properties, the comparative frames in their co-text often demand careful scrutiny for simile identification. For this reason, one has to consider the figures with some caution. Although we have provided frequency information, it is important to bear in mind that with figurative expressions it is the local rhetorical effect that matters more than exact quantitative data.

This study shows that similes make up phraseological units which vary in figurative impact according to the comparative structure used. Some patterns are more innovative than others, i.e. *is like a(n)* and V *like a(n)*. At the same time V *like a(n)* and *as* Adj *as*-sequences tend to involve a high proportion of lexicalized uses. Similes, then, resemble metaphors in the sense that they form a cline from lexicalized units to more imaginative expressions. Data from the BNC provides evidence of such a cline as well as differences in both quantitative and qualitative figurative usage. The use of corpus data allows us to examine similes in new ways, going beyond stereotypical examples to authentic use. In this way corpus data can provide a lot of empirical material to throw more light on figurative comparisons.

References

Aristotle (1926 [337 BC]) (transl. H. Treece). *The Art of Rhetoric.* Cambridge, MA: Loeb Classical Library.

Aston, G. & L. Burnard (1998). *The BNC Handbook: Exploring the British National Corpus with SARA.* Edinburgh: Edinburgh University Press.

Biber, D., S. Johansson, G. Leech, S. Conrad & E. Finnegan (1999). *Longman Grammar of Spoken and Written English*. London: Longman.

Black, M. (1981 [1955]). Metaphor. In Johnson, M. (ed.) *Philosophical Perspectives on Metaphor*, 63–82. Minneapolis MN: University of Minnesota Press.

The British National Corpus, World Edition (abbr. BNC).

Cacciari, C. (1998). Counterpoint commentary. In Katz, A. N., C. Cacciari, R. W. Gibbs & M. Turner (eds.) *Figurative Language and Thought*, 186–188. Oxford: Oxford University Press.

Carston, R. (2002). *Thoughts and Utterances. The Pragmatics of Explicit Communication*. Oxford: Blackwell.

Chiappe, D. L. & J. M. Kennedy (2001). Literal basis for metaphor and simile. *Metaphor and Symbol* 16(3–4): 249–276.

Cambridge International Dictionary of English (1995). Cambridge: Cambridge University Press. (abbr. CALD).

Glucksberg, S. (2001). *Understanding Figurative Language*. Oxford: Oxford University Press.

Goatly, A. (1997). *The Language of Metaphors.* London: Routledge.

Grady, J. (1999). A typology of motivation for conceptual metaphor. Correlation vs. resemblance. In Gibbs, R. W. & G. J. Steen (eds.) *Metaphor in Cognitive Linguistics*, 9–100. Amsterdam: John Benjamins.

Grice, H. (1975). Logic and conversation. In Cole, P. & J. Morgan (eds.) *Syntax and Semantics* 3: *Speech Acts*, 41–58. New York, NY: Academic Press.

Kövecses, Z. (2002). *Metaphor. A Practical Introduction.* Oxford: Oxford University Press.

Lakoff, G. & M. Johnson (1980). *Metaphors We Live By.* Chicago, IL: University of Chicago Press.

Lee, D. Y. W. (2001). Genres, registers, text types, domains and styles: Clarifying the concepts and navigating a path through the BNC jungle. *Language Learning and Technology* 5(3): 37–72. Also at <http://llt.msu.edu/vol5num3/lee>.

Leech, G. (1969). *A Linguistic Guide to English Poetry*. London: Longman.

Lodge, D. (1977). *The Modes of Writing. Metaphor, Metonymy, and the Typology of Modern Literature*. London: Arnold.

The Longman Dictionary of Contemporary English. 3rd edn. (2003). London: Longman.

Miller, G. A. (1993). Images and models, similes and metaphors. In Ortony, A. (ed.) *Metaphor and Thought*, 2nd edn, 357–400. Cambridge: Cambridge University Press.

Moon, R. (1998). *Fixed Expressions and Idioms in English*. Oxford: Clarendon Press.

Moon, R. (2005). 'White as snow, red as blood, black as ebony.' Conventionalized English similes and corpus evidence. In Cosme, C., C. Gouverneur, F. Meunier & M. Paquot (eds.) *Phraseology 2005. The Many Faces of Phraseology. An Interdisciplinary Conference*, 269–271. Louvain-la-Neuve: Université catholique de Louvain.

New Oxford Dictionary of English (1998). Oxford: Clarendon Press.

Nowattny, W. (1962). *The Language Poets Use.* London: The Athlone Press.

Ortony, A. (1993). The role of similarity in similes and metaphors. In Ortony, A. (ed.) *Metaphor and Thought*, 2nd edn, 342–56. Cambridge: Cambridge University Press.

Quirk, R., S. Greenbaum, G. Leech, & J. Svartvik (1985). *A Comprehensive Grammar of the English Language.* London: Longman.

Searle, J. R. (1981 [1979]). Metaphor. In Johnson, M. (ed.) *Philosophical Perspectives on Metaphor*, 248–285. Minneapolis, MN: University of Minnesota Press.

Steen, G. (1999). Metaphor and discourse. Towards a linguistic checklist for metaphor analysis. In Cameron, L. & G. Low (eds.) *Researching and Applying Metaphor*, 81–104. Cambridge: Cambridge University Press.

Wales, K. (2001). *A Dictionary of Stylistics*, 2nd edn. Harlow: Longman.

Wikberg, K. (2004). A corpus study of similes. Poster presentation at the Euro Conference on the Processing of Metaphor and Metonymy – From Computers to Neuropsychology. Granada, 24–29 April, 2004.

Foot and *mouth*

The phrasal patterns of two frequent nouns

Hans Lindquist and Magnus Levin

In this paper concepts from cognitive linguistics are combined with methods from corpus linguistics to study the phraseology formed around the frequent body part nouns FOOT and MOUTH. The material consists of The British National Corpus accessed through Fletcher's (2003/2004) database *Phrases in English* supplemented with British, American and Australian newspapers on CD-ROM. In more than half of the occurrences in the BNC the single word forms *foot*, *feet*, *mouth* and *mouths* were used in phrases, where furthermore their meaning had often been extended metonymically or metaphorically. The frequent lemmas FOOT and MOUTH are thus frequent at least partly because they occur in conventionalized phrases. Body parts are frequently mapped onto topographical phenomena in phrases like *the foot of the mountain* and *the mouth of the river*. Apart from being used in such phrases MOUTH is often connected to conventional ways of describing eating, drinking, speaking and the experience and expression of emotions. FOOT more often refers to location, and also occurs in phrases expressing other meanings, such as measurement. Metonymy and metaphor play a major role in the creation and extension of new phrasal patterns. Metonymic links are frequent because a physical reaction connected to the body part is used to represent the underlying emotion. In many cases these physical reactions have become such a conventionalized way of expressing the emotion that the reaction alone can stand for the emotion. The relative transparency of some phrases such as *down in the mouth*, *stamping one's foot* and *foaming at the mouth* is likely to facilitate their learning in spite of the fact that they are not very frequent in themselves. Phrases are often manipulated in various ways, so that they occur in non-canonical forms and in word play. The use of word play shows that the borderline between literal and nonliteral meanings is fuzzy, and that both a literal and a nonliteral meaning can be available to speakers simultaneously, although at any given moment one is usually more salient than the other.

1. Introduction

1.1 Background

This paper combines two strands in the study of phraseology. First, the work inspired by H. E. Palmer's (1933) notion of collocations leading up to Sinclair's (1991) open choice principle and idiom principle, Moon's (1998) work on fixed expressions and idioms and Stubbs's (2001) work on corpus semantics. Second, work in cognitive grammar and cognitively based theories about metaphor (Langacker 1987, 1991a, 1991b; Lakoff & Johnson 1980; Lakoff 1990). These different approaches have spawned a rich terminology; Wray (2002: 8–10) found 50 different terms for phrases or formulaic sequences in the literature. We will use the term 'n-gram' for the recurring strings (with or without linguistic integrity) that we find in the corpora and 'phrase', 'pattern' or 'construction' for meaningful, linguistically structured recurring sequences of words.

In a number of recent papers, Stubbs (2002, 2007a, forthcoming) has developed methods for corpus-based investigations of phrasal patterns in English. Referring to Sinclair (1998, 1999), Stubbs argues that "many words are frequent because of their strong constructional tendency" (2007a) and that "[m]any phrases [...] are conventional ways of expressing common meanings" (ibid.). He suggests two main approaches: looking for very frequent patterns or looking for patterns with frequent words. This paper is an exercise of the second type.

1.2 Aims

We want to (a) test Stubbs's hypothesis that frequent words are frequent partly because they have "strong constructional tendencies", (b) describe the phrasal patterns that form around some frequent nouns and (c) draw some general conclusions about the nature of phraseology. Among the questions we are interested in are the following:

What is the relation between literal and figurative senses of phrases?
What kinds of meaning development can be seen in the phrases?
What is the role of metonymy and metaphor?
What are the 'bodily motivated' differences and similarities of meaning between phrases with FOOT and MOUTH?

1.3 Scope

For this exploratory study we have chosen to investigate two lemmas, FOOT and MOUTH. These nouns can be used for reference to body parts in everyday situations and medical contexts, but they also occur frequently with extended and metaphorical meanings. Body part nouns are in fact a major source domain for metaphors (cf. e.g. Lakoff & Johnson 1980; Goossens 1990; Gibbs et al. 2004). Even when they refer literally to body parts, FOOT and MOUTH often occur in more or less fixed phrases. We

are aware of course that the terms 'literal' and 'nonliteral' are problematic; e.g. Ariel (2002: 362) has argued that meanings can be literal in at least three ways: linguistically, psycholinguistically and interactionally. For our present purposes, however, we will assume one concept of literalness and talk about more or less literal and more or less nonliteral or figurative meaning, seeing these as end-points on a scale rather than absolutes.

Although both FOOT and MOUTH relate to the body, the two words differ in that the foot is a fairly straightforward, 'mechanical' part of the body whereas the mouth is more complex with a number of functions: eating, tasting, breathing, talking and kissing, to name a few. The various functions of the foot and the mouth can be expected to be reflected in the phraseological patterns of the nouns, since their metonymic and metaphorical extensions are based on these functions. This would be in accordance with the basic tenet of cognitive linguistics that human cognition is embodied, i.e. that sensorimotor experiences are projected onto abstract concepts (Lakoff & Johnson 1980; Langacker 1987, 1991a, 1991b). Or, as Sweetser (1990: 9) has it: "Words do not randomly acquire new senses [...]". We are therefore interested to see whether these differences in real-world functions are mirrored in the phrasal properties of FOOT and MOUTH. The study will start from the individual lexical items, investigating in what phrases they occur and what meanings they express. This is the opposite of the standard approach of many studies of metonymy and metaphor, which tend to start from a particular semantic field. For instance, Kövecses (2000) begins with an emotion and investigates the various ways it can be expressed, e.g. LUST IS A PHYSICAL FORCE (ibid. 29). There are exceptions to this rule, however. Kövecses & Szabó (1996) make a detailed study of "idioms that have to do with the human hand" (1996: 337) and Simon-Vandenbergen (1991) combines the two approaches in a study of 175 body part metaphors referring to "linguistic action expressions". In both these studies, however, the data consist of types found in dictionaries, while the present study is based on tokens found in corpora.

FOOT and MOUTH are used in a number of increasingly abstract areas of meaning, all central to the human experience: location, posture, locomotion, measurement, extent, experience/expression of emotion and communication. In all these areas, there are more literal and more figurative meanings. The present paper will be based on the occurrence of the lexical items under study in n-grams and phrases in large corpora of authentic text.

2. Method and material

The method we use has been called "from lexis to n-grams" by Stubbs (2007b). To a certain extent it is corpus-driven (Tognini-Bonelli 2001) since the analytical categories are based on the returns from the computer searches and since the choice of FOOT and MOUTH was based on their overall frequency. But corpus-drivenness has its limits: the choice of body part nouns was theory-driven in that it was based on

N + + + + + + +	
+ N + + + + + +	
+ + N + + + + +	flex the **foot** and swing the leg forward
+ + + N + + + +	
+ + + + N + + +	be displayed at the **foot** of the page
+ + + + + N + +	
+ + + + + + N +	to put his money where his **mouth** is
+ + + + + + + N	born with a silver spoon in his **mouth**

Figure 1. The extraction of 8-grams

the notion of embodiment within cognitive linguistics. To investigate whether *foot/feet* and *mouth/mouths* occur primarily on their own or in phrases we searched for random instances in the BNC, while for the main investigation we used Fletcher's (2003/2004) database *Phrases in English (PIE)* which includes all n-grams with a length between 1 and 8 words which occur 3 times or more in the BNC. This means that the initial frequency cut-off point for a particular phrase is 3 tokens in the total corpus of 100 million words. However, for the purpose of analysis and discussion, occasional reference will be made to phrases with lower frequencies as well. The searches were made in a number of consecutive passes, so that n-grams of different lengths and with the key word in different positions were detected. In this way all cases of 8-grams including *foot/feet* and *mouth/mouths* were retrieved as illustrated in Figure 1 (where N stands for *foot*, *feet*, *mouth* and *mouths*, and + for any other word), and then the same procedure was carried out for 7-grams down to 2-grams. The output consisted of lists which were scanned manually for instances of phrases with linguistic integrity. For instance, *pull your feet in and at* (9 occurrences) would not be a candidate for further analysis, whereas *a few feet above the ground* (4 occurrences) might be.[1]

Using *PIE,* which is derived from the BNC, brings with it the advantages of a large, well-documented standard corpus. However, for lexically based studies like this one, even larger corpora are sometimes desirable in order to obtain reliable results. To address the problem of size, we have therefore complemented *PIE* with *The Independent* (1990, 1995, 2000), *The Times* (2000), *Sydney Morning Herald* (1992–1995) and *The New York Times* (1990, 1994–1997, 2000) on CD-ROM (in all 500+ million words).

Partington (1998: 107–108) suggests that one of the distinguishing features of genres is the types of metaphors that are found in them, which means that results from the total corpus cannot be generalized to the language as a whole without qualifications. The written component of the BNC (90 million words) consists of 22 per cent imaginative writing, i.e. literary texts and the like, and the spoken component (10 million words) contains a considerable amount of everyday private conversation; it is likely

1. Obviously, this procedure to some extent involves the use of intuition and subjective judgement by the analyst.

that these genres include more literal references to human mouths and feet, and other types of nonliteral phrases than e.g. texts about commerce and politics. However, it is not within the scope of the present paper to compare different registers or genres.

3. Results

3.1 Singular and plural

In the BNC, the lemma FOOT occurs 214 times per one million words and the lemma MOUTH 99 times per one million words (Leech et al. 2001). The proportions of singular and plural forms, however, are quite different: *foot* 73, *feet* 141, *mouth* 93 and *mouths* 6, a state of affairs which is likely to be at least partly due to the fact that humans are normally endowed with two feet and only one mouth. Regardless of this, singular and plural forms often behave differently and occur in different kinds of phrases (as pointed out by Sinclair 2003: 167–172 in relation to *eye* and *eyes*). We will return to this phenomenon below in connection with the different phraseological patterns with *foot* and *feet*.

3.2 Occurrence in phrases

A look at 100 random instances of the single word forms *foot*, *feet*, *mouth* and *mouths* in the BNC shows that in more than half of the cases they are used in phrases. This lends support to Stubbs's hypothesis that frequent words are frequent partly because they occur in frequent phrases. The overall figures are given in Table 1.

Table 1 shows that there are rather small differences between singulars and plurals in their use for body parts, units of measurement and other meanings. However, as will be seen in the discussion below, within the 'Other meanings' category there are specialized meanings which are expressed primarily or exclusively with either plural or singular forms.

Table 1. The uses of the words *foot*, *feet*, *mouth* and *mouths* in a random sample of 100 instances of each form in the BNC

	Human body part	Animal body part	Measurement	Other meanings	Total
Foot	38	4	27	31	100
Feet	33	2	29	36	100
Mouth	51	7	–	42	100
Mouths	45	4	–	51	100

3.3 From n-grams to phrases

Most of the 8-grams with *foot* were chance recurrences from the same text like *flex the foot and swing the leg forwards*. This illustrates the need for taking range as well as total frequency into account when analysing corpus data. With *mouth*, a few proverbs and idioms turned up in the 8-gram category: *born with a silver spoon in his mouth* (5 tokens), *never look a gift horse in the mouth* (3) and *to put his money where his mouth is* (3).[2] However, it was only at the 5- and 4-gram levels that the more frequent patterns began to show up. Working down from 8-grams to 2-grams we became aware that it is far from obvious where one phrase ends and the next begins – in fact they often blend into each other (cf. Altenberg 1998: 107–109 for a discussion of this).

As mentioned above, the phrases with FOOT and MOUTH fell into a number of semantically-based categories: location, posture, locomotion, measurement, extent, emotion and communication. Space does not allow us to cover more than two of these in this paper, viz. location and experience/expression of emotion.

3.3.1 *Location and metonymic and metaphorical extensions of locational meaning*

It has been observed that very frequent phrases express place, time, cause and intention (Stubbs forthcoming). Both FOOT and MOUTH are used in phrases expressing location, e.g. with the preposition *at*, but *foot* and *feet* are much more frequent in such phrases than *mouth/mouths*. In *at x's foot*, the reference is occasionally to a human foot, but more often to the lower end of things like rocks and other topographical features, trees and various man-made structures as in (1).

(1) [...] the coastline consists of high inaccessible cliffs [...] with many small inlets and detached rocks **at their foot**. (H0C)[3]

However, the phrase *at the foot of x* is more common than the corresponding possessive construction, which can be explained by the fact that phrases with singular *foot* normally have non-human referents and consequently favour the *of* construction, cf. (2)

(2) Wengen is **at the foot of** the Jungfrau range. (CBC)

The 3-gram *the foot of* (584 tokens) usually refers metaphorically to natural phenomena or artefacts which are similar to the human form (trees, crosses, pillars etc.) or to objects which are less human-like (mountains, walls etc.). *Foot* has also taken on the meaning 'most distant part of' as can be seen in phrases like *at the foot of the lake/loch* and *at the foot of the field/garden*.

---

2. It should be noted, however, that colourful phrases ('idioms') which are often cited in handbooks and the literature on metaphor and metonymy are quite rare in our corpora (cf. Moon 1998: 57–74; Deignan & Potter 2004: 1239).

3. The code in brackets gives the source of the example. A combination of three letters or figures refers to a text in the BNC, three letters plus a year refers to a newspaper CD-ROM.

Table 2. Referents in 59 cases of *at the feet* in the BNC

Referent	N
Single individual, concrete meaning	20
Single individual, metaphorical meaning	13
Several individuals, concrete meaning	3
Several individuals, metaphorical meaning	12
Collective, metaphorical meaning	3
Concrete, inanimate, extended meaning	1
Abstract, metaphorical	2
Other	3
Total	59

The plural *feet* is frequently used in expressions describing the location of something as being close to or below a person's feet in 3-grams like *at my/your/her/his/our/their feet* (475 tokens). These phrases are used with three main types of meaning, 'location' as in (3), 'movement towards a location' as in (4) and metaphorical meanings like 'submission' in (5).

(3) Kathleen Lavender saw herself, for a moment standing vacantly in the middle of the dining room, the vacuum cleaner **at her feet.** (**AD1**)

(4) There was fighting in the gangways and a young girl beat the bouncers to throw herself **at his feet**. (A8F)

(5) Germany wants all the European currencies **at her feet**, and she is on the point of abusing her dominant position. (AMK)

Note that there is an obvious path from the literal meaning of (3) to the metaphorical meaning of (5). Similarly, the physical action of directing your gaze towards your own feet as in (6) below has over time achieved the additional metonymic meaning of showing timidity or shame, and this meaning is now the salient one in most uses of the phrase (for a discussion of salient meanings, cf. Giora 2003: 13–38).

(6) "Because. . ." Sarah stopped and looked down **at her feet**. (GWC)

Similar metonymic meaning developments where bodily reactions represent the experience and expression of emotions are seen in Sections 3.3.2 and 3.3.3 below. Such phrases are good illustrations of Giora's claim that the literal meanings of phrases are not always uppermost in language users' minds.

The 59 cases of *at the feet of* show a rather different picture than the phrases with the singular form. The referents in the examples could be divided into the categories shown in Table 2.

Of the 59 cases, 30 have metaphorical meaning, i.e. they do not exclusively express location. Furthermore, among the metaphorical meanings there are three distinctive

types which we have labelled 'blaming or praising' (7), 'being a disciple of' (8) and 'being a victim of' (9):

(7) Tschwete lay [sic] the blame firmly **at the feet of** South Africa's white rugby administrators. (CEP)

(8) For it's no use looking to this novel for even a premonition of a cleansed Russia sitting **at the feet of** Jesus.[4] (A18)

(9) At the basis of his outrageous talent and inevitably the root of his temperamental insecurities, was his diminutive height and the almost vindictive treatment he received **at the feet of** much bigger and stronger opponents.[5] (B1L)

That *feet* occurring in phrases expressing the sense 'blaming or praising' is beginning to lose its original semantic content is shown by its use with an abstract noun, a plural mass noun and a concrete noun referring to clearly footless things in (10):

(10) This was because of the smaller number of cases reported (99 by June 1984, with 5 deaths) and the fact that the blame was laid mainly **at the feet of** absorbency, synthetic fibres and deodorising tampons (much more common in the States than here). (HSL)

Similar to *foot*, *mouth* is frequently used to describe location, in this case usually metaphorically in relation to a topographical feature which is likened to a human mouth, as in (11).

(11) We landed the boats **at the mouth of** the second river, then began to climb Spyglass Hill. (FSJ)

As regards *the mouth of*, around two-thirds of all tokens refer to topographical phenomena, the most frequent by far being rivers. *Mouth* is also frequently mapped onto the end points of other elongated topographical conduit-like phenomena such as tunnels, valleys and streets. Similarly, the openings of cavities (holes, caves, tombs etc.) and tube-like artefacts (bottles, guns etc.) also occur commonly with this 3-gram. Croft & Cruse (2004: 19) point out that in distinction to humans, caves can have several mouths. This holds for tunnels and streets as well, but it seems likely that at any moment, a speaker will normally conceptualize only one particular orifice as *the mouth* of a cave or a tunnel.

---

4. This example is clearly metaphorical, while in some others, like the following, cotextual clues indicate that there is a basic, literal meaning, which, of course, in its turn can have metaphorical overtones: *Yet soon afterwards he sat at the feet of Jesus clothed and in his right mind* (Mark 5:15). (CEJ)

5. Since this is about soccer, a literal interpretation also lies near at hand and blends with the metaphorical one. At the same time, this is clearly a play on the much more common phrase *suffer at the hands of*.

In 3-grams with *mouth*, *in* is the most frequent locative preposition (*in his mouth* 219; *in her mouth* 132; *in your mouth* 89). Most of these instances are literal, but a few are metonymic and metaphorical (e.g. 13 out of 100 random instances of *in his mouth*). Some of the nonliteral 4-, 5- and 6-grams will be discussed in Sections 3.3.2 and 3.3.3 below. Of the more literal instances of *in his/her/your mouth* there are some where the object is entirely contained in the mouth, as exemplified in (12). More frequently, however, only a portion of the object is contained within the mouth, as illustrated in (13).

(12) When questioned, 17-year-old James Graham, now of Allen Close, Alton, was found to be hiding a cannabis leaf **in his mouth**. (BM4)

(13) He lights a cigarette for me, holding it **in his mouth**. (FP6)

The latter example is an instance of synecdoche where the whole object is taken to represent a part of it: it is only the end of the cigarette which is inside the mouth. There is usually no need to specify if the object is entirely or only partially contained within the mouth: our world knowledge and recognition of conventionalized phraseology enables us to construe the situation. Langacker (1991b: 192–193) points out that there is no need for full specificity as regards prepositional use. This enables speakers to avoid clumsy locutions such as *Susan has the end of a cigarette in her mouth*.

Phrases with other prepositions include *into x's mouth* which only rarely occurs in metaphors and metonymies (6 in 100 cases of *into his mouth*), and *to x's mouth*. *Into his mouth* mainly deals with eating (e.g. *pushed forkfuls of food into his mouth*). Perhaps surprisingly, the most common verb used with *into his mouth* was *pop*, an example of which can be seen in (14). It occurred 13 times in 100 instances and is used about small discrete items like cookies, pills etc. In general such phrases metonymically represent the process of eating. Readers know that food which is popped or pushed into one's mouth is intended to be consumed.

(14) He scooped up a handful of the cakes, and **popped** one **into his mouth**. (G17)

With *to his mouth*, a different set of verbs are used than with *into his mouth*; *lift* (13 tokens), as seen in (15), and *raise* (7 tokens) account for more than a quarter of the 74 examples.

(15) Absently, Lachlan **lifted** the bottle **to his mouth** and drank long and deep [...] (APW)

3-grams such as in *into his/her mouth* and *to his/her mouth* are frequent in a corpus because they are conventionalized ways of expressing common meanings such as someone starting to eat or drink.

One phrase which is frequently used with metaphorical meaning is *foot in x's mouth*. There were 6 instances of *foot in his/her/the mouth* in the corpus, 2 out of which were unclear and 1 which was used literally (*being photographed with her foot in his*

mouth). The remaining 3 were used metaphorically, although the jocular extension in (16) shows that the literal meaning remains quite salient.

(16) Peter was famous for nothing besides denying the Lord and **putting his foot in his mouth** *up to the kneecap* on several occasions, but he was chosen. (BND)

Example (16) also supports Nerlich and Clarke's (2001) thesis that polysemy is often exploited in everyday discourse, so that both a literal and a metaphorical meaning can be kept simultaneously in the minds of interlocutors.

Of the 42 instances of the related 3-gram *foot in it* in the corpus, 3 occurred in the phrase *set foot in*, 3 were used literally and 1 was used in the phrase *have a foot in*, meaning 'have a part in'. The remaining 34 (81%) were used in the phrase *put my/his/her/their foot in it*, i.e. saying something inappropriate, as in (17).

(17) Some wives or husbands quite enjoy the times their pompous partners **put a foot in it**. (FS5)

In a paper entitled *Foot* and *mouth*, it is hard not to mention foot and mouth disease. This term occurred 11 times in the corpus, 10 times with its ordinary veterinary meaning and one with a metaphorical meaning as in (18).

(18) Another hero with **foot and mouth disease**, feet of clay and a mouth less than squeaky clean. (HJG)

Clearly this relates to the phrase *foot in mouth* discussed above, and in fact there is a common joke built on the two phrases: *foot in mouth disease*. This was not found in the corpus but it is very frequent elsewhere: a search on the Internet gave 84,900 hits.

3.3.2 *Experience of emotion*

When it comes to the experience of emotion, the mouth is obviously more central than the feet, but there is at least one phrase with *feet* where a physical sensation related to fear, viz. losing body temperature, is exploited metonymically to express an emotion: *cold feet*. This phrase occurred 37 times, 18 out of which clearly had emotive meaning, as in (19).

(19) He gets **cold feet** and phones his bank manager asking him to stop the cheque. (C8V)

In the majority of the metaphorical cases (11/18) the phrase was *get cold feet*, but there were also occasional instances of *have/develop cold feet*, *attack of cold feet, etc*.

Most phrases with *mouth* referring to the experience of emotions are metaphorical, in contrast to those chiefly metonymic phrases which refer to the expression of emotions (cf. 3.3.3 below). Some of the metaphors dealing with the experience of emotion can also retain their more literal meaning. The metaphors *make one's mouth water*, *mouth-watering* and *leave a taste in one's mouth* are all based on the conceptual metaphor IDEAS ARE FOOD, which in its turn is based on the metaphors IDEAS ARE OBJECTS and THE MIND IS A CONTAINER (Lakoff & Johnson 1980). The IDEAS ARE FOOD

metaphor with its extension FEELINGS ARE FOOD are pervasive in English, as can be seen in 2-grams and 3-grams such as *food for thought* (84 instances in the corpus), *swallowed his/her pride* (10) and *insatiable curiosity (9)*.

Make one's mouth water and the adjective *mouth-watering* occur in similar phrasal patterns regardless of whether they are used literally or metaphorically as in (20) and (21):

(20) Such a woman, such a body! It fair makes your **mouth water**. (H98)

(21) The tournament provides the **mouth-watering** prospect of top continental sides Sporting Lisbon and Real Sociedad playing against Newcastle and Middlesborough. (K4T)

The emotions expressed with this metaphor are exclusively positive, which follows from the original literal meaning – only food that looks and smells good makes your mouth water. Both *mouth water* and *mouth-watering* are mainly used in their literal sense in the BNC: only one out of 20 (5%) *mouth(s) water* and 23 out of 95 (24%) *mouth-watering* were used metaphorically. This rather small proportion of metaphors was compared to the distribution in our newspaper corpora. In those texts, 16 out of 27 (59%) of *mouth(s) water* and 46 out of 100 random instances of *mouth-watering* were metaphorical. This metaphor is particularly frequent in the contexts of sports and in financial reporting, which may account for the greater proportion of metaphors in the newspaper material.

The phrase *a(n)* ADJ *taste in x's mouth* is very different, however. First of all, the constructions used with the literal and the metaphorical meanings are different, as exemplified in (22) and (23) below. For the non-metaphorical use, there are a number of phrasal patterns, including existentials, as seen in (22), and use in subject position (*the taste in her mouth was slightly sour*), while the metaphorical meaning only occurs in the phrase *leave a(n)* ADJ *taste in x's mouth*, where *bad*, *nasty*, *bitter* and *sour* are almost exclusively the adjectives used, as in (23). These adjectives show the second major difference to the *mouth-watering* examples above, namely that *leave a taste in x's mouth* is exclusively negative.[6]

(22) There was *a filthy taste* **in his mouth** and his lips felt as if they had been stung. (APU)

(23) What some people in the game said at the time **left a bad taste in my mouth**. (K4T)

6. It is noteworthy that most of the n-grams relating to the experience and expression of emotions relate to negative emotions (fear, anger, sadness, etc.). This is also true of most n-grams with *mouth* referring to speech (*mouth shut*, *all mouth*, *big mouth*). The tendency of associating certain body parts with negative connotations is discussed in connection with the concept of semantic prosody (Sinclair 1991; Stubbs 2001; Channell 1999; Louw 1993; Partington 2004) in Levin & Lindquist (2007).

All 22 instances of *leave a(n)* ADJ *taste in x's mouth* were metaphorical while the 25 instances without *leave* were all non-metaphorical.

Some phrases with *mouth* denoting the experience of emotions are metonymic in nature. This is seen for instance with *mouth go/went dry*. Like *stamp(ing) one's foot* in Section 3.3.3 below, this phrase typically occurs in connection with the emotions of women; *mouth go/went dry* refers to the experience of fear and usually occurs with the addition of other physical reactions such as *her legs tremble in (24)*.

(24) What she saw made her **mouth go dry** and her legs tremble. (HHA)

Here, the meaning of *made her mouth go dry* is conventionalized to be something like 'made her experience fear'.

3.3.3 *Expression of emotion*

Emotions can be experienced through body parts, but they can also be expressed by means of the same or different body parts. Whether this expression is seen as voluntary or involuntary is sometimes indicated by the context, in phrases such as *his mouth betrayed an inner fury*, while in many other cases it is unclear. Emotions rarely seem to be expressed with the feet, but there is at least one phrase which is an example of this: *stamp one's foot*.

Stamp(ing) his/her foot occurred 19 times in the corpus. Of these, 15 referred to females, so this particular action is more often ascribed to women than to men. In 15 of the instances there were emotional overtones concerning the expression of anger or frustration. In some cases this is indicated in the immediate context with specifications such as *in frustration*, while in others, like (25), it is taken for granted that the emotional meaning of the phrase is conventionalized.

(25) "I'll never live it down!" Louise Butler **stamped her foot**. Her mother Doreen grinned. (FAB)

In phrases with *stamping one's feet* (in the plural) the distribution between literal and metonymic uses was the reverse: only 17 out of 58 instances had a figurative meaning, and there was no clear gender bias. The literal cases were mostly about stamping one's feet to keep warm.

The mouth is much more often seen as a vehicle for the expression of a person's feelings. When the mouth is used in phrases about expressing emotions, it mainly seems to involve negative, or at least serious, emotions, as will be seen in the following. A typical example of an n-gram that is used to signify the expression of emotions is the 4-gram *down in the mouth* ('depressed') of which there were 10 instances. Here, as in other cases of emotions, the physical responses to the emotions are taken to stand metonymically for the emotion itself (Kövecses 2000: 133–134). Nevertheless, in some of the examples, as noted earlier in this paper, the literal meaning is still available to speakers, cf. (26).

(26) PENSIONER Ivy Groves was **down in the mouth** yesterday – because Government red tape will leave her without false teeth for weeks. (CH6)

The meanings of many phrases are conventionalized so that the phrases can be interpreted immediately to represent the feeling. This can be seen with the phrase *mouth fell/dropped open*. 43 out 49 instances mean 'suddenly become surprised'. Most examples (29 out of 43), such as (27), express the surprise only by reference to the opening of the mouth and do not use any other overt indication of the feeling, which indicates that there is a fairly strong conventionalized meaning component to the phrase. Some occurrences, however, contain further indications of surprise, often in the form of other bodily reactions such as *her eyes widened* in (28). Other indications of feelings are even more explicit as *in shock* in (29).

(27) His **mouth dropped open** when he saw me standing in front of him, carrying a gun. (G15)

(28) Alyssia's **mouth dropped open** and her eyes widened. (H8H)

(29) Her **mouth fell open** in shock, but more shock was to come. (JY1)

Mouth also occurs in phrases expressing anger or illness. *Foaming/frothing at the mouth* is based on the fact that people and animals sometimes produce froth at the mouth when they are very angry or ill. In the material there is a slight preference for *foaming at the mouth* to express anger (9 tokens), as exemplified in (30), in comparison to illness (5 tokens), as exemplified in (31).

(30) "Get inside, the lot of you, or I'll call the police," shouted the prebendary, **foaming at the mouth**. (HA0)

(31) By then his friend was sweating, breathing rapidly and **foaming at the mouth**, and finally had a fit. (K41)

It should be noted that the prebendary's anger in (30) is only expressed by the phrase *foaming at the mouth*, which is conventionally taken to express such an emotion. This is true of most instances where these 3-grams express anger. When *foaming/frothing at the mouth* expresses illness, however, it usually occurs with other symptoms as well, as in (31). *Frothing at the mouth* is only used with human subjects, and this 4-gram expresses illness (6 tokens) slightly more often than anger (4 tokens). Similar results for *foaming at the mouth* were found in the newspaper corpora, where most examples describe anger. It is noteworthy that the newspaper corpora contain some examples where *foaming at the mouth* neither refers to anger nor to illness, but rather excitement or enthusiasm and political extremism, as in (32). Such examples seem to be connected to the adjective *rabid*, which can be used to describe both enthusiasm and fanaticism.

(32) Asked why he thought the Conservatives were still unpopular with young people, he said: "The thing I've learned since the last election is that you mustn't come across as **foaming at the mouth**. Perhaps we gave people to [sic] many reasons not to like us." (*Ind* 2000)

The metonymic and conventionalized extension of the meaning of physical reactions like *foaming at the mouth* to describe feelings thus has a certain element of vagueness or ambiguity about it as long as the physical reaction has several possible causes.

4. Summary and conclusion

The frequent lemmas FOOT and MOUTH are frequent at least partly because they occur in conventionalized phrases that express common meanings, such as location, units of measurement, experience and expression of emotions, rather than just referring to physical feet and mouths. We found that the phraseology connected with these common words is extremely rich, and that, indeed, at least half of the tokens in the main corpus (BNC accessed through *PIE*) occurred in phrases.

Apart from being used in phrases referring to, for instance, topographical phenomena, *mouth* is often connected to conventional ways of describing eating, drinking, speaking and the experience and expression of emotions. *Foot* and *feet* more often than *mouth(s)* refer to location, and generally occur in phrases expressing other meanings, such as measurement.

Looking at a subset of phrases expressing location, experience of emotions and expression of emotions, we have also seen that metonymy and metaphor play a major role in the creation and extension of new phrasal patterns. Body parts are frequently mapped onto topographical phenomena in phrases like *the foot of the mountain* and *the mouth of the river*. Metonymic links are also frequent because a physical reaction connected to the body part is often used to represent the underlying emotion. In many cases these physical reactions have become such a conventionalized way of expressing the emotion that the reaction alone can stand for the emotion (e.g. *foaming at the mouth*, *cold feet*). Some phrases such as *down in the mouth* and *stamping one's foot* are (fairly) transparent semantically and refer to well-known physical reactions. Because of this, these phrases can be learnt and remembered easily in spite of the fact that they are not very frequent in themselves.

The use of punning shows that the borderline between literal and nonliteral meanings is fuzzy, and that both meanings are available to speakers simultaneously, although at any given moment one is usually more salient than the other.

In this paper we have only just begun to look at the metonymic and metaphorical processes involving two body part nouns chosen almost at random. In other studies we have focused on evaluation in phrases with *nose* (Levin & Lindquist 2007) and symmetrical constructions with identical body nouns like *head to head* and *toe to toe* (Lindquist & Levin forthcoming). The field of body part phraseology is extremely rich and promises to yield interesting results illuminating the mental processes behind the acquisition, production and reception of phraseological sequences.

References

Altenberg, B. (1998). On the phraseology of spoken English: The evidence of recurrent word-combinations. In Cowie, A. P. (ed.) *Phraseology*, 101–122. Oxford: Clarendon Press.

Ariel, M. (2002). The demise of a unique concept of literal meaning. *Journal of Pragmatics* 34: 361–402.

Channell, J. (1999). Corpus-based analysis of evaluative lexis. In Hunston, S. & G. Thompson (eds.) *Evaluation in Text. Authorial Stance and the Construction of Discourse*, 38–55. Oxford: Oxford University Press.

Croft, W. & D. A. Cruse (2004). *Cognitive Linguistics*. Cambridge: Cambridge University Press.

Deignan, A. & L. Potter (2004). A corpus study of metaphors and metonyms in English and Italian. *Journal of Pragmatics* 36: 1231–1251.

Fletcher, W. (2003/4). *PIE: Phrases in English*. <http://pie.usna.edu>.

Gibbs, W. R. Jr., P. L. Costa Lima & E. Francozo (2004). Metaphor is grounded in embodied experience. *Journal of Pragmatics* 36: 1189–1210.

Giora, R. (2003). *On our Mind. Salience, Context, and Figurative Language*. Oxford: Oxford University Press.

Goossens, L. (1990). Metaphtonomy: The interaction of metaphor and metonymy in expressions for linguistic action. *Cognitive Linguistics* 1: 323–340.

Kövecses, Z. (2000). *Metaphor and Emotion: Language, Culture, and Body in Human Feeling*. Cambridge: Cambridge University Press.

Kövecses, Z. & P. Szabó (1996). Idioms: A view from cognitive semantics. *Applied Linguistics* 17(3): 326–355.

Lakoff, G. (1990). The invariance hypothesis: Is abstract reason based on image schemas? *Cognitive Linguistics* 1(1): 39–74.

Lakoff, G. & M. Johnson (1980). *Metaphors We Live By*. Chicago, IL: Chicago University Press.

Langacker, R. W. (1987). *Foundations of Cognitive Grammar.* Vol. 1: *Theoretical Prerequisites*. Stanford, CA: Stanford University Press.

Langacker, R. W. (1991a). *Foundations of Cognitive Grammar.* Vol. 2: *Descriptive Application*. Stanford, CA: Stanford University Press.

Langacker, R. W. (1991b). *Concept, Image, and Symbol. The Cognitive Basis of Grammar.* Berlin: Mouton de Gruyter.

Leech, G., P. Rayson & A. Wilson (2001). *Word Frequencies in Written and Spoken English: Based on the British National Corpus.* London: Longman. <http://www.comp.lancs.ac.uk/ucrel/bncfreq/flists.html>. Accessed 15 August 2005.

Levin, M. & H. Lindquist (2007). Sticking one's nose in the data. Evaluation in phraseological sequences with *nose*. *ICAME Journal* 31: 63–86.

Lindquist, H. & M. Levin (forthcoming). The syntactic properties of recurrent phrases with body part nouns: The N_1 *to* N_1 pattern. Paper presented at the ELeGI conference, Hanover 5–7 October 2006.

Louw, B. (1993). Irony in the text or insincerity in the writer? – the diagnostic potential of semantic prosodies. In Baker, M., G. Francis & E. Tognini-Bonelli (eds.) *Text and Technology: In Honour of John Sinclair*, 157–76. Amsterdam: John Benjamins.

Moon, R. (1998). *Fixed Expressions and Idioms in English: A Corpus-Based Approach*. Oxford: Clarendon Press.

Nerlich, B. & D. D. Clarke (2001). Ambiguities we live by: Towards a pragmatics of polysemy. *Journal of Pragmatics* 33: 1–22.

Palmer, H. E. (1933). *Second Interim Report on English Collocations.* Tokyo: Kaitakusha.

Partington, A. (1998). *Patterns and Meanings. Using Corpora for English Language Research and Teaching*. Amsterdam: John Benjamins.

Partington, A. (2004). Utterly content in each other's company. *International Journal of Corpus Linguistics* 9(1): 131–156.

Simon-Vandenbergen, A.-M. (1991). The body in talking about language. A study of metaphors. In Granger, S. (ed.) *Perspectives on the English Lexicon. A Tribute to Jacques van Roey* [Cahiers de l'Institut de Linguistique de Louvain 17], 51–63. Louvain-la-Neuve: Peeters.

Sinclair, J. (1991). *Corpus Concordance Collocation*. Oxford: Oxford University Press.

Sinclair, J. (1998). The lexical item. In Weigand, E. (ed.) *Contrastive Lexical Semantics*, 1–24. Amsterdam: John Benjamins. Reprinted in Sinclair, J. & R. Carter (eds.) *Trust the Text. Language, Corpus and Discourse*, 131–148. London: Routledge.

Sinclair, J. (1999). A way with common words. In Hasselgård, H. & S. Oksefjell (eds.) *Out of Corpora*, 157–79. Amsterdam: Rodopi.

Sinclair, J. (2003). *Reading Concordances. An Introduction*. London: Longman.

Stubbs, M. (2001). *Words and Phrases*. London: Blackwell.

Stubbs, M. (2002). Two quantitative methods of studying phraseology in English. *International Journal of Corpus Linguistics* 7(2): 215–44.

Stubbs, M. (2007a). On very frequent phraseology in English: Structures, distributions and functions. In Facchinetti, R. (ed.) *Corpus Linguistics Twenty-Five Years On*. Amsterdam: Rodopi. <http://www.uni-trier.de/uni/fb2/anglistik/Projekte/stubbs/icame-2004.htm>.

Stubbs, M. (2007b). Quantitative data on multi-word sequences in English: The case of the word *world*. In Hoey, M., M. Mahlberg, M. Stubbs & W. Teubert *Text, Discourse and Corpora*, 163–189. London: Continuum.

Stubbs, M. (forthcoming). Quantitative data on multi-word sequences in English: The case of prepositional phrases. Lecture given at the Berlin-Brandenburgische Akademie der Wissenschaften, 3 November 2006.

Sweetser, E. (1990). *From Etymology to Pragmatics: Metaphorical and Cultural Aspects of Semantic Structure*. Cambridge: Cambridge University Press.

Tognini-Bonelli, E. (2001). *Corpus Linguistics at Work*. Amsterdam: John Benjamins.

Wray, A. (2002). *Formulaic Language and the Lexicon*. Cambridge: Cambridge University Press.

The Good Lord and his works

A corpus-driven study of collocational resonance

Geoffrey C. Williams

The idiom principle outlined by John Sinclair has shown how much language consists of reused formulae of a collocational and colligational nature. Resonance seeks to look at the usage of words and expressions that have retained strong semantic prosodies from earlier usage, prosodies of which the current user may not necessarily be aware. It appears here as a very diffuse form of intertextuality with an initial move from contextual to restricted collocation followed by a gradual move to the purely formulaic. This chapter illustrates this by exploring certain key words from the New Testament to see how they have been used in the works of Shakespeare, the other most cited source in the English language, and finally how these expressions are used in the British National Corpus. Although having become almost purely formulaic, these expressions seem to retain sufficient religious resonance to give them their force.

1. Introduction

Intertextuality is essentially a literary tool seeking to see how the influence of an author may be found in later texts. It deals with elements that are largely what might be termed quotable, that is to say relatively large formulae that can be traced back through earlier works and assigned to an author. In addition to these intertextual elements, language also has a large share of other clearly identifiable items such as idioms and proverbs, but, as Sinclair (1991) has amply shown, the idiom principle is a dominant factor in our language output, which can only mean that a plethora of other reusable units exist.

The idiom principle covers a variety of repeatable forms, including what Moon (1998) has termed 'Fixed Expressions and Idioms' (FEIs). Of the FEIs, collocations are particularly insidious as they pervade language, but remain on a cline from free to restricted combinations and are far from easy to tie down. This is because collocation may take two forms: the restricted collocations that can be classified and put into dictionaries, and the contextual collocations which demonstrate thematic relationships within language. The two obviously overlap, the restricted collocation does not appear *ex nihilo*, but is formed by usage, the contextual collocation may be the source of restricted collocations later consecrated by usage, or remain purely thematic. Whilst

restricted collocations form variable units of language, contextual collocations can be seen to form networks of relations, collocational networks (Williams 1998) that display the essence of a text or corpus. Given that these networks illustrate a meaning environment it may be surmised that they will help in elucidating another form of intertextuality, collocational resonance.

Resonance may be defined as the reuse of lexical items or units of language in other contexts with a subconscious carry-over of meaning from an earlier collocational environment to another. The idea of resonance is that a particularly strong concept, expressed in a single word or a word complex, may enter the sub-conscious to such an extent that it becomes part of our repertoire of daily language without our realising it. We call the phenomenon 'resonance' because the initial word or unit can be seen within a collocational network that echoes through a widening lexical network of words and concepts, in which the inner node will influence its outer satellites. The semantic weight of the node will echo through the widening circles like a stone thrown into a pool with spreading ripples of meaning until the effect wears off at the outer limits. When the central node is found in another context it will retain some of its original contextual meaning, but the wider contextual environment will obviously change, thereby altering the value of both the node and its co-occurrents. An example of this can be seen with the following statement.

> A bat's hooked teeth are thought to be specifically designed to provide the young with a secure attachment while the mother is in flight. (Scott 2005: 112–113)

The writer is a known zoologist who, like most modern scientists, works in a neo-Darwinian framework wherein evolution occurs through some form of selection. Normally a scientist defending a Darwinian view of evolution would be careful to use the verb 'evolve' rather than 'design', a creationist term. It is not that Scott has turned creationist, it is simply that in the UK the creation-evolution debate is a dead one and that Scott is simply unaware of the resonance carried over from a biblical reading of creation.

Although the main thrust of this ongoing research is into the nature of resonance within essentially scientific texts, the above quotation shows that a study of the text which is fundamental to western civilisation, the Bible, may well be of interest. Boswell (1791: 476) quotes an avowed atheist as saying that "I was born in the wilds of Christianity, and the briars and thorns still hang about me". In other words, the biblical resonance continues to affect his speech, even when in contradiction with his desired meaning. This then is the stuff of resonance,[1] the briars and thorns of our subconscious.

In this chapter, the New Testament has been chosen as the point of departure as, although we consciously quote extracts that may be seen as intertextual in everyday life,

---

1. *Stuff of* is also a form of resonance linking back to Shakespeare's Prospero "We are such stuff As dreams are made on, and our little life Is rounded with a sleep" (Shakespeare's *The Tempest*, 1611).This contextual collocation is now formulaically frequent as in *the stuff of dreams*.

its language in the forms of 'briars and thorns' of meaning is much more pervasive. The aim of this text will be to see whether examples of resonance can be found in another fundamental of British culture, the works of Shakespeare, and in the modern context of the British National Corpus (BNC).

2. Source corpora and textual resources

The Bible is not a corpus; it fulfils none of the criteria of either a linguistic or literary corpus in terms of homogeneity and authorship. However, its authority, size and the easy availability of electronic versions make it a tempting area of study. This is all the more true given that it is also one of the first works to have been studied through a concordance, albeit a manual one (Kennedy 1998). Nevertheless, the Bible has attained the status of a single work in the western collective conscience, its history and the history of its translations giving it a form of folk unity. It also has a unique position in our civilisation as being considered as much a literary work as a religious one. The edition used here is King James, or Authorised Version, as this is possibly the most read in the United Kingdom and, although better translations have been made since, this translation, and its style of language, remains very much a reference. The edition used in this study is the electronic one that can be downloaded freely from the University of Virginia electronic text archive (http://etext.lib.virginia.edu/).

The second source is the works of Shakespeare. This is for two reasons: first, the works are another central corpus in British literary tradition and are much studied in schools, thereby creating an additional source of resonance, and second, at this time, religion played a very central role in society. It must be stressed here that the methodology adopted is a corpus-driven exploration of New Testament collocational resonance and not a study of the influence of the Bible on the works of Shakespeare. The intertextual aspects of biblical usage and Shakespearian texts are the subject of literary and historical studies, but here they represent only a stage in the changing resonance values between an English translation of the Bible and current usage. The electronic version used here comes from the Oxford Text archive and is supplied with the BNC Baby[2] which, like the full BNC, can be obtained from the Oxford University website (http://www.natcorp.ox.ac.uk).

The third source used here is a genuine reference corpus, that is to say one that has been built using clearly defined criteria. The British National Corpus is a representative sample of modern British usage of English. It has been thoroughly classified so that concordance lines can be traced to different registers and genres.

2. The BNC Baby is a subset of the full corpus and consists of four sub-corpora, each of 1 million words in size. The subcorpora concern fiction, newspapers, academic discourse and spoken conversation.

The three 'corpora' have been analysed using three tools: WordSmith Tools, a suite of corpus analysis programmes developed by Mike Scott; SARA, the SGML-aware corpus exploration tool developed for the BNC; and XAIRA, the current XML aware concordancer used with the BNC and other XML marked up corpora.

3. Collocational resonance

Resonance draws on two theories of collocation; the lexical priming of Hoey (2005) and the collocational networks of Williams (1998). Lexical priming is described as the pervasive and subversive aspect of collocation in that our knowledge of words develops with the cumulative effects of encounters with other words to create expectancies that can shift over time.

Collocational networks show those encounters *in situ* in corpora such that the use of one word in a given context leads to expectations as to the surrounding lexical field. In this contextual theory of collocation we no longer talk of base and collocate in a relation of domination as all the lexical words in a network influence each other so that we have a relationship of modified to modified. It is important to note that collocational networks draw on a contextual approach to collocation and are more concerned with the thematic environment than with the restricted collocation of dictionaries. Networks are built using statistical means to extract significant co-occurrence. In scientific texts, mutual information (MI) is used as it is the rarer terminological elements that are sought, in literary texts z-score or simple raw frequency are often more appropriate. This phenomenon can be illustrated with the word *God* in the BNC where the MI rating finds *offler* and *Turhan* as the most significant collocates. The former occurs only twice and the second fives times representing *Offler the crocodile God* and *The Great God Turhan Bey*. Such rare forms are of little use here whereas the top-scoring z-score items of *sake* and *bless,* occurring respectively 424 and 99, are of more immediate phraseological use.

Once a network has been extracted, the pairings of node and collocate are then explored for compounds, collocations and phraseological units which are used to explore the textual reality of the network, and can be entered in a glossary or dictionary depending on the needs of the project.

A collocational network is the first stage in analysing resonance as it shows the spreading influence of the node within a single-corpus environment. In resonance studies, the weight of the node is such that its meaning will be felt beyond immediate relationships, retaining some of its initial meaning even when expressed in very different contexts. It is thus necessary to look at the node, and some of the collocational units it forms, in other corpora so as to see the effect of these on other textual environments.

In the original studies on collocational networks, these were analysed in scientific texts and used for lexical extraction and the sub-categorisation (Williams 2002). The current work is part of ongoing studies that apply this approach to other types

of texts and seeks to see how contextual meaning in an authoritative text can influence meanings in other texts, with the user often unaware of the fact. This form of resonance pervades language and can even be felt in straightforward terminological practice where, within a given topic-specific discourse community, scientists from different disciplines will disagree over meanings. Such resonance may also, through more subtle forces of educational background, carry over meanings that are largely unconscious in all our language use.

3.1 New Testament resonance

In order to see the effects of resonance of New Testament formulae, we shall start with the central node of *God*. The first stage is to build a collocational network for a key or high frequency word, in this case *God*. Figure 1 shows the collocates of what is, unsurprisingly, the most frequent lexical unit in the New Testament.

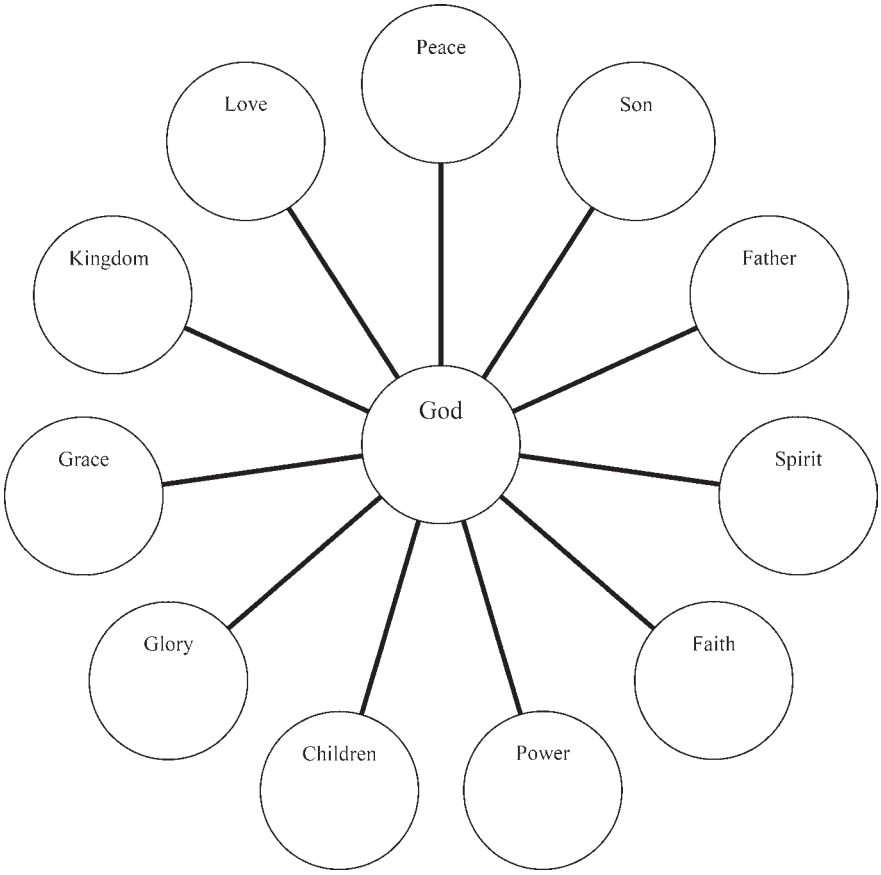


Figure 1. Immediate collocational network for *God* in the New Testament

Table 1. Possible categorisation of immediate collocates of 'God' in the New Testament

Gifts	Relationships	Authority	Spirituality
Peace	Father	Kingdom	Spirit
Love	Son	Power	Glory
Grace	Children		Faith

A thematic analysis can be carried out by categorising the lexical relationships between the central node and its contextual collocates. In this way we could, for instance, establish a categorisation on the attributes of God or the relationship with his son. This analysis gives us the thematic environment that is necessary to understand resonance. The results can be categorised in various ways to show, for example, important themes such as the gifts of God, relationships between God and others, his authority and notions of spirituality (Table 1).

In the theory of collocational networks, it is necessary to go beyond the immediate collocates to build a bigger thematic picture. The diagram in Figure 2 shows how this second degree of collocates reinforces and develops the themes found through a categorisation of the immediate collocates. It is obvious that at this level there will be linkage between categories, but at the same time we find enlargement of themes as in the addition of the gift of mercy and a reinforcement of the notion of identity which leads to a new level of authority. The *son of God* is also the *son of David*, therefore in the terrestrial line of the Kings of Israel. Similarly *Kingdom* takes us beyond the notion of authority to a place which may be entered. Through *spirit* we name the son of God and give the authority of *Lord*. This complexification of the categorisation leads to a more complete picture of the meaning environment that must then be traced in other texts. The exploration of the relationships between lexical units also takes us beyond the contextual environment leading to the discovery of multi-word units such as *God the Father* or *Son of God*.

To understand these it is not sufficient to just list them, they must be put into a textual environment so as to see how they are being used rather than just that they are used. As we shall see later, the development of collocation means that in some cases we

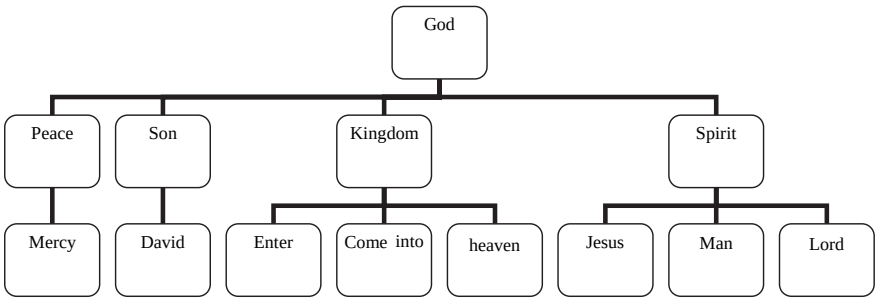


Figure 2. Diagram showing a development of four immediate collocates of *God*

shall also find that contextual collocations have developed into restricted collocations with the change of cultural environment.

Let us start, as in other contexts, with the local network for *God* in Shakespeare, followed by that of the British National Corpus.

3.2 Shakespearian resonance

Shakespeare may not have read the King James Version, but he certainly had access to the Bible in English. Britain was, by then, a thoroughly protestant country with preaching in the vernacular. Protestant Christianity was the religion of the state, but by this time the absolute fear of hell and damnation had given way to a slightly more relaxed view of religion: to be an atheist would have been dangerous, but the Lord's name could be taken more in vain than in earlier periods. This means that a Shakespearian viewpoint of resonance will be of interest before moving towards contemporary usage.

The immediate network for *God* (Figure 3) already shows differences with that found in the New Testament. If we try and categorise these collocates (Table 2) we find only one adjective, the rest consists of verbs for which it is not the actions of God that are described, but more invocations of the deity. The adjective is a title of *God* reinforcing the power of the deity, an idea present in the New Testament collocates with the formula *Lord God Almighty.* Looking at the concordances shows that whereas in the Bible the collocational environment could be expanded, it is not the case here; we are already in the use of formulae.

Table 3 shows how the resonance has decreased in some cases, but remains stable in others. Compared to their New Testament usage, in Shakespearian texts, there is a clear move from a request for direct intervention to an invocation of God as assistance, protection or as witness. God is directly present in the New Testament; in Shakespeare the deity represents an external force. God is still present, but the resonance is weaker. The most used formulae are much closer to what we might see as modern usage; *thank God* is an expression of relief that something has or has not happened, *God forbid* is a desire that something will not happen. How exactly these are used in modern parlance will be seen later.

The same move away from direct involvement of God may be seen if we look at the main collocates from the New Testament shown in Table 4.

Table 2. Categorisation of collocates of 'God' in the works of Shakespeare

Action of invoker	Actions of God	Naming Adjective
Thank	Defend	Almighty
Pray	Forbid	
	Knows	
	Bless	
	Save	

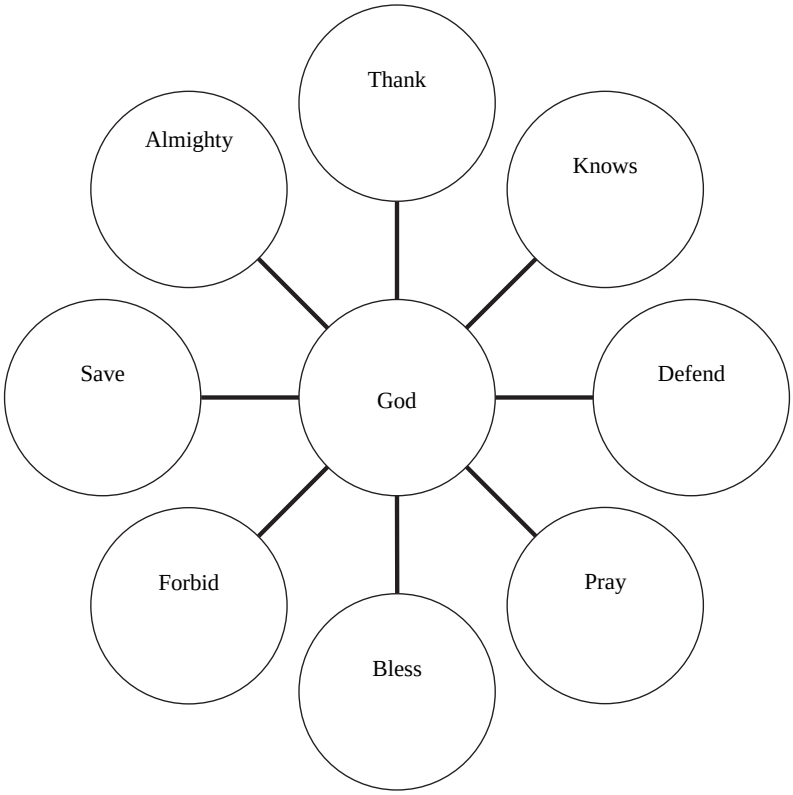


Figure 3. *God* in Shakespeare

Table 3. Comparison between the main formulae from Shakespearian use in the New Testament

	Bible	Sense	Shakespeare	Sense
Thank + God	I thank (my) God	Literal thanks	(I) thank God	Relief
Pray + God	I pray God (that)	Something will not happen	'Pray God'	Invoking God as witness
Defend + God	None		'God defend me'	Invoking God for protection
Forbid + God	God forbid (that)	May something not happen	'God forbid' (that)	May something not happen
Know(s) + God	Know not God	'We' know God	God knows	Invoking God as witness
Bless + God	Bless we God	'We' thank God	God bless (them)	Invoking blessing from God
Save + God	Save God	Save = except	God save (you/the King)	Invoking God for protection

Table 4. Comparison between the main formulae from the New Testament in Shakespeare

	Bible	Sense	Shakespeare	Sense
Peace + God	'peace from God'	Internal peace as a gift	Make peace with God	Settle your moral situation
Love + God	God loves us We love God	Reciprocal love as affection	'For the love of God'	Imploring someone to do, or not do, sth.
Grace + God	'By the grace of God'	Gift	'by the grace of God'	Putting oneself at his mercy
Faith + God	Faith in God	Gift	Break faith to God	Revolt

Table 5. From contextual to restricted collocation in the New Testament and Shakespeare

Shakespeare	New Testament
God have mercy	Association of God as source of mercy 13x
God be thanked	Association of God as source of bounty 30x
God of love	Association of God and love of people for God and love of God for his people. 39x

As can be seen in Table 4, only four of the 11 collocates are to be found. These are far from frequent in Shakespearian texts, but once again God is simply invoked as an outside force, whereas in the biblical context we are more concerned with the gifts of God to man.

Another interesting factor in looking at the collocates in Shakespeare and the Bible is the move from the contextual to the restricted collocation. The latter are often treated as if created *ex nihilo*, whereas the former seems to be more a move from a strong contextual link to a semantic attraction between the two elements. In the case of resonance, this also seems to lead to a lessening of the original semantic strength of the individual elements and a move to the purely formulaic (Table 5).

In the case of *God be thanked* we are at the intermediary stage to the formula *thank God*, where the real reference to God as opposed to a simple expression of relief is well underway.

The works of Shakespeare, like the Bible, are not a corpus in the sense of corpus linguistics. We can only see how God is portrayed in Shakespeare, but have no means of knowing how the deity was spoken of in daily life. However, it is obvious that the usage found in Shakespeare was acceptable to his audience, and that we can therefore surmise that the formulae in daily use simply invoked God, with a more or less strong feeling of religiosity. To see whether we are moving to the purely formulaic we can turn to the British National Corpus.

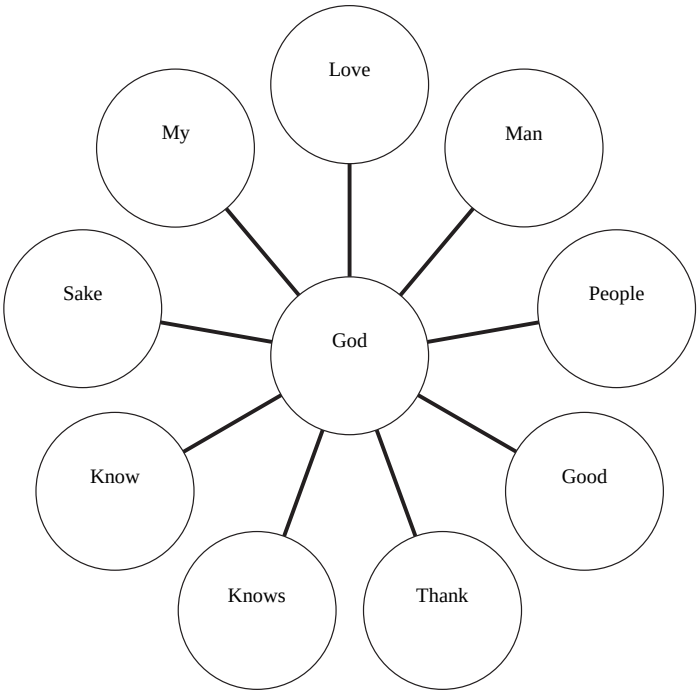


Figure 4. *God* in the BNC

3.3 God in the BNC

The BNC is a fully fledged reference corpus, which means that it is possible to sub-categorise to see not only how a formula is being used, but by whom. The immediate network for 'God' has however been calculated for the entire corpus (Figure 4).

What is immediately notable is the dominance of the formulaic; *Good God*, *My God*, *Thank God*. The other forms seem to be more contextual collocation in nature (Table 6).

The biblical context of these collocations is largely contextual; with the exception of *My God* there is little in the way of formulaic expression. With *good* + *God* we have a clear move from the contextual to restricted collocation, *good* moves from being an attribute or God or any act that is desirable to being a formula of surprise with little religious connotation. However, what is also clear is that we have two groups of collocation, those that remain collocational, *love, man, people,* and those, such as with *good*, *sake* and *thank*, which have developed into formulaic expressions. The former all retain a religious connotation in the BNC whereas the latter have little religious resonance.

We shall look in more detail at four formulae; two from the BNC context, *My God*, *Good God* and two that are shared by the New Testament and the Shakespearian context, *God forbid* and *thank God*.

Table 6. Collocates of *God* in the BNC

	Bible	Sense	BNC	Sense
Good God	He that doeth good is of God	Contextual. Do good or have good done	He had to burn them, Good God.	Expression of great surprise or relief
Know + God	They know God	Literal. Have knowledge and recognise.	God knows how/what...	Expression of inability to understand
Love + God	Love of God	Reciprocal love as affection	God's love, love of God	Predominantly religious context
Man + God	If any man love God	Relationship of man to God	Man of God, Man and God	Predominantly religious context
My + God	I thank my God	Relationship of man to God	Oh My God!	Expression of surprise, horror or complaint
People + God	People of God	Relationship of man to God	People of God	Predominantly religious context
Sake + God	For God/Jesus's sake	To act in the interest of God	For God's sake	Expression to show annoyance
Thank + God	I thank (my) God	Literal thanks	Thank God nothing more by me needs be said.	Expression to show relief

My God

> *"Oh, my God, Charlie she screamed"*

My God is a frequent expression expressing surprise, horror or complaint. It is generally the negative aspect that dominates which collocates, pointing to problems of all types. This formula occurs 1136 times in the corpus as a whole with an even split between spoken and written contexts. The contexts are however invariably dialogue with the written sources being almost exclusively reported dialogue. The expression takes a thematic role.

Good God

> *"'Good God' we said, 'Are you serious?"*

Good God is far less frequent occurring only 335 times. Usage is evenly divided between spoken and written types. The sense is generally one of surprise, or great surprise with the addition of *almighty*. The situation is generally an undesirable one, but not in all cases. The spoken type is dialogue and in a majority of cases rhematic, the problem is outlined and the utterance ends with the expression. The written types are also dialogue, with an even mix of thematic and rhematic usage.

God forbid

> *"God forbid that they should seek to exercise control over AMB"*

God forbid occurs only 66 times in the entire corpus. Its use may be rhematic to show that the situation described is potentially disastrous or thematic with the formula *God forbid that*. . . . Usage is both written and spoken, but the former represents the majority of cases in the BNC. Written usage is invariably to argue a case and to persuade the reader, the style of writing being closer to the spoken. The writer is clearly developing a dialogue with the reader.

Thank God

> *"Thank God it's Saturday!"*

Thank God is a relatively frequent formula occurring some 560 times. The general sense is that there is relief that something untoward has not happened. The vast majority of examples are of spoken language-style, although the texts are mostly written sources. This would imply that, whilst *Thank God* is a largely spoken formula, it is widely seen as characteristic of current language and as such made use of in novels.

In all four situations God is invoked as a source of aid, but the contexts do not show any expectation of real assistance. We are in the realm of spontaneous or literary spontaneous speech with highly formulaic language. The degree of religious resonance is minimal in these forms, but remains strong in written-to-be-read sources. The Bible is still a much read and discussed text in contemporary English, resonance in its most basic forms comes through with the use of fixed formulae. Further research would be needed to locate the more subtle forms of the briars and thorns of resonance.

4. Diminishing resonance: Shakespeare and the BNC

In the New Testament the formula *God Almighty* is used only to name God in collocation with *Lord*. In Shakespeare, it has three uses: a call for assistance – *God Almighty help me* (*Henry VI, Part 2*), invoking the name of God – *He wills you, in the name of God Almighty* (*The Life of Henry V*) and a form of salutation, also in *The Life of Henry V – Good morrow, Brother Bedford, God Almighty*. It may well be that, apart from very precise situations, an Elizabethan playwright would refrain from invoking the name of God. This is not so in modern occurrences where it is purely an exclamation of surprise. If the surprise is great then *good* is added as in *Good God Almighty*. This is a serious loss of resonance leaving a largely empty spoken formula.

However, with *God forbid* we do not have so severe a loss of resonance, although the invocation of God may be purely formulaic. The formula is essentially thematic implying that something is so bad it should not be envisaged. The same thematic usage and connotation is found in all texts from the New Testament through to the BNC, it is simply that in more recent usage the contextual environment shows no real appeal

to God. In Shakespeare's *Timon of Athens* the formula retains a real appeal to God as shown by the use of *heaven* in the co-text.

(1) "From heaven! Alas, sir, I never came there: God forbid I should be so bold to press heaven in" (*Timon of Athens)*.

In the BNC, we find the expression:

(2) "God forbid anything should happen to their children, never mind their house or their car". (Book on Social Sciences)

The list of priorities is interesting. This textual environment has little religious connotation and the priorities are strictly earthly.

The last example is that of thanking God. This is an interesting move from a contextual collocation to a lexical unit. In the Bible the formula *Thank God* is rare, and does not appear at all in Shakespeare. In the Bible the literal meaning of giving thanks is carried by the pronouns *I* and *We*, someone is literally thanking for some blessing received. In Shakespeare we have *God, I thank*, but this occurs only once, in the *Merchant of Venice*. In the BNC however, the formula is frequent, but given that the degree of resonance is low, this is purely a formulaic expression of relief that something has or has not happened. Once again the nearest we get to the biblical sense is in the use of the personal pronoun, but even here the idea that it is God who is really being thanked is unlikely. The formula is largely empty, the resonance virtually zero.

Why has the religious resonance largely disappeared? Could it be that familiarity breeds contempt, and that this underlies the decline in religiosity? While it is true that there has been a decline in church attendance since 1945, it is also true that the underlying influence of protestant Christianity remains a factor in British identity (Weight 2002). It could then just be that the Bible and hymn singing have been part of Anglo-Saxon culture since the reformation, which means that there is no contempt, just familiarity. It could also be that a bigger corpus, such as the full BNC or the Bank of English, is needed. These are, however, tracks that I leave open for someone else to explore, someone, who, perhaps, wants to use resonance as a factor in the study of British civilisation.

However, although we have diminishing resonance in many cases, it is apparent that the religious collocational value, and hence resonance, does remain; without it these locutions would have lost all value. There is however a clear diminishing of resonance over time. Is this a diachronic factor?

It would be presumptuous to say that as we move further in time from the missionary days of early Christianity the religious factor inevitably lessens. If a true diachronic study was carried out we would find, throughout time, periods of religious revival, and more secular periods. It is possible that our current comfortable world, in Western Europe, has led to a decline in religiosity, but it is also possible that a crisis could bring a return to these values. The only way to get a clear picture is to be able to control all the parameters, context of culture and context of situation, through a well-constructed diachronic corpus. This does not exist, and it is not even possible to build one as we

cannot go back in time to seek the lost lay texts and spoken data to complement that which we possess for modern corpora. We can of course do more in-depth studies in which we clarify the contexts as much as possible, but that is not the task I have set myself here. The important factor in this study is that we go beyond literary intertextuality to attempt to gain an insight into this very much more diffuse area of collocational resonance, where meanings are carried over time at both conscious and subconscious levels. In the same way that a well-constructed corpus can give access to 'langue' in the Saussurean sense (Tognini Bonelli 2001: 159), so a corpus can show our subconscious use of meanings, the strength and origins of which we may be unaware and which carry values to which we may not consciously subscribe.

However, in undertaking any corpus analysis, it is vital to bear in mind the nature of the corpora being used. As underlined earlier, the Bible is not a corpus, the works of Shakespeare, neither, only the BNC can be truly considered as a corpus in corpus linguistic terms. The other problem with this heterogeneous collection of sources is that we often lack the context of culture. Only a full knowledge of the conditions of creation and the environment of use will allow a complete understanding of a text, and consequently of any changes of meaning that may occur over time. In a true intertextual study such background research is generally carried out, what is generally missing is the corpus-driven analysis that will reveal aspects of lexico-grammatical usage that will not be clearly visible to the eye.

Corpus linguistics is not about purely quantitative studies, the qualitative element remains essential; there should be a constant movement between the corpus and the text so as to gain a fuller insight. In combining classical intertextuality with the corpus-driven notion of collocational resonance, we might have a tool that will open new vistas in understanding the link between language and collective memory.

The problem with collocation is its ordinariness. We have an acknowledged continuum from free to bound, the bound end is amply studied, the middle is much discussed for dictionary and teaching use, but always with an emphasis on the most bound parts. This leaves the greater part of collocation as too ordinary for study; however, the commonplace may be too easily overlooked and contain important clues as to language use. Thus *God forbid* may be standard use in English, but why and how has it developed and what meaning content does it hold?

5. Conclusion

It is clear that the value given to the units varies over time and context of situation. The first fundamental difference with other contexts is that in the Bible we have the revelation of the relationship of God with his son, who is deemed literally present; in Shakespeare the invocation of God is through the mediation of Christianity. However, we also find that the invocation of the deity may already in some cases be purely mechanical. *God forbid*, for instance, can simply mean that something is not considered desirable. By the time we get to the BNC, these oral expressions are purely formulaic,

which does not necessarily mean that all religious meaning has gone, but that the ready invocation of a deity is no longer part of our current means of expression, at least in British English.

It must be borne in mind that this is ongoing research so the conclusions are far from definitive. However, it is clear that contextual collocation is an important factor in the organisation of language. Collocational knowledge can act as a stimulus for information retrieval, but with the value given to the collocational unit varying with factors of time and context. Collocational resonance does imply, though, that we may not always be aware of meanings that we carry over from earlier learnt contexts. This study has looked at a biblical source. It will be interesting to see what this influence had on the language of Darwin as a next step into the study of collocational resonance in science texts.

The final word however concerns the Bible, and must go to a child: "A Christian charity is sending a film about the Christmas story to every primary school in Britain after hearing a young boy who asked why Mary and Joseph had named their baby after a swear word."[3] 'Out of the mouth of babes', but that is idiom, not resonance.

Acknowledgements

I wish to thank the anonymous reviewers for their instructive comments and location of the far too numerous typographical errors.

References

Boswell, J. (1791). *The Life of Samuel Johnson, LLD*. London: MacMillan (Globe Edition 1906).
Hoey, M. (2005). *Lexical Priming: A New Theory of Words and Language*. London: Routledge.
Kennedy, G. (1998). *An Introduction to Corpus Linguistics*. London: Longman.
Moon, R. (1998). *Fixed Expressions & Idioms in English*. Oxford: Clarendon Press.
Sinclair, J. (1991). *Corpus, Concordance, Collocation*. Oxford: Oxford University Press.
Scott, J. (2005). Scott's Natural History. *The Field*, August 2005: 112–113.
Tognini Bonelli, E. (2001). *Corpus Linguistics at Work*. Amsterdam: John Benjamins.
Weight, R. (2002). *Patriots: National Identity in Britain 1940–2000*. London: MacMillan.
Williams, G. (1998). Collocational networks: Interlocking patterns of lexis in a corpus of plant biology research articles. *International Journal of Corpus Linguistics* 3(1): 151–171.
Williams, G. (2002). In search of representativity in specialised corpora: Categorisation through collocation. *International Journal of Corpus Linguistics* 7(1): 43–64.

Software

Wordsmith Tools version 4: http://www.lexically.net
XAIRA: http://www.oucs.ox.ac.uk/rts/xaira/

3. *Catholic Church no longer swears by truth of the Bible*. The Times. Wednesday October 5th 2005, p. 4.

Fixed expressions, extenders and metonymy in the speech of people with Alzheimer's disease

Margaret Maclagan, Boyd Davis and Ron Lunsford

In this chapter we demonstrate how fixed expressions, extenders and metonymy help maintain the appearance of competence for speakers with Alzheimer's disease. We compare impaired and unimpaired speakers from two different countries (the United States of America and New Zealand) to show that speakers with Alzheimer's disease use fixed expressions, extenders and metonymy similarly to unimpaired speakers. Initially, speakers with Alzheimer's disease can retrieve the full range of meanings referenced by the extender or metonymy. As the disease progresses, extenders and metonymy are still used appropriately, but only the conversational partner is able to retrieve the full set of meanings. The use of fixed expressions, extenders and metonymy helps speakers with Alzheimer's disease to continue to position themselves as competent interactors and to participate in conversations.

1. Introduction

In this chapter we demonstrate how fixed expressions, extenders and metonymy help maintain the appearance of competence for speakers with Alzheimer's disease. In the early stages, speakers with Alzheimer's disease (DAT – Dementia of the Alzheimer's Type) use these devices in ways similar to the normal population. As time goes on, and the disease progresses, the usage appears to be habitual, even automated. However, the speaker's self-positioning as competent is sustained as the listener assumes that the speaker intends the full range of meanings indexed by the fixed expressions, extenders and instances of metonymy. To demonstrate how speakers with DAT maintain the appearance of competence, we use case studies that typify the larger groups of speakers within four corpora, (a) the Charlotte Conversation and Narrative Collection (CNCC), now included in (b) the New South Voices (NSV) digital resource, which also includes the WSOC Oral History collection; (c) the Intermediate Archive of the Origins of New Zealand English (ONZE) project and (d) a longitudinal collection of conversations with speakers having cognitive impairment, predominantly Alzheimer's

disease. The longitudinal corpus contains a subcorpus of speakers recorded in New Zealand.[1]

We use the term 'fixed expressions' (Moon 1998) to refer to the use of almost invariant expressions to talk about favourite topics. Some of the phrases are common sayings or popular expressions like *have a cup of tea, only the good die young* or *that's the way it goes*; many are colloquial, and different ones are idiosyncratic to each individual. They are used in relatively fixed contexts to refer to similar subjects whenever they are used (see Wray 2002 for definitions of formulaic phrases and fixed expressions). Other fixed expressions may be multi-word expressions (MWE) that are very like what Biber et al. (1999) call 'lexical bundles', or formulaic sequences identified by frequency from a corpus: for example, one speaker habitually uses *a + minimizer* phrases such as *a little bit*. Extenders are phrases such as *things like that, something like that, all that sort of thing* and similar expressions (Overstreet 1999). The speaker uses extenders in the hope that the listener can identify the larger construct of which the extender is a part, mentally finish the set referenced by the extender and signal cultural affiliation/congruence in their response. Metonymy is the "substitution of some attributive or suggestive word for what is actually meant" (Corbett & Connors 1999). In all three devices, the listener understands more than the speaker actually says. When normal speakers and listeners without Alzheimer's disease share assumptions and reference sets, the intended meaning is usually retrieved without difficulty. This is true for DAT speakers at the start of the disease. As the disease progresses, listeners assume that the DAT speaker is still referring to this shared reference set, even when the DAT speaker can no longer access the whole of the set. The fixed expressions, extenders and metonymy thus allow DAT speakers the appearance of competence even when they can no longer retrieve all the content of the referenced material. Very little analysis has so far been carried out on the spontaneous speech of DAT speakers. Davis (2005) found intact use of discourse markers *oh*, *well* and *so* in the speech of three DAT speakers and Davis & Bernstein (2005) considered formulaic phrases and general terms (like *thing*) in their consideration of reference and politeness. The present investigation represents a new area in the study of the speech of speakers with Alzheimer's disease.

1. The work done by members of the ONZE project in preparing the data, making transcripts and obtaining background information for the Intermediate Archive speakers is gratefully acknowledged. Interviews with Walters, Wilcox, and Mason are part of the ALZ NC corpus; unimpaired US interviews are accessible online at New South Voices, http://newsouthvoices.uncc.edu. We gratefully acknowledge the assistance of UNC Charlotte oral archives librarians and Krystal Blanton, project manager for ALZ NC.

2. Methodology

2.1 Corpora

We have examined material from four corpora.

a) The Charlotte Conversation and Narrative Collection (CNCC), a satellite corpus to, and part of, the 11.5 million-words in the first release of the American National Corpus (ANC). In addition to being a part of the ANC, it is now included in the New South Voices (NSV) digital resource housed at the University of North Carolina at Charlotte Library. The speakers' data were collected between 1995 and 2005, with data collection ongoing, from Charlotte, North Carolina, its surrounding county, Mecklenburg, and other, nearby counties. There are four age cohorts (young children, youth, working adults, and older adults/over-55), with 523 narrative conversations transcribed by the end of Winter 2005 and mounted with audio at NSV. The original media for the CNCC is partly analogue tape and partly digital (Sony digital and minidisk recorders). The recordings have been digitized and/or remastered in wav and mp3 formats without cleaning or enhancements. The corpus is available online in NSV [http://newsouthvoices.uncc.edu]. The sample analyzed as a basis for this discussion consists of 29 older men and women, Euro-, African-, Native and Hispanic American, aged 55–90.

b) The WSOC-TV Oral History Collection (WSOC) is a monitor corpus, that is, a collection of conversations recorded in 1979 with regional citizens whose age, gender, ethnicity, oldtimer–newcomer status, and amount of education match proportions in the 1970 Census. The speakers come from Charlotte/Mecklenburg and surrounding counties. The interviews were collected on reel to reel recorders in the lobby of Charlotte's main library over a three day period; library construction can be heard in the background. The tapes have not been cleaned or enhanced; they have been digitized and transcribed, and are being placed online within the NSV. We examined the full component as our sample, 75 older men and women, Euro- and African-American, aged 55–95.

c) The Intermediate Archive is part of the Origins of New Zealand English Project (ONZE) at the University of Canterbury (see Gordon et al. 2007). The whole archive contains approximately 140 recordings of people born between 1890 and 1930. The sub-section used in the present analysis consists of 35 oral history recordings of people born between 1914 and 1930. The recordings were made in the 1990s by Rosemary Goodyear. They focus on accounts of childhood, both schooling and childhood games. Each recording is up to three hours long, but only one hour per recording has been fully transcribed. They were recorded on analogue tape and are available on-line to researchers at the University of Canterbury.

d) The Alzheimer's corpus (ALZ) consists of a longitudinal, protocol-restricted, collection of consented conversations with speakers having cognitive impairment, predominantly Alzheimer's Disease. The material used for this discussion was collected

between 1999–2005, with data collection ongoing, from Charlotte, North Carolina and Mecklenburg County. It is soon to include surrounding counties and similar sections of South Carolina. Interviews are longitudinal, quarterly or more often as speaker health and inclination permits. Individuals start at different dates and recording continues as long as possible. To date there are interviews in natural settings (non-clinical) with 50+ men and women, diagnosed with dementia of the Alzheimer's type. The subjects are predominantly Euro-American, with some African American. The tapes have not been cleaned or enhanced and are not yet available to researchers. For this chapter, material is largely drawn from 50 conversations with one man (Robbie Walters), European-American, aged 82. ALZ also contains a subcorpus of interviews from New Zealand. All five speakers in this subcorpus were analysed, and one woman (Kitty Turner, born 1918 and aged 87) is used as a case study.

2.2 Analysis methods

Our analysis is corpus-based (Berber-Sardinha 2002). We used the web-based software tool for corpus processing, WMATRIX (Rayson 2001), to identify multi-word expressions. WMATRIX allows the use of a web browser to access the CLAWS 7 part-of-speech tagger and a semantic tagger (see Rayson 2005 for a discussion of semantic tagging and of CLAWS 7, and Phillip 2005). The WMATRIX software tags words and phrases in corpora and creates frequency profiles and concordances. The software also allows comparisons at domain and word level across larger corpora or across data the user uploads, and provides some calculation of potential significance (see http://ucrel.lancs.ac.uk/wmatrix, which includes a partial bibliography including work in phraseology). For the present analysis, we applied the Stanford Multi-word Expression Reading Group working definition of a multi-word expression as "any phrase that is not entirely predictable on the basis of standard grammar rules and lexical entries" (http://mwe.stanford.edu/reading-group.html), such as *kind of*, *a little bit*, *last time*, *at all* or *in that*. We note that this definition works well with that by Wray (2002:9) for a formulaic sequence, viz. "a sequence, continuous or discontinuous, of words or other elements which is, or appears to be, prefabricated". The USAS list of Multi-word Expressions[2] is incorporated into the WMATRIX software, which identifies and tabulates the MWE. In addition, we also used the third version of WordSmith Tools (Scott 1999) to extract the appropriate data for each part of the analysis. For each speaker, we also listed any fixed or formulaic phrases that had been noted by the interviewers and searched for all occurrences of these phrases (or of close variations). To locate extenders, we used WordSmith Tools concordances for *all this/that/these/those*

2. The USAS list is obtainable at the main website, http://www.comp.lancs.ac.uk/wmatrix; and publications about combining semantic and POS taggers, such as Piao et al. 2005, from a special issue on Multi-word Expressions in *Computer Speech and Language*, are linked at http://www.comp.lancs.ac.uk/computing/users/paul/public.html

kind(s) of/sort(s) of thing(s), some-/any-/everything, thing, and *stuff*. We then used the concordances to suggest areas in each text that were likely to include metonymy.

3. Analysis

3.1 Fixed expressions

Most of the speakers in the ALZ corpora (both US and NZ speakers) have fixed expressions that they use when they talk about favourite topics. These expressions vary from speaker to speaker. Glory Mason,[3] aged 88–92 during the 4 years of conversation, for example, often talked about her father who was a Baptist preacher. She used the phrase *hell hot and heaven beautiful* to describe the content of his preaching. Examples (1) and (2) were taken from conversations with Davis. (1) is from a conversation on 31 May, 2000 and (2) is from a conversation almost two years later on 30 April, 2002:

(1) GM: Yes. He was a preacher that preached *hell hot and heaven beautiful*! (They both laugh)
BD: Heaven beautiful
GM: Yes. *Hell hot and heaven beautiful*! That was one of his messages. I don't know ... he preached all right. He was an evangelistic-type preacher. (5-31-2000)

(Within extracts, . . . signifies a pause, not omission of material.)

(2) BD: Well tell me, what does he preach about?
GM: *Hell hot and heaven beautiful*. That's all I know.
BD: Well that's true.
GM: He done evangelistic work. (4-30-2002)

Robbie Walters, diagnosed with late moderate dementia and aged 82, did not employ particular fixed expressions as much as he used particular kinds of multi-word expressions (MWE). A search of his conversations in 2000–2002 employing the online corpus searching tool WMATRIX (Rayson 2001) offers a good example of the retention and manipulation of MWE (see Biber et al. 1999, for 'lexical bundles'). Walters' MWEs for the year 2000 beginning with *a/an*, for example, which are shown in Table 1, almost always insert a minimizer or maximizer. These MWEs were usually used appropriately in context. For example in a relatively late conversation (May 2003) Walters said 'I'm, uh, a little bit tired.'

We take Kitty Turner, a New Zealander, born in 1918, as another case study to illustrate the use of fixed expressions. Despite being diagnosed with moderate dementia of the Alzheimer's type, Kitty is still living in her own home. She still holds conversations with Maclagan, though the conversations seem to lack content much of the

3. All names have been changed to protect privacy.

Table 1. Multi-word expressions used by Robbie Walters in all conversations in 2000 (his total words = 4262)

	MWE	No.	% of total words	Type	Tokens
Minimizer	a_couple	3	0.07		
	a_little_while	1	0.02		
	a_little_bit	3	0.07		
	a_little	3	0.07		
Total				4	10
Maximizer	a_great_deal	1	0.02		
	a_lot	5	0.12		
Total				2	6

Table 2. Fixed expressions used by Kitty Turner from April 2004 to April 2005. The numbers in the columns indicate the number of times the phrase was used in that session

Fixed phrase	4-19-04	8-21-04	12-20-04	4-7-05
the way it is/goes, (isn't it?)	8	5	6	12
it'd/it'll be nice		1	5	1
the good die young	1	3		1
go (in)to town	3	6	2	5
have a cup of tea		7	2	4
have a (wee) look (a)round		6	3	3
have a nosey (a)round		3	3	3
keep/get moving	5		1	5
sit(ting) out/(a)round there	4	1	7	6
keep it/that/things tidy	1	1	1	2

time. Kitty willingly answers questions, with relatively repetitious content. The most marked feature of her conversation is the number of fixed expressions that she uses. Table 2 summarizes the most frequent of these expressions over four conversations between Kitty and Maclagan. The conversations were four months apart, and each lasted approximately 20 minutes.

Kitty often uses fixed expressions to end her conversational turn, often using several in one utterance, as in (3). All the fixed expressions that Kitty Turner uses are appropriate in context. The interviewer responds to Kitty's fixed expressions with similar or repeated fixed expressions, producing 'quilting' routines (Moore & Davis 2002) as in (4).

(3) KT: That's the way it goes, happy days they say, don't they? (12-20-04)

(4) KT: So that's the way it goes, I'm afraid
MM: Yeah
KT: Yeah
MM: Yeah, but that's life.

KT: Not to worry
MM: Exactly (08-03-05)

Conversations with individual unimpaired speakers are often replete with MWEs, fixed expressions, formulaic sequences, and the like, so that many analysts (cf. Hoey 2005) note that MWE of all sorts make up a large proportion of any stretch of language. However, conversational contributions by unimpaired speakers do not normally contain multiple repetitions of the same phrases or fixed expressions. Table 3 presents the result of a search for Kitty's fixed expressions, throughout the transcripts of somewhat longer conversations with five unimpaired New Zealand women of approximately the same age as Kitty Turner (all are born between 1918 and 1921).[4] Two of the women do not use any of Kitty's fixed expressions. The other women use one or two examples of two or three of the phrases. We did not attempt to extract all the fixed expressions used by the unimpaired speakers. However, as we read through the interview transcripts, we did not notice repetitive phrases that parallel Kitty Turner's usage. We present Table 3 to show that none of these unimpaired speakers repeated phrases to the extent that Kitty does. What distinguishes Kitty's use of fixed expressions from that of non-DAT speakers is the sheer number of such expressions she uses and the numerous repetitions of the same fixed phrase within each twenty minute conversation. The fixed expressions do not add semantic content; instead, they facilitate her continuing to take an active part in conversation.

Table 3. Examples of Kitty Turner's fixed expressions used by five New Zealand women residents, born between 1914 and 1930

	MB 1919	MA 1921	JA 1919	MD 1921	M H 1918
the way it is/goes, (isn't it?)	0	0*	0	0	0
It'd/it'll be nice	0	0	0	0	0
the good die young	0	0	0	0	0
go (in)to town	1	0	0	1	2
have a cup of tea	0	0	0	0	0
have a (wee) look (a)round	1	0	0	1	0
have a nosey (a)round	0	0	0	0	0
keep/get moving	0	0	0	0	0
sit(ting) out/(a)round there	1	0	0	2	1
keep it/that/things tidy	0	0	0	0	0

* MA uses two examples of *in a way*

4. MB and Kitty Turner are speakers in the ALZ NZ corpus. The unimpaired conversation with MB was recorded at least 10 years before the onset of her dementia, and represents unimpaired speech. Unimpaired interviews are part of the Intermediate Archive of the Origins of New Zealand English project (ONZE).

3.2 Extenders and small stories

Like metaphors (Koller 2003; Skelton et al. 2002), extenders are multifunctional. Used as evaluators (e.g., Moon 1998: 252), extenders offer one way to identify places where small stories may be found in conversation and conversational narrative. And it is this feature, the use of an extender to flag what Bamberg (2004a, b, c) calls 'small stories', that gives us greater insight into Alzheimer discourse, which is filled with indefinite constructions and formulaic phrases (Wray 2002; Davis & Bernstein 2005). Bamberg explains that small stories are "the ones we tell in passing, in our everyday encounters with each other, and which I considered the real stories of our lived lives" (2004c); elsewhere he characterizes them as "Short narrative accounts, embedded in everyday interactions, unnoticed as 'stories' by the participants, unnoticed as 'narratives' by researchers, but highly relevant for identity formation processes" (2004b).[5] While Bamberg's interest is in how people use small stories to build identity, we reference his work here for its explanatory power in displaying retained identity.

In (5) we give an example of extender use from Larry Wilcox, who is in the late stages of Alzheimer's disease. He is no longer able to remember, or even to find out, his daily schedule, or the schedule of other people where he is living, but his use of the extender *nothin' like that* is totally appropriate. By contrast, his use of the fixed phrase *that part* is less appropriate. Nevertheless, this fixed phrase still helps him present himself as a competent communicator and maintain social relations.

(5) BD: I will be back next week and we can talk again then?
LW: Now I can't give you no dates or *nothin' like that* I don't know her schedule.
BD: Well I thought I would call ahead before I came.
LW: (Au-) that sounds I like *that part* about it.

We turn now to a consideration of the use of extenders in fifty conversations with Robbie Walters. During these conversations, Walters was in his early eighties with late moderate (Stages 5 and 6) Alzheimer's Disease. Two examples, (6) and (7), typify the ways he positioned extenders appropriately within the phrase. Extenders that are often metonymic in normal conversations may, in Alzheimer's discourse, be more useful for maintaining social relationships than for suggesting propositional meaning. However, since, at this stage of the disease, Walters retains and uses other features of language to monitor hearer response, sustain interaction, and mask holes in the retrieval of lexical items, beyond the use of extenders, his extenders are appropriately metonymic (Papagno 2001).

(6) RW: Yeah, well, I don't have a good memory, I don't know, because I don't have a great memory I recall events of any size *and all that* . . . and like back then, well, who was in the plant management level *and that sort*

5. Each phrase was originally bulleted in Power Point format.

of thing but . . . not a great a lot of detail to me, there wasn't too much exciting (2-23-00)

In (6), *all that* and *that sort of thing* indicate that there is something beyond big, if unnamed, events, and important, if unnamed, people that Walters can remember. The extenders ask for common ground and affiliation, by being part of an apology for not being able to remember more detail, and bound the items in his list. Some small stories might surface about events that were large and of a general nature (such as the office Christmas party). However, the conversational partner is less likely to elicit small stories about specific people.

(7) RW: it's not all the same kind of service . . . and then . . . . . so I spent a few years you know . . . . . you kind of forget all the details end of it but . . .
BD: Well, you do. . .
RW: and, but . . . I spent, you know, quite a bit of time . . . over there before. I believe I must have been overseas about . . . four years, maybe, I think as much as that.
BD: Well with a six year enlistment, you would be, where were you, Germany maybe?
RW: Yeah, I was in Germany . . . and . . . of course you move around some. I think I might've been in France for a wh- a little bit and so I don't know. My brother missed *all that*. (2-23-00)

In (7), what the brother missed is not specified. It was not just moving around between two countries. It seems, rather, to have been an entire complex of events clustered around the war. RW's brother didn't have a war. Another 'small story' is surfacing, the listener supplies the larger construct of which the extender is a part, and Walters seems to be moving toward metonymy.

3.3 Metonymy

Extenders can also be metonymic: we begin with three examples from unimpaired speakers in the CNCC. Mrs. Thomas (8) uses metonymy to signal her community standing through social involvement. Mr. Douglas (9) uses it to signal community standing through political involvement. The Grogans (10), father and daughter, signal their shared knowledge and common ground, with an extender used to close a topic in a co-constructed conversation.

(8) Thomas: So I agreed to help her, and we gathered 10 or 12 women who were interested in the same thing. And the first year, we worked without an organization. We had Easter egg hunts for the children. We had Halloween parties. We had Christmas parties, and *all kinds of things*.

All the women are interested in the same *thing*. But what is that thing? It is not stated. Instead, we get a metonymy – *Easter egg hunts* . . . . *Halloween parties* . . . . *Christmas*

parties. The *all kinds of things* signals that 1) there were many other ways to quantify this quality, and 2) the quality is not going to be named.

(9) Douglas: Raleigh would be the second largest city. It ought to be the largest because they have got so much there in schools and politics and government and *all the things* that they have naturally, it ought to be the largest city in the state, but it isn't.

In Douglas's example, *schools and politics and government* act as metonymies (quantitative) for an un-nameable quality. If one has these – whatever they are – one has *so much*. But so much *what*? We cannot really name the quality, so the schools, politics and government act as metonymies for this un-nameable quality; *and all the things* stands for that quality that has been indicated but not specified.

(10) Bob: . . . Sharon Road was two lanes and
Pam: not much traffic, it was a small town
Bob: not much traffic a-tall and ah never worried about the kids being picked up
Pam: you didn't hear about crime
Bob: crime – *anything like that*

Again, an extender – *anything like that* – signals additional, unnamed qualities, beyond the literal description of a specific Road as having two lanes. The extender follows an echo, and suggests how paradise-like the hometown area was in an earlier time.

In the next example, Robbie Walters, DAT speaker, uses metonymy to position himself as modest:

(11) GN: What did your house look like?
RW: Well, about like most Metro Buffalo houses.
GN: Did it have a nice porch?
RW: Well, I'm not going to get involved in how good a house, or *anything like that*. (3-19-01)

In this example, the extender lets Walters back off from bragging; as a metonymy, it represents other specific attributes that one could list if he wanted to imply that his house belonged in a certain class of houses.

4. Final comments

Early in the course of Alzheimer's disease, use of fixed expressions and extenders by DAT speakers is similar to their use by unimpaired speakers, with the DAT speakers able to retrieve the whole meaning that is implied by the phrase or extender. Metonymic use of extenders is appropriate. As the disease progresses, DAT speakers continue to use fixed expressions and extenders even though they can no longer them-

selves retrieve the full implied meanings. Alzheimer's disease, being diffuse, is highly variable, and this variability is reflected in the great variation in the number of fixed expressions and extenders used by different speakers and by variation within the one speaker (see Maclagan & Mason 2005). Example (12) from Kitty Turner demonstrates this variability. The extender *that sort of thing* in the first turn seems to convey the same set of meanings that the interviewer would convey. In the following turn, it is less clear what the full set referenced by *and that* is intended to be. In the next turn, *yes mm . relax and sit around and*, Kitty does not actually use the extender and the interviewer agrees without any pause, again indicating that the full meaning is understood. The extract ends with one of Kitty Turner's favourite fixed expressions.[6]

(12) KT: you look forward to your . whole holiday don't you <you do> going away and *that sort of thing* <yeah> yeah so no it's just a break away you know
MM: and this is the time when everyone goes away
KT: yes . mm . yeah they do . well people are working *and that* and they . look forward to it don't they . you got to have a holiday
MM: did . yes you do . you have to have one
KT: yes mm . relax and sit around and
MM: yes
KT: [*laughs*] yes so . that's the way it is I'm afraid (12-20-04)

DAT speakers continue to use fixed expressions, extenders and, on occasion, metonymy in pragmatically appropriate ways even when they cannot themselves retrieve the full referential sets implied. Their use sustains the appearance of flow and offers some validation to the impaired speaker, by continuing to keep the non-impaired speaker involved with the larger effort of co-constructing meaning and understanding (cf. Hamilton 1994). An understanding of the social-interactional functions for discourse features such as extenders can contribute to the current discussion of the social construction of dementia and can help professional and family caregivers and care providers to avoid what Sabat et al. (2004) call the "malignant social positioning" that limits the DAT speaker to being seen merely as a patient.

References

Bamberg, M. (2004a). Narrative discourse and identities. In Meister, J. C., T. Kindt, W. Schernus & M. Stein (eds.) *Narratology beyond Literary Criticism*, 213–237. Berlin: Walter de Gruyter.

Bamberg, M. (2004b). A narrative-discursive approach to everyday stories. *Memory@ Research Conference in Huddersfield, UK*. <http://www.clarku.edu/~mbamberg/future.html>.

6. Phrases in angle brackets <yeah> indicate brief feedback comments from the interviewer and stops '.' indicate brief pauses.

Bamberg, M. (2004c). Talk, small stories, and adolescent identities. *Human Development* 47: 366–369.

Berber-Sardinha, T. (2002). Metaphor in corpora: A corpus-driven analysis of dissertations. *International Conference on Metaphor in Language & Thought*. <http://www2.lael.pucsp.br/~tony/>.

Biber, D., S. Johansson, G. Leech, S. Conrad & E. Finegan (1999). *Longman Grammar of Spoken and Written English*. London: Longman.

Corbett, E. & R. Connors (1999). *Classical Rhetoric for the Modern Student*, 4th edn. Oxford: Oxford University Press.

Davis, B. (2005). So, you had two sisters, right? Functions for discourse markers in Alzheimer's talk. In Davis, B. (ed.) *Alzheimer Talk, Text, and Context: Enhancing Communication*, 128–145. London: Palgrave.

Davis, B. & C. Bernstein (2005). Talking in the here and now: Reference and politeness in Alzheimer conversation. In Davis, B. (ed.) *Alzheimer Talk, Text, and Context: Enhancing Communication*, 60–87. London: Palgrave.

Gordon, E., M. Maclagan & J. Hay (2007). The ONZE corpus. In Beal, J. C., K. P. Corrigan & H. Moisl (eds.) *Models and Methods in the Handling of Unconventional Digital Corpora:* Vol. 2, *Diachronic Corpora*, 82–104. London: Palgrave.

Hamilton, H. (1994). *Conversations with an Alzheimer's Patient*. Oxford: Oxford University Press.

Hoey, M. (2005). *Lexical Priming*. London: Routledge.

Koller, V. (2003). Metaphor clusters, metaphor chains: Analyzing the multifunctionality of metaphor in text. Accessed 2–24–05 online at < http://metaphorik.de/05/>.

Maclagan, M. & P. Mason (2005). Bad times and good times: Lexical variation over time in Robbie Walters' speech. In Davis, B. (ed.) *Alzheimer Talk, Text and Context: Enhancing Communication*, 146–166. London: Palgrave.

Moon, R. (1998). *Fixed Expressions and Idioms in English: A Corpus-Based Approach*. Oxford: Clarendon Press.

Moore, L. & B. Davis (2002). Quilting narrative: Using repetition techniques to help elderly communicators. *Geriatric Nursing* 23(5): 1–5.

Overstreet, M. (1999). *Whales, Candlelight, and Stuff like That: General Extenders in English Discourse*. Oxford: Oxford University Press.

Papagno, C. (2001). Comprehension of metaphors and idioms in patients with Alzheimer's disease: A longitudinal study. *Brain* 124: 1450–60.

Phillip, G. (2005). Identifying multi-word units in context. Bologna: AMS Acta. <http://amsacta.cib.unibo.it/archive/00001133/01/MultiWordUnits.pdf>.

Piao, S., P. Rayson, D. Archer & T. McEnery. (2005). Comparing and combining a semantic tagger and a statistical tool for MWE extraction. *Computer Speech and Language* (Special issue on multi-word expressions) 19(4): 378–397.

Rayson, P. (2001). WMATRIX: A web-based corpus processing environment. Software demonstration presented at *ICAME 2001* conference, Université catholique de Louvain, Belgium. May 16–20, 2001.

Rayson, P. (2005). Right from the word go: identifying multi-word-expressions for semantic tagging. Invited talk at *BAAL Corpus Linguistics SIG/OTA Workshop: Identifying and Researching Multi-Word Units*, Oxford University Computing Services. April 21, 2005. Slides available May 2006 at <http://www.comp.lancs.ac.uk/computing/users/paul/public.html>.

Scott, M. (1999). *WordSmith Tools* version 3. Oxford: Oxford University Press.

Sabat, S. R., L. Napolitano & H. Fath. (2004). Barriers to the construction of a valued social identity: A case study of Alzheimer's disease. *American Journal of Alzheimer's Disease and Other Dementias* 19(3): 177–185.

Skelton, J., A. Wearn & F. Hobbs (2002). A concordance-based study of metaphoric expressions used by general practitioners and patients in consultation. *British Journal of General Practice* 52: 114–118.

Wray, A. (2002). *Formulaic Language and the Lexicon*. Cambridge: Cambridge University Press.

SECTION III

Phraseology across languages and cultures

Cross-linguistic phraseological studies

An overview

Jean-Pierre Colson

Cross-linguistic research on phraseology covers a wide range of challenging topics, from the simple comparison of idioms or metaphors in two languages, to the systematic contrastive study of all categories of set phrases across different languages. Current research demonstrates that phraseology in the broad sense is one of the key components of language and is probably universal. However, the theoretical debate on the definition of set phrases and the interaction between culture, meaning, syntax, figurative language and phraseology is not yet completely settled. Two major theoretical approaches have so far yielded promising results: the first is more semantic and is often associated with cognitive linguistics, while the second can be described as cross-linguistic corpus linguistics. The cognitive approach to phraseology across languages lays stress on metaphors and images as the constituent principles of set phrases, and provides interesting information about the intriguing interplay between universal cognitive principles, culture and phraseology. Cross-linguistic studies based on corpora, on the other hand, offer a statistical analysis of the various categories of set phrases as well as a very reliable methodology. Cross-linguistic phraseology is closely linked to translation studies. Active collaboration between multilingual corpus linguistics, contrastive phraseology and natural language processing may offer insightful perspectives on translation practice.

1. Introduction

In recent years, phraseology in the broad sense has become a unifying theme for an increasing number of theoretical and practical linguistic studies. Among this broad palette of investigations into the meaning, structure or use of set phrases, cross-linguistic research is one of the major and most fascinating topics.

An Englishman may sleep like a log, but a Frenchman will, among other possibilities, sleep like a marmot (*dormir comme une marmotte)*, a Dutchman like a rose (*slapen als een roos)*, a German like a stone (*schlafen wie ein Stein*) and a speaker of the Bété language (Ivory Coast) like a python (*Ô honhoun glibi yèrè*, Zouogbo 2003). This list might be extended to all the languages of the world and would reveal the amazing

richness and diversity of language. The famous Danish linguist Hjelmslev pointed out as early as 1961 that there is a difference between the form and the substance of language, and he argued that this dichotomy was also applicable to the level of content, so that the whole semantic organisation of the lexicon and its interaction with the real world varies substantially from one language to another.

This is undoubtedly a possible starting point for carrying out research on phraseology across languages. Is there no rhyme or reason to the unbridled imagination underlying set phrases in different languages, or is it possible to discover some universal principles? Will set phrases enable researchers to gain information about the cultural patterns and ways of life prevailing in other parts of the world? Can we improve translation practice or theory by a systematic comparison of set phrases across languages? These are just a few examples of the very wide range of approaches involved in cross-linguistic and contrastive phraseology. The language peculiarities as illustrated by concrete examples are only the tip of the iceberg.

It might be quite interesting to investigate the diversity of phraseology by concentrating on specific cases across languages. This could, however, create the impression that comparing languages from the point of view of their set phrases is only a practical matter, and that no thorough theoretical grounding is necessary. Nothing could be less true, as the very starting point of the research, the mere existence of a separate linguistic domain called 'phraseology', remains controversial. In this chapter, I shall briefly mention a few theoretical and practical issues that arise when set phrases are analysed in a variety of languages.

2. Does phraseology exist in all the world's languages?

Set phrases in the broad sense (see Burger et al. 1982) have been identified in many languages. It is well known that the phraseological tradition originated in Russia and Germany (Vinogradov 1946). As a result, Russian and German were among the first languages to be fully described from the point of view of phraseology, although the movement later extended to English, French and most European languages.

It became clear at an early stage that a comparison between set phrases in two or more languages was of crucial importance for discovering the theoretical principles underlying phraseology, as well as its contextual use. As the European Society for Phraseology (cf. http://www.europhras.org) is mainly composed of German-speaking researchers, it is no wonder that the lion's share of cross-linguistic phraseology has consisted of comparing German with other languages including Russian (Dobrovol'skij 1997), Slovakian (Durco 1994), Hungarian (Hessky 1987), Japanese (Rothkegel 2003; Ueda 2004), Spanish (Piñel López 2003), Lithuanian (Budvytyte 2003), Rumanian (Zaharia 2003), French (Gréciano 1989; Dalmas 1999; Valentin 1999), Finnish (Korhonen 1989), Dutch (Piirainen 1995) and Icelandic (Sverrisdóttir 1987).

However, there has been a growing interest in comparing English with one or more other languages; examples include Arabic (Awwad 1990), German (Gläser 1984), Ger-

man & Polish (Paszenda 2003), French (Gläser 1999), Spanish (Marín-Arrese 1996; Mena Martinez 2003), Hebrew (Newman 1988), Latvian (Veisbergs 1992) and Malay (Charteris-Black 2003). Dobrovols'kij & Piirainen (2005), in a major contribution to which I shall refer later in this chapter, analysed figurative language, an important component of phraseology, in English, German, Dutch, Swedish, French, Russian, Lithuanian, modern Greek, Finnish, Japanese, and a low German dialect.

We are now clearly moving towards more comprehensive coverage of phraseology in all languages, but there is still a long way to go before we can claim that phraseology as we know it in European languages is a universal phenomenon. A number of the studies already mentioned were based on non-Indo-European languages (Bété, Japanese, Arabic, Finnish, Malay), and can be considered valuable clues. The common features in these contrastive studies can be summarised as follows:

– In all the languages, there are many examples of a wide variety of constructions that meet the general definition of phraseology (Burger et al. 1982; Burger 1998). Phraseology in the broad sense meets the criteria of 'polylexicality' and 'fixedness', whereas phraseology in the narrow sense requires the additional criterion of 'idiomaticity'. It is not yet clear, however, that the proportions of the various categories of set phrases are universal. Indeed there are many indications that some language families or subfamilies will have clear preferences for the use of verbal vs. nominal set phrases, or metaphorical vs. opaque set phrases, to mention just two examples.
– In spite of the various definitions of phraseology proposed in the literature, the analysis of phraseology in many languages regularly poses the initial question of the interaction between syntax, phraseology and semantics. On the basis of European syntax, we may have a slightly biased view of what phraseology looks like in other language families. Metaphor seems to be a key element in the phraseology of all languages, but some languages may prefer simple metaphors to complex set phrases.
– There is a close link between culture and phraseology (Sabban 2007 and this volume). This is best revealed by proverbs and fully idiomatic set phrases, because they tend to rely heavily on images, traditions or habits that are characteristic of a given culture. It is no easy matter, however, to draw a line between images that are related to more or less universal aspects of the human mind, and features of a specific culture. There is also a common idiomatic heritage to all European languages, originating from biblical or Latin and Greek expressions.
– A distinction between maritime and continental cultures seems to be relevant for the description of phraseology in the world's languages. English and Dutch, for instance, have a larger proportion of set phrases deriving from the sea (Jeans 2004).

From an ethno-linguistic point of view, extending the study of set phrases to the language families that are considered to be the most ancient ones on the basis of both archaeology and biology would be a very welcome development. Recent studies have shown that the Khoisan language family (spoken in southern Africa, among others

by the Bushmen) may very well be the most ancient language family, as archaeological evidence goes back some 60,000 years ago. The Khoisan languages have only recently been studied extensively by linguists (Westphal 1971; Treis 1998). As with other languages from distant parts of the world, a number of distinctive features have been noted, but they do not contradict the universal principles of syntax, semantics, pragmatics and culture and contain a great number of set phrases.

If confirmed by further research, the findings available for a broad array of languages show that phraseology, just like syntax, is one of the key components of human language. This inevitably poses a more general question: why is that so? I will show below that the theoretical underpinnings of cross-linguistic research on phraseology are quite diverse.

3. Contrastive phraseology across theories

The weak theoretical background of research on phraseology has been criticised by Čermák (2001). Anyone studying set phrases across languages needs to be aware that several interpretations of the term 'contrastive' are possible (Dobrovol'skij & Piirainen 2005; Korhonen 2007). One possible approach is to consider 'contrastive' and 'cross-linguistic' as synonyms. In that case, any kind of comparison of set phrases between languages is considered as contrastive phraseology. However, 'contrastive' in the narrow sense implies a truly systematic comparison between two or more languages, on the basis of all their differences and similarities. Finally, a more restricted interpretation of 'contrastive' is also possible, in which only the differences between languages are taken into account. This is more than simply a terminological issue. Mentioning a few examples taken from a number of languages may be interesting from a cross-linguistic point of view, but a truly contrastive study presupposes an in-depth analysis of the phraseology of two or more languages.

Apart from these methodological issues, cross-linguistic and contrastive phraseology are based on examples, but these only make sense if they are interpreted in a theoretical framework. As pointed out by Čermák (2001) and Dobrovol'skij & Piirainen (2005), one of the main flaws of many traditional studies in contrastive phraseology is that they are little more than description or comparison of examples, with no particular attention being paid to the theoretical implications. A number of very valuable contributions have already considered theoretical questions related to phraseology, including problems of classification and terminology, and interaction with context (Burger et al. 1982; Burger 1998; Cowie 1998; Gläser 1984, 1985). There is, however, no global theory of phraseology, in the sense that the origin of set phrases, their relative importance in language and their interaction with syntax, semantics and pragmatics remain largely controversial. If set phrases turn out to be a major aspect of language, both for their frequency and their semantic connections, a subtheory of language must be developed to explain them.

In the absence of such a theory, at least two main linguistic schools, viz. cognitive semantics and corpus linguistics, already provide a theoretical foundation for cross-linguistic or contrastive phraseology.

3.1 Cognitive semantics

Cognitive semantics (Lakoff 1988) and cognitive linguistics (Langacker 1999, 2000; Taylor 2002) have stressed the role of metaphor as a cornerstone of language. From a cognitive point of view, metaphors play a crucial role in most set phrases, especially idioms, and there are abstract concepts underlying metaphors, such as 'GOOD IS UP; BAD IS DOWN' (Lakoff & Johnson 1980; Chun 2002). As cognitive semantics has historical links with generative linguistics, it comes as no surprise that these abstract structures are presented as more or less universal. Checking the validity of universal metaphors in a variety of languages is therefore one of the sources of inspiration for contrastive phraseology (see, for example, Kempcke 1989; Marín-Arrese 1996; Charteris-Black 2003; Piñel López 2003; Piirainen 2007 and this volume).

Although there are obvious similarities between metaphors and set phrases, using a cognitive framework for the analysis of phraseology raises a number of problems. In the first place, not all set phrases correspond to metaphors. Most pragmatic or communicative set phrases such as routine formulae are not metaphorical. On the other hand, many metaphors are closely related to set phrases and there are numerous borderline cases. An *angel* can be considered as a one-word metaphor referring to a very kind person, but the imperative form *Be an angel and...* is considered by most dictionaries as a set phrase. To use another metaphor, we are really getting to the heart of the matter here. Is an *angel* really a metaphor or has this meaning become so common (in many European languages) that it is a simple case of polysemy? How can we distinguish between metaphors and idioms? What is the exact relationship between idiomaticity, figurativeness, motivation, opacity, convention and culture, both for metaphors and set phrases? And, for that matter, how do we define meaning? The absence of a universally recognised semantic theory makes the whole cognitive approach very complex. It may also be criticised from the point of view of the reliability of the data and the replicability of the experiments, two key features of any scientific method. Indeed, defining the underlying cognitive structures of metaphors or set phrases relies extensively on the intuition of the linguist, and different cognitive linguists will inevitably come to different analyses of the same structures. The methodology is largely deductive, in much the same way as generative linguistics used to be.

The Conventional Figurative Language Theory (CFLT) proposed by Dobrovol'skij & Piirainen (2005) is not a general theory of phraseology. Nevertheless, it can be seen as a major theoretical breakthrough in understanding the cognitive foundations of both metaphors and idioms, as well as their mutual relationship. It applies to figurative language as defined by the authors according to two basic criteria: image requirement (a conceptual structure mediating between the lexical structure and the actual meaning) and additional naming (figurative language is not the only way of expressing a

specific idea). Unlike most cross-linguistic and contrastive studies on phraseology, Dobrovol'skij & Piirainen consider the many theoretical assumptions that can be derived from the observation of the diversity of languages. Their theory stresses the image component as a specific conceptual structure underlying figurative units and a relevant element of their meaning. They also claim that some restrictions on the use of figurative units are directly attributable to this image component.

This is obviously one of the key issues. The image component is an interesting theoretical construct providing an improved account of the interaction between form and meaning in figurative units such as metaphors and idioms, but it is no more than a cognitive hypothesis if the linguistic data provide no corroborative evidence. Dobrovol'skij & Piirainen give a few interesting examples. They show that the set phrase *(to be) caught between a rock and a hard place* displays a number of usage restrictions that can be traced back to the image component. The general meaning of this set phrase is *to be in a very difficult position*, but they point out that it cannot be used in all situations in which someone is in a difficult position, because it involves "the mental picture of being between two obstacles, i.e. the idea of a 'lack of freedom of movement'" (Dobrovol'skij & Piirainen 2005: 15). If this view is accepted, it is evidence for the cognitive approach to set phrases, and especially for the image component. The example shows how interesting, and at the same time how complex, a semantic approach to set phrases in the world's languages can be, all the more so as the cognitive approach is not the only possible way. The same example could also be analysed from a purely pragmatic point of view, with restrictions due to context or speaker. The interaction between figurative meaning, cognitive principles and literal meaning is also problematic. For example, it is not quite clear to what extent the literal meaning of *a rock and a hard place* also contributes to some usage restrictions.

More generally, a comparison between figurative units in several languages undoubtedly unveils a number of interesting cognitive and semantic principles. At the same time, the image component is influenced by the culture of a specific language, and can therefore yield a lot of information about differences in culture, especially when very remote languages are the object of investigation.

However, as brilliantly demonstrated by Dobrovol'skij & Piirainen, comparing figurative language in several languages does not cover all set phrases, because many phrases (e.g. grammatical or pragmatic phrases, phrasal verbs, routine formulae, and many collocations) are not figurative. There is obviously a need for additional contrastive work to determine the exact place of phraseology within general linguistic theory. If we claim that phraseology is just one aspect of figurative language, we will disregard the great bulk of set phrases. If, on the other hand, co-occurrence is used as the only principle underlying set phrases, the semantic link between metaphors and idioms will be ignored.

Across the diversity of studies on set phrases in several languages, another major theoretical issue centres around the following question: what is the central category of set phrases? A lot of attention has traditionally been devoted to fully idiomatic set phrases, the well-known 'idioms'. In many respects, these can be considered as ex-

treme cases of phraseology, especially when they are opaque or non-compositional. Dobrovol'skij & Piirainen (2005: 39) call them the "central and most important class of phrasemes". Comparing idioms in several languages is particularly useful for analysing cultural phenomena, and idioms are open to several types of modification, variation and literal reinterpretation (see Burger 1998). They can also create stylistic effects in various registers of language, including literature. Thus, when Dickens writes in his Christmas Carol that *Old Marley was dead as a doornail*, he is preparing the reader for the scene where Scrooge glances at the knocker of his door and sees Marley's face.

3.2 Corpus linguistics

If idioms are considered to be the essence of phraseology, then only the cognitive or semantic aspect of language is taken into account. If, on the other hand, the relative frequencies of the various categories of set phrases are considered (cf. Moon 1998; Colson 2003, 2004, 2007), the very low incidence of idioms is striking. Pragmatic set phrases such as routine formulae are much more frequent than idioms, both in written and in spoken language. This is a fundamental issue to address in further research on phraseology across languages. Many studies have been devoted to idioms, but as revealed by corpus studies, idioms are fairly marginal from a purely quantitative point of view: most of them have a frequency that is lower than one occurrence per million words.

If it is claimed that idioms are the central category of set phrases, this leads to the conclusion that phraseology is a marginal phenomenon, because idioms are rather rare in corpora. This seems to be confirmed by the semantic and cognitive research on figurative language, which is characterised by 'additional naming' (Dobrovol'skij & Piirainen (2005: 18), indicating that figurative language is not the primary way of expressing an idea. This restricted view of phraseology is also consistent with the traditional interpretation of phraseology as an ornamental or rhetorical feature of language. However, a comparison between languages (especially in the case of translation) reveals just the opposite: phraseology turns out to be a major aspect of all languages. Taken in the broad sense, phraseology is present at all levels of linguistic production and comprehension, because native speakers will assemble lexical elements according to a wide variety of existing patterns that may have little to do with grammar.

Like in other sciences, linguistics may have to find a unifying principle behind apparently contradictory theories. Linguistic co-occurrence in various linguistic samples should be studied in more depth, and this is precisely where corpus linguistics comes in. John Sinclair's (1991) idiom principle implies that set phrases in the broad sense are responsible for at least half of the constructions that are found in most texts. From the very beginning of corpus-based research, it has been clear that co-occurrence phenomena, especially collocations, are a major linguistic issue.

There are many useful ways of combining corpora with the study of phraseology across languages. The frequency issue should rather be analysed on very large corpora (Moon 1998; Colson 2003), because the more idiomatic set phrases tend to be rather infrequent. This remains to some extent problematic, as it has so far not been possible

to determine the precise frequency levels for phraseology. For all their interest and importance, semantic classifications of set phrases are no more than hypotheses, and hard evidence is very difficult to find. This implies that one semantic classification can always be replaced by another, and that this may go on for some time. Further research might, on the other hand, focus on corpus evidence from various languages that would point to the existence of set phrases, the criteria for recognising and classifying them, as well as frequency limits that would help differentiate specific categories of set phrases.

A related topic is that of frequency differences across languages. It is not at all clear, for instance, that all languages use set phrases in the same proportions. The relative importance of the noun category may vary from language to language and will therefore interfere with the importance of verbal expressions, one of the main categories of set phrases. Identifying a phraseological profile for various languages on the basis of large corpora would be very useful for both language learners and translators, because many errors are due to an insufficient or incorrect mastery of phraseology.

Corpus-based approaches to lexical problems (Johansson & Oksefjell 1998; Altenberg & Granger 2002; Johansson 2007) play a very important role in documenting the actual use of lexis in context, and its many interactions with phraseology. Across the diversity of languages, it is becoming increasingly clear that a very detailed analysis, both manual and automated, of lexical and co-occurrence phenomena in corpora is particularly useful for solving underlying theoretical issues, such as the role of semantics and syntax and their interplay with phraseology. Prepositions are very interesting in this respect, because they can often be positioned on the borderline between syntax and phraseology (see Gilquin 2000; Cosme & Gilquin this volume). The frequency issue again plays a significant part in this interaction. In traditional linguistics, prepositions, as well as adverbs, connectives and articles have been regarded as essentially grammatical parts of speech, but their behaviour in large corpora seems to point in the opposite direction. As noted by Sinclair (1991), most grammatical constructions are largely dependent on the use of lexical elements. In other words, the choice between prepositions may often be influenced by phraseology. This may even apply to the choice between the definite and indefinite article in many European languages, as this choice is largely dependent on both grammatical and usage principles.

As far as connectives are concerned, interesting research has recently been devoted to cross-linguistic differences and their origins (Degand 2005; Altenberg 1998, 1999). The use of causal connectives in different languages, for instance, reveals both similarities and striking differences. This is an interesting meeting point between contrastive phraseology and pragmatics. Obviously, the choice of connectives is motivated by semantic and pragmatic principles, but connectives are often part of larger units such as clichés, routine formulae or grammatical phrases, all of which are part of phraseology in the broad sense (see Blumenthal 1999). Future research on large corpora may therefore benefit from a combination of linguistic approaches, including phraseology.

Within this field, pioneering work has been done by researchers in French/German contrastive phraseology. Gréciano (1997) and Dalmas (1999), among others, have investigated the use of phrasemes (in the sense of fully idiomatic set phrases) in com-

bination with discourse particles. Many examples taken from French and German corpora point to the frequent association between phrasemes and German particles (e.g. *doch, übrigens, überhaupt, ja denn auch, ganz* used in combination with a phraseme), whereas French often seems to moderate or introduce the use of a phraseme with a variant of the verb 'to say' or 'to name', as in the following example: "*cela ne ressemblait en rien à* ***ce qu'on appelle*** *un coup de chapeau*" (Gréciano 1997:458). The interesting theoretical point made by Gréciano is "qui se ressemble s'assemble" (like attracts like). In other words, discourse particles and set phrases share many common features, and it should therefore be no surprise that they often co-occur. It is not obvious, however, what percentage of set phrases (and what category) will regularly be accompanied by particles, or to what extent this is language dependent.

Another closely related issue is whether set phrases across languages are regularly accompanied by introducers. Some researchers (see Čermák 2002) have pointed out that many set phrases, especially verbal idioms, are often accompanied by syntactic constructions or specific words that seem to introduce or moderate them. A typical example is the English adjective *proverbial*, as in *to spill the proverbial beans*. The same holds true of Dutch (with the adjective *spreekwoordelijk*) and German (*sprichwörtlich*), which is in itself an interesting starting point for a more thorough contrastive analysis of this phenomenon. It is still unclear to what extent the use of such types of introducers in combination with set phrases relates to rhetorical or pragmatic principles. Obviously, it is always possible to combine pragmatic modifiers with set phrases, but the case of *proverbial* associated with verbal idioms rather suggests that languages such as English, German and Dutch have recourse to conventionalised patterns.

4. Contrastive phraseology and translation

Studying phraseology in a wide range of languages inevitably leads to a consideration of translation. Indeed, translation is often a way of detecting phraseology. Many set phrases, and especially verbal idioms, cannot be translated literally, even in closely related languages. Thus, a phrase like *down the hatch* is easily recognised as a set phrase by French-speaking learners, because no literal translation is possible. There are, however, notable exceptions to this principle, because a large number of set phrases are common to several languages. This is particularly the case with the many phrases that European languages have borrowed from Greek, Latin and Hebrew. In addition to this practical use of translation as a way of recognising set phrases, the interaction between phraseology and translation also raises a number of theoretical issues. If, as many researchers within corpus linguistics and phraseology have pointed out, set phrases constitute a major aspect of any language, it is clear that translating from one language to another will mean being confronted with a very difficult task twice: establishing the meaning of the source text while taking figurative language and phraseology into account, and

then trying to find an equivalent formulation in the target language. In other words, phraseology will be one of the major pitfalls of translation.

Strangely enough, 'phraseology and translation' is not such a common research field. Apart from a conference in German (Sabban 1999) and a few articles on this subject (Roberts 1998; Poirier 2003; Rojo 2003; Koller 2007), the very concept of phraseology still seems to be largely absent from studies on translation theory or practice. Delisle (2003), one of the best reference books on translation theory and practice, does not mention the domain of phraseology. Set phrases are treated as expressions and whilst their importance is not underestimated, they are only treated as part of the lexicon.

In several respects, phraseology and translation constitutes a highly interdisciplinary research field. Sabban (1999) illustrates the rich cultural diversity underlying any attempt to translate a set phrase from one language into another. As pointed out by several researchers, a widely held misconception about set phrases is that a set phrase from L1 should be translated into a corresponding set phrase in L2. Foreign language teachers and learners are often faced with the practical problem of trying to make set phrases correspond across languages and, as in the case of lists of idioms, they tend to reinforce this misconception.

A more dynamic view of the translation of set phrases takes into consideration a number of cultural and linguistic principles. Once again, phraseology is the meeting point of conflicting theories about form, meaning and culture in language. Poirier (2003) analyses the arbitrary and conventional nature of the translation of set phrases from the point of view of semiotics. He emphasises its arbitrary character which stems from the fact that a semantic paraphrase is always possible if the idiomatic aspect is ignored (for instance, *spill the beans* can be translated into other languages by simple constructions meaning *reveal a secret*). On the other hand, the translation of set phrases is conventional in the semiotic sense of the word, because of the conventional relations between lexical units, and as a result of the conventional nature of the notion of equivalence.

This interesting theoretical approach to the translation of set phrases points to the complex interplay between phraseology, semiotics and translation. Because of their special status, somewhere between lexicon and syntax, set phrases are particularly adept at revealing both the strong and the weak points of current linguistic theories. Cognitive semantics insists upon metaphors, but many set phrases cannot be reduced to metaphors, and this principle is of little help for their practical translation. Corpus linguistics, on the other hand, lays stress on the numerous contextual examples derived from a corpus, but the many intricate facets of a given phrase are also governed by semantic principles, and cannot easily be inferred from its linguistic context.

As already mentioned, Delisle (2003) does not use the term 'phraseology', but he insists on the thorny problems posed by the translation of various categories of expressions. His very informative and useful handbook is corroborated by the experience of many a translation teacher: phraseology hampers the translation of most texts, be they general and informative, or technical and scientific. In the latter case,

phraseology often combines with terminology because many disciplines or technical domains create their own set phrases and multi-word terms. Future research might aim at testing a number of hypotheses about the exact role of phraseology in translation practice. What is the impact of phraseology on the overall pattern of translation processes? What kind of psychological or cognitive activities does phraseology require from translators and interpreters? Are there universal translation techniques for set phrases, or is the solution dependent on each and every language pair? From a theoretical point of view, it remains for future research to determine whether phraseology deserves its own place among the underlying principles of translation, as well as in the list of major translation errors.

There may also be an interesting link between phraseology and the research on 'translationese', the intermediate language resulting from an imperfect translation (Tirkkonen-Condit 2002). A number of studies have compared translated and non-translated monolingual corpora (Hansen 2003; Laviosa 1998; Puurtinen 2003). Baroni & Bernardini (2006) used an automated method to recognise translationese and claim that the computer achieved better results than human evaluators. If confirmed by other studies, this might open the door to better identification of translationese, and to better automated evaluation of translations. Baroni & Bernardini's method is based on support vector machines; it highlights the importance of function words and morpho-syntactic categories as possible clues to the identification of translationese and to translation quality assessment. As the results are partly derived from n-gram extraction, it comes as no surprise that the authors mention collocational and colligational patterns as other interesting aspects of translationese that should be investigated further. Assessing the overall quality of translation in an automated way has become a necessity for large companies providing translation services (De Sutter 2005). Because of time constraints, and in view of the very large number of language combinations, the evaluation of translators is already partly automated, but the existing methods need to be improved. Phraseology may be one of the key factors in evaluating the quality of a translation, and it may be a new challenge for natural language processing and machine learning algorithms to extract set phrases from translated corpora and to compare them with the original texts.

5. Concluding remarks

Phraseology can be seen as the linguistic repository of a number of cultural traditions that are specific to a given language. It is therefore quite natural to study phraseology across languages, because this will elucidate the origins of many of these linguistic and cultural habits. A number of them may turn out to be more or less universal, which would reveal fascinating aspects of human cognition. However, some caution is needed in pursuing an analysis of this sort. Until now, the focus of research has been primarily on European languages, and a comparison with other language families is necessary before we can draw any firm conclusions about the universality of phraseology.

From a theoretical point of view, cross-linguistic phraseology is a highly multidisciplinary field. It has strong links with contrastive lexicology, syntax, pragmatics and semantics, but also with semiotics and translation theory. The wide diversity of linguistic theories underpinning phraseology across languages can be an advantage, but the downside is that no single agreed methodology has been developed. Cognitive linguists largely rely on their intuition, while corpus linguists have recourse to large corpora. A widely accepted view is that there is some truth in every theory, and future research may therefore benefit from meeting points between various linguistic schools.

Phraseology across languages also has important consequences for translation theory and translation practice. The technological evolution in translation assessment should greatly benefit from new insights into the structure and functioning of set phrases.

References

Altenberg, B. (1998). Connectors and sentence openings in English and Swedish. In Johansson, S. & S. Oksefjell (eds.) *Corpora and Cross-Linguistic Research: Theory, Method, and Case Studies*, 115–143. Amsterdam: Rodopi.

Altenberg, B. (1999). Adverbial connectors in English and Swedish: semantic and lexical correspondences. In Hasselgård, S. & S. Oksefjell (eds.) *Out of Corpora. Studies in Honour of Stig Johansson*, 249–268. Amsterdam: Rodopi.

Altenberg, B. & S. Granger (eds.) (2002). *Lexis in Contrast. Corpus-Based Approaches* [Studies in Corpus Linguistics 7]. Amsterdam: John Benjamins.

Awwad, M. (1990). Equivalence and translatability of English and Arabic idioms. *Papers and Studies in Contrastive Linguistics* 26: 57–67.

Baroni, M. & S. Bernardini (2006). A new approach to the study of translationese: Machine-learning the difference between original and translated text. *Literary and Linguistic Computing* 21(3): 259–274.

Blumenthal, P. (1999). Clichés dans les commentaires de presse: analyse contrastive. *Nouveaux Cahiers d'allemand* 17: 577–593.

Budvytyte, A. (2003). Der axiologische Aspekt der deutschen und litauischen Somatismen. In Burger, H., A. Häcki Buhofer & G. Gréciano (eds.) *Flut von Texten – Vielfalt der Kulturen. Ascona 2001 zu Methodologie und Kulturspezifik der Phraseologie*, 255–266. Baltmannsweiler: Schneider Verlag Hohengehren.

Burger, H. (1998). *Phraseologie. Eine Einführung am Beispiel des Deutschen.* Berlin: Erich Schmidt.

Burger, H., A. Buhofer & A. Sialm (eds.) (1982). *Handbuch der Phraseologie.* Berlin: de Gruyter.

Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) (2007). *Phraseologie / Phraseology. An International Handbook of Contemporary Research*. Berlin: Mouton de Gruyter.

Čermák, F. (2001). Substance of idioms: Perennial problems, lack of data or theory? *International Journal of Lexicography* 14(1): 1–20.

Čermák, F. (2002). Text introducers of proverbs and other idioms. In Földes, C. & J. Wirrer (eds.) *Phraseologismen als Gegenstand sprach- und kulturwissenschaftlicher Forschung*, 27–46. Baltmannsweiler: Schneider Verlag Hohengehren.

Charteris-Black, J. (2003). A prototype based approach to the translation of Malay and English idioms. In Granger, S., J. Lerot & S. Petch-Tyson (eds.) *Corpus Based Approaches to Contrastive Linguistics and Translation Studies*, 123–140. Amsterdam: Rodopi.

Chun, L. (2002). A cognitive approach to *Up/Down* metaphors in English and *Shang/Xia* metaphors in Chinese. In Altenberg, B. & S. Granger (eds.) *Lexis in Contrast. Corpus-Based Approaches* [Studies in Corpus Linguistics 7], 151–174. Amsterdam: John Benjamins.

Colson, J.-P. (2003). Corpus linguistics and phraseological statistics: a few hypotheses and examples. In Burger, H., A. Häcki Buhofer & G. Gréciano (eds.) *Flut von Texten – Vielfalt der Kulturen. Ascona 2001 zu Methodologie und Kulturspezifik der Phraseologie*, 47–59. Baltmannsweiler: Schneider Verlag Hohengehren.

Colson, J.-P. (2004). Phraseology and computational corpus linguistics: From theory to a practical example. In Bouillon, H. (ed.) *Langues à niveaux multiples. Hommage au Professeur Jacques Lerot à l'occasion de son éméritat*, 35–45. Louvain-la-Neuve: Peeters.

Colson, J.-P. (2007). The World Wide Web as a corpus for set phrases. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseologie / Phraseology. An International Handbook of Contemporary Research*, Vol. 2. 1071–1077. Berlin: Mouton de Gruyter.

Cowie, A. P. (ed.) (1998). *Phraseology. Theory, Analysis, and Applications*. Oxford: Clarendon Press.

Dalmas, M. (1999). Les particules et autres mots de la communication dans les formules et routines: Degrés de tolérance. *Nouveaux Cahiers d'allemand* 17: 437–449.

Degand, L. (2005). De l'analyse contrastive à la traduction: le cas de la paire puisque – aangezien. In Williams, G. (ed.) *La linguistique de corpus*, 155–168. Rennes: Presses Universitaires de Rennes.

Delisle, J. (2003). *La traduction raisonnée*. Ottawa: Presses de l'Université d'Ottawa.

De Sutter, N. (2005). Automated translation quality control. *Communicator* Summer 2005: 22–25.

Dobrovol'skij, D. (1997). Kontrastive Idiomatik Deutsch-Russisch. Zur lexicographischen Konzeption. In Gréciano, G. & A. Rothkegel (eds.) *Phraseme in Kontext und Kontrast*, 45–59. Bochum: Brockmeyer.

Dobrovol'skij, D. & E. Piirainen (2005). *Figurative Language. Cross-Cultural and Cross-Linguistic Perspectives.* Amsterdam: Elsevier.

Durco, P. (1994). *Probleme der allgemeinen und kontrastiven Phraseologie: Am Beispiel Deutsch und Slowakisch.* Heidelberg: Groos.

Gilquin, G. (2000). The integrated contrastive model. Spicing up your data. *Languages in Contrast* 3(1): 95–123.

Gläser, R. (1984). The translation aspect of phraseological units in English and German. *Papers and Studies in Contrastive Linguistics* 18: 123–134.

Gläser, R. (1985). Idiomatik und Sprachvergleich. *Sprache und Literatur in Wissenschaft und Unterricht* 16: 67–73.

Gläser, R. (1999). Zur Wiedergabe von Phraseologismen in englischen und französischen Übersetzungen ausgewählter Prosawerke von Christa Wolf. In Sabban, A. (ed.) *Phraseologie und Übersetzen*, 99–118. Bielefeld: Aisthesis Verlag.

Gréciano, G. (1997). Qui se ressemble s'assemble: locutions, particules et compères. *Nouveaux Cahiers d'Allemand* 17: 451–460.

Gréciano, G. (1989). Auf dem Weg zur vergleichenden Phraseologie: Deutsch-Französisch. In Gréciano, G. (ed.) *EUROPHRAS 88: Phraséologie contrastive: Actes du Colloque International Klingenthal-Strasbourg, 12–16 mai 1988*, 155–163. Strasbourg: Université des Sciences Humaines.

Hansen, S. (2003). *The Nature of Translated Text*. Saarbrücken: Saarland University.

Hessky, R. (1987). *Phraseologie. Linguistische Grundlagen und kontrastives Modell deutsch-ungarisch*. Tübingen: Max Nimeyer.

Hjelmslev, L. (1961). *Prolegomena to a Theory of Language*. Madison, WI: University of Wisconsin Press.

Jeans, P. D. (2004). *Ship to Shore. A Dictionary of Everyday Words and Phrases Derived from the Sea*. London: International Marine/McGraw-Hill.

Johansson, S. (2007). *Seeing through Multilingual Corpora. On the Use of Corpora in Contrastive Studies*. Amsterdam: John Benjamins.

Johansson, S. & S. Oksefjell (eds.) (1998). *Corpora and Cross-Linguistic Research. Theory, Method and Case Studies*. Amsterdam: Rodopi.

Kempcke, G. (1989). Struktur und Gebrauch der somatischen Phraseme mit den Bedeutungskomponenten Kopf und tête. In Gréciano, G. (ed.) *EUROPHRAS 88: Phraséologie contrastive: Actes du Colloque International Klingenthal-Strasbourg, 12–16 mai 1988*, 225–232. Strasbourg: Université des Sciences Humaines.

Koller, W. (2007). Probleme der Übersetzung von Phrasemen. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseologie / Phraseology. An International Handbook of Contemporary Research,* Vol. 1, 605–613. Berlin: Mouton de Gruyter.

Korhonen, J. (1989). Zur syntaktischen Negationskomponente in deutschen und finnischen Verbidiomen. In Gréciano, G. (ed.) *EUROPHRAS 88: Phraséologie contrastive: Actes du Colloque International Klingenthal-Strasbourg, 12–16 mai 1988*, 253–264. Strasbourg: Université des Sciences Humaines.

Korhonen, J. (2007). Probleme der kontrastiven Phraseologie. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseologie / Phraseology. An International Handbook of Contemporary Research,* Vol. 1, 574–589. Berlin: Mouton de Gruyter.

Lakoff, G. (1988). Cognitive semantics. In Eco, U., M. Santambrogio & P. Violi (eds.) *Meaning and Mental Representations*, 119–154. Bloomington, IN: Indiana University Press.

Lakoff, G. & M. Johnson (1980). *Metaphors We Live By*. Chicago, IL: University of Chicago Press.

Langacker, R. W. (1999). Assessing the cognitive linguistic enterprise. In Janssen, T. & G. Redeker (eds.) *Cognitive Linguistics: Foundations, Scope and Methodology*, 13–59. Berlin: Mouton de Gruyter.

Langacker, R. W. (2000). *Grammar and Conceptualization*. Berlin: Mouton de Gruyter.

Laviosa, S. (1998). Core patterns of lexical use in a comparable corpus of English narrative prose. *Meta* 43(4): 557–570.

Marín-Arrese, J. I. (1996). To die, to sleep. A contrastive study of metaphors for death and dying in English and Spanish. *Language Sciences* 18(1): 37–52.

Mena Martinez, F. (2003). Creative modifications of phraseological units in English and Spanish. In Burger, H., A. Häcki Buhofer & G. Gréciano (eds.) *Flut von Texten – Vielfalt der Kulturen. Ascona 2001 zu Methodologie und Kulturspezifik der Phraseologie*, 169–182. Baltmannsweiler: Schneider Verlag Hohengehren.

Moon, R. (1998). *Fixed Expressions and Idioms in English*. Oxford: Clarendon Press.

Newman, A. (1988). The contrastive analysis of Hebrew and English dress and cooking collocations: Some linguistic and pedagogic parameters. *Applied Linguistics* 9(3): 293–305.

Paszenda, J. (2003). A lexicographic corpus of religious phraseological internationalisms in English, German and Polish. Criteria for the selection of data. In Burger, H., A. Häcki Buhofer & G. Gréciano (eds.) *Flut von Texten – Vielfalt der Kulturen. Ascona 2001 zu Methodologie und Kulturspezifik der Phraseologie*, 239–254. Baltmannsweiler: Schneider Verlag Hohengehren.

Piirainen, E. (1995). Niederländische und deutsche Phraseologie: Zu einem kontrastiven Beschreibungsansatz. *Neerlandica Wratislaviensia* 8: 193–217.

Piirainen, E. (2007). Phrasemes from a cultural semiotic perspective. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseologie / Phraseology. An International Handbook of Contemporary Research,* Vol. 1, 208–219. Berlin: Mouton de Gruyter.

Piñel López, R. M. (2003). Der Tod und das Sterben in der deutschen und spanischen Phraseologie: ein interkultureller Vergleich. In Burger, H., A. Häcki Buhofer & G. Gréciano (eds.) *Flut von Texten – Vielfalt der Kulturen. Ascona 2001 zu Methodologie und Kulturspezifik der Phraseologie*, 229–238. Baltmannsweiler: Schneider Verlag Hohengehren.

Poirier, E. (2003). Conséquences didactiques et théoriques du caractère conventionnel et arbitraire de la traduction des unités phraséologiques. *Meta* 48(3): 402–410.

Puurtinen, T. (2003). Genre-specific features of translationese? Linguistic differences between translated and non-translated Finnish children's literature. *Literary and Linguistic Computing* 18(4): 389–406.

Rojo, J. (2003). Las lenguas de Europa: estudios de fraseología, fraseografía y traducción. *International Journal of Lexicography* 16(4): 449–452.

Roberts, R. (1998). Phraseology and translation. In Fernández Nistal, P. & J. M. Bravo Gozalo (eds.) *La traducción: Orientaciones lingüísticas y culturales*, 61–78. Valladolid: SAE.

Rothkegel, A. (2003). Zur Komposition sprachlicher Bilder – ein Ansatz zum kulturellen Vergleich Deutsch-Japanisch. In Burger, H., A. Häcki Buhofer & G. Gréciano (eds.) *Flut von Texten – Vielfalt der Kulturen. Ascona 2001 zu Methodologie und Kulturspezifik der Phraseologie*, 277–288. Baltmannsweiler: Schneider Verlag Hohengehren.

Sabban, A. (ed.) (1999). *Phraseologie und Übersetzen*. Bielefeld: Aisthesis Verlag.

Sabban, A. (2007). Culture-boundness and problems of cross-cultural phraseology. In Burger, H., D. Dobrovol'skij, P. Kühn & N.R. Norrick (eds.) *Phraseologie / Phraseology. An International Handbook of Contemporary Research,* Vol. 1, 590–604. Berlin: Mouton de Gruyter.

Sinclair, J. (1991). *Corpus, Concordance, Collocation*. Oxford: Oxford University Press.

Sverrisdóttir, O. (1987). *Land in Sicht. Eine kontrastive Untersuchung deutscher und isländischer Redensarten aus der Seemanssprache*. Frankfurt: Peter Lang.

Taylor, J. R. (2002). *Cognitive Grammar*. Oxford: Oxford University Press.

Tirkkonen-Condit, S. (2002). Translationese – a myth or an empirical fact? *Target* 14(2): 207–220.

Treis, Y. (1998). Names of Khoisan languages and their variants. In Schladt, M. (ed.) *Language, Identity, and Conceptualization among the Khoisan*, 463–503. Köln: Rüdiger Köppe.

Ueda, Y. (2004). Kontrastive Phraseologie – idiomatische Wendungen mit Tierbezeichnungen als Hauptkomponenten im Deutschen und Japanischen. In Földes, C. (ed.) *Res humanae proverbiorum et sententiarum, Ad honorem Wolfgangi Mieder*, 351–364. Tübingen: Gunter Narr Verlag.

Valentin, P. (1999). Les lexèmes nominaux composés: Équivalences et traductions entre allemand et français. *Nouveaux Cahiers d'Allemand* 17: 533–540.

Veisbergs, A. (1992). English-Latvian borrowing of idioms. *Contrastive and Applied Linguistics* (University of Latvia) 1: 36–48.

Vinogradov, V. V. (1946). Osnovnye ponjatija russkoj frazeologii kak lingvističeskoj discipliny. In Vinogradov, V. V. (1977) *Izbrannye trudy. Leksikologija I leksikografija*, 118–139. Moscow: Nauka.

Westphal, E. O. J. (1971). The click languages of Southern and Eastern Africa. In Sebeok, T. A. (ed.) *Current Trends in Linguistics 7: Linguistics in Sub-Saharan Africa*, 367–420. Berlin: Mouton.

Zaharia, C. (2003). Quellen der idiomatischen Ausdrücke im Deutschen und Rumänischen. In Burger, H., A. Häcki Buhofer & G. Gréciano (eds.) *Flut von Texten – Vielfalt der Kulturen. Ascona 2001 zu Methodologie und Kulturspezifik der Phraseologie*, 267–276. Baltmannsweiler: Schneider Verlag Hohengehren.

Zouogbo, J.-Ph. (2003). Bété-Sprichwörter im Vergleich mit dem Deutschen und Französischen. In Burger, H., A. Häcki Buhofer & G. Gréciano (eds.) *Flut von Texten – Vielfalt der Kulturen. Ascona 2001 zu Methodologie und Kulturspezifik der Phraseologie*, 289–302. Baltmannsweiler: Schneider Verlag Hohengehren.

Figurative phraseology and culture

Elisabeth Piirainen

This chapter seeks to outline the main trends in research on cultural aspects of conventional figurative language (phraseology in particular) and to describe the connection between figurative units (such as idioms, proverbs, etc.) and culture as it becomes manifest in phraseological data from several languages. Numerous studies have highlighted the fact that phrasemes are not only units of a sign system, language, but also carriers of cultures, pointing out the necessity for modern phraseological research to turn to cultural phenomena. What exactly constitutes the connection between figurative phrasemes and culture, however, has never been examined systematically. This chapter uses a typology of the cultural phenomena which underlie phrasemes, along with the relevant phraseological types, to explore this connection. In doing so, one has to consider both the literal and the figurative readings of phrasemes, as well as the different levels of describing phrasemes, since there are various ways in which the cultural aspects may become manifest.

1. Preliminary remarks

In earlier stages of phraseology research, the interest in cultural phenomena was varied. Until quite recently, topics like idiom syntax, idiom semantics, idiom pragmatics, including sociolinguistic and psycholinguistic perspectives, text-related modifications of idiom structure and cross-linguistic research, have figured more prominently in many studies from (Western) Europe than cultural topics. Most current studies on phraseology, however, regard culture as a fundamentally acknowledged constant in phraseology (cf. e.g. Telija 1998; Cowie 1998; Gréciano 2002; Sabban 2004, 2007 and this volume; Dobrovol'skij & Piirainen 1997, 2005; Wierzbicka 2007).

This turn towards cultural phenomena is connected not least to cognitive approaches to conventional figurative language (phraseology). The cognitive perspective allows us to put aside any strong distinction between a synchronic and a diachronic level of analysis and instead place the relevant knowledge structures which underlie phrasemes (conceptual structures such as frames and scripts, textual knowledge, symbolic knowledge, i.e. a wide range of cultural knowledge) at the centre of phraseological research. There is no cohesive notion of culture suitable for all phraseological studies yet. We will have a brief look at attempts to define this notion from other

linguistic sub-disciplines (Section 2). I will then outline the main types of cultural aspects which seem to underlie phrasemes (Section 3).

The present study employs a wide conception of *phraseology*, a conception that most European phraseology researchers agree on today (cf. Burger et al. 2007). Phraseology is understood here as the totality of fixed multi-word units of a language, i.e. formulaic expressions that are elements of the lexicon and that go beyond the level of a single word but do not go beyond sentence level. From this point of view, phraseology is a subset of the more general complex of *formulaic language*, which includes text sequences or complete texts.[1]

The term *phraseme* is used as a hyperonym for all kinds of fixed word groups, while *idiom* is a term used for members of one subgroup, most of which are also figurative (idiomatic). To ascertain the significance of culture within the heterogeneous set of phrasemes, we have to distinguish between non-figurative and figurative units. Figurative phrasemes can be affected by cultural aspects with respect to both their literal reading (the source concept) and their 'actual' meaning (the target concept). From this follows that different types of phrasemes are related to cultural phenomena in different ways (Section 4). We will later look at some individual studies on specific conceptual domains (Section 5).

The notion that phraseology is in some way a mirror of a national culture or a worldview is problematic, and this question should be touched upon briefly. Similar issues apply to cross-linguistic comparisons (Section 6). Historical or etymological studies on phrasemes, most of which deal intensively with cultural aspects of phrasemes, have been carried out from the very beginning up to the present, independently of linguistic trends (Section 7).

The various sections mentioned above seek to combine two objectives: to describe the connection between figurative phraseology and culture as it becomes manifest in phraseological data from several languages and to outline main trends in research on cultural features of phraseology.

2. Attempts at a definition of *culture*

Although there is general agreement on the fact that culture plays an important role for most phraseological issues, only a few studies have actually treated the relation

1. Formulaic text sequences such as ritual speech, epic sung poetry, nursery rhymes, chants, etc. are of course rooted in culture as well (cf. e.g. Stolz & Shannon 1976; Jarrett 1984; Pawley 2007), but they are not the topic of this chapter. The same is true for formulae in poetic language, cf. the detailed study on Indo-European poetic text fragments involving myth and ritual by Gamkrelidze & Ivanov (1995: 731–740) or modern social rituals, formulaic small talk or formulaic sequences used by auctioneers, sportscasters etc. (e.g. Kuiper 1991, 1996). On the other hand, recent English-language studies, especially on language acquisition, use the term *formulaic language* primarily in the sense of phraseology (e.g. Wray 2002; Schmitt 2004; Skandera 2007).

between phraseology and culture in detail. While it is true that titles of phraseological studies rarely use the words *culture* or *cultural* (judging from relevant bibliographies), the studies themselves make ample use of these terms (and words such as *culture-specific, culture-based, culturally marked, culturally significant, culturally tainted, culturally bound, cultural connotations* etc.). Due to the vagueness of the term *culture-specifity*, Sabban (2007: 590 and this volume) proposes replacing it with the term *culture boundness.* It should be added that the term *cultural foundation* has been applied when describing cultural elements that underlie phrasemes (Dobrovol'skij & Piirainen 2005: 216ff.).

Most studies concerned with *culture* or terms derived from it make no attempt to define these terms. To understand them better, let us take a look at neighbouring disciplines and their treatment of the term *culture*. Notions of culture seem to fluctuate between a wide and a narrower concept.[2] Wierzbicka (1992, 1996) for example, favours a wide concept, pointing out that almost everything in language reveals a certain degree of cultural specifics. She states that the meanings of most words differ from language to language because they are cultural artefacts, reflecting aspects of the cultures in which they were created. According to Wierzbicka (1996: 15), the concept underlying a word like German *Seele* is not identical to the concepts underlying the English word *soul* or the Russian word *duša*, etc. because these concepts are unique and culture-specific configurations of semantic features. It is not incorrect to use the term *culture* for these cases; however, such a broad definition almost renders the term semantically empty. In such cases we deal with a kind of *language specific* which has to be separated from *culture specific.*

A wide concept of culture is also used in the field of cultural anthropology. Here, *culture* refers to the broad fields of human behaviour and social interaction, cf. Tylor's (1871: 1) often-quoted definition of culture as "that complex whole which includes knowledge, belief, art, morals, law, custom, and any other capabilities and habits acquired by man as a member of society". This classic anthropological notion of culture has been criticised, modified and developed further over the last decades. Central to this conception of culture is the idea of *cultural models*, which are shared by members of a given community and which make up their entire cultural knowledge (see also Shore 1996). Within the broad field of cultural models studied by anthropologists, *linguistic cultural models* are of particular interest for phraseology, since collectively shared cultural norms, attitudes or values can manifest themselves in presuppositions underlying proverbs and other prefabricated stereotypes. A notion of culture close to

2. For an impression of the diversity of the concepts of culture let me point to Kroeber & Kluckhohn (1952) who list as many as 164 definitions of culture from popular and academic sources. According to Duranti (1997: 23ff.) six principal understandings of culture have been identified: 1. culture as distinct from nature, 2. culture as knowledge, 3. culture as communication, 4. culture as a system of mediation, 5. culture as a system of practices, 6. culture as a system of participation. As it turned out, point 2 and 3 are particularly relevant to the issue of phraseology and culture.

that of anthropology has also been applied to the description of linguistic formulae in a wider sense, emphasizing "the shared way of life of a group of people" (Sabban 2004: 403).

Other principal characteristics of culture come to the fore in the field of semiotics, particularly in the so-called semiotics of culture. The notions of culture in this field include both a wide concept of culture and a narrower one (cf. e.g. Portis-Winner 1994). Central to the attempts at defining culture from a semiotic viewpoint is the human predisposition to create signs and to give significance to all things that surround us; culture is viewed as a system of symbols or meaningful signs. Some ideas of this semiotic view of culture, mainly elaborated by the Moscow-Tartu school, have been adopted by Dobrovol'skij & Piirainen (2005: 213ff.) to describe cultural phenomena underlying figurative language, above all cultural symbols in phraseology (see Section 3). Crucial are the various parts of cultural knowledge that stand behind figurative units. Culture can be seen as the sum of all ideas about the world (including fictional, mythological etc. ideas) that are characteristic of a given community.

3. Typology of cultural aspects underlying phrasemes

This section tackles two questions: how is culture actually revealed in figurative phraseology, and how can we categorize the cultural knowledge structures that underlie phrasemes? For this purpose, the distinction between a synchronic and a diachronic (etymological) level of description can be left out of consideration. However, it is convenient to concentrate on those phrasemes where there is little doubt about the kinds of cultural knowledge involved. We can use the typology of principal cultural phenomena that occur in figurative phrasemes developed by Dobrovol'skij & Piirainen (2005: 214ff.). Relying on idiomatic material from many different languages as the starting point for research, one can note the emergence of structures of cultural phenomena that are strikingly similar to those established by semioticians, often resulting in triads such as social, material, and mental culture. The category 'mental culture', however, seems unsuitable for describing linguistic phenomena because language as a whole is a mental phenomenon. It is therefore replaced by the three types (a) textual dependence, (b) pre-scientific conceptions of the world and (c) cultural symbols. Complementing these types of cultural phenomena are the two types (d) aspects of material culture and (e) aspects of culture-based social interaction.

(a) An important type of cultural foundation of phraseology can be labelled *textual dependence*. This group consists of phrasemes whose image components can be traced back to an identifiable textual source. There can be direct references to particular texts, original quotations, which gradually developed into idioms or proverbs, or there can be allusions to an entire text, summarizing a certain situation or the gist of a text. Phrasemes of this type have been explored very well for several European languages. Large groups are made up of items related to the Bible (see Williams this volume) or to

fables (e.g. *the black sheep; sour grapes*); other groups make reference to works of belles-lettres, fairy tales, narratives, movies or even titles of books, films, TV shows, etc. (e.g. *to gild the lily; with seven-league boots; a happy-hunting ground*). Some idioms of this type display lexical irregularities that are 'inherited' from the source text, cf. *to cast pearls before swine*, using the rather uncommon word *swine* instead of the more usual *pig*. It would go beyond the scope of this chapter to list textual sources significant in other cultural areas, as for example the Koran for Arabic (Baccouche 2007), the Beijing Opera for Chinese phraseology (cf. Ying & Erh-li 1996: 49ff.) or Chinese literature, tales and legends for the phraseologies of several East Asian languages.

(b) Phrasemes that are based conceptually on *pre-scientific conceptions of the world* make up another culturally relevant group. However, they have never been explored as systematically as have the links between phrasemes and well-known texts. Subgroups of this type include, among other things, phrasemes whose underlying fictive concepts belong to folk belief (*enough to make the angels weep*), superstitions (*to thank one's lucky stars*) or old folk medicine (rejected in the course of later scientific developments). 'Humoral pathology' – the doctrine of the four fluids of the body, or humors, that determine the four prototypical temperaments – was of great influence on the phraseologies of European languages. It can be recognized in idioms such as the French *se faire du mauvais sang, se faire de la bile* (to be worried) or the German *jmdm. läuft die Galle über* (sb.'s bile overflows; sb. becomes very angry). The cultural specifics of these idioms become even more comprehensible when we turn our attention to languages of 'distant' cultural areas. Yu (2003) explains the concept GALL BLADDER in the Chinese culture and phraseology, a concept which is deeply anchored in the theory of internal organs in traditional Chinese medicine. According to this edifice of medical ideas, which dates back thousands of years, the gall bladder serves to make judgments and decisions and determines the degree of a person's courage. A wealth of Chinese idioms reflects this pre-scientific concept.

(c) In conventional figurative units such as idioms and proverbs, *cultural symbols* manifest themselves chiefly in one single key constituent that contains the relevant cultural knowledge (as opposed to the phraseme as a whole). The motivational link between the literal and figurative readings of these constituents is established by semiotic knowledge about the symbol in question, about its meaning in culturally relevant sign systems other than language (e.g. in mythology, religions, popular customs, fine arts). The symbol undergoes a semantic reinterpretation: it is a sign whose primary content is used as a sign for denoting another content. For example, the primary meaning of 'white' in the idiom *whiter than white* has shifted metonymically to meanings such as 'honest', 'true' or 'morally pure'. WOLF assumes symbolic functions such as 'danger' (*cry wolf*) or 'economic despair' (*keep the wolf from the door*), which are recurrent in figurative language and supported by other cultural codes. This goes back to the semiotisation of the wolf as a dangerous, greedy, man-devouring demon in various cultural codes, from the Bible to fairy tales and modern comics (see Dobrovol'skij & Piirainen 1997: 215, 2005: 253ff. for details).

(d) The cultural foundation of a large number of phrasemes can be ascribed to *aspects of material culture*, which are embedded in everyday life of the present or the past. All kinds of artefacts can be part of literal readings of phrasemes. Several idioms, spread across many languages, use achievements of modern technological culture such as motorized transport or telecommunications as their source frames (*to see the light at the end of the tunnel, to be on the same wavelength as sb.*). Idiosyncratic aspects of material culture in European standard languages seem to be rare. Possibly, the concept SAUNA, an essential part of the traditional culture of Finland, is a source frame that is unique to Finnish figurative phraseology (*lisätä löylyä* 'to increase sauna steam' i.e. to cause a conflict to become more intense). In contrast to that, the phraseology of dialects, or language varieties close to dialects, reveals an abundance of idioethnic realia. Luxemburgish phraseology, for instance, is imbued with cultural realia of winegrowing and winemaking as source concepts (a productive constituent is *Hatt* 'a basket which winegrowers carried on the back for picking grapes or transporting soil to the vineyards' (Filatkina 2005: 348, 2006: 119)). Certainly the further one moves away from the relatively unified European cultures, the more elements of culture-specific artefacts can be found in phraseologies. The image components underlying several Japanese idioms, for example, reveal the traditional Japanese dwelling house with its veranda, sliding paper screens or *tatami*-straw mats (cf. Piirainen 1999; Dobrovol'skij & Piirainen 2005: 193ff.).

(e) *Aspects of culture-based social interaction* can be used as an umbrella term for a heterogeneous group of phrasemes whose underlying cultural knowledge chiefly goes back to knowledge about social experiences and behaviours within a given community. This means that some shared knowledge about culture-based phenomena in a society is involved in the processing of these phrasemes. Sub-categories include, among other things, semiotised gestures (*to take off one's hat* 'to show one's admiration for sb./sth.'), gender specifics (*sb. wears the trousers (at home, in the family)* 'it is the wife rather than her husband or partner who is the dominant person in a household') and bans and taboos (*to be under the influence* 'to be drunk'). The cultural foundation of the latter lies in the need of the speech community to avoid direct naming and instead employ strategies of glossing over. Thus, all phrasemes revealing cultural models belong to this group, above all proverbs, regarded as giving information about which values are upheld in a given culture and expressing generally applicable rules governing social behaviour (cf. e.g. White 1987). Finally, routine formulae are part of *social interactions* and therefore belong to this type of cultural foundation as well (see Sections 4 (iv) and (v)).

The goal here is not to assign each and every phraseme unambiguously to a particular type but to structure the cultural knowledge that shapes these units and lies at the very heart of phraseology as a whole. It is therefore not necessary to draw sharp lines between these types, which tend to overlap and interrelate. The idiom *cry wolf* (type (c)) also belongs to type (a), as it goes back to an Aesopian fable. The idiom *the green-eyed monster* meaning 'extreme jealousy' is a quotation from Shakespeare's

tragedy *Othello, the Moor of Venice* and as such belongs to type (a). The connection between the colour green (or yellow) and emotions such as anger and envy, however, goes back to the pre-scientific 'humoral pathology' (type (b)), which in Shakespeare's days was still alive and appears throughout his popular dramas. Cultural symbols such as GOLD, LILY or the symbolic number SEVEN can be found in idioms of intertextual origin such as *to gild the lily; with seven-league boots.*

4. Types of phrasemes and their relevance to culture

Phraseology is concerned with very heterogeneous entities, and researchers are used to dividing them into smaller, relatively homogeneous subtypes. Here we will adopt a division into five more or less commonly accepted types[3] and for each of them try to outline at what semantic level culturally significant features can present themselves. As far as figurative phrasemes are concerned, their correlation with cultural aspects can be due to both their literal reading (their 'inner form' or source concept) and their phraseological meaning (their actual meaning or target concept).

(i) *Idioms* are regarded as the central and most irregular category of phrasemes. What makes idioms stand out from other phrasemes is their high degree of *idiomaticity*, which manifests itself in semantic reinterpretations and/or semantic opacity (cf. Baranov & Dobrovol'skij 2005; Dobrovol'skij 2007 for a definition and further details). First, let us look at an example where elements of culture are located on the level of the source domain. The idiom *when hell freezes over* (something will never happen) is motivated on the basis of shared cultural knowledge about HELL as a hot place that will never freeze over. The literal reading of the idiom refers to cultural concepts of mythology or folk belief, passed on throughout the centuries. Let us compare this to the idiom *to be left on the shelf* said of a woman who is past an age at which she might expect to get married. Cultural aspects of the source domain (e.g. *shelf* as an object of material culture) can be disregarded here. Rather, the cultural dimension becomes apparent at the level of the actual meaning. The gender specifics are a stable component of the semantic structure of this idiom; there is no male counterpart. The idiom conveys the idea of conforming to the standard that only a woman (but not a man) should have married by a certain age and thus reveals a cultural model in contemporary society.

(ii) *Similes* should be mentioned here first of all because many phraseological studies,[4] especially cross-linguistic ones, point to their cultural implications (cf. Section 6).

3. Cf. Burger et al. (2007: XIIf.). Other classifications which, for instance, subsume proverbs under the category of idioms (e.g. in Skandera 2007) seem unsuitable for the task at hand.

4. Judging from relevant bibliographies, phraseology research has paid considerable attention to this group from its very beginnings up to the present day and produced some hundred articles on this topic.

Similes stand out by their specific structure of comparison, consisting of the *tertium comparationis* and the *comparatum* (the vehicle of comparison), which are connected via a particle (*as, like*) (cf. e.g. Burger 1973: 48ff. for defining criteria; see also Wikberg this volume). Aspects of culture are mainly fixed in the literal readings of the vehicle. Melchers' (1997) study of similes in English dialects denoting STUPIDITY gives many examples like *as daft as a besom, a grindstone, a scuttle, a shovel, a wagon-horse, turnips, muck*, etc. All of these vehicle words denote objects specific to the particular rural material culture and tend to reflect salient concepts of a given community, including idioethnic realia that may stand out for their cultural connotations. In addition, similes can convey cultural symbols; cf. *to eat like a wolf.* The image of the wolf eating voraciously is not supported by actually observing the animal but strongly supported by cultural symbolism, cf. various narrative traditions that establish the conventional wisdom about the wolf as a gluttonous animal.

(iii) The term *collocation* has no unified meaning or definition in linguistic studies (cf. e.g. Sinclair 1991; Wray 2002; Burger et al. 2007). It is either understood as the co-occurrence of words in general, irrespective of their fixedness or convention of use, or as a word combination that co-occurs habitually and therefore belongs to phraseology. Here, the term *restricted collocation* is used for the latter and thus for phrasemes such as French *se brosser les dents* (to clean one's teeth), consisting of a (collocative) base, used in its literal sense (*dents* in this example), and a collocate, which is to a certain extent arbitrary (*brosser* is the only appropriate verb here). Most restricted collocations of this type are not figurative and hardly affected by aspects of culture. However, there is another structural type of collocations which has to be regarded as partly figurative, cf. *a busy bee* (a very busy person). While the collocate *busy* is used in its literal meaning, the base *bee* has been semantically reinterpreted to denote a (female) person rather than the animal. It is this characterisation of the bee that reveals cultural implications. Since antiquity and up to the present day bees have been used as a basis of comparison for prototypical DILIGENCE and BUSYNESS.

(iv) From the viewpoint of folklore studies, *proverbs* are elements of a code of folk culture; they are the object of investigation of paremiology. From the viewpoint of linguistics, proverbs are a central type of phrasemes. Despite the extensive literature on proverbs,[5] a generally acknowledged definition has not yet been arrived at (cf. e.g. Mieder 2004: 2ff., 2007; Kleiber 1989, 2005). Many proverbs are figurative and have far-reaching cultural significance. They reveal all of the types of cultural phenomena outlined in Section 3. There are not only aspects of material culture (e.g. by means of constituents denoting culture-specific realia), but many proverbs are also directly interrelated with other culturally relevant texts. However, proverbs are most significantly connected with aspects of culture-based social interaction – a fact that is due to their special semiotic, semantic and pragmatic characteristics. One outstanding property of

5. Cf. Mieder's (1982–2001) bibliographies and *Proverbium. Yearbook of International Proverb Scholarship*.

proverbs is the existence of a *universal quantifier* (or *all-operator*) in the content plane. Proverbs are general statements that are believed to express a universal truth, i.e. they refer to allegedly shared knowledge about rules governing social behaviour. Of course they do not allow the drawing of conclusions about attitudes and values of an entire language community but only of special groups at a special time. Besides, proverbs can have the illocutionary force of 'recommendation/recommending'. They can provide moral support for an argument or action by referring to a generalised proposition and thus give advice on how to behave in certain situations. Proverbs quote socially approved ideas that can be used instead of an argumentation; they can reveal traces of social concepts (of special groups and/or former times) and hand them down to future generations. An example of such a model is the proverb type *Women have long hair and short brains*, which is widespread over many European and Oriental languages; the idea is that women have to accept a subordinate position in society because of their alleged intellectual deficiency.

(v) Terms like *routine formulae, communicative phrasemes* (cf. e.g. Coulmas 1979; Aijmer 1996) or *pragmatic idioms* (Burger 1973: 58ff.) cover a large continuum of phrasemes. While greetings, expressions of thanks, excuses, congratulations, etc. belong to the core elements, the boundaries of this type are not always clear, especially in demarcating formula of comments (e.g. *that's where the shoe pinches*) from idioms proper (cf. Wotjak 2005 for an overview). Only a few routine formulae are figurative in the sense that elements of culture can be found in their source domain. The German *Hals- und Beinbruch!* (broken neck and leg!) is a jocular formula used in order to wish someone good luck. Its origin is explained by the ancient concept of superstition or folk belief that an unconcealed wish of good luck brings misfortune and one can outwit fate by wishing something bad. For a detailed analysis of the relationship between linguistic routine and culture, see Sabban (2004). In contrast to the phraseme types discussed above, the cultural link of routine formulae is almost exclusively restricted to the pragmatic level. Routine formulae are tools of communication; their most important function is the constitution of speech acts. They are therefore part of a larger complex of stereotyped action patterns and social interaction (comparable to specific gestures like bow and handshake). From this perspective, all communicative formulae reveal aspects of culture-based social interaction (cf. type (e) in Section 3 above). The cultural boundness becomes particularly visible when languages spoken by members of 'distant' cultures are taken into account, for example in the inconsistent use of English and Chinese formulae in Singapore (Kuiper & Tan 1989). There are also a number of communicative formulae in Japanese which have no equivalents in European languages, e.g. formulae used when leaving the house or coming home, together with their appropriate replies (cf. Coulmas 1981).

5. Manifestations of culture in different conceptual domains

The majority of figurative phrasemes such as idioms are semantically motivatable (albeit with hindsight). The motivation can involve different types of cultural knowledge, e.g. knowledge of folk theories, frames and scripts, cultural symbols, etc. This means that most figurative phrasemes can be analysed both at the level of their literal reading (source concept) and at the level of their phraseological meaning (target concept). In order to comprehensively capture the cultural phenomena that underlie phraseology, it seems appropriate not to be concerned merely with some individual phrasemes that just happen to attract attention but to deal with entire conceptual domains. In this section, we would like to look at some work that covers larger portions of the complete phraseological inventory, i.e. entire source concepts (image-based frames and scripts),[6] on the one hand and complete semantic fields on the other.

The taurine phraseology, which is exclusive to Spanish, is an impressive example of how conventional figurative language can be interwoven with elements of a culturally important source concept. The concept of bullfighting, the 'national feast', is assumed to have contributed about 500 idiomatic expressions to the Spanish language. As Luque Durán & Manjón Pozas (1999) have shown, this concept, at whose centre is the corrida, the bullfight ceremony, not only has produced such a large number of common phrasemes, but also is a kind of cultural coding that creates an intricate universe of metaphors. This fully-fledged concept is an all-pervasive reality in the minds of Spanish speakers; "no one in Spain can really escape the influence of bullfighting folklore and myths, as they have become embedded in Spanish language and culture throughout the ages" (ibid. p. 34). According to Yangüela (1998), the concept of the bullfight has entered the language in two ways: it is present and frequent in everyday language, while the special taurine language (a jargon and a technical language at the same time), for its part, originates from the common language. A culturally loaded idiosyncratic source concept of this kind and scale is, to our knowledge, unparalleled by other European standard languages.

Source concepts referring to religious and folk beliefs have been favourite phraseological research topics for several languages and inspired studies such as Ángeles Calero Fernández' (1998) work on Spanish and Catalan phraseology. With the help of 925 phrasemes containing the concepts GOD and DEMON, the author analyses the impact of religious thoughts on language. The ideas of Christianity, of the divinity and of diabolic forces, are indeed significant cultural components in the constitution of Spanish and Catalan figurative language. In the author's opinion, language helps to canalise the religious feeling of a people and conserves traditions that are transmitted to sub-

6. Cultural symbols in phraseology (cf. Section 3 (c)) also have a tendency to occur in groups, viz. in symbolic domains. For lack of space, we will not discuss the literature on this subject matter here. See Dobrovol'skij & Piirainen (1997, 2000, 2005, 2006) for a discussion of the domains of colour symbolism, number symbolism and animal symbolism in the phraseologies of several languages.

sequent generations (ibid. p. 157). One should not, however, draw direct conclusions about individual culture-specific traits of a language community from the fact that these source concepts appear so frequently. It is sufficiently well known that the concepts GOD and DEMON were very productive in other languages as well, e.g. in French, English or German phraseology. The Russian phraseology of the 18th century, in particular, was imbued with ideas from the domains of religion and folk belief, reflecting the significance of folk piety and popular belief of former times (Bierich 2005: 130ff.).

Earlier phraseology research used terms like 'thematic groups' or 'image donor domains' (German 'Sachgruppen' or 'Bildspenderbereiche') approximately in the sense of the term 'source concepts' used by modern cognitive approaches to phraseology. Several research papers on thematic groups of historical or more modern times succeeded in revealing cultural aspects that underlie the phraseology of current German, including studies on knightly combat and the mediaeval legal system or modern technology and traffic and radio and telecommunication (see Fleischer 1997: 182ff. for an overview). Similarly, the phraseology of an old Low German dialect spoken by a rural community has been analysed in view of its source domains. From a cognitive semantic view, sources such as POPULAR CUSTOMS, including atavistic funeral customs, HORSE AND CARRIAGE and THE ANCIENT FARMHOUSE, were found to reflect cultural knowledge on the part of the remaining native speakers of this dialect (cf. Piirainen 1999, 2000, Vol. 1: 207ff., 2004: 52ff.).

In what follows, we will have a look at studies that start from an entire semantic field (a target concept). First of all, we have to find out at what level of description cultural aspects are expected to appear. Idioms that are metaphorically motivated can be described either on the superordinate level of the conceptual metaphor or on the basic level (in the sense of Rosch 1975) of the "rich image" (cf. Lakoff 1987: 406), depending on the current research questions and objectives. As it turns out, it is only on the latter level that cultural implications can readily be detected.

Let us demonstrate this with the help of the often-discussed ANGER metaphors analysed in the framework of the Cognitive Theory of Metaphor (CTM, cf. Lakoff (1987); Lakoff & Kövecses (1987)). Some general and productive ANGER metaphors have been found to exist in various languages (e.g. ANGER IS THE HEAT OF A FLUID IN A CONTAINER), a fact that has been ascribed to the concept of 'embodiment', the idea that body experiences underlie metaphors. Due to the 'sameness' of human beings and their same physiological mode of operation across different cultures, conceptual metaphors have been regarded as ubiquitous in all cultures, if not universal.[7] Reactions provoked by postulates of the CTM triggered a number of studies on similar

7. Some weaknesses have been identified insofar as the CTM tends to construct many ad hoc metaphors and does not distinguish between novel metaphors and conventionalised metaphors (such as idioms). In contrast to that, Iñesta Mena & Pamies Bertrán (2002) start from phraseology (idioms from 23 languages) and come to similar results as the CTM, namely that many universal conceptions (culturally independent cross-linguistic similarities) can be uncovered on the abstract superordinate level of description.

topics, among them contributions to emotion concepts in non-European languages like Zulu, Chinese, the Tsou language or Tunisian Arabic (see Maalej 2004 for an overview). Thanks to these studies, we know a lot more about cultural implications of idioms, all of which emerge at the basic level of rich image.

An analysis of Tunisian Arabic provides significant evidence for the cultural foundation of idioms denoting ANGER. As Maalej (2004) explains in detail, ANGER idioms do not describe physiological change to the body but, among other things, refer metaphorically to customs and rituals of the Arabian culture. One highly elaborated source concept is the custom for Muslims to sacrifice a sheep every year at the occasion of pilgrims visiting Mecca. A wealth of idioms reveal, in their literal readings, the scenario of slaughtering and butchering a sheep, from breaking the sheep's bones, opening and cleaning its stomach, to skinning it by using a special technique of inflating the animal. According to Maalej (2004: 72), these idioms are hyperboles, using an *as if* mode: an angry person is conceptualised *as if* he/she was a slaughtered sheep with all its painful experiences. In view of such a rich imagery it would be hard to formulate conceptual metaphors at the superordinate level of categorisation.

Besides ANGER, the target concept DEATH, TO DIE has been a popular topic of phraseology research (cf. Piirainen 2000, Vol. 1: 176, 2002: 216f. for an overview). The outstanding phraseological activity in this field and its elaborate structuring in many languages have often been mentioned. As one example, let us look at Anders's (1995) substantial work on idioms denoting 'death' and 'to die' in current English and medieval English literature. It gives insight into the many cultural aspects that combine to constitute this semantic field, from underlying cultural knowledge structures (e.g. textual and symbolic knowledge) to pragmatic and psychological aspects. The specifics of the field TO DIE have deeper psychological causes. Dealing with the existential threat, this field is connected with fear and inhibitions, which linguistically manifests itself in a large number of euphemistic and dysphemistic circumlocutions.

A comparable case is that of the semantic field PREGNANCY in the Low German dialect mentioned above (Piirainen 2000, Vol. 1: 157ff., 165ff.). This dialect has no word with the meaning 'pregnant' but more than 30 idioms that can be used instead. As it turns out, pregnancy is a taboo topic for speakers of the dialect, a topic which one must not mention directly, while prefabricated units offer indirect ways of talking about it. Many idioms are ambiguous in the sense that the literal reading does not immediately allow conclusions about the target concept. Besides, only very negative conceptualisations of PREGNANCY manifest themselves in the dialectal phraseology. PREGNANCY is presented as a physical pain, misfortune, distress or impediment to motion, which is clearly in contrast to conceptualisations known from the major standard languages analysed so far in this regard. Here, we will content ourselves with the observation that the semantic-pragmatic specifics of this field may reflect some aspects of former attitudes and values of the dialectal speech community and not jump to the conclusion that the dialect speakers themselves share a worldview or mentality predetermined by their language.

This leads us to the question of whether conceptualisations detected in phraseology can reveal certain views or hierarchies of values of a speech community, as has often been claimed in phraseological research (see Section 6). The semantic field INACTIVITY, LEISURE has been used to investigate such questions. This field has been studied independently for three European languages. Telija (1996: 231) discusses the Russian idiom *bit' bakluši* (to laze about), whose image component is guided by the scenario of manufacturing wooden spoons, where the idiom points to a very simple and very unimportant activity. Since this idiom and many others of the field of INACTIVITY, LEISURE reveal very negative assessments, the author concludes that the whole concept INACTIVITY, LEISURE is negatively marked in Russian. Using a model developed within the framework of the project "Ethic concepts and mental cultures: virtues and vices in the mirror of the language", Skog-Södersved & Stedje (1997) study German and Swedish fixed expressions connected with LAZINESS. The authors come to the conclusion that in Swedish, the concept TO LAZE ABOUT is closer to that of LEISURE and less negatively connoted than in German. However, such statements should be made with all due caution, and they do not allow conclusions about the worldview or mentality of the speakers in question.

6. National culture, worldview and culture in cross-linguistic research

The assumption that phrasemes, particularly idioms and proverbs, allow access to a collective way of thinking of a language community belongs to the more general and highly complex issues of the relation between language, cognition, and culture.[8] The idea that the world is principally perceived through the medium of language, which determines the speakers' worldview (proposed by W. v. Humboldt and later reformulated by Sapir and Whorf as the theory of linguistic relativity) can be found in several directions of phraseological research. This conception usually is accompanied by postulates that the analysis of phraseology allows insight into the speech community's own culture and mentality, if not into its national character.

First, let us mention a direction of phraseological research called linguo-cultural studies, mainly carried out by a group of researchers in Russia,[9] who start directly from Humboldt's or Sapir and Whorf's idea of linguistic relativity (cf. e.g. Telija 1996, 1998; Bragina 1998; Telija et al. 1998). For this group, data of the linguo-cultural analysis of phraseology "can serve as an empirical basis for verifying the linguistic relativity

8. Compare Langacker's (1994: 26ff.) model to assess the relation of language and culture not as separate entities, but as overlapping facets of cognition.

9. Some international conferences held in Moscow have been devoted to "Phraseology and the Context of Culture". Topics included, among other things, the methodological foundation of linguo-cultural analysis of phrasemes, cultural stereotypes or the special worldview, which can be actualised by contrastive analysis (cf. Bragina 2000).

hypothesis" (Telija 1998: 792). An important question is, among other things, how cultural comments on phrasemes can be placed in dictionaries (Telija et al. 1998). Some ideas go back to the hypothesis of earlier Soviet research that phrasemes are cultural signs per se, which are unique to the one language in question, and thus for the most part untranslatable. From this view, phrasemes are generally culture-specific, and therefore, cultural components encoded in the semantic structure of phrasemes may be regarded as a mirror of the national culture or a national mentality.[10]

Numerous studies have since been carried out on the presupposition that analysing phraseology in terms of cultural components provides the basis for uncovering a specific cultural worldview. As a result, various Russian phrasemes have been referred to as typical of Russia's national mentality. One example is the Russian *zerno istinui* (a grain of truth) which, according to Telija et al. (1998: 66), is "loaded with cultural associations: grain is part of the rite of sowing connected with the archaic concept of Birth and Rebirth". This is not to deny the two-sided fact that language forms part of culture and culture manifests itself in the phraseology of any given language. However, the supposition that the analysis of phrasemes can contribute to uncovering information about a specific mentality or worldview of a language community is largely disputed. The main objections come from the fact that most of the empirical data are taken from one single language (Russian) and, above all, that terms like *national culture* or *national mentality* lack any operational definitions in phraseology. According to Dobrovol'skij (1997, 2000), the assumption that most idioms are to a certain extent marked in terms of national culture is rooted in mixing up different linguistic phenomena. As a rule, many idioms have no absolute equivalents in other languages. The reason for this is not grounded in any cultural or national specifics, however, but in the fact that different languages go different ways with respect to semantic reinterpretation, i.e. in creating figurative meanings on the basis of literal ones.

Contrastive phraseology research has followed ideas that are similar to those outlined above for the mostly monolingual studies, and there is a rich tradition of cross-linguistic comparisons of phrasemes. Since the 1970s and early 1980s, a wealth of mostly bilingual comparative studies have been carried out, including, among others, works on Russian and German idioms (Rajchštejn 1980: 23–56) and German and Hungarian idioms (Hessky 1987). While earlier studies gave priority to issues of morphosyntactics and equivalence relations, many of the more recent studies seek to discover cultural differences (or similarities) between the language communities

10. Similar ideas can be found in Wierzbicka (1997: 13ff.). Although not concerned with phraseology proper, the author refers to Russian proverbs and sayings to demonstrate that in the case of 'love of truth' the Russian national character can be captured by means of the phraseolexicon: "[...] the view that the 'full truth' must be loved, cherished, and respected like a mother, is at variance with Anglo cultural norms, which value 'tact,' 'white lies,' 'minding one's own business,' and so on" (ibid. 15).

in question. Terms like *culture-specifics, culture-markedness* etc. are used repeatedly, although no definitions are given for these terms.

Almost stereotypically, various bilingual studies begin with remarks about the tasks of their cross-linguistic approach. For example, a comparative study on somatisms in the Lithuanian and German phraseologies aims at uncovering differences with respect to cultural patterns of behaviour and mentality (Budvyte 2003: 258), while a Spanish-German contrastive study assumes that phrasemes are reliable indicators for the connection between worldview and figurative language, claiming that every people, in fact every social group with its own culture, also has its own conception of the world (Balzer et al. 2004: 253).

Cross-linguistic studies on similes, in particular, point to their relevance for the worldview. As a negative example, let us consider Rakusan's (2000) article on Czech and English similes with animal constituents, which starts with the assertion that "[p]eople in all cultures perceive the world through culturally shared models transmitted largely by language" (ibid. p. 265). As it turns out, *dog* and *pig* are highly popular constituents in both languages, while Czech similes also use other farm animals whereas English similes contain several wild birds. Rakusan's cultural interpretation is that the Czech language reveals traces of a former agrarian culture whereas English does not, since England was a country of hunting. Similarly, the conception of DOG differs slightly. In Czech similes, the dog appears exclusively as an inferior creature, while English similes also provide some positive connotations (truth, friend). The author labels this as a culturally distinctive feature, related to extralinguistic aspects. According to the author, the English language has adapted a more modern conception of the DOG in the sense that pets in England are treated more like fellow humans, a development which has no place in the Czech community yet (ibid. p. 277f.).

At this point of discussion, Hessky's (1989) article on German and Hungarian similes seems to be helpful. According to Hessky, results that are purely based on the analysis of the subgroup of similes cannot claim general validity because languages have always several options to express a particular concept. Thus, any concrete linguistic realisation given in one language is independent of the concrete linguistic means used to express the same concept in another language (ibid. p. 201).

This chapter could only outline the issue of worldview as it arises in various studies on comparative phraseology. Studies on the translation of phrasemes have to cope with this set of problems as well, and in quite similar ways, for example when dealing with the translation of culturally connoted phrasemes, unique to one special language, into other languages (cf. Eismann 1995; Segura García 1997). For lack of space, we will not go further into this discussion here.

7. Historical and etymological research on phraseology

The aims of historical comparative approaches to phraseology are manifold,[11] ranging from uncovering the etymological origin or the initial form of single phrasemes to reconstructing former stages in the development of a phraseological system or whole domains that have been culturally significant in former times. This branch of research has to rely on cooperation with culturally oriented academic disciplines other than linguistics (e.g. folklore, mythology research, ecclesiastical history) and incorporate dialectal and historical language varieties as well as comprehensive extra-linguistic material.

Let us look at some examples. Since the 1960s, several Slavic studies have been exploiting dialectal phrasemes as a source of etymological and cultural-historic research. Ethnographic and dialectal material was used to reconstruct the phraseology of a Proto-Slavic variety (Tolstoj 1973) and aspects of early religious and mythological concepts or folk culture. Using variants or (quasi)-synonyms of a given phraseme in many different dialects and related languages as a starting point, Mokienko (1973, 1989) develops structural semantic models. This approach finds expression in further etymological and diachronic studies, above all in the historical-etymological dictionary of Birich et al. (2003).

As an example of many other culturally relevant studies, we would like to make particular mention of Eckert's (1991) book on the former significance of forest beekeeping in the Baltic-Slavic countries, which was reconstructed from phrasemes of different language varieties of this area. In this context, phrasemes are treated on the same level as other small linguistic forms of folklore, e.g. folk songs. Another study by Eckert (1998) on such culturally relevant elements contributes, among other things, to reconstructing an ancient cult of the snake in Baltic and Slavic traditions.

For German and related languages, Röhrich's (1995) culturally and historically significant work should be mentioned, as it gives insight into the etymology, older variants and meanings of numerous current phrasemes, cf. also Spalding (1959ff.). Much attention has also been paid to medieval legal practice as a source frame of many German phrasemes. Many idioms, and binomials in particular, can be traced back to gestures once performed in court, together with ancient wordings of laws (cf. e.g. Schmidt-Wiegand 1991, 1993, 2002; Schowe 1993). Recently, Wanzeck (2003) has given a coherent description of the etymology of phrasemes containing colour words in historical and current language. Starting from the actual meaning of the

11. Here we cannot go into the details of other branches of historical phraseology, most of which are occupied with earlier periods of one individual language. However, two large-scale projects should be mentioned, both strongly connected to aspects of culture: the historio-linguistic project "Formulaic Language and Traditions of Formulating" (Filatkina 2007; www.histphras.uni-trier.de), which investigates the development of German idiomatic expressions from 800 to 1700, and Knappe's (2004) investigation of English phraseology, which centres on the contexts and ways in which phrasemes have been analysed and commented upon by English language scholars between approx. 1440 and 1800.

phrasemes, the study centres on the question of how the colour adjectives obtained their phraseological meanings. Exhausting the written sources from the very beginning and considering their cultural and historical contexts, the author succeeds in clarifying the true etymology of many phrasemes that became obscure in the course of history (cf. Williams's notion of resonance in this volume).

In recent times, there have also been voices calling for a look at the etymological origins of phrasemes in order to describe their semantics. Using German and French material, Gréciano (2002:433ff.) discusses etymological explanations of phrasemes with respect to philological aspects and thematic groups of constituents, which allows her to gain relevant cultural and semantic insights into the motivation of phrasemes.

In these latter studies, the uncovering of the etymological origins of phrasemes is by no means an end in itself; rather, they are to be seen as a way of discovering and understanding the processes of how phrasemes come into being and thus contribute to a theory of phraseology. Fragments of world knowledge, i.e. cultural knowledge, are integral components of such current research.

8. Conclusion

In this chapter, we have attempted to approach the complex of figurative phraseology and culture from various angles. After a brief look at efforts to define the term *culture*, the main types of cultural knowledge underlying figurative phrasemes have been examined, followed by an outline of the types of phrasemes in view of their connections with cultural aspects. A look at entire conceptual domains has then shown that cultural phenomena are determinable at the levels of complete source concepts and semantic fields. Finally, the (im)possibility of capturing aspects of a cultural worldview by means of the analysis of cultural components and cross-linguistic comparisons has been touched upon briefly, as have etymology and historical phraseology. To summarise, phrasemes as conventional figurative multi-word units that are passed on from generation to generation through continual repetition turn out to be especially suitable for revealing cultural relevant concepts.

References

Aijmer, K. (1996). *Conversational Routines in English.* London: Longman.

Anders, H. (1995). *Never Say Die – Englische Idiome um den Tod und das Sterben.* Frankfurt: Peter Lang.

Ángeles Calero Fernández, M. (1998). Acerca de Dios y del demonio en la fraseología española y catalana. In Wotjak, G. (ed.) *Estudios de fraseología y fraseografía del español actual*, 155–194. Vervuert: Iberoamericana.

Baccouche, M. G. (2007). Arabic phraseology. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseology. An International Handbook of Contemporary Research*, Vol. 2, 752–758. Berlin: de Gruyter.

Balzer, B., C. Moreno, R. Piñel, M. Raders & M. L. Schilling (2004). Kulturelle Besonderheiten in der Kontrastiven Phraseologie. In Brdar-Szabó, R. & E. Knipf-Komlósi (eds.) *Lexikalische Semantik, Phraseologie und Lexikographie. Abgründe und Brücken*, 253–272. Frankfurt: Peter Lang.

Baranov, A. N. & D. O. Dobrovol'skij (2005). Zum Idiombegriff. In Breiteneder, E. & D. O. Dobrovol'skij (eds.) *Dostoevskij in Focus*, 35–48. Wien: Verlag der Österreichischen Akademie der Wissenschaften.

Bierich, A. (2005). *Russische Phraseologie des 18. Jahrhunderts. Entstehung, Semantik, Entwicklung*. Frankfurt: Peter Lang.

Birich, A. K., V. M. Mokienko & L. I. Stepanova (2003). *Slovar' russkoj frazeologii. Istoriko-ètimologičeskij spravočnik*, 3rd edn. Moscow: Astrel'.

Bragina, N. (1998). Lexical collocations: A dialogue between language and culture. In Eismann, W. (ed.) *EUROPHAS 95. Europäische Phraseologie im Vergleich: Gemeinsames Erbe und kulturelle Vielfalt*, 55–66. Bochum: Brockmeyer.

Bragina, N. (2000). Phraseology and the context of culture. *International Journal of Lexicography* 13(1): 29–34.

Budvyte, A. (2003). Der axiologische Aspekt der deutschen und litauischen Somatismen. In Burger, H., A. Häcki Buhofer & G. Gréciano (eds.) *Flut von Texten – Vielfalt der Kulturen. Ascona 2001 zur Methodologie und Kulturspezifik der Phraseologie*, 255–265. Baltmannsweiler: Schneider Verlag.

Burger, H. (1973). *Idiomatik des Deutschen.* Tübingen: Niemeyer.

Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (2007). 1b. Phraseology: Subject area, terminology and research topics. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseology. An International Handbook of Contemporary Research*, Vol. 1, 10–18. Berlin: de Gruyter.

Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) (2007). *Phraseology. An International Handbook of Contemporary Research.* Berlin: de Gruyter.

Coulmas, F. (1979). On the sociolinguistic relevance of routine formulae. *Journal of Pragmatics* 3: 239–266.

Coulmas, F. (1981). *Routine im Gespräch. Zur pragmatischen Fundierung der Idiomatik*. Wiesbaden: Athenaion.

Cowie, A. P. (1998). Introduction. In Cowie, A. P. (ed.) *Phraseology. Theory, Analysis, and Applications*, 1–20. Oxford: Oxford University Press.

Dobrovol'skij, D. O. (1997). Nacional'no-kul'turnaja specifika vo frazeologii (I). *Voprosy jazykoznanija* 6: 37–48.

Dobrovol'skij, D. O. (2000). La específicidad nacional y cultural en fraseología. In Pamies Bertrán, A. & J. D. Luque Durán (eds.) *Trabajos de lexicografía y fraseología contrastivas*, 63–77. Granada: Universidad de Granada.

Dobrovol'skij, D. O. (2007). Cognitive approaches to idiom analysis. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseology. An International Handbook of Contemporary Research*, Vol. 2, 789–818. Berlin: de Gruyter.

Dobrovol'skij, D. O. & E. Piirainen (1997). *Symbole in Sprache und Kultur.* Bochum: Brockmeyer. (2nd. edn, 2002).

Dobrovol'skij, D. O. & E. Piirainen (2000). Sobre los símbolos: Aspectos cognitivos y culturales del lenguaje figurativo. In Pamies Bertrán, A. & J. D. Luque Durán (eds.) *Trabajos de lexicografía y fraseología contrastivas*, 29–53. Granada: Universidad de Granada.

Dobrovol'skij, D. O. & E. Piirainen (2005). *Figurative Language: Cross-Cultural and Cross-Linguistic Perspectives.* Amsterdam: Elsevier.

Dobrovol'skij, D. O. & E. Piirainen (2006). Cultural knowledge and idioms. *International Journal of English Studies* 6(1): 27–41.

Duranti, A. (1997). *Linguistic Anthropology*. Cambridge: Cambridge University Press.

Eckert, R. (1991). *Studien zur historischen Phraseologie der slavischen Sprachen unter Berücksichtigung des Baltischen*. München: Sagner.

Eckert, R. (1998). On the cult of the snake in ancient Baltic and Slavic tradition (based on language material from the Latvian folksongs). *Zeitschrift für Slawistik* 43: 97–100.

Eismann, W. (1995). Pragmatik und kulturelle Spezifik als Problem der Äquivalenz von Phraseologismen. In Baur, R. S. & C. Chlosta (eds.) *Von der Einwortmetapher zur Satzmetapher*, 95–119. Bochum: Brockmeyer.

Filatkina, N. (2005). *Phraseologie des Lëtzebuergeschen. Empirische Untersuchungen zu strukturellen, semantisch-pragmatischen und bildlichen Aspekten*. Heidelberg: Winter.

Filatkina, N. (2006). Theory revisited: Aspekte der Bildlichkeit im Lëtzebuergeschen. In Häcki Buhofer, A. & H. Burger (eds.) *Phraseology in Motion I. Methoden und Kritik. Akten der Internationalen Tagung zur Phraseologie (Basel, 2004)*, 115–128. Baltmannsweiler: Schneider Verlag.

Filatkina, N. (2007). Formelhafte Sprache und Traditionen des Formulierens (HiFoS): Vorstellung eines Projekts zur historischen formelhaften Sprache. *Sprachwissenschaft* 32(2): 217–242.

Fleischer, W. (1997). *Phraseologie der deutschen Gegenwartssprache.* Tübingen: Max Niemeyer. (1st edn, 1982).

Gamkrelidze, Th. V. & V. V. Ivanov (1995). *Indo-European and the Indo-Europeans. A Reconstruction and Historical Analysis of a Proto-Language and Proto-Culture,* Vol. 1: *The Text*. Berlin: Mouton de Gruyter.

Gréciano, G. (2002). Semantik und Herkunftserklärungen von Phraseologismen. In Cruse, D. A., F. Hundsnurscher & D. Werkmüller (eds.) *Lexicology. An International Handbook on the Nature and Structure of Words and Vocabularies*. Vol. 1, 433–441. Berlin: de Gruyter.

Hessky, R. (1987). *Phraseologie. Linguistische Grundlagen und kontrastives Modell deutsch-ungarisch.* Tübingen: Max Niemeyer.

Hessky, R. (1989). Sprach- und kulturspezifische Züge phraseologischer Vergleiche. In Gréciano, G. (ed.) *Europhras 88: Phraséologie contrastive. Actes du Colloque International Klingenthal-Strasbourg, 12–16 mai 1988*, 165–204. Strasbourg: USHS.

Iñesta Mena, E. M. & A. Pamies Bertrán (2002). *Fraseología y metáfora: Aspectos tipológicos y cognitivos*. Granada: Granada Lingvistica.

Jarrett, D. (1984). Pragmatic coherence in an oral formulaic tradition: I can read your letters/sure can't read your mind. In Tannen, D. (ed.) *Coherence in Spoken and Written Discourse*, 155–171. Norwood, NJ: Ablex.

Kleiber, G. (1989). Sur la définition du proverbe. In Gréciano, G. (ed.) *Europhras 88: Phraséologie contrastive. Actes du Colloque International Klingenthal-Strasbourg, 12–16 mai 1988*, 233–252. Strasbourg: USHS.

Kleiber, G. (2005). Petite sémantique des proverbes avec une vue spéciale sur leur statut de dénomination. In Almela, R., E. Ramón Trives & G. Wotjak (eds.) *Fraseología contrastiva con ejemplos tomados del alemán, español, francés e italiano*, 19–38. Murcia: Universidad de Murcia.

Knappe, G. (2004). *Idioms and Fixed Expressions in English Language Study before 1800. A Contribution to English Historical Phraseology.* Frankfurt: Peter Lang.

Kroeber, L. A. & C. Kluckhohn (1952). *Culture: A Critical Review of Concepts and Definitions* [Papers of the Peabody Museum of Archaeology and Ethnology, Vol. 47(1)] (reprint). Cambridge, MA: Peabody Museum Press.

Kuiper, K. (1991). The evolution of an oral tradition: Racecalling in Canterbury, New Zealand. *Oral Tradition* 6: 19–34.

Kuiper, K. (1996). *Smooth Talkers. The Linguistic Performance of Auctioneers and Sportscasters.* Mahwah, NJ: Lawrence Erlbaum.

Kuiper, K. & D. Tan (1989). Cultural congruence and conflict. Acquiring formulae in second language learning. In Garcia, O. & R. Otheguy (eds.) *English across Cultures and Cultures across English*, 281–304. Berlin: Mouton de Gruyter.

Lakoff, G. (1987). *Women, Fire, and Dangerous Things: What Categories Reveal about the Mind.* Chicago, IL: The University of Chicago Press.

Lakoff, G. & Z. Kövecses (1987). The cognitive model of anger inherent in American English. In Holland, D. & N. Quinn (eds.) *Cultural Models in Language and Thought*, 95–221. Cambridge: Cambridge University Press.

Langacker, R. W. (1994). Culture, cognition, and grammar. In Pütz, M. (ed.) *Language Contact and Language Conflict*, 25–53. Amsterdam: John Benjamins.

Luque Durán, J. D. & F. J. Manjón Pozas (1999). Phraseology and lexical productivity: Bullfighting as a metaphoric referent in the Spanish language. *Acta linguistica. Ekonomická fakulta Univerzity Mateja Bela v Banskej Bistrici* 3: 33–46.

Maalej, Z. (2004). Figurative language on anger. Expressions in Tunisian Arabic: An extended view of embodiment. *Metaphor and Symbol* 19(1): 51–75.

Melchers, G. (1997). As smart as a carrot: Proverbial similes in English dialects. In Ramisch, H. & K. Wynne (eds.) *Language in Time and Space. Studies in Honour of Wolfgang Viereck on the Occasion of his 60th Birthday*, 164–173. Stuttgart: Franz Steiner.

Mieder, W. (1982–2001). *International Proverb Scholarship: An Annotated Bibliography.* New York, NY: Garland (1982, 1990 and 1991; Supplement II 1993), New York: Peter Lang (Supplement III 2001).

Mieder, W. (2004). *Proverbs. A Handbook*. London: Greenwood Press.

Mieder, W. (2007). Proverbs as cultural units or items of folklore. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseology. An International Handbook of Contemporary Research*, Vol. 1, 394–414. Berlin: de Gruyter.

Mokienko, V. M. (1973). Istoričeskaja frazeologija: ėtnografija ili linguistika? *Voprosy jazykoznanija* 2: 21–34.

Mokienko, V. M. (1989). *Slavjanskaja frazeologija*, 2nd edn. Moscow: Vysšaja škola.

Pawley, A. (2007). Developments in the study of formulaic language since 1970: A personal view. In Skandera (ed.) *Phraseology and Culture in English*, 3–45. Berlin: Mouton.

Piirainen, E. (1999). Dat sitt in de Pöste. Das niederdeutsche Hallenhaus als bildliche Domäne westmünsterländischer Idiome. Kognitive und kultursemiotische Aspekte der dialektalen Phraseologie. *Jahrbuch des Vereins für niederdeutsche Sprachforschung* 122: 115–142.

Piirainen, E. (2000). *Phraseologie der westmünsterländischen Mundart.* Vol. 1: *Semantische, kulturelle und pragmatische Aspekte dialektaler Phraseologismen.* Vol. 2: *Lexikon der westmünsterländischen Redensarten.* Baltmannsweiler: Schneider.

Piirainen, E. (2002). Er zahlt keine Steuern mehr. Phraseologismen für 'sterben' in den deutschen Umgangssprachen. In Piirainen, E. & I. T. Piirainen (eds.) *Phraseologie in Raum und Zeit. Akten der 10. Tagung des Westfälischen Arbeitskreises "Phraseologie/Parömiologie" Münster 2001*, 213–238. Baltmannsweiler: Schneider.

Piirainen, E. (2004). Cognitive, cultural and pragmatic aspects of dialectal phraseology – exemplified by the Low German dialect "Westmünsterländisch". *Dialectologia et Geolinguistica* 12: 46–67.

Portis-Winner, I. (1994). *Semiotics of Culture. "The Strange Intruder".* Bochum: Brockmeyer.

Rajchštejn, A. D. (1980). *Sopostavitel'nyj analiz nemeckoj i russkoj frazeologii*. Moscow: Vysšaja škola.

Rakusan, J. (2000). Language constructs of animals and men in two cultures: Czech vs. English similes with animals in comparatum. *Multilingua. Journal of Cross-Cultural and Interlanguage Communication* 19: 265–279.

Röhrich, L. (1995 [1977]). *Das große Lexikon der sprichwörtlichen Redensarten*, 5 Vols. Freiburg: Herder.

Rosch, E. (1975). Cognitive representations of semantic categories. *Journal of Experimental Psychology: General* 104: 192–233.

Sabban, A. (2004). Wege zu einer Bestimmung der Kulturspezifik sprachlicher Formeln. In Palm-Meister, C. (ed.) *EUROPHRAS 2000. Internationale Tagung zur Phraseologie vom 15.–18. Juni 2000 in Aske/Schweden*, 401–416. Tübingen: Stauffenburg.

Sabban, A. (2007). Culture-boundness and problems of cross-cultural phraseology. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseology. An International Handbook of Contemporary Research*, Vol. 1, 590–605. Berlin: de Gruyter.

Schmidt-Wiegand, R. (1991). Mit Hand und Mund. Sprachgebärden aus dem mittelalterlichen Rechtsleben. *Frühmittelalterliche Studien* 25: 283–299.

Schmidt-Wiegand, R. (1993). Sprichwörter und Redensarten aus dem Bereich des Rechts. In Buchholz, S., P. Mikat & D. Werkmüller (eds.) *Überlieferung, Bewahrung und Gestaltung in der rechtsgeschichtlichen Forschung. Festschrift für Ekkehard Kaufmann*, 277–296. Paderborn: Schöningh.

Schmidt-Wiegand, R. (ed.) (2002). *Deutsche Rechtsregeln und Rechtssprichwörter: ein Lexikon.* München: Beck.

Schmitt, N. (ed.) (2004). *Formulaic Sequences: Acquisition, Processing and Use.* Amsterdam: John Benjamins.

Schowe, U. (1993). *Mit Haut und Haar. Idiomatisierungsprozesse bei sprichwörtlichen Redensarten aus dem mittelalterlichen Strafrecht.* Frankfurt: Peter Lang.

Segura García, B. (1997). Kulturspezifische Phraseologismen in literarischen Texten und ihre Interferenzen beim Übersetzen vom Spanischen ins Deutsche. In Sabban, A. (ed.) *Phraseme im Text: Beiträge aus romanistischer Sicht*, 221–236. Bochum: Brockmeyer.

Shore, B. (1996). *Culture in Mind: Cognition, Culture, and the Problem of Meaning.* New York, NY: Oxford University Press.

Sinclair, J. (1991). *Corpus, Concordance, Collocation.* Oxford: Oxford University Press.

Skandera, P. (ed.) (2007). *Phraseology and Culture in English*. Berlin: Mouton.

Skog-Södersved, M. & A. Stedje (1997). *Der Arbeit aus dem Wege gehen.* Zur Beschreibung und Bewertung der Faulheit im Deutschen und Schwedischen. In Skog-Södersved, M. (ed.) *Ethische Konzepte und mentale Kulturen* 1. *Auf der Suche nach ethischen Konzepten in Wortschatz und Phraseologie*. Umeå: Umeå University.

Spalding, K. (1959ff.). *An Historical Dictionary of German Figurative Usage*. Oxford: Blackwell.

Stolz, B. A. & R. S. Shannon (eds.) (1976). *Oral Literature and the Formula.* Ann Arbor, MI: Center for the Coordination of Ancient and Modern Studies.

Telija [Teliya], V. N. (1996). *Russkaja frazeologija. Semantičeskij, pragmatičeskij i lingvokul'turologičeskij aspekty*. Moscow: Škola "Jazyki russkoj kul'tury".

Telija [Teliya], V. N. (1998). Phraseological entities as a language of culture (methodological aspects). In Eismann, W. (ed.) *EUROPHAS 95. Europäische Phraseologie im Vergleich: Gemeinsames Erbe und kulturelle Vielfalt*, 783–794. Bochum: Brockmeyer.

Telija [Teliya], V. N., N. Bragina, E. Oparina & I. Sandomirskaya (1998). Phraseology as a language of culture: Its role in the representation of a collective memory. In Cowie, A. P. (ed.) *Phraseology. Theory, Analysis, and Applications*, 55–75. Oxford: Oxford University Press.

Tolstoj N. I. (1973). O rekonstrukcii praslavjanskoj frazeologii. In *Slavjanskoe jazykoznanie. VIII Meždunarodnyj s'ezd slavistov. Doklady sovetskoi delegacii*, 272–294. Moscow: Nauka.

Tylor, E. B. (1871). *Primitive Culture. Researches into the Development of Mythology, Philosophy, Religion, Language, Art, and Custom* (2 Vols.), Vol. 1, 4th edn. London: John Murray.

Wanzeck, C. (2003). *Zur Etymologie lexikalisierter Farbverbindungen. Untersuchungen anhand der Farben Rot, Gelb, Grün und Blau.* Amsterdam: Rodopi.

White, G. M. (1987). Proverbs and cultural models. An American psychology of problem solving. In Holland, D. & N. Quinn (eds.) *Cultural Models in Language and Thought*, 151–172. Cambridge: Cambridge University Press.

Wierzbicka, A. (1992). *Semantics, Culture, and Cognition. Universal Human Concepts in Culture-Specific Configurations.* Oxford: Oxford University Press.

Wierzbicka, A. (1996). *Semantics: Primes and Universals.* Oxford: Oxford University Press.

Wierzbicka, A. (1997). *Understanding Cultures through Their Key Words. English, Russian, Polish, German, and Japanese*. Oxford: Oxford University Press.

Wierzbicka, A. (2007). Reasonably well: Semantic metalanguage as a tool for the study of phraseology and its cultural underpinnings. In Skandera, P. (ed.) *Phraseology and Culture in English*, 49–78. Berlin: Mouton.

Wotjak, B. (2005). Fórmulas rutinarias en los diccionarios didácticos. In Luque Durán, J. D. & A. Pamies Bertrán (eds.) *La creatividad en el lenguaje: colocaciones idiomáticas y fraselogía*, 331–349. Granada: Granada Lingvistica.

Wray, A. (2002). *Formulaic Language and the Lexicon*. Cambridge: Cambridge University Press.

Yangüela, M. T. (1998). Consideraciones lingüístico-pragmáticas acerca del trasvase de las expresiones fijas del lenguaje taurino al código general. In Wotjak, G. (ed.) *Estudios de fraseología y fraseografía del español actual*, 129–153. Vervuert: Iberoamericana.

Ying, C. & P. Erh-li (1996). *Guanyongyu. Idiome des modernen Chinesisch. Eine Lehr- und Lernhilfe.* Heidelberg: Julius Groos.

Yu, N. (2003). Metaphor, body, and culture: The Chinese understanding of *gallbladder* and *courage*. *Metaphor and Symbol* 18: 13–31.

Critical observations on the culture-boundness of phraseology

Annette Sabban

This chapter argues for a more rigorous description of the culture-boundness of phrasemes, making a number of terminological proposals; also, for a clear distinction to be made between linguistic analyses and categorizations on the one hand, and claims to modes of thinking of the present-day speaker on the other, the latter generally being seen as belonging to the cognitive dimension of culture. A parallelism with two fundamental modes of existence of a group's 'cultural memory' is identified.

1. Introduction

It is generally agreed that multi-word units, or phrasemes, may be 'culturally marked' to a significant extent, more so than single word units. There have been quite a number of studies, especially within the last fifteen years, investigating the relationship between phrasemes and culture. These differ as to their theoretical background, the languages and cultures involved, or the types of phrasemes investigated.[1] In the following, I shall address some of the questions that arise from a consideration of these studies. The questions relate to some of the terminology used, and to some proposals for the interpretation of phrasemes in terms of culture.

2. Terminology

2.1 The terms *culture* and *culture-specific*

A first point concerns the terms *culture* and *culture-specific*. It appears that these are not always used in a consistent way. It is not so much that there are different concepts of culture involved, but rather that there is a more general kind of inconsistency, or ambiguity, in the application of the terms. This ambiguity parallels general usage, where there are essentially two, metonymically related, meanings of the word *culture*. Thus, in

1. For an account of the state of the art see Sabban (2007a) and Piirainen this volume.

(1) behind the headlines lurks a grimy *subculture* of publicity agents (The Times April 7, 2004: 16),

the word *(sub-)culture* means a group of people. By contrast, in

(2) a *culture* of accepted violence [...] in all areas of society (The Times April 7, 2004: 8)

culture refers to the rules of behaviour and underlying value orientations (of a particular group). These are important ingredients of culture according to, for example, the classical anthropological notion of culture as the "shared way of life of a group of people", which also includes shared patterns of behaviour (cf. Berry et al. 2002: 225ff.).

In phraseology, the terms *culture* and *culture-specific* are sometimes used only in the sense of group, just as in (1). This is the case, for example, when an Australian English idiom such as *to have a kangaroo loose in the top paddock*, i.e. 'to be crazy', is labelled culture-specific because its image component refers to the kangaroo, which is part of the "unique cultural setting" of Australians (Gläser 1999: 167).[2] 'Cultural setting' in this case refers to the natural habitat of a particular group, providing the context for human development and action, including characteristic plants, animals, and other objects. It does not refer to some dimension of culture, such as norms and patterns of behaviour, as in (2) above.

Similarly, gestures accompanying a particular phraseme have been labelled 'culture-specific' (Baur et al. 1998: 6ff.), but without any attempt to define the notion of culture or to specify how culture can be said to manifest itself in the way the gestures are carried out.

When used as a synonym of *group*, as in (1), the fact of being specific to a group, or culture, is simply posited. The task for the linguist boils down to delimiting the group which uses certain phrasemes – which is a classical procedure in the study of linguistic varieties. By contrast, any investigation of the relation between language and culture requires a closer definition of the term *culture*, for example by referring to a concept of culture or by specifying a relevant cultural dimension.[3] This corresponds to the need for operationalizing the term (cf. Dobrovol'skij 1999; Sabban 2004).

2.2 The terms *culture-specific* and *culturally bound*

A second point concerns the terms *culture-specific* and *culturally bound* (or *culture bound*). These are both used, though often only as stylistic variants, the term *culture-specific* being more widespread. Still, it seems that it would be preferable to use the less

2. This idiom is a stock example in descriptions of Australian English, both in linguistic accounts (e.g. Gläser 1999) as well as writings of a more general nature, such as M. Reif, "No *worries, mate* – An absolute beginner's guide to the Aussie Lingo", http://www.tuchemnitz.de/phil/leo/rahmen.php?seite=r_kult/reif_aussie.php.

3. For a comprehensive account of approaches to culture see for example Berry et al. (2002).

frequent of the two, i.e. *culturally bound*, and for the following reasons: firstly, it tends to raise the question in which way a particular word or concept is culturally bound, and this lends weight to the need to define the term *culture* more closely. Secondly, its use presupposes that language and culture can be considered separately, although they are obviously closely interrelated. For the same reason, the term seems preferable to *culturally marked,* since this latter term seems to imply a one-sided view of language as a dependent variable marked by culture as an independent variable. Finally, the term *culturally bound* is less exclusive than the term *culture-specific* in that it allows for similarities in a comparison of phraseologies, and this in turn allows for a consideration of cultures being related, for example in terms of family resemblances.[4]

3. Some problems in the interpretation of phrasemes in terms of culture

Generally speaking, phrasemes may be considered culturally bound in different ways, depending on the following considerations:[5]

1. the type of phraseme and its associated dominant semiotic function, such as
   a. referential phrasemes, e.g. idioms *(to let off steam)*
   b. communicative phrasemes, e.g. routine formulae *(I am (so) sorry)*[6]
2. the semantic level under consideration, in particular
   a. the phraseological level or entire target domains
   b. the literal level or entire source domains
   c. concepts and types of motivation
3. the definition of culture or a particular dimension of culture, such as
   a. modes of conceptualizing and evaluating
   b. modes of interaction and social behaviour and underlying value orientations
   c. a culture's 'texts', or cultural codes, as in the framework of cultural semiotics[7]
4. the synchronic or diachronic orientation.

---

4. The nominal form *culture-boundness* may not be aesthetically pleasing to all native speakers of English. The term is not particularly frequent according to Google (47 occurrences in October 2005, some of which are in articles written by non-native speakers of English or in translations). In phraseology, the term is encountered e.g. in Bragina (1996: 199) or in Gläser (1999: 156). The adjectival phrase *culturally bound*, however, is widely used (more than 13,000 occurrences according to Google in September 2005).

5. For more details see Sabban (2007a).

6. For a classification of phrasemes according to major semiotic functions see Burger (2003: 36).

7. Cf. Dobrovol'skij & Piirainen (1997, 1998, 1999).

Phraseology research has favoured the study of idioms, which are an important subgroup of referential phrasemes (1a). They serve to categorize and to evaluate the world. It is generally agreed that modes of conceptualizing and evaluating as offered by language, and more specifically by idioms, may contribute to the way in which the world is experienced. Hence, referential phrasemes are closely associated with the cognitive dimension of culture and with language as an instrument of thought (see 3a). As such, they may ultimately contribute to the expression as well as the formation of the world-view of the group in question.

It is different with communicative phrasemes (1b), which are closely associated with language as an instrument of action (as in linguistic pragmatics). A major function of an important subgroup of communicative phrasemes is to perform speech acts; they are therefore an immediate part of behavioural and interactive routines, i.e., of modes of social behaviour, including interactional (or: communicative) styles (see 3b). These, too, are generally considered an important feature of a culture (cf. Berry et al. 2002: 52ff.).

With very few exceptions (e.g. Kuiper & Tan Gek Lin 1989), the potential of communicative phrasemes for an explanation in cultural terms has largely passed unnoticed by phraseologists proper. Instead, these phrasemes have more often been studied in the context of other subdisciplines, in particular contrastive pragmatics and discourse analysis. The phrasemes have been set against the background of preferred modes of action and interaction, such as task- versus person-orientedness in conversation, or the degree of routine made use of in typical exchanges, all of which may themselves be related to value orientations that are characteristic of a group, or culture.[8]

The remainder of this paper will be focussed on idioms. Investigating their culture-boundness has typically involved establishing explicit links between the literal level or entire source domains on the one hand, and aspects of culture on the other. An example might be a national sport like bullfighting, a source domain for many Spanish idioms, as in

(3) *mirar los toros desde la barrera,* lit. 'watch the bulls from [behind] the barriers (in the arena)', i.e. watch without intervening (further examples in Tecedor 1998),

or judicial practices and ritual gestures in courtroom proceedings of the Middle Ages, which originally gave rise to a number of German idioms still in use today, such as

(4) *stehenden Fußes*, lit. 'standing on one's foot' (requirement of remaining upright while protesting against a sentence which had just been pronounced); i.e. immediately (further examples in Schmidt-Wiegand 1991).

8. Studies of this kind include: Blum-Kulka et al. (1989); House (1998); Suszcyńska (1999).

It has also involved looking for patterns such as recurring concepts, or more generally, metaphoric models, and interpreting these patterns in terms of culture (e.g. Piirainen 1999 on concepts of a HOUSE exploited in idioms of two Germanic varieties, or Filatkina 2005: 249ff. on wine-growing etc. in Lëtzebuergesch).

In the following, a number of interpretative proposals from different studies will be considered, with respect to the assumptions underlying them, leading to a number of conclusions of a more general nature.

3.1 Cultural knowledge and speakers' motivation of idiomatic meaning

A first problem can be illustrated by the following idiom and the analysis provided for it:

(5) *to make a silk purse out of a sow's ear,* i.e. to succeed in making sth good out of material that does not seem very good at all.[9]

The idiom is discussed by Dobrovol'skij & Piirainen (2005: 40), whose approach is synchronic and cognitive. Its meaning is said to be "based on" some "shared cultural knowledge", namely "the knowledge of SILK as something valuable and of a sow (or PIG) as something inferior or dirty" (ibid.). This piece of knowledge is assumed to be the basis for the various speakers' subjective account of the meaning of the idiom, this being part of the speakers' world-view and therefore of the cognitive dimension of culture (see above, 3a). This explanation, though rather convincing at first glance, raises a few questions.

First of all, the term *knowledge*, widely used in cognitive studies, is unfortunate for several reasons. It implies factivity – or knowing that something is the case – and could misleadingly be understood as describing a general truth, in this case that silk *is* something valuable and that pigs *are* inferior or dirty. But what is at stake is a particular value judgement, bound to a particular culture and to a particular time. Furthermore, cultural knowledge is understood by the authors in a rather specific sense: it is knowledge of the way the sow or pig is conceived of in other cultural codes, such as fables (see above, 3c). This way of operationalizing the term yields numerous insights into patterns of conceptualization across different cultural codes. However, if knowledge of such correspondences can justly be assumed to be the basis for the speakers' account of meaning – for this is what the authors set out to describe –, this presupposes a high degree of cultural awareness on the part of the ordinary speakers of the language. Statements of this sort would seem to call for empirical investigation – in supplement to the correspondences established, by the linguist, between different kinds of systems.

---

9. Oxford Advanced Learner's Dictionary of Current English, [6]2000, s. v. *silk*. Similarly: 'to make something good out of something that is bad quality', Longman Dictionary of Contemporary English, 2003, s.v. *silk*. The expression very often occurs in the form of a proverb: *You can't make* ..., see the examples given in the Oxford English Dictionary, [2]1989, s. v. *silk*.

As pointed out by Geeraerts & Grondelaers (1995), it is important to distinguish, from a methodological point of view, between patterns of concepts in language on the one hand, and present-day ways of conceptualizing by speakers of that language on the other, or, more generally speaking, between conventionalized ways of talking and ways of thinking. Due to the historical nature of language, the two need not coincide.[10] Finally, it should also be borne in mind that knowledge, at least in the way it is normally understood outside the framework of cultural semiotics, is not limited to views expressed in language and other cultural codes. According to other sources of knowledge,[11] the sow or pig is not inferior or dirty at all.

A second point concerns the adequacy of the account itself. Does one really have to resort to the idea of the dirty pig to explain the meaning of this idiom? Why should such a value judgement necessarily apply in this particular case? Alternative interpretations spring to mind and are indeed offered by native speakers. One of these is based on a different kind of knowledge; it also involves considering the *sow's ear* as a whole and in the context of the full phrase, instead of isolating it. In the past, the *sow's ear* represented an item of at least some value; it provided a small amount of meat which peasants could afford – in contrast to the silk purse, which was an attribute of nobility, the rich and the courtly world. In this account, it is not at all necessary to invoke the idea of the sow or pig being dirty or an inferior creature.

This explanation of the phraseme's motivational base is etymological in nature.[12] From a purely synchronic standpoint, it would normally be rejected, and it would be argued that etymology is stored in the archives, so to speak, in order to keep it accessible, but that it would not be relevant to speakers' accounts of the motivation of idioms. However, following Assmann (1988: 13), conserving cultural knowledge and keeping it accessible is only one of two fundamental modes of existence of a society's

---

10. In their highly instructive paper, Geeraerts & Grondelaers (1995) discuss the relics of the old medical doctrine of the four humors in present-day vocabularies of some European languages, and the methodological conclusions to be drawn from this kind of lexical legacy. In doing so, they question a number of assumptions made in competing accounts written within the cognitive linguistics paradigm.

11. See for example the observations by ethologists and zoologists, presented in the context of an exhibition on the natural and cultural history of the pig at the Westphalian Museum of Natural History in Münster 2003.

12. The etymological explanation offered is based on general knowledge of cultural history and as such is only highly plausible. There is no way of checking on its correctness with the help of the works of reference available. It should be stressed that is not the purpose of this paper to offer a "correct" etymological explanation, but rather to point out that there are a number of different explanations that can be envisaged, among them an account in terms of cultural history. Indeed, there are even more possibilities: It may suffice to interpret the opposition between a *silk purse* and a *sow's ear* as the opposition between items of high and low value, and this may, in turn, be based on considerations of a *purse* being more valuable that an animal's ear because it is man-made as opposed to natural, thus reflecting a widespread value judgement.

collective memory, or more strictly, cultural memory. What matters here is that some such knowledge is actually *accessed:* the type of explanation offered is an instance of interpretation, or reconstruction, from the (changing) present perspective, based on some knowledge of cultural history, or, more precisely, on what speakers consider to be relevant facts of cultural history. Accessing knowledge of the past corresponds to the second fundamental mode of existence, or mode of dealing with, a society's cultural memory (ibid.).

Such knowledge about cultural history – which is not necessarily conserved in cultural texts proper – should not be neglected, even in a synchronic account which sets out to describe speakers' subjective motivations of idioms. For the interpretation is based on knowledge of cultural history which is still available to these speakers, or which they think is available to them. Availability and accessibility of knowledge is a culturally relevant fact in its own right. It is an important prerequisite for transmitting cultural knowledge. From this perspective, idioms are relevant to culture in that they potentially evoke and transmit a piece of cultural knowledge.

More generally speaking, it follows from these considerations that there may be different explanations that are all plausible and culturally relevant in different ways. One need not have recourse to the idea of a dirty or inferior pig in order to provide the idiom with a subjective motivation. It is therefore important to make a clear distinction between (1) correspondences between concepts in language and in other cultural codes, on the one hand, and (2) speakers' synchronic motivation of idiomatic expressions on the other.

Emerging patterns of concepts across cultural codes should be considered insights in their own right. Speakers' motivation, by contrast, should be established on independent grounds; they cannot be immediately inferred from the concepts in language[13] or from conceptual patterns across different codes. If one identifies the one too closely with the other, this amounts to asserting psychological reality on the sole basis of a description of systemic correspondences, set up by the investigator. Furthermore, the procedure of establishing correspondences between concepts in language and other 'cultural codes' may involve the risk of looking for supportive evidence only.[14]

3.2 Conceptual metaphors: cultural relevance

A second question is that of the cultural relevance of conceptual metaphors, or *metaphoric* and *metonymic models.*[15] This somewhat complex term seems preferable to the more widely used *conceptual metaphor* for two reasons: firstly, it explicitly takes into

13. See also Geeraerts & Grondelaers (1995: 174).

14. See the cogent arguments presented by Keysar & Bly (1999), strongly summed up in the statement that "idioms cannot provide evidence for conceptual structures because they cannot provide negative evidence" (p. 1565).

15. The single term *metaphoric model* is used for example in Burger (2003: 85).

account metonymic models, which are often overlooked or sometimes erroneously labelled as metaphoric; secondly, the term is more likely, at least in principle, to preclude implications as to the cognitive status (psychological reality) of the models proposed prior to this having been demonstrated.

It is often assumed in cognitive studies that the repeated exploitation of a particular source domain in the conceptualization of similar target concepts may be an indicator of cultural differences and cognitive differences between cultures (cf. Section 2.1). Such a line of argument can be found, for example, in Liu's comparative study (2002) of metaphoric models (viz. dominant metaphors) in American English and Chinese. According to this study, idioms in American English draw heavily on sport as a source domain, with baseball and American football figuring prominently, as in

(6) AmE: *to hit a home run/ to hit a homer*, i.e. to achieve a great success

(7) AmE: *to touch base with s. o.,* i.e. to consult with someone (Liu 2002: 124ff.)

Such idioms are shown to be particularly frequent in texts on governmental affairs. By contrast, in Chinese, metaphors and idiomatic expressions in comparable domains of discourse typically exploit the family as a source domain (Liu 2002: 27ff., 39ff.). These observations are immediately interpreted in cognitive terms: it is assumed that Americans conceive of government as a competitive affair, whereas Chinese conceive of it as a family affair. More precisely still, these are assumed to be patterns of thought of synchronic relevance, a mirror of the speakers' current consciousness, a living ingredient in a world-view, or, as summed up by the author, "a collective conceptual system shared by most speakers" (Liu 2002: 19). Here again, conventionalized ways of talking are identified with synchronic ways of thinking, although with respect to the higher level of dominant metaphors (or metaphoric models) abstracted from the data. The analysis is supported by considerations of the comparative importance of the family in China.

Liu's conclusions fit in well with the results of a linguistic study from a different methodological perspective, that of Kuiper & Tan Gek Lin (1989), who show that certain communicative formulae in Singapore English, as used by Chinese, reflect a high esteem for the family. One should however draw attention to the lack of a clear distinction made between linguistic concepts and the patterns of thought in the actual formulation. Rather than discuss the details of Liu's analysis, I would like to focus on two general considerations.

The first point concerns a prior stage in the analysis, i.e. the adequacy of a metaphoric (or metonymic) model as an account of an idiom's meaning, or its linguistic relevance proper. Establishing a metaphoric model is an instance of categorization. Very often, this is done in a way which, in the last resort, resembles that of classical componential analysis, the validity of which has been questioned in other areas of linguistic description, for example in lexical semantics. The procedure consists of selecting a part of the motivating link, a metaphoric component, or semantic feature in the broadened sense. This is assumed to capture an essential part of the meaning and to be of similar importance, or 'weight', for all the idioms assigned to the same cate-

gory. This may, however, run counter to usage, and the model might therefore turn out not to be descriptively adequate, as may be illustrated by an example from German:

(8) Der Eurokurs *macht eine Berg- und Talfahrt,* lit. 'the course of the Euro *is on an up- and downhill ride*', i.e. on a roller coaster.

This idiom has been assigned to the well-known metaphoric model MORE/MUCH IS UP, LESS/FEW IS DOWN (Dobrovol'skij & Piirainen 2005: 93). If, however, one looks at corpus data – a total of 87 occurrences of the idiom in the COSMAS II corpus of the German language[16] –, the following picture emerges. The idea of a (figurative) up-and-down movement is not the only idea relevant to its use. In fact, it is closely tied to either

– the idea of repetitive changes of such a kind, as in (9), explicitly signalled by the plural of the complementing nouns, as in (10), or to be inferred from the time span given, as in (11):

(9) bei dieser Berg-und-Tal-Fahrt [...]: *einen Tag* bin ich verrückt, *den nächsten* völlig normal
lit. 'this up- and downhill ride [...] *one day* I am crazy, *the other* completely normal'

(10) unsere Berg-und-Tal-Fahrt mit Heimsieg*en* und Auswärtsniederlag*en*
lit. 'our up- and downhill ride with home win*s* and away defeat*s*'

(11) Harald Juhnke ... über *sein Leben* als Berg- und Talfahrt
lit. 'Harald Juhnke on *his life* as an up- and downhill ride'

– or the idea of rapid movements and unrest, which may turn hectic, as in (12) and (13):

(12) In der Liebe ist man zurzeit *eher* auf Berg-und-Tal-Fahrt, *als dass man zur Ruhe kommt.*
lit. 'Love at the moment is more of an up- and downhill ride *rather than something relaxed and restful.*'

(13) "Yo-Yo-Spiel", "Achterbahn", "Berg-und-Tal-Fahrt" nannten Kommentatoren die *hektischen* Kursbewegung*en* zu Recht.
lit. "'Like a yo-yo", a "roller coaster", an "up- and downhill ride"' are the labels commentators aptly gave to the *hectic* price movement*s* in the stock market.'

None of these ideas, which in some instances may even be more prominent than the idea of metaphoric ups and downs, are captured by ascription of the idiom to the model MORE/MUCH IS UP, LESS/FEW IS DOWN. The categorization therefore turns out

16. COSMAS II (= Corpus Search, Management and Analysis System, second version), Institute of the German Language, Mannheim (http://www.ids-mannheim.de/), consulted in August 2005.

to be selective and somewhat arbitrary. In the final analysis, it may even be regarded as another instance of the pointed criticism voiced by Keysar & Glucksberg (1992: 652) and taken up again in Keysar & Bly (1999: 1563): "although many metaphorical expressions appear to instantiate a conceptual mapping, this mapping does not contribute to the expression's meaning".[17] The procedure may be justified as a way of organizing data, but again it is highly questionable whether one may claim cognitive relevance and psychological reality for such models without providing independent evidence.

The second point concerns the question as to whether we can justly assume that metaphoric and metonymic models correspond to a living ingredient of a world-view. This applies equally to the image component of a single idiom. The focus in this case is not so much on the claim of psychological reality as such, but on the synchronic relevance of such a claim. This question is closely related to the more general question of whether or not to separate diachronic and synchronic considerations in studies related to cultural matters. This question has already been touched on in Section 2.1., albeit from a different angle.

It seems that, in the case of idioms, claiming psychological reality of synchronic relevance is highly problematical. First of all, there is generally a time-lag between the time of creation and the time of use; people may be more aware of an idiom's image base at the moment when it comes into being, thus making the metaphor psychologically more real at that particular point in time. The metalinguistic comment in the following example from the Steam Age, when many idioms based on the imagery of steam came into being, may be taken as an indication of such a heightened awareness.

(14) 1857 (by Hughes): [...] then joining in some chorus of merry voices; in fact, *blowing off his steam*, *as we should now call it.* (The Oxford English Dictionary ^{2}1989, s. v. *steam*)

Secondly, with regard to idioms which are well established in usage, it is a commonplace of phraseology that they may be used holistically, without the user understanding the image or even being aware that there is an image involved. Idioms, as well as other figurative expressions, may not be semantically transparent to speakers, and even if they are transparent, using them does not require that the speaker actually subscribes to the views or beliefs on which the idioms are based. Generally speaking, the vocabulary in any language bears the impact of former ways of thinking (as in the often quoted example of the *rising sun*), and the words and expressions continue to be used even though the beliefs and assumptions which gave rise to them are no longer part of the speakers' world-view.

This is in no way incompatible with the fact that the literal base may be activated and exploited in many different ways and for different purposes in particular

17. Keysar & Bly (1999: 1563) give the following example: "[...] the metaphor *our marriage is nothing but a filing cabinet* seems to be an instantiation of LOVE IS A CONTAINER. Yet, this mapping is irrelevant to the actual meaning, which is an expression of the lack of spontaneity of emotional involvement in the marriage."

circumstances and modes of usage: the literal level of idioms may be activated in certain circumstances, but need not always be. This broadly parallels the discussion of the two fundamental modes of existence of a group's cultural memory referred to in Section 2.1. The first mode, i.e. storage, is like a linguistic repertoire of signs which is available to the speakers and which includes the modes of construction of semiotically complex signs such as idioms. The second mode, i.e. actually accessing this repertoire from the present perspective, is like using idioms in texts. More particularly, accessing their mode of construction corresponds to exploiting the literal base for the purposes of construction of that text by various processes of modification, frequently discussed in the literature,[18] thus allowing both for continuation and re-interpretation of an idiom's literal base.

4. Conclusions

This chapter has argued for a more rigorous description of the culture-boundness of phrasemes, making a number of terminological proposals. In addition, it has argued for a clear distinction to be made between linguistic analyses and categorizations, on the one hand, and claims to modes of thinking of the present-day speaker, on the other. A parallelism with two fundamental modes of existence of a group's cultural memory was postulated. This amounts to distinguishing between concepts in language (which are the result of modes of thinking that need no longer be relevant to the users of the language) and (current) concepts of thought.

Concepts in language (or linguistic concepts proper) are stored in the linguistic signs of a speech community. Current conceptual structures and modes of thinking need not coincide with them. It is hardly surprising that the investigation of these two kinds of concepts requires different and independent methods and approaches. In particular, current ways of thinking cannot be immediately inferred from the concepts accounting for the make-up of linguistic signs.

Nevertheless, after arguing for a separation of these two aspects, we would like to conclude with a suggestion that reintegrates them: Recurring patterns of conceptualization as manifest in linguistic expressions – as well as possibly other cultural codes – can be looked upon as traditions of conceptualization, these being themselves part of a culture and its legacy. Adopting this approach highlights the need to include a diachronic perspective in linguistic studies with a cultural orientation.

18. For an account of modifications of phrasemes and their relevance to the construction of texts see Sabban (2007b).

Acknowledgements

I would like to thank my colleague Francis Jarman for his constructive assistance in the final drafting of this paper.

References

Assmann, J. (1988). Kollektives Gedächtnis und kulturelle Identität. In Assmann, J. & T. Hölscher (eds.) *Kultur und Gedächtnis*, 9–19. Frankfurt: Suhrkamp.

Baur, M., R. S. Baur & C. Chlosta (1998). 'Ras le bol = Mir steht's bis hier?' Phraseogesten im Französischen und im Deutschen. In Hartmann, D. (ed.) *'Das geht auf keine Kuhhaut' – Arbeitsfelder der Phraseologie. Akten des Westfälischen Arbeitskreises Phraseologie/Parömiologie 1996*, 1–35. Bochum: Brockmeyer.

Berry, J. W., Y. H. Poortinga, M. H. Segall & P. R. Dasen (2002). *Cross-Cultural Psychology: Research and Applications*. Cambridge: Cambridge University Press.

Blum-Kulka, S., J. House & G. Kasper (eds.) (1989). *Cross-Cultural Pragmatics: Requests and Apologies*. Norwood, NJ: Ablex.

Bragina, N. (1996). Restricted collocations: cultural boundness. In Gellerstam, M., J. Järborg, S. G. Malmgren, K. Norén, L. Rogström & C. Röjder Papmehl (eds.) *Euralex '96 Proceedings I–II. Papers Submitted to the Seventh EURALEX International Congress on Lexicography in Göteborg 1996*, vol. 1, 199–207. Göteborg: Göteborg University.

Burger, H. (2003). *Phraseologie. Eine Einführung am Beispiel des Deutschen*, 2nd rev. edn. Berlin: Erich Schmidt.

Dobrovol'skij, D. (1999). Kulturelle Spezifik in der Phraseologie: Allgemeine Probleme und kontrastive Aspekte. In Sabban, A. (ed.) *Phraseologie und Übersetzen*, 41–58. Bielefeld: Aisthesis.

Dobrovol'skij, D. & E. Piirainen (1997). *Symbole in Sprache und Kultur. Studien zur Phraseologie aus kultursemiotischer Perspektive*. Bochum: Brockmeyer.

Dobrovol'skij, D. & E. Piirainen (1998). On symbols. Cognitive and cultural aspects of figurative language. *Lexicology* 4: 1–34.

Dobrovol'skij, D. & E. Piirainen (1999). 'Keep the wolf from the door'. Animal symbolism in language and culture. *Proverbium* 16: 61–93.

Dobrovol'skij, D. & E. Piirainen (2005). *Figurative Language: Cross-Cultural and Cross-Linguistic Perspectives*. Amsterdam: Elsevier.

Filatkina, N. (2005). *Phraseologie des Lëtzebuergeschen. Empirische Untersuchungen zu strukturellen, semantisch-pragmatischen und bildlichen Aspekten*. Heidelberg: Winter.

Geeraerts, D. & S. Grondelaers (1995). Looking back at anger: cultural traditions and metaphorical patterns. In Taylor, J. R. & R. E. MacLaury (eds.) *Language and the Cognitive Construal of the World*, 153–179. Berlin: de Gruyter.

Gläser, R. (1999). Indigenous idioms and phrases in Australian and New Zealand English. In Carls, U. & P. Lucko (eds.) *Form, Function and Variation in English. Studies in Honour of Klaus Hansen*, 155–168. Frankfurt: Peter Lang.

House, J. (1998). Kontrastive Pragmatik und interkulturelle Kompetenz im Fremdsprachenunterricht. In Börner, W. & K. Vogel (eds.) *Kontrast und Äquivalenz*, 162–189. Tübingen: Narr.

Keysar, B. & B. M. Bly (1999). Swimming against the current: Do idioms reflect conceptional structure? *Journal of Pragmatics* 31: 1559–1578.

Keysar, B. & S. Glucksberg (1992). Metaphor and communication. *Poetics Today* 13(4): 633–658.

Kuiper, K. & D. T. Gek Lin (1989). Cultural congruence and conflict in the acquisition of formulae in a second language. In García, O. & R. Otheguy (eds.) *English across Cultures: Cultures across English*, 281–304. Berlin: de Gruyter.

Liu, D. (2002). *Metaphor, Culture, and Worldview. The Case of American English and the Chinese Language.* Lanham, MD: University Press of America.

Piirainen, E. (1999). 'Dat sitt in de Pöste': Das niederdeutsche Hallenhaus als bildliche Domäne westmünsterländischer Idiome. *Jahrbuch des Vereins für niederdeutsche Sprachforschung* 122: 115–142.

Sabban, A. (2004). Wege zu einer Bestimmung der Kulturspezifik sprachlicher Formeln. In Palm-Meister, C. (ed.) *EUROPHRAS 2000*, 401–416. Tübingen: Stauffenburg.

Sabban, A. (2007a). Culture-boundness and problems of cross-cultural phraseology. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseologie/Phraseology. Ein internationales Handbuch zeitgenössischer Forschung / An International Handbook of Contemporary Research*, §50, 590–605. Berlin: de Gruyter.

Sabban, A. (2007b). Textbildende Potenzen von Phrasemen. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseologie/Phraseology. Ein internationales Handbuch zeitgenössischer Forschung / An International Handbook of Contemporary Research*, §18, 237–253. Berlin: de Gruyter.

Schmidt-Wiegand, R. (1991). Mit Hand und Mund. Sprachgebärden aus dem mittelalterlichen Rechtsleben. *Frühmittelalterliche Studien* 25: 283–299.

Suszcyńska, M. (1999). Apologizing in English, Polish and Hungarian: Different languages, different strategies. *Journal of Pragmatics* 31: 1053–1065.

Tecedor, Y. M. (1998). Consideraciones lingüístico-pragmáticas acerca del trasvase de las expresiones fijas del lenguaje taurino al código general. In Wotjak, G. (ed.) *Estudios de fraseología y fraseografía del español actual*, 129–154. Frankfurt: Vervuert.

Phraseology in a European framework

A cross-linguistic and cross-cultural research project on widespread idioms

Elisabeth Piirainen

Widespread idioms (WIs) are idioms that have the same lexical structure (or at least very similar structures) and the same figurative core meaning in various languages, including genetically unrelated and geographically distant languages. The project *Widespread Idioms in Europe and Beyond* to be outlined here is aimed at systematically investigating into potential WIs. Questionnaires that have been sent back for more than 70 languages are producing first results: different attempts at explaining cross-linguistic similarities of idioms (be it their genetic affiliation, spontaneous metaphorization in various languages, the common cultural European heritage or the increasing influence of English on other languages) can be qualified to a certain extent.

1. Preliminary remarks

Out of the two invariable dimensions, space and time, phraseology research has paid considerably more attention to the dimension of time than to that of space. Reliable phraseological accounts – both theoretical or practical, e.g. phraseographic, ones – usually incorporate the diachronic aspect of their subject matter by documenting how a phraseme becomes obsolete or, conversely, comes into vogue. Diatopic variation, however, is rarely taken into account. Phraseological research as related to historical developments is met with more interest than the areal aspects of this discipline. Often there is little awareness of the fact that phrasemes may be used in only a small, limited area, but also in large areas crossing individual languages.

Classical phraseology research usually starts from the concept of a homogeneous set of phrasemes, i.e. from equating (a national) language with 'space', namely the area where that individual language is spoken. This concept, however, is not tenable, for two reasons: on the one hand, it does not take account of the regional restrictions of many phrasemes within one individual language. This is the case e.g. with German, a language historically based on various dialects, which have left clear traces in the German phraseology (that is, geographical restrictions of various idioms). On the other hand, there is the observation that certain idioms are used across much larger linguistic areas

than that of one individual language, an observation akin to the discovery of the 'European linguistic area (Sprachbund)' in typology research. The aim of this chapter is to discuss the set of problems to do with the second observation, i.e. it deals with phraseology research in a European framework and beyond, and to introduce a research project on this matter. For the sake of simplicity, the chapter will be confined to idioms.

2. On the terms *Europeanism* and *internationalism*

The literature on phraseology research occasionally draws attention to certain cross-linguistic similarities of idioms in some of the European standard languages. Rather pretentious terms have been used for this phenomenon, which is basically just an accidental observation in a few culturally related standard languages, including *Europeanism* or *internationalism*, and in places even *universalism*. Lists of so-called Europeanisms or internationalisms are usually based on only three or four languages. An example would be Mokienko's (1998) listings of idioms that happen to be similar in German, English, French and/or some Slavonic languages. There is a striking lack of any theoretical explanation why precisely these languages should be considered representative of such a comprehensive concept. The central problem of what idioms actually constitute the so-called 'phraseological uniformity of Europe' has never been studied methodically.

The term *Europeanism* is well established in lexicology, especially in the field of vocabulary and loanword research. A wide range of lexical borrowings, loan translations, calques, and borrowed word meanings have been classified as Europeanisms, independently of the genetic affiliation of the languages to which these words belong. Most of them have been adopted from Latin over two thousand years of history (cf. Munske & Kirkness 1996). We can often recognise a Eurocentric perspective here, as many of these so-called Europeanisms have actually been studied only for a few major European languages. The same holds true for *internationalism*, a term originally used in work on technical language. According to Hengst's definition (1978: 467), an internationalism is a loanword spread across several languages with phonetic and morphologic adaptation and a constant meaning. The Eurocentric view has also been criticised because studies on internationalisms have almost completely excluded non-European languages (Matta 2002/2003). Phraseological studies occasionally use the term *internationalism*; Rajchštejn (1980: 36, 125) speaks of *ФЕ-интернационализмы* (*FE-internatsionalizmy*) in view of idioms like *to worship the golden calf*, *a Trojan Horse* or *to sweep up the Augean stables*.

In sum, cross-linguistic lexical-semantic similarities of the idioms of some languages have now and then come to the notice of phraseology researchers. This phenomenon seems to be one of the many faces of phraseology. However, as terms in idiom research, Europeanism or internationalism are merely of an accidental nature and therefore unsuitable. What is missing is systematic empirical research, a working definition of the terms, a catalogue of criteria by which idioms can be defined as

Europeanisms/internationalisms, as well as a theoretical foundation of how to operationalise their compilation. From a theoretical viewpoint, all the crucial questions as to how relevant Europeanisms/internationalisms can be singled out and determined remain unanswered. To answer these questions, a large-scale research project is needed, aimed at systematically investigating the similarities among idioms in as many languages as possible.

3. European languages and Eurolinguistics

The first step of such a research project would be to clarify the linguistic situation in Europe. The number of languages estimated to exist in Europe varies and partly depends on where to draw the line between languages and dialectal varieties. Haarmann (2002: 30) speaks of 143 languages, but according to Stolz's (2001: 227) estimate there are more than 200. Let us have a quick look at the linguistic map of Europe, which is made up of at least five completely unrelated phyla and one isolated language:

1. *The Indo-European language family*, including the Germanic, Romance, Celtic, Baltic, Slavonic languages as well as Albanian, Greek, Armenian and some (Indo-)-Iranian languages (e.g. Ossetic, Kirmanji, Talysh and Tati, located in the Caucasus region, belong to the European area).
2. *The Uralic language family*. As opposed to the traditional division into the Samoyed group (of which only one language, Tundra Nenets, can be found in Northeast Europe) and the main group of Finno-Ugric languages, there has recently been a preference for another classification (cf. Jazyki Mira 1993), distinguishing the Ugric branch (Hungarian and Mansi), the North-Finnic branch (Finnish, Estonian, Karelian, Vepsian, Livonian, Votic, among other things), the Permic branch (Udmurt, Komi, Permyak), the Volgaic branch (Mordvin, Mari) as well as other Uralic languages in Europe (Sámi languages, Tundra-Nenets).
3. *The Altaic language family*, traditionally divided into the subfamilies of the Mongolian (of which only Kalmyk is spoken in Europe, in a region to the west of the lower Volga, northwest of the Caspian Sea) and the Southwestern Common Turkic languages of Europe (e.g. Azerbaijani, Bashkir, Chuvash, Crimean Tatar, Gagauz, Karaim, Kumyk, Noghay, Tatar, Turkish, cf. Jazyki Mira 1997).
4. *The autochthonous Caucasian languages*, traditionally divided into three families, the Nakh-Daghestanian (Northeast Caucasian) languages, the Abkhazo-Adyghian (West Caucasian) languages and the Kartvelian (South Caucasian) languages, of which Georgian has been studied quite well. These groups are also regarded as three different language families.
5. *The Semitic languages*: first and foremost, Maltese should be mentioned here; other Semitic languages spoken in (parts of) Europe include Cypriotic Arabic, New Assyrian and New Aramaic.
6. *Basque*, the only European language that is completely isolated.

Eurolinguistics is a current, promising research concept striving to examine European languages across individual philologies and independently of their genetic relationship. The Eurocentric ideas sometimes connected with it (cf. Reiter 1999; Leuschner 2001), however, must be avoided. Two comprehensive research projects should be mentioned here, whose methodological range and view of Europe may serve phraseological research well. Both projects use Europe's geographic frontiers as starting points, extending eastwards as far as to the Ural mountain range and including the Caucasus region.

Firstly, there is the *Atlas Linguarum Europae* (ALE), the first continental linguistic atlas founded in 1970. It contains the six language families with a total of 22 language groups which, in turn, consist of many individual languages. This large geolinguistic research project is aimed at interpreting the heterogeneous data collected from 2,631 localities and putting them on symbol maps in order to gain insight into early cultural-historic layers of the European lexicon and Europe's cultural past (Viereck 2000). Secondly, there is the *Typology of Languages in Europe* (EUROTYP) project; see e.g. Haspelmath et al. (2001). The aim of this large-scale project is to find structural features that are common to many European languages but cannot be due to their genetic affiliation – that is, to verify the hypothesis of Europe as a linguistic area ('Sprachbund') – and to study these phenomena for as many European languages as possible, also against the background of non-European languages. Properties that are valid for a large number of European languages have been formulated as 'Euroversals', although this term did not gain general acceptance. With the help of so-called 'isopleths' or 'quantified isoglosses' (lines showing the geographical distribution of languages that share the same number of features, cf. Dahl (2001: 1458); Haspelmath (2001: 1505)), congruencies of many languages can be mapped onto the map of Europe.

In contrast to these substantial research projects, idiom research throughout Europe has so far been concerned with only a few standard languages, all of which belong either to the Indo-European family or to Finno-Ugric, a subfamily of the Uralic languages. Apart from at best 20–25 easily accessible languages, most European languages have hardly ever been the object of idiom research. For many languages, there are no usable idiom dictionaries available. The same holds for corpus data (whether using standard corpora or using the web as corpus) as these resources only exist for some of the major languages of Europe. As a result, questioning of as many native speakers as possible remains the best source of data for a large-scale multilingual research project on idioms. Things are different with regard to proverb studies since paremiology has been practised on a more comprehensive, multi-language scale from the very beginning. There is no tradition of Europe-wide or worldwide idiom studies that would be comparable to the prosperous tradition of international cooperation in the field of proverb research, a gap that was not even recognised until recently.

Attempts to demarcate Europe from other cultural and linguistic areas by means of distinctive features are doomed to failure. Culturally, it is impossible to establish clear boundaries. Rather, the so-called European Occidental culture has its roots in the Middle East as well as in North Africa (cf. the Hellenistic area, extending beyond

the European continent). The linguistic situation is quite similar. One of the major language families spoken in Europe, called Indo-European, has substantially more representatives in Asia than it has in Europe. Moreover, a number of languages do not stop at any virtual border within the Eurasian continuum. Therefore, Europe should be understood in terms of its geographic boundaries. If we speak of the languages of Europe, we should keep in mind that we deal with 143–200 languages. In phraseology research, expressions such as 'all languages of Europe' can be found when it is really only some European major standard languages that are being talked about, but this is seriously misleading – and such a metonymical use of 'Europe(an)' is to be avoided.

4. *Widespread Idioms in Europe and Beyond* – a research project

4.1 Pre-conditions and aims

The first ideas to establish a research project of supposedly widespread idioms go back to the EUROPHRAS conference held in Basel, Switzerland in August 2004 (see Piirainen 2006a). The starting point was the notion that today, in a uniting Europe, it should really be possible to find competent collaboration for many languages of Europe – or at least for all languages spoken in the democratically governed European states not affected by civil wars. The project *Widespread Idioms in Europe and Beyond*, which is supported by the European Society of Phraseology, is aimed at creating a comprehensive inventory of factually existing idioms widespread across many languages in order to answer questions such as the extent to which phraseology takes part in the assumed uniformity of the languages of Europe, based on reliable empirical data and no longer on purely accidental observations.

It has been put forward that the term *Europeanism* should be replaced since a Eurocentric view is to be avoided. Although it is true that the idioms focused upon centre around European languages, this does not mean that they do not occur in non-European languages (cf. the example in the appendix below). Thus, for idioms supposedly widespread across European languages and beyond, the term *widespread idiom* (or WI for short) has been suggested as a working term. WIs are idioms that – when their particular cultural and historical development is taken into account – have the same or a similar lexical structure and the same figurative core meaning in various different languages, including geographically and genetically distant languages (working definition).

A systematic investigation into the phenomenon of WIs makes it necessary to include not only as many European languages as possible but also non-European languages functioning as a contrast group. Therefore, such an investigation has to rely on the assistance of many native speakers and linguists interested in phraseology. The proper place of this research project is not in contrastive linguistics in the traditional sense. It is not primarily concerned with relationships of equivalence between idioms of different languages (e.g. with respect to all their diasystematic features or their

syntactic, pragmatic, and/or textual behaviour). Rather, the project is aimed at either corroborating or refuting the oft-claimed uniformity of European phraseologies, allegedly based on the 'common European cultural heritage' (see Piirainen 2005, 2006a, b, 2008).

4.2 Methodological approaches

Studies carried out within both the atlas project ALE and the EUROTYP project have demonstrated how linguistic features can be compared across an enormous number of languages spoken in Europe, including minor and minority languages, and how the results can be based on a wide variety of linguistic data. WI research could benefit greatly from these methodological approaches. As these projects have shown, the first step was to identify suitable linguistic units, that is, to subdivide the total number of idioms into smaller and therefore comparable entities and thus operationalise the compilation of potential WIs. Idiom research, however, does not deal so much with lexical and structural phenomena as the above-mentioned projects, but, to a considerable degree, with culture. Most idioms are based on knowledge of various cultural aspects. Therefore, the decisive step to discovering WIs was the development of a tool that enabled us to compare given parts of idiomatic language with respect to their cultural foundations. To compare as many languages as possible along the same lines, classifying idioms according to their underlying cultural phenomena was a suitable way.

Idioms were grouped according to their cultural foundations into five main types: (a) textual dependence, (b) pre-scientific conceptual domains, (c) cultural symbols, (d) aspects of material culture, and (e) aspects of social culture (cf. Dobrovol'skij & Piirainen 2005: 205ff.). Each of these individual types can be divided into subcategories and studied across a large number of languages to find WIs. Type (a), for example (which proved to provide the largest number of WIs), can easily be subdivided into smaller and therefore comparable categories based on the sources of the texts (the Bible, fables, fairy tales, works of classical literature, etc.). Idioms of this type have been explored very well for many European languages; in its attempt to create an inventory of WIs, the project could start from this preliminary work.

The first criterion for WI candidates is whether an idiom occurs not only in some major Central European languages but also in geographically more distant and genetically unrelated languages. Accordingly, the next step was to carry out preliminary tests on the 'widespreadness' of individual idioms that had already been observed to occur in several languages. Finnish, Estonian, Hungarian, Greek, Russian, Bulgarian, Croatian, Polish, Latvian, Romanian and Spanish served as test languages. Let us have a look at some examples of type (d) material culture: Equivalents of the idioms *to be in the black/to be in the red; (like) a piece of cake; to put all one's eggs in one basket* or *to hit the nail on the head* can be found in several genetically and/or geographically related languages; however, all of them proved to be inappropriate in the pre-tests. Preliminary tests produced about 250 idioms supposedly to be found in many different European languages.

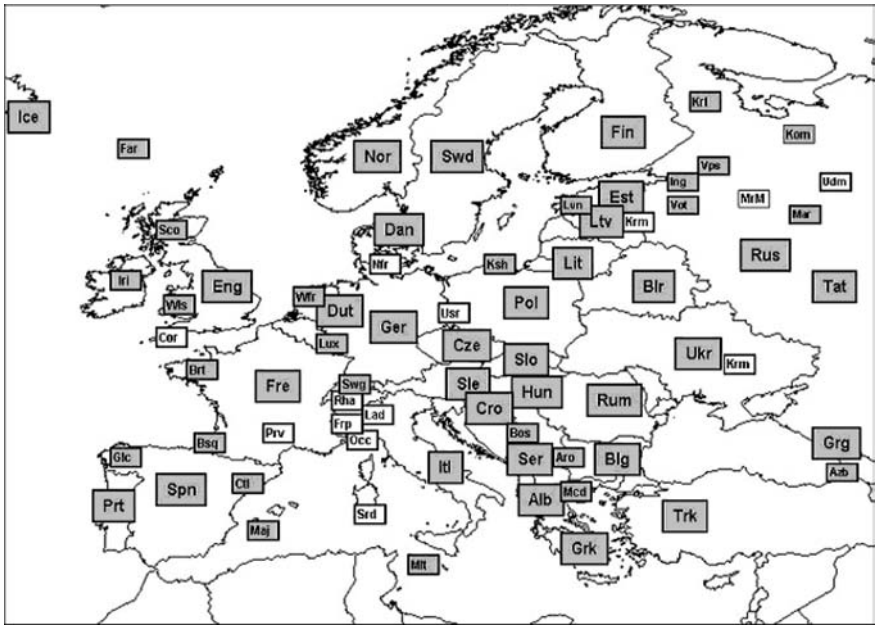


Figure 1. Map of the idiom *to swim against the stream/tide* in European languages

The next step was to carry out surveys among experts of as many languages as possible, and questionnaires were designed for this purpose. Because of the different target groups, the questionnaires and covering letters were designed individually in different languages and versions. Questionnaires with a total of 200 supposedly widespread idioms were sent to competent colleagues, covering about 80 languages and asking addressees to look into the questions and answer them based on both their own competence with regard to idioms and discussions within their circle of colleagues.

To date (i.e. December 2006), I have received data for more than 70 languages. Several answers are still pending. Indo-European is represented almost exhaustively with 54 of the languages accessible to research. The same holds for Maltese and Basque. Unfortunately, results for the Uralic and Altaic languages are not as comprehensive. Of the Caucasian languages, only Georgian takes part in the project. For the languages represented in the project, see Figure 1 and the example in the appendix. For further details, see http://www.piirainen.homepage.t-online.de/.

4.3 Presentation of the data

There are two main ways of presenting the empirical data collected: either in the form of maps or in the form of a dictionary. Since the WI project is partly connected to areal linguistics, a projection of the data onto the map of Europe is called for. Because of the

geographical overlap between many languages, the maps we can draw must necessarily be schematic. These maps do not show the spreading of languages as such but their occurrence, marked by the abbreviations (three letter short forms). The map for the idiom *to swim against the stream/tide* shows how this idiom occurs across the languages of Europe: Equivalents do exist in those languages whose abbreviations are given on a grey background, but have not been proven for those languages whose abbreviations are given on a white background.

Like idioms themselves, the WI project is at the same time connected with cultural phenomena as well, and therefore the presentation of WIs in form of a dictionary is also appropriate. The following four main topics should be treated consistently throughout each individual article on widespread idioms:

1. Idioms of all languages (to the extent to which they are currently available through our informants) are presented in a standardised dictionary form accompanied by exact descriptions of their literal and figurative meanings. Variants of an idiom can be placed next to the prototypical form. If a given idiom does not exist in a particular language, it is marked by (-). As the appendix shows, our example exists not only in many European languages but also in the Asian languages used as contrast group, small languages and dialects.
2. Then, there will be an analysis of specifics of the lexical and morpho-syntactic structure, i.e. potential special features of the inner form of individual idioms. Only some idioms from our example go their own way at the level of literal reading. Irish has a variant *to swim against the waterfall*, while Basque has a calque from French and a correspondence of *to go against the air* which is quite different. Several languages use verbs meaning 'to row' or 'to walk, go' instead of 'to swim'. Obviously, the idioms provide the same conceptual basis for a similar figurative meaning.
3. In the next section, we look at potential peculiarities regarding the phraseological (figurative) meanings. This section deals with the problem of potential false friends and subtle semantic differences. In the course of its history, the meaning of one individual idiom may develop into different directions, prioritising certain relevant features of the target concept while suppressing others, so that the 'associative aura' may change.
4. Finally, there will be a general interpretation of the individual data. As far as the lexical structure of the equivalents of our example is concerned, nothing can be said about the causes of its widespreadness and the ways of borrowing or about the donor languages (as is possible in other cases). Rather, the idiom's biblical origin should be taken into account, i.e. there is the possibility that different languages may have drawn from the same textual source independently, although its unobtrusive image is not of specific biblical content.

5. First results produced by the WI-Project

5.1 The existence of idioms spread across a large number of languages

One important result is the discovery that widespread idioms do indeed exist, a fact that until now could not have been stated this clearly with respect to idiom research. Many of the 200 idioms examined so far in the questionnaires reveal a pattern similar to the example *to swim against the stream/tide*. In earlier research, the occurrence of an idiom in three, four or five languages was considered 'conspicuous'. In view of the results of the WI survey, there may be 30, 40, 50 or more languages revealing the very same idiom. It is therefore rather the non-existence of an idiom in a given language that should be considered conspicuous, and reasons should be ascertained why a certain language does not have one particular idiom.

Let us look at an extreme case, the equivalents of the English *(to weep/shed) crocodile tears* '(to show) sadness that is not sincere; (to shed) hypocritical tears'. According to the traditional grouping of idioms into thematic groups, several cross-linguistic studies subsume this unit under idioms with animal constituents and come to the result that *crocodile* – a wild animal not native to Europe – can be found nevertheless in the phraseology of some European languages (see Sabban 2007: 592 for examples). Very probably, the idiom is to be traced back to common narrative traditions passed on from antiquity and via the *Pantchatantra* (partly from the 2nd century BC) into various Indian, Arabic and European areas. However, there has not been any research on the route that this idiom has taken to get into the individual contemporary languages, whether through the old traditional legends or based on a more recent layer of borrowing. The idiom has been proven to exist not only in 45 European languages (including Maltese, Turkish and Georgian), but also in languages of Africa and Asia. The idiom is well-known in the Arabic languages, for example in the dialects of Morocco, Tunisia and Egypt, as well as in Swahili, in Persian and in various languages of India (e.g. Telugu, Malayam and Hindi); furthermore it has been recorded for Thai, Vietnamese, Chinese, Mongolian, Aklanon and Tagalog. Presumably, there are still many more languages providing this idiom.

5.2 Genetic affiliation as cause of the wide spread?

There have been different attempts at explaining cross-linguistic similarities of idioms, among them an earlier view that genetic affiliation could play an important role, e.g. for some similarities between Hungarian and Finnish idioms. This genetic criterion can still be found in recent studies (e.g. Ostapovič 2006: 147f.). As the example *to swim against the stream/tide* has shown, the linguistic affiliation is of no importance, since the idiom is represented in Indo-European, Finno-Ugric and Turkic languages, in Maltese (a Semitic language), in Georgian (a Caucasian language) and in Basque, whereas some of the smaller languages (e.g. Cornish, Provençal, Rhaeto-Romance,

Ladin, Francoprovençal, Occitan, Sardinian, Macedonian, Udmurt) do not possess the idiom.

Although not all of the European language families are represented in every individual WI, it is obvious that the genetic affiliation does not influence the spreading of idioms. None of the WIs is restricted to one phylon (e.g. Indo-European). Gaps in the spreading rather concern the lesser used or declining languages. Contrary to expectation, the languages at the periphery of Europe (e.g. Icelandic, Finnish, Estonian, Maltese or Greek, even Turkish and Tatar) are not marginal figures but central to the 'European uniformity', while several minor and minority languages in Central Europe, as well as Irish and Sámi languages, clearly are outside. In view of the WI *to weep/shed crocodile tears* mentioned in Section 5.1, small languages such as North Frisian, West Frisian, Irish, Cornish, Breton, Ladin, Francoprovençal, Upper Sorbian, Karelian, Udmurt or Basque do not provide the idiom. Thus, it is crucial for European idiom research to get away from the 'genetic' view of idioms and instead gain an overview of the real linguistic situation of Europe.

5.3 The common cultural heritage as cause of the wide spread?

Similarities of the phraseologies of the European languages are often ascribed to a so-called 'common cultural European heritage', which, for its part, would require a precise explanation. Subsumed under this term are cultural traditions from antiquity, Christianity, biblical and Medieval Latin literature to renaissance, humanism and the Enlightenment up to works of classical literature. This rather nebulous concept can be stated somewhat more precisely thanks to the questionnaires that have been sent back.

As was to be expected, the greatest part of widespread idioms is made up by WIs originating from identifiable textual sources, among them numerous idioms going back to the Bible or to different fable traditions. Things are different for classical poetry, works of world literature, as postulated sources of WIs. Among the large number of pre-tested idioms and the 200 idioms tested via the questionnaires, only few WIs of this kind could so far be identified, e.g. *to tilt at windmills*, an allusion to Cervantes' novel *Don Quixote* (1605). Except for *much ado about nothing* and *(this is) the beginning of the end*, neither idioms going back to quotes from Shakespeare, however, so ubiquitous in the English language and often popularised in other languages as well through translations or performances of his plays, nor other idioms of classic literary origin could be found among our WIs.

Instead, there are quite different textual domains effective in constituting WIs. For example, one important source are conceptions of the way of life of Native Americans as they have been conveyed to Europeans through stories of J. F. Cooper (1789–1863) and are kept alive by film productions. We can prove a wide spread across many European languages for quite a number of idioms of this origin (e.g. French *fumer le calumet de la paix avec qn* 'to smoke the peace pipe with someone'; *le dernier des Mohicans* 'the last of the Mohicans'; *être sur le sentier de la guerre (avec qn)* 'to be on the war-path with someone' or *enterrer la hache de guerre* 'to bury the hatchet'). It may

be regarded as a new discovery that there are textual domains effective in constituting widespread idioms that are quite different from the so-called 'European cultural heritage of classical and Christian provenance'.

5.4 Spontaneous metaphorization in various languages?

Explanations of the similarities of idioms in two or three languages, which are detected more or less by chance, often revolve around another assumption, namely that metaphorization has occurred independently in several languages, predominantly based on common human experience or perceptions of the world. The occurrence of idioms in a very uniform lexical structure across some dozens of (geographically and culturally related or distant) languages seems to suggest that there is a connection between the idioms and that spontaneous coinages of the same metaphor can be ruled out. It is more probable that all these idioms go back to one and the same textual source, even though this source has not been recognised in every individual case.

Thus, the idioms mentioned in Section 5.3 enable us to refute speculations on the origins of idioms and to give a more accurate account. These idioms are not, as has often been claimed, based on direct contact with the Native American population of North America. In contrast, they are products of textual dependence, as can be seen from their occurrence in many languages whose speakers were never in contact with American Indians. Furthermore, we can qualify the view often held that idioms come about predominantly on the basis of everyday life experiences. Many widespread idioms show that knowledge about texts (e.g. about elements of foreign cultures in films featuring Indians) are an important factor in our world knowledge contributing to the origin of new idioms.

The same is true for the idiom *the tip of the iceberg*. Equivalents of this idiom are spread across 40 European languages, and likewise can be found in Japanese, Korean, Mongolian, Chinese and Aklanon. Again, the assumption of a spontaneous metaphorization in various languages, on the basis of everyday perceptions of the world, can be excluded. None of the speakers of these languages have 'experiences' with icebergs. Rather, the diffusion of this idiom, too, is due to knowledge about common texts, be it the modern textual source of film producing (i.e. filming of the sinking of the Titanic).

5.5 English as the donor language?

There is even more to be learnt from these examples: quite a number of idioms that go back to the world of Native Americans do not exist in contemporary English. This is an indication that English might not have been the donor language for these idioms in the first place. Individual languages may adapt these idioms independently of English.

It has been claimed that the idiom *the tip of the iceberg* is also a borrowing from English into other languages (e.g. by Mokienko 1998: 541). This claim can only be corroborated by careful analyses within the philologies of all these individual languages.

Another example is the English *to add fuel to the flame(s)* (or *to add fuel to the fire, to fuel the flames, to fuel the fire, to add fuel*). The idiom has been adopted by at least 56 European and various Asian languages. In fact, only the English idiom has the *fuel* in its lexical structure. All other languages have constituents denoting 'oil' (following the Latin version *oleum addere camino*) or 'firewood', 'straw' and 'peat'. This difference clearly indicates that English cannot be the giving language. As these examples have shown, English is far from always representing the most important donor language. The role that the English language plays in the uniformity of European phraseology has often been overrated.

6. Outlook

It is only during the course of the project outlined here that we will gain insight into the causes of the many common features so far observed in European phraseologies – be they parallel formations based on the same textual source or borrowings resulting from cultural and geographical contacts. As far as the wide spread of idioms, especially from educated language, is concerned, we need to consider that it can be attributed not so much to oral language contact but much rather to the common educational code of the literate European social stratum. Intensive exchange of ideas across many different languages was only to be achieved via reading and writing, mostly using Latin as a lingua franca. Recently, however, there are other media, such as film producing, favouring the diffusion of new idioms beyond the boundaries of individual languages. The role of textual dependence seems to be more important than has so far been assumed by phraseology research.

The results emerging from the WI-project can in many cases give a more detailed account of hitherto vague ideas about the phraseological similarities between European languages and 'European cultural heritage' as their origin, as well as theoretical insight to phraseology research. An inventory of the widespread idioms in the form of maps and dictionary articles could be used for various further research projects. First of all, however, some intensive work in a circle of international phraseological researchers is required to reach this goal.

References

Dahl, Ö. (2001). Principles of areal typology. In Haspelmath, M., E. König, W. Oesterreicher & W. Raible (eds.) *Language Typology and Language Universals. An International Handbook*, Vol. 2, 1456–1470. Berlin: Walter de Gruyter.

Dobrovol'skij, D. O. & E. Piirainen (2005). *Figurative Language: Cross-Cultural and Cross-Linguistic Perspectives.* Amsterdam: Elsevier.

Haarmann, H. (2002). *Sprachen-Almanach. Zahlen und Fakten zu allen Sprachen der Welt.* Frankfurt: Campus.

Haspelmath, M. (2001). The European linguistic area: Standard Average European. In Haspelmath, M., E. König, W. Oesterreicher & W. Raible (eds.) *Language Typology and Language Universals. An International Handbook*, Vol. 2, 1492–1510. Berlin: Walter de Gruyter.

Haspelmath, M., E. König, W. Oesterreicher & W. Raible (eds.) (2001). *Language Typology and Language Universals. An International Handbook*, Vol. 2. Berlin: Walter de Gruyter.

Hengst, K. (1978). Zur Semantik fachsprachlicher Internationalismen. *Wissenschaftliche Zeitschrift der Humboldt-Universität zu Berlin*, *Ges.-Sprachwiss*, Reihe 77: 467–469.

Jazyki, M. (1993). *Языки мира: уральские языки*. Moscow: Наука.

Jazyki, M. (1997). *Языки мира: тюркские языки*. Moscow: Индрик.

Leuschner, T. (ed.) (2001). *Linguistik Online* 8(1) (Sprachkontaktforschung und Areallinguistik: Europa. Language contact and areal linguistics: Europe).

Matta, H. (2002/2003). Internationalismen-Forschung und Internationlismen-Theorie aus der Perspektive des Arabischen. Eine kritische Auswertung vorliegender Internationalismen-Sammlungen. *Kairoer Germanistische Studien* 13: 181–203.

Mokienko, V. M. (1998). Phraseologisierung von Europäismen oder Europäisierung von Phraseologismen? Divergente und konvergente Prozesse in phraseologischen Systemen europäischer Sprachen. In Eismann, W. (ed.) *Europhas 95. Europäische Phraseologie im Vergleich: Gemeinsames Erbe und kulturelle Vielfalt*, 539–555. Bochum: Brockmeyer.

Munske, H. H. & A. Kirkness (eds.) (1996). *Eurolatein. Das griechische und lateinische Erbe in den europäischen Sprachen*. Tübingen: Niemeyer.

Ostapovič, O. (2006). Kontrastive transkulturelle Analyse von mentalen Konzepten in der Kognitiven Phraseologie unverwandter Sprachen. In Häcki Buhofer, A. & H. Burger (eds.) *Phraseology in Motion I. Methoden und Kritik. Akten der Internationalen Tagung zur Phraseologie (Basel, 2004)*, 146–154. Baltmannsweiler: Schneider.

Piirainen, E. (2005). Europeanism, internationalism or something else? Proposal for a cross-linguistic and cross-cultural research project on widespread idioms in Europe and beyond. *Hermes. Journal of Linguistics* 35: 45–75.

Piirainen, E. (2006a). Widespread idioms: Cross-linguistic and cross-cultural approaches. In Häcki Buhofer, A. & H. Burger (eds.) *Phraseology in Motion I. Methoden und Kritik. Akten der Internationalen Tagung zur Phraseologie (Basel, 2004)*, 155–173. Baltmannsweiler: Schneider.

Piirainen, E. (2006b). Cultural foundation of phraseology. A comparison of standard languages and dialect. In Kürschner, W. & R. Rapp (eds.) *Linguistik International. Festschrift für Heinrich Weber,* 321–336. Lengerich: Pabst Science Publishers.

Piirainen, E. (2008). Phraseology from an areal linguistic perspective. In Álvarez de la Granja, M. (ed.) *Beiträge zur Phraseologie aus kontrastiver Sicht*. Hamburg: Kovač.

Rajchštejn, A. D. (1980). *Sopostavitel'nyj analiz nemeckoj i russkoj frazeologii*. Moscow: Vysšaja škola.

Reiter, N. (ed.) (1999). *Eurolinguistik: Ein Schritt in die Zukunft. Beiträge zum Symposion vom 24. bis 27. März 1997 im Jagdschloß Glienicke (bei Berlin)*. Wiesbaden: Harrassowitz.

Sabban, A. (2007). Culture-boundness and problems of cross-cultural phraseology. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseology. An International Handbook of Contemporary Research*, 590–605. Berlin: de Gruyter.

Stolz, T. (2001). Minor languages and general linguistics (with special focus on Europe). In Stolz, T. (ed.) *Minor Languages of Europe. A Series of Lectures at the University of Bremen, April–July 2000*, 211–242. Bochum: Brockmeyer.

Viereck, W. (2000). The *Atlas Linguarum Europae* and its insights into the cultural history of Europe. In Gottlieb, H., J. E. Mogensen & A. Zettersten (eds.) *Symposium on Lexicography X. Proceedings of the Tenth International Symposium on Lexicography May 4–6, 2000 at the University of Copenhagen,* 19–29. Tübingen: Max Niemeyer.

Appendix

Example TO SWIM AGAINST THE TIDE/STREAM

A. Languages of Europe:

I. INDO-EUROPEAN LANGUAGES IN EUROPE

1. Germanic

Icelandic	*að synda/berjast/standa/... móti straumnum* 'to swim/fight/stand/... against the stream'
Faroese	*ganga ímóti streyminum* 'to go against the stream'
Norwegian	*være/gjøre noe mot strømmen* 'to be/do something against the stream', nynorsk *gå mot straumen* 'to go against the stream'
Swedish	*gå/simma mot strömmen* 'to go/swim against the stream'
Danish	*gå mod strømmen* 'to go against the stream'
English	*to swim against the tide/stream*
Scots	*tae suim agin the tide* 'to swim against the tide'
Dutch	*tegen de stroom (in) zwemmen* 'to swim against the stream'
North Frisian (Sylt)	*töögen di Stroom swum* 'to swim against the stream'
West Frisian	*tsjin de stream op roeie/tsjin de stream yn wolle* 'to row against the tide/want (to go) against the stream'
German	*gegen den Strom schwimmen* 'to swim against the stream'
Luxembourgish	*géint de Stroum schwammen* 'to swim against the stream'

Individual German dialects:

Low German (Schleswig Holstein)	*gegen dat Water answümmen* 'to swim up against the water'
Moselle Franconian (Lorraine)	*gént de Stroum schwammen* 'to swim against the stream'
Alemannic (Alsace)	*gége de Strom schwimme* 'to swim against the stream'
Highest Alemannic (Aosta Valley)	*goa gegen dem loufene Wasser* 'to go against the running water'
Swiss German	*gegene Strom schwümme* 'to swim against the stream'

2. Celtic

Irish	*dul in aghaidh an tsrutha* 'to go against the stream/current' *snámh in aghaidh (an) easa* 'to swim against a waterfall'
Welsh	*nofio yn erbyn y llif* 'to swim against the current'
Cornish	(-)
Breton	*mont a-enep/a-benn da'n dour/da'r red-dour* 'to go against/ toward the stream of the water'

3. Romance

French	*nager/aller contre le courant/à contre-courant* 'to swim/go against the stream/ upstream'
Provençal	(-)
Occitan	(Pragelato/Torino) (-)
Italian	*andare/remare/nuotare contro corrente* 'to go/row/swim against the stream'
Sardinian	(-)
Ladin	(-)
Rhaeto-Romance	(-)
Francoprovençal	(Aosta Valley) (-)
Spanish	*nadar/ir contra corrente* 'to swim/go against the stream'
Catalan	*nedar a contracorren/nedar contra la corrent* 'to swim upstream/ to swim against the stream'
Galician	*nadar/ir contra corrente* 'to swim/go against the stream'
Mallorquin	*anar contra corrent* 'to go against (the) stream'
Portuguese	*ir contra a corrente* 'to go against the current' *remar contra a maré* 'to row against the tide'
Romanian	*a înnota împotriva curentului* 'to swim against the stream'
Aromanian/Vlach	*s-ljea cu vintul/s-acatsã cu vintul* 'to go/swim against the stream'

4. Baltic

Latvian	*peldēt pret straumi* 'to swim against the stream'
Lithuanian	*prieš srovę paplaukti* 'to swim against the stream'

5. Slavonic

Russian	*плыть против течения* 'to swim against the stream'
Belorussian	*плыць супраць течення* 'to swim against the stream'
Ukrainian	*пливти проти течії* 'to swim against the stream'
Czech	*plavat/jít proti proudu* 'to swim/go against the stream'
Slovakian	*plavať proti prúdu* 'to swim against the stream'
Polish	*iść/jechać/płynąć/... pod prąd* 'to go/swim/... against the stream'
Kashubian	*płënąc pòd żoch/wòdã* 'to swim against the stream/water (literary, polonism)
Upper Sorbian	*přećiwo prudej/wodźe płuwać* 'to swim against the stream/water'
Slovene	*plavati proti toku* 'to swim against the stream'
Croatian	*plivati protiv struje* 'to swim against the stream'
Bosnian	*plivati protiv struje/uzvodno* 'to swim against the stream/upstream'
Serbian	*plivati protiv struje* 'to swim against the stream'
Macedonian	(-)
Bulgarian	*плувам срещу течението* 'to swim against the stream'

6. Albanian

të ecësh kundër rrymës/te shkosh kundra rrymes 'to walk/go against the stream'

7. Greek

πάω κόντρα/αντίθετα προς το ρεύμα 'to walk/go against/contrary to the stream'

II. Uralic (finno-ugric) languages in Europe

1. Ugric languages

Hungarian	*árral szemben úszik* 's/he swims against the water'

2. North-Finnic languages

Finnish	*kulkea/uida vastavirtaan* 'to walk/swim upstream'
Karelian	*uija vaste virdoa* 'to swim against the stream'
Estonian	*vastuvett/vastuvoolu minema/ujuma* 'to go/swim against the tide'
Vepsian	*vastvedele ujuta* 'to swim against the tide'
Ingrian	*vast virtaa ujjuua* 'to swim against the stream'
Votic	*vassaa virtaa ujua* 'to swim against the stream'
Livonian	*vastõ õuvvõ voigõ* 'to swim against the tide'

3. Permic languages

Udmurt	(-)
Komi-Zyrian	*ва паныд катны* 'to swim against the stream'

4. Volgaic languages

Mari	*jogən vaštaгəš ijaš* 'to swim upstream'
Mordvin	(-)

III. Altaic languages in Europa

Turkic languages

Karaim	(-)
Turkish	*akıntıya karşi kürek çekmek* 'to row against the current'
Tatar	*агымга каршы йозу* 'to swim against the stream'
Azerbaijani	*axına qarşıüzmək* 'to swim against the stream'

IV. (Autochthonous) Caucasian languages

Georgian	*dinebis sazinaamdebod zurwa* 'to swim against the direction of the river'

V. Semitic languages in Europe

Maltese	*jgħum kontra l-kurrent* 'to go against the stream'

VI. Basque	*korrontearen kontra joan* 'to swim against the stream' (a calque from French) [different: *aizearen kontra aurka doa/ibili* 'to go against the air']

B. Non-European languages (by way of comparison):

Persian	*bar-aleieh jarian shena kardan* 'to swim against the stream'
Telugu	*e:tiki eduri:dinatlu* 'like swimming against the stream'
Mongolian	*урсгал сөрж сэлэх (ursgal surj seleh)* 'to swim against the current'
Chinese	*nìliú er shàng* 'against the current and upwards'
Korean	*sorui-reul geoseureu-da* 'to swim against the tide/currant'
Japanese	*jiryu ni sakaratte oyogu* 'to swim against the stream of the time (the prevailing trend)'
Vietnamese	*bói ngúóc dòng* 'to swim against the stream'
Aklanon	*nag-eangoy kuntra sa sueog* 'to swim against the current'

C. Esperanto *naghi kontraù la fluo* 'to swim against the stream'

Free and bound prepositions in a contrastive perspective

The case of *with* and *avec*

Christelle Cosme and Gaëtanelle Gilquin

The starting point of this chapter is the observation that the French preposition *avec*, which is generally considered as the translational equivalent of *with*, is rarely used as a translation of *with* in parallel corpus data. A more detailed analysis of the corpus data reveals that polysemy partly explains this situation (*with* is more polysemous than *avec*, having developed more metaphorical senses than its French equivalent), but that another important factor is phraseology. Contrary to expectations, free uses of the preposition *with* are less often translated by means of *avec* than bound uses. This, however, takes no account of the fact that some bound uses with a word-for-word translation in French occur repeatedly in the data. It also conceals the fact that it is mainly nominal bound uses which often correspond to *avec*.

1. Introduction

Because they fall somewhere in between lexicon and grammar, prepositions tend to be treated inadequately in the literature, if at all. Many traditional grammars ignore them, on the grounds that they are not, strictly speaking, a grammatical phenomenon,[1] and lexicographers often regard them as "lexically empty" (Lindstromberg 2001: 80), portraying them inaccurately in dictionaries.[2] Yet, prepositions possess many interesting characteristics, as revealed by some recent studies (e.g. Zelinski-Wibbelt 1993a; Tyler & Evans 2003). One such characteristic, especially prevalent in English (see Downing & Locke 1992), is their largely phraseological nature. While some prepositions are selected for the independent meaning they carry, others are mainly chosen because they are phraseologically 'bound' to another word in the sentence.

---

1. Two major exceptions are Quirk et al. (1985), Leech & Svartvik (1994) and Biber et al. (1999).

2. This is the case both in monolingual (see Lindstromberg 2001) and bilingual (see Schmied 1998) dictionaries.

Using Biber et al.'s (1999) distinction between free and bound prepositions, we set out to compare the English preposition *with* and its intuitive equivalent in French, *avec*. On the basis of corpus data, we will show that *with* and *avec* actually display a surprisingly low degree of correspondence and we will seek to explain this lack of congruence by means of two hypotheses, one involving polysemy and the other having to do with the phraseological uses of the two prepositions. This corpus analysis will be preceded by an overview of the contrastive approaches to prepositions, with special emphasis on *with* and *avec*.

2. Contrastive approaches to prepositions

2.1 The study of prepositions

Although prepositions rank among the most frequent words of a language and play an important role in discourse, they have been discarded as "an annoying little surface peculiarity" (Jackendoff 1973: 345) for a long time. Lately, however, there has been a revival of interest in prepositions. So much so, actually, that we are now "witnessing a veritable plethora of investigations into the semantics of prepositions" (Zelinski-Wibbelt 1993b: 1). There are several reasons for this renewed interest. One is the emergence of the field of lexico-grammar (see Sinclair 1991), which recognises the existence of a continuum between lexicon and grammar. Another reason is the claim that metaphor is an all-pervasive phenomenon (cf. Lakoff & Johnson 1980), which has resulted in the popularity of metaphors in general and prepositional metaphors in particular (Lindstromberg 2001: 81).

Lexico-grammar mainly owes its success to the advances of corpus linguistics, which has shown that lexicon and grammar interact at every step and are therefore ultimately inseparable (Francis 1993: 142–146). Of particular interest here is the notion of collocation, which refers to the co-occurrence of words along the syntagmatic axis (see e.g. Tognini-Bonelli 2001). In some cases this co-occurrence is essentially preferential. The noun *employment*, for instance, prefers the lexical company of words such as *conditions*, *contract*, *discrimination* or *rights* (Stubbs 1995: 254). In other cases restrictions are imposed on the words that can occur together. Thus, **wide shoulders* and **to turn old* are excluded and should be replaced by *broad shoulders* and *to get/grow old* respectively (Van Roey 1990: 72). Such collocational restrictions are very common with prepositions. One can laugh **at** someone, but not **of** someone, and while you may be faithful **to** someone, you have faith **in** them. Prepositions which have little independent meaning and whose choice depends on some other word (*laugh*, *faithful* and *faith* in the above examples) are referred to as "bound prepositions" by Biber et al. (1999: 74). Bound prepositions are contrasted with "free prepositions", which have an independent meaning and whose choice is not dependent on any specific word in the context (ibid.).

Unlike corpus linguistics, which focuses on the way prepositions combine with other words in the sentence, cognitive linguistics is mainly interested in the meaning of prepositions in their own right and more precisely, how the different meanings of a preposition relate to each other. Prepositions are said to have a core, 'prototypical' meaning, which is normally anchored into the spatial domain (Brala 2002). In addition, they have more abstract, metaphorical meanings, which have developed from the core meaning through a process known as 'meaning extension' (Zelinski-Wibbelt 1993b). Cognitivists claim that meaning extensions are motivated in that they follow a "path of gradually increasing abstractions" (Dirven 1993:76), where each meaning has an obvious link with the previous one. It is therefore possible to organise the different meanings of a preposition in a 'radial network', with the prototypical meaning at the centre and the extended meanings surrounding it. The distance between the prototypical meaning and an extended meaning in the network reflects the degree of metaphorical abstraction of the extension.

Both the phraseological approach of corpus linguistics and the more semantic approach of cognitive linguistics will have a role to play in this analysis, although the focus will be on the former. These two approaches are all the more relevant here, in fact, since they tend to present different pictures across languages, as will be shown in the next section.

2.2 Prepositions across languages

Prepositions, like most high-frequency words, are interesting to study in a cross-linguistic perspective. While they are often regarded as having clear translation equivalents in most languages, a detailed analysis usually reveals a large number of language-specific uses (see e.g. Viberg 1998, 2002, 2004 or Altenberg 2001 for verbs, and Schmied 1998 for prepositions). These language-specific uses are often said to belong to one of the domains described above, namely bound uses and metaphorical extensions.

The lack of cross-linguistic correspondence for bound uses of words has regularly been underlined in the literature. Nesselhauf (2003:236), for instance, in her investigation of verb-noun collocations in English and German, notes that "more restricted combinations [are] less often congruent than freer combinations". Similarly, Lewis (1997:64) points out that common words which individually carry very little meaning and are elements in multi-word lexical items usually have no word-for-word translation. The bound uses of a preposition, therefore, are unlikely to be translated by the 'obvious' equivalent in the other language. This, according to e.g. Brala (2002), explains why the idiomatic usage of prepositions is particularly troublesome for learners of a foreign language.

Metaphorical extensions too tend to be problematic from a cross-linguistic perspective. Boers (cited in Lindstromberg 2001:82) notes that, in languages that are closely related to English, the central meanings of prepositions often have direct equivalents, but "[o]nce you move into the metaphoric extensions [...] deviations are likely to occur" (see also Brala 2002). He gives the example of the preposition *on* which, in

its spatial meaning, can usually be translated into French by *sur*, but whose figurative meanings often require a different translation. This is because each language, in its meaning extensions, has followed different chains of meanings and has exploited differently the "common cognitive endowment" of prepositions (Zelinski-Wibbelt 1993b: 20). As Dirven (1993: 96) puts it, "languages, even the most related ones, have conceptualised the links between spatial and mental domains of experience in slightly or markedly different ways".

Although English and French are related languages, one can therefore expect a number of differences in their preposition usage. As will appear from the next section, however, these differences are not just a question of bound or metaphorical uses. There is also a more fundamental difference between the two languages with respect to prepositions, namely the more prominent place they occupy in English as compared to French.

2.3 Prepositions in English and French

Van Hoof (1989: 40–41) notes the important role played by prepositions in English. Their kinetic value, he argues, is such that they can even fulfil the function of a verb. French, comparatively, uses fewer prepositions. As a consequence, translations of English prepositions into French often require supplementation, as illustrated by the following examples, taken from Vinay & Darbelnet (1995: 108ff.):

(1) **Supplementation by a noun:**
The news was announced in headlines that extended clear across the front page of *The Clyde Herald*.
= La nouvelle fut annoncée par un titre qui occupait toute la largeur de la première page du *Clyde Herald*.

(2) **Supplementation by a verb:**
He stopped at the desk for his mail.
= Il s'arrêta au bureau pour prendre son courrier.

(3) **Supplementation by an adjective or past participle:**
A dash had been put in the space for the holder's profession.
= On avait mis un tiret dans l'espace réservé à la profession du titulaire.

(4) **Supplementation by a relative clause:**
The courtiers around him...
= Les courtisans qui l'entouraient...

Notice that this lack of equivalence between English and French comes on top of the problems caused by bound and metaphorical uses. In sentence (4), for example, the preposition is free and clearly has its core, spatial meaning, but its translation does not involve the corresponding French preposition, *autour de*.

While there is a general tendency for prepositions to be more common in English than in French, this tendency is even stronger for the phraseological uses of preposi-

tions. Thus, Downing & Locke (1992: 580) note that "[a] notable feature of the English language is the extremely wide lexico-grammatical use it makes of prepositions". In other words, bound prepositions are more likely to occur in English than in French.

Now that we have outlined the main differences between English and French prepositions, let us turn to the prepositions *with* and *avec*.

2.4 *With* and *avec*

Avec is traditionally assumed to be the intuitive translational equivalent of *with* (cf. Vreck 2000). The two prepositions are also given as first translations of each other in bilingual dictionaries, e.g. Collins-Robert Dictionary (Atkins et al. 1998). Yet, Vreck (2000) notes that they are not often translationally equivalent. Like many English prepositions, *with* is more frequent than its French counterpart *avec*. It is also used in a wider variety of contexts than *avec* (Vreck 2000: 111). Consequently, *with* often has to be translated by alternative equivalents, as appears from Vreck's (2000) review of the various translations of *with* into French. Besides the direct equivalent *avec* (5), it can be translated by other prepositions, cf. *de* in (6), *à* in (7) and *en* in (8).[3] It can also be translated by other means, such as transposition (e.g. the prepositional phrase becomes an adverb, as exemplified by (9)), supplementation ('étoffement', see also Vinay & Darbelnet 1995) as in (10), complete reorganisation of the sentence ('refonte'), e.g. (11), or disappearance ('effacement'), cf. (12).

(5) I had asked her to come down with me, to look over the old place.
= Je lui avais demandé de venir avec moi jeter un coup d'œil à la vieille propriété.

(6) We quivered with fear.
= Nous frissonnions de peur.

(7) With the arrival of younger recruits, the group changed.
= À l'arrivée des jeunes recrues, le groupe évolua.

(8) The branch broke with a crack.
= La branche se brisa en craquant.

(9) The horse pulled the plough with ease.
= Le cheval tirait la charrue aisément.

(10) We thanked them when we could with windfall apples.
= Nous les remerciions quand nous le pouvions en leur donnant des pommes tombées.

3. Note that in (8), the noun following the preposition *with* becomes a present participle in French. This change can be related to the fact that the English noun *crack* is itself morphologically derived from a verb (Vreck 2000).

(11) We'll have to make do with it.
= Il faudra bien s'en contenter.

(12) Do not walk with your hands in your pockets.
= Ne marchez pas Ø les mains dans les poches.

Coming back to the two distinctions introduced in Section 2.1, it appears that both *with* and *avec* have free and bound uses, as well as core and metaphorical meanings. Compare the pairs of sentences in (13) and (14). In (13), the choice of the preposition is not dependent on any specific word in the sentence. In (14), on the other hand, it depends on the preceding verb (*part* and *concorder*). As regards meaning extension, both prepositions prototypically denote accompaniment (see Dirven 1993 for English and Jayez & Mari 2000 for French), cf. (15a) and (15b).

(13) a. But the only other thing perhaps, he'll go with one of the kids, that's a possibility. (Biber et al. 1999: 74)
b. Vous allez vous attirer des ennuis avec votre franchise. (Hanse 1987: 143)

(14) a. They've got to be willing to *part* with that bit of money. (Biber et al. 1999: 74)
b. Ses idées *concordent* avec les nôtres. (Hanse 1987: 261)

(15) a. John is sitting with Mary. (Dirven 1993: 75)
b. Paul se promène avec Marie. (Jayez & Mari 2000)

Each preposition has then developed further meanings, as illustrated by Dirven (1993) for English, with the spatial use of *with* arguably giving rise to several metaphorical extensions, among which area (e.g. *deal with a problem*), manner/instrument (e.g. *with great precision*), circumstance (e.g. *I cannot do it with everybody laughing*) and cause (e.g. *tremble with fear*). All these abstract meanings, Dirven (1993: 81) claims, involve a metaphorisation of the notion of accompaniment and are therefore linked to the prototype:

> [I]n the domain of area, the thematic field is seen as a companion one has to deal with; manner or instrument are likewise seen as companions to fulfil an assignment; circumstance is a companion standing by and cause a feeling or emotion leading to visible effects.

The exact proportions of free and bound uses and prototypical and metaphorical meanings for each preposition, as well as the extent to which they overlap (or not) in the two languages, will appear from the corpus analysis.

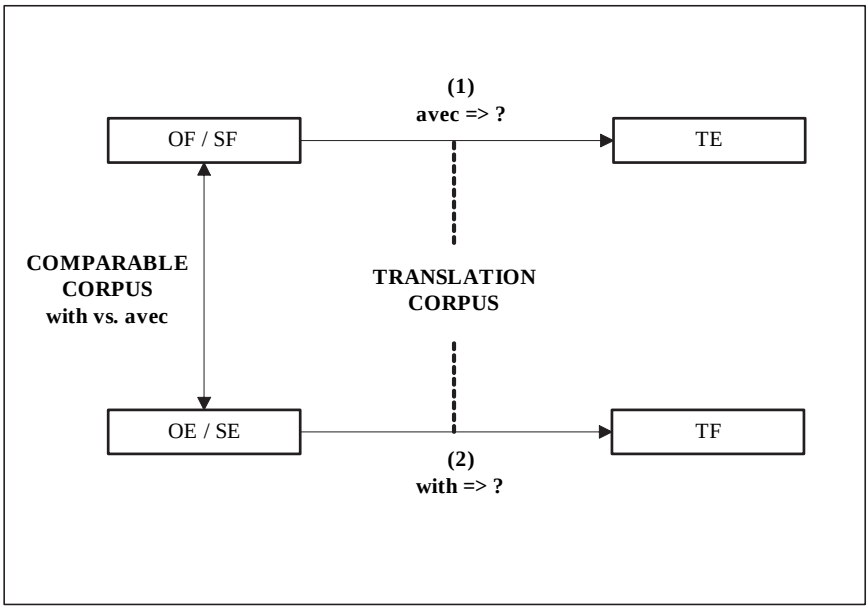


Figure 1. PLECI (*Poitiers-Louvain Échange de Corpus Informatisés*)

3. Corpus study

3.1 Data and methodology

The present analysis was carried out using English-French corpus data coming from PLECI (*Poitiers-Louvain Échange de Corpus Informatisés*), a bilingual corpus made up of journalistic and fictional texts.[4] The great strength of the corpus is that, as shown in Figure 1, it can function both as a comparable and parallel corpus. Not only does it allow for a comparison between original texts in English (OE) and similar original texts in French (OF), but it also allows for a comparison between source texts in French (SF) and their translations into English (TE) on the one hand (see (1) in Figure 1) and source texts in English (SE) and their translations into French (TF) on the other (see (2) in Figure 1). Both types of data proved useful for the present purposes. Using comparable corpus data, we were able to compare the use of *with* in English original texts and the use of *avec* in French original texts, while resorting to bidirectional parallel corpus data enabled us to gain insight into (i) how *avec* is translated into English and (ii) how *with* is translated into French.

Since *with* and *avec*, like most prepositions, are frequent phenomena, it is reasonable to use only a small sample of PLECI. It was therefore decided to limit the

4. See Gilquin (2000/2001), Cosme (2006) and Chuquet & Chuquet (2006) for studies based on PLECI.

Table 1. Breakdown of the PLECI sample used in the study: word counts

Original texts	**OE**	49,006	**OF**	50,081
Translated texts	**TF**	53,128	**TE**	48,552

Table 2. Mutual translatability of *with* and *avec*

OE ⇒ TF		OF ⇒ TE		MT
# with	*# with = avec*	*# avec*	*# avec = with*	
310 (100%)	94 (30.32%)	130 (100%)	77 (59.23%)	38.86%

analysis to the journalistic component, from which we selected a 50,000-word sample in each language and its translation in the other language (see Table 1 for the exact word counts).

Both the comparable corpus analysis and the parallel corpus analysis necessitate the use of a concordancer to retrieve the words under study. For the former, we made use of the text-retrieval package *WordSmith Tools* (Scott 1996). The latter was carried out with the help of the bilingual concordancer *Multiconcord* (Woolls 1997), which makes it possible to view concordance lines displaying a sentence in one language together with its equivalent in the other language.

3.2 Mutual translatability

Preliminary observation of the parallel corpus data reveals that, despite being obvious candidates for translational equivalence (cf. bilingual dictionaries), *with* and *avec* actually display a low mutual translatability (MT) of under 40% (see Altenberg 1999 on the concept of mutual translatability).[5] This result, however, irons out an important difference between the two translation directions, as *with* is translated by *avec* in only 30% of the cases, whereas *avec* is translated by *with* in 60% of the cases (see Table 2).

Two hypotheses can be put forward to explain this discrepancy, both of which were tested against the corpus data. The first hypothesis is that *with* is more polysemous than *avec*, as suggested by e.g. Vreck (2000: 111), and more precisely, that it has developed more metaphorical, non-prototypical senses than *avec*. The second one is that *with* has more bound uses than *avec* (cf. Downing & Locke 1992: 580) and that these bound uses cannot, as a rule, be translated by *avec* (see Section 2.2 on the regular lack of congruence for bound uses of words). Each of these two hypotheses will be examined in turn in what follows.

5. Fairly low mutual translatability scores have also been obtained for other language pairs, e.g. Schmied (1998) for *with* and German *mit* and Váradi & Kiss (2001) for *with* and Hungarian *-val/-vel*.

3.3 Polysemy

In order to test the number of senses of each preposition, we examined all the occurrences of *with* and *avec* in the corpus (original language) and identified a number of clusters of meaning for each preposition. It turns out from this review that *with* has indeed more senses than *avec*, and that these additional senses tend to be more metaphorical. Thus, for *avec*, the prototypical sense of accompaniment (16) and the related sense of mutual relation (17) represent almost 50% of all the occurrences of the preposition, while the other senses are comparatively much rarer.[6] *With*, on the other hand, is less often used in a prototypical sense (34.5%), but has other peripheral senses such as 'occupation' (e.g. *to deal with* (18), *to cope with*), 'knowledge' (e.g. *familiarity with* (19), *experience with*) or 'containment' (e.g. *filled with* (20), *peopled with*), which never occur in the corpus with *avec*.[7]

(16) Au fil des années, il a fignolé la maison en brique qu'il avait construite avec sa mère en 1972 [...].[8]

(17) Jusqu'en 1990, l'Ostpolitik visait d'abord à normaliser les relations avec les voisins orientaux.

(18) But it fails to discuss more immediate problems such as how to deal with the misdeeds of a local business that advertises in your paper.

(19) The familiarity of American readers with current world literature is no less abysmal.

(20) It is a community often filled with those local business owners whom Gannett designates as the first beneficiaries of "new journalism".

As a result, the translational equivalence is much higher for the prototypical uses than for the other uses. In the sense of mutual relation, for instance, *with* is rendered by means of *avec* in 67% of the cases (21). For a more peripheral sense such as 'exchange of goods' (e.g. *to replace with*), the translational equivalence falls to 14% only. When a sense cannot be expressed by *avec*, French uses another preposition (containment, for

6. Note that the distinction between prototypical and metaphorical senses is not a clear-cut one. Although the prototype of *with* and *avec* is said in the literature to express accompaniment, a sense such as 'mutual relation' is still very close to the notion of accompaniment (much closer, for example, than the notion of 'knowledge') and will therefore be considered as prototypical here.

7. It is important to realise that the sense of the bound preposition is actually construed through the combination of a verb, noun or adjective with the preposition (for example, *with* only means 'occupation' because it is bound to the verb *to deal*). For the sake of convenience, however, we will assign the sense to the preposition itself.

8. Starting from here, all the examples quoted come from PLECI.

instance, is almost always expressed in French by *de*, cf. *filled with* = *plein de* in (22)) or a different formulation altogether (e.g. *to deal with* = *gérer* in (23)).

(21) [...] Saudi Arabia and Egypt must weigh the merits of a distinctly junior partnership with the US compared with that of Israel.
⇒ [...] ils doivent évaluer les avantages d'un partenariat, en position mineure par rapport à Israël, avec les États-Unis.

(22) Its curving, open-ceilinged halls are filled with working video equipment.
⇒ Les salles, très hautes de plafond, sont pleines d'équipements vidéo.

(23) The terminology implies an appeal to the goodwill of the protagonists in the field [...] to "deal with" things, [...].
⇒ Si ce registre fait appel à la bonne volonté des acteurs pour "gérer", [...].

3.4 Phraseology

In an attempt to determine the influence of phraseology on translational equivalence (cf. our second hypothesis), we calculated the number of bound and free uses of both *with* and *avec*. The comparable corpus data reveal that *with* has significantly more bound uses than *avec* (62% vs. 48%, $\chi^2 = 7.61$, $p \leq 0.01$).[9] This, together with the higher polysemy of *with* (see Section 3.3), seems to explain the low mutual translatability between *with* and *avec*, and more precisely, the fact that *with* is less frequently translated by *avec* than the other way round. If, as is often assumed in the literature (see above), bound combinations are more difficult to translate word by word than freer combinations, then it is only normal that *with*, which predominantly occurs in bound combinations, should rarely be translated by its direct equivalent *avec*. This, undoubtedly, would demonstrate that phraseology is a decisive factor in the choice of a translation.

However, things are more complex than this. As appears from Table 3, and contrary to expectations, the parallel corpus data show that translational equivalence is in fact significantly higher for the bound uses of *with* than for its free uses (34% vs. 24%, $\chi^2 = 3.92$, $p \leq 0.05$). In other words, the English preposition is more likely to be translated by its direct French equivalent if it is part of a restricted multi-word expression. Thus, the corpus data contain a large number of cases where *with* is used as a bound preposition and translated by *avec* in French (24). Cases where *with* is used as a free

9. Interestingly, this phraseological difference between *with* and *avec* does not exist in the translated part of the corpus (i.e. English texts translated from French and French texts translated from English), as *with* and *avec* display a roughly similar proportion of bound uses (65% and 61% respectively). The high proportion of bound uses of *avec* in translated French might be a manifestation of so-called 'translationese', i.e. influence of the source language. Such a hypothesis would, however, require further investigation.

Table 3. Translational equivalence for bound vs. free uses of *with* (tokens)

	# with	*# with = avec*
Bound uses	192 (100%)	66 (34.37%)
Free uses	118 (100%)	28 (23.73%)
TOTAL	310 (100%)	94 (30.32%)

preposition and translated by *avec*, by contrast, are less frequent, and most of the time, it is another translation that is chosen, as in (25).[10]

(24) [...] the Iranians are not yet prepared to move *relations* with Israel to a different plane.
⇒ [...] les Iraniens ne sont pas encore prêts à reconsidérer leurs *relations* avec Israël.

(25) With ethnic tensions thus exacerbated, the UN's task will remain even more daunting [...].
⇒ Compte tenu de l'exacerbation des tensions interethniques, la tâche des Nations unies s'annonce gigantesque [...].

This somewhat surprising finding, however, should be put into perspective and considered together with two important caveats. The first one has to do with the type/token ratio. As is typical of idioms and other phraseological expressions, the bound uses of *with* tend to be repeated in the corpus (one type, but several tokens), e.g. *agreement with* (6 times), *relation with* (5 times) or *with the help of* (3 times). Several of these repeated combinations have a word-for-word equivalent in French (*accord avec*, *relation avec*, *avec l'aide de*), as shown in examples (26) to (28). The free combinations, by contrast, are rarely repeated. The recurrence of bound combinations (as opposed to free combinations) therefore has the effect of inflating the translational equivalence for the bound uses of *with*.

(26) If an *agreement* with Syria fails to materialise, Israel will need another rationale to be able to profit from this extraordinary manna.
⇒ Si un tel *accord* avec Damas échouait, Israël devra trouver d'autres arguments pour pouvoir, quand même, bénéficier de cette extraordinaire manne.

(27) One of the central arguments made in Israel's internal deliberations on its *relations* with Iran is the need not to reduce the momentum in the US [...].
⇒ L'un des principaux arguments dans les délibérations internes israéliennes

10. See Chuquet & Chuquet (2006) on the difficulties involved in translating sentences such as (25), where *with* has a causal value.

Table 4. Translational equivalence for the syntactic categories of the bound uses of *with* (tokens)

	# with	*# with = avec*
Noun + *with*	69 (100%)	44 (63.77%)
Adj. + *with*	30 (100%)	0 (0.00%)
V + *with*	66 (100%)	13 (19.70%)
Complex prep.	27 (100%)	9 (33.33%)
Total bound uses	192 (100%)	66 (34.37%)

sur les *relations* avec l'Iran est la nécessité de ne pas freiner l'élan pris aux États-Unis [...].

(28) Its mission, as described by its president, is "to develop ideas and get them into mainstream circulation - with *the help of* the 'media food chain'".
⇒ Sa mission, explique son président, est "de développer des idées et de les mettre en circulation auprès du grand public" avec, précise-t-il, *l'aide de* la "chaîne alimentaire des médias".

The second caveat concerns the syntactic analysis, which, as pointed out by Schmied (1998), tends to be neglected in contrastive analyses. If we make a distinction – as shown in Table 4 – between the different bound uses of *with* according to the nature of the word on which *with* is dependent (a noun, e.g. *relation with*; an adjective, e.g. *comparable with*; a verb, e.g. *comply with*; or a preposition, in the case of complex prepositions such as *along with*), it appears that the translational equivalence is much higher with nouns (64%) than with complex prepositions (33%), verbs (20%) or adjectives (no *avec*-equivalents). In other words, the high translational equivalence for the bound uses of *with* is mainly due to the nominal bound uses, with equivalences such as those illustrated by (29) to (31).

(29) Can real *agreement* be reached with him in the absence of any glimmer of democratisation in Palestine?
⇒ Peut-on signer un *accord* avec lui en l'absence du moindre signe de démocratisation en Palestine?

(30) And Tehran would be deprived of its strategic *relationship* with Syria.
⇒ L'Iran devait être privé de sa *relation* stratégique avec la Syrie.

(31) *Contacts* with Indonesian military officers [...] were "very significant factors in determining the favourable orientation of the new Indonesian political elite".
⇒ Des *contacts* avec les officiers indonésiens [...] ont été "des facteurs très

significatifs dans le choix d'une orientation favorable par les nouvelles élites politiques".

This factor, of course, combines with the preceding one, with many of these nominal uses being recurrent in the corpus. It also combines with the results of the semantic analysis (cf. our first hypothesis), since many of these nouns express the prototypical sense of mutual relation, rather than the more metaphorical senses which are characteristic of *with*. The syntactic aspect can therefore be said to complement – and sometimes override – the phraseological and semantic aspects.

4. Conclusion

It appears from this study that prepositions are not the annoying little surface peculiarity that they were once thought to be. In fact, they are worth investigating, preferably on the basis of corpus data, and their investigation can produce surprising results – in our case, the fact that *with* and *avec*, which look like direct equivalents, actually display a very low mutual translatability. Furthermore, thanks to the examination of the contrastive corpus data, it is possible to explain this lack of equivalence (e.g. by the higher polysemy of *with*) and to show in which cases equivalence is more likely to occur (e.g. with nominal bound uses).

Such findings have important implications for bilingual lexicography and foreign language teaching (FLT). Several scholars such as Brala (2002) have underlined the largely unsatisfactory treatment of prepositions in dictionaries. More precisely, this study points to two weaknesses of this treatment. First, we have seen that the translation mentioned first in dictionaries and presented as the most obvious equivalent is not necessarily the most frequent one. Second, as Schmied (1998: 271) himself suggests, syntactic information should not be neglected, as it turns out to be of great importance in translation.

As for FLT, it should also take the results of studies such as this one into account. While it is certainly true that "collocations do deserve a place in language teaching" (Nesselhauf 2003: 238), they should not automatically be seen as a source of problems for learners. Granted, there are bound uses which have no direct equivalent in learners' mother tongues and these should be brought to their attention (when possible, with the help of semantic clusters equivalence, e.g. CONTAINMENT = *with* / *de*). But there are also bound uses which have a word-for-word translation in the other language and hence may be acquired effortlessly by learners.[11] In other words, phraseology does not only lead to negative, but also to positive transfer. In addition, many free uses are in fact not directly translatable into learners' mother tongues and should therefore be included in the curriculum (see also Nesselhauf 2003). Adverbial *with* followed by a

11. But need not, cf. Nesselhauf (2003), who shows that mistakes can also be made when collocations are congruent.

non-finite or verbless clause (see example (25) in Section 3.4) is a case in point. Because such constructions do not normally have a word-for-word equivalent in French, they are likely to be problematic for French-speaking learners of English. We argue that learners' attention should be directed to any uses, free or bound, which have no direct equivalent in their mother tongues. Adapting foreign language teaching accordingly will eventually help ease the "colossal task" (Brala 2002: 1) of mastering the use of prepositions.

Acknowledgements

We wish to thank Sylviane Granger as well as two anonymous reviewers for their comments on an earlier version of this chapter. We are also grateful to the Belgian National Fund for Scientific Research (FNRS) for funding our research.

References

Altenberg, B. (1999). Adverbial connectors in English and Swedish: Semantic and lexical correspondences. In Hasselgård, H. & S. Oksefjell (eds.) *Out of Corpora. Studies in Honour of Stig Johansson*, 249–268. Amsterdam: Rodopi.

Altenberg, B. (2001). Contrasting delexical English *make* and Swedish *göra*. In Aijmer, K. (ed.) *A Wealth of English. Studies in Honour of Göran Kjellmer*, 195–219. Göteborg: Acta Universitatis Gothoburgensis.

Atkins, B. T., A. Duval & R. C. Milne (1998). *Collins Robert Unabridged French-English, English-French Dictionary.* 5th edn. Glasgow: HarperCollins.

Biber, D., S. Johansson, G. Leech, S. Conrad & E. Finegan (1999). *Longman Grammar of Spoken and Written English*. London: Longman.

Brala, M. (2002). *Understanding and Translating (Spatial) Prepositions: An Exercise in Cognitive Semantics for Lexicographic Purposes* [Working paper of the Research Centre for English and Applied Linguistics]. Cambridge: Cambridge University.

Chuquet, H. & J. Chuquet (2006). De la « circonstance » à la « cause » : les prédications en *with* et leur traduction en français. In Paillard, M. & H. Chuquet (eds.) *Causalité et contrastivité: études de corpus*, 189–210. Rennes: Presses Universitaires de Rennes.

Cosme, C. (2006). Clause combining across languages. A corpus-based study of English-French translation shifts. *Languages in Contrast* 6(1): 71–108.

Dirven, R. (1993). Dividing up physical and mental space into conceptual categories by means of English prepositions. In Zelinski-Wibbelt, C. (ed.) *The Semantics of Prepositions: From Mental Processing to Natural Language Processing*, 73–97. Berlin: Mouton de Gruyter.

Downing, A. & P. Locke (1992). *A University Course in English Grammar*. Hertfordshire: Prentice Hall.

Francis, G. (1993). A corpus-driven approach to grammar. Principles, methods and examples. In Baker, M., G. Francis & E. Tognini-Bonelli (eds.) *Text and Technology. In Honour of John Sinclair*, 137–156. Amsterdam: John Benjamins.

Gilquin, G. (2000/2001). The integrated contrastive model: Spicing up your data. *Languages in Contrast* 3(1): 95–123.

Hanse, J. (1987). *Nouveau dictionnaire des difficultés du français moderne. Deuxième édition mise à jour et enrichie.* Louvain-la-Neuve: Duculot.

Jackendoff, R. (1973). The base rules for prepositional phrases. In Anderson, S. R. & P. Kiparky (eds.) *A Festschrift for Morris Halle*, 345–356. New York, NY: Holt, Rinehart and Winston.

Jayez, J. & A. Mari (2000). *Avec*: la solidarité par les canaux. Unpublished manuscript. Available at <http: //perso.wanadoo.fr/jjayez/avec/avec.pdf> (retrieved on 22 March, 2006).

Lakoff, G. & M. Johnson (1980). *Metaphors We Live By*. Chicago, IL: The University of Chicago Press.

Leech, G. & J. Svartvik (1994). *A Communicative Grammar of English*. London: Longman.

Lewis, M. (1997). *Implementing the Lexical Approach. Putting Theory into Practice*. Australia: Thomson Heinle.

Lindstromberg, S. (2001). Preposition entries in UK monolingual learners' dictionaries: Problems and possible solutions. *Applied Linguistics* 22(1): 79–103.

Nesselhauf, N. (2003). The use of collocations by advanced learners of English and some implications for teaching. *Applied Linguistics* 24(2): 223–242.

Quirk, R., S. Greenbaum, G. Leech & J. Svartvik (1985). *A Comprehensive Grammar of the English Language*. London: Longman.

Schmied, J. (1998). Differences and similarities of close cognates: English *with* and German *mit*. In Johansson, S. & S. Oksefjell (eds.) *Corpora and Cross-Linguistic Research. Theory, Method and Case Studies*, 255–274. Amsterdam: Rodopi.

Scott, M. R. (1996). *WordSmith Tools*. Oxford: Oxford University Press.

Sinclair, J. (1991). *Corpus, Concordance, Collocation*. Oxford: Oxford University Press.

Stubbs, M. (1995). Corpus evidence for norms of lexical collocation. In Cook, G. & B. Seidlhofer (eds.) *Principle and Practice in Applied Linguistics*, 245–256. Oxford: Oxford University Press.

Tognini-Bonelli, E. (2001). *Corpus Linguistics at Work*. Amsterdam: John Benjamins.

Tyler, A. & V. Evans (2003). *The Semantics of English Prepositions. Spatial Scenes, Embodied Meaning, and Cognition*. Cambridge: Cambridge University Press.

Van Hoof, H. (1989). *Traduire l'anglais: théorie et pratique*. Louvain-la-Neuve: Duculot.

Van Roey, J. (1990). *French-English Contrastive Lexicology. An Introduction*. Louvain-la-Neuve: Peeters.

Váradi, T. & G. Kiss (2001). Equivalence and non-equivalence in parallel corpora. *International Journal of Corpus Linguistics* 6: 155–166.

Viberg, Å. (1998). Contrasts in polysemy and differentiation: Running and putting in English and Swedish. In Johansson, S. & S. Oksefjell (eds.) *Corpora and Cross-Linguistic Research. Theory, Method and Case Studies*, 343–376. Amsterdam: Rodopi.

Viberg, Å. (2002). Polysemy and disambiguation cues across languages. The case of Swedish *få* and English *get*. In Altenberg, B. & S. Granger (eds.) *Lexis in Contrast. Corpus-Based Approaches*, 119–148. Amsterdam: John Benjamins.

Viberg, Å. (2004). Physical contact verbs in English and Swedish from the perspective of crosslinguistic lexicology. In Aijmer, K. & B. Altenberg (eds.) *Advances in Corpus Linguistics. Papers from the 23rd International Conference on English Language Research on Computerized Corpora (ICAME 23), Göteborg 22–26 May 2002*, 327–352.

Vinay, J.-P. & J. Darbelnet (1995). *Comparative Stylistics of French and English. A Methodology for Translation*. Translated and edited by Sager, J. C. & M.-J. Hamel. Amsterdam: John Benjamins.

Vreck, F. (2000). *Entraînement à la version anglaise*. Paris: Ophrys/Ploton.

Woolls, D. (1997). *Multiconcord. The Lingua Multilingual Parallel Concordancer for Windows*. Birmingham: CFL Software Development.

Zelinski-Wibbelt, C. (ed.) (1993a). *The Semantics of Prepositions: From Mental Processing to Natural Language Processing*. Berlin: Mouton de Gruyter.

Zelinski-Wibbelt, C. (1993b). Introduction. In Zelinski–Wibbelt, C. (ed.) *The Semantics of Prepositions: From Mental Processing to Natural Language Processing*, 1–24. Berlin: Mouton de Gruyter.

Contrastive idiom analysis

The case of Japanese and English idioms of anger

Priscilla Ishida

Past approaches to contrastive idiom analysis have often focused on shared characteristics of groups of idioms or taken for granted correspondences between the meanings of individual L1/L2 idioms. This chapter presents a method for the contrastive analysis of idioms that focuses on the L1/L2 semantic networks, as well as on the way that idioms are actually used in text and discourse. The target of analysis is Japanese and English verb phrase idioms of anger, including *hara ga tatsu* 'one's belly rises up', *blow one's stack/top,* etc. Corpus data and co-occurrence tests are used to identify a number of semantic features that function to distinguish the meanings of idioms in the respective L1/L2 lexical fields (e.g. <continuative>, <instantaneous>, <time before realisation>, <other-oriented>, <expressive>). Results of analysis show that, while many Japanese and English anger idioms overlap partially in meaning, only a few have full semantic correspondence. This chapter also points out some non-semantic factors relevant to the question of idiom translation in real contexts, including register and frequency of occurrence.

1. Introduction

Idioms present many difficulties for learners and teachers of foreign languages, for translators, and for editors of bilingual dictionaries. One question of particular importance is how to judge whether an L1 and L2 idiom with similar meaning are truly equivalent, or whether they simply overlap partially in meaning and usage.

The aim of this chapter is to present a method for the contrastive analysis of idioms with similar meaning in different languages. The target of study is Japanese and English verb phrase idioms that express the meaning of anger, as listed in (1) and (2) respectively. Details on the selection of these idioms can be found in Section 3.

(1) *atama ni kuru* ('[it] comes to one's head'), *kanshaku o okosu* ('have a fit of anger'), *kanninbukuro no o ga kireru* ('the cord of one's patience-bag snaps'), *hara ga tatsu* ('one's belly rises up')/*hara o tateru* ('raise up one's belly'), *hara ni suekaneru* ('be unable to keep [it] in one's belly'), *harawata ga niekurikaeru* ('one's guts boil');

(2) *bite s.o.'s head off, blow/let off steam, blow one's stack/top, fly off the handle, get/be hot under the collar, one's blood boils/make one's blood boil, see red*.[1]

Many definitions of 'idiom' can be found in the literature. Some are based on a single criterion, such as non-compositionality (Fraser 1970; Makkai 1972) or conventional co-occurrence (Fernando 1996). Others are based on multiple criteria, including semantic, syntactic, and/or lexical stability (Barkema 1996; Itō 1997b) as well as figuration, institutionalisation, evaluative meaning, and so on (Nunberg et al. 1994; Moon 1998). The lack of scholarly agreement on the definition of idiom is a question that requires further discussion; however, such discussion is beyond the scope of this chapter. Here I will use the following multi-dimensional definition, which draws on past research on both Japanese and English idioms:

Idioms are multi-word expressions with the following three properties:

i. formal frozenness: idioms usually do not allow the replacement or deletion of constituents or changes in phrase structure (Fernando 1996; Itō 1997b; Ishida 1998; Moon 1998).
ii. syntactic frozenness: idioms are resistant to grammatical operations (passivisation, adjunction of adnominal modifiers, etc.) that ordinary collocations with the same syntactic structure would allow (Fraser 1970; Gibbs & Gonzales 1985; Ishida 2000).
iii. semantic frozenness: the phrasal meaning of an idiom is not the sum total of the meanings of its individual constituents (Nunberg et al. 1994; Barkema 1996; Ishida 2004). For example, the meaning 'get angry' is not predictable from the usual meanings of *atama* 'head,' *ni* 'to', and *kuru* 'come'.

Of course, individual idioms vary with respect to the degree of each of these properties (Gibbs & Gonzales 1985; Gibbs 1990; Nunberg et al. 1994; Barkema 1996; Ishida 2004). Idioms with a relatively high degree of these properties (e.g. *atama ni kuru*) are prototypical; idioms with a relatively low degree (e.g. *me o mukeru*, 'turn one's eyes [to]') are borderline, i.e. close to the border between idioms and ordinary collocations. For a detailed discussion of these properties and how they can be measured, see Ishida (1998, 2000 and 2004).

Also, although idioms share the three properties listed above with other multi-word expressions – including proverbs/maxims, compounds, and phrasal verbs – this chapter excludes the latter categories from consideration. This is because these categories can be distinguished from idioms on the basis of lexico-grammatical structure and discoursal function (Miyaji 1985; Moon 1998), and because it is necessary to limit an analysis of this nature to a homogenous target of investigation.

1. *Blow/let off steam* can be used to refer to the release of stress or pent-up energy, and *get/be hot under the collar* sometimes refers to sexual arousal. This chapter, however, limits its focus to uses referring to anger.

A brief word of explanation is required regarding pairs such as *hara ga tatsu/hara o tateru* and *one's blood boils/make one's blood boil*. These are institutionalised idiom variants that share some, but not all, of the same lexical items and syntactic structure (Ishida 1998; Moon 1998). In (1) and (2), institutionalised lexico-syntactic variants are separated by a slash mark. These variants share a common core of meaning; however, as the following analysis shows, some also reflect important semantic distinctions.

2. Past approaches to the cross-linguistic analysis of idioms

Past approaches to the cross-linguistic analysis of idioms represent a variety of perspectives. In a series of papers on German and Japanese idioms, Itō presents a typological analysis of the constituents, syntactic structure, and meanings of L1/L2 idioms (1992); compares the concrete images associated with the literal meanings of L1/L2 anger idioms (1997a); and compares the figurative meanings of selected idiom constituents, including *Blut/chi* ('blood') and *Hand/te* ('hand') (1997b, 1999). Ueda (1994, 2003) classifies German idioms according to the 'fields' to which their kernel constituents belong (physical actions, animals, colours, etc.) and then analyses sub-groups of idioms with the same kernel constituent (*Arm, Wolf, grün,* etc.). He points out cases of semantic correspondence between German and Japanese idioms, as well as cases of 'false friends' and lack of a corresponding idiom.

These approaches provide useful frameworks for the comparison of L1/L2 idioms. However, a difficulty with both the typological approach (Itō 1992) and the 'field' approach (Ueda 1994, 2003) is that correspondence between the meanings of L1/L2 idioms is taken for granted, with no analysis of the network of semantic relationships between idioms with similar meaning in each language, or of how each idiom is actually used in text and discourse. The 'kernel constituent' approach is useful to clarify similarities and differences in the range of meanings associated with corresponding L1/L2 constituents. However, it does not address the question of semantic differences between idioms of the same language classified into the same group, or the question of semantic correspondence between individual L1/L2 idioms.

Matsuki (1995) uses a cognitive approach to identify Japanese expressions of anger that reflect the conceptual metaphors and metonymies given by Kövecses (1986) for English. She also proposes, for Japanese alone, the metaphor ANGER IS (IN THE) *HARA* ('belly') and argues that the prototypical scenario for anger in Japanese includes movement from *hara* to *mune* ('chest') and finally up to *atama* ('head').[2] Charteris-Black (2003) combines the conceptual metaphor approach with a corpus-based approach, proposing a model for the comparison of English and Malay idioms based on patterns of correspondence between the lexical constituents and conceptual basis of

2. McVeigh (1996) and Hasada (2002) also show that *hara* represents a locus of emotional activity in Japanese, including decision-making and intention as well as anger.

L1/L2 idioms. While Matsuki focuses on correspondences between L1/L2 conceptual metaphors and similarities between groups of idioms, Charteris-Black also identifies correspondences between individual L1/L2 idioms.

Dobrovol'skij (1998) focuses on the combinatorial properties of Russian and German idioms and shows that L2 equivalents for an L1 idiom vary depending on the class of noun with which the L1 idiom co-occurs. His analysis provides persuasive evidence for his assertion that few, if any, idioms have one-to-one equivalents in other languages. However, Dobrovol'skij's approach focuses on identifying different L2 equivalents for an L1 idiom in different contexts and leaves unaddressed the question of the L1 semantic network.

3. The present analysis: Theoretical framework and method of analysis

This chapter takes the view that, in order to identify semantic correspondences between individual L1/L2 idioms, it is necessary to look at both the L1/L2 semantic networks (Coseriu 1968; Coseriu & Geckeler 1981) and the way that idioms are actually used in text and discourse (Firth 1968; Sinclair 1987). Corpus data is indispensable for contrastive analysis, because it reveals patterns of use that reflect semantic distinctions – patterns that are not always revealed by native-speaker introspection alone. The method of analysis in this chapter is based on the following assumptions:

a. Contrastive analysis is synchronically oriented and aims to identify both similarities and differences between L1/L2 idioms (Takada 1974; Dobrovol'skij 1998).
b. Idioms have the property of 'lexicality.' This means that although they are phrases composed of two or more words, their structure and constituents are stable, and they function in sentences much like single words (Muraki 1985; Itō 1997b). In other words, VP idioms function as verbs, NP idioms as nouns, etc.[3]
c. Idioms belong to lexical fields in which they function in opposition to each other as well as to single words. A lexical field is a group of lexical units that share a common semantic feature but also have distinctive semantic features that set them apart from each other (Coseriu & Geckeler 1981; Ishiwata & Takada 1990).
d. The meaning of an idiom can be analysed by applying co-occurrence tests developed from naturally-occurring data and comparing the idiom's range of use to that of other idioms/words in the same lexical field (Ishida 2003). Co-occurrence restrictions reflect semantic constraints and provide evidence for the distinctive semantic features of individual expressions.

3. The fixed correspondence between an idiom phrase and its idiomatic meaning does not obviate the possibility that in some cases individual constituents contribute semantically to idiomatic meaning (Ishida 2004: 49). For example, it can be thought that the verb *kireru* ('break, snap') contributes the feature <instantaneous> to the meaning of *kanninbukuro no o ga kireru* (4.1).

Based on the preceding assumptions, this chapter proposes the following method for contrastive analysis and presents an analysis of Japanese and English anger idioms (Section 4) that was carried out using this method:

– **Step 1: Delimitation of L1/L2 lexical fields.** Substitution tests were used to delimit the lexical fields of Japanese (1) and English (2) idioms of <<anger>>.[4] These fields contain other idioms and verbs as well (e.g. (1) *okoru* 'get angry'; (2) *hit the ceiling*). However, the aim of this chapter is not to carry out an exhaustive analysis; it is to present a method of analysis. Thus focus is limited to a number of idioms used relatively frequently in spoken and written language. The benchmark for inclusion in this analysis was a minimum frequency of between 0.1 and 0.3 occurrences per million words (see 'Data sources').
– **Step 2: L1 analysis.** Examples of Japanese idioms were retrieved from newspaper databases and analysed in order to develop a number of co-occurrence tests. Questionnaires (based on newspaper examples) were prepared for each test and five native speaker informants were interviewed to obtain acceptability judgments. Questionnaire results were then analysed and used to identify distinctive semantic features of L1 idioms.
– **Step 3: L2 analysis.** Step 2 was repeated for English idioms. In this case, acceptability judgments were provided by the researcher and two other informants, all native speakers of North-American English.
– **Step 4: Contrastive analysis.** Similarities and differences between L1/L2 distinctive features and semantic correspondences between L1/L2 idioms were identified.

In practice, the steps listed above are not entirely discrete. During L1 analysis it is important to consider the applicability of L1 semantic features to L2 idioms. During L2 analysis it is necessary to test the applicability of L1 semantic features and to consider the applicability of any newly-identified L2 features to L1 idioms. Some contrastive analysis thus takes place concurrently with L1/L2 analysis.

Section 4 (below) describes six co-occurrence tests used in this analysis and five sets of distinctive features identified as a result.

4. In the sentence *When John saw my bumper sticker, he {flew off the handle/saw red}*, both *flew off the handle* and *saw red* express that 'John' got angry. Substitutability in one context does not imply complete semantic correspondence; it does, however, show that the idioms in question share a common semantic feature – in this case, <<anger>>. (Double brackets (<< >>) indicate common semantic features and single brackets (< >) indicate distinctive semantic features.)

4. Contrastive analysis: co-occurrence tests and distinctive features

4.1 <continuative (c)>, <instantaneous (i)>, <stative (s)>, <continuative/stative (c/s)>

One way that both Japanese and English anger idioms differ from each other is in terms of their aspectual properties – i.e., how they present anger situations in relation to time. This is evident, first of all, from patterns of use with progressive forms. Some Japanese anger idioms have progressive interpretation in the *-te iru* form,[5] and some English idioms have progressive interpretation in the *be V-ing* form.

(1) *Uchi no mae de no meiwaku-chūsha ni hara o tatete iru*. ('I am raising my belly at the illegal parking in front of my house.')

(2) The season hasn't even started yet and Chiba Lotte Marines manager Bobby Valentine is already getting hot under the collar.

The anger situation expressed by (1) *hara o tatete iru* is interpreted as being in progress at the present moment. It is a psychological activity that is dynamic and extends over a period of time. This idiom thus has an affinity with so-called 'continuative verbs.' The same can be said about (2) *get hot under the collar*.

Other idioms, however, do not allow progressive interpretation in the *-te iru/be V-ing* forms.

(3) *Musume no renjitsu no asagaeri ni kanninbukuro no o ga {kireta/??kirete iru}*. ('The cord of my patience bag {snapped/??has snapped} at my daughter's staying out all night so many times in a row.')

(4) #The season hasn't even started yet and Chiba Lotte Marines manager Bobby Valentine is already blowing his top.

As we see in (3), *kanninbukuro no o ga kireru* is resistant to use in the *-te iru* form. Progressive interpretation is impossible, and even resultative interpretation is highly unlikely. We can thus infer that this idiom expresses an anger event that takes place instantaneously and ends immediately after realisation, without any succeeding resultative state. On the other hand, (4) shows that *blow one's top* allows the *be V-ing* form, but not with progressive interpretation (the hash symbol indicates a sentence that is acceptable to native speakers but does not express the meaning under consideration).

---

5. The *-te iru* construction has two possible readings, depending on the type of verb to which it attaches. The first expresses an activity in progress at the present moment (e.g. *hashitte iru* '[he] is running'). The second expresses a state of affairs that results from an instantaneous event (e.g. *kaeru ga shinde iru* 'the frog is dead'). Japanese verbs with progressive interpretation in the *-te iru* form are classified as 'continuative verbs'; those with resultative interpretation are 'instantaneous verbs' (Kindaichi 1950; Machida 1989).

Here the interpretation is iterative: the subject is experiencing repeated episodes of anger over a period of time, and each episode takes place at a point in time.

Based on the preceding examples, I will propose here that Japanese and English anger idioms can be differentiated by the semantic features <continuative> and <instantaneous>. Idioms that allow progressive interpretation in the *-te iru/-ing* forms have the feature <continuative> (e.g. *blow/let off steam, see red*); idioms that do not allow progressive interpretation in these forms are <instantaneous> (e.g. *bite s.o.'s head off, fly off the handle*).[6]

Not all anger idioms, however, are analysable as either <continuative> or <instantaneous>; some have <stative> properties. One clear example is *be hot under the collar,* which contains the stative verb 'be' and, predictably, has present interpretation in the simple present only ('John {is/*is being} hot under the collar').

Japanese stative verbs (*aru* 'be,' *dekiru* 'be able to,' etc.) have present interpretation in the *-ru* (plain) form only and are never used in the *-te iru* form. None of the Japanese idioms under consideration here is analysable as purely <stative>, because none has present interpretation in the *-ru* form only. Some, however, allow present interpretation in both the *-ru* and the *-te iru* forms.

(5) *Fan, senshu o mushi shita ōnā-tachi ni hara ga {tatsu/tatte iru}.* ('My belly {rises up/is rising up} at the owners, who ignored the fans and the players.')

In (5), both *hara ga tatsu* (the *-ru* form) and *hara ga tatte iru* convey the meaning that the speaker was angry at the time he uttered this statement. Because this idiom has present interpretation in both of these forms, it can be said to have both continuative and stative properties (<continuative/stative>). Several other Japanese idioms have this feature too, including *atama ni kuru, hara ni suekaneru,* and *harawata ga niekurikaeru.*[7]

Are there any English anger idioms that have present interpretation in both the simple present and the progressive *-ing* form?

(6) 'It {makes/is making} my blood boil,' Clinton told police officials here, as he blasted the 225–210 procedural vote that blocked final consideration of the $33.2 billion anti-crime measure.

---

6. Labeling for the features proposed in this section borrows from Kindaichi's (1950) terminology for the aspectual properties of Japanese verbs. However, this chapter follows the view that the classes 'stative,' 'continuative,' and 'instantaneous' for Japanese verbs correspond more or less to 'states,' 'activities,' and 'achievements' for English verbs (Tsujimura 1996).

7. Baba (2001: 164ff.) also points out that *hara ga tatsu* etc. have an affinity with stative verbs, based on their potential for use in the *-ru* form with present meaning. Idioms analysed in Section 4.1 as purely <continuative> (*hara o tateru* etc.) have present interpretation in the *-te iru/be -ing* form only. When used in the *-ru* form, these usually have habitual or conditional interpretation.

In (6), 'It makes my blood boil' conveys the meaning that the speaker experiences the anger emotion every time he thinks about the referent of 'it' – including the time of utterance of this statement. Here we have present interpretation in the simple present, as well as in the present progressive. This shows that *make one's blood boil,* like (5) *hara ga tatsu*, has both continuative and stative properties. However, this is the only English idiom under consideration here that is analysable as <continuative/stative>.[8]

4.2 <time before realisation>

Both Japanese and English anger idioms vary with respect to how they present the time preceding realisation of the anger situation. This is evident from the kinds of adverbs with which they co-occur:

(7) *Sono chichi wa, musume no renjitsu no asagaeri ni tsuini kanninbukuro no o ga kireta.* ('Finally the cord of the father's patience-bag snapped at his daughter's staying out all night so many times in a row.')

(8) *Ramuzuferudo wa 'Bagudaddo wa muhō-chitai' to iu hōdō o kiki, {??suguni/??kyūni} kanninbukuro no o ga kireta.* ('When Rumsfeld heard the report "Baghdad is a lawless area," *{??immediately/??suddenly}* the cord of his patience-bag snapped.')

(9) After putting up with John's rude and inappropriate remarks all evening, I finally {let off some steam/blew my stack}.

(10) When John saw my bumper sticker, all of a sudden he {*let off some steam/?blew his stack}.

Kanninbukuro no o ga kireru and *let off steam/blow one's stack* are very natural in (7) and (9) respectively, where the co-text includes reference to a series of triggering events and the adverbs *tsuini* ('finally') and *finally.* On the other hand, these idioms are unacceptable in (8) and (10), where the co-text and the adverbs *suguni* ('immediately')/*kyūni* ('suddenly') and *all of a sudden* indicate that there is little or no time between a triggering event and the anger event. These examples suggest that the idioms in question present anger events whose realisation is preceded by a period of 'build-up': the anger event takes place only after a series of triggering events, or a situation that develops over time.

In contrast to (9) and (10), some English idioms are incompatible with *finally* but compatible with *immediately/suddenly.*

---

8. Other causative and causative/transitive anger expressions in English also have present interpretation in the simple present (e.g. *make s.o. angry/mad, piss s.o. off,* etc.). The link between causative form and present interpretation requires further investigation; however, such investigation is beyond the scope of the present chapter.

(9') ...I finally {?flew off the handle/??saw red}.

(10') ...all of a sudden he {flew off the handle/saw red}.

(9') and (10') show that *fly off the handle* and *see red* present anger events that are realised abruptly and without warning. These idioms do not include in their meaning any reference to the time preceding realisation of the anger event.

From examples (7)–(10'), it is possible to argue that Japanese and English anger idioms can be differentiated with the semantic feature <time before realisation>. (This is an abbreviated version of the feature identified by Nagashima (1982) for the adverbs *tsuini/tōtō* 'finally'.[9]) Japanese idioms that co-occur with *tsuini/tōtō* but not *suguni/kyūni* and English idioms that co-occur with *finally* but not *immediately/suddenly* are analysed as <+time before realisation> (e.g. *blow/let off steam*). English idioms that co-occur with *immediately/suddenly* but not *finally* are <–time before realisation> (e.g. *fly off the handle)*.

None of the Japanese idioms under consideration here is analysable as <–time before realisation>, because none co-occurs with *suguni/kyūni* but not *tsuini/tōtō*. However, some co-occur with all of these adverbs: for instance, *{tsuini/suguni} atama ni kita* ('{finally/immediately} [it] came to my head'). Depending on the context, *atama ni kuru* may express an anger event that takes place after a period of build-up over time, or one that takes place abruptly, with little or no warning. This idiom and others like it are analysed as neutral with respect to the length of time required before realisation (<±>).

No English idioms are analysed as <±time before realisation>, because none co-occurs with both *finally* and *immediately/suddenly*. Several idioms, however, are resistant to use with all of these adverbs – e.g. '{??finally/??all of a sudden} my blood boiled'. The key here is the focus that all three adverbs have on the moment of realisation of an event. They co-occur with verbs presenting situations that are realised **at a point in time**. Resistance to use with these adverbs suggests that *one's blood boils* presents an event that is not realised at a point in time. This question will be pursued in the following section; here it is sufficient to say that the feature <time before realisation> does not apply to *one's blood boils,* or to other English idioms that resist co-occurrence with *finally* and *immediately/suddenly.* The same can be said for Japanese idioms that do not co-occur with *tsuini/tōtō* or *suguni/kyūni* (e.g. *hara ga tatsu)*.

4.3 <gradual realisation>

Japanese and English anger idioms vary not only with respect to how they present time before realisation of the anger situation, but also with respect to how they present realisation itself. See, for example, the following uses of *hara ga tatsu* and *one's blood boils:*

9. *Tsuini* is usually interchangeable with *tōtō*, although *tōtō* is more natural in spoken Japanese.

(11) *Dandan hara ga tatte kita. Seiji wa nani o shiteru.* ('My belly gradually started to rise up. What's the government doing?')

(12) 'When I went to the supermarket and walked down the cold-cereal aisle, my blood gradually started to boil,' Schumer said. 'The prices are just way out of line.'

The *-te kuru* ('start/begin to') construction in (11) and *start to*~ in (12) indicate the inception of change in the experiencer from a situation of non-anger to one of anger. The adverbs *dandan* ('gradually') and *gradually* specify that this change is realised in stages over time.

Other idioms, however, do not co-occur with these constructions/adverbs. **Dandan kanninbukuro no o ga kirete kita* (*'Gradually the cord of my patience-bag began to snap') and *'I gradually started to see red,' for example, are anomalous. Resistance to use with *dandan/gradually* and *-te kuru/start to* shows that *kanninbukuro no o ga kireru* and *see red* express situations that are realised not in stages but at an undivided point in time. The compatibility of these idioms with the adverbs in Section 4.2 supports this view (cf. (7), (10')).

The examples above suggest that Japanese and English anger idioms can be differentiated using the semantic feature <gradual realisation>. Idioms that behave like *kanninbukuro no o ga kireru* and *see red* are <–gradual realisation> (e.g. *kanshaku o okosu; bite s.o.'s head off*); idioms like *hara ga tatsu* and *one's blood boils* are <+gradual realisation> (e.g. *harawata ga niekurikaeru; get hot under the collar*). What is interesting is that all the idioms analysed here as <+gradual realisation> are also 'n/a' with respect to <time before realisation> (4.2). In other words, compatibility with expressions referring to gradual realisation is accompanied by incompatibility with expressions focusing on the moment of realisation of an event.

It should be noted that the feature <gradual realisation> does not apply to <stative> idioms, which present static situations with no reference to a beginning or endpoint. Further, it is predictable that <instantaneous> idioms are also <–gradual realisation>, because these idioms, by definition, present situations realised at a point in time.

4.4 <other-oriented>

The examples given in Sections 4.1–4.3 suggest that Japanese and English anger idioms usually present situations in which the experiencer's anger is oriented toward an object (target or subject matter) or cause outside of himself (for a discussion of these terms, see Bandō & Matsumura 2001: 80–83). However, some idioms may also present situations in which the experiencer's anger is oriented toward himself or his own behaviour. As we see in (13) and (14), the Japanese idiom *hara ga tatsu* co-occurs naturally with the complements *jibun ni*~ ('~ at myself') and *jibun no*~ *ni*~ ('~ at my own~').

(13) *Chīmu no fun'iki o kaerarenai jibun ni hara ga tatta.* ('My belly rose up at myself, [who was] unable to change the team's mood.')

(14) *'Ēsu nanbā o moratte ita no ni, nani shiteru n da.' Jibun no shūtai ni hara ga tatta.* (' "What am I doing, when I used to be Number 1." My belly rose up at my own wretched performance.')

Atama ni kuru, kanshaku o okosu, and *hara o tateru* also take *jibun ni*~ or *jibun no*~ *ni*~ complements. Since these idioms can be used to indicate both self- and other-oriented anger, they are analysed here as having the semantic feature <± other-oriented>.[10]

Other idioms, however, do not take *jibun ni*~ or *jibun no*~ *ni*~ complements. For example, neither ??*jibun ni harawata ga niekurikaetta* (?'My guts boiled at myself') nor **jibun no shūtai ga hara ni suekanete*... (*'I couldn't keep my own wretched performance in my belly...') is acceptable. Since these idioms are restricted to use with external objects and causes, they are analysed as <+other-oriented>.

What about English idioms? As (15) suggests, most are resistant to co-occurrence with complements indicating self-directed anger (~*at myself,* ~*about/at/over my [own]*~) and are used solely with external objects and causes.[11] *Blow/let off steam* (16), however, is an exception.

(15) Warren flew off the handle at {his wife/*himself/the receiver's fumble/*his own fumble}.

(16) After the game, Dave blew off some steam over {his star player's/his own} wretched performance on the playing field.

As we see in the example above, *blow/let off steam* can be used with complements indicating that the experiencer's own behaviour is the subject matter of his anger. This idiom cannot be used with experiencer-objects (*Dave blew off some steam at himself}), and in this respect it differs somewhat from Japanese idioms such as *hara ga tatsu*. However, the fact that *blow/let off steam* can be used to express at least some self-oriented anger situations distinguishes this idiom from the other English anger idioms in this analysis.

---

10. It would be possible to stipulate another feature <self-oriented> along with <other-oriented>. However, since no idioms under consideration here are analysable as <+self-oriented>, it is sufficient to stipulate a single feature <other-oriented> and analyse idioms that can also be used in self-oriented contexts as <±other-oriented>.

11. *Bite s.o.'s head off* – unlike any of the other idioms under consideration here – has a slot for an external object built into its syntactic structure. An experiencer-object is thus structurally impossible (I bit {my husband's/*my own} head off), and an experiencer-related cause is semantically anomalous (John$_1$ bit my head off because {I/*he$_1$} made a few errors in the infield).

4.5 <expressive>

Some Japanese and English anger idioms include in their meaning indication of angry behaviour on the part of the experiencer – verbal output, physical action, etc. – and some do not. Compare *get hot under the collar* and *fly off the handle* in the examples below:

(17) Michael Johnson seemed to {get hot under the collar/*fly off the handle} when told the last hour of his daytime talk program would be cut.

(18) Michael {got hot under the collar/*flew off the handle}, but he held his tongue.

Get hot under the collar co-occurs with conjectural expressions such as *seem to* (17) and *look as if,* which indicate a degree of uncertainty on the part of the observer with respect to the anger situation in question. This idiom is also compatible with clauses like ~*but he held his tongue* (18), which specifies a lack of overt angry behaviour. *Fly off the handle,* on the other hand, is incompatible with both of these contexts. This shows that *fly off the handle* includes in its meaning some kind of observable expression of anger (<+expressive>), while *get hot under the collar* presents the internal anger emotion only (<–expressive>). Other <+expressive> idioms include *bite s.o.'s head off*[12] and *blow one's stack/top;* <–expressive> idioms include *make one's blood boil* and *see red*.

Japanese idioms can be differentiated in a similar manner, using tests for co-occurrence with the conjectural expressions ~*rashii/yō da* ('seem to/appear to') and with clauses such as ~*ga, jibun de wa kuchi ni dasenai* ('~but he himself can't talk about [it]').

(19) *Otōto wa nitchō kyōgi no shinten no nasa ni naishin {harawata ga niekurikaette ita/*kanshaku o okoshite ita} rashii.* ('My brother looked as if inwardly {his guts were boiling/*he was having a fit of anger} at the lack of progress in the Japan-Korea talks.')

(20) {*Harawata ga niekurikaette iru/*Kanshaku o okoshite iru} ga, jibun de wa kuchi ni dasenai to iu hito*... ('People {whose guts are boiling/*who are having a fit of anger} but they themselves can't talk about [it]...')

---

12. *Bite s.o.'s head off* differs from other <+expressive> idioms in that it presents a situation in which anger is expressed exclusively by verbal output. Compare this idiom to *fly off the handle* in the examples below:

a) He {flew off the handle/bit off his trainer's head}, ranting and raving like a maniac.

b) He flew off the handle, throwing the bat and physically abusing his trainer.

b') *He bit off his trainer's head, throwing the bat and physically abusing him.

Both of these idioms are semantically compatible with the verbal expression of anger indicated by 'ranting and raving' (a). However, only *fly off the handle* is compatible with contexts specifying the physical expression of anger (b).

The compatibility of *harawata ga niekurikaeru* with these contexts shows that this idiom presents an anger situation that is purely internal (<–expressive>). Further testing shows that this idiom – along with *atama ni kuru, hara ga tatsu, hara o tateru,* and *hara ni suekaneru* – actually requires the use of conjectural expressions like ∼*rashii/yō da* in contexts with a third-person experiencer. Use without such expressions suggests an omniscient narrator and is not usual in everyday discourse.

On the other hand, the incompatibility of *kanshaku o okosu* with the test items in (19) and (20) shows that this idiom presents a situation in which the experiencer exhibits his anger physically and/or verbally (<+expressive>). Use with ∼*rashii/yō da* is anomalous (19) or interpreted as hearsay ('Apparently he was having a fit of anger'). This resistance to use with conjectural expressions shows that *kanshaku o okosu* expresses angry behaviour that is readily observable by others.

5. Semantic correspondence and the problem of translation

Figure 1 shows that, while many Japanese and English anger idioms overlap in meaning, few have full semantic correspondence.

One exception is the pair *blow one's stack/top* and *kanninbukuro no o ga kireru,* which share all of the semantic features identified in this analysis. We might thus

	<c>,<s>, <c/s>,<i>,	**<time before realisation>**	**<gradual realisation>**	**<other-oriented>**	**<expressive>**
atama ni kuru	<c/s>	±	–	±	–
hara ga tatsu	<c/s>	n/a	+	±	–
get hot under the collar/	<c>	n/a	+	+	–
be hot under the collar	<s>	n/a	n/a	+	–
harawata ga niekurikaeru	<c/s>	n/a	+	+	–
make one's blood boil/	<c/s>	n/a	+	+	–
one's blood boils	<c>	n/a	+	+	–
hara ni suekaneru	<c/s>	+	–	+	–
kanshaku o okosu	<c>	±	–	±	+
blow/let off steam	<c>	+	–	±	+
hara o tateru	<c>	±	–	±	–
see red	<c>	–	–	+	–
fly off the handle	<i>	–	–	+	+
blow one's stack/top	<i>	+	–	+	+
kanninbukuro no o ga kireru	<i>	+	–	+	+
bite s.o.'s head off	<i>	–	–	+	+

<c>=<continuative>, <s>=<stative>, <c/s>=<continuative/stative>, <i>=<instantaneous>; n/a=not applicable

Figure 1. Correspondence between Japanese and English anger idioms

expect that these idioms will be translation equivalents in many contexts, including the following:

(8') *Ramuzuferudo wa 'Bagudaddo wa muhō-chitai' to iu hōdō o kiki, kanninbukuro no o ga kireta.*
'When Rumsfeld heard the report that "Baghdad is a lawless area," he blew his stack.'

The same can be said about *harawata ga niekurikaeru* and *make one's blood boil/one's blood boils*. These examples show that L1/L2 semantic correspondence is sometimes – but not always – accompanied by correspondence in lexical constituents.

Many of the L1/L2 idioms in this analysis exhibit partial semantic correspondence (Figure 1). We can expect, therefore, that although these idioms overlap in meaning, they are not interchangeable in all contexts. In (14'), for instance, *hara ga tatta* cannot be translated as '*I got hot under the collar*' because the feature <+other-oriented> blocks use of the English idiom in contexts of self-oriented anger.

(14') *...Jibun no shūtai ni hara ga tatta.*
*'...I got hot under the collar at my own wretched performance.'

Of course, it would be misleading to suggest that L1/L2 idioms with complete semantic correspondence are automatic translation equivalents in all contexts. Factors such as frequency, register, and stylistic constraints should also be taken into account. It may not be appropriate to translate an L1 idiom using an L2 semantic equivalent when the former has a much higher or lower frequency of occurrence than the latter. In some cases, L1 idioms might be more or less colloquial or formal than semantically corresponding L2 idioms. There may be demetaphorisation or wordplay in the L1 context that makes it difficult to use an L2 idiom with unrelated lexical constituents. Gender or age constraints on usage may also come into play; for instance, *blow your top* usually has a man as its grammatical subject and *atama ni kuru* is used more frequently by younger than older people.

An additional consideration is that, in some cases, an L1 idiom may be semantically closer to an L2 word than it is to an L2 idiom. Although *fly off the handle* and *kanninbukuro no o ga kireru* share most of the same semantic features, the former is <–time before realisation> while the latter is <+time before realisation>. This means that *fly off the handle* might be better translated using the verb *kireru* ('snap/lose control'), which is <±time before realisation>.

The considerations described above are important; however, sound semantic analysis is also important. The method of contrastive analysis presented in this chapter is one way to capture the complexity of semantic relationships between L1/L2 idioms and to identify semantic distinctions that are relevant for language education, translation, and lexicography.

6. Questions for further consideration

One question for further consideration is expansion of the present analysis to include other Japanese/English idioms and verbs of anger (see Section 3, Step 1). Also, it should be noted that although the list of distinctive features presented in this analysis is sufficient to distinguish the idioms under consideration, it is not exhaustive. It may be possible, for instance, to distinguish *blow/let off steam* from other English anger idioms on the basis of <positive evaluation>, because this idiom is compatible with contexts such as *Fortunately, I had a chance to blow off some steam after the game.* (It is doubtful that any Japanese anger idioms would be compatible with parallel contexts.) Identification of additional features, however, is a task to be undertaken at another opportunity. A final question is that of the universality of the semantic features identified in this analysis. Do they function in lexical fields other than those considered here? Do they function in languages other than English and Japanese? These questions can be answered only after extensive analysis of a variety of lexical fields in different languages.

References

Baba, N. (2001). Ikari o arawasu dōshi(ku)' no bunrui to sono tokuchō. *Journal of Japanese Grammar* 1(1): 159–176.

Bandō, M. & H. Matsumura (2001). Shinri dōshi to shinri keiyōshi. In Kageyama, T. (ed.) *Nichi-ei Taishō: Dōshi no Imi to Kōbun*, 69–97. Tōkyō: Taishūkan Shoten.

Barkema, H. (1996). Idiomaticity and terminology: A multi-dimensional descriptive model. *Studia Linguistica* 50(2): 125–160.

Charteris-Black, J. (2003). A prototype based approach to the translation of Malay and English idioms. In Granger, S., J. Lerot & S. Petch-Tyson (eds.) *Corpus-Based Approaches to Contrastive Linguistics and Translation Studies*, 123–140. Amsterdam: Rodopi.

Coseriu, E. (1968). Les structures lexématiques. In Elwert, W. T. (ed.) *Zeitschrift für Französische Sprache und Literatur,* Beiheft 1 (Neue Folge), 3–16. Wiesbaden: Steiner.

Coseriu, E. & H. Geckeler (1981). *Trends in Structural Semantics*. Tübingen: Narr.

Dobrovol'skij, D. (1998). Russian and German idioms from a contrastive perspective. In Weigand, E. (ed.) *Contrastive Lexical Semantics*, 227–242. Amsterdam: John Benjamins.

Fernando, C. (1996). *Idioms and Idiomaticity*. Oxford: Oxford University Press.

Firth, J. R. (1968). Linguistic analysis as a study of meaning. In Palmer, F. R. (ed.) *Selected Papers of J. R. Firth 1952–59*, 12–26. London: Longman.

Fraser, B. (1970). Idioms within a transformational grammar. *Foundations of Language* 6(1): 22–42.

Gibbs, R. W. (1990). Psycholinguistic studies on the conceptual basis of idiomaticity. *Cognitive Linguistics* 1(4): 417–451.

Gibbs, R. W. & G. Gonzales (1985). Syntactic frozenness in processing and remembering idioms. *Cognition* 20: 243–259.

Hasada, R. (2002). 'Body part' terms and emotion in Japanese. *Pragmatics & Cognition* 10(1/2): 107–128.

Ishida, P. (1998). Kanyōku no hen'ikei ni tsuite: Keishiki-teki koteisei o megutte. *Tsukuba Journal of Applied Linguistics* 5: 43–56.

Ishida, P. (2000). Dōshi-kanyōku ni taisuru tōgo-teki sōsa no kaisō kankei. *Japanese Linguistics* 7: 24-43.

Ishida, P. (2003). Kanyōku no imi o bunseki suru hōhō. *Japanese Language and Literature* 37: 13–26.

Ishida, P. (2004). Dōshi-kanyōku no imi-teki koteisei o hakaru hōhō: Tōgo-teki sōsa o shudan toshite. *Studies in the Japanese Language* 55(4): 42–56.

Ishiwata, T. & M. Takada (1990). *Taishō Gengogaku*. Tōkyō: Ōfūsha.

Itō, M. (1992). Kanyōku taishō kenkyū: Nichi-doku kanyōku no taiō kankei. *Studies in Languages and Cultures* 36: 155–169.

Itō, M. (1997a). Nichi-doku kanyōku no gushōsei to imi kinō. *Rhodus Zeitschrift für Germanistik* 13: 118–130.

Itō, M. (1997b). Gengo no gushōsei, hiyusei, judōsei: Nichi-doku kanyōku o megutte. In University of Tsukuba Society of Modern Linguistics (ed.) *Voisu ni kansuru hikaku gengo-teki kenkyū*, 249–297. Tōkyō: Sanshūsha.

Itō, M. (1999). Kōsei yōso no hiyuteki imi ni tsuite: Nichi-doku kanyōku no shintai bui o chūshin ni. In *Report II of the Special Project for Typological Studies of East-West Languages and Cultures (University of Tsukuba)*, 763–788. Tsukuba: University of Tsukuba.

Kindaichi, H. (1950). Kokugo dōshi no ichibunrui. Reprinted in Kindaichi, H. (ed.) (1976) *Nihongo Dōshi no Asupekuto*, 5–26. Tōkyō: Mugi Shobō.

Kövecses, Z. (1986). *Metaphors of Anger, Pride, and Love*. Amsterdam: John Benjamins.

Machida, K. (1989). *Nihongo no Jisei to Asupekuto*. Tōkyō: Aruku.

Makkai, A. (1972). *Idiom Structure in English*. The Hague: Mouton.

Matsuki, K. (1995). Metaphors of anger in Japanese. In Taylor, J. & R. MacLaury (eds.) *Language and the Cognitive Construal of the World*, 137–151. Berlin: Mouton.

McVeigh, B. (1996). Standing stomachs, clamoring chests and cooling livers: metaphors in the psychological lexicon of Japanese. *Journal of Pragmatics* 26: 25–50.

Miyaji, Y. (1985). Kanyōku no shūhen: Rengo, kotowaza, fukugōgo. *Japanese Language Studies* 1985(1): 62–75.

Moon, R. (1998). *Fixed Expressions and Idioms in English: A Corpus-Based Approach*. Oxford: Oxford University Press.

Muraki, S. (1985). Kanyōku, kinō-dōshi ketsugō, jiyūna goketsugō. *Japanese Language Studies* 1985(1): 15–27.

Nagashima, Y. (1982). Yatto, yōyaku, tsuini, tōtō. In Kunihiro, T. (ed.) *Kotoba no Imi 3: Jisho ni Kaite nai Koto*, 170–177. Tōkyō: Heibonsha.

Nunberg, G., I. A. Sag & T. Wasow (1994). Idioms. *Language* 70(3): 491–538.

Sinclair, J. McH. (1987). Collocation: A progress report. In Steele, R. & T. Threadgold (eds.) *Language Topics: Essays in Honour of Michael Halliday*, 319–331. Amsterdam: John Benjamins.

Takada, M. (1974). Taishō gengogaku: Genjō no gaikan to kenkyū no wakugumi no sobyō. *Language Life* 279(12): 37–45.

Tsujimura, N. (1996). *An Introduction to Japanese Linguistics*. Oxford: Blackwell Publishers.

Ueda, Y. (1994). Nichi-doku ideiomu taishō no shiten. *Hiroshima University Studies-Faculty of Letters* 54: 193–210.

Ueda, Y. (2003). *Doitsu-go Ideiomu Gakushū/Kyōjuhō ni kansuru Sōgōteki Kenkyū: Nichi-doku Ideiomu Hikaku-taishō Kenkyū no Shiten kara*. Kanagawa: Gendai Tosho.

Data sources

Japanese idioms: *Asahi Newspaper* Digital News Archives for Libraries, 1990–2004; *Yomiuri Newspaper* Online Database, 2004.

English idioms: North American News Text Corpus (Linguistic Data Consortium), *Los Angeles Times & Washington Post* 1994–1997, *New York Times* 1994–1996 (approximately 225 million words).

Automatic extraction of translation equivalents of phrasal and light verbs in English and Russian

Olga Mudraya, Scott S. L. Piao, Paul Rayson, Serge Sharoff, Bogdan Babych, and Laura Löfberg

This chapter aims at bridging the functionalist theoretical perspective on word usage with corpus-based studies. We are dealing with the issue of construction of reliable lists of what is called 'phraseological units' in general linguistics literature or 'multi-word expressions' (MWEs) in literature on computational linguistics. The two groups of constructions under investigation in this chapter are phrasal verbs and light verb constructions. Another distinguishing feature of this study is its multilingual aspect. Previous computational approaches to MWEs have mainly focussed on English, and there has been little research on computational approaches to MWEs in other languages. In this chapter, we examine phrasal verbs in English and their translation equivalents in Russian, and compare English-Russian/Russian-English translation equivalents of selected light verb constructions in two case studies. Our study reveals some interesting cross-language structural divergences between the languages under consideration and shows that a phraseological expression in a language may have equivalent expressions in other languages with different morpho-syntactic structures and semantic properties. However, our investigation not only reveals marked differences between English and Russian, but also discovers some general corresponding structural patterns between them; for example, English phrasal verbs usually have single-word translation equivalents in Russian. Moreover, our study of phrasal and light verbs demonstrates that corpus-based resources can provide an invaluable help to a practising translator, as dictionaries do not cover a large variety of real-life language examples.

1. Introduction

John Sinclair (1991) proposed a useful dichotomy of approaches to lexicological research distinguishing between the open-choice and idiom principles. The open-choice principle assumes that words have their own meanings and can be combined according to the rules of grammar and selectional restrictions. On the other hand, the idiom

principle (advocated by Sinclair himself) assumes that the majority of words acquire a specific meaning only in more or less fixed combinations. Traditional dictionaries are mostly word-oriented; they take into account some word combinations but rarely cover the variety of options possible in a given language. For example, in the *Oxford Russian Dictionary* (ORD), the entry for *strong* does not mention such combinations as *strong* {*feeling, field, opposition, sense, voice*}, all of which frequently occur in the *British National Corpus* (BNC). Furthermore, dictionaries tend to favour colourful idiomatic expressions, like *strong as a horse*, which are much less frequent according to corpus data. Similarly, many natural language processing (NLP) applications operate with lexicons based on single word forms, assume that meanings are encoded in words, and pay much less attention to word combinations.

WordNet (Miller et al. 1990) is a particularly influential approach that uses the idea of the lexical matrix and is designed to include most senses that exist in English and to map them onto most word forms. An extension of the *WordNet* model to other languages (Vossen 1998) is also based on the assumption of mapping between a finite set of words and senses. The opposite theoretical perspective assumes that words are used in their context to deliver meanings appropriate for the communicative purposes of the speaker (Sharoff 2005). However, the problem with function-oriented studies is that we cannot extract meanings from collections of texts and we have to rely on lists of word forms to make any corpus-based studies.

This chapter aims at bridging the functionalist theoretical perspective on word usage with corpus-based studies. We are dealing with the issue of construction of reliable lists of what is called 'phraseological units' in general linguistics literature (Cowie 1998: 4) or 'multi-word expressions' (MWEs) in literature on computational linguistics (Bond et al. 2003). The two groups of constructions under investigation in this chapter are phrasal verbs and light verb constructions (LVCs), as reported in Sections 2 and 3 respectively.

Another distinguishing feature of this study is its multilingual aspect. Previous computational approaches to MWEs have mainly focussed on English, and there has been little research on computational approaches to MWEs in other languages – a few exceptions are studies on Chinese (Piao et al. 2006) and Russian (Sharoff 2004). However, a phraseological expression in a language may have equivalent expressions in other languages with different morpho-syntactic structures and semantic properties.

In Section 2 on phrasal verbs we examine phrasal verbs in English and their translation equivalents in Russian. In Section 3 on light verb constructions we consider detection of MWEs and translation equivalents in Russian and compare the results to English. This work is driven at a practical level by a larger research effort on porting an existing semantic field lexicon for English (Rayson et al. 2004) to the Finnish (Löfberg et al. 2005) and Russian (Sharoff et al. 2006a; Mudraya et al. 2006) languages.

2. English phrasal verbs and their translation equivalents in Russian

Multi-word verbs form an important part of English vocabulary. Quirk et al. (1985) made a distinction between phrasal verbs, prepositional verbs and phrasal-prepositional verbs. Jackson & Zé Amvela (2000: 64–65) develop Quirk's classification further, distinguishing phrasal verbs from other types of multi-word verbs by examining the notion of transitivity, the position of the direct object and the number of particles following the main verb. According to the *Longman Grammar of Spoken and Written English* (Biber et al. 1999), phrasal verbs are multi-word units consisting of a verb followed by an adverbial particle with a spatial or locative meaning (e.g. *find out*, *run away*, *catch up*, etc.); when combined together, they represent single semantic units with extended meanings that cannot be derived from the individual meanings of the parts. However, Jackson and Zé Amvela (2000: 65) point out that phrasal verbs vary in the extent to which the combination preserves the individual meanings of the verb and particle, and in some cases the meaning of a phrasal verb can be derived from that of its constituents (e.g. *sit down*, *look up*, etc.). In our work, we use Jackson and Zé Amvela's definition and description of phrasal verbs as the guidelines for phrasal verb extraction and analysis.

2.1 Procedure

In this section, we will compare some frequently occurring English phrasal verbs with their translation equivalents in Russian, both in terms of morpho-syntactic structures and semantic properties. Our study focuses on the comparison between a list of English phrasal verbs, which share frequently occurring morpho-syntactic structures in the form of part-of-speech (POS) patterns, and their equivalent expressions in Russian.

Firstly, for the purpose of comparison, a frequency list of MWE POS patterns is extracted from the semantic MWE lexicon of the English semantic tagger (EST), developed at Lancaster University (Rayson et al. 2004), which contains nearly 19,000 MWE template entries. From the pattern-frequency list, the most frequent phrasal verb patterns occurring ten or more times in the list are selected. As a result, eight phrasal verb POS patterns whose frequencies range from 103 to 10 are extracted, as listed in Table 1.

Table 1. Frequent phrasal verb POS patterns[1] extracted from USAS MWE lexicon

Phrasal Verb POS Pattern	Frequency
VV* {Np/P*/R*} RP	103
VV* {R*} RP	34
VVD {Np/P*/R*} RP	29
VVD {R*} RP	15
VV* {R*} RL	13
VVD {R*} II NN1	11
VV* {Np/P*/R*} RL	10
VVD {R*} RL	10

Explanation
VV* = a form of a verb
VVD = verb past tense
Np = noun phrase
NN1 = singular noun
P* = any pronoun
R* = any adverb
RP = prep. adverb
RL = locative adverb
II = general preposition

Secondly, the English phrasal verbs sharing each of the patterns are retrieved, as shown in the following sample:

Pattern: VV* {Np/P*/R*} RP	Pattern: VV* {Np/P*/R*} RL
break_VV* {Np/P*/R*} in_RP	bring_VV* {Np/P*/R*} together_RL
find_VV* {Np/P*/R*} out_RP	hold_VV* {Np/P*/R*} together_RL
lace_VV* {Np/P*/R*} up_RP	hide_VV* {Np/P*/R*} away_RL
pay_VV* {Np/P*/R*} up_RP	keep_VV* {Np/P*/R*} away_RL
wake_VV* {Np/P*/R*} up_RP	leave_VV* {Np/P*/R*} behind_RL
wipe_VV* {Np/P*/R*} out_RP	scrape_VV* {Np/P*/R*} together_RL
	take_VV* {Np/P*/R*} apart_RL
Pattern: VV* {R*} RP	**Pattern: VV* {R*} RL**
die_VV* {R*} down_RP	move_VV* {R*} ahead_RL

Next, these English phrasal verbs are compared with their equivalent expressions in Russian, using dictionaries and corpus resources. As mentioned earlier, we carried out the comparison both in terms of morpho-syntactic structures and semantic features. By doing so, we attempted to generalise differences in syntactic structure and typical cases of semantic diversities.

2.2 Cross-language comparison

Our study has revealed some interesting cross-language structural divergences between the languages under consideration (see also Mudraya et al. 2005). The most marked divergence is that Russian has no phrasal verbs at all; however, it employs a variety of grammatical mechanisms to convey the meanings expressed by English phrasal verbs.

We have observed that the Russian translation equivalents of the English phrasal verbs following the *VV* {Np/P*/R*} RP/RL* pattern tend to show a distinctly different structure from English morpho-syntactic structure. In Russian, the *RP/RL* elements in

1. For a full list, see http://ucrel.lancs.ac.uk/claws7tags.html

this type of English phrasal verbs are generally expressed by means of verbal prefixes. For instance, English phrasal verbs with *RP* elements, which on their own would be function words such as *in*, *on*, *out*, *up*, *down*, etc., are usually translated into Russian as single verbs with an additional prefix, as in the following examples:

die down – замирать
find out – выяснить
lace up – зашнуровать
pay up – оплатить
wake up – разбудить
wipe out – вытирать

The *RP/RL* elements in English phrasal verbs denote the general spatial direction of the action or express its qualitative or quantitative characteristics, such as beginning, duration, completion, intensity, etc. Likewise, Russian verbal prefixes (*вы-*, *за-*, *на-*, *о/об-*, *раз-*, etc.) can indicate various qualities of actions and states, thus closely resembling the semantic functions of the *RP/RL* elements in English phrasal verbs. To illustrate, the Russian equivalent for *wipe out* is literally 'outwipe' (*вытирать*).

On the other hand, English phrasal verbs with *RL* elements that independently may act as content words, such as *forward*, *ahead*, *behind*, *apart*, *together*, etc., are often expressed as verb + adverb in Russian, e.g. *bring together – сводить вместе*, *leave behind – оставлять позади*. However, this should be regarded more as a tendency rather than a rule, as some phrasal verbs in the second group can be translated by means of prefixation (e.g. *take apart – разбирать*; *scrape together – наскрести*) whilst some others allow both means. For example, *move ahead* can be translated into Russian as either *продвигаться* (verb with prefix) or *двигаться вперёд* (MWE).

Furthermore, if an English phrasal verb is highly idiomatic, i.e. its meaning is unpredictable from the sum of its constituents' meanings, it will be rendered in Russian either with an idiomatic expression that may have a very different lexicosemantic composition or with a lengthy explanation. Yatskovich (1999) argues that it is almost impossible to create a consistent rigid system of lexical correspondences between English adverbial particles and Russian prefixes without encountering numerous problems.

By way of illustration, Yatskovich shows that in the English sentence *He liked to break in his assistants slowly*, neither the context nor the RP element hints at the real meaning of the phrasal verb *break in*, which according to the latest edition of the *Longman Phrasal Verbs Dictionary* (2000: 36) means 'to make someone get used to doing a job or activity, especially by letting them do it a little and then gradually making them do it more'. An earlier Russian edition of this dictionary (1997) defines this phrasal verb with a Russian idiomatic expression *вводить (кого-л.) в курс (новой работы и т.п.)*, literally meaning 'to introduce (sb) to (a new job, etc.)'.

In addition, English phrasal verbs with multiple senses are usually translated into different Russian verbs with different meanings:

- *die down* – замирать [become calm], затихать [become quiet], увядать [wither], угасать [weaken];
- *hold together* – сплачивать(ся) [unite]; держать(ся) [hold]; держаться вместе [keep together]; выдерживать [withstand];
- *wipe out* – вытирать [clean], утирать [wipe]; смывать [wash off], уничтожать [destroy], падать [fall down].

It is important to note that the Russian dictionary translations of English phrasal verbs do not always correspond to those we find in actual translations of texts such as fiction. The following comparison of dictionary and literary translations of *die down* will demonstrate this.

ABBYY Lingvo 10 English-Russian Electronic Dictionary (2004), which is the most up-to-date dictionary of six million words, provides three translations for *die down*: 1) увядать ~ wither; 2) падать в обморок ~ faint; 3) замирать (о звуке) ~ become quiet (about sound). It also gives translation examples of three sentences containing *die down* with completely different translations:

- The fire is dying down, put some more wood on! = Костер гаснет [~ is going out], подложи-ка дров!
- The wind died down at last and all was quiet. = Ветер стих [~ calmed down] и установилась тишина.
- When the hunt has died down we shall be able to leave our hiding place. = Когда за нами перестанут [~ has stopped] охотиться, мы сможем покинуть наше убежище.

Furthermore, when checked against the aligned English-Russian/Russian-English parallel corpus of fiction (http://corpus.leeds.ac.uk/serge/fiction/), even more translation equivalents of *die down* were found. Below are two translation examples from William Golding's *Lord of the Flies:*

- Then Piggy was standing cradling the great cream shell and the shouting died down. = Хрюша покачал на руках большую розовую раковину, и крики улеглись [~ settled down].
- As the fire died down so did the excitement. = Костер угасал [~ was going out], а с ним вместе и оживленье.

That gives us five translation equivalents – *гаснуть, угасать, стихать, переставать, улечься* – in addition to the three dictionary definitions *увядать, падать в обморок* and *замирать*. This example demonstrates the importance of the corpus resources to a translator. Dictionaries alone cannot account for all possible ways of translating a particular expression. Corpora, on the other hand, could be much more useful to a practising translator, as has been proved by Bowker in several studies (cf. Bowker 2003), for they present different translation equivalents in different contexts. In particular, translators may find especially helpful corpus-based translation equivalents of such language-specific phenomena as phrasal verbs and light verbs, which will be discussed in Section 3.

3. Light verbs

3.1 Theoretical background and operational procedure

The concept of light verbs was introduced by Jespersen (1933); it characterises instances of idiomatic (non-compositional) usage of some verbs, where part of their predication (i.e. some of their propositional semantic relations) is expressed by complements of these verbs, not just by verbs themselves. Light verbs in such constructions are used figuratively, while their complements are typically used in their literal sense (Cowie 1998: 5). Prototypical examples of light verb constructions are *take action, take part, put pressure, make a decision,* etc., where objects of the verbs are logical predicates (i.e. they open new valencies, or presuppose other actants). These objects are typically names of actions, activities, states, properties, relations, that put forward some situational propositions (Mel'čuk 1998: 37; Piñango et al. 2006). They often have one-word verb synonyms: *to act, to participate, to decide,* etc.

Light verbs are a cross-linguistic phenomenon and are found in genealogically and typologically different languages (Butt 2003). For this reason, light verbs become an interesting and important problem for translation technology, including Machine Translation (MT) and Machine-Aided Human Translation. The fact is that even frequent light verb constructions (LVCs) are often missed by the developers of MT systems and by authors of bilingual dictionaries. For dictionaries and MT systems developed with corpus data, there is also a need for a clear procedure for identifying and mapping LVCs in different languages. Finding a systematic solution to the problem of their automatic identification and translation across languages could in a principled way extend dictionary coverage of this type of expression. It can also improve the quality of the state-of-the-art MT systems, since LVCs require specific translation strategies – their complements are usually translated more or less literally while verbs require oblique translation. For example, *put pressure* is translated into Russian as *оказывать давление* [~ exert pressure] and not *класть давление* [*класть* is a literal translation of *put*]; likewise, *make a decision* is translated as *принимать решение* [~ take a decision] and not *делать решение* [*делать* = *make*], etc. Potential improvement in MT quality could be akin to the improvement achieved by Named Entity Recognition, which is yet another cross-language technology that identifies a special type of constructions that require specific translation strategies (Babych & Hartley 2003; Somers 2003).

LVCs are a type of MWE, namely 'collocations', following the terminology of Mel'čuk (1998: 30). Their meaning goes beyond what is expected for normal word combination of this type according to general rules of syntax (i.e. 'lexical meaning' + 'syntactic meaning, e.g. instrument, of action, etc.'). According to the traditional linguistic classification proposed by Vinogradov, as discussed in Cowie (1998), MWEs (or phraseological units in Vinogradov's terms) consist of:

1. compositional phraseological units (e.g. *washing machine*), which are re-constructed in speech and are part of the 'mental lexicon';

2. idioms in the broad sense, i.e. non-compositional phraseological units (meaning/usage is restricted, specific to smaller classes of word combinations, or to individual words), which include:
   2.1. phraseological fusions ('idioms in the narrow sense' where relation to meaning is not motivated, e.g. *spill the beans*);
   2.2. phraseological unities (motivated by metaphorical extension of some originally neutral sense, e.g. *blow off steam*);
   2.3. phraseological combinations, or collocations (motivated, but one component is used figuratively, e.g. *meet the demand*) – all LVCs belong here.

Phraseological combinations, including LVCs (Vinogradov's category 2.3), are called idioms mainly under Wittgenstein's assumption that meaning is use; they are semantically asymmetric, with one of their components, namely the verb, phraseologically bound, which in its depleted sense can be combined only with a restricted set of complements, whilst the other component, the complement, is free, preserving its usual sense, in which it is used with a wider range of words. Most importantly, the complement cannot be freely replaced by its synonyms: *meet the challenge/demand/need* but not *meet the necessity/deficiency/shortage/requisite*. This is different from *meet a friend/colleague/buddy/chum/classmate/companion/compatriot/comrade* where there are no such restrictions. If close synonyms do not exist, we can rely only on external criteria, such as whether *meet* has the same meaning in its figurative and non-figurative use.

Thus, restrictions on usage imply that there are some restrictions on meaning, so the LVCs are idioms in a broad sense, but not in the narrow sense. As the following patterns fit this matrix of restricted usage, they are included into the list of LVCs, even though the boundaries are rather unclear and difficult to define:

put to death (put to departure/dying?)
put to flight (put to take-off?)
put to good use (put to service/employment/function?)
put to shame (put to embarrassment/dishonour/humiliation/indignity?)

This phraseological nature distinguishes light verbs from other types of complex predicates; for example, *start talking* is a complex predicate but is not considered a LVC. However, for a LVC it is not sufficient to be an idiom – phrases like *give a hand*, *get the cream*, *run out of steam* are not LVCs since their complements are not used in their literal sense and are not logical predicates.

In our study of light verbs, we started with a list of verbs that most frequently occur in LVCs in English and Russian, filtered out patterns that a priori cannot be used as LVCs (for example, incomplete phrases such as *weapon of mass, student from the poor, run a completely ethical*), and identified real LVCs in the list. We then compared the coverage of LVCs in two dictionaries: *ABBYY Lingvo 10* and *ORD*. Section 3.2 presents a Russian case study and Section 3.3 describes an English case study; both case stud-

ies examine and compare English-Russian/Russian-English translation equivalents of selected light verb constructions.

3.2 A case study of Russian light verbs and their English translation equivalents

For this study we have initially selected a list of nine Russian light verbs, namely *брать* [= take], *вести* [= carry/lead], *давать* [= give], *делать* [= do], *иметь* [= have], *нести* [= carry], *положить* [= put (down)], *принимать* [= take/get] and *ставить* [= put (up)]. POS configuration filter was used to identify lists of potential LVCs with these verbs, extracted from the 70-million-word corpus of Russian newspapers[2] (Sharoff et al. 2006b). This produced lists of MWEs based on relatively frequent n-grams with more than nine occurrences in the corpus. Real LVCs were then hand-annotated in these lists. Table 2 presents statistics on the MWEs that passed through the POS filter (with precision figures and n-gram length) and were further annotated as LVCs.

It can be seen from Table 2 that average precision of the POS filters and frequency threshold is around 30%, but different words vary considerably with respect to their productivity, precision and n-gram length. As the n-gram length increases, both the absolute number of extracted entities and relative figures of LVC precision fall. However, for individual verbs the situation can be different; for instance, the verb *давать* [= give] is the most stable across n-grams of different length in terms of precision, with the raw numbers of extracted items falling, in accordance with the general tendency. However, the absolute numbers of LVCs with the verb *принимать* [= take/get] peak at N=2, although the precision falls according to the established pattern. In general, the proposed method of LVC identification is relatively accurate, but obviously requires a manual annotation stage.

We checked dictionary coverage of the identified LVCs for three Russian verbs *вести, нести,* and *ставить*, using *ABBYY Lingvo 10* electronic dictionary, which is one of the most complete and constantly updated lexicographic resources for translators into/from Russian. The results are presented in Table 3.

Table 3 illustrates that our corpus-based method of identifying LVCs can be an efficient lexicographic tool. Firstly, it gives a reasonable recall of about 60% for dictionary entries; however, it is important to keep in mind that some dictionary entries are specialised terms and are not frequently used in the language of news media, so to be fair, the recall on most common dictionary entries is actually greater. Secondly, the method efficiently identifies LVCs which are not covered by the dictionary. Surprisingly, we found that about 75% of Russian LVCs in the news corpus were not in the

2. The Russian news corpus contains 70 million words from several major Russian newspapers, such as *Izvestia* (2001–2003), *Trud* (2000–2004) and *Strana* (2001–2004). It is available at http://corpus.leeds.ac.uk/ruscorpora.html.

Table 2. Statistics on MWEs with selected Russian verbs

	N=2		N=3		N=4		All	
	MWEs	LVCs	MWEs	LVCs	MWEs	LVCs	MWEs	LVCs
брать	39	*13*	19	*15*	7	*5*	65	*33*
= take		*(33%)*		*(79%)*		*(71%)*		*(51%)*
вести	147	*69*	94	*27*	24	*2*	265	*98*
= carry/lead		*(47%)*		*(29%)*		*(8%)*		*(37%)*
давать	312	*111*	169	*29*	27	*13*	508	*153*
= give		*(36%)*		*(38%)*		*(48%)*		*(30%)*
делать	105	*50*	49	*13*	9	*2*	163	*65*
= do		*(48%)*		*(27%)*		*(22%)*		*(40%)*
иметь	190	*65*	186	*78*	75	*25*	451	*168*
= have		*(34%)*		*(42%)*		*(33%)*		*(37%)*
нести	26	*15*	25	*6*	7	*4*	58	*25*
= carry		*(58%)*		*(24%)*		*(57%)*		*(43%)*
положить	4	*2*	2	*0*	0	*0*	6	*2*
= put (down)		*(50%)*		*(0%)*		-		*(33%)*
принимать	176	*50*	300	*34*	241	*13*	717	*97*
= take/get		*(28%)*		*(11%)*		*(5%)*		*(14%)*
ставить	37	*13*	29	*16*	11	*2*	77	*31*
= put (up)		*(35%)*		*(55%)*		*(18%)*		*(40%)*
All	1036	*388*	873	*218*	401	*66*	2310	*672*
		(37%)		*(25%)*		*(16%)*		***(29%)***

Table 3. Comparison of light verb expressions extracted from the Russian news corpus and *ABBYY Lingvo 10 English-Russian Electronic Dictionary*

Light verb expressions	Russian news corpus		ABBYY Lingvo 10		
	Number of LVCs	Overlap			Number of LVCs
		%	freq.	%	
вести = carry/lead	98	21%	21	52%	40
нести = carry	25	12%	3	37%	8
ставить = put (up)	31	48%	15	83%	18
Overall	**154**	**25%**	**39**	**59%**	**66**

dictionary. Lists of such missing items can be presented to lexicographers in order to extend the dictionary coverage.

3.3 A case study of English light verbs and their Russian translation equivalents

For this case study, we used as a starting point the work by Tan et al. (2006) who have predefined a list of seven light verbs for their experiment, namely, *do, get, give, have, make, put* and *take*, all of which have been previously recognised as light verbs in other related studies. We have identified three more verbs – *carry, find* and *run* – that can be used as light verbs with certain complements. For the purposes of this chapter, we have

Table 4. Comparison of light verb expressions extracted from the British news corpus and *ABBYY Lingvo 10 English-Russian Electronic Dictionary*

Light verb expressions	British news corpus	ABBYY Lingvo 10			
	Number of LVCs	Overlap			Number of LVCs
		%	freq.	%	
carry	35	43%	15	75%	20
put	81	47%	38	58%	65
run	9	78%	7	47%	15
Overall	**125**	**48%**	**60**	**60%**	**100**

selected three light verbs *carry, put* and *run* for further investigation. As in previous studies, we have limited our selection of examples to the basic V + N morpho-syntactic structure, but allowed embedded elements such as articles, pronouns, determiners and adjectives.

We began our investigation with generating lists of light verb expressions from the 217-million-word corpus of British newspapers[3] (Sharoff et al. 2006b). After eliminating metaphorical expressions, in which complements were not used in their literal sense (e.g. *put weight, put to death, carry too much weight, run down the clock*) and were not logical predicates (e.g. *carry water, run a company, put his money*), and all the expressions, in which verbs *carry, put* and *run* were not used as light but in their literal sense (e.g. *carry a gun, put the knife, run for charity*), we were left with 35 expressions with the verb *carry*, 81 expressions with the verb *put* and 9 expressions with the verb *run*. These expressions were looked up in the *ABBYY Lingvo 10 English-Russian Electronic Dictionary* (2004) in order to find their Russian translations if available. Overall, we identified 20 light verb expressions with the verb *carry*, 65 light verb expressions with the verb *put* and 15 light verb expressions with the verb *run* in *ABBYY Lingvo 10*. However, there was only partial overlap between the light verb expressions generated from the news corpus and the ones found in ABBYY Lingvo 10. As presented in Table 4, of the 20 light verb expressions with the verb *carry* in *ABBYY Lingvo 10*, only 15 matched the corpus-based examples; of the 65 light verb expressions with the verb *put* in the electronic dictionary, only 38 corresponded to the corpus-based expressions; and of 15 light verb expressions with the verb *run* in the dictionary, 7 agreed with the corpus-based ones.

Table 4 shows an interesting overall picture: in our case study, only 48% of the corpus-based expressions with the selected light verbs have found their way into the translation dictionary. This means that the majority of the expressions are not registered in the dictionary at all (neither should we expect them to be[4]). On the other

3. The British news corpus contains 217 million words from a collection of major British newspapers, such as *Guardian, Observer, Times, Sunday Times, Independent* and *Telegraph*, published in 2004.

4. Cf. Salkie's (2002) interesting discussion of this issue.

hand, only 60% of the dictionary examples with the verbs under examination are represented in corpora, which means that the other 40% of the dictionary examples are too infrequent in their modern usage to filter in corpora. Consequently, if translators had no access to corpus-based resources, they would not be able to find translations of a large proportion of modern language data in the dictionary.

Another noteworthy finding that emerged out of this case study is that many of the English light verb constructions under investigation are translated into Russian light verb constructions. For example, *carry a resolution* = *выносить/принимать резолюцию; выносить/принимать решение*, where *выносить* and *принимать* are Russian light verbs corresponding to English light verbs *carry* and *take*. This finding supports the idea formulated earlier in this chapter that light verbs are a cross-linguistic phenomenon and can be found in genealogically and typologically different languages (Butt 2003). Characteristically, often Russian translation equivalents of English light verb expressions also include single verbs, as in the examples below, which corroborates the observation expressed earlier that light verbs often have one-word verb synonyms:

> *carry into effect* – осуществлять, приводить в исполнение, приводить в действие, проводить в жизнь
> *put the kibosh on* – положить конец, покончить; прикончить
> *put the squeeze* – оказать давление, надавить
> *put in [good] order* – привести в порядок; упорядочивать
> *put in motion* – привести в движение; пустить
> *put in force* – вводить в действие, осуществлять, проводить в жизнь
> *run a temperature* – иметь повышенную температуру, температурить

A rather unexpected outcome of this investigation is that some phrasal verbs can be used as light verbs. The examples of such phrasal verbs with *carry, put* and *run* are *carry on, carry out, put forth, put forward, put in, put on, put out, put up, run into* and *run up*, as in the following expressions: *carry on negotiations, carry out business, put forth an effort, put forward a proposal, put in a claim, put on an act, put out bait, put up a good show, run into difficulty* and *run up a debt*. In our case study, 73 expressions with these phrasal verbs, used as light verbs, were extracted from the British news corpus, and 58 expressions were found in ABBYY Lingvo 10; however, only 40 of them overlapped (see Table 5). These figures result in 55% coverage of the corpus-based occurrences in the translation dictionary and 69% coverage of the dictionary examples in corpora, which is slightly better than the coverage of light verbs (48% and 60% respectively), discussed earlier. On the whole, however, the figures are consistent and remain rather low.

To conclude this section, we would like to stress again that corpus-based resources can provide an invaluable help to a practising translator, as dictionaries do not cover a large variety of real-life language examples. Our study of phrasal and light verbs has sufficiently demonstrated this discrepancy.

Table 5. Comparison of phrasal verbs used as light verbs in the British news corpus and *ABBYY Lingvo 10 English-Russian Electronic Dictionary*

Phrasal verbs	British news corpus	ABBYY Lingvo 10			
	Number of LVCs	Overlap			Number of LVCs
		%	freq.	%	
carry on	2	100%	2	50%	4
carry out	27	44%	12	92%	13
put forth	–	–	–	–	1
put forward	5	80%	4	100%	4
put in	6	83%	5	56%	9
put on	19	37%	7	54%	13
put out	–	–	–	–	1
put up	6	100%	6	75%	8
run into	6	33%	2	67%	3
run up	2	100%	2	100%	2
Overall	**73**	**55%**	**40**	**69%**	**58**

4. Discussion

Dealing with MWEs is an important requirement in a number of multilingual technologies, including Machine Translation and Machine-Aided Human Translation. Significant results have already been achieved in named entity recognition. Elsewhere, in cross-lingual information retrieval, Ballesteros & Croft (1998) report that phrase loss results in 20–25% of loss of effectiveness. Our study of English/Russian LVCs and their Russian/English translation equivalents in particular lends support to Ballesteros and Croft's finding.

LVCs are currently receiving much attention in linguistics literature (Culicover & Jackendoff 2005) and in literature on computational linguistics (Stevenson et al. 2004; North 2005; Tan et al. 2006). Interestingly, they have been analysed mostly from a structural perspective, and their functional nature has not been sufficiently acknowledged. Most studies focused on small lists of verbs combined with appropriate complements, often in limited morpho-syntactic configurations (usually V + N), and did not attempt any comprehensive analysis of LVCs beyond the scope of their structural patterns. As a result, the scope of the LVC phenomenon still remains unclear. It is not known how many verbs in any language allow light usage, or if there exists an exhaustive list of complements and context features which license such usage. It is certain, however, that the LVC phenomenon goes beyond structural patterns typically used to identify LVCs.

Studies of LVCs in computational linguistics typically follow the fixed-list and fixed-pattern approaches, even if they miss many important configurations of LVCs. Firstly, constructions like *exert pressure* or *lend support* are next to never covered by computational linguistics studies of light verbs. Secondly, even for the verbs which

are covered, fixed patterns (e.g. [V (+DET) +N]) miss out many relevant cases as outlined below:

– Phrasal verbs may often have light usage: *give away a penalty, run into trouble,*[5] *take the pressure off, get your hopes up*, etc. In general, there is greater variability in the verbal part of an LVC than is usually envisaged by such search patterns.
– A complement of the light verb can be used within a prepositional phrase, and it does not have to be a direct object: *take into consideration, get into trouble,*[5] etc. Again, there can be some variability in the structure of complements in LVCs, as in *take a free kick* (note that *take a kick* is not an LVC).
– A wider range of constructions (possibly underspecified for some morpho-syntactic and lexical features) can also have light usage and form analytic LVCs, e.g. *put [someone] in charge* (meaning *to appoint [someone]*), *take pride in [something], take [someone] by surprise, get in contact with [someone], make history by [something], put [someone] at risk, put [someone] under pressure, etc.*

In our research we attempted to go beyond fixed patterns and study a larger number of constructions. However, unlike in theoretical linguistics, we cannot start our study from functions performed by LVCs. For instance, Mel'čuk (1998) links LVCs to his notion of lexical functions (namely Oper_i) and gives examples such as *deal a blow* or *meet resistance,* which are beyond the scope of our study. As an extension of this study we envisage a compilation of a database of about 500-1,000 light verbs in English and Russian and of their Russian/English translations, to become part of the English semantic lexicon (Rayson et al. 2004) and the Russian semantic lexicon (Sharoff et al. 2006a).

5. Conclusion

A deeper understanding of complex cross-language relations between phraseological expressions is important for various tasks such as language learning, translation, automatic bilingual/multilingual lexicon extraction, etc. In this chapter, we have presented our comparative study of English phrasal verbs and their equivalent expressions in Russian, as well as two case studies on English and Russian light verbs. Our investigation not only reveals marked differences between English and Russian, but also discovers some general corresponding structural patterns between them; for example, English phrasal verbs usually have single-word translation equivalents in Russian.

5. Boundaries between phrasal verbs and verbs combined with prepositional phrases are often vague, so we adopted a distinction based on how these patterns are tagged by the Tree Tagger: where the function word is tagged as a preposition, we assume the verb-complement construction, and where it is tagged as a particle, we assume the phrasal verb construction. For example, *run into trouble* is an example of the LVC with the phrasal verb, and the expression *get into trouble* the LVC with a prepositional complement, according to the Tree Tagger annotation.

Despite the limited depth and scale of our study, it helps us gain a deeper insight into the morpho-syntactic structural relations between equivalent expressions across languages, which can benefit various research areas including language teaching, contrastive linguistics and multilingual lexicon extraction.

Acknowledgements

This work is supported by the EU funded Benedict Project (IST-2001-34237) and the UK-EPSRC funded ASSIST Project (EP/C004574 at Lancaster and EP/C005902 at Leeds).

References

Babych, B. & A. Hartley (2003). Improving Machine Translation quality with automatic Named Entity recognition. In *Proceedings of the 7th International EAMT Workshop at EACL 2003 (Budapest, Hungary, April 13th)*, 1–8.

Ballesteros, L. & W. B. Croft (1998). Resolving ambiguity for cross-language retrieval. In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval (Melbourne, Australia, August. 24–28)*, 64–71. New York, NY: ACM Press.

Biber, D., S. Johansson, G. Leech, S. Conrad & E. Finegan (1999). *Longman Grammar of Spoken and Written English*. Harlow: Pearson Education.

Bond, F., A. Korhonen, D. McCarthy & A. Villavicencio (2003). Preface. In *Proceedings of the ACL 2003 Workshop on Multiword Expressions: Analysis, Acquisition and Treatment Held in Conjunction with ACL 2003, 41st Annual Meeting of the Association for Computational Linguistics (Sapporo, Japan, July 12)*. Morristown, NJ: ACL.

Bowker, L. (2003). Corpus-based applications for translator training: Exploring the possibilities. In Granger, S., J. Lerot & S. Petch-Tyson (eds.) *Corpus-Based Approaches to Contrastive Linguistics and Translation Studies*, 169–183. Amsterdam: Rodopi.

Butt, M. (2003). The light verb jungle. *Harvard Working Papers in Linguistics* 9: 1–49. <http://ling.uni-konstanz.de/pages/home/butt/harvard-work.pdf>.

Cowie, A. P. (1998). Introduction. In Cowie, A. P. (ed.) *Phraseology: Theory, Analysis, and Applications*, 1–20. Oxford: Oxford University Press.

Culicover, P. W. & R. Jackendoff (2005). *Simpler Syntax*. Oxford: Oxford University Press.

Jackson, H. & E. Zé Amvela (2000). *Words, Meaning and Vocabulary: An Introduction to Modern English Lexicology*. London: Cassell.

Jespersen, O. (1933). *Essentials of English Grammar*. London: Allen and Unwin.

Löfberg, L., S. L. Piao, P. Rayson, J.-P. Juntunen, A. Nykänen & K. Varantola (2005). A semantic tagger for the Finnish language. In *Proceedings of the Corpus Linguistics 2005 Conference (Birmingham, UK, July 14–17)*. <http://www.corpus.bham.ac.uk/PCLC/cl2005_fst_fullpaper_final.doc>.

Mel'čuk, I. (1998). Collocations and lexical functions. In Cowie, A. P. (ed.) *Phraseology: Theory, Analysis, and Applications*, 23–54. Oxford: Oxford University Press.

Miller, G., R. Beckwith, C. Fellbaum, D. Gross & K. J. Miller (1990). Introduction to WordNet: An on-line lexical database. *International Journal of Lexicography* 3(4): 235–244.

Mudraya, O., S. L. Piao, L. Löfberg, P. Rayson, & D. Archer (2005). English-Russian-Finnish cross-language comparison of phrasal verb translation equivalents. In Cosme, C., C. Gouverneur, F. Meunier & M. Paquot (eds.) *Proceedings of Phraseology 2005: An Interdisciplinary Conference (Louvain-la-Neuve, Belgium, October 14)*, 277–281.

Mudraya, O., B. Babych, S. L. Piao, P. Rayson & A. Wilson (2006). Developing a Russian semantic tagger for automatic semantic annotation. In *Proceedings of the International Conference on Corpus Linguistics* (St. Peterburg, Russia, October 10–14), 290–297.

North, R. (2005). Computational Measures of the Acceptability of Light Verb Constructions. MSc thesis, University of Toronto: Department of Computer Science. <http://www.cs.toronto.edu/~ryan/ryannorth-lvcs.pdf>.

Piao, S., G. Sun, P. Rayson & Q. Yuan (2006). Automatic extraction of Chinese multiword expressions with a statistical tool. In *Proceedings of the EACL 2006 workshop on Multi-Word-Expressions in a Multilingual Context (Trento, Italy, April 3)*, 17–24. <http://www.aclweb.org/anthology-new/W/W06/W06-2403.pdf>.

Piñango, M., J. Mack & R. Jackendoff (2006). Semantic combinatorial processes in argument structure: Evidence from light verbs. In *32nd Annual Meeting of the Berkeley Linguistics Society*. <http://www.linguistics.berkeley.edu/BLS/abstracts/bls32_pinangoetalLA.pdf>.

Quirk, R., S. Greenbaum, G. Leech & J. Svartvik (1985). *A Comprehensive Grammar of the English Language*. London: Longman.

Rayson, P., D. Archer, S. L. Piao & T. McEnery (2004). The UCREL semantic analysis system. In *Proceedings of the Workshop on Beyond Named Entity Recognition Semantic Labelling for NLP Tasks in Association with LREC 2004 (2004, Lisbon, Portugal, May 25)*, 7–12. <http://www.comp.lancs.ac.uk/computing/users/paul/publications/usas_lrec04ws.pdf>.

Salkie, R. (2002). Two types of translation equivalence. In Altenberg, B. & S. Granger (eds.) *Lexis in Contrast*, 51–72. Amsterdam: John Benjamins.

Sharoff, S. (2004). What is at stake: A case study of Russian expressions starting with a preposition. In *Proceeding of the ACL 2004 Workshop on Multiword Expressions: Integrating Processing (Barcelona, Spain, July, 2004)*, 17–23. <http://www.aclweb.org/anthology-new/W/W04/W04-0403.pdf>.

Sharoff, S. (2005). The communicative potential of verbs of away-from motion in English, German and Russian. *Functions of Language* 12(2): 205–240.

Sharoff, S., B. Babych, P. Rayson, O. Mudraya & S. L. Piao (2006a). ASSIST: Automated Semantic Assistance for Translators. In *Proceedings of the 11th EACL Conference (Trento, Italy, April 3–7)*, 139–142. <http://www.comp.lancs.ac.uk/computing/users/paul/publications/eacl06_sharoffetal.pdf>.

Sharoff, S., B. Babych & A. Hartley (2006b). Using collocations from comparable corpora to find translation equivalents. In *Proceedings of the LREC 2006 (Genoa, Italy, May 2006)*, 465–470.

Sinclair, J. (1991). *Corpus, Concordance and Collocation*. Oxford: Oxford University Press.

Somers, H. (2003). Machine Translation: Latest developments. In *The Oxford Handbook on Computational Linguistics*, 512–544. Oxford: Oxford University Press.

Stevenson, S., A. Fazly & R. North (2004). Statistical measures of the semi-productivity of light verb constructions. In *Proceedings of the ACL 2004 Workshop on Multiword Expressions: Integrating Processing (Barcelona, Spain, July, 2004)*, 1–8. <http://www.cs.toronto.edu/~ryan/mwe04-ref.ps>.

Tan, Y. F., M.-Y. Kan & H. Cui (2006). Extending corpus-based identification of light verb constructions using a supervised learning framework. In *Proceedings of the EACL 2006 Workshop on Multi-Word-Expressions in a Multilingual Context (Trento, Italy, April 3)*, 49–56.

Vossen, P. (1998). *EuroWordNet: A Multilingual Database with Lexical Semantic Networks*. Dordrecht: Kluwer.

Yatskovich, I. (1999). Some ways of translating English phrasal verbs into Russian. *Translation Journal* 3(3). <http://accurapid.com/journal/09russ.htm>.

SECTION IV

Phraseology in lexicography and natural language processing

Dictionaries and collocation

Rosamund Moon

This chapter discusses the treatment of phraseology in dictionaries: its specific focus is on collocation and on English. It first describes the collocational behaviour of three words (*river, rivet, riven*), as observed in a large corpus of current English. It then considers the ways in which their collocates and patterning are represented in monolingual dictionaries for native speakers and for EFL/ESL learners, followed by bilingual French-English dictionaries and dictionaries of English collocations, assessing how far these dictionaries provide reasonable accounts of collocational phenomena.

1. Introduction

Phraseology presents a major challenge for lexicographical practice. The very nature of dictionaries, with their linear organization, forces words to be considered as isolates: a series of individual headwords where each item becomes, in turn, the most important element within its co-texts. Even in electronic dictionaries, the notion of headword still prevails. However, the evidence of texts, including corpora, shows clearly that words are not isolates, but rather connected through phraseological patterning and given meaningfulness from context of use. Furthermore, models of language production point to its chunked and phraseological nature: compare Sinclair's idiom principle (1987), Hoey's dynamic model of the lexicon in terms of 'lexical priming' (2005), Wray's work on formulaic language (e.g. 2002; see also Wray & Fitzpatrick 2008), as well as analogies drawn by Jackendoff between musical phrasing and text (1988).

This does not mean that lexicographers are unaware of phraseology and the interdependence of phraseology with meaning – far from it. Lexicographers confront phraseology word by word, sense by sense, and make decisions about the status and significance of recurrent patterns: whether these represent fully lexicalized multi-word items, or collocations worth recording, and if so, where and how they should be recorded. Dictionaries are therefore knowing compromises between a need to respect the phraseological evidence of continuous text and a need to explain words one by one.

Lexicography is not simply a sophisticated, ultimately flawed, branch of applied linguistics or publishing. One of its great strengths is that it presents its own challenge to linguistic theory, showing up the inadequacy of generalization: a point that has been made by others, including John Sinclair and František Čermák. Dictionaries have to be

comprehensive, to account for all words and phrases in a target lexicon, and to examine their combinatorial behaviour in context: they do not have the luxury of looking only at selected items with salient properties which seem interesting to an observer. It is not simply a matter of saying that verbs X, Y, and Z collocate with A, B, and C, but saying what the collocates are for a thousand or ten thousand verbs: in the case of bilingual lexicography, saying whether translations occupy similar semantic and phraseological spaces. Similarly, it is not a matter of saying that idioms such as *kick the bucket*, *bury the hatchet*, and *spill the beans* exhibit particular transformational behaviours, but how every single idiom in the dictionary text behaves. In consequence, there is tension between phraseological theory, the evidence of texts, and the lexicographical description of phraseological phenomena. It is this tension which underpins the present chapter, where my main focus will be on the treatment of English collocations in general dictionaries.

While this chapter is concerned only with British approaches to collocation in lexicography, it is important to pay tribute to the extensive work that has been carried out in other languages and that follows other traditions, including those which treat phraseology in terms of valency or lexical functions. For valency, special mention should be made of the German tradition: realized, for example, in the *Wörterbuch zur Valenz und Distribution deutscher Verben* (1980, etc.), and, with respect to English, in Herbst et al.'s corpus-based *A Valency Dictionary of English: A Corpus-Based Analysis of the Complementation Patterns of English Verbs, Nouns and Adjectives* (2004). For lexical functions, Mel'čuk's work is well-known, and his lexicographical output in French includes the *Dictionnaire explicatif et combinatoire du français contemporain: Recherches lexico-sémantiques* (1984): see Mel'čuk 1988 for discussion. Two major projects, DAFA (= *Dictionnaire d'apprentissage du français des affaires*) and DAFLES (= *Dictionnaire d'apprentissage du français langue étrangère ou seconde*), also make special features of treating collocation and valency: see Selva et al. (2002), Binon & Verlinde (2004). Another substantial project, ongoing, is that of Blumenthal, where the aim is to create a dictionary of French collocations (for a description of the project, see http://www.romanistik.uni-koeln.de/home/blumenthal). See also the special issue of *Langue Française* edited by Blumenthal & Hausmann (2006), which addresses issues of collocation, corpora, and lexicography, and Tutin (2005) for evaluation of dictionaries of French collocations by Beauchesne and Mel'čuk & Polguère, alongside two electronic editions of major French dictionaries. There are of course many other important phraseological/lexicographical projects in Europe and elsewhere, and many other significant contributions to metalexicography, all of which deserve mention, but space here is, unfortunately, too limited to discuss them.

2. The contribution of corpora

Corpus data is the great facilitator for the description of phraseology, and its use in dictionary-making has heavily influenced the ways in which lexicographical attitudes

towards phraseology have developed over the last twenty-five years. The whole thrust of British and British-influenced lexicography now is to provide not a description of an ideal language, but an idealized description of language in use. Phraseology becomes inescapably part of that description, because corpus evidence makes it impossible to ignore. How successfully it is described in dictionaries is evaluated in the following sections. First, I want to establish a base line for comparison by looking at three English words which happen to begin with the sequence *riv-*: *river, rivet, riven*. They represent the middle segment of a meaningfulness continuum which extends from semantically depleted, context-dependent items (*of, take, situation*) to semantically restricted items such as technical terms (*carburettor, dahlia, deuterium*). Corpus data is drawn from the 450-million word Bank of English (BoE), created by Cobuild at the University of Birmingham.

2.1 *River*

River is by far the commonest of the four words, amongst the 1200 most frequent lemmas in English. Overall, singular *river* is five times as common as its plural. Nearly 40% of tokens of the lemma in BoE occur as *River* in names: structures such as *the River Thames, Severn, Shannon* in British English; *the Hudson, Mississippi, Colorado River* in American English.[1]

General collocation listings for non-capitalized singular *river* contain predictable items, many referring to likely co-occurrents in the real world: *bank(s), water, side, valley, bridge, fish*. Collocates for the plural *rivers* include many other plural forms and co-hyponyms – *lakes, streams, water, creeks, canals, sea; mountains, forests* – as well as items such as *flow/ing, fish, salmon*.

In terms of syntagmatic patterning, singular *river* often occurs in the structure PREPOSITION + *the river*: *on, across, into, in, from, down, by, along* (etc.) *the river*. *River* also recurs in complex noun groups with *of*: *the other side of the river, the banks/mouth of the river, views of the river*, and so on. Other words preceding *of* in this structure include *stretch, head, middle, reaches, section, tributary*. Such noun groups themselves tend to be part of locatives, as in *along the bank of the river*.

River occurs as the first element in a number of nominal compounds, mainly transparent and variously hyphenated. These include *river bank, river bed, river blindness, river system, river mouth, river boat*. *River/s* also occurs as a partitive in the structure *river of* -, and many realizations are figurative or allusive: *river(s) of blood, fire, dreams, life, light, tears, time*, and so on.

Adjectives immediately premodifying singular, non-capitalized *river* mainly refer to physical characteristics such as size, position, or an aspect of the water: *great, major, little, small, deep, wide; underground, nearby, local; swollen, raging, tidal, polluted,*

1. More commonly, though, rivers are named in discursive text simply as *the Thames*, etc.

muddy, according to T-score statistics.[2] Mutual information foregrounds a different, though overlapping, set of adjectives: *navigable, subterranean* as well as *polluted, swollen, tidal, raging, muddy*, and items which are more descriptive and evaluative (*mightiest, forested, woody, icy, sinuous, placid, murky*). Sets of adjectives associated with *rivers* are broadly similar, though *large* and *coastal* move up the T-score rankings, *rancid, freshwater, glacial* up those for mutual information.

Collocating verbs should represent what people do to rivers and what rivers do (at least in corpus texts). The following lists give those collocates foregrounded in BoE: where multiple forms of a lemma occur as collocates, this is indicated by * following the base form. T-score shows *go** and *cross** as verbs preceding *river/rivers*, along with *overlooking, find*, throw*. Mutual information produces a more colourful set, referring to specific activities such as *punting, rafting, kayaking, wade*, paddle*, drown*; ford*, recross*, navigate*; dam*, dredge**, and so on. There are similarly differences between measures with verbs following *river/rivers*. T-score shows that rivers *run*, flow**, or less often *go**, and they *burst* their banks from time to time. Mutual information provides further verbs such as *overflow*, empty*, foaming, divide*, recede*, gushing*.

River occurs in few idioms. *Sell someone down the river* 'betray, exploit' occurs roughly one hundred times in BoE, mainly in British print journalism or books: one-third of tokens are passive, and those betraying or exploiting are typically people in a continuing position of power within a political or other public arena. American English *(send) someone up (the) river* 'send to prison' occurs just once in the predominantly British BoE. There are a very few tokens of variations on the proverb *all rivers run to the sea, all rivers flow into the sea/ocean*.

2.2 *Rivet*

The noun *rivet* ranks just inside the top 30,000 lemmas in BoE: the plural is roughly twice as common as the singular. One pattern is its use in semilexicalized combinations such as *rivet hole*, *rivet head*, *pop rivet*, and *rivet gun*. Typical adjectives premodifying *rivet* refer to materials (*copper, brass, steel, metal, aluminium*). Other collocates refer to manufacturing processes and artifacts (*seams, zip, screws, bolt(s) machine, drill; overstitched, reinforced*). BoE does not provide much evidence of what people do with rivets, but verbs such as *fasten, insert, hold* recur.

The verb *rivet* is more frequent than the noun, within the top 15,000 lemmas in BoE. Its dominant uses are figurative, referring to deep interest and focussed attention, in structures such as

> *SOMEONE is riveted by ...*
> *ONE's attention is riveted on/upon ...*

2. See Hunston (2002) for a discussion of the respective merits of T-score and mutual information. Both have their applications: crudely, T-score seems to foreground typical patterning, mutual information to foreground specificity of semantic space.

SOMETHING rivets ONE's attention / the attention of ...
ONE's eyes are riveted on/upon/to ...
(parenthetically) ..., *(ONE's) eyes riveted on/upon/to/by ...*
ONE's gaze is riveted on ...
ONE's mind is riveted on ...

Other less frequent figurative uses have human subjects and fixed locatives:

SOMEONE is riveted to SOMEONE's chair, etc.
SOMEONE is riveted to SOMEONE's television
SOMEONE is riveted to the spot

The last can be compared to phrasal *be rooted to the spot*, which is roughly 25 times as frequent.

Over half of the verbal tokens occur as *riveting*, typically used adjectivally in relation to artistic or cultural activities: *a riveting performance, account, story, drama, book, spectacle; riveting reading, viewing, television; the story* (etc.) *is riveting*. Adverbial modifiers of this use are *absolutely, utterly, most, not exactly*.

There are relatively few examples of verbal uses 'fasten by means of rivets'. Dominant phraseological patterns here are:

SOMETHING is riveted to SOMETHING
SOMEONE rivets SOMETHING to SOMETHING

2.3 *Riven*

BoE has almost no evidence for forms of the verb *rive* 'split, divide' other than *riven* and the compound *riving knife*. *Riven* itself has a frequency of slightly over 1/million words: just inside the 25,000 commonest lemmas.

Only a handful of tokens in BoE have physical meanings (*riven stone/slate/flagstone; riven with gorges; a face riven with smile-lines*). Other uses are figurative, mainly found in the pattern *be riven by/with*, where the subject slot is realized by nouns such as *society, parties, families, communities, countries*, and the agent is a noun which denotes conflict (*dissent, disputes, infighting, conflicts, controversy, war, feuds, strife, factionalism, rivalries*, etc.) or something which causes disturbance to norms and status quo (*drugs, corruption, injury*, etc.) This sets up a semantic prosody (Louw 1993), so that even in neutral or positive contexts, we infer damage or an unsettling effect:

These either/or names are typical of Batman, which is a tale riven by double identity.

Riven occurs occasionally as a combining-form, typically with reference to social division (*faction-riven, conflict-riven, colour-riven*) and also physical terrain (*gulley-riven, wind-riven)*.

3. Phraseology and dictionaries for native speakers

The next step is to see how lexicographers record this kind of evidence of phraseological patterning. Before looking at monolingual dictionaries for native speakers, in this case of English, it is important to consider why it should be recorded at all. Certainly adult users should be familiar with *river* and *rivet* (at least) and competent in producing idiomatic and well-formed utterances. Is phraseological information redundant?

Of course, idioms and other fixed phraseologies, where meaning is dependent on form, must be treated: a primary role of a native-speaker dictionary is to list and explain the lexical items of a language. There is also a need to show phraseology where senses or items are restricted co-textually, for example where verbs are followed by specific prepositions or associated with particular kinds of object: another primary role of a dictionary is to provide information on 'correct' or standard usage. Phraseology, too, has a function in clarifying sense differentiation, whether information appears as part of the definition or within an illustrative example: in the latter case, it may also clarify the definition itself. This function relates as much to the decoding of a dictionary as of a language.

A further reason for including phraseological information can be characterized as linguistic: a wish to create an inventory of lexical behaviour as part of a complete and integrated description of a language. However, only very large dictionary projects, with generous funding, would be in a position to do this for all words. Besides, average users are unlikely to find the information useful enough to be worth the extra work, while interactive corpus tools now provide the information more economically and effectively for researchers. The compilers of lexica for language engineering work might well take a different view of the value of phraseological information, but the target users and functional constraints here are radically different from those of commercial publishers.

3.1 Historical aspects

Early monolingual dictionaries of English were concerned with 'hard words': words adopted from French, Latin, and Greek which were problematic in terms of meaning, spelling, or both. Phraseology was not a concern, and few multi-word items of any kind were recorded. Information on selectional/collocational restrictions was at most implied within definitions, as in the following from Cawdrey's *A Table Alphabeticall* (1604):[3]

> *nauigable* – where ships may safely passe, or that may be sailed vpon.
> *vegetable* – springing, or growing, as herbes.

3. Typography in dictionary entries cited in this chapter is not reproduced exactly.

Over the course of the 17th century in England, monolingual dictionaries expanded into early inventories of a more complete lexicon, including common words. While compounds and other multi-word items were treated, there was some ambivalence towards colloquial expressions such as idioms and proverbs.[4] Perhaps most interesting for phraseology, however, is the way in which words were occasionally recorded and explained within phraseological structures: that is, the definiendum or entry word was a collocation rather than a simplex word or multi-word item. For example, Nathan Bailey's *Dictionarium Britannicum* (1736) has the following:

> *To grow* PALE, to grow wan, or white looked.
> *To* RIPPLE *Flax*, to rub or wipe of the seed-vessels.
> RISING *of the Sun*, its appearing above the horizon.
> *To* RISK] to venture, to hazard.
> *To run a* RISK]
> *A good* ROAD [with *Sailors*] a place where neither sea nor wind has much power over the ship.

Samuel Johnson instituted a much greater degree of rigour in his *Dictionary of the English Language* (1755), in particular pioneering the systematic treatment of phrasal verbs: see Osselton (1986) for discussion. Apart from compounds, he treated other types of multi-word item more sporadically, though there is evidence of grammatical items (*at all, (in) spite of, as well*), speech formulae (*pardon me, I pray, to be sure*), and metaphorical idioms (*the coast is clear, to lead by the nose, thick and thin, to turn the tables*). Phraseological patterning of headwords and senses was largely shown implicitly, within the citations which accompanied definitions in order to demonstrate good usage. For example, those for different senses of the verb *rivet* happen to include prepositional choices:

> This man
> If all our fire were out, would fetch down new,
> Out of the hand of Jove; and *rivet* him
> To Caucasus, should he but frown.
> *Benj. Johnson.*

> In *rivetting,* the pin you *rivet* in should stand upright to the plate you *rivet* it upon; for if it do not stand upright, you will be forced to set it upright, after it is *rivetted.*
> *Moxon.*

> You were to blame to part with
> A thing stuck on with oaths upon your finger,
> And *rivetted* with faith unto your flesh.
> *Shakesp.*

4. See Moon (2000) for a brief overview of the treatment of phraseology in early dictionaries.

But just as Johnson observed the impossibility of defining verbal elements of phrasal verbs in isolation from particles, so there are many other cases where, like Bailey, he defined words within phraseological structures:

> FEMALE *Rhymes.*
> Double rhymes so called, because in French, from which the term is taken, they end in *e* weak or feminine.
>
> HOLD *of a Ship.*
> All that part which lies between the keelson and the lower deck.
>
> *It* RAINS.
> The water falls from the clouds.
>
> *To* SERVE *a warrant.*
> To seize an offender, and carry to justice.
>
> *To cut the* THROAT.
> To murder; to kill by violence.

The *Oxford English Dictionary* (1884–) has a much more extensive coverage of multi-word items: for example, the main article for **river** alone has around 160 combinations beginning with *river-*, including the names of creatures and plants, and words related to geography and transportation. It also shows phraseological structures within definitions. Those in its article for the verb **rivet** demonstrate two typical techniques, with selectional/collocational restrictions stated in parentheses, or with collocating particles mentioned explicitly:

1. *trans.* To secure (a nail or bolt) by hammering or beating out the projecting end of the shank into a head or knob; to clinch. Also with *down*.
2. To secure or fasten with or as with rivets. Also with *down, in, together*.
3. Const. *to* or *into* something.
4. To fix intently (the eye or the mind); to command or engross (the attention).

See Hanks (1987) for discussion of ways in which definitions show phraseology.

3.2 Current practice

Amongst current monolingual dictionaries, the treatment of phraseology varies according to size of the text. The following looks at two large British dictionaries, *Collins English Dictionary* (2003: hereafter CED) and the *New Oxford Dictionary of English* (1998: NODE). Both deal with compounds as individual headwords, and define idioms and other fixed phrases as embedded items within the body of the entry for a component word, though CED treats phrasal verbs as headword items, while NODE treats them in secondary sequences following senses for the simplex verbs.

3.2.1 *The treatment of river*

CED and NODE say little about the phraseology of the core literal sense of *river*. CED lists uses of *river* as modifier and in combination ('*river traffic, a river basin; riverside, riverbed*'), while NODE comments on its use in animal and plant names ('*river dolphin, river birch*') the first of which is also a headword in its own right. Both record an extended or figurative use 'quantity, flow', and imply structure through examples:

> [CED] any abundant stream or flow: *a river of blood*.
>
> [NODE] a large quantity of a flowing substance: *great rivers of molten lava* | figurative *the trickle of disclosures has grown into a river of revelations.*

Though the *of* -structure is shown, neither dictionary really does much to explain usage from decoding and encoding perspectives, nor do they delimit the range of realizations which follow *of* .

Both dictionaries treat the idiom *sell (someone) down the river*, though somewhat perfunctorily: neither exemplifies it nor gives information on its phraseological patterning. NODE also treats *up the river*, and gives the origins of both idioms. See Moon (2007) for an overview of the treatment of idioms in general monolingual dictionaries.

3.2.2 *The treatment of rivet*

It is the verbal uses of *rivet* which are more interesting phraseologically. CED gives two literal (or physical) senses, followed by one which represents figurative uses, though not expressly limited to the figurative. Only this last sense is exemplified.

> 1 [noun sense]
> 2 to join by riveting.
> 3 to hammer in order to form into a head.
> 4 (*often passive*) to cause to be fixed or held firmly, as in fascinated attention, horror, etc.: *to be riveted to the spot*.

The three senses are explicitly labelled as transitive, but valency and other structures are not shown, nor is there direct mention of the very strong collocation between *attention, eyes, gaze* etc. and *rivet*, as evidenced in BoE (used by Collins in dictionary-making, though this entry is unchanged from CED's first, pre-corpus edition of 1979).

NODE gives one literal and three figurative uses:

> [with obj.] join or fasten (plates of metal) with a rivet or rivets: *the linings are bonded, not riveted, to the brake shoes for longer wear*.
> ☐ fix (someone or something) so as to make them incapable of movement: *the grip on her arm was firm enough to rivet her to the spot*.
> ☐ attract and completely engross (someone): *he was riveted by the newsreels shown on television* | [as adj. **riveting**] *a riveting story*.
> ☐ (usu. **be riveted**) direct (one's eyes or attention) intently: *all eyes were riveted on him*.

NODE's entry certainly seems to represent more adequately the semantic and phraseological behaviour of the word. The question remains, though, as to how readers use the information.

3.2.3 *The treatment of riven*

Both CED and NODE treat *riven* within the entry for *rive*, but state clearly that the dominant form of the verb is passive. CED has three senses, of which the first is literal, the third figurative, and the second indeterminate:

> 1 to split asunder: *a tree riven by lightning*.
> 2 to tear apart: *riven to shreds*.
> 3 *Archaic*: to break (the heart) or (of the heart) to be broken.

While CED's entry can be mapped onto BoE evidence, it scarcely represents the phraseological patterns found. BoE in fact has just one example of *rive*/riven* in collocation with *lightning* (a literary quotation), one with *heart* (an archaic reference), and none with *to* or *shreds*. Altogether missing is the striking pattern where societal groups are riven by conflict.

NODE also has three senses: a figurative one, given first, followed by two literal:

> (usu. **be riven**) split or tear apart violently: *the party was riven by disagreements over Europe*. | figurative *he was riven with guilt*.
> ☐ archaic split or crack (wood or stone): *the wood was riven with deep cracks*.
> ☐ [no obj.] archaic (of wood or stone) split or crack: *I started to chop furiously, the dry wood riving and splintering under the axe*.

The first shows clearly the dominant pattern observed in BoE and missing from CED; sequencing of senses reflects their relative frequencies. Perhaps missing, though in part accountable through the literal uses, is a more generalized sense 'fissured', relating to terrain or lined faces, and with its own sets of collocates.

4. Phraseology and learners' dictionaries

In contrast to the situation with dictionaries of English for native speakers, those aimed at EFL users have from the start paid special attention to phraseology, particularly to the phraseology of common words. This has been motivated by practical pedagogical concerns, including learners' needs to encode in idiomatic English, and an awareness of asymmetries between first and foreign languages in the phraseologies and semantics of individual words. Hornby et al.'s *Idiomatic and Syntactic English Dictionary* (ISED: 1942), the first monolingual learners' dictionary, established a model for the treatment of collocations and structures, as in:

> [at **rise**, noun] an increase in value, amount, degree, etc., as *a rise in prices* [*temperature, social position, etc.*].

> [at **rivet**, verb] 3 (fig.) fix or concentrate (the eyes, the attention) on. 4 (fig.) secure; attract (attention).

All major British learners' dictionaries continue to prioritize information on phraseology and collocation, particularly for central vocabulary items.

The present era of learners' dictionaries, the corpus era, began with the Cobuild project and its first dictionary, the *Collins Cobuild English Language Dictionary* (1987). Lexicographically, this has led to a special focus on corpus evidence and the typical lexicogrammatical patterns revealed. The following sections review coverage of the three *riv-* words in recent editions of the *Oxford Advanced Learner's Dictionary* (OALD: 2005, 7th edition), the *Longman Dictionary of Contemporary English* (LDOCE: 2003, 4th edition), and the second edition of CCED (1995), which was based on the Bank of English. See Landau (2001: 308ff.) for discussion of lexicographical methodology; Klotz (1999) and Mittmann (1999) for methodical comparisons of the treatment of complementation and collocation in the 1995 editions of four learners' dictionaries; Cowie (1999) for discussion of the historical development of learners' dictionaries; Siepmann (2006: 13ff.) for discussion of the presentation of collocational information in dictionaries, including both monolingual learners' and bilingual dictionaries; and Čermák (2006) for a discussion of collocation in relation to corpus evidence and LDOCE.[5]

4.1 The treatment of *river*

As a high frequency noun in English, *river* is treated in detail in OALD and LDOCE. ISED's entry was just 21 words long, excluding the combinations *river basin* and *river bed*, and its only example was for a figurative use, *a river of lava*. In contrast, entries in OALD and LDOCE are both around 120 words, six times the length, with much of their entries consisting of the exemplification of phraseological structures of literal *river* 'waterway', variously glossed and highlighted. OALD gives the following:

> *the River Thames*
> *the Hudson River*
> *on the banks of the river* (= the ground at the side of a river)
> *to travel* **up/down river** (= in the opposite direction to/in the same direction as the way in which the river is flowing)
> *the mouth of the river* (= where it enters the sea/ocean)
> *Can we swim* ***in the river****?*

5. Discussion here is limited to the treatment of phraseology in entries for individual words, but dictionaries also include a variety of special features, including articles in front, middle, and end matter on general phraseological phenomena. Some dictionaries add listings of collocates for selected words: for example, the *Macmillan English Dictionary* (2002) lists for *road* the verb collocates *branch, cross, descend, dip, fork, lead, narrow, run, widen, wind.*

a boat on the river
They have a house on the river (= beside it)

LDOCE has:

the Mississippi River
the River Thames
on a river *a boat on the river*
along a river *We went for a walk along the river*
up/down (a) river *a ship sailing up river* | *They drifted slowly down river.*
across a river *a bridge across the river*
on the south ***bank of the river*** (=the land on one side of a river)
trees on the ***river bank*** (=the land next to a river)
the ***River*** *Elbe* ***flows*** *through the Czech Republic.*
at the ***mouth of the river*** (=where a river joins the sea)

Thus both indicate prepositional selections which precede *river*, pattern structures of river names (though without comment on British/American distinctions), and collocating nouns *bank* and *mouth*. LDOCE draws specific attention to the verb collocate *flow*, while OALD only does so by implication in its gloss of *down river*. Neither deals with adjective collocates. Both dictionaries use typography to distinguish levels and types of information here, and presumably fixedness and frequency of collocation, though the distinctions may be oversubtle in places for users who have not familiarized themselves fully with the front matter and its explanations.

Both OALD and LDOCE include, more cursorily, the extended or figurative use of *river*, explicitly labelling its occurrence in the pattern *river of*. Both give a single, similar, example:

(OALD) *Rivers of molten lava flowed down the mountain*
(LDOCE) *a river of hot lava flowing from the volcano*

CCED's entry is by comparison disappointingly thin, just over 30 words long. Only the literal use is treated, while collocates are shown by implication, within examples:

a chemical works on the banks of the river
boating on the River Danube

Nothing is foregrounded here: neither variation in names (the grammar code 'oft in names before n' ignores American usage), nor collocating prepositions other than *on*, nor lexical collocates other than *bank*. The centrality of *river* as a lexical item is flagged up through its frequency marker, and CCED and Cobuild in general had a special interest in corpus-driven phraseology, but there seems to have been a decision that the word needed no further description.

4.2 The treatment of *rivet*

The phraseological interest, as before, relates to verbal uses, particularly figurative ones, which are prioritized over the literal in each case. CCED treats only the figurative, incorporating a canonical phraseology into its definition, as is its conventional practice:

> If you **are riveted** by something, it fascinates you and holds your interest completely.

Phraseology is further shown in the subsequent examples:

> *As a child I remember being riveted by my grandfather's appearance*
> *He was riveted to the John Wayne movie*
> *The scar on her face had immediately riveted their attention*

This covers two typical prepositional patterns, but not collocation with *eyes, gaze*, etc.

OALD labels the use as usually passive, and builds phraseological information into definition and examples:

> to hold sb's interest or attention so completely that they cannot look away or think of anything else: *I was absolutely riveted by her story.* ◊ *My eyes* ***were riveted on*** *the figure lying in the road.*

This shows a collocation with *eyes*, *by*, *on*, and (arguably) *attention*. A collocation with *interest* may be implied, by analogy with *attention*; however, there are no examples of its occurring in BoE. OALD treats separately as an idiom **be riveted to the spot/ground** (the latter variant is not attested in BoE). Phraseological information for the literal sense 'fasten with rivets' is limited to an example, *The steel plates were riveted together*.

LDOCE is more overt than either of the other dictionaries, showing phraseology through citation form, sentential definition, and example:

> **be riveted on/to/by sth** if your attention is riveted on something, you are so interested or so frightened that you keep looking at it: *All eyes were riveted on her in horror.*

This shows the three dominant prepositional structures, and noun collocation with *attention* and *eyes*. *Be riveted to the spot* is treated as a fixed phrase. No phraseological information is given for the literal sense.

All these dictionaries have separate entries for *riveting* as an adjective. None provide collocational information within their definitions, and OALD gives no example. CCED's example shows only a less common phraseology, *find ... riveting:*

> *I find snooker riveting though I don't play myself*

LDOCE gives two examples:

> *a riveting performance* | *His story makes riveting listening*

These show the typical semantic range of collocates found in BoE: the set of nouns such as *performance, story*, and collocation with a verbal noun, though BoE attests *reading, viewing* more than *listening*.

4.3 The treatment of *riven*

ISED had treated *riven* within an entry for the verb *rive*, exemplified in an example *riven rocks*. However, all three of the recent dictionaries examined omit the original verb and treat *riven* as an adjective – in keeping with corpus evidence. OALD and LDOCE both include a literal sense 'split', though as secondary and without exemplification: both their definitions specify 'an object' as selection restriction here. CCED includes only the figurative sense.

Corpus evidence, as described in Section 2.3, shows a very strong phraseological pattern in terms of grammar and selectional restrictions: social groups are riven, and they are riven by conflict. All three dictionaries build such information into definitions and grammatical coding, and show phraseology and collocates further in examples:

> (CCED) If a country or organization is **riven** by conflict, its unity is torn apart by a violent disagreement between its people.*The four provinces are riven by deep family and tribal conflicts... The Communist movement has been riven with factional fighting.*
> (in grammar column) usu v-link ADJ *by/with* N
>
> (OALD) (**by/with sth**) (of a group of people) divided because of disagreements, especially in a violent way: *a party riven by internal disputes.*
>
> (LDOCE) if a group of people are riven, they are divided by disagreements, especially in a violent way: [+**by/with**] *a community riven by religious differences.*

CCED's examples demonstrate the register and flavour of collocates more fully – or, unsurprisingly, they are consistent with BoE; however, they may be more difficult for users.

4.4 The treatment of *sell down the river*

Learner's dictionaries do not generally deal with metaphorical idioms in much detail. Most are considered receptive items, and they are often not exemplified. LDOCE, for example, gives an extended explanation:

> to do something that harms a group of people who trusted you, in order to gain money or power for yourself

This shows, though through implication, the power relationship evident in corpus evidence. CCED has a longer entry:

> If someone **sells** you **down the river**, they betray you for some personal advantage. *He has been sold down the river by the people who were supposed to protect him.*

Its example shows the passive structure and implies the power relationship between betrayer and betrayed, less obvious in the definition.

Only dictionaries of idioms really have the scope to give fuller accounts of the forms, variations, structures, meanings, pragmatics, and usage of such minor phraseological items as this one. For example, the *Longman Idioms Dictionary* (1998) has:

> to do something that harms a group of people who trusted you, in order to get an advantage for yourself: *The government has sold the fishermen right down the river, by making deals with other countries that don't protect our fishing rights.* | *The Nationalists believe that they have been sold down the river by their former allies, the Democrats.*

The *Cobuild Dictionary of Idioms* extends the entry in CCED:

> If someone **sells** you **down the river**, they betray you for some personal advantage. *He said he could not agree to measures which would sell British farmers down the river in order to keep smaller, less efficient farms in production. / He has been sold down the river by the people who were supposed to protect him. It has a devastating effect on his health.*

In both cases, examples demonstrate power structures and show passive uses, but little more. In contrast is the pioneering *Oxford Dictionary of Current Idiomatic English* (Volume 1, 1975; Volume 2, 1983).[6] Rooted in the ISED tradition, it makes a special feature of indicating phraseological patterning explicitly in terms of both structure and typical collocates or valency:

> **sell down the river** [B2 pass] (informal) betray the interests of one's own people, of members of one's party, trade union etc. **S:** minister, leader; delegate, representative. **O:** country; union, member... *At a mass meeting of car workers called by shop stewards the official leadership was accused of* ***selling*** *the rank and file* ***down the river.***

The text can be criticized for its accessibility for users, especially for the notional target group of learners, but no other recent English dictionary of idioms compares with it for delicacy of information about the phraseology of multi-word items. See Moon (1999) for further discussion of issues.

---

6. The two volumes were later republished as the *Oxford Dictionary of Phrasal Verbs* and the *Oxford Dictionary of Idioms*.

5. Phraseology and bilingual dictionaries

Bilingual dictionaries as such did not appear in England until the 16th century, and were mainly unidirectional, typically decoding into English from the other language, though Palsgrave's dictionary *Lesclarcissement de la Langue Françoyse* (1530) encoded into French. They were far more sophisticated than the first monolingual English dictionaries of the following century (see Stein 1986, 1997 for discussion): more comprehensive, more inclusive, and showing more awareness of phraseology because their different aims and functions necessitated a different approach. For example, Cotgrave's French-English dictionary of 1611 included many compound items, phrases, and proverbs, and also translated or glossed collocations which were or are, perhaps, less fully lexicalized:

> [sv **rien**]
> Elle le haït sur tout rien. *She hates him extreamely, aboue all things...*
>
> [sv se **rigoler de**]
> Se rigoler au soleil. *To sport, solace, or ioyfully to bleake, or spread himselfe in the Sunne...*
>
> [sv **risque**]
> Ie le prens à ma risque. *Hab or nab* [= have or have not], *at my perill be it, happen how it will.*
>
> [sv **rive**]
> La rive d'un bois. *The skirt, edge, or side of a wood.*
> La rive d'un pain. *The side of a loafe.*
>
> [sv **roder**]
> Roder les rues. *To iet, walke, trot up and downe the streets (especially anights).*

These foreshadow the definitions of words as collocations, found a century later in Bailey and Johnson: see Section 3.1 above.

Entries in large bilingual dictionaries are now characterized by arrays of translated examples, showing typical phraseological patterning, and oriented towards lexis, grammar, or both. When pairs of languages have comparable status in terms of market, such as English and French or Spanish, bidirectional dictionaries have four sets of user – English speakers decoding from French/Spanish and encoding into French/Spanish, French/Spanish speakers decoding from English and encoding into English. Each half of the dictionary has to provide suitable information for two sets of users with quite different goals and linguistic knowledge, and translated examples have to fulfil multiple purposes. They assist encoding in general, whether from or into a first language, though they are more likely to be useful for those encoding into a second/foreign language. They also indicate sense in the case of polysemous words, and the appropriateness of translations where there are multiple equivalents (this type of information is also likely to be recorded formulaically in parenthesis or through la-

bels). For further discussion, see, for example, Svensén 1993, passim; Atkins 1996; and Siepmann 2006: 16f., 33ff.

The following reviews practice in two large bilingual French-English dictionaries: the *Collins Robert French-English English-French Dictionary* (CRFED: 2002, 6th edition), and the *Oxford Hachette French Dictionary* (OHFD: 2001, 3rd edition). Its focus is on *river* and figurative *rivet*, along with French equivalents or cognates *fleuve/rivière* and *river/rivé*. Neither dictionary offers more for *riven* than a translation *déchiré (par)*, without mention of typical collocates and context; on the other side, *déchirer* is a far more general verb.

5.1 The treatment of *river* and *rivière/fleuve*

To begin with English *river*, much of the phraseological information relates to a range of compound formations, which are translated individually. For example, both CRFED and OHFD include *river basin* and *river mouth*, and their translations of the first and third coincide, respectively *bassin fluvial* and *embouchure* (though CRFED also gives *bouche d'une rivière / d'un fleuve*). Both dictionaries give translations of *up river* and *down river* at **river**, though they also appear as items at **up-** and **down-**. Phraseological patterning of river names is indicated in CRFED simply by translating *the river Seine, the Seine river*, with relevant variety labels, as *la Seine*; OHFD, however, has a special boxed feature which includes more detailed information about the *fleuve/rivière* distinction, and a series of translated phraseologies which relate to river names: hence *to go down the Rhine, the course of the Danube*, are glossed as *descendre le Rhin, le cours du Danube*. In both dictionaries, figurative uses of *river* are given the translation *fleuve*, but neither provides general information about structure or collocation, other than for the specific item *rivers of blood* and its translation, *des fleuves de sang*. The only translated example in either dictionary is in CRFED, which has

> the accident has resulted in several rivers being polluted
> à la suite de cet accident, plusieurs cours d'eau ont été pollués

While users might find the additional translation of *river* helpful, they could also find other elements distracting. The idiom *sell someone down the river* is simply glossed in both OHFD and CRFED as 'trahir qn'.

Entries for *rivière* are sparse in both dictionaries: the equivalent *river*, without explanation of its different range of reference, and then specific translations for its use in horse-riding/racing (*water jump*) and the combination *rivière de diamants*. OHFD includes the saying *les petits ruisseaux font les grandes rivières* with a conceptual equivalent *great oaks from little acorns grow*; CRFED includes it at **petit**, with a literal translation *little streams make big rivers*. This translation is unlabelled, and it is quite possible that an unwary user might assume it to be lexicalized in English, which it is not. No further encoding information for *rivière* is given.

Only OHFD gives any phraseological information for the literal sense of *fleuve*: a single translated example (*au bord d'un fleuve*, 'by a river') and a translation of the

boxed feature which appeared at *river* on its English side. Both dictionaries treat the specific collocation *fleuve de larmes* 'flood of tears', though neither contextualize it further, for example by showing collocating verbs or prepositions. OHFD treats other uses of *fleuve* as a second sense

(flot de boue, lave, sang) river (**de** of)

and a glossed phrase

fleuve(s) humain(s) stream of humanity

CRFED treats extended uses as a translated collocation:

fleuve de boue, de lave river of mud/of lava

and adds separately

fleuve de sang river of blood

sa vie n'a pas été un long fleuve tranquille (*hum*) his life hasn't been a bed of roses[7]

Postnominal uses of *fleuve* as in *discours/procès/manifestation fleuve* are treated separately: both dictionaries give collocates which suggest semantic type, but do not delimit range (typically a noun referring to verbal output or artistic performance).

All these entries are functional, focussed, and clear, but it is disappointing to realize just how little general phraseological information is included. There is almost nothing to indicate typical prepositional patterning, nor typical adjectives or verbs associated with *fleuve* and *rivière*. It thus suggests a translation-driven approach to encoding in these dictionaries (which are marketed to advanced learners as well as language professionals). They provide appropriate translations for English verbs such as *flow* or *run* in the context of rivers, but at those headwords and not necessarily elsewhere. This compares unfavourably with the situation with monolingual learners' dictionaries of English, or monolingual dictionaries of French in the *petit Robert* series, which list collocations.

5.2 The treatment of *rivet* and *rivé/river*

Discussion is limited to the figurative sense of the English verb *rivet* and cognate French *river*: it omits literal uses of *rivet* in French/English, or other senses of French *river* and metaphorically parallel *clouer*. CRFED treats *rivet* on the English side in two translated examples

7. Like the earlier example with 'les petits ruisseaux...', the translation here raises a question of the appropriateness of offering semantically parallel idioms. In these cases, are they simply intended to help English speakers decoding from French to appreciate the clichéed status and pragmatics of the source language expression?

it riveted our attention ça nous a fascinés
riveted with fear rivé *or* cloué sur place par la peur

BoE shows *attention* as a strong collocate of *rivet*, though more typically passive than active, but has no examples at all of *riveted with fear*, and almost none of *rivet* with ...*. There is a separate entry for adjectival **riveting** with the broader equivalent *fascinant* and no example.

OHFD, which has an entry for **riveting** like CRFED's, treats other uses in two senses, preceding a literal sense, and giving collocational and selectional information:

1 (captivate) **to be riveted by** être captivé par [performance]

2 (fix) to be **riveted** on [eyes, gaze] être rivé sur; **to be riveted to the spot** [person] être cloué sur place

While this seems clearer than CRFED's coverage, it implies a semantic distinction (captivate/fix) which perhaps misrepresents the way figurative uses shade into one another. *Attention* as a collocate is not mentioned.

On the French side, CRFED has separate entries for adjectival *rivé* and *river*, though the figurative uses relating to cognitive engagement appear at the first. They are treated through translated examples and specified collocations:

rivé à [+ *bureau, travail*] tethered *ou* tied to; [+ *chaise*] glued *ou* riveted to
les yeux rivés sur moi/la tache de sang (with) his eyes riveted on me/the bloodstain
rester rivé sur place to be *ou* stand riveted *ou* rooted to the spot
rivé à la télé glued to the TV

OHFD has a single entry **river**, and three translated examples for figurative uses, which are broadly consistent with CRFED:

être rivé à qch FIG to be tied to [*travail, famille*]; to be glued to [*télévision*]
je suis restée rivée sur place par la surprise I stood riveted *ou* rooted to the spot with surprise
avoir les yeux rivés sur to have one's eyes riveted on

Rivet/riveted and *river/rivé* are certainly not direct equivalents, and it is entirely right that other verbs should be used in the English translations. However, taking into account typical and dominant English phraseologies, it is perhaps questionable that *tethered* should be offered before *tied* as a translation in *rivé au bureau* in CRFED, or that both dictionaries offer *riveted to the spot* as a translation before the much more common *rooted to the spot*, thus prioritizing lexical parallelism over idiomaticity: moreover, *stand* riveted...* occurs just once in BoE (*stand rooted* is a more significant collocation).

6. Collocations dictionaries

My main concern in this chapter has been the treatment of collocation in general dictionaries, but it would be inappropriate not to refer to specialist dictionaries of collocations. Cowie (1998: 209–228) provides an overview of dictionaries of phraseology, situating them within the context of contrasting traditions in Europe, and in particular the different approaches of British and Russian/Soviet lexicography. The following is limited to brief discussion of some recent monolingual English texts, all oriented towards non-native speakers. Two were written without corpora (the *BBI Dictionary of English Word Combinations*, BBI: 1997, 2nd edition, and LTP's *Dictionary of Selected Collocations*, DSC: 1997). In contrast, the *Oxford Collocations Dictionary* (OCD: 2002) drew on evidence in the British National Corpus: see Lea & Runcie (2002) for an account of its compilation.

In contrast to simple corpus listings of collocates, all these texts filter collocational information by organizing it in terms of grammatical structure and semantics: for example, giving adjectives separately from verbs, and listing collocates within subsets. BBI is the least explicit:

> **river** *n.* 1. to cross; ford a ~ 2. to dam; drag; dredge a ~ 3. a broad, wide; deep ~. 4. a ~ floods; flows (into the sea); narrows; overflows (its banks); recedes; rises; widens 5. the bank; course; mouth, source of a ~ 6. down ~; up ~. 7. (misc.) to sell down the ~ ("to betray")

DSC uses syntactic labels rather than numbers:

> V: bridge, cross, dam, dredge, ford, navigate ~
> V: ~ dried up, forks, flooded, flows, meanders, overflowed/broke its banks, rose, runs, winds
> A: broad, deep, navigable, salmon, shallow, sluggish, swift, tidal, wide ~

Most of the collocates listed in these dictionaries show up in BoE evidence, but do not necessarily correspond with significance listings as identified by BoE's software: BBI does not show *run*, neither text lists *burst its banks* or adjectives such as *great, major*, and *small*. In contrast, there is a far better correspondence between BoE's collocates and those listed in the much longer, more explicit entry for *river* in OCD, which also includes examples for selected collocates. The following gives three short extracts from the entry:

- ADJ. **broad, great, large, long, mighty, wide** *the mighty River Nile ◊ The river was too wide to swim across comfortably...*
- VERB + RIVER **cross, ford, get across** *We crossed the river by ferry ◊ How are we going to get across the river?...*
- PREP. **across a/the** ~ *There's a bridge across the river.* **along a/the** ~ *We walked along the river...*

This is an impressive text: its principal disadvantage is that it only covers 9000 core items. Neither OCD nor DSC, for example, give any information for *rivet*, though BBI has entries for both noun ('to drive a ~ (into metal)') and verbal adjective *riveted* ('~ on, to (all eyes were ~ on the door; she stood ~ to the spot').

7. Conclusion and implications

This chapter has presented only a partial look at the treatment of phraseology in current printed dictionaries. Most of the texts have been monolingual, all concern English, and there has been only limited reference to phraseological phenomena other than collocation, or to specialist dictionaries of phraseological items.[8] Yet from even this narrow focus, there are clearly important points to consider, apart from the quality, depth, and range of information provided. Particularly crucial is the function of phraseological information in relation to the needs and interests of the target users. The lexicographical task here is to second-guess what users might want to know about the phraseology of an individual lemma, form, or sense, as well as identifying which patterns to record.

I have also neglected discussion of electronic products and resources, though see Heid & Gouws (2006) for discussion of a 'multifunctional' electronic dictionary of collocations. The challenge here has been to move from a position where the retrieval and delivery of phraseological information is designed from the perspective of linguistic research, including provision of data for lexicography, to one where users' needs are prioritized. Yet it is arguably even more difficult to identify what these are than in the case of print dictionaries. Earlier tools simply provided raw corpus data, encouraging users to work empirically, observing patterns for themselves; however, there are disadvantages, including time factors, and difficulties with interpreting the evidence found. Furthermore, large corpora are too large to use effectively, small corpora may well be too small, and all corpora are subject to skewing from constituent texts – especially relevant where phraseological patterning varies according to genre. It seems obvious that tools have to be dynamic and provide filtered data, organized in terms of significance, word class, syntagmatic positioning, meaning, and genre, and equally that products have to be constructed around such data, but the filtering and prepackaging processes themselves may rely on second-guessing the nature of tasks, questions, and competence of users: overly filtered data may be as restrictive and misleading as underinformative entries in print dictionaries.

Much of this chapter has been taken up with criticizing dictionaries, and my discussion of any shortcomings should be set alongside the fact that these are by and large fine dictionaries: skilfully constructed and executed, often innovative, and highly

8. See Siepmann (2006) for discussion of an ongoing project to create a multilingual onomasiological dictionary of collocation.

sensitive to both object language and users. At the outset, I mentioned the way in which lexicography challenges linguistic theory, just as linguistics challenges lexicographical practice. In dealing with a lexicon word by word and sense by sense, lexicographers deal with individual cases and know that there are few watertight categories and rules. Phraseology in particular has to be discussed at the level of the individual item: *river, rivet, riven; rival, rivalry, riverine, riviera, rivulet,* and thousands of other items in turn. Words can be explored in classes or sets (*river* with co-hyponyms and plesionyms, such as *stream, brook, rill, tributary, canal*, or *rival* with *enemy, competitor*, and so on) rather than in alphabetical runs, but because each item is individual, it is unlikely to be tractable to any but the grossest of generalizations. Dictionaries may be unsuccessful for many reasons, but not an inability to generalize, which is not their role.

There is a final point to address with respect to evidence, linguistics, and the lexicographical treatment of phraseology. Just as words have to be treated individually in dictionaries, so they may require different kinds of evidence. It is not yet obvious that any one collocational measure, or corpus type, can supply everything necessary to create useful dictionary entries for all items, regardless of individual frequency and coreness: lexicographers still have to use intuition and judgement in selecting, interpreting, and setting out the evidence, rather than simply relaying it to the user as quasi-scientific truth. Meanwhile, the nature of collocational meaning is only partly understood. Thus it is not the case that lexicographers fail to appreciate phraseology, whether from textual, lexicogrammatical or semantic points of view; but at the present time they perhaps appreciate only too well that the nature of the dictionary is changing, and that descriptive phraseology as a subdiscipline has not yet fully matured.

Acknowledgements

I am indebted to colleagues, to an anonymous reviewer, and to Sylviane Granger for their helpful comments on an earlier version of this chapter.

References

Atkins, B. T. S. (1996). Bilingual dictionaries: Past, present, and future. In Gellerstam, M. (ed.) *EURALEX '96 Proceedings*, 515–546. Gothenburg: Göteborg University.

Binon, J. & S. Verlinde (2004). Les collocations: Clef de voûte de l'enseignement et de l'apprentissage du vocabulaire d'une langue étrangère ou seconde. *Romaneske* 4: 15–29.

Blumenthal, P. & F.-J. Hausmann (eds.) (2006). *Langue française: Collocations, corpus, dictionnaires* [Langue française 150]. Tübingen: Niemeyer.

Čermák, F. (2006). Collocations, collocability and dictionary. In Corino, E., C. Marello & C. Onesti (eds.) *Proceedings XII EURALEX International Congress*, 929–937. Alessandria: Edizioni dell'Orso.

Cowie, A. P. (ed.) (1998). *Phraseology: Theory, Analysis, and Applications*. Oxford: Oxford University Press.

Hanks, P. (1987). Definitions and explanations. In Sinclair, J. (ed.) *Looking Up: An Account of the COBUILD Project in Lexical Computing*, 116–136. London: HarperCollins.

Heid, U. & R. H. Gouws (2006). A model for a multifunctional dictionary of collocations. In Corino, E., C. Marello & C. Onesti (eds.) *Proceedings XII EURALEX International Congress*, 979–988. Alessandria: Edizioni dell'Orso.

Hoey, M. (2005). *Lexical Priming*. Abingdon: Routledge.

Hunston, S. (2002). *Corpora in Applied Linguistics*. Cambridge: Cambridge University Press.

Jackendoff, R. (1988). *Semantics and Cognition*. Cambridge, MA: The MIT Press.

Klotz, M. (1999). Word complementation in English learners' dictionaries – A quantitative study of CIDE, COBUILD2, LDOCE3 and OALD5. In Herbst, T. & K. Popp (eds.) *The Perfect Learners' Dictionary(?)*, 33–43. Tübingen: Niemeyer.

Landau, S. (2001). *Dictionaries: The Art and Craft of Lexicography*, 2nd edn. Cambridge: Cambridge University Press.

Lea, D. & M. Runcie (2002). Blunt instruments and fine distinctions: A collocations dictionary for students of English. In Braasch, A. & C. Povlsen (eds.) *Proceedings of the Tenth EURALEX International Congress*, 819–829. Copenhagen: Center for Sprogteknologi.

Louw, B. (1993). Irony in the text or insincerity in the writer? – The diagnostic potential of semantic prosodies. In Baker, M., G. Francis & E. Tognini-Bonelli (eds.) *Text and Technology: In Honour of John Sinclair*, 157–76. Amsterdam: John Benjamins.

Mel'čuk, I. (1988). Semantic description of lexical units in an explanatory combinatorial dictionary: Basic principles and heuristic criteria. *International Journal of Lexicography* 1(3): 165–88.

Mittmann, B. (1999). The treatment of collocations in OALD5, LDOCE3, COBUILD2 and CIDE. In Herbst, T. & K. Popp (eds.) *The Perfect Learners' Dictionary(?)*, 101–111. Tübingen: Niemeyer.

Moon, R. (1999). Needles and haystacks: Idioms and corpora. In Herbst, T. & K. Popp (eds.) *The Perfect Learners' Dictionary(?)*, 265–281. Tübingen: Niemeyer.

Moon, R. (2000). Phraseology and early English dictionaries: The growth of tradition. In Heid, U., S. Evert, E. Lehmann, & C. Rohrer (eds.) *Proceedings of the Ninth EURALEX International Congress, EURALEX 2000*, 507–516.

Moon, R. (2007). Phraseology in general monolingual dictionaries. In Norrick, N., H. Burger & D. Dobrovol'skij (eds.) *International Handbook of Phraseology*. Berlin: Mouton de Gruyter.

Osselton, N. (1986). Dr Johnson and the English phrasal verb. In Ilson, R. (ed.) *Lexicography: An Emerging International Profession*, 7–16. Manchester: Manchester University Press.

Selva, T., J. Binon & S. Verlinde (2002). Le DAFLES, un nouveau dictionnaire électronique pour apprenants du français. In Braasch, A. & C. Povlsen (eds.) *Proceedings of the Tenth EURALEX International Congress*, 109–208. Copenhagen: Center for Sprogteknologi.

Siepmann, D. (2006). Collocation, colligation, and encoding dictionaries. Part 2: Lexicographical aspects. *International Journal of Lexicography* 19(1): 1–39.

Sinclair, J. M. (1987). Collocation: A progress report. In Steele, R. & T. Threadgold (eds.) *Language Topics: Essays in Honour of Michael Halliday*, II, 319–331. Amsterdam: John Benjamins.

Stein, G. (1986). Sixteenth-century English-Vernacular dictionaries. In Hartmann, R. R. K. (ed.) *The History of Lexicography*, 219–228. Amsterdam: John Benjamins.

Stein, G. (1997). *John Palsgrave as Renaissance Linguist*. Oxford: Oxford University Press.

Svensén, B. (1993). *Practical Lexicography: Principles and Methods of Dictionary-Making*. Oxford: Oxford University Press.

Tutin, A. (2005). Le dictionnaire de collocations est-il indispensable? *Revue Française de Linguistique Appliquée* 10(2): 31–48.
Wray, A. (2002). *Formulaic Language and the Lexicon*. Cambridge: Cambridge University Press.
Wray, A. & T. Fitzpatrick (2008). Why can't you just leave it alone? Deviations from memorized language as a gauge of nativelike competence. In Meunier, F. & S. Granger (eds.) *Phraseology in Foreign Language Learning and Teaching*. Amsterdam: John Benjamins.

Dictionaries cited

Bailey, N. (1736). *Dictionarium Britannicum*, 2nd edition.
BBI Dictionary of English Word Combinations (1997), 2nd edition. Amsterdam: John Benjamins.
Beauchesne, J. (2001). *Dictionnaire des Cooccurrences*. Montréal: Guérin.
Cawdrey, R. (1604). *A Table Alphabeticall of Hard Learned Words*.
Collins Cobuild Dictionary of Idioms (1995), 1st edition. London & Glasgow: HarperCollins.
Collins Cobuild English Language Dictionary (1987), 1st edition. London & Glasgow: Collins.
Collins Cobuild English Dictionary (1995), 2nd edition. London & Glasgow: HarperCollins.
Collins English Dictionary (2003), 6th edition. London & Glasgow: HarperCollins.
Collins Robert French-English English-French Dictionary (2002), 6th edition. London & Glasgow: HarperCollins.
Cotgrave, R. (1611). *A Dictionarie of the French and English Tongues*.
Dictionary of Selected Collocations (1997). Hove: Language Teaching Publications.
Dictionnaire d'apprentissage du français des affaires (2000). Paris: Didier.
Dictionnaire d'apprentissage du français langue étrangère ou seconde (<http://www.kuleuven.ac.be/dafles>).
Dictionnaire explicatif et combinatoire du français contemporain: Recherches lexico-sémantiques, I (1984). Montréal: Les Presses de l'Université de Montréal.
Idiomatic and Syntactic English Dictionary (1942). Tokyo: Kaitakusha.
Johnson, S. (1755). *Dictionary of the English Language*.
Longman Dictionary of Contemporary English (2003), 4th edition. Harlow: Longman.
Longman Dictionary of Idioms (1998). Harlow: Longman.
Macmillan English dictionary (2002). Oxford: Macmillan.
Mel'čuk, I. & A. Polguère (2007). *Lexique actif du Français*. Brussels: De Boeck.
New Oxford Dictionary of English (1998). Oxford: Oxford University Press.
Oxford Advanced Learner's Dictionary (2005), 7th edition. Oxford: Oxford University Press.
Oxford Collocations Dictionary (2002). Oxford: Oxford University Press.
Oxford Dictionary of Current Idiomatic English (1975, 1983). Oxford: Oxford University Press.
Oxford English Dictionary (1884–). Oxford: Oxford University Press.
Oxford Hachette French Dictionary (2001), 3rd edition. Oxford: Oxford University Press.
Palsgrave, J. (1530). *Lesclarcissement de la Langue Françoyse*.
A Valency Dictionary of English: A Corpus-Based Analysis of the Complementation Patterns of English Verbs, Nouns and Adjectives (2004). Berlin: Mouton de Gruyter.
Wörterbuch zur Valenz und Distribution deutscher Verben (1980), 5th edition. Leipzig: VEB Bibliographisches Institut.

Computational phraseology

An overview

Ulrich Heid

This chapter describes computational linguistic work which deals with phraseological units. This includes both computational support for human-use tools and resources in the field of phraseology, such as electronic learners' dictionaries, and work in natural language processing aimed at the automatic treatment of phraseological units, for example in the analysis of texts, or in machine translation. It points to a number of issues which have received attention from computational linguists, but space prevents a detailed account of them. The issues are grouped around the phenomena targeted (Section 2), questions of how to represent phraseological units in lexicons and corpora (Section 3), as well as methods for automatically identifying and/or classifying phraseological units in texts (Section 4). The chapter gives an overview of the current approaches in these fields and points to relevant literature.

1. Introduction

This chapter gives an overview of computational work on phraseology. Inspired by the term 'computational lexicography', we call the activities we report on 'computational phraseology'. By analogy with computational lexicography, we mean computational phraseology to refer to at least two kinds of computational activities which have to do with phraseological units:

- computing and computational linguistic methods applied to phraseology (i.e. the computer supporting humans who describe, (re-)present or classify phraseological units); and
- methods of automatic language processing (natural language processing, NLP) that can handle phraseological units in their analysis of texts (i.e. the computer processing of texts which contain phraseological units).

The first aspect (the 'tool' use) can serve to support lexicographers, linguists, and developers of NLP techniques, i.e. those who conceive procedures pertaining to the second use of the term.

There are many strands and trends in computational linguistics and its applications in NLP. Work on phraseology tends to cut across them, rather than being part of one approach or subdiscipline. This is because phraseological units need to be identified, described, classified, represented and manipulated with respect to all levels of linguistic description, such as morphology and morphosyntax, syntax, semantics, pragmatics and, last but not least, with respect to contrastivity between languages. As this chapter is organized around phraseological phenomena, their representation and automatic identification in texts, some of the different approaches to NLP are necessarily mentioned in more than one place.

In fact, many problems analysed in computational phraseology can be tackled with a variety of techniques, and with a variety of approaches or theoretical models in mind. More precisely, computational phraseological work, like any other computational linguistic work, can be oriented either towards a computational modelling of linguistic phenomena which is intended as an explanatory model, or, alternatively, towards a given NLP application. Moreover, different applications have different needs, and the main 'clients' of computational phraseology are as diverse as corpus analysis and annotation, syntactic parsing (i.e. finding a representation for a sentence) and the generation of texts (from an abstract representation to idiomatic speech or texts), question answering in natural language, information extraction from large text streams, information retrieval and machine translation. This last application focuses on the contrastive or possibly multilingual dimension, whereas many of the others are primarily monolingual.

Each of these applications has its own methods and techniques, but from a more coarse-grained perspective, it makes sense, for many of them, and for computational linguistics as a whole, to distinguish between (mainly) symbolic procedures, (mainly) statistical ones, and hybrid, combined procedures. This distinction is evident in much of the work on phraseology.

Writing a chapter (of this length) on computational phraseology in general is, in principle, almost impossible. The different strands of computational linguistics and NLP, the different NLP applications, viewpoints and tasks open up a multidimensional grid which is hard to fill evenly. Thus, this chapter has to be partial, in both senses of the word: it is of necessity incomplete, covering only part of the major research tasks and trends; and it is partial in that it favours certain aspects, simply because I am more familiar with some of them than with others. Thus, I will cover some lexicographic applications and symbolic NLP work in more technical detail than some of the statistical approaches currently under development. However, the objective is to give, as far as possible, a panorama of ongoing work in computational phraseology and to hint at some of the challenges this research field is facing.

There are not many overviews of this kind available, at least to my knowledge. The recent international handbook edited by Burger et al. 2007 contains two articles which summarise the field: Rothkegel (2007) and Heid (2007). Otherwise, one of the best overviews of ongoing research in the field is provided by the series of workshops on multi-word expressions organised under the aegis of the Association for Compu-

tational Linguistics (ACL).[1] The publications from these workshops, together with other conference contributions, constitute the bulk of the material summarised in the present chapter.

In the remainder of this chapter, I will first review the notion of 'multi-word expression' commonly used in computational phraseology (Section 2.1) and point to those fields of NLP where computational phraseological work has a tradition (Section 2.2). Section 3 is devoted to the representation of phraseological units, in terms of both their formal aspects and, in particular, their idiomaticity (Sections 3.1 and 3.2). Ways of accounting for the frequency and productivity of phraseological units are briefly addressed in Section 3.3. Automatic and semi-automatic methods of extracting data on multi-word expressions from text corpora and classifying them are discussed in Section 4. Finally, I present my conclusions in Section 5.

2. Types of phenomena – types of issues

2.1 The notion of multi-word expression

The term 'computational phraseology' is not much used in NLP research. Researchers working on aspects of this field prefer to speak of 'multi-word expressions' (MWE) and their processing. With a few exceptions, the notion of multi-word expression can be equated with that of phraseological units, phrasemes, or phraseologisms, for which Moon (1998) says:

> ...there is no unified phenomenon to describe, but rather a complex of features that interact in various, often untidy ways and represent a broad continuum between non-compositional (or idiomatic) and compositional groups of words.

Sag et al. (2002), in a paper which has become a classic, call MWEs a "pain in the neck for NLP", and they describe some of the phenomena which contribute to this designation. Among these are the fact that MWEs are decomposable into two or more single words, as well as their idiosyncrasies at the lexical, syntactic, semantic and/or pragmatic levels. These idiosyncrasies may come with a fair amount of flexibility and/or variability, and with complex interrelationships between the MWE components.

The term 'multi-word expression' is closely related to the rather simplistic notion of word which is standard in NLP, viz. a string of characters surrounded by blanks. Consequently, any meaningful unit made up of more than one string of characters

1. The first workshop, held in 2003, addressed the analysis, acquisition and treatment of phraseological units (Bond et al. 2003). Its successor in 2004 was subtitled 'Integrating processing', and 2006 saw two workshops, one on 'identifying and exploiting underlying properties', and the other on multilingual aspects. The 2007 workshop was devoted to a broader perspective on multi-word expressions; work towards evaluation of tools and the comparison of procedures used to identify different types of multi-word expressions is the topic in 2008.

qualifies as an MWE. In this sense, the term comes close to Sinclair's (1991) notion of phraseology and Granger's (2005) suggestion to keep track of "the whole range of co-occurrence patterns". A similarly comprehensive view is also advocated by Gross (1996) and in the computational work of his research group on French *expressions figées* (fixed expressions).

Below, I list a few types of multi-word expressions which have been analysed by computational phraseologists. The list shows that formal aspects (e.g. in binomials) and semantic aspects (e.g. in idioms), but also the category of the expression as a whole (e.g. in multi-word function words), its pragmatic use and relevance (e.g. in stereotyped comparisons or proverbs, quotes and sayings) play a role as classification criteria; in other words, the list is rather heterogeneous.

1. MW named entities, e.g. *New York, Buenos Aires, Badische Anilin- und Sodafabriken AG, George W. Bush,* etc.;
2. MW function words, e.g.
   – prepositions: FR *en fonction de* ('depending on'), EN *in spite of*, IT *in merito a* ('with respect to'), NL *in plaats van* ('instead of');
   – adverbs: EN *at large*, *by and large*; *ad hoc*; DE *so gut wie* ('almost');
   – conjunctions: FR *au cas où* ('if'), *sans que* ('without' (conj.)); DE *geschweige denn* ('let alone');
3. MW adjectives: DE *aus dem Häuschen sein* ('be excited'), EN *be up to date*;
4. Verb + particle constructions: EN *call (someone) up, take off*;
5. Nominal compounds: FR *pomme de terre* ('potato'), CAT *porta-documents* ('folder(s)'), FR *fichier client* ('client file');
6. MW verbal expressions and complex predicates: FR *faire de son mieux pour* ('do one's best'); DE *jmdn nicht riechen können* ('dislike sb'); IT *essere in grado* ('be able'); FR *faire partie de* ('be part of');
7. Collocations, including light verb constructions (or: support verb constructions): EN *take a walk, give a speech*; DE *eine Frage stellen* ('ask a question'), *Unterstützung finden* (lit.: 'find support': 'be supported');
8. Stereotyped comparisons: DE *wie Feuer und Wasser sein* ('be diametrically opposed'), *lügen wie gedruckt* (lit. 'lie like printed');
9. Idiomatic expressions: EN *pull one's leg*; DE *jmdm stehen die Haare zu Berge* ('sb's hair stands on end');
10. Binomials: EN *salt and pepper*, *bed and breakfast*; DE *(wie) Hund und Katz(e)* ('diametrically opposed'), *Feuer und Flamme (sein)* ('(be) enthused');
11. Proverbs, quotes and sayings: EN *an apple a day keeps the doctor away*, DE *Morgenstund hat Gold im Mund*;
12. Spoken language particle clusters: NL *maar even*; DE *gerade mal* ('just');
13. Routine formulae of conversation: EN *never mind*; FR *ne quittez pas* ('please hold the line').

This list includes a few phenomena (such as named entities or particle clusters), which have traditionally not been at the centre of interest of linguistic phraseology, but which are relevant to NLP and its applications.

Work on named entities includes their recognition in texts and their markup, e.g. as an input to information extraction programs. Multi-word prepositions have been discussed, with a view to their recognition in texts, by Bouma & Villada Moirón (2002), and multi-word adjectives by Seelbach (2000). French noun compounds have been addressed, with a focus on their form and internal structure (plural formation, morphosyntax, etc.), in the framework of, inter alia, lexicon grammar (see the description in Gross 1996). Work on verbal expressions and collocations, especially on support verb constructions, includes their extraction from text (see Section 4), their representation (cf. e.g. Keil 1997; Krenn 2000, Heid & Gouws 2006) and specific issues, both monolingual and contrastive (see below for details). With idioms, one of the main issues is how to automatically tell them apart from compositional constructions (see, for example, Fazly & Stevenson 2006). A similar issue arises with the extraction of binomials from text (see Widdows & Dorow 2005). Work on items towards the end of the list is mainly corpus-based descriptive work, such as research on the identification and linguistic properties of particle clusters or routine formulae (Mittmann 2004).

2.2 NLP fields with a research tradition in computational phraseology

There are a few recent publications devoted in particular to the handling of MWEs in one of the NLP applications mentioned above. Examples include Doucet & Ahonen-Myka (2004) for information retrieval, Venkatsubramanyan & Perez-Carballo (2004) for ontology building, Baldwin & Tanaka (2004) or Váradi (2006) for machine translation. In addition to English and some of the Romance and Germanic languages, first attempts at dealing with other, typologically different languages have also been made: Kaalep & Muischnek (2006) consider MWEs in Estonian, an inflective language; similarly, Alegria et al. (2004) propose a representation of MWEs for Basque, and Oflazer et al. (2004) deal with Turkish MWEs and with ways of integrating MWEs into the morphological and morphosyntactic processing of an agglutinative language.

In fact, machine translation and natural language generation were historically the first NLP applications in which MWEs were discussed in detail. As early as in the 1960s, the SYSTRAN machine translation (MT) program contained a multi-word lexicon, and a specific subroutine to identify and translate multi-word phrases. This was necessary because the system analysed text strings more or less by chunks, i.e. by cutting the text up into meaningful short word sequences, and it was important to identify MWEs at an early point in the analysis. Today's symbolic MT systems, such as the respective component in Verbmobil (cf. Emele et al. 2000), contain sizeable MWE lexicons and use these to integrate MWE analysis with the deep syntactic and semantic analysis of the sentences or dialogue turns to be translated. Obviously, to do so, the lexicon has to provide considerable linguistic detail about the MWEs. With statistical MT systems, the issue of recognising MWEs becomes less hard, insofar as these systems are trained

on corpora which typically include MWEs, and thus the MWEs are part of the language models used by the systems. Nevertheless, the actual translation of MWEs (be it as MWEs or as single words) requires specific devices.

Automatic natural language generation is the second field of NLP with a historical affinity to computational phraseology. In fact, text generation implies the production of fluent, idiomatic language, for example through the correct selection of collocation partners. As early as 1975, Becker suggested using a 'phrasal lexicon' in a generation system. Cumming (1995: 181–200) listed the needs of generation with respect to the lexicon, capitalising on phraseology. Designers of generation systems discuss phraseologisms in the context of lexicalisation and grammaticalisation. Their focus is on the following issues: the selection of collocational co-occurrents, the possibility of using idioms in text generation, and the interaction between contextual, syntactic and collocational constraints in the determination of the form of an utterance (cf. Reiter & Dale 2000: 124–132 or Wanner 1996 for an overview).

Since the late 1990's, broad coverage NLP grammars have been under construction, especially for English. In the light of these developments, there is also massive interest in the description and computational analysis of MWEs. A good example is the LinGO project (Copestake et al. 2004), which has developed a format for representing MWEs (Copestake et al. 2002). Similarly, large NLP dictionaries, such as the Danish STO dictionary (cf. Braasch & Olsen 2000) and FrameNet (cf. Ruppenhofer et al. 2002), include descriptions of, and representation formats for, MWEs. For the LinGO lexicon, see Villavicencio et al. (2004).

2.3 Issues related to MWEs

There are a few types of issues related to all kinds of MWEs which can be considered as central topics of computational phraseology. These are concerned with the form and idiomaticity of MWEs, which have an effect on the classification of MWEs, their representation in text corpora and electronic lexicons, and the tools used to identify or manipulate them in text within a given grammar. A third aspect relevant for the representation of phraseological units is their productivity, i.e. the potential for building ad hoc new expressions on the basis of existing phraseological patterns. Form-related issues have to do with the (obvious) fact that MWEs consist of several word forms. Issues related to the idiomaticity of MWEs derive from their unpredictable aspects at one or more levels of linguistic description.

The classification of MWEs (or phraseologisms) qualifies, at first sight, as a genuine task for linguistic phraseology, and it could therefore be expected to play an equally important role in computational phraseology. However, what is most crucial for NLP is not a particularly fine-grained subdivision into specific kinds of multiword phrases, but, in the first place, a classification into compositional and non-compositional (or only partly compositional) expressions. This distinction is most relevant for any computational treatment which targets semantics: it allows a distinction to be drawn between the standard, rule-based construction of semantic rep-

resentations and non-standard ways of dealing with the semantics of phraseological expressions (cf. Section 4.3). This simplistic binary distinction is close, by the way, to that which a dictionary user who reads a text and stumbles over a MWE would operate. For him or her, the most important question is not to know what type of MWE he or she is facing, but where in the text the MWE starts and ends, and what it means (cf. Tarp 2006: 284–286). Representational issues thus include the way in which MWEs are signalled in corpora, as well as the form and content of lexical entries for NLP systems.

Tools for identifying multi-word expressions may serve different purposes. The main distinction is between the task of locating new MWEs (i.e. MWEs so far not contained in the system's dictionary) in text, and the task of identifying illustrative examples, linguistic properties, etc. of multi-word expressions which are already listed. Some tools undertake both tasks at the same time. In the following sections, I will address in turn the representation (Section 3) and the identification and classification (Section 4) tasks of computational phraseology in more detail.

3. Representational issues in computational phraseology

3.1 Representing the form of phraseological units

Items such as the French multi-word preposition *en fonction de* ('depending on') consist of several words, which can all also occur independently, outside the MWE, unlike 'cranberry' words which are only found in a given MWE, such as *au fur et à mesure* ('gradually'). In a corpus, the sequence *en fonction de* often has its prepositional function, as in (1) below, but it may also be part of the noun compound *mise en fonction de*, as in (2).

(1) *Le prix d'une oeuvre varie en fonction de sa taille et ...*
(the price of a piece varies according to its size and ..., *Le Monde 1992/93)*

(2) *... pourquoi en matière de sécurité et de surveillance des immeubles [...] la mise en fonction du système de surveillance mixte est envisagée ...*
(... why with respect to security and surveillance of buildings the installation of the mixed surveillance system is envisaged ..., *European Parliament Debates)*

Early proposals for the annotation of multi-word expressions in corpora suggested considering them as 'words with spaces'. Obviously, this only works for absolutely fixed sequences (e.g. *ad hoc*, *New York*) which do not overlap with other multi-word sequences (unlike *en fonction de*). This suggestion also leads to problems with morphological marks within the multi-word expression (cf. e.g. the plural marks in *pommes de terre* ('potatoes'), and *bottiglie da vino* ('wine bottles'); see Sag et al. 2002 for a discussion of the words-with-spaces approach). Current thinking favours a multilayered view of corpus annotation: for example, each of the elements of the multi-word sequence *en fonction de* could be annotated with its standard word class (*en*$_{\text{Prep}}$ *fonction*$_{\text{N}}$

de$_{\text{Prep}}$), and a second layer of annotation introduced in which the sequence as a whole is annotated as a preposition (cf. Pianta & Bentivogli 2004). Such an approach would permit yet another layer of annotation that would cater for the alternative bracketing needed for the analysis of sentence (2): (*mise en fonction (de X)*). Current multi-layer architectures for corpus annotation represented in XML allow for such solutions.

Similar issues also arise in the lexicon: it must also be possible to indicate the extension of MWEs, and whether or not they can be interrupted by non-MWE material. A typical example is the various classes of English particle verbs. Inflectional variation and variability at the level of modification also need to be taken into account: DE *Frage + stellen* ('ask + question') accepts an attributive adjective of *Frage*, whereas the word *Frage* in the phrase *in Frage stellen* ('[to] question') does not.

3.2 Representing idiomaticity

3.2.1 *Idiosyncrasy and variation*

As mentioned above, idiomaticity translates itself into the two aspects of deviance and idiosyncratic variation. Both play a role at almost all levels of linguistic description.

Lexical idiosyncrasy. At the lexical level, collocations are a prime example of idiosyncrasies: one has to learn which nouns and adjectives go together (e.g. DE *tiefer Schlaf* (lit. 'deep sleep'), FR *peur bleue* ('terrible fear')) and which verbs and their objects (*ask + question*, *raise + question*, *take + shower*, but not *make + shower*, unlike IT *fare la doccia*). Collocational preferences may vary across regional variants of a language. An example is Brazilian Portuguese which uses *levar + susto* (lit. 'raise + fright') to mean 'get + fright', whereas Portuguese from Portugal uses *apanhar + susto* (lit. 'pick + fright') for the same meaning. Moreover, collocational preferences only partially follow the semantic classes of the base lexemes involved (cf. Mel'čuk & Wanner 1994; Tutin et al. 2006). These properties make it necessary to describe in detail the possible lexical combinations in a dictionary, be it for NLP purposes or for human use.

Lexical variation. On the other hand, it is well known that some idiomatic expressions allow a certain degree of lexical variation (see, for example, the corpus study by Cignoni & Coffey 1998). For *da stehen jemandem die Haare zu Berge* ('that makes sb's hair stand on end'), I found *da steht ihm das Zöpfchen zu Berge*; *da stehen ihr die strubbeligen Härchen zu Berge*; *da stehen dem Eber die Borsten zu Berge*. An exploration into which parts of common idioms can be lexically modified requires very large corpora (because this phenomenon is quite rare), and rather sophisticated search tools, and has not yet been undertaken on a large scale. Fellbaum's corpus-based data collection of German verbal idioms contains hints to such variation (cf. Fellbaum et al. 2006), and Kwaśniak (2006) uses corpus data from Fellbaum's project to analyse the way in which the sentential idiom *da haben wir den Salat* ('now we're in a right mess') develops over time into a verb phrase idiom (with subjects other than *wir*). It is hard to cover such phenomena in detail in an NLP dictionary, even though this would be desirable. For the time being, however, it seems more realistic to try to identify mechanisms by means of which lexical variation of idioms can be investigated.

Morphosyntactic idiosyncrasy. Morphosyntactic idiomatisation mainly concerns preferences at the level of number (e.g. *have high hopes*: typically in the plural), active/passive (e.g. DE *das Angebot wird <gerne...> angenommen* ('the offer is <gladly, etc.> accepted') mainly passive in the language of social services), etc. These are again subject to change across regions: the AC/DC corpus of Portuguese from Portugal (*Linguateca*) shows an 11:1 preference for the plural *ter ciúmes* (lit. 'have jealousies', 'be jealous') over the singular, whereas there is a 1:1 ratio of singular to plural in the Brazilian corpus of *Linguateca.* Related issues are the modifiability of nouns in noun + verb collocations and idioms, such as DE *eine <ADJ> Frage stellen* ('ask a <ADJ> question') compared to *etw. in Frage stellen* ('call into question'), and distributional preferences (e.g. *high hopes* vs. *hopes are high that ...* as opposed to *heavy drinking* vs. **his drinking is heavy)*.

Morphosyntactic variation. The phenomenon of morphosyntactic idiomatisation has received comparatively little attention in computational phraseology. Such idiomatisations are preferential in nature (no categorical constraints), and detailed electronic dictionaries should account for them (cf. Krenn 2000; Odijk 2004; Heid & Gouws 2006). To this end, it is necessary to extract the respective preferential values from text (cf. Heid & Ritz 2005; Ritz 2006). Simple frequency counts of the respective form variants may not be sufficient for this, as not all sentences analysed may provide clear evidence due to morphological syncretism (different morphological analyses for one surface form) and occasionally to errors in automatic processing. Evert et al. (2004) suggest a way of counting and statistics which lead to confidence intervals for each morphosyntactic value considered (e.g. singular as opposed to plural). They take all unclear or ambiguous evidence as counting against the value under consideration, which leads to percentages which can be interpreted as the minimum of evidence clearly suggesting this value. Corpus-related percentages extracted in this way have been used to represent preferences in an electronic dictionary (Spohr & Heid 2006). Obviously, the morphosyntactic variation potential of MWEs can also be represented as a disjunction of alternatives. This is, among other approaches, supported by PhraseManager (Pedrazzini 1994), a lexical representation system based on WordManager (Domenig & ten Hacken 1992). WordManager is used to represent morphological data on individual lexical items and to analyse and generate inflected forms. PhraseManager extends the WordManager principles to the representation of analytic tense forms and MWEs. Tschichold (2000 and this volume) has illustrated the PhraseManager approach by modelling a large set of English MWE types.

Syntactic idiosyncrasy. A well-known aspect of the syntactic idiosyncrasy of MWEs is the fact that certain multi-word function words are composed of sequences of categories that do not exist in compositional syntax (e.g. by_{prep} and_{conj} $large_{\text{Adj}}$) and for which it is difficult to find a well-formed phrasal category: what category would be used for *by and large*, and what rules would allow us to derive it? Such phenomena cause problems in many NLP lexicons and grammars. Many system designers find it more convenient to only assign a word class label to the multi-word as a whole. Another aspect of syntactic idiosyncrasy is related to the valency properties of certain

types of noun + verb collocations and idioms. It has long been observed that a description of the subcategorization properties of verbal idioms is necessary (cf. Burger 1998; Keil 1997). Keil discusses, for example, DE *einen Bären aufbinden* ('tell sb. a lie'), which requires a subject and an indirect object. Unlike idioms, most collocations share their valency with the noun they contain (compare *proposal to leave* and *make a proposal to leave)* and often with the verbal predicate from which the noun is derived (*propose to leave*). But some collocations have valency properties which cannot be explained by either those of the noun or those of the verb: DE *zur Sprache bringen* ('mention') takes a *daß*-clause, which neither of its components do. Similar cases are DE *zu Protokoll geben, daß* ('state') or *zum Ausdruck bringen, daß* ('express') which take a complement clause although their components do not allow for one. This brings these collocations close to idiomatic expressions with non-predictable valency, such as IT *essere in grado di* + infinitive ('be able to + infinitive') and its French and German equivalents *être en mesure de* or *in der Lage sein*. A corpus-based preliminary analysis of both collocations and idioms in German is described by Lapshinova & Heid (2007), yet proposals for the lexical representation of valency properties of collocations and idioms are still missing. Indeed, most formal grammars only provide mechanisms for those collocations where the subcategorisation properties of the noun are preserved (cf. Krenn & Erbach 1994). Obviously, these are more numerous than collocations whose properties are idiosyncratic, and so the majority of cases are covered.

Syntactic variation. MWEs not only allow for morphosyntactic variation, for example in terms of modifiability (see above); many of them also vary in the extent to which they permit passivisation, certain extractions, etc. Odijk (2004) has proposed a descriptive program for Dutch idioms which is based on syntactic patterns and on lexical lists, and which also accounts for the syntactic variation potential of the MWEs. This inventory has been checked against the SAID database (Kuiper et al. 2003): it covers over 80% of SAID's over 14,000 individual MWEs. A more implicit analysis of the syntactic variation potential, for English and French, underlies the work by Sereţan et al. (2004) on multilingual collocation extraction. There, the objective is, however, to identify collocation candidates in as many syntactic contexts as possible, whereas Odijk's proposal is aimed at lexical representation and sharing.

Semantic idiosyncrasy. The semantic specificity of MWEs is a major issue not only for their lexical representation, but also for the annotation of corpora. In the lexicon, the meaning of MWEs has to be captured, and their insertion into compositionally built semantic representations ensured. Representational proposals are discussed by, among others, Pulman (1993), Jackendoff (1995), Sag et al. (2002) and Nunberg et al. (1994).

3.2.2 *Representing idiomaticity in interactive electronic dictionaries*

From the point of view of the representation of lexical data for human users, detailed suggestions for a database for interactive use have been made by Keil (1997) and in the recent work by Hallsteinsdóttir (2003), with a view to a bilingual Icelandic-German electronic dictionary of idioms. These proposals include an informal account

of all levels of linguistic description, from (morpho-)syntax to semantics, oriented towards human users. The same is true of the online idiom dictionary produced by the lexicography research centre of Aarhus university, *Dansk Idiomordbog*. It is intended for interactive consultation, providing explanations of the meaning of individual idiomatic expressions, notes on the syntax of the idioms, as well as links to related expressions and to idioms which share lexical items. Another type of electronic dictionary for idiomatic expressions are those tools which allow the reader of an electronic text to point the computer mouse to a given word which belongs to an MWE and to display the corresponding MWE dictionary entry. Such tools (the first one was Locolex, cf. Bauer et al. 1995) rely on the lexical material contained in an idiom, check for its presence (and possibly sequencing) in the sentence under analysis and then provide the corresponding entry from a human use dictionary. Since about the year 2000, such functions have been included in several commercial general language dictionaries.

In addition, there are several online learners' dictionaries which address collocations in considerable detail. Those with the richest internal representation (most descriptive detail, large number of data categories) are the ones based on Mel'čuk's *Meaning ⇔ Text-Theory* (cf. Mel'čuk et al. 1995). Polguère & Mel'čuk (2006) give details of the data model of their DiCo/LAF dictionary; it not only caters for French collocations, but also for their interrelationships with single words (e.g. paraphrase relations) and for certain kinds of morphological relations. The DiCo/LAF model (and the online search interface *dicouèbe*) are mainly intended for advanced users who know elements of explanatory and combinatorial lexicography. The combination of phraseological and morphological data makes DiCo/LAF a real text production dictionary.

The DiCE dictionary (*Diccionario de colocaciones del Español*), a model of a learner's collocation dictionary of Spanish (with about 25 sample entries so far), is more clearly pedagogically oriented, again using Mel'čuk's descriptive model. Another step towards non-expert users, the French learners' dictionary, DAFLES/BLF (*Dictionnaire actif du français langue étrangère ou seconde*; *Base lexicale du français*, cf. Verlinde et al. 2006) also draws its inspiration from Mel'čuk's model, but it 'translates' the names of Lexical Functions into meaning paraphrases of collocations (e.g. instead of 'oper', it says 'carry out', etc.).

What is common to all these collocation dictionaries is that they explicitly describe lexical idiosyncrasies. Morphological and syntactic idiosyncrasies are usually mentioned in the examples, but rarely presented explicitly, for example through features and warnings such as 'mostly in the plural!') or as search criteria. The dictionaries provide different devices for the semantic classification of collocations, at a minimum explaining the collocations, and more rarely enabling a search for the semantic classifications.

3.2.3 *Representing idiomaticity in text corpora*

A prominent example of a semantically annotated corpus is the SALSA corpus (cf. Burchardt et al. 2006), a collection of about 20,000 sentences of German newspaper text, annotated at the levels of (morpho-)syntax and lexical semantics. The semantic

annotation makes use of Frame Semantics (cf. Baker et al. 2003 for an overview of the FrameNet project), i.e. it comprises, among other things, the identification of predicates as well as the semantic roles (frame elements) of their arguments and of some modifiers. This semantic annotation relies on the syntactic annotation provided, in terms of constituency and grammatical functions, by the TIGER treebank (Brants et al. 2002). SALSA also has devices to account for idiomaticity.

If the standard frame semantic annotation is sufficient for compositional semantics, additional devices are, however, needed to account for phraseology. In fact, support verb constructions (SVCs), idioms and metaphorical expressions each receive a specific treatment in SALSA (cf. Burchardt et al. 2006, Section 5): support verb constructions (e.g. *Abschied nehmen*, 'say good-bye', lit. 'take leave') are singled out in the annotation, and their verbal element (*nehmen*) co-occurs with a nominal one which carries the role of 'supported' (pointing to *Abschied*). This allows for later refinement of the annotation and signals the fact that the SVC should not be confused with 'normal', compositional constructions. In the framework of the Berkeley FrameNet, Ruppenhofer et al. (2002) have made an alternative proposal for annotating SVCs: they simply took over the annotation used for verbal predicates and added a feature to indicate the presence of a support verb construction, including the lemma of the support verb. In SALSA, idiomatic expressions, including discontinuous ones (e.g. *in Kauf nehmen*, 'accept', cf. *er nimmt dieses Problem in Kauf*) are annotated as multi-word predicates. Metaphorical expressions receive a double treatment, i.e. they are annotated with a source frame (literal) and with a transferred target frame which represents their figurative meaning. This double annotation is applied to both single word and multi-word metaphors. Obviously, this kind of annotation requires a multi-layer annotation system, very much like the one discussed above, in connection with the form of discontinuous multiwords (Section 3.1). SALSA thus allows annotators to keep track of the interaction between (morpho-)syntax and semantics in the treatment of multi-word expressions. This is a major step forward compared to most other annotated corpora.

3.3 Accounting for the frequency and productivity of phraseology

Since NLP researchers have started to work on phraseological multi-word expressions, they have been interested in quantifying the phenomenon. Jackendoff (1995: 156) assumes that the lexicon of a language available to speakers in everyday situations contains at least as many multi-word expressions as single words. In WordNet, roughly half of the entries are MWEs. Current research suggests that these figures are rather low estimates, and that the number of MWEs in our languages is much greater than that of single words.

Obviously, specialised languages and their terminologies are even more massively composed of multi-word expressions. In many such languages, compound nouns come as MWEs, and many adjective + noun groups are equally considered as terms. Furthermore, certain sublanguages, such as the juridical and administrative termi-

nology, are very rich in verbal phraseology (cf. Anderson 2006). The frequency of multi-word groups and of the co-occurrence of (possibly distant) words is exploited in tools for extracting MWEs from text (cf. Section 4). If words co-occur more frequently than one would expect, this co-occurrence may be due to lexicalization as a multi-word expression. Most co-occurrence measures are based on word pairs, and thus many extraction tools determine the significance of the co-occurrence of two items and propose as multi-word candidates those pairs (bigrams) whose co-occurrence is higher than a certain threshold.

Despite the quantitative importance of MWEs, no detailed data about their frequency is available, at least to my knowledge. When general dictionaries provide frequency data, they do so for lemmata, not for their uses or the word combinations in which they occur. As many MWEs are not adjacent, we need more than simple window-based tools to extract (and count) them in corpus text. Currently available collections and frequency lists mainly provide data for word sequences (e.g. texts from the British National Corpus or those accessed by the Google search engine).

The notion of productivity, as it is known from the field of morphology for example, implies the presence of rules or patterns which allow speakers to build ad hoc new linguistic objects from an existing inventory of basic elements. Typical examples are derivational processes such as those involved in the nominalization of verbs (for a quantitative approach, cf. Baayen 2001). Lexical productivity has implications for the dictionaries used in NLP: a fixed list of lexical items cannot cover productively built items. Instead, the lexicon must be combined with a rule system to account for ad hoc items. This is how many morphology systems account for productive word formation.

A similar phenomenon can be observed with idiomatic expressions which show variability. In Section 3.2, I mentioned the exploitation of idiomatic patterns, such as *DE jemandem stehen die Haare zu Berge.* Lexical variation and the productive exploitation of idiomatic patterns are a challenge for both the lexical representation of idioms and their identification in textual data: which elements of an MWE can be modified or replaced, without compromising the idiomatic meaning, and how can we search for productively modified MWEs in text?

3.4 Towards standards for the representation of MWEs

Work towards shared resources and the reuse of existing NLP tools with large amounts of text data from different sources presupposes agreed and shared representations. This is why several multinational initiatives are involved in the development of standards for corpora, lexicons and related tools. Even though many of the problems connected with the representation of MWEs are still far from being solved, first steps towards generic and sharable description and representation schemes for MWEs have been taken. Besides the proposals made in the framework of broad coverage grammars, examples of such initiatives include the ISLE project (International Standards for Language Engineering) and the Xmellt project (Calzolari et al. 2002). The latter came up with proposals for the representation of support verb constructions and complex nomi-

nals. Odijk (2004) made proposals for a standardized syntactic description of idioms, also as a basis for sharing of data and tools. Current proposals for lexicon representation standards, such as the ISO proposal for a *Lexical Markup Framework, LMF*, make reference to MWEs. The need for consensual representations is expected to increase as the behaviour of MWEs is analysed in more detail.

4. Issues and methods for the (semi-)automatic identification and classification of MWEs

The identification and classification of MWEs may be concerned with either finding MWEs in texts (identification proper) or with finding illustrative examples or data from which a description of the properties of MWEs can be derived. In the following sections, I will discuss the main trends in these fields in more detail.

4.1 Extracting MWE candidates with association measures

A basic goal of extraction in computational phraseology is to find candidate word pairs or word tuples that qualify as MWEs. Many tools to carry out this task rely on the recurrence of word pairs or word groups and in particular, on the co-occurrence frequency of the candidates.

Depending on the extraction objective, raw text corpora, part-of-speech (POS)-tagged and lemmatised corpora, or chunked and possibly parsed corpora may be used. On all kinds of corpora, the co-occurrence of word forms is counted and compared with the expectation for such co-occurrence. This is what most association measures (AMs) do: they count pairs of two words – for example, *pay* and *attention* (observed frequency O_{11}) – and compare the count with other contexts of *pay* + nouns and of *attention* + verbs (O_{12}, O_{21}), as well as with the overall number of noun + verb pairs in the corpus (O_{22}). These four figures, usually represented in a 2*2 contingency table, can be compared with statistical expectations for how frequent a pair like *pay* + *attention* would be if the words happened to be evenly distributed. The result is either a measure of the extent to which the observed figures deviate from the expected ones (significance of the association or collocational strength), or a measure of the amount of evidence available to falsify, for the word pair under consideration, the hypothesis of a statistically even (e.g. Gaussian) distribution. Details on the mathematical foundations of association measures can be found in Evert (2005), who has analysed over 40 different measures which have been proposed in the literature for the purpose of collocation extraction. The first experiments with association measures for MWE extraction go back to the 1960s and to systemicist and contextualist views on collocations, i.e. the observation that the elements of a collocation co-occur significantly more than expected by chance. In computational linguistics, the work of Church &

Hanks (1990) started a wave of massive interest in word pair extraction by means of association measures (cf. the overview in Evert 2005).

The association measures provide a numeric value for each word pair, and one use of these values is to sort the list of all word pairs in a text according to them. Intuitively, one would expect to find at the top of such a list the word pairs which most prominently co-occur, and which are thus most clearly phraseological. The underlying assumption is that phraseological units are lexically fixed, i.e. usually co-occur in the same form.

If carried out on a language like English, which has little morphological variation, the results of such a numerical comparison may be more revealing than for an inflective language. As the counts for a larger number of different forms will be lower, the co-occurrence data will be more diluted, and fewer combinations will reach a significant co-occurrence value (sparse data problem). As a result, lemmatised and POS-tagged corpora have been used and associations calculated for lemmas rather than word forms. Word class tagging also allows for filtering in terms of co-occurrence types: one may select only pairs of adjectives and nouns, for example, or, if chunking or parsing is available, noun + verb collocations (e.g. verb + object pairs, subject + verb pairs, etc).

Thus, preprocessing (corpus tagging and lemmatisation, with or without syntactic analysis), word class based selection (by means of patterns) and association based selection (lists of decreasing association values) currently constitute a standard triple of tool components for extracting word pairs as MWE candidates. Architectures differ with respect to the sequencing of the grammatical (POS) and the statistical (AM) filter. Smadja (1993) first determines statistically significant pairs of any kind, and then applies a POS-filter. Heid (1998), Krenn (2000) and Sereţan et al. (2004) extract syntactically homogeneous word pair candidates first and then filter them according to their association values. Similar mechanisms are part of the Sketch Engine (Kilgariff et al. 2004; Kilgariff & Tugwell 2001), a tool for lexicographers which extracts significant co-occurrence data from corpora and presents them in a double classification: by word class combination type and, within each type, by decreasing significance.

Even though the extraction tools mentioned above are most commonly called 'collocation extraction tools', not all word pairs which receive high association figures and are thus extracted need to be collocations in (one of) the phraseological sense(s). The association may accord with the lexicographical intuition about collocations described by, for example, Hausmann (2004), Bartsch (2004: 76) and Grossmann & Tutin (2003), but it may also be due to the frequency of a trivial word combination which is fully compositional and predictable (e.g. *new results*), or it may be a name (*Rotes Kreuz*, 'Red Cross'), part of a larger idiomatic expression or an artefact of the formulaic expressions of the language under review (e.g. *DE die Polizei teilt mit, daß ...,* 'the police informs that' comes out prominently from certain kinds of German news stories, cf. Ritz 2006). Thus manual selection of candidates is necessary, if the results of the application of association measures are to be used for lexicographical purposes. In fact,

the frequency phenomena captured by association measures are not unique to lexical collocations; they are a necessary, but not sufficient, element of collocation extraction.

Evert & Krenn (2001) have suggested ways of evaluating the output of association measures for both precision (how many true positives are there in the candidate list?) and recall (which percentage of all true positives in the text to be analysed are found by the extraction system?). They work against a lexicographical gold standard. Their comparison of different association measures shows that there is no single 'best measure' and that the selection of a measure may depend on the kinds of phraseological units the researcher wants to extract (cf. Evert 2005).

The association measures are mathematically well-defined for bigrams (*pay* + *attention*, *keep* + *tabs*). However, to extract groups of more than two content words, experiments with an iterative application of the word pair measures have been made, as contingency tables for three and more elements get mathematically rather complex. Other approaches just rely on the frequency of recurrent word sequences.

4.2 Extracting and classifying MWE candidates according to their form

In Section 3, I mentioned the aspects of idiosyncrasy and restricted variation which characterise many phraseological expressions. Both of these aspects concern the lexical combinability as well as the morphosyntactic and syntactic properties of MWEs. Many discovery procedures for MWEs therefore rely on the degree to which lexical combinations, morphosyntactic forms or the syntactic distribution of word combinations are restricted or fixed.

Association measures can be understood as measuring the fixedness of a lexical combination. In a similar way, one may measure the fixedness of morphosyntactic preferences (singular/plural, determination, etc.), the fixedness of syntactic modifications and the MWE's potential for modification, passivisation, etc. A symbolic approach to this has been proposed by Heid & Ritz (2005) and Ritz (2006): they extract German noun + verb pairs and note several morphosyntactic features of each occurrence of each pair of lemmas considered. Summing over lemma pairs, this procedure yields figures for the distribution of the evidence with respect to each feature value. For example, 96 % of the corpus sentences illustrating the collocation *DE Protokoll schreiben* ('take notes') occur without an article. From this, the authors conclude that *Protokoll schreiben* preferentially occurs without article.

In a follow-up step, a set of such formal preferences or idiosyncrasies can be determined and their presence considered as a sign of idiomaticity: all candidate pairs displaying such preferences could then be classified as idiomatic MWEs, on the assumption that trivial word combinations would show more variation. A similar reasoning underlies the extraction of idiomatic binomials by Widdows & Dorow (2005), such as *(rain) cats and dogs*, *bed and breakfast*, *Punch and Judy*, etc. These MWEs are characterised by their syntactic form and by the fixedness of the order of the two nouns.

Like methods to identify lexical fixedness, extraction work based on (morpho-)-syntactic fixedness is used in combination with statistical calculations. Fazly & Stevenson (2006) combine morphosyntactic fixedness data of different kinds into a vector expressing the overall idiomaticity of the targeted MWEs. They are less interested in a detailed lexicographical description of the idiomatic preferences of a given MWE than in classifying it, on the basis of its fixedness, into compositional or non-compositional (i.e. in terms of idiomaticity). This classification can be seen as a first step in automatic dictionary building for MWEs, as it helps to single out the idiomatic candidates which require a special treatment.

4.3 Identifying idiomatic MWEs by means of distributional semantics

Not all idiomatic MWEs are lexically or formally deviant, and thus the methods discussed so far do not capture all kinds of idiomatic expressions. In some cases, only the semantic interpretation of the context allows a distinction to be made between the literal and the idiomatic use of a candidate. For example, *ins Wasser fallen* may mean '[to] be cancelled' (if idiomatically used) or, literally '[to] fall into (the) water'.

Approaches from distributional semantics are being explored to address these issues. The underlying hypothesis of distributional semantic approaches is that items with similar contexts share meaning components (cf. Lin 1998). Thus, nouns denoting pet animals (*dog*, *cat*, *hamster*) will occur with similar verbs (e.g. *keep* or *feed)*. To identify *hot dog* as not being a kind of 'dog' (and thus being a non-compositional expression), one may check for typical verbs showing up with *hot dog* and compare its typical contexts with those of other words; verbs like *eat*, *cook*, *serve* will be shared by *hot dog*, *meal*, *burger*, *sandwich*, etc.

Analyses of the idiomatic use of words and word combinations in terms of their distributional semantics rely on existing thesauri (or on thesaurus-like resources created automatically from the texts under analysis), or on clustering approaches such as latent semantic analysis. All these approaches can be used with ranking mechanisms, so as to sort candidates on a compositionality scale and to single out non-compositional ones. The more deviant the contexts of a candidate with respect to the contexts of related words (*hot dog* vs. *dog*, *cat*, etc.), the higher the probability that the candidate is non-compositional. Recent experiments have addressed, inter alia, light verbs, verb+object pairs and particle verbs in this way (McCarthy et al. 2003). The procedures rely on (partially) parsed corpora, extraction and clustering techniques.

A related strand of research uses parallel corpora and word alignment to automatically distinguish literal meaning (i.e. compositional use) from non-compositional use. The assumption is that single words used compositionally will usually be consistently translated by one equivalent or one of very few alternatives. Word groups built from such items should contain the same 'standard' equivalents, whereas non-compositional use would result in a situation where translated material does not contain these standard equivalents (cf. Villada Moirón & Tiedemann 2006). By locating equivalents of MWEs in translations and comparing them with standard equivalents

of the MWE components, decisions can be made about the compositionality of the source language MWEs.

5. Conclusion

In this chapter, I have tried to summarise some of the issues, methods and trends in computational phraseology. Many of the issues have been addressed in detail by linguistic phraseology research, and the computational methods to approach them in large text corpora are the actual innovation of NLP. On the other hand, the attempt to deal with phraseological units in automatic systems for the analysis of texts has shed a new light on numerous research questions addressed by phraseology.

The field of computational phraseology is rapidly evolving, as NLP researchers and developers have become aware of the prevalence of non-compositional uses of lexical items in texts: it is impossible to create large scale NLP applications without ways of handling phraseological phenomena of all kinds. There is clearly a need for more research in this field. The availability of large syntactically annotated corpora, as well as syntactic-semantic lexicons and thesauri, and parallel corpora opens up new approaches which combine knowledge from different levels of linguistic description to capture the properties of non-compositional uses of language. Although such approaches allow us to detect and possibly classify phraseological expressions, more work still needs to be done on the integration of MWEs into the syntactic and semantic representation of the sentences and texts in which they occur.

Acknowledgements

Much of the experience the author can draw on is due to collaborative work with Stefan Evert, Katerina Lapshinova, Julia Ritz, Dennis Spohr and Marion Weller, as well as with the colleagues from the project on "Collocations en Contexte" (financed in part by the Agence Universitaire pour la Francophonie, réseau LTT), namely Amalia Todirascu, Christopher Gledhill and François Rousselot (Strasbourg), Dan Tufiş and Dan Ştefanescu (Bucarest). Obviously, all misconceptions and misrepresentations are my own.

References

Alegria, I., O. Ansa, X. Artola, N. Ezeiza, K. Gojenola & R. Urizar (2004). Representation and treatment of multiword expressions in Basque. In Tanaka, T., A. Villavicencio, F. Bond & A. Korhonen (eds.) *Second ACL Workshop on Multiword Expressions: Integrating Processing, Barcelona, Spain*, 48–55. Morristown, NJ: ACL.

Anderson, W. J. (2006). *The Phraseology of Administrative French. A Corpus-Based Study* [Language and Computers 57]. Amsterdam: Rodopi.

Baayen, R. H. (2001). *Word Frequency Distributions*. Dordrecht: Kluwer.

Baker, C., C. J. Fillmore & B. Cronin (2003). The structure of the FrameNet database. *International Journal of Lexicography* 16: 281–296.

Baldwin, T. & T. Tanaka (2004). Translation by machine of complex nominals: Getting it right. In Tanaka, T., A. Villavicencio, F. Bond & A. Korhonen (eds.) *Second ACL Workshop on Multiword Expressions: Integrating Processing, Barcelona, Spain*, 24–31. Morristown, NJ: ACL.

Bartsch, S. (2004). *Structural and Functional Properties of Collocations in English, A Corpus Study of Lexical and Pragmatic Constraints on Lexical Co-Occurence.* Tübingen: Narr.

Bauer, D., F. Segond & A. Zaenen (1995). *Locolex: The Translation Rolls off Your Tongue*. Grenoble: Xerox.

Becker, J. D. (1975). The phrasal lexicon. In *Proceedings of the 1975 Workshop on Theoretical Issues in NLP*, 60–63. Morristown, NJ: Association for Computational Linguistics.

Bond, F., A. Korhonen, D. McCarthy & A. Villavicencio (eds.) (2003). *ACL-03 Workshop on Multiword Expressions: Analysis, Acquisition and Treatment*. East Stoudsburg, PA: ACL.

Bouma, G. & B. Villada Moirón (2002). Corpus-based acquisition of collocational prepositional phrases. In Theune, M., A. Nijholt & H. Hondorp (eds.) *Computational Linguistics in the Netherlands 2001, Selected Papers from the Twelfth CLIN Meeting (2001)*, 23–37. Amsterdam: Rodopi.

Braasch, A. & S. Olsen (2000). Formalised representation of collocations in a Danish computational lexicon. In *Proceedings of the Euralex International Congress 2000. Stuttgart, Germany*, 475–487.

Brants, T., S. Dipper, S. Hansen, W. Lezius & G. Smith (2002). The TIGER treebank. In *Proceedings of the Workshop on Treebanks and Linguistic Theories*. <http://www.bultreebank.org/proceedings/paper03.pdf>.

Burchardt, A., K. Erk, A. Frank, A. Kowalski, S. Padó & M. Pinkal (2006). The SALSA corpus: A German corpus resource for lexical semantics. In *Proceedings of the Linguistic Resources and Evaluation Conference 2006, Genova*. Paris & Pisa: ELRA/CNR. [CD-ROM]

Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) (2007). *Phraseologie – Phraseology. Ein internationales Handbuch der zeitgenössischen Forschung – An International Handbook of Contemporary Research*. Berlin: Walter de Gruyter.

Burger, H. (1998). *Phraseologie. Eine Einführung am Beispiel des Deutschen*. Berlin: Erich Schmidt.

Calzolari, N., C.J. Fillmore, R. Grishman, N. Ide, A. Lenci, C. McLeod & A. Zampolli (2002). Towards best practice for multiword expressions in computational lexicons. In *Proceedings of the Linguistic Resources and Evaluation Conference 2002. Las Palmas de Gran Canaria, Spain*, 1934–1940.

Church, K. & P. Hanks (1990). Word association norms, mutual information and lexicography. In *Computational Linguistics* 16(3): 22–29.

Cignoni, L. & S. Coffey (1998). A corpus-based study of Italian idiomatic phrases: From citation forms to 'real-life' occurrences". In *Proceedings of the Euralex International Congress 1998, Liège, Belgium*, 291–300.

Copestake, A., F. Lambeau, A. Villavicencio, F. Bond, T. Baldwin, I. A. Sag & D. Flickinger (2002). Multiword expressions: Linguistic precision and reusability. In *Proceedings of the Linguistic Resources and Evaluation Conference 2002. Las Palmas de Gran Canaria, Spain*, 1941–1947.

Copestake, A., F. Lambeau, B. Waldron, F. Bond, D. Flickinger & S. Oepen (2004). A lexicon module for a grammar development environment. In *Proceedings of the Linguistic Resources and Evaluation Conference 2004. Lisboa, Portugal*, 1111–1114.

Cumming, S. (1995). The lexicon in text generation: progress and prospects. In Walker, D. E., A. Zampolli & N. Calzolari (eds.) *Automating the Lexicon. Research and Practice in a Multilingual Environment*, 171–206. Oxford: Oxford University Press.

Domenig, M. & P. ten Hacken (1992). *WordManager: A System for Morphological Dictionaries*. Hildesheim: Olms.

Doucet, A. & H. Ahonen-Myka (2004). Non-contiguous word sequences for information retrieval. In Tanaka, T., A. Villavicencio, F. Bond & A. Korhonen (eds.) *Second ACL Workshop on Multiword Expressions: Integrating Processing, Barcelona, Spain*, 88–95. Morristown, NJ: ACL.

Emele, M. C., M. Dorna, A. Lüdeling, H. Zinsmeister & C. Rohrer (2000). Semantic-based transfer. In Wahlster, W. (ed.) *Verbmobil: Foundations of Speech-to-Speech Translation*, 359–377. Berlin: Springer.

Evert, S. (2005). *The Statistics of Word Cooccurrences – Word Pairs and Collocations*. Stuttgart: University of Stuttgart, IMS. <http://www.collocations.de/phd.html>.

Evert, S., U. Heid & K. Spranger (2004). Identifying morphosyntactic preferences in collocations. In *Proceedings of the Linguistic Resources and Evaluation Conference 2004, Lisboa, Portugal*, 907–911.

Evert, S. & B. Krenn (2001). Methods for the qualitative evaluation of lexical association measures. In *Proceedings of the 39th Annual Meeting of the Association for Computational Linguistics*, 188–195. Toulouse: ACL.

Fazly, A. & S. Stevenson (2006). Automatically constructing a lexicon of verb phrase idiomatic combinations. In *Proceedings of the Workshop on Multiwords in a Multilingual Context, 11th Conference of the European Chapter of the Association for Computational Linguistics, Trento, Italy, April 2006*, 337–344.

Fellbaum, Ch., A. Geyken, A. Herold, F. Körner & G. Neumann (2006). Corpus-based studies of German idioms and light verbs. *International Journal of Lexicography* 19(4): 349–360.

Granger, S. (2005). Pushing back the limits of phraseology: how far can we go? In Cosme, C., C. Gouverneur, F. Meunier & M. Paquot (eds.) *Proceedings of Phraseology 2005. An Interdisciplinary Conference*, 165–168. Louvain-la-Neuve: Université catholique de Louvain.

Gross, G. (1996). *Les expressions figées en français. Noms composés et autres locutions*. Gap & Paris: Ophrys.

Grossmann, F. & A. Tutin (2003) (eds.). *Les collocations – analyse et traitement* [Travaux et recherches en linguistique appliquée E1]. Amsterdam: De Werelt.

Hallsteinsdóttir, E. (2003). Vom Wörterbuch zum Text zum Lexikon. In Fix, U., G. Lerchner & M. Schröder (eds.) *Zwischen Lexikon und Text – lexikalische, textlinguistische und stilistische Aspekte*, 325–337. Leipzig: Verlag der Sächsischen Akademie der Wissenschaften zu Leipzig.

Hausmann, F. J. (2004). Was sind eigentlich Kollokationen? In Steyer, K. (ed.) *Wortverbindungen – mehr oder weniger fest* (Institut für Deutsche Sprache, Jahrbuch 2003), 309–334. Berlin: de Gruyter.

Heid, U. (1998). Building a dictionary of German support verb constructions. In *Proceedings of the 1st International Conference on Linguistic Resources and Evaluation, Granada, May 1998*, 69–73.

Heid, U. (2007). Computational linguistic aspects of phraseology II. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) (2007). *Phraseologie – Phraseology. Ein internationales Handbuch der zeitgenössischen Forschung – An International Handbook of Contemporary Research,* Vol. II, 1036–1044. Berlin: Walter de Gruyter.

Heid, U. & R. Gouws (2006). A model for a multifunctional electronic dictionary of collocations. In *Proceedings of the XIIth Euralex International Congress. Torino (2006)*, 979–989.

Heid, U. & J. Ritz (2005). Extracting collocations and their contexts from corpora. In Kiefer, F., G. Kiss & J. Pajzs (eds.) *Papers in Computational Lexicography – COMPLEX 2005*, 107–121. Budapest: Linguistics Institute, Hungarian Academy of Sciences.

Jackendoff, R. (1995). The boundaries of the lexicon. In Everaert, M., E. J. van der Linden, A. Schenk & R. Schreuder (eds.) *Idioms: Structural and Psychological Perspectives*, 133–165. Hillsdale, NJ: Lawrence Erlbaum.

Kaalep, H.-J. & K. Muischnek (2006). Multi-word verbs in a flective language: The case of Estonian. In *Proceedings of the Workshop on Multiwords in a Multilingual Context, 11th Conference of the European Chapter of the Association for Computational Linguistics, Trento, Italy, April 2006*, 57–64.

Keil, M. (1997). *Wort für Wort. Repräsentation und Verarbeitung verbaler Phraseologismen (Phraseolex)*. Tübingen: Niemeyer.

Kilgarriff, A., P. Rychly, P. Smrz & D. Tugwell (2004). The sketch engine. In Williams, G. & S. Vessier (eds.) *Proceedings of the XIth EURALEX International Congress, Lorient, Université de Bretagne Sud*, 105–116.

Kilgarriff, A. & D. Tugwell (2001). WASPBENCH: A lexicographic tool supporting WSD. In *Proceedings of the ACL-SIGLEX SENSEVAL Workshop, Toulouse, 2001*, 151–154.

Krenn, B. (2000). *The Usual Suspects: Data-Oriented Models for the Identification and Representation of Lexical Collocations*. Saarbrücken: DFKI, Universität des Saarlandes.

Krenn, B. & G. Erbach (1994). Idioms and support verb constructions. In Nerbonne, J., K. Netter & C. Pollard (eds.) *German in Head Driven Phrase Structure Grammar*, 297–340. Stanford, CA: CSLI Publications.

Kuiper, K., H. McCann, H. Quinn, T. Aitchison & K. van der Veer (2003). *SAID: A Syntactically Annotated Idiom Dataset*. University of Pennsylvania: Linguistic Data Consortium.

Kwasniak, R. (2006). Wer hat nun den Salat? – Now who's got the mess? Reflections on phraseological derivation: From sentential to verb phrase idiom. *International Journal of Lexicography* 19(4):459–478.

Lapshinova, E. & U. Heid (2007). Syntactic subcategorization of noun+verb-multiwords: description, classification and extraction from text corpora. In *Proceedings of the 26th Conference on Lexis and Grammar, Bonifacio, 2–6 October 2007*.

Lin, D. (1998). Automatic retrieval and clustering of similar words. In *Proceedings of COLING-ACL 98, Montreal, 1998*, 768–774.

McCarthy, D., B. Keller & J. Carroll (2003). Detecting a continuum of compositionality in phrasal verbs. In *Proceedings of the ACL-SIGLEX Workshop on Multiword Expressions, 2003*. <http://acl.ldc.upenn.edu/W/W03/W03-1810.pdf>.

Mel'čuk, I. & L. Wanner (1994). Lexical co-occurrence and lexical inheritance. Emotion lexemes in German: a lexicographic case study. *Lexikos* 4: 86–161.

Mel'čuk, I. A., A. Clas & A. Polguère (1995). *Introduction à la lexicologie explicative et combinatoire*. Louvain-la-Neuve: Duculot.

Mittmann, B. (2004). *Mehrwort-Cluster in der englischen Alltagskonversation* (Language in Performance 30). Tübingen: Narr.

Moon, R. (1998). *Fixed Expressions and Idioms in English. A Corpus-Based Approach*. Oxford: Clarendon Press.

Nunberg, G., I. Sag & T. Wasow (1994). Idioms. *Language* 70: 491–538.

Oflazer, K., Ö. Çetinoğlu & B. Say (2004). Integrating morphology with multi-word expression processing in Turkish. In Tanaka, T., A. Villavicencio, F. Bond & A. Korhonen (eds.) *Second ACL Workshop on Multiword Expressions: Integrating Processing, Barcelona, Spain*, 64–71. Morristown, NJ: ACL.

Odijk, J. (2004). A proposed standard for the lexical representation of idioms. In *Proceedings of the 11th Euralex International Congress 2004, Lorient, France*, 153–164.

Pedrazzini, S. (1994). *PhraseManager: A System for Phrasal and Idiomatic Dictionaries*. Hildesheim: Olms.

Pianta, E. & L. Bentivogli (2004). Annotating discontinuous structures in XML: The multiword case. In Witt, A., U. Heid, H. S. Thompson, J. Carletta & P. Wittenburg (eds.) *XML-Based Richly Annotated Corpora, Proceedings of the Linguistic Resources and Evaluation Conference 2004, Postconference Workshop, Lisboa, Portugal*, 30–37.

Polguère, A. & I. Mel'čuk (2006). Dérivations sémantiques et collocations dans le DiCo/LAF. In *Langue française* 150: 66–83.

Pulman, S. G. (1993). The recognition and interpretation of idioms. In Cacciari, C. & P. Tabossi (eds.) *Idioms. Processing, Structure and Interpretation*, 249–270. Hillsdale, NJ: Lawrence Erlbaum.

Reiter, E. & R. Dale (2000). *Building Natural Language Generation Systems*. Cambridge: Cambridge University Press.

Ritz, J. (2006). Collocation extraction: Needs, feeds and results of an extraction system for German. In *Proceedings of the Workshop on Multiwords in a Multilingual Context, 11th Conference of the European Chapter of the Association for Computational Linguistics, Trento, Italy, April 2006*. <http://acl.ldc.upenn.edu/W/W06/W06-2406.pdf>.

Rothkegel, A. (2007). Computerlinguistische Aspekte der Phraseme I. In Burger, H., D. Dobrovol'skij, P. Kühn & N. R. Norrick (eds.) *Phraseology – Phraseology. Ein internationales Handbuch der zeitgenössischen Forschung – An International Handbook of Contemporary Research*, Vol. II, 1027–1035. Berlin: Walter de Gruyter.

Ruppenhofer, J., C. F. Baker & C. J. Fillmore (2002). Collocational information in the FrameNet database. In *Proceedings of the Euralex International Congress 2002. Copenhagen, Denmark*, 359–369.

Sag, I. A., T. Baldwin, F. Bond, A. Copestake & D. Flickinger (2002). Multiword expressions: A pain in the neck for NLP. In Gelbukh, A. (ed.) *Computational Linguistics and Intelligent Text Processing: Third International Conference: CICLing-2002*, 1–15. Berlin: Springer-Verlag. <http://lingo.stanford.edu/pubs/WP-2001-03.pdf>.

Seelbach, D. (2000). Zur Verwendung und Übersetzung von Prädikatausdrücken im Rahmen der Lexikongrammatik. In *Proceedings of the Euralex International Congress 2000. Stuttgart, Germany*, 529–545.

Sereţan, V., L. Nerima & E. Wehrli (2004). A tool for multi-word collocation extraction and visualization in multilingual corpora. In *Proceedings of the 11th Euralex International Congress, Lorient*, Vol. 2, 755–766.

Sinclair, J. M. (1991). *Corpus, Concordance, Collocation*. Oxford: Oxford University Press.

Smadja, F. (1993). Retrieving collocations from text. *Computational Linguistics* 19(1): 143–177.

Spohr, D. & U. Heid (2006). Modeling monolingual and bilingual collocation dictionaries in Description Logics. In *Proceedings of the EACL Workshop on Multiwords and Multilinguality, in Connection With EACL-2006, Trento, Italia, 2006*, 65–72.

Tarp, S. (2006). *Leksikografien i grænselandet mellem viden og ikke-viden. Generel leksikografisk teori med saerlig henblik på lørnerleksikografi*. Århus: HHÅ Center for Leksikografi. [to appear in English, in spring 2008]

Tschichold, C. (2000). *Multi-Word Units in Natural Language Processing* [Informatik und Sprache 6]. Hildesheim: Olms.

Tutin, A., I. Novakova, F. Grossmann & C. Cavalla (2006). Esquisse d'une typologie des noms d'affect à partir de leurs propriétés combinatoires. *Langue française* 150: 32–49.

Váradi, T. (2006). Multiword units in an MT lexicon. In *Proceedings of the Workshop on Multiwords in a Multilingual Context, 11th Conference of the European Chapter of the Association for Computational Linguistics, Trento, Italy, April* 2006, 73–77.

Venkatsubramanyan, S. & J. Perez-Carballo (2004). Multiword expression filtering for building knowledge. In Tanaka, T., A. Villavicencio, F. Bond & A. Korhonen (eds.) *Second ACL Workshop on Multiword Expressions: Integrating Processing, Barcelona, Spain*, 40–47. Morristown, NJ: ACL.

Verlinde, S., T. Selva & J. Binon (2006). The Base Lexicale du Français (BLF): A multifunctional online database for learners of French. In Corino, E., C. Marello & C. Onesti (eds.) *Proceedings – XII Euralex International Congress, Torino, Italy, 2006*, 471–482. Alessandria: Edizioni dell'Orso.

Villada Moirón, B. & J. Tiedemann (2006). Identifying idiomatic expressions using automatic word-alignment. In *Proceedings of the Workshop on Multiwords in a Multilingual Context, 11th Conference of the European Chapter of the Association for Computational Linguistics, Trento, Italy, April 2006*, 33–40.

Villavicencio, A., A. Copestake, B. Waldron & F. Lambeau (2004). Lexical encoding of MWEs. In Tanaka, T., A. Villavicencio, F. Bond & A. Korhonen (eds.) *Second ACL Workshop on Multiword Expressions: Integrating Processing, Barcelona, Spain*, 80–87. Morristown, NJ: Association for Computational Linguistics.

Wanner, L. (1996). Lexical choice in text generation and machine translation. *Machine Translation* 11:3–35.

Widdows, D. & B. Dorow (2005). Automatic extraction of idioms using graph analysis and asymetric lexicosyntactic patterns. In Baldwin, T., A. Korhonen & A. Villavicencio (eds.) *ALC-SIGLEX 2005 Workshop on Deep Lexical Acquisition, Proceedings*, 48–56. Ann Arbor, MI: University of Michigan.

Online Resources

A Broader Perspective on Multiword Expressions, 2007, URL (11.09.2007) http://www.let.uu.nl/~Nicole.Gregoire/personal/ACL07-MWE/

ACL workshop on Collocation: Computational Extraction, Analysis and Exploitation, 2001, URL (11.09.2007): http://www.cs.unca.edu/~bruce/acl01/collocation.html

BLF [=Base lexicale du français], URL (11.09.2007): https://www.kuleuven.be/ilt/blf/

BNC: British National Corpus, URL (22.02.2007): http://www.natcorp.ox.ac.uk/

Corpus of *Digitales Wörterbuch der deutschen Sprache (DWDS-Kernkorpus),* URL (22.02.2007) http://www.dwds.de/textbasis/kerncorpus

DAFLES [= Dictionnaire d'apprentissage du français langue étrangère ou seconde], URL (07.07.2005): http://www.kuleuven.ac.be/dafles

Dansk Idiomordbog, URL (22.02.2007): http://www.idiomordbogen.dk

DiCE, URL (11.09.2007): http://www.dicesp.com/

DiCo/LAF dicouèbe, URL (11.09.2007): http://olst.ling.umontreal.ca/dicouebe/

LinGO project: URL (22.02.2007): http://lingo.stanford.edu/

LMF, URL (11.09.2007): http://tagmatica.fr/doc/ISO24613cdRev9.pdf

Multiword Expressions: Analysis, Acquisition and Treatment, 2003, URL (11.09.2007) http://www.cl.cam.ac.uk/~alk23/mwe/mwe.html

Multiword Expressions: Integrating Processing, 2004, URL (11.09.2007) http://www.cl.cam.ac.uk/~alk23/mwe04/mwe.html

Multiword Expressions: Identifying and Exploiting Underlying Properties, 2006, URL (11.09.2007): http://www.inf.ufrgs.br/~avillavicencio/mwe-acl06.html
Pargram Project: URL (22.02.2007): http://www2.parc.com/istl/groups/nltt/pargram/
SALSA, Saarbrücken Lexical Semantics Annotation Project: URL (22.02.2007) http://www.coli.uni-saarland.de/projects/salsa/

A computational lexicography approach to phraseologisms

Cornelia Tschichold

The cycle of lexicographic and linguistic work involved in compiling a computational phraseological database is divided into three phases and described in relation to the specific challenges multi-word expressions (MWEs) pose for a lexical database. Data collection is a process that is far from complete for the MWEs found in English, with the variability of some phrases making identification of all occurrences in large corpora a major challenge. Formalization of the form and variability of MWEs is an interrelated process which can improve tools for data collection and other applications. Increased use of the phraseological lexical database in NLP applications can ultimately lead to further insights into the nature of MWEs and to improvements in the database. Due to the volume of lexicographic data on MWEs that still needs to be collected, analysed and formalized, and the cyclical nature of the work, the resulting lexical database should be reusable in as many applications as possible. *WordManager-PhraseManager*, the lexical resource described in the second part of the chapter, can capture the variability of MWEs in a way that allows for maximum reusability of lexical data.

1. Introduction

An interdisciplinary approach to phraseology can reveal a number of interesting contact points. Computational linguistics and its applied branch, natural language processing (NLP), can hardly be called natural associates of phraseology as traditionally there has been relatively little contact between phraseologists and computational linguists. One area where this gap between the two fields manifests itself is that of terminology. While in linguistics, and the field of phraseology more specifically, an idiom is generally defined as a multi-word lexeme whose meaning is not a compositional function of the meanings of the component words, in computational linguistics the term 'idiom' is used in a much looser sense (cf. e.g. McEnery (1992:64ff.) or other descriptions of the CLAWS program, which contains a subroutine called IDIOMTAG for tagging completely fixed expressions such as *by and large* with a single tag). The items that interest phraseologists so much are often seen by computational linguists as annoying problems, a "pain in the neck", as Sag et al. (2002) called it, and too often remain outside the focus of the field as whole. For a considerable period, the emphasis

in NLP research was on statistical (rather than symbolic or linguistic) methods, and multi-word expressions (MWEs) did not fit well into such a perspective. But large-scale NLP applications, e.g. automatic translation systems, information retrieval systems or text summarization tools, normally have the aim of dealing with real text, so their developers have been forced to acknowledge that they cannot ignore the problem of what to do with these expressions in the long term (Villavicencio et al. 2005). The fact that MWEs pose serious problems for NLP systems that rely (too) heavily on statistical methods can easily be seen when, say, a newspaper article is fed through one of the machine translation systems freely available on the internet. MWEs of just about any kind almost invariably lead to mistranslations because the system's lexicon has failed to recognize the MWE in the original text. The importance of being able to identify and parse MWEs adequately has now been recognized in the computational linguistics community. NLP systems developed for dealing with field-specific texts in particular need to be able to recognize and treat MWEs. Technically and commercially viable NLP systems often have to deal with texts full of terminology, e.g. domain-related compounds, that needs to be identified and translated or otherwise dealt with correctly. The pressure of large-scale, real-life applications has therefore led to the recognition that MWEs and – crucially – their variations need to be integrated into the lexical databases used by these NLP systems.

For their part, phraseologists have expected little help from the tools developed in computational linguistics and have been more cautious than lexicographers to embrace computers for anything more than storage of their objects of study. However, the contact zone between the two linguistic fields of phraseology and computational lexicography has great potential for fruitful cooperation. At this intersection of computational linguistics and phraseology, there are three fields of study or processes that – at least theoretically – can be distinguished as three logical steps in a sequence, and where increased contact between computational linguists and phraseologists could lead to progress on both sides of the intersection of their fields. All three potential points of contact involve computers and methods from computational linguistics dealing with lexemes consisting of more than one word. Firstly, there is data collection, when the multi-word lexemes are identified (a process called lexical acquisition), discussed in Section 2. Once the required lexemes have been identified, analysed and sorted, the second step is the formalization and entering of the resulting lexicographic data into a computational database, described in Section 3. Finally, the last process is the use and subsequent improvement of the resulting lexicographic database in computational linguistics applications, such as corpus query systems, machine translation systems, or computer-assisted language learning programs. In practice, these three steps are not usually neatly separated, but tend to be applied in a cyclical fashion, with more and more items being added to the database in order to improve the results of the NLP application for which the lexical database is being built. As long as the lexical database is set up in a way that satisfies the requirements for the formalization and analysis of any newly added expressions, such a cyclical procedure is an advantage rather than a problem. For phraseologists, the resulting lexical database would make for an interesting

collection of items for further study, especially where terminological aspects are part of the focus. Traditional phraseology has had a tendency to study linguistically and culturally interesting, but relatively rare MWEs, while compounds and other linguistically well-formed phrases are considerably more frequent and much more numerous in most corpora. Modern corpus linguistics now makes it possible to study these phenomena as well as the traditional idioms. And more studies on MWEs and the precise extent of their variability are still greatly needed before the lexical databases can be said to be anywhere near complete.

Due to the number of items that need to be processed, the first two procedures, i.e. data collection and formalization, require enormous efforts in terms of human labour, all with the long-term aim of eventually producing a fully-fledged lexicographic database that includes significant numbers of MWEs. Exactly how significant these numbers are is still unclear as the estimates for the number of MWEs in our vocabulary seem to increase with the rise of corpus linguistics and the increase in the volume of texts that can be used to study them. Despite early hints about the quantity of idioms in our lexicon such as those in Healey (1968),[1] the general view among linguists, including computational linguists, was for a long time that idioms were a marginal phenomenon, an anomaly in the lexicon. But with the increased interest in the lexicon as a whole, these 'anomalies' have also received more attention. Moon (1998b) comes to the conclusion that the phrasal lexemes she examines in an 18 million word corpus are indeed quite rare, or at least too infrequent to get a balanced lexicographic view, while Jackendoff (1997: 156) gives a cautious estimate of "the same order of magnitude" as the number of single word items in our lexicon. Sag et al. (2002) consider this an underestimate, especially if items of specialized terminology are included in the lexicon. Even if we make a distinction between the number of types that need to be listed in the dictionary and the number of tokens that occur in a text, it is probably safe to say that both these numbers are considerably larger than was thought a generation ago.

Compared to the corpora needed to study single word lexemes, the size of corpora needed to study multi-word lexemes in depth runs into the hundreds if not thousands of millions of words, and such corpora have only recently become available. A problem that remains however is the lack of good computational tools to study the variability of these lexemes in such large corpora.

At the moment, while computer programs can help us identify candidates for inclusion (cf. Mudraya et al. this volume), the ultimate decision for or against inclusion

1. "Several problems face the lexicographer as to how to incorporate idioms into a dictionary. And these are not small problems, as he may well be faced with twice as many idioms as there are main entries" (Healey 1968: 95).

in a phraseological dictionary has to be taken by a human.[2] This restriction also applies to the coding of items (or groups of items) for variability. Because the sheer amount of human work involved is so large, it is important to ensure that the outcome is a lexical database that can be used in as many NLP applications as possible, in order to avoid unnecessary duplication of these huge efforts. The computational lexicon system described in Section 5 was designed with this aim in mind. But let us first return to the problem of identifying MWEs in corpora.

2. Data collection

The first point of contact between computational linguistics and phraseology is the area of lexical data collection, also called lexical acquisition or lexical knowledge acquisition (Matsumoto & Utsuro 2000) if the process is automated to some degree. The two linguistic fields can benefit from each other's expertise in relation to any question where lexemes consisting of more than one text word are concerned, but in this area there seems to have been particularly little contact, with computational linguistics taking an approach largely inspired by frequency-driven corpus linguistics, and phraseologists tending to concentrate on the rather less frequently-occurring strongly idiomatic expressions, identified mainly on the basis of their intuition (Wray 2002). Neither of these approaches has so far come up with wholly satisfying results in terms of applicability to NLP. Knowledge about phraseology and the typical patterns found in MWEs needs to inform the computational methods used in lexical acquisition on the one hand, and frequency aspects need to take on a more important role in the study of traditional phraseology on the other.

As long as no even remotely complete lists of idioms, collocations and other multi-word lexemes exist in electronic form, lexical data collection clearly remains a major aim at this first stage. This is obviously one of the aims of much of the work done within the framework of computer corpus lexicography. Much has already been achieved, as illustrated by Moon (1998a), in the identification of frequent word combinations in English once the corpora used as the raw material for this work started to be large enough. Statistics-based algorithms can be run on any given corpus in order to identify frequently co-occurring word combinations, and although simple lists of frequent word combinations are a useful result of this process, more information is needed for each of the strings identified in this way. If frequency and probability of co-occurrence alone are used as criteria for inclusion in the list of word combinations, we end up with a list which includes many items of little linguistic interest (Wray 2002) or practical use in NLP. Before any such list can be put to good use in an application,

2. While many researchers in the area of NLP would probably agree that purely statistical approaches cannot produce excellent large-scale applications, the quest for fully automated lexical acquisition tools capable of finding all types of MWEs goes on despite this insight.

a selection will need to be made. As Moon (1998a: 51) points out, we have no signs today of computers soon being able to identify MWEs automatically "because in so many cases FEIs [fixed expressions and idioms] are not predictable, not common, not fixed formally, and not fixed temporarily".

In addition to simple frequency in a given corpus, the question of institutionalization or lexicalization (Bauer 1983) should therefore be considered as a factor for inclusion in the lexical database. All corpora, even very large ones, have their limits in terms of coverage, and therefore information derived from a corpus should always be filtered and complemented with linguistic assessment by speakers of the language before a word combination is added to the list of MWEs that are to be entered into a computational dictionary. This clearly goes beyond the possibilities of automatic lexical acquisition and is part of the labour-intensive stages in computational lexicography, where it makes sense to take stringent measures in order to ensure both consistency within the database and sufficient reusability of the final outcome.

A second major complication is the fact that many MWEs show some degree of flexibility in their syntactic behaviour. The range goes from completely frozen, invariable expressions, such as *by and large*, or *in front of* and other multi-word prepositions, to almost fully flexible combinations of words, as are found among verbal collocations. Possible modifications include the paradigmatic variation of one or more of the components (e.g. the inflection of a verb), the interruption by elements not belonging to the multi-word lexeme, and various transformations. In the context of natural language processing, it is vital first to know which modifications are possible for each idiom, as this will determine whether both an idiomatic and a literal reading are possible, and then to express these acceptable variations in an appropriate formalism. In contrast to paper dictionaries, where the first major problem is under which headword the MWE should be listed, the main difficulty for electronic dictionaries lies with the strict formal requirements of the medium. Paper dictionaries such as the ODCIE (Cowie & Mackin 1993) have developed a more or less intricate system of grammatical coding. For the user who is interested enough to read through the relevant sections of the introduction, the ODCIE's codes are relatively easy to interpret. If we look up the phrase *pay attention (to)* we will find the code [Vn.pr pass emph rel] representing the structure 'verb-noun-preposition' where the three transformations 'passive', 'emphatic', and 'relative' are allowed. These transformations are explained with the help of many examples in the detailed description of the dictionary.

In order to make maximum use of MWEs in a computational lexicon, the exact variability for each lexeme needs to be established and formally coded in such a way that NLP applications can make full use of it. The ODCIE's codes are a helpful starting point, but further specification is needed for an electronic lexicographic database. Many phraseologists would argue that it is not possible to describe an idiom exhaustively in this way, that there is no fixed set of modifications or transformations that can or cannot be applied to such expressions, and that the creative use of language allows little or no variation in one context and more variation in another context. For the sake of finding a viable solution, however, a practical way of handling this continuum

of variability will have to be found. It can easily be argued that a lexicographic database that incorporates idioms and collocations and that gives a relatively conservative description of their variability is a vastly more useful database for any NLP application than a lexicon without any information on MWEs, or with multi-word lexemes in their canonical forms only.

Once we have used computers to produce a list of potential multi-word lexemes which can then be weeded out, the problem of coming up with enough corpus data to decide on the frequency and variability remains, precisely because of the variability of many of these expressions. We are faced with a chicken and egg problem: we can only find all instances of an expression in a corpus if we know the precise extent of its variability and are able to formalize this into a sufficiently stringent formalism to be useful in an NLP tool, but we can only find out about the variability of an expression if we have a sufficient number of corpus occurrences to study the expression in depth.

The criteria of frequency, institutionalization and lexicalization can be used as guiding principles for the inclusion of an expression in the database, but other considerations such as the needs of a particular application might also fruitfully be taken into account. Another potentially useful criterion is that of translational needs (see Mudraya et al, this volume). These days, no end user with a realistic view of the field would expect a phraseological dictionary that could be called complete in any way. So the pragmatic solution is to include what is known and needed, and to work from this as a basis for further expansion of the database. Once the decision has been taken to include a specific MWE in the database, attempts should be made to find and analyse at least its most common variations, so that the database can be used to handle the more common text occurrences of the lexemes that are included. Lexical data collection should be seen as an incremental process which requires supplementing and updating on a regular and ongoing basis.

3. Lexicographic work

The lists of MWEs compiled from corpora in the process of lexical acquisition will necessarily be incomplete, but this is inherent in the reality of natural language and should certainly not be a reason not to embark on the next step. The English language, which is used here as an example, may not offer much of a challenge to computational lexicography from the point of view of (single-word) morphology, but it is certainly a language very rich in MWEs. Not only does English have a large stock of idioms and collocations, there are also many phrasal and prepositional verbs, multi-word prepositions and adverbs. Even its verbal inflection is largely periphrastic rather than truly morphological, thus necessitating a computer analysis spanning several words. Clearly, in order to have a reasonably complete lexicon, a methodology for including these complex lexemes has to be developed. One of the first questions this raises is that of the access route.

As computational linguistics typically treats the orthographic word as its basic unit (just like traditional lexicography), it is convenient to access the look-up process for MWEs via their component single words as well. For some expressions this is a relatively simple process, but for those MWEs that show considerable variation in form, especially for those with variable word order of the component words, a formalism for access that is linguistically transparent and at the same time technically feasible has to be found. The challenge is to find a formalism that can cope with the full range of variability found in the different kinds of MWEs (Villavicencio et al. 2005). A successful formalism will be able to exclude those readings which cannot be idiomatic, while at the same time making sure all possible idiomatic readings are included. In other words the lexical database needs to be designed in such a way that the sentence *The bucket was kicked* is not analysed as the passive version of the idiom *kick the bucket*, but that only the literal sense is given in the analysis.

Compilation of a dictionary of MWEs therefore starts with the collection and – more problematic perhaps – the detailed formalization of available phraseological information on the variability of each expression. For invariable lexemes of the type *by and large*, this information on form will be very simple, but for other types of lexemes, like *take steps*, more wide-ranging information on the morphological and syntactic behaviour is needed. Formalizing this information seems to be a major problem at this stage, as corpus-based research on phraseologisms is still in its beginning phase, especially when compared to what we know about single words with their virtually complete description of morphology.

Attempts at formalizing the variability of MWEs immediately lead to difficult decisions about analyses at different linguistic levels and the interaction of these levels during the processing of text. Humans use their pragmatic, semantic, syntactic and phonological knowledge simultaneously when processing language, but the corresponding components of large NLP systems are much less integrated and tend to work more sequentially (and typically bottom-up, starting with word segmentation and lexical look-up). Given today's NLP methods, it is therefore problematic to assume that the lexical database can make use of semantic or even syntactic context, as the components dealing with these linguistic levels typically start their analysis only after the look-up stage, when the text has been analysed into words, drawing on the knowledge stored in the lexical database. A highly reusable lexical database should therefore use a formalism for encoding the variability of MWEs that uses only very basic and commonly accepted grammatical categories and only where strictly necessary. An additional reason for such a procedure is the fact that correct identification of MWEs at an early stage in the process can help both the syntactic and semantic analysis, i.e. parsing and word sense disambiguation (Villavicencio et al. 2005).

4. Using the lexicographic database

Assuming phraseologists and computational linguists have successfully worked together to compile and code a list of MWEs, we can move on to a third point of contact between the two fields. At this stage, possible applications of the resulting lexical database in all types of complex NLP programs can finally be developed, from natural language query systems to machine translation programs. Such applications have diverging needs as regards the depth of syntactic and semantic analysis. For a spell checker, a relatively shallow analysis is likely to be sufficient, while a large machine translation system will need a much more in-depth analysis of the source text. A lexical database should therefore be both as theory-neutral and as application-neutral as possible in order to avoid a rewriting or recoding of the whole lexicographic database every time the grammar formalism is modified, or a new application requires different lexicographic information. The compilation of a lexicon involves such large amounts of human coding work that the results of efforts in this area need to be reusable in as many applications as possible.

For a computational lexicon, the approach necessarily has to be different from that of paper-based dictionaries such as the ODCIE, which are compiled with human users in mind. If we accept the principle that a computational lexicon should offer a high degree of reusability, the lexicon should remain neutral with respect to the types of applications it is used in and the various syntactic theories and parsing mechanisms employed there. This in turn implies that in a natural language application, potential MWEs should be recognized at the stage where text words are looked up in the dictionary, i.e. before parsing has been carried out. In order to make this possible, recognition needs to be string-based. Such an approach will allow the recognition of a superset of all potential MWEs. If we specify as precisely as possible how flexible each MWE is, this superset can be kept as small as possible.

The lexical database system described below was created on the basis of the task analysis described above. It consists of two major components, one for mapping single text words onto their lexemes, called *WordManager*™ (*WM*), and one for mapping MWEs found in running text onto their canonical forms. The *WordManager* core system deals with single text words. It will map a verb form like *wrote* onto its lexeme *write* and identify its tense (and any other relevant morphological features). The extension, called *PhraseManager*, links the lexemes encoded in *WordManager* to the MWEs and to the specific information needed for the recognition of potential idioms in a sentence. This second part will recognize, for example, *keeping a stiff upper lip* as a token of *to keep a stiff upper lip*. The focus here is on *PhraseManager* (*PM*).

5. *PhraseManager*

In *PhraseManager* (*PM*), the descriptive framework for MWEs can be used for a variety of idioms and other multi-word lexemes, including phrasal and prepositional

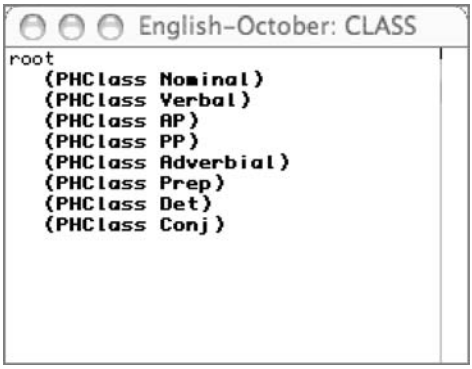


Figure 1. *Phrase Manager*: Class window

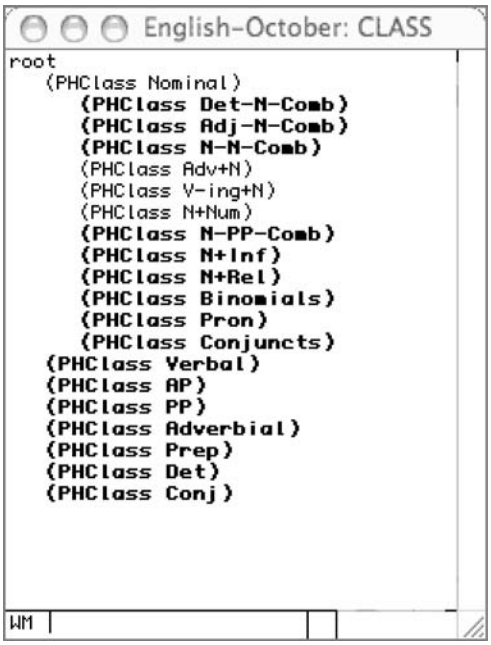


Figure 2. Nominal classes

verbs and collocations, and other not necessarily semantically opaque expressions. These MWEs display a range of flexibility from completely frozen to (almost) fully flexible combinations of words. *PM* offers a transparent formalism to handle the formalization of the canonical form and all of the modifications of these complex lexemes and to specify which are possible for each expression. Within the database, MWEs are grouped strictly according to their syntactic class and form. The variability is used as a second criterion for classification after syntactic class and internal form have been

taken into account. Classes of MWEs are organized in a hierarchical structure in a tree window, as seen in Figures 1 and 2.

In the lexical database shown in Figure 1, the first group of phrasal classes (PHClass Nominal) includes all those subgroups and classes that can be called nominal phrases, e.g. classes of compound nouns of various types (as illustrated in Figure 2, where the node for the PHClass Nominal has been expanded by one level), followed by the large second group of verbal multi-word lexeme classes. Bold names of classes indicate those classes that have subclasses or groups not visible in the present view. Non-bold names of classes which do not have further subclasses give access to the specification window of that phrasal class.

The other groups seen in Figures 1 and 2 are for adjectival and prepositional phrases, adverbials, multi-word prepositions, determiners and conjunctions, respectively. All these groups are further subdivided into subgroups and classes, so that all occurring form types can be described. The formal specifications of the lexemes are only made at the lowest level where actual phrases can be found, not at the level of phrasal groups. For the classes seen in Figure 2, the label for the phrasal class Adv+N thus gives access to the specification of a class that includes the phrase *dearly beloved*, whereas the label 'PHClass N-N-Comb' gives access to the full list of classes within that group, e.g. N+N, Det+N+N, etc. and only indirectly to specification windows (described in the following sections). A class of phrases thus provides the formal model for all phrases that share the same phrase structure tree in their canonical form.

5.1 Invariant phrases

To illustrate the coding of phrases in *PhraseManager*, we will first consider the simplest possible case of a phrase, using the example of the multi-word preposition *in front of*. Figure 3 shows the specification for this class.

This expression can be said to have the word class of a preposition (if taken in its entirety) and to consist internally of a preposition followed by a noun followed by another preposition. This information is expressed in the line below the label SYNTAX-TREE, with the first value in the parentheses giving the category of the expression as a whole, followed by the list of parts of speech of the individual words making up the expression. In this case the syntax tree has been kept flat, as the internal structure of the expression is not relevant, given that this multi-word lexeme does not show any variation whatsoever.

The next few lines specify that no modifications (insertions of optional elements) or transformations (resulting in changes of word order) are allowed for this phrase. This information is followed by an example of a phrase in this class and the details for the words making up that phrase. All the individual words used to make up this complex preposition need to be listed and encoded in *WM* before they can be used for the definition of a MWE in *PM*. Once the class has been defined and an example given with the complete specification, further MWEs that follow the same phrasal pattern can be entered, e.g. *in charge of, in view of, in accordance with, by means of, in addition*

```
SYNTAX-TREE
(Prep Prep N Prep)

MODIFICATIONS
-

TRANSFORMATIONS
-

EXAMPLE
in front of

in      (Cat Prep)
front   (Cat N)
of      (Cat Prep)
-
```

Figure 3. Specification of a simple phrasal class

to, etc. All of these complex prepositions can be said to consist of the sequence Prep + N + Prep, where each of these slots is filled with a word that has a lexicographic description in the *WM* part of the database. The information stored with the single words is percolated to the MWE through the description shown in Figure 3. The word *front* cannot appear in its pluralized form within the complex preposition because no inflectional variation has been allowed (cf. the word *drop* in Figure 5 below).

The second example of a fully specified phrase is that for the class of *by and large*, a class that contains only one single member. This expression is completely frozen and syntactically ill-formed if we accept the assumption that the word *by* is a preposition, as it would be in other contexts. In principle, the lexicographer could also code this expression as consisting of the sequence 'adverb – conjunction – adverb', but this would entail having single-word entries listing both *large* and *by* as adverbs. Figure 4 shows the specification, which otherwise is similar to that seen in Figure 3 for the preposition *in front of*. Such completely frozen invariable units have been called "long words" or "words with spaces" (Sag et al. 2002) in the NLP literature. Their form is of mainly historical interest to phraseologists, but they are nevertheless part of the problem MWEs in general pose for NLP. A parser that tries to analyse a sentence containing *by and large* without knowledge of its phraseological character will inevitably come up with a wrong or at best an incomplete parse. Even adding only such invariable MWEs to the lexical database will significantly improve the results of the parser, as Nivre & Nilsson (2004) show, because the MWEs can be identified and given a unified tag in a preprocessing step before the parser is set to work.

```
SYNTAX-TREE
(AdvP Prep Conj Adj)

MODIFICATIONS
-

TRANSFORMATIONS
-

EXAMPLE
by and large

by      (Cat Prep)
and     (Cat Conj)
large   (Cat Adj)
-
```

Figure 4. Specification of *by and large*

5.2 Phrases with modifications and transformations

We now turn to a type of MWE that exhibits somewhat more variability. *Phrase Manager* allows for modifications that can be specified for the level of the whole class or alternatively for that of individual phrases only, in order to make a flexible description of each expression possible. To illustrate the type of variation formalized in modifications, a longer phrase is used as the next example. Among others, the class Det+Adj+N+PP illustrated here includes the expressions *a red rag to a bull* and *the thin edge of the wedge.* The syntax tree for this phrasal class shows a more complex pattern with nested phrases. While the class as a whole does not allow modifications or transformations, the example given in Figure 5 shows the amount of specification that can be given for individual phrases.

The angle brackets around the determiner *the* and the two adjectives in the example indicate that both of these elements are optional. This ensures recognition of the phrase even if it occurs in the form *no arm of the law (is long enough to reach . . .)*. The two adjectives given in the example indicate a choice of adjectives for this expression. At the bottom of the specification window in Figure 5, an individual modification for *the strong/long arm of the law* can be seen. The left-pointing arrow next to the word *law* causes a modification to be allowed left of this word, so that variations such as *the long arm of the French law* can also be recognized as instances of the phrase defined here.

Such an approach can take care of much of what Sag et al. (2002) call the "lexical proliferation problem", i.e. the failure to capture the generalization that can be made for groups of MWEs such as structurally similar collocations, because each expression is coded individually in the lexical database and receives a separate description of its variability. Both a finite and an open-ended list of adjectives to choose from, as in the example above, can be formalized in *PM*. Individual variations such as the choice of adjectives and the optionality of the determiner can be expressed adequately as indi-

```
SYNTAX-TREE
(NP Det Adj N (PP Prep (NP Det N)))

MODIFICATIONS
-

TRANSFORMATIONS
-

EXAMPLE
[the] [strong, long] arm of the law

the     (Cat Det)
strong  (Cat Adj)(Comparison Absolute)
arm     (Cat N)(Number Sg)
of      (Cat Prep)
the     (Cat Det)
law     (Cat N)(Number Sg)

law <
```

Figure 5. Specification of a phrasal class with variations

```
SYNTAX-TREE
(VP+pass V (NP Det N))

MODIFICATIONS
V <

TRANSFORMATIONS
Passive

EXAMPLE
<drop> [a] hint
drop        (Cat V)
a           (Cat Det)
hint        (Cat N)

hint <
```

Figure 6. Specification of a verbal phrasal class

vidual modifications without having to create new lexical entries for each version of the phrase.

In addition to the modifications described so far, some phrases, especially those with verbs, need more complex specifications. Figure 6 shows the class PH-Class V+Det+N. For expressions in this phrase class that includes *drop a hint*, it can be specified that the verb and the noun can be inflected in the usual way (as shown by the pointed brackets < > in the text window), that it can be modified by an adverb

(*He would never drop a hint*) or an adjective (*I only dropped a very subtle hint*), and that the whole expression can undergo passivization (*A few hints were dropped*). This whole formal description is based on the text occurrences of the component words and not tied to semantic properties, as can be seen in Figure 6. Passivization is described centrally, elsewhere in the formalism, and strictly in terms of the word order changes this transformation engenders. Such a procedure offers the considerable advantage of being independent of any specific grammar formalism that might conflict with the formalism used in an application making use of the lexical database. This independence of a grammar formalism brings with it a small risk of overgeneration, i.e. the risk that some strings might be falsely analysed as being an instance of the MWE in question, but the *WM/PM* program always keeps the alternative 'literal' analysis as well, so a parser can subsequently reject the phrasal analysis and choose the non-phraseological output instead.

All MWEs that conform to this V + NP pattern and that can be passivized are entered into the lexical database under this PHClass V+Det+N. This addresses the lexical proliferation problem on the one hand and also captures the psycholinguistic difference between such decomposable idioms (Sag et al. 2002) as *spill the beans*, which can be passivized without losing its idiomaticity, and the group of non-decomposable idioms of the type *kick the bucket*. The latter type is in a different phrasal class that does not allow passivization. In other words, (1) has an idiomatic reading whereas (2) does not.

(1) The beans about Jane's promotion were spilled at this morning's meeting.

(2) The bucket was kicked by Jane this morning.

Figure 7 finally shows the document window from where the linguist creating the database can access all the details of the database. The phrase class hierarchy (Figures 1 and 2) and the specification windows (Figures 3 to 6) described here are accessed via the 'class' button, while transformations are described in the window accessed via the

Figure 7. *Word Manager* – main document window

'transf' button. For a full description of an English database, see Tschichold (2000) and the webpages given in the references section.

6. Conclusion

The *WM/PM* system thus offers a lexical database and matching formalism which allows the linguist and lexicographer to specify the internal syntactic structure of MWEs to the desired extent without imposing any specific grammar formalism on the applications it is used in. The system is language-independent and has been tested for several (European) languages. It thus goes some way towards Villavicencio et al.'s (2005) call for a lexical database system that can deal with variability of MWEs and also towards Sag et al.'s (2002) call for a system where the MWEs inherit the lexical properties of their component words. In *PhraseManager*, the part of speech, along with inflectional information, gender, and other morphological features are automatically percolated to the MWE that makes use of the single word entries. The important types of variability found among multi-word lexemes can be handled sufficiently well, while remaining independent of a specific grammar formalism and thus making the resulting database as reusable as possible.

References

Bauer, L. (1983). *English Word-Formation*. Cambridge: Cambridge University Press.

Cowie, A. P. & R. Mackin (1993). *Oxford Dictionary of Current Idiomatic English*. Oxford: Oxford University Press.

Healey, A. (1968). English idioms. *KIVUNG* 1(2): 71–108.

Jackendoff, R. (1997). *The Architecture of the Language Faculty*. Cambridge, MA: The MIT Press.

Matsumoto, Y. & T. Utsuro (2000). Lexical knowledge acquisition. In Dale, R., H. Moisl & H. Somers (eds.) *Handbook of Natural Language Processing,* 563–610. New York, NY: Marcel Dekker.

McEnery, T. (1992). *Computational Linguistics: a Handbook and Toolbox for Natural Language Processing.* Wilmslow: Sigma Press.

Moon, R. (1998a). *Fixed Expressions and Idioms in English*. Oxford: Clarendon Press.

Moon, R. (1998b). Frequencies and forms of phrasal lexemes in English. In Cowie, A. P. (ed.) *Phraseology: Theory, Analysis, and Applications,* 79–100. Oxford: Clarendon Press.

Nivre, J. & J. Nilsson (2004). Multiword units in syntactic parsing. In Dias, G., J.G.P. Lopes & S. Vintar (eds.) *MEMURA 2004 – Methodologies and Evaluation of Multiword Units in Real-World Applications*, 39–46. LREC 2004. Lisbon, Portugal, 25 May 2004.

Sag, I., T. Baldwin, F. Bond, A. Copestake & D. Flickinger (2002). Multiword expressions: A pain in the neck for NLP. In Gelbukh, A. F. (ed.) *Computational Linguistics and Intelligent Text Processing, Third International Conference, CICLing 2002, Mexico City, Mexico, February 17–23, 2002, Proceedings* [Lecture Notes in Computer Science 2276], 1–15. Berlin: Springer.

Tschichold, C. (2000). *Multi-Word Units in Natural Language Processing*. Hildesheim: Olms.

Villavicencio, A., F. Bond, A. Korhonen & D. McCarthy (2005). Introduction to the special issue on multiword expressions: Having a crack at a hard nut (editorial). *Computer Speech and Language* 19(4): 365–377.

Wray, A. (2002). *Formulaic Language and the Lexicon*. Cambridge: Cambridge University Press.

For more information on *WordManager*

Morphological analyzer: http://www.canoo.com
and http://pages.unibas.ch/LIlab/projects/wordmanager/wordmanager.html

Extracting specialized collocations using lexical functions

Brigitte Orliac

This chapter presents a method for extracting specialized verb + noun collocations based on lexical functions, the formal device developed within the framework of the Meaning-Text theory to represent collocations. Although based on a linguistic analysis of collocations, our program (*Colex*) uses statistical measures to distinguish true collocations from free combinations. We argue that log-likelihood is a better measure of precision in this domain than mutual information. We are now developing the semantic encoding of the combinations extracted by *Colex*. We believe that a filter based on measuring a combination's ability to be encoded as a lexical function would achieve higher precision than a purely statistical filter.

1. Introduction

This chapter presents a method for automatically extracting specialized lexical combinations from corpora based on lexical functions, the model developed for the representation of collocations within the framework of the Meaning-Text theory (Mel'čuk et al. 1995: 125–152; Mel'čuk 1996). Among the habitual expressions of a specialized language, we focused on verb + noun combinations: *[to] run a program*, *[to] open a file*, *[to] load into memory*, *[to] process data*, *[to] configure software*, etc. Except for a few bilingual dictionaries (Cohen 1986; Caignon 2000; Meynard 2000; Binon et al. 2000), these expressions are seldom represented in terminologies. The automatic extraction of collocations from texts offers many advantages for terminology and terminography, by encouraging the development of terminologies that take into account the syntactic behavior of terms.

Lexical functions allow for a systematic description of verbal collocations at the syntactic and semantic levels. At the syntactic level, they represent the three major grammatical relations: subject + verb, verb + direct object, verb + indirect object. At the semantic level, they describe the meanings of the most prominent verbal collocates of terms. The lexical function **Real**, for example, describes the prototypical use of an artefact: *[to] run a program*, *[to] edit a file*, *[to] fill a disk*. A more detailed description of **Real**, and of other lexical functions, is given in Section 3.

Lexical functions define collocations first as syntactic relations between lexical units. We planned to extract collocations from the analyses of sentences of texts from a specific domain. We analyzed a corpus of computer science texts of 600,000 word tokens with the Logos machine translation system.[1] We then defined rules that identify, in the analyses of the corpus sentences, the syntactic structures that are characteristic of English verb phrases and extract the verb and one of its syntactic dependents from each paired structure. We defined three types of rules for the three types of grammatical relations.

Lexical functions also express semantic relations between words that correspond to a small set of prototypical situations and are realized idiomatically. Because they are produced semi-automatically, they tend to appear regularly in sentences and lend themselves well to statistical analysis.

We therefore applied statistical methods to isolate collocations in the set of combinations extracted on syntactic grounds. Statistical tests measure the strength of association between two lexical units and help discriminate real collocations (strong association) from freely formed combinations (weak association). In order to compare the performance of the two most frequently used measures in automatic extraction of collocations (mutual information and log likelihood), we selected 397 collocations from the list of combinations extracted for the 10 most frequent terms in our corpus. The majority of the selected collocations illustrated prototypical uses of the 10 terms in the field of computer science. Evaluation of the statistical filter was then based on semantic criteria, the best filter being able to isolate the largest number of typical expressions in the list of syntactically-extracted combinations.

Before presenting, in Section 5, our collocation extraction system (*Colex*), we offer a formal description of collocations, with examples taken from the field of computer science (Section 2). In Section 3, we present lexical functions, the model for describing collocations on which we have based our methodology. Section 4 offers a summary of other programs developed for extracting collocations. In the final section, we consider the prospect of semantic encoding of the combinations extracted by *Colex*. We argue that a combination's ability to express one of the general meanings associated with lexical functions would represent a better test of its collocational nature than the statistical test.

2. Syntactic and semantic characteristics of special language collocations

Whereas a number of dictionaries document general language collocations (cf. for example the *BBI Combinatory Dictionary of English* (*BBI*) (Benson et al. 1997) or the

1. We wish to thank GlobalWare AG for allowing us to use the Logos system throughout the development of this project. We would also like to thank the anonymous reviewers for the numerous and valuable comments which have contributed greatly to the quality of this paper.

Collins COBUILD Advanced Learner's English Dictionary (2001) for English), very few such works exist for the set phrases of a specialized language, other than adjective + noun or noun + noun compounds. From the beginning, terminology has sought to describe the specific knowledge of a scientific or technical domain. In this approach to terminology, terminological descriptions must facilitate the comprehension of the notions designated by terms and the rational organization of specialized knowledge. Terminologies built along these principles are rich in encyclopedic information and place great emphasis on the description of the structuring relations between terms (hyponymy and hyperonymy). They rarely concern themselves with the description of terms' behavior in texts.

This classical approach to terminology has been challenged for some time by a number of researchers, terminologists and terminographers, who recommend an analysis of the linguistic, particularly semantic, properties of terms (Frawley 1988; Heid 1992; L'Homme 2007). Changes advocated by these researchers include formal descriptions of specialized meanings and of the collocations terms can enter into.

As with general language, specialized collocations describe regular associations between terms in a particular field. The following are examples of verbal collocations in the field of computer science:

(1) *[to] run a program*
[to] execute a command
[to] press a key
[to] store on a disk
The program runs
The command works

Together with terms, the combinations in the above examples represent the conventional means of expression of a specialized discourse, used by specialists to communicate. Like idiomatic expressions, to which they are related, collocations associate lexical units in a pre-established manner that must be learned. They are distinguished from completely idiomatic expressions by their semantics. Whereas the meaning of an idiomatic expression is largely incomprehensible when heard for the first time (cf. the expressions *[to] run the gauntlet* or *[to] run the gamut*), the meaning of a collocation is easier to deduce, although less directly than that of a free combination. The semantic transparency of a collocation is caused by the lexical unit that keeps its meaning inside the combination and is considered the base of the collocation (the noun in the above examples). The idiomatic meaning is expressed by the collocate, in the context of a particular base, and may be, for that reason, partially obscure (cf. the verb *[to] run* in the expression *[to] run a program)*.

The difference in interpretation between the two units that form a collocation reflects the difference in their selection. Whereas the base is chosen freely, entirely for its meaning, the collocate is chosen on the basis of the base to express a specific meaning: the verbs in the above examples all denote a prototypical operation of the nouns (objects) to which they are associated.

Restricted selection of the collocate word, more than opacity of its semantic content, truly defines a collocation. Restrictions imposed by the nouns on the verbs chosen to express a particular meaning are illustrated in the examples below. In each example, two semantically close nouns select different verbs to express the same idiomatic meaning (represented in single quotes after each example):

(2) *[to] start a program* vs. *[to] launch an application* ('cause to operate')

(3) *[to] issue a command* vs. *[to] give an instruction* ('cause to exist')

(4) *The command works* vs. *The instruction executes* ('operates')

Several researchers (Heid 1992; L'Homme 2000; L'Homme & Bertrand 2000) have pointed out the many regularities that exist between the base and the collocate in the field of terminology: nouns of a certain semantic type tend to select the same verbs. Still, the high number of idiosyncratic combinations – even among semantically related nouns – makes it necessary to describe specialized collocations in terminologies.

3. The lexical functions of the Meaning-Text theory

Lexical functions were developed within the framework of explanatory and combinatorial lexicology to model all the relations that obtain between the lexical units of a language and are called upon in the production of texts, along the paradigmatic and syntagmatic axes. They are functions, in the mathematical sense, that define, for each lexical unit, the set of lexical units that are semantically and syntactically related to it. Lexical functions are represented formally by the formula $\mathbf{f}(x) = y$ where x – the argument of **f** – is a lexical unit and y is the set of lexical units returned by the application of **f** to x.

Lexical functions describe very general relations between words, observed in all the languages that have been modeled within the framework of the Meaning-Text theory (see Mel'čuk 1996 for examples of lexical functions in different languages). They can be thought of as patterns of relations – realized in each context by individual lexical units – whose meanings can only be approximated: 'use (object) as intended' or 'operates as intended' in the case of the computer verbs in example [1] above.

In the case of syntagmatic functions, this general meaning, close to a semantic primitive, is associated with a specific syntactic relation. The lexical function **Real**, which describes the prototypical use of an object, also represents a verb + direct object relation:

Real$_1$(*key*)	=	*[to] press*
Real$_1$(*drive*)	=	*[to] operate*
Real$_1$(*disk*)	=	*[to] fill*

Lexical functions such as **Real** also encode the syntactic role of the semantic arguments of their nominal base (the participants in the situation denoted by the base, appearing

as variables in the base's lexicographic definition). Thus **Real**$_1$ describes a verb that takes the first semantic argument of the base as subject – the user of the artefact – and the base itself as direct object. The same meaning – of operating properly or realizing the purpose of an object – is expressed by the other two 'realization' functions, **Fact** and **Labreal**, which represent respectively a verb + subject and verb + indirect object relation:

Fact$_0$(*disk*)	=	*rotates*
Labreal$_{12}$(*disk*)	=	*[to] store* [N = Y *on* ~]

While '0' attached to **Fact** indicates an intransitive verb, '12' at the end of **Labreal** signifies a realization verb that takes the first argument of the base as subject, the second argument of the base as direct object and the base itself as indirect object.

In total, there are twenty-five functions to represent the meanings of the verbal collocates of terms. Lexical functions also exist to describe the three phases of a state or event (**Incep**, **Fin** and **Cont**) and the three types of causation: **Caus** ('cause'), **Liqu** ('liquidate') and **Perm** ('permit'). The phasal and causative functions are most often combined with other lexical functions to form complex functions. The process of combining lexical functions is very productive and further increases the descriptive power of the model:

CausReal$_2$(*disk*)	=	*[to] place* [N=Y *on* ~]
LiquReal$_2$(*disk*)	=	*[to]delete* [N=Y *from* ~]
PermReal$_2$(*disk*)	=	*[to] save* [N=Y *on* ~]

Causative and related functions (cf. the lexical function **Prepar** or 'prepare') figure prominently in the description of computer and other technical field collocations. Complex lexical functions that combine causative with other, more general, meanings account for the most of the verbal collocates of noun *file*:

CausFunc$_0$(*file*)	=	*[to] create*
LiquFunc$_0$(*file*)	=	*[to] delete, [to] erase*
PermFunc$_3$(*file*)	=	*[to] save* [~ *on* N=Z]
CausFunc$_3$(*file*)	=	*[to] copy* [~ *to* N=Z]
PreparReal$_{13}$(*file*)	=	*[to] load* [~ *on* N=Z]
PreparReal$_1$(*file*)	=	*[to] access*
CausIncepReal$_1$(*file*)	=	*[to] open*
LiquReal$_1$(*file*)	=	*[to] close*

This presentation only addresses the standard lexical functions (which express very general semantic relations). Collocations that describe more specific actions, involving a few terms only, will be described with another type of lexical function, the non-standard lexical function.

4. Related work on the automatic extraction of collocations

The first programs developed for the automatic extraction of collocations used only statistics to identify the habitual combinations of a language (Berry-Rogghe 1973; Choueka et al. 1983; Church & Hanks 1989). These programs worked on word-forms that appeared together in a small text window and implemented different measures to determine the strength of the association between two co-occurring words. Systems developed following these earlier programs showed their limitations: pairs extracted on a purely statistical basis were not all collocations, but described other types of associations: associations between a lexical unit and a grammatical word (article, preposition) or between two semantically related words (see the first examples in Church & Hanks (1989) of the pairs formed with the word-forms *doctors* and *nurses* or *doctors* and *dentists*).

To address these problems and increase the relevance of the extracted combinations, researchers have added linguistic methods to the statistical measures used to identify collocations. In one of the first programs that combine linguistic and statistical methods, Smadja (1993) used linguistic criteria to isolate the true collocations from the combinations acquired on strictly statistical grounds. Evaluation of the combinations extracted by the first step of the program, which relied on traditional statistical methods, showed that only 40% of the combinations were collocations.

More recent approaches apply linguistic processing first and base the extraction of collocations on linguistically enriched corpora. Lin (1998) was among the first to use a morphosyntactic analyzer to label syntactic relations in a corpus of newspaper articles. Labeled relations that yield interesting combinations are verb-object, subject-verb, adjective-noun and noun-noun relations. Similarly, Kilgariff & Tugwell (2001) extracted combinations of two (or three) lexical units from a previously annotated corpus, the British National Corpus (BNC). Combinations extracted from the BNC described all the relations that exist between the lexical units in the corpus, collocations and other types of relations (governed prepositions, etc.). Mutual information was then used to order the combinations. Finally, Goldman et al. (2001) used a deep syntactic parser to identify the dependency relations representative of collocations. The authors employed a statistical measure (log likelihood) to rank the results.

Although they base the collocation extraction on linguistic data – the syntactic relation between two lexical units – the systems mentioned above cannot provide a semantic interpretation of the extracted combinations. The combinations manifest the different constraints – semantic, lexical and syntactic – that govern the selection of lexical units in context and must always be analyzed. Also, the precision of the statistical filter used to isolate collocations in the list of syntactically-extracted combinations is never evaluated.

5. *Colex*: A hybrid system for the extraction of specialized collocations

5.1 Introduction

Colex, the collocation extraction system presented here, is a hybrid system that combines symbolic and statistical methods to automatically extract collocations from texts. Like the extraction systems just surveyed, our program is based on a linguistic analysis of collocations, more specifically on the lexical functions, the model elaborated within the Meaning-Text theory (MTT) to represent collocations. Following MTT, we consider a collocation first as a linguistic expression, the realization, at the syntactic level, of a specific lexical relation. We have therefore based the extraction of collocations on the analyses, by the Logos machine translation system,[2] of the sentences of a corpus of computer science texts of 600,000 word tokens.

We have defined rules that identify the syntactic structures characteristic of verbal collocations in the trees produced by the translation system's parser. The collocation extraction rules have been modeled on the syntactic parser rules, using the same symbolic language. They are organized in three different grammars, for each of the grammatical relations represented by the verbal lexical functions:

- Subject + verb: *The program executes*
- Verb + direct object: *[to] close a file*
- Verb + indirect object: *[to] load into memory*

When an extraction rule matches a sentence tree, the pair formed by the verb and one of its arguments (the pair corresponding to a particular grammatical relation) is extracted from the tree. The program also calculates the frequency of the extracted pairs (the number of times a particular combination has been extracted from the corpus).

We then apply a statistical test – using the frequencies previously obtained – to isolate the collocations in the list of combinations extracted by *Colex*. Statistical measures evaluate the strength of the association between the two members of a combination and can help to discriminate between collocations (strong association) and free combinations (weak association).

In the next section, we examine the first step in the extraction of collocations: the analysis, by the syntactic parser of the Logos system, of the sentences from which the verb + noun combinations will be extracted. Section 5.3 presents the extraction system itself. We describe the structure and the operation of the rules used to extract verbal combinations from the analyses produced by Logos and evaluate the results of the syntax-based extraction (before application of the statistical tests). Section 5.4 deals with the statistical filter implemented to eliminate free combinations from the list of

2. The Logos system is now available as an Open Source license model through a collaboration between GlobalWare AG and the German Research Center for Artificial Intelligence (Deutsche Forschungszentrum für Künstliche Intelligenz (DFKI) GmbH). More information on OpenLogos Machine Translation can be found at http://logos-os.dfki.de/.

extracted combinations. We describe the program developed to calculate the score of a combination according to either of the two statistical measures considered (mutual information and log likelihood) and evaluate the performance of these two measures in filtering out unwanted combinations.

5.2 Corpus analysis

During the first step in the extraction of collocations, the corpus analysis produces the representations from which the verb + noun combinations of the computer field will be extracted: these consist of trees representing the syntactic structure of the corpus sentences. The trees are produced by the syntactic parser of the Logos machine translation system.

The Logos syntactic parser is a deep syntactic parser that incorporates semantic and syntactic analyses of sentences through SAL (Semantico-syntactic Abstraction Language), the structured set of semantico-syntactic categories developed by the creators of the system to represent the lexical units of a language (Scott 2003). The ontological and structural foundation of the Logos system, SAL, is an abstract language that encodes, for each lexical unit stored in the dictionary, three types of information: the first type is the lexical unit's part of speech, which determines also its semantico-syntactic and morphosyntactic characteristics.

Analysis of a sentence by Logos is done incrementally, through four modules corresponding to four analysis levels (TRAN1-4). The modules work together to build the syntactic tree, starting with the low-level constituents, and moving up to the sentence's governing nodes (nouns, verbs, sentence boundaries, etc.). In each of the first three modules, syntactic dependents identified by the module are concatenated with their governor (they do not appear in the tree produced by that module). The final

If files are simply copied to a disk, they may fill the disk until there is no more room for the next file ...

[Punc	if	sub
]		
[NP	file	pl
]		
[Aux	are	
]		
[V	copied	pass
]		
[NP	disk	to
]		
[Punc	COMMA	comma
]		

Figure 1. Syntactic tree produced at the end of the analysis

tree is produced by the last analysis module (TRAN3). It consists of the list of the sentence's governing nodes in the form of SAL elements. We give an example of the final tree produced for the first clause of a sentence from our corpus in Figure 1.

As illustrated in Figure 1, the final representation of a sentence analysis only keeps the main constituents of the sentence: the verb and its syntactic dependents.

Trees produced by the Logos parser at the end of the analysis phase represent the surface syntactic structure of English sentences (the verb in the tree in Figure 1 is in the passive). The extraction of verb + noun combinations presented below is based on these trees. Extraction rules developed to identify the syntactic structures representative of verbal collocations will therefore rely on the Logos schema for representing lexical units, and particularly on the morphosyntactic characteristics of the SAL elements in the trees produced at the end of analysis.

5.3 Extraction of verb + noun combinations

In order to extract the verbal collocations of the computer field, *Colex* uses rules that represent the surface syntactic structures of English verb phrases and that specify, for each structure, the verb + noun combination to be extracted from the paired subtree. *Colex* rules are specialized for the extraction of the three types of combinations that correspond to verbal collocations: combinations made of the verb and its subject, direct object and indirect object respectively. They consist of two parts: a condition part or morphosyntactic pattern to be paired with a syntactic tree and an action part that specifies the combination to be extracted from the tree. The rule action also calculates the frequency (the number of occurrences) of the extracted combination, this information being used in the statistical test that follows.

The morphosyntactic pattern in the rule condition is defined using elements that are similar to the SAL elements produced by Logos at the end of analysis. It can be thought as a matrix of values to unite to each input tree. While the x axis gives the ordered sequence of elements to be matched with the tree's governing nodes, the y axis lists the values associated with each element in the sequence (corresponding to the values of the SAL elements). Pairing of an extraction rule to the tree in Figure 1 is illustrated in Figure 2 (the rule is for a passive structure).

As illustrated in Figure 2, when the elements in the rule are matched with the nodes in the candidate tree, the verb + noun combination that the rule specifies is extracted. Here the program extracted a verb + direct object combination (when extracting the combination, the rule restores the canonical order of the verb's complements).

Each of the three grammars used to extract verb + noun combinations from the trees produced by Logos is made of a few dozen rules: 112 rules extract verb + direct object relations, 34 rules extract subject + verb relations, and 65 rules extract verb + indirect object relations. The relatively high number of rules in each grammar is needed to specify the various morphosyntactic features of the verb: active, passive, imperative, etc.

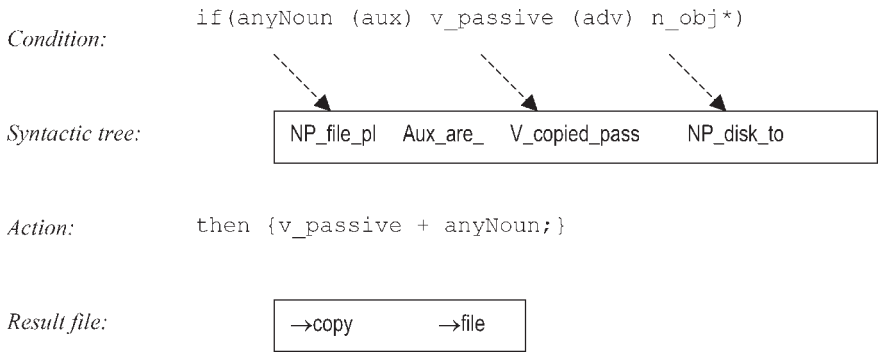


Figure 2. Extraction of the verb combination *copy + file*

Applying the three grammars of *Colex*, we were able to extract a little over 40,000 verb combinations from the corpus of computer science texts. We then evaluated a representative sample of these combinations and measured the precision of the extraction system (the number of correct combinations). First, we considered only the syntactic properties of the combinations extracted by *Colex* and retained the combinations that illustrated one of the three major grammatical relations.

When measured with this criterion only, the precision of *Colex* is 85.15%. The precision of the extraction system is much lower when one considers also the meaning of the retained combinations. If 85% of the combinations extracted by *Colex* represent dependency relations between the verb and its arguments, only a little more than half (57.03%) are true collocations. Along with verbs which describe prototypical uses of terms (*[to] boot a computer*, *[to] install memory*, *[to] configure software*), we find a similar number of verbs with which they combine freely: *[to] buy a computer*, *[to] ship memory*, *[to] describe software*.

These findings demonstrate the limitations of an extraction method based solely on morphosyntactic criteria. They motivated the development and implementation of the statistical filter presented in the next section.

5.4 Statistical processing

The statistical processing presented below aims at isolating true collocations from the combinations extracted by *Colex*. Statistical analysis of a text – or of a body of texts – reveals the usual associations between lexical units, which manifest the constraints at play in collocation formation. One of the first tests used to identify the common phrases of a language is the frequency test. In her research on the automatic extraction of terms, Daille (1994) ranked frequency as the second most accurate test in identifying terminological phrases in a list of candidate terms (log likelihood was the best performing measure). Similarly, Krenn & Evert (2001) used the frequency curve of the collocations found in a newspaper corpus to evaluate the precision of various statisti-

Table 1. List of the 20 most frequent combinations for *file*

f:	46	be	file
f:	21	save	file
f:	21	copy	file
f:	20	open	file
f:	18	download	file
f:	17	create	file
f:	16	use	file
f:	13	send	file
f:	12	find	file
f:	12	include	file
f:	12	back up	file
f:	11	see	file
f:	10	move	file
f:	10	store	file
f:	9	locate	file
f:	9	parse	file
f:	9	read	file
f:	9	delete	file
f:	8	get	file
f:	8	access	file

cal measures. The authors concluded that none of the measures – save for a specifically designed test – was better than simple frequency at identifying collocations.[3]

We also tried to use frequency as a filter for the combinations extracted by *Colex*. Table 1 lists the twenty most frequent combinations for *file*.

A first examination of the most frequent combinations extracted for *file* reveals that many of them represent actions commonly associated with that term (*[to] save*, *[to] copy*, *[to] open*, *[to] create*, etc.). Among the twenty most frequent combinations, one also finds atypical verbs such as *[to] be* (the most frequent verb) or *[to] use*, *[to] include*, *[to] see*. These results show that frequency alone cannot successfully discriminate collocations from other combinations extracted by *Colex*. We also note that several combinations have the same frequency.

Collocations are not only recurrent combinations; they manifest the lexical affinities within the vocabulary of a science or technology that found specialized discourse. Because they appear frequently in our corpus, *[to] be* and *file* will also combine frequently, while the combination may not indicate a particular affinity between the two words.

We therefore chose to apply two statistical measures to the problem of identifying collocations in the combinations extracted by *Colex* and to compare their performance in order to determine which measure would make the better filter. The two measures

3. Evert (2005) explores the possible correlation between different association measures and linguistically-defined subtypes of lexical collocations.

Table 2. Comparison of the highest ranking verbs for *file*

	Logl		MI
copy	20,29	tune	5,75
download	13,23	**upload**	5,09
open	12,89	parse	4,92
save	12,32	**back up**	4,33
back up	11,78	**copy**	4,28
parse	10,89	*designate*	4,16
upload	9,03	*double-click on*	4,16
delete	5,39	rename	4,16
name	4,84	exchange	4,01
locate	4,81	overwrite	3,94
move	4,1	write to	3,75
tune	4,08	**erase**	3,63
attach	3,8	**download**	3,59
have	3,8	*swap*	3,58
double-click on	3,67	**name**	3,47
update	3,39	**open**	3,30
modify	2,99	**delete**	3,16
highlight	2,79	**modify**	3,16
rename	2,75	**save**	3,11

evaluated – mutual information and log likelihood – figure prominently in a number of works which, like our own, base the extraction of collocations on morphosyntactic criteria. Both measures evaluate the strength of the association between two lexical units by comparing their joint frequency (their frequency together or frequency of co-occurrence) with their marginal frequencies (the frequency of occurrence of each independently of the other). The ratio of the two frequencies gives a score which increases with the strength of the association.

We developed a program which computes the association score of each combination extracted by *Colex* according to one or the other of the two statistical measures and use the score to rank the combinations, from the combination with the highest score (or the strongest association) to the combination with the lowest. We present in Table 2 the first twenty collocates of *file* according to the log likelihood ratio (*Logl*) and mutual information (*MI*). Collocates that are the values of standard lexical functions are in bold whereas freely combined verbs are in italics; non standard collocates do not have special characters.

As illustrated in Table 2, the log likelihood ratio offers the better filter for the combinations extracted by *Colex* by ranking higher the greatest number of prototypical collocates of *file*. These results support results in other studies (Daille 1994) which also note the better correlation of log likelihood with frequency. Mutual information does not perform as well because of the tendency to overestimate rare combinations: the highest ranking pair in Table 2 is also the least frequent one.

6. Conclusion

Statistical measures have enabled us to improve significantly the precision of the automatic extraction of verb collocations from computer texts: with the log likelihood ratio, 71% of the combinations extracted by *Colex* represent true collocations. These results were computed using a list of manually vetted collocations which represented, for the most part, standard lexical functions. The semantic criteria used to validate the ranking of combinations by log likelihood explain in part the average performance of the measure. Log likelihood also ignored a number of specific expressions, which did not appear regularly in the corpus: *[to] format file*, *[to] capture data*, *[to] translate data*, *[to] look for information*, *[to] return information*, etc.

Filtering out free combinations remains an issue in systems built for acquiring collocational data automatically. This problem finds an original solution in the work of Wanner (2004) and Wanner et al. (2005) on the automatic encoding of Spanish verb combinations. Using machine learning techniques, these researchers modeled the meaning of several lexical functions, using collections of sample collocations for each of the modeled meaning. They then use the model to automatically encode new combinations. This method also allowed them to eliminate free combinations. In a future project, we plan to apply similar techniques to the combinations extracted for English by *Colex* and use semantic encoding – the ability of a combination to express one of the general meanings associated with lexical functions – to filter out uncharacteristic combinations and replace the statistical test.

References

Benson, M., E. Benson & R. Ilson (1997). *The BBI Dictionary of English Word Combinations*. Amsterdam: John Benjamins.

Berry-Rogghe, G. (1973). The computation of collocations and their relevance in lexical studies. In Aitken, A. J., R. W. Bailey & N. Hamilton-Smith (eds.) *The Computer and Literary Studies*, 103–112. Edinburgh: Edinburgh University Press.

Binon, J., S. Verlinde, J. Van Dyck & A. Bertels (2000). *Dictionnaire d'apprentissage du français des affaires*. Paris: Didier.

Caignon, P. (2000). *Essential Lexicon in Accounting*. Saint-Laurent, Quebec: Fides.

Choueka, Y., S. T. Klein & E. Neuwitz (1983). Automatic retrieval of frequent idiomatic and collocational expressions in a large corpus. *Journal of the Association for Literary and Linguistic Computing* 4: 34–38.

Church, K. W. & P. Hanks (1989). Word association norms, mutual information, and lexicography. In *Proceedings of the 27th Annual Meeting of the Association for Computational Linguistics*, 76–83. Morristown, NJ: ACLs.

Cohen, B. (1986). *Lexique de cooccurrents Bourse Conjoncture économique*. Montréal: Linguatech.

Collins COBUILD Advanced Learner's English Dictionary. (3d edn, 2001). London: HarperCollins.

Daille, B. (1994). Approche mixte pour l'extraction de terminologie: Statistique lexicale et filtres linguistiques. Thèse de doctorat en informatique fondamentale, Université Paris 7.

Evert, S. (2005). The Statistics of Word Cooccurrences – Word Pairs and Collocations. Stuttgart: University of Stuttgart, IMS. Available from http://www.collocations.de/phd.html.

Frawley, W. (1988). New forms of specialized dictionaries. *International Journal of Lexicography* 1(3): 189–213.

Goldman, J. P., L. Nerima & E. Wehrli (2001). Collocation extraction using a syntactic parser. In *Proceedings of COLLOCATION: Computational Extraction, Analysis and Exploitation, ACL-EACL 2001*, 61–66. Toulouse.

Heid, U. (1992). Décrire les collocations. *Terminologie et Traduction* 2–3: 523–548.

Kilgarriff, A. & D. Tugwell (2001). Word sketch: Extraction, combination and display of significant collocations for lexicography. In *Proceedings of COLLOCATION: Computational Extraction, Analysis and Exploitation, ACL-EACL 2001*, 32–38. Toulouse.

Krenn, B. & S. Evert (2001). Can we do better than frequency? A case study on extracting PP-verb collocations. In *Proceedings of COLLOCATION: Computational Extraction, Analysis and Exploitation, ACL-EACL 2001*, 39–46. Toulouse.

L'Homme, M. C. (2000). Understanding specialized lexical combinations. *Terminology* 6(1): 86–110.

L'Homme, M. C. (2007). Using explanatory and combinatorial terminology to describe terms. In Wanner, L. (ed.) *Selected Lexical and Grammatical Topics in the Meaning-Text Theory. In Honour of Igor Mel'cuk*. Amsterdam: John Benjamins.

L'Homme, M. C. & C. Bertrand (2000). Specialized lexical combinations: Should they be described as collocations or in terms of selectional restrictions? In Evert, S., U. Heid, E. Lehmann & C. Rohrer (eds.) *Proceedings of the Ninth EURALEX International Congress, EURALEX 2000*, 497–506. Stuttgart: EURALEX

Lin, D. (1998). Extracting collocations from text corpora. In *Proceedings. of the First Workshop on Computational Terminology, COLING-ACL'98*, 57–63.

Mel'čuk, I. (1996). Lexical functions: A tool for the description of lexical relations in a lexicon. In Wanner, L. (ed.) *Lexical Functions in Lexicography and Natural Language Processing*, 37–102. Amsterdam: John Benjamins.

Mel'čuk, I., A. Clas & A. Polguère (1995). *Introduction à la lexicologie explicative et combinatoire*. Louvain-la-Neuve: Duculot.

Meynard, I. (2000). *Internet : Répertoire bilingue de combinaisons lexicales spécialisées*. Brossard, Québec: Linguatech.

Scott, B. (2003). The Logos model: An historical perspective. *Machine Translation* 18(1): 1–72.

Smadja, F. (1993). Retrieving collocations from text: Xtract. *Computational Linguistics* 19(1): 143–177.

Wanner, L. (2004). Towards automatic fine-grained semantic classification of verb-noun collocations. *Natural Language Engineering* 10(2): 95–143.

Wanner, L., B. Bohnet, M. Giereth & V. Vidal (2005). The first steps towards the automatic compilation of specialized collocation dictionaries. *Terminology* 11(1): 143–180.

Combined statistical and grammatical criteria for the retrieval of phraseological units in an electronic corpus

José-Manuel Pazos Bretaña and Antonio Pamies Bertrán

The aim of this study is to refine and optimise the mainly statistical and distributional approach to the automatic extraction of phraseological units (PUs) from text corpora, by introducing minimal linguistic elements (lemmatisation and grammatical tagging). These operations were first tested using the same corpora as in our previous research (Pamies & Pazos 2003 & 2004). This provided us with a new set of results, which we compared with the previous ones. We found that the detection ability had improved substantially, especially when dealing with verb + noun and verb + adjective collocations. This methodology was then applied to a larger corpus. Again, the results were encouraging, with phraseological densities up to 64.5% for the verb + noun category.

1. Introduction

Modern lexicology calls 'phraseological units' a wide and heterogeneous set of word combinations which coincide more or less with the 'fixed expressions' as described by Moon (1998). This broad category subsumes idioms (*between the Devil and the deep blue sea*), collocations (*to pay attention; to fall in love*), stereotyped similes (*as light as a feather*), proverbs (*barking dogs never bite*), and routine formulae (*nice to meet you*). All these word combinations share the formal features of lexical and grammatical fixedness, which is described as a restriction of the combinatory and morphosyntactic properties of the words when they are a part of such combinations. Many, though not necessarily all, such phrases also share the semantic property of idiomaticity, which implies that the global meaning of the whole combination cannot be deduced from the meaning of its components (e.g. *to kick the bucket*, *to mess around)*.

Halliday (1966: 158) defined collocations as "the syntagmatic association of lexical items, quantifiable, textually, as the probability that they will occur at n removes (a distance of n lexical items) from an item x, the items a, b, c...". This notion is based only on distributional and statistical factors and harbours a more formal conception than Firth's original 'discovery' of collocations. This idea, which could also be used for the other kinds of phraseological units, was taken over by other researchers who

put these theoretical postulates into practice using computational tools and statistical processing techniques. In the 1960s, J. M. Sinclair applied statistical calculations to the study of linear co-occurrence of words in large corpora. Jones & Sinclair (1974) defined significant collocations as those composed of two lexical items whose frequency of co-occurrence is higher than expected from the raw frequencies of the constituents (node and collocate) in the text. In a later study Sinclair (1991: 115) introduced the concept of upward and downward collocations, refining the earlier general statement by taking into account the relative relevance of the collocation for each constituent: "[w]hen two words of different frequencies collocate significantly, the collocation has a different value in the description of each of the two words. If word a is twice as frequent as word b, then each time they occur together is twice as important for b as it is for a".

These broad conceptions of collocation hint at the possibility of automatic retrieval of collocations in electronic text corpora using only statistics. In order to perform such an analysis it is necessary to establish which statistical measure is best suited to identifying relevant combinations. Studies of this type include those by Berry-Roghe (1973), Church et al. (1989, 1991) Clear (1993), Dunning (1993), Stubbs (1995), Oakes (1998) and Manning & Schütze (1999). In theory, this kind of approach can be carried out on any raw corpus (Aguilar 1994; Kraif 1997; Pamies et al. 1998). That is, pre-processing of the corpus can be limited to changes in the file format imposed by the computational tools.

Our intent is to evaluate the strengths and weaknesses of this kind of purely statistical approach and test the influence of different methodologies on the results. Eventually this could lead to a critical revision of the initial hypothesis (Pamies & Pazos 2004c: 191–203). The implementation of the method is described in Section 2. In Section 3 we discuss the results of a combined quantitative and qualitative methodology. In Section 4 we conclude and offer some avenues for future research.

2. Extraction of phraseological units from raw corpora

After contrasting different statistical measures the results showed that statistical criteria (in particular, log-likelihood) are effective by themselves because their implementation makes it possible to extract numerous phraseological units (PUs) from a raw corpus without prior linguistic processing. In our first experiments (Pamies & Pazos 2003, 2004a, 2004b) we used a small corpus since we had to check the efficiency of the processing and retrieval manually. This means that we printed out and read the listings that contained all the bigrams ordered by the values of the statistical measures so as to identify and count the number of phraseological combinations. We adhered to the broad definition of PU given above and made no distinction between the different classes (collocations, idioms, stereotyped similes, proverbs and formulae) within the phraseological paradigm. The pre-processing was limited to the removal of all lexical units representing grammatical morphemes (determiners, conjunctions, prepositions, personal pronouns, auxiliaries, etc.), since combinations like Art + N

[e.g. *el* + *perro* (the + dog) etc.] tend to appear in head positions in the listings due to the high co-occurrence frequency of clitic and autonomous words. In a tagged corpus this operation would not be necessary; in an untagged one, the easiest way to remove this kind of morphemes is by means of a stop list, i.e. a filter word list that contains all articles, all possessive, demonstrative, personal, relative and interrogative pronouns, and all conjunctions.

Three figures suffice for a purely statistical approach: total frequency (the number of occurrences of a particular word in the corpus), the frequency of co-occurrence of two words within a certain span, and the expected frequency (or random co-occurrence). The simplest distributional model is the normal or random distribution, which implies that words will occur randomly around others in the same proportions throughout the text. Statistical measures like z-score and t-score are based on these considerations. However, Dunning (1993) argued that the idea of a normal distribution was not realistic given the considerable variations observed in corpora of different sizes and types. He proposed a different measure, log-likelihood, which is not affected by the size of a corpus and does not rely on normal distributions.

We tested the efficiency of these three measures, i.e. z-score, t-score and log-likelihood, in a relatively small corpus (Cervantes' *Don Quixote* in raw format). The software used in this experiment was TACT[1] and the SPSS[2] package. Our initial hypothesis was that, once the calculations had been performed and the obtained listings arranged in decreasing order according to the value of each measure, the PUs would be concentrated in the upper areas of the lists, and could thus be distinguished from random combinations. However, the results showed that the upper areas were also populated with non-phraseological recurrent combinations. Some of these random combinations (e.g. *ama* + *sobrina* (housekeeper + niece) were even more frequent[3] than PUs in these areas. The data we analysed were:

a. the number of recurrent bigrams (BG) and the number of phraseological units (PU);
b. what we call 'phraseological density', which we define as the ratio of phraseological units to bigrams in a certain area [henceforth denoted %PU/BG];
c. what we call the 'phraseological volume', defined as the ratio of PUs in a certain area to PUs in the whole corpus [%PU/TOT]; and
d. the 'co-occurrence volume', defined as the ratio of bigrams in an area to the total number of recurrent bigrams in the corpus [%BG /TOT].[4]

1. Bradley, J. et al. (1996) TACT 2.1 http://www.chass.utoronto.ca/tact/

2. Statistical Package for the Social Sciences, SPSS Inc. SPSS v. 11 (2001)

3. We extracted a total of 10,716 recurrent bigrams of which only 723 were PUs (6.75%). Checks were done manually.

4. Distinct PU *types* (the number of *tokens* is much higher).

z-score	PU	BG	%PU /BG	%PU /TOT	%BG /TOT
>200	14	69	20,29	1,98	0,64
>150	21	83	25,30	2,97	0,77
>100	27	199	13,57	3,82	1,86
>50	81	534	15,17	11,46	4,98
>25	154	1184	13,01	21,78	11,05
>10	229	2614	8,76	32,39	24,39
>9	29	365	7,95	4,10	3,41
>8	40	423	9,46	5,66	3,95
>7	33	481	6,86	4,67	4,49
>6	29	552	5,25	4,10	5,15
>5	18	671	2,68	2,55	6,26
>4	10	689	1,45	1,41	6,43
>3	14	826	1,69	1,98	7,71
>2	4	826	0,48	0,57	7,71
>1	3	656	0,46	0,42	6,12
>0	1	404	0,25	0,14	3,77
<0	0	140	0,00	0,00	1,31

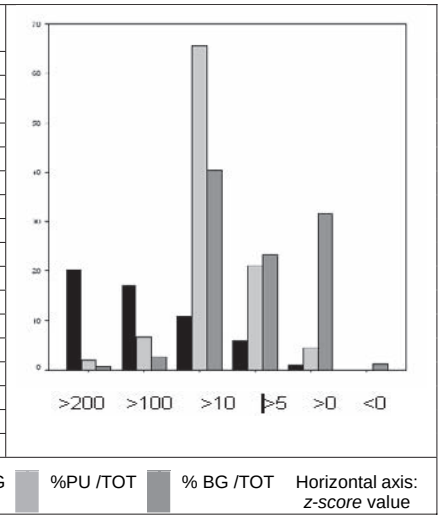


Figure 1. Z-score (Corpus *Don Quixote* in raw format)

Our comparison of the different statistical measures showed that the best results were achieved by the log-likelihood measure (as proposed by Dunning 1993). This measure performed far better than z-score and t-score in discriminating PUs from other types of bigrams, since the phraseological density is much higher at the top of the output list for log-likelihood. Figures 1, 2 and 3 illustrate the results obtained using these three statistical measures.[5]

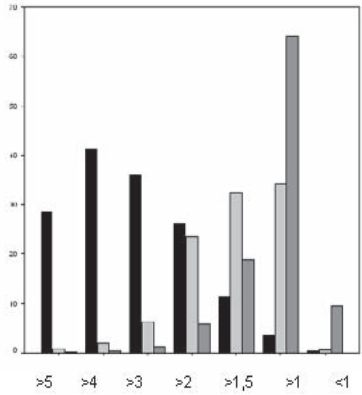
The items in the listings (see Appendix for a sample of the listings) were segmented according to the absolute value of the statistical measures (e.g. log-likelihood >50, >20 etc.). However, in order to be able to compare the results from corpora of different sizes, we had to resort to relative values when defining the areas of analysis. Once each segment contained an equal number of bigrams (for example 10% of the list), the listings were ordered according to the relative values of the statistical measures. Figure 4 shows the results of this re-segmentation for *Don Quixote*. It is clear that a higher value of the statistical measure is, in general, associated with a higher phraseological density (especially in the upper segments). As appears from Figure 2, log-likelihood performs better than the other measures as 302 PUs (viz. 42,7% of the total) appear in the first interval of the listing (phraseological density of 28%).

As the highest density is only 28%, it is clear, however, that this purely statistical approach leaves much to be desired. It proves to lead to the extraction of a great number of recurrent non-phraseological bigrams that also comply with the imposed contour conditions. Among the bigrams that fall into the 'noise' category were

5. Note also that *t-score* performs best in areas of low phraseological density, thus reducing its practical utility.

many unexpected combinations such as frequent situational associations and rhetorical reduplications.

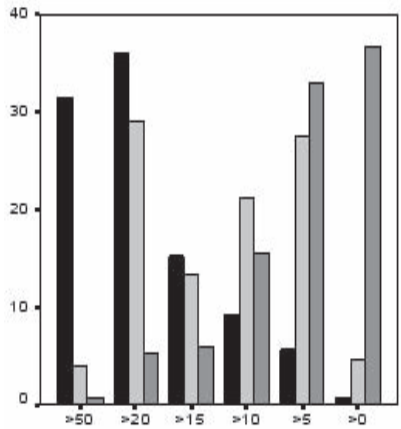
t-score	PU	BG	%PU /BG	%PU /TOT	%BG /TOT
>5	6	21	28,57	0,85	0,20
>4	14	34	41,18	1,98	0,32
>3	45	125	36,00	6,36	1,17
>2	167	638	26,18	23,62	5,95
>1,5	229	2019	11,34	32,39	18,84
>1	242	6862	3,53	34,23	64,04
<1	4	1017	0,39	0,57	9,49



%PU /BG %PU /TOT % BG /TOT Horizontal axis: *t-score* value

Figure 2. T-score (Corpus *Don Quixote* in raw format)

log likelihood	PU	BG	%PU /BG	%PU /TOT	%BG /TOT
>100	6	28	21,43	0,83	0,26
>50	23	64	35,94	3,18	0,60
>40	29	66	43,94	4,01	0,62
>30	38	122	31,15	5,26	1,14
>20	134	394	36,29	19,78	3,68
>15	95	637	15,23	13,42	5,94
>10	151	1668	9,17	21,16	15,57
>9	23	520	4,42	3,18	4,85
>8	33	689	4,79	4,56	6,43
>7	38	792	4,80	5,26	7,39
>6	62	831	7,82	8,99	7,75
>5	41	964	5,91	5,67	6,48
>4	16	1002	1,60	2,21	9,35
>3	8	977	0,82	1,11	9,12
>2	7	887	0,79	0,97	8,28
>1	2	634	0,32	0,28	5,92
>0	1	441	0,23	0,14	4,12



%PU /BG %PU /TOT % BG /TOT Horizontal axis: *log-likelihood* value

Figure 3. Log-likelihood (Corpus *Don Quixote* in raw format)

These results point to the need for a critical revision of the very definition of a collocation based solely on statistics. The source of the undesired side effects is the broad nature of the contour conditions: they are necessary but not sufficient to discriminate

collocations (and PUs in general) from other non-phraseological combinations that are due to extralinguistic factors. Such non-phraseological combinations include:

a. proper nouns and toponyms: e.g. *Alejandro + Magno, Julio + César, Amadis + Gaula, Miguel + Cervantes, San + Pedro*; *Vélez + Málaga;*
b. recurrent contextual combinations specific to the chosen corpus: e.g. *ingenioso + hidalgo* (inventive + nobleman)*; barbero + cura* (barber + priest); *gobierno + ínsula* (government + island)*; bacia + barbero* (bowl + barber);
c. conceptual free combinations due to ontological links between the components: e.g. *come + bebe* (eat + drink); *hambre + sed* (hunger + thirst); *silla + sentado* (chair + sitting); *leer + libros* (to read + books); *mano + derecha* (right + hand); *lado + izquierdo* (left + side); *rayos + sol* (rays + sun)*; pan + queso* (bread + cheese); *jarro + agua* (jug + water); *tronco + árbol* (trunk + tree); *hojas + árboles* (leaves + trees);
d. rhetorical reduplications, for:
    a. emphasis: *Ladrones, ladrones!* (thieves, thieves!); *Cásese, cásese* (Marry, marry); *...refranes y más refranes* (proverbs and more proverbs);
    b. paralellism: *de mesón en mesón y de venta en venta* (from hostel to hostel and inn to inn);
    c. antithesis: *porque yo me case o no me case* (for me to marry or not to marry);
    d. pleonasm: *Carlomagno es Carlomagno* (Charlemagne is Charlemagne);
    e. anadiplosis: *no todos los caballeros pueden ser cortesanos ni todos los cortesanos pueden ni deben ser caballeros andantes* (not all knights can be courtiers and not all courtiers can be knights);
    f. puns: *no osa la voz entrar por tan estrecho estrecho*.[6]

BG	PU		
	log-likelihood - (%PU/BG)	*t-score*	*z-score*
0 - 10%	302 (28%)	288	164
10 - 20%	115 (11%)	128	139
20 - 30%	73 (7%)	71	115
30 - 40%	47 (4%)	65	81
40 - 50%	58 (5%)	41	87
50 - 60%	72 (7%)	57	69
60 - 70%	19 (2%)	37	27
70 - 80%	9 (1%)	7	15
80 - 90%	9 (1%)	9	6
90 - 100%	3 (0,3%)	4	4

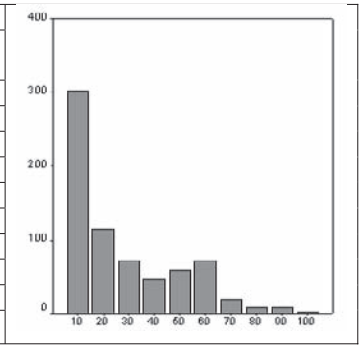


Figure 4. Phraseological density for log-likelihood (Corpus *Don Quixote* in raw format resegmented)

6. Untranslatable non-phraseological pun: *estrecho* means 'narrow' but also 'strait'.

3. Extraction of phraseological units from tagged corpora

Having reached the limits of strictly distributional methods, we decided to evaluate the efficiency of a combined quantitative and qualitative approach including a non-statistical operation, i.e. lemmatisation. However, this did not result in an improvement of the results when compared to the purely statistical approach. Noise increased to almost exactly the same extent as the relevant combinations extracted. We analysed the short novel *La vida de Pascual Duarte* (C. J. Cela). The results showed that the phraseological density in the first interval was 47% using a lemmatised text compared to 49% using the raw text. This was a disappointing result, even taking into account the fact that lemmatisation doubled the number of PUs extracted. Random recurrent bigrams benefited from lemmatisation to the same extent as relevant combinations. For example, random combinations such as *sangrar* + *pensar* (bleed + think), *orar* + *clavo* (pray + nail) and thematical combinations such as *fumar* + *pitillo* (smoke + cigarette), *encender* + *luz* (turn on + light) were as frequent as phraseological combinations in the first interval.

Since lemmatisation by itself did not significantly improve the efficiency of the method, we tested a more balanced combined approach similar to that used by other researchers for analysing German and English (Smadja 1993; Evert & Krenn 2001; Krenn & Evert 2001; Krenn 2000a, 2000b; Zinsmeister & Heid 2002; Evert & Kermes 2003a, 2003b; Evert et al. 2004; Granger et al. 2006). Data based on frequency was combined with the linguistic operations of lemmatisation, grammatical filtering and corpus part-of-speech (POS)-tagging. In order to examine the consequences of this shift in methodology we used as tertium comparationis the same corpus as in our previous experiments.

The computational tools for this experiment were *Freeling 1.2: Open Source Suite of Language Analyzers* (Carreras et al. 2004)[7] and *NSP 0.71: Ngram Statistics Package* (Banerjee & Pedersen 2003).[8] This software enabled us to overcome some of the difficulties we had encountered. The first stage of the procedure (lemmatisation) allowed the corpus to be filtered according to morphosyntactic criteria (i.e. the grammatical category of the components) with a view to reducing the noise level by restricting the search to some grammatical combinations that are more likely to be collocational such as V+N (*prestar atención,* pay attention) and Adj+N (*error garrafal,* terrible mistake), etc. The removal of undesired elements and consideration of only these (potentially PU richer) targeted categories should result in an increase in precision. For example, the noise due to rhetorical repetitions and proper names (whose word components have the same grammatical category: N+N, V+V, etc.) is eliminated by such filters.

In a second phase, we subjected the filtered corpus to statistical treatment (using *NSP 0.71*) and compared the results with those obtained in the previous analyses.

7. http://garraf.epsevg.upc.es/freeling/

8. http://www.d.umn.edu/~tpederse/nsp.html

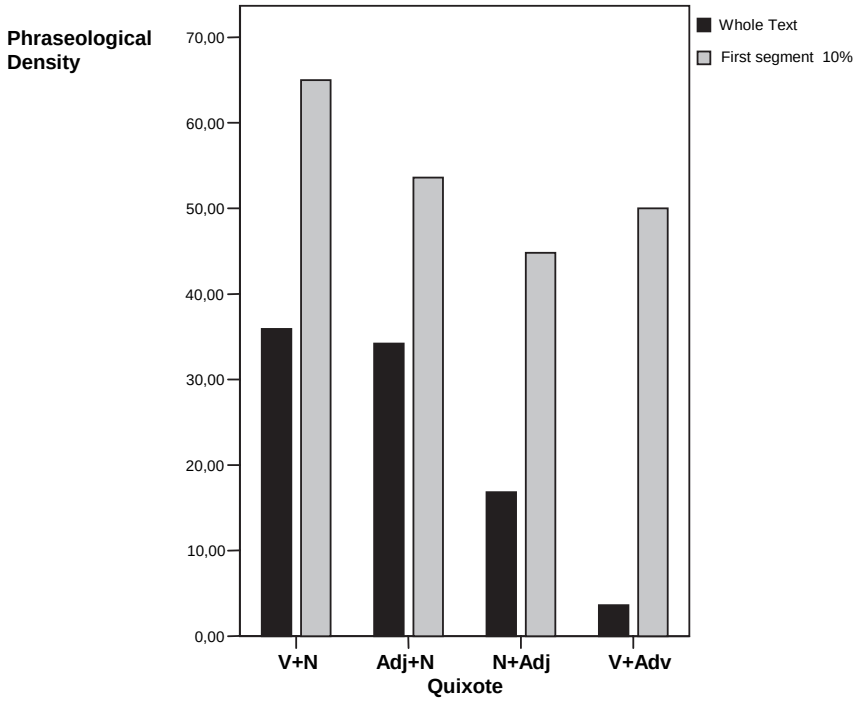


Figure 5. Phraseological density of POS bigrams
(Corpus *Don Quixote* lemmatized and tagged)

We used log-likelihood as the only statistical measure for the calculations, since it had provided the best overall results. The elements we analysed were composed of items belonging to the following categories: V+N, N+Adj., Adj+N, and V+Adv (Figure 5). The results for these major groups (although they do not belong to the most populated) were better than those obtained with purely statistical methods, with the exception of V+Adv. For example, *Don Quixote* contains 713 recurrent V+N bigrams, 72 of which are in the initial segment; of these, 42 are PUs (phraseological density of 65%) [e.g. *tener licencia* *have[9] permit: to be allowed]. This value of the phraseological density is far better than the 28% obtained in the same interval using purely statistical approaches. The disappointing results in the V+Adv category (phraseological density of 3,6%) might be due to the fact that recurrent combinations containing a component with a high raw frequency (e.g. *bien,* well; *mal,* badly; *mucho,* a lot; *poco*, a little; *más,* more; *menos,* less) rank poorly using the log-likelihood measure. Adverbs are a reduced paradigm and therefore prone to repetition (*conocer* + *bien*, know + well, *ver* + *nunca,* see + never). For the Adj+N category we counted 561 bigrams, of which 192

9. The asterisk indicates a literal translation.

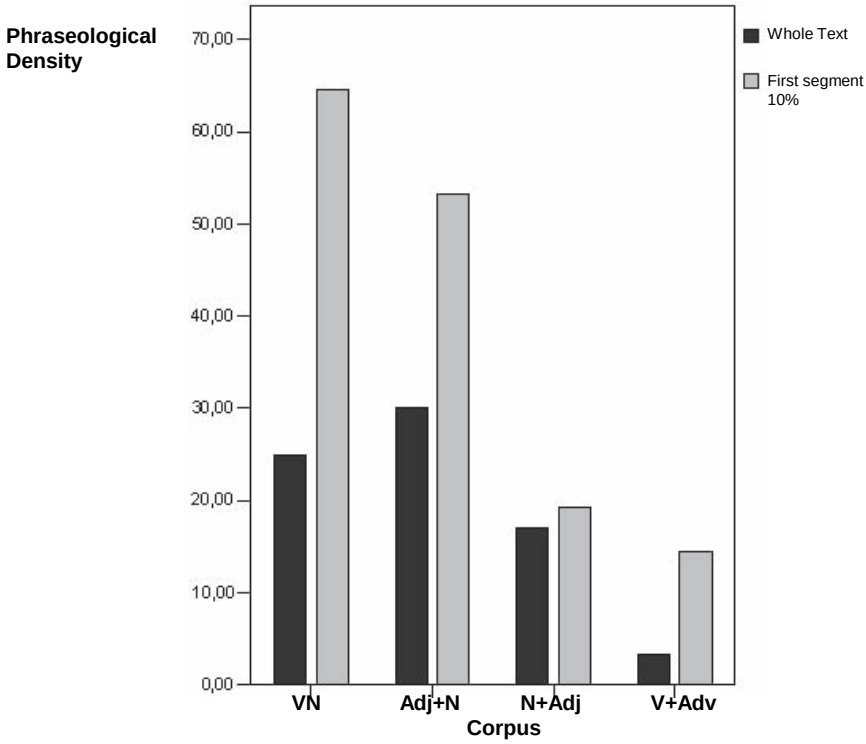


Figure 6. Recurrent bigrams by grammatical categories (Modern literary Corpus)

(34,2%) are PUs. Overall, the phraseological density in the first segment was 53.6% (58 bigrams and 30 PUs). The results for the N+Adj category were 49 PUs from a total of 291 bigrams (16.8%), but in the first segment the values were significantly higher: 49 bigrams and 13 PUs, giving a phraseological density of 44,8%. Figure 5 shows the phraseological density for each category. The dark bars indicate the total number of BGs (that is the whole text), and the light bars the 10% of BGs with the highest log-likelihood value.

After testing this method, we applied it to a bigger and more varied corpus, containing. 1,137,323 words and comprising 14 well-known modern Spanish novels.[10] We retrieved a total of 355,503 recurrent bigrams that we classified according to the grammatical categories of their constituents. As expected, the increased size of the corpus resulted in an improvement in the quality of the results (cf. Figure 6). For the

10. *Niebla* and *San Manuel Bueno mártir* by Miguel de Unamuno, *Las inquietudes de Shanti Andía* by Pío Baroja, *Réquiem por un campesino español* by Ramón J. Sénder, *El hereje* and *Cinco horas con Mario* by Miguel Delibes, *La leyenda del César visionario* by Francisco Umbral, *La vieja sirena* by José Luis Sampredro, *La pasión turca* by Antonio Gala, *El oro del rey, El capitán Alatriste, Limpieza de sangre, La carta esférica* and *El sol de Breda* by Arturo Pérez Reverte.

V+N group we found 1,375 bigrams, of which 347 (25%) were phraseological - generally collocations (e.g. *ajustar cuentas*, make even; *pegar fuego*, set on fire; *entablar combate*, start fighting). The list of bigrams ordered according to their log-likelihood value shows a dense PU grouping in the first interval (64,5%). The results are also encouraging for the N+Adj group: 1,060 bigrams of which 321 (30%) are PUs. The phraseological density in the first interval was 53,3%. As for the V+Adv group, the results were even worse than those obtained in the calibration test with the smaller corpus: the phraseological density was only 3,25%. The Appendix contains a sample of the listings for these categories.

4. Conclusion

Our intention throughout this series of analyses was to test different statistical measures and techniques in order to develop an efficient method of retrieving and extracting phraseological units from a corpus with the least human intervention possible. The key value to measure the efficiency of such a method is the phraseological density (%PU/BG) in the head segment of the bigram listings, ordered according to the value of the statistical measure. In other words, our determinant of success was the number of phraseological combinations in this portion of the list as a proportion of all the bigrams extracted. The higher this value, the more efficient the method. The measures tested were z-score, t-score and log-likelihood. In the first analysis, log-likelihood proved to always be the most efficient, so we discarded z-score and t-score in the subsequent analyses, when we used a refined combined statistical and qualitative methodology to improve the results. The purely statistical approach, while confirming the initial hypothesis, turned out to be insufficient to effectively discriminate PUs from other non-phraseological combinations. Our study therefore demonstrates that additional processing of the corpus is a necessary step.

The decrease in the noise level (irrelevant combinations) we observed with a larger corpus shows that this combined approach – lemmatisation, tagging, category filtering and log-likelihood ranking – has reached a level of effectiveness that makes it suitable for productive use in large corpora (at least for groups like V+N and N+Adj). This improvement is evident in the phraseological densities of around 65% and 53% respectively, which compare favourably to the best results for z-score (25,30%) and t-score (41,18%).[11] It is also worth noting that it is possible to increase the phraseological density in the output by considering only those bigrams that occur more than a certain number of times, although the exact quantification of this requisite is yet to be calculated. In future research we intend to apply this method to a very large multilin-

11. These results were obtained in intermediate portions of the listings thus reducing the usefulness for retrieval purposes, while the best values achieved by the log-likelihood measure were located in the top portion of the listings.

gual (French, German, English and Spanish) aligned corpus (more than 150 million words) in order to obtain an immediate comparison of collocational patterns in pairs of languages.

References

Aguilar-Amat, A. (1994). Colocaciones en un corpus: detección y aplicaciones. In Martín Vide, C. (ed.) *Lenguajes naturales y lenguajes formales: Actas del X congreso de lenguajes naturales y formales*, 327–334. Barcelona: PPU.

Banerjee, S. & T. Pedersen (2003). The design, implementation, and use of the Ngram statistics package. In Gelbukh, A. F. (ed.) *Computational Linguistics and Intelligent Text Processing. 4th International Conference, CICLing 2003, Mexico City, Mexico, February 16–22, 2003. Proceedings* [Lecture Notes in Computer Science 2588], 370–381. Berlin: Springer.

Berry-Roghe, G. L. (1973). The computation of collocations and their relevance in lexical studies. In Aitken, A. J., R. Bailey & N. Hamilton-Smith (eds.) *The Computer and Literary Studies*, 103–112. Edinburgh: Edinburgh University Press.

Bradley, J., L. Presutti & M. Stairs (1996). *TACT 2.1* (Computer software). <http://www.chass.utoronto.ca/tact/>.

Carreras, X., I. Chao, L. Padró & M. Padró (2004). FreeLing: An open-source suite of language analyzers. In *Proceedings of the 4th International Conference on Language Resources and Evaluation (LREC'04)*. Lisbon, 26–28 May 2004.

Church, K., W. Gale, P. Hanks & D. Hindle (1989). Parsing, word associations and typical predicate-argument relations. *Proceedings of the International Workshop on Parsing Technology '89*, 389–398. Morristown, NJ: ACL.

Church, K., W. Gale, P. Hanks & D. Hindle (1991). Using statistics in lexical analysis. In Zernik, U. (ed.) *Lexical Acquisition: Exploiting On-Line Resources*. Hillsdale, NJ: Lawrence Erlbaum Associates.

Clear, J. (1993). From Firth principles: Computational tools for the study of collocation. In Baker, M., G. Francis & E. Tognini-Bonelli (eds.) *Text and Technology*, 271–292. Amsterdam: John Benjamins.

Dunning, T. (1993). Accurate methods for the statistics of surprise and coincidence. *Computational Linguistics* 19(1): 61–74.

Evert, S. & B. Krenn (2001). Methods for the qualitative evaluation of lexical association measures. *Proceedings of the 39th Annual Meeting of the Association for Computational Linguistics*, 188–195. Morristown, NJ: ACL.

Evert, S. & H. Kermes (2003a). Experiments on candidate data for collocation extraction. In *Companion Volume to the Proceedings of the 10th Conference of the European Chapter of the Association for Computational Linguistics*, 83–86. Morristown, NJ: ACL.

Evert, S. & H. Kermes (2003b). Annotation, storage, and retrieval of mildly recursive structures. In *Proceedings of the Workshop on Shallow Processing of Large Corpora (SProLaC 2003)*, 23–33. Lancaster, 27 March 2003.

Evert, S., U. Heid & K. Spranger (2004). Identifying morphosyntactic preferences in collocations. In *Proceedings of the 4th International Conference on Language Resources and Evaluation (LREC 2004)*, 907–910. Lisbon, 26–28 May 2004.

Granger, S., M. Paquot & P. Rayson (2006). Extraction of multi-word units from EFL and native English corpora. The phraseology of the verb 'make'. In Buhofer, A. H. & H. Burger (eds.) *Phraseology in Motion* I, 57–68. Baltmannsweiler: Schneider.

Halliday, M. A. K. (1966). Lexis as a linguistic level. In Bazell, C. E., J. C. Catford, M. A. K. Halliday & R. H. Robins (eds.) *In Memory of John Firth*, 148–162. London: Longman.

Jones, S. & J. M. Sinclair (1974). English lexical collocations. A study in computational linguistics. *Cahiers de Lexicologie* 24: 15–61.

Kraif, O. (1997). Modèles probabilistes pour le traitement automatique de corpus textuel. *Travaux du L.I.L.L.A* 2: 81–100.

Krenn, B. (2000a). Empirical implications on lexical association measures. In Evert, S., U. Heid, E. Lehmann & C. Rohrer (eds.) *Proceedings of the Ninth EURALEX International Congress, EURALEX 2000*, 359–371. Stuttgart: EURALEX.

Krenn, B. (2000b). Collocation mining: Exploiting corpora for collocation identification and representation. In Schukat-Talamazzini, E.G. & W. Zühlke (eds.) *Proceedings of KONVENS 2000*. Berlin: VDE.

Krenn, B. & S. Evert (2001). Can we do better than frequency? A case study on extracting PP-verb collocations. In *Proceedings of the ACL Workshop on Collocations.* Toulouse, 21 July 2001.

Manning, C. D. & H. Schütze (1999). *Foundations of Statistical Natural Language Processing*. Cambridge, MA: The MIT Press.

Moon, R. (1998). *Fixed Expressions and Idioms in English*. Oxford: Oxford University Press.

Oakes, M. P. (1998). *Statistics for Corpus Linguistics*. Edinburgh: Edinburgh University Press.

Pamies, A., J. M. Guirao & J. Bolívar (1998). Criterios para la detección de fraseologismos en corpus real. In Luque, J. d. D. & A. Pamies (eds.) *Léxico y fraseología*. Granada: Método.

Pamies, A., J. M. Guirao & J. Bolívar (1998). Critères pour la détection automatisée de phraséologismes en corpus réel. *Travaux du L.I.L.L.A* 3: 117–130.

Pamies, A. & J. M. Pazos (2003). Acceso automatizado a fraseologismos y colocaciones en corpus no etiquetado. *Language Design* 5: 39–50.

Pamies, A. & J. M. Pazos (2004a). El método estadístico en la detección automatizada de colocaciones y fraseologismos. In Cano López, P. (ed.) *Actas del VI congreso de lingüística general, Santiago de Compostela, 3–7 de mayo de 2004*, 893–900. Madrid: Arco Libros.

Pamies, A. & J. M. Pazos (2004b). On automatic retrieval of collocations and idioms in written corpora, paper presented at *Europhras 2004,* Basel, 26–29 August (unpublished). Spanish version ("Extracción automática de colocaciones y modismos"). In Luque, J. D. & A. Pamies (eds.) (2005) *La creatividad en el lenguaje: colocaciones idiomáticas y fraseología*. Granada: Método.

Pamies, A. & J. M. Pazos (2004c). Extracción automática de colocacións e modismos. *Cadernos de Fraseoloxia Galega* 6: 191–203.

Sinclair, J. (1991). *Corpus, Concordance, Collocation*. Oxford: Oxford University Press.

Smadja, F. (1993). Retrieving collocations from text: Xtract. *Computational Linguistics* 19(1): 143–177.

Stubbs, M. (1995). Collocations and semantic profiles: On the cause of trouble with quantitative studies. *Functions of Language* 2: 23–55.

Zinsmeister, H. & U. Heid (2002). *Collocations of complex words: implications for the acquisition with a stochastic grammar.* In *Proceedings of the International Workshop on Computational Approaches to Collocations*. Vienna, 22–23 July 2002. <http://www.ofai.at/~brigitte.krenn/colloc02/zinsmeister_heid.pdf>.

Appendix

This appendix reproduces a small sample of the initial section of the listings, containing the results of the last analysis filtered according to the grammatical categories. Corpus: 1,137,323 words, lemmatised and tagged.

Key:

A: Position of the bigram in the unfiltered listing (all listings are ordered according to the log-likelihood value)
B: First word of the bigram
C: Grammatical category tag of B
D: Second word of the bigram
E: Grammatical category tag of D
F: Rank of the bigram in the unfiltered listing[12]
G: Log-likelihood value
H: Number of occurrences of the bigram in the corpus
I: Total number of bigrams with B in initial position
J: Total number of bigrams with D in initial position

Table 1. V + N bigrams

A	B	C	D	E	F	G	H	I	J
49	dar	VM	cuenta	NC	49	2465.882357571450484	271	3181	537
574	hacer	VM	falta	NC	570	396.608349438734933	55	5119	152
620	tener	VM	razón	NC	616	365.391495954226798	63	4696	348
621	tener	VM	miedo	NC	617	363.561187199859091	56	4696	226
632	dar	VM	vuelta	NC	628	359.346711275885355	54	3181	287
749	hacer	VM	cargo	NC	744	312.667302481902482	41	5119	96
867	ser	VS	verdad	NC	862	277.950641006501087	93	18377	563
961	formar	VM	parte	NC	954	249.378929951635229	28	198	757
992	guardar	VM	silencio	NC	983	242.361242333924622	26	243	461
1149	hacer	VM	daño	NC	1139	209.515820781326397	27	5119	60
1161	prestar	VM	atención	NC	1151	206.907012880478987	16	65	191
1525	caber	VM	duda	NC	1514	166.353911890727488	15	87	316
1583	hacer	VM	caso	NC	1570	162.142830943514980	40	5119	473
1760	pedir	VM	perdón	NC	1746	147.539616900544530	12	422	32
1782	echar	VM	mano	NC	1767	145.826972237334274	31	679	1938
1824	estar	VM	muerto	NC	1808	142.761989542275870	33	5702	307
1848	abrir	VM	paso	NC	1828	141.245740517963100	22	783	498
1865	meter	VM	mano	NC	1845	140.432860120008513	26	401	1938
1914	tragar	VM	saliva	NC	1894	137.683116207366965	8	56	16
2014	tomar	VM	nota	NC	1992	131.876001328198555	14	676	86
2034	dar	VM	miedo	NC	2012	130.396096619680833	24	3181	226
2124	parecer	VM	mentira	NC	2100	125.659162487418271	14	1623	48
2138	tener	VM	tiempo	NC	2114	124.999826162163714	49	4696	1393
2366	tener	VM	sentido	NC	2338	112.862572412982331	22	4696	166
2389	dar	VM	importancia	NC	2361	112.262746584652703	17	3181	92
2658	hacer	VM	además	NC	2622	102.489519110621330	16	5119	62
2677	beber	VM	vino	NC	2641	102.092558686071612	12	291	257

12. The term *rank* groups together all bigrams which have the same log-likelihood value. For instance, if a bigram's rank is 2 this means that it belongs to the group of bigrams which share the second highest log-likelihood value.

Table 2. Adj + N bigrams

A	B	C	D	E	F	G	H	I	J
158	vuesa*	AQ	merced	NC	156	1098.446244412980832	68	70	343
518	mayor	AQ	parte	NC	514	432.047950037222279	50	406	757
981	tal	AQ	vez	NC	974	243.595652796014946	21	22	2841
986	largo	AQ	rato	NC	977	243.013583408692483	28	569	290
1296	pobre	AQ	mamá	NC	1285	188.259293860180406	17	305	103
1475	verdadero	AQ	religión	NC	1464	170.986324540978188	14	181	82
1999	gran	AQ	cosa	NC	1977	132.897918028902211	25	519	1494
2057	real	AQ	gana	NC	2035	129.493269884909893	12	204	121
2074	malo	AQ	suerte	NC	2051	128.655937937931157	15	412	221
2127	buen	AQ	humor	NC	2103	125.415750884066441	14	1198	63
2315	gentiles*	AQ	hombre	NC	2288	115.015093307201752	10	12	2107
2481	duro	AQ	pena	NC	2448	108.939739143462958	11	201	172
2534	ancho	AQ	ala	NC	2500	106.920524472071051	9	126	86
2559	alto	AQ	funcionario	NC	2524	106.340246669581305	11	622	65
2741	largo	AQ	trago	NC	2703	99.625875275379826	10	569	56
2795	vuesas*	AQ	merced	NC	2757	97.353249877299419	6	6	343
3040	pobre	AQ	hombre	NC	2993	90.810654096897451	18	305	2107
3067	mediano	AQ	edad	NC	3020	90.231181130262044	6	13	141
3101	el*	AQ	beneficio	NC	3051	89.423045576043393	7	101	54
3146	largo	AQ	silencio	NC	3096	88.111807622838413	14	569	461
3193	malo	AQ	noticia	NC	3140	86.701204853774655	10	412	140
3270	buen	AQ	parte	NC	3215	84.716934960656644	19	1198	757
3341	mechón	AQ	albino	NC	3278	83.058554988575409	3	3	3
3536	buen	AQ	fe	NC	3470	78.613491424152130	12	1198	167
3650	nero	AQ	correndo	NC	3580	76.328438318986656	3	5	3
3704	viejo	AQ	amigo	NC	3632	75.382440661086662	13	637	490
3850	malo	AQ	intención	NC	3775	72.586250110375843	8	412	92

* The lemmatiser and POS tagger experienced some problems, and minor errors occurred, especially when dealing with archaic or unusual lexical items.

Table 3. N + Adj bigrams

A	B	C	D	E	F	G	H	I	J
105	día	NC	siguiente	AQ	104	1576.607917459072041	149	1686	274
239	voz	NC	bajo	AQ	236	783.413673552527712	77	820	294
503	mano	NC	derecho	AQ	499	442.730953609701828	47	1938	117
556	mañana	NC	siguiente	AQ	552	407.694805509719629	41	490	274
581	secretario	NC	real	AQ	577	393.764776699674371	29	87	204
705	ojo	NC	azul	AQ	700	326.543732824265874	43	1512	294
847	mano	NC	izquierdo	AQ	842	282.456451615749131	34	1938	134
882	ojo	NC	claro	AQ	877	274.210266415133844	36	1512	243
919	cárcel	NC	secreto	AQ	912	262.121391286334017	19	122	81
991	garrote	NC	vil	AQ	982	242.476013497889483	12	35	14
1043	enano	NC	melancólico	AQ	1034	231.757963429619338	14	33	65
1051	ser	NC	humano	AQ	1042	230.965516822436996	18	94	154
1052	carta	NC	náutica	AQ	1043	230.798247129108631	18	442	38
1057	guardia	NC	civil	AQ	1048	229.218494198398048	17	217	49
1070	tierra	NC	firme	AQ	1061	226.205170950297514	22	573	107
1092	voz	NC	alto	AQ	1082	221.479548987143232	33	820	622
1267	cabeza	NC	redondo	AQ	1256	192.020827055967544	21	1052	98
1373	lado	NC	izquierdo	AQ	1362	180.611337089732899	20	750	134
1420	sargento	NC	mayor	AQ	1409	175.494840835653463	15	53	406
1452	línea	NC	recto	AQ	1441	172.727535369353802	13	166	51
1480	puente	NC	levadizo	AQ	1469	170.061933100970407	9	94	9
1554	embajador	NC	inglés	AQ	1543	164.183632297708840	10	19	88
1591	correo	NC	urgente	AQ	1578	161.216651793366111	10	39	44
1613	palo	NC	mayor	AQ	1600	158.731185291052185	16	126	406
1701	día	NC	anterior	AQ	1687	151.439975967102356	20	1686	124
1705	sangre	NC	frío	AQ	1691	151.173478637736594	16	373	171
1723	calzas	NC	abotonadas	AQ	1709	149.886552143097191	7	29	7

Table 4. V + Adv bigrams

A	B	C	D	E	F	G	H	I	J
198	ser	VS	mucho	RG	195	871.436757665268033	344	18377	2648
412	estar	VM	allí	RG	408	513.901147275686299	116	5702	1032
421	ser	VS	más	RG	417	506.316940503454134	366	18377	5433
423	ser	VS	tanto	RG	419	504.729644675194720	235	18377	2222
703	estar	VM	aquí	RG	698	326.886077992508831	79	5702	810
710	estar	VM	mucho	RG	705	325.324811595818574	121	5702	2648
759	estar	VM	bien	RG	754	310.156881265717516	92	5702	1396
831	dejar	VM	atrás	RG	826	285.775024187654594	40	1605	319
953	mirar	VM	alrededor	RG	946	250.624927664211384	34	2025	196
1225	mirar	VM	fijamente	RG	1214	198.410086456233898	17	2025	20
1308	seguir	VM	adelante	RG	1297	186.885974813872053	22	1374	108
1323	ser	VS	demasiado	RG	1312	185.409222300295994	60	18377	345
1666	conocer	VM	bien	RG	1653	153.885299410304441	33	1022	1396
1695	tener	VM	más	RG	1681	151.958507823636808	102	4696	5433
1913	estar	VM	tanto	RG	1893	137.763013215987741	69	5702	2222
2185	ser	VS	así	RG	2161	121.972859784853483	95	18377	1494
2243	quedar	VM	allí	RG	2217	118.697331055747298	27	1294	1032
2271	ser	VS	precisamente	RG	2245	117.241005463361972	28	18377	96
2367	hacer	VM	más	RG	2339	112.830475130541402	93	5119	5433
2665	venir	VM	aquí	RG	2629	102.354594221480312	22	1175	810
2768	tener	VM	delante	RG	2730	98.551669837218085	22	4696	228
2869	estar	VM	cerca	RG	2830	95.221904605203349	27	5702	376
2932	ser	VS	sólo	RG	2892	93.596288711442654	81	18377	1385
3014	se	VM	dijo	RG	2973	91.413091333181697	7	109	44
3044	quedar	VM	atrás	RG	2997	90.773365640086496	16	1294	319
3091	encontrar	VM	allí	RG	3042	89.641092174745978	20	914	1032
3249	estar	VM	seguro	RG	3194	85.262093703288855	15	5702	73

ENVOI

The phrase, the whole phrase, and nothing but the phrase

John Sinclair

1. The phrase

Phrases have never had a proper status in linguistic theory, and, as a consequence, are anomalous in descriptions.[1] The reason for their omission from theory is that received theories require grammar and lexis to be separated from the outset, and they are then described without reference to each other.[2] But since the central notion of a phrase entails *coselection*, the simultaneous selection from both grammar and lexis, there is no place for phrases in descriptions that rely for their architecture on the received theories.

The strategy of describing grammar and lexis separately rests on the assumptions that (a) the word is the primary unit of meaning, and (b) words are chosen from paradigmatic lists, independently of their surroundings. If these assumptions hold, then grammar and lexis can be described separately without risk of distortion. Structures can be described in terms of an ordering of elements, each of which is ultimately one word. In such circumstances there is no need to postulate the existence of phrases except as a small list of marginal cases.

Received theories accept the above position as broadly correct, and the applications of the theories that we find in dictionaries and grammars show it in operation.

---

1. I mean here the word 'phrase', unmodified. Early transformational-generative grammar attempted to fix the terms 'noun phrase' and 'verb phrase' for primary clause-elements, but the word has retained its general utility value despite that. I suggest that it is best left as a pre-theoretical term, because it is very useful when writing on grammar to have a term for a string of words whose status is not determined. Also, its use as a technical term is doubtful because the meaning of 'more than one word', which 'phrase' entails, is not helpful in language description. All the language groupings that I know of are expressed in terms of one or more than one of the units specified, or in rare (and often doubtful) cases, one only. I know of no case where all exponents of a class consist of more than one of the unit next below (Halliday 1961).

2. Grammars which attempt to build in lexical selections once the grammatical framework has been articulated without reference to lexical structure are even less sensitive to lexical considerations than the earlier ones which at least allowed 'separate development' to the lexis.

Dictionaries are almost exclusively focused on the word, so much so that it is difficult to treat a phrase adequately in a dictionary and difficult to find it once it is there. Grammars, with the notable exception of the Pattern Grammars (Francis et al. 1996, 1998), find it sufficient to illustrate grammatical classes with one or two typical exponents.

With models of this kind imposed on them, phrases just do not have a chance. Before even being identified as phrases, they are saddled with a grammatical analysis and each word is allocated a meaning; such assignments are irrelevant and often misleading, and give the researcher a task which is completely unnecessary – to correlate these irrelevant assignments with the accepted meaning of the phrase.

It is a waste of time to subject a phrase to conventional grammatical analysis, indeed to any analysis at all except where the phrase shows internal variation. Grammatical meaning is created by choice, and where there is no choice, there is no meaning. It is thus quite useless to assign structural roles to the fixed word-components of a phrase. Also, the individual words often contribute to the meaning in a way which does not relate to their meanings as we find them in dictionaries; once again we are starting the description at the wrong place. Most of the descriptive work on phrases consists of attempts to supply 'explanations' for assignments of meaning that should not have been made in the first place, and which in many cases stretch the credibility of the reader.

A theoretical framework that is more suited to what we know about phrases is taking shape. It is an elaboration of Firth's contextual theory of meaning, building in cotextual evidence from corpus studies. So the topmost stratum consists of *contextual settings*. Here the assignment of the personal pronouns is made, and the assignment of other fixed reference points such as *now* and *here*. The conventions of genre are also set out.

The second stratum is the phraseological, which deals with the "collocational frameworks" (Renouf & Sinclair 1991), which are the linguistic skeletons of full-bodied phrases, composed typically of a sequence of small closed word classes and/or individually-specified members of such classes. These frameworks are typically discontinuous, leaving room for choices from open classes to combine with them. I have recently taken to calling these *phraseological items* (Sinclair 2004) to associate them with, and distinguish them from, *lexical items*, which are specified at the next stratum.

The third stratum is where lexis and grammar divide. On the grammatical side, the phraseological units are reconciled with the full choice sets of the grammar, in a domain where the abstract patterns take precedence over the meaning; while on the lexical side the specification of lexical items is detailed according to the meaning they create.

The phrase is thus quite central and pivotal in the description. It is the place where structures are engineered to allow meanings to take shape. Our attention should be focused here and not in the more abstract realm of 'pure' grammar.

2. The whole phrase

One of the major contributions of the early years of corpus linguistics has been the information on which accurate accounts of the interdependence of items sharing a cotext have been based. Previous scholars managed to identify the *core* of a number of multi-word items because the core is almost invariable, and so its repetitions were easily noticed. However, corpus work shows us quite clearly that it is unusual for the core to be completely independent of its cotext, that there are other components of a meaningful phrase lurking around, less obvious because more variable, but playing a number of important roles in the creation of meaning.

If we make the meaning the focus of our quest, it must be the total meaning of an entire document or encounter that is the final aim. We know that such meanings are exceedingly complex and subtle, and are created by the selections and juxtapositions of words and phrases as the text unfolds. In such circumstances it is highly unlikely that smaller units are selected independently of each other – indeed the opposite is almost certainly the case. The initiator of an utterance has something in mind to say or write, and this dominates his or her behaviour throughout. Individual meaningful units, even if not specifically coselected, are likely to be co-ordinated, contrasted or otherwise chosen with respect to each other. The studies of "keywords" (Scott, e.g. 2001) and "aboutness" (Phillips 1989) show such tendencies over entire artefacts, so we must expect everything from strong collocations to general homogeneity of vocabulary within a single document or speech event.

These less obvious patterns of meaning-governed selection are not usually retrievable for a language user, or even a language scholar to examine, though when they are pointed out, they suddenly appear to be obvious (this is Bill Louw's "20-20 hindsight", Louw 1993). The role of the computer here is crucial, and its contribution to the elucidation of text structure is already impressive. Ambiguity all but disappears, multi-word units of meaning seem to be the norm. We have come a long way from the presumption that, mostly, the word is the unit of meaning. First, we note that, however we circumscribe the unit of meaning, there will be connections like tentacles stretching out to the surrounding cotext, supporting or modifying the selection. Then, we have to concede that the normal primary carrier of meaning is the phrase and not the word; the word is the limiting case of the phrase, and has no other status in the description of meaning.

3. Nothing but the phrase

There are, of course, other items in a language than phrases; this subheading refers to the difficulty of deciding where one meaningful unit ends and another begins. It is a common enough problem in linguistic description, less obvious in written language, which tends to be segmented, and in alphabetical script, which is clearly segmented and often has word separators and punctuation marks as well as letters.

But the meaningful units do not coincide regularly with the units of the orthography; while meaningful units normally terminate at word boundaries, and rarely cross the boundaries of punctuation units, their behaviour can only be predicted statistically.

Since meaningful units are formed and shaped by the persistent recurrence and coselection of words, it is to be expected that the boundaries between them will be indicated by a falling off or an absence of such patterns. The simplest models show meaningful items in a simple sequence, but we have to allow for the possibility of overlap, discontinuity, embedding and other familiar descriptive complications. As research efforts identify and describe the meaningful items, these can replace the simple word-tokens in a corpus, and become single units for statistical purposes. This process reduces the inaccuracies of the initial counting of tokens, and, applied cyclically, could be the basis of a self-organising model which would eventually produce an exhaustive lexicon of the language.

References

Francis, G., S. Hunston & E. Manning (1996). *Collins Cobuild Grammar Patterns* 1: *Verbs*. London: HarperCollins.

Francis, G., S. Hunston & E. Manning (1998). *Collins Cobuild Grammar Patterns* 2: *Nouns and adjectives*. London: HarperCollins.

Halliday, M. A. K. (1961). Categories of the theory of grammar. *Word* 17(3): 241–292.

Louw, W. (1993) Irony in the text or insincerity in the writer? In Baker, M., G. Francis & E. Tognini-Bonelli (eds) *Text and Technology*, 157–176. Amsterdam: John Benjamins

Phillips, M. (1989). *Lexical Structure of text* [Discourse Analysis Monographs 12]. Birmingham, University of Birmingham, Department of English.

Renouf, A. & J. Sinclair (1991). Collocational Frameworks in English. In Aijmer, K & B. Altenberg (eds) *English Corpus Linguistics: Studies in Honour of Jan Svartvik*, 128–143. London: Longman.

Scott, M. (2001). Comparing corpora and identifying key words, collocations, and frequency distributions through the WordSmith Tools suite of computer programs. In Ghadessy, M., A. Henry & R. L. Roseberry (eds) *Small corpus studies and ELT: theory and practice*, 47–67. Amsterdam: John Benjamins.

Sinclair, J. (2004). Language and computing, past and present. In Rogers, M. & K. Ahmad (eds) *New Directions in LSP Studies. Proceedings of the 14th European Symposium on Language for Special Purposes: Communication, Culture, Knowledge, 18–22 August 2003*, University of Surrey, Guildford, 1–12. URL for LSP Proceedings: http://portal.surrey.ac.uk/computing/news/lsp2003

Author index

A
Aijmer 215
Alegria 341
Allerton 30, 45
Altenberg 17, 35, 39, 148, 198, 261, 266
Anders 218
Anderson 349
Ángeles Calero Fernández 216
Ariel 86, 145
Aristotle 130
Assmann 234
Aston 129
Atkins 263, 329
Awwad 192

B
Baayen 349
Baba 281
Babych xxv, 293, 299
Baccouche 211
Baker 348
Baldwin 341
Ballesteros 305
Bally 27
Balzer 221
Bamberg 182
Bandō 284
Banerjee 397
Baranov 213
Barkema 32, 90, 276
Barnbrook 41
Baroni 201
Barth 20, 39
Bartsch 41, 351
Bauer D. 347
Bauer L. 365
Baur 230
Beardsley 86
Becker 342
Benson xvi, 33, 43, 378
Bentivogli 344
Berber-Sardinha 178
Bernardini 201
Bernstein 176, 182
Berry 230, 232
Berry-Rogghe 382, 392
Bertrand 380
Biber 35, 39, 41, 112, 122, 139, 176, 179, 259, 260, 264, 295
Bierich 217
Bindi 101
Binon 314, 377
Birich 222
Black 86, 131
Blum-Kulka 232
Blumenthal 198, 314
Bly 235, 238
Boers 70, 85, 261
Bond 294, 339
Boswell 160
Bouma 341
Bowker 298
Braasch 342
Bradley 393
Bragina 219, 231
Brala 261, 271, 272
Brants 348
Brenier 15
Brinton 112
Budvytyte 192
Burchardt 347
Burger xx, 37, 38, 42, 45, 192–194, 197, 208, 213–215, 231, 235, 338, 346
Burnard 129
Butt 299, 304
Bybee 13, 111

C
Cacciari 139
Caignon 377
Calzolari 349
Cantos 21
Carreras 397
Carston 131
Carter 61
Channell 153
Čermák 194, 199, 313, 323
Charteris-Black 193, 195, 277, 278
Chiappe 131, 132
Choueka 382
Chun 195
Chuquet 265, 269
Church 20, 173, 350, 382, 392
Cignoni 101, 103, 344
Clark 112
Clarke 152
Clear 392
Coffey 101, 103, 344
Cohen 377
Colson xxi, xxiv, 191, 197
Connors 176
Conrad 39
Copestake 342
Corbett 176
Coseriu 278, 289
Cosme xxv, 198, 259, 265
Coulmas 215
Coulson 67, 69
Cowie xvi, 11, 27–29, 31, 34–38, 45, 194, 207, 294, 299, 323, 332, 365
Croft 12, 56, 150, 305
Cruse 56, 150
Culicover 11, 12, 305
Cumming 342

D
Dahl 246
Daille 386, 388
Dale 342
Dalmas 192, 198
Darbelnet 262, 263
Davis xxiv, 175, 176, 179, 180, 182
De Cock 35, 39

De Sutter 201
DeCarrico 35
Degand 198
Deignan 148
Delibegović 74, 76
Delisle 200
Dias 21
Dirven 261, 262, 264
Dobrovol'skij 71, 192–197, 207, 209–213, 216, 220, 230, 231, 233, 237, 248, 278
Domenig 345
Dorow 341, 352
Doucet 341
Downing 259, 263, 266
Dunning 392–394
Duranti 209
Durco 192
Dzierżanowska xvi

E
Eckert 222
Eeg-Olofsson 17
Eismann 221
Emele 341
Erbach 346
Erh-li 211
Erman 82, 111
Evans 259
Evert 20, 29, 40, 345, 350–352, 354, 386, 387, 397

F
Far 83
Fauconnier 67–69
Fazly 341, 353
Feilke 57
Fellbaum 344
Feretti 67
Fernando 276
Ferragne xxiii, 111, 123
Filatkina 212, 222, 233
Fillmore 12, 14, 51, 52
Firth xvi, 16, 31, 278, 289, 391, 408
Fitzpatrick 313
Fleischer 217
Francis 5, 17, 18, 20, 97, 98, 101, 260, 408
Fraser 6, 10, 11, 276, 289
Frawley 379
Fry 123

G
Gaatone 29
Gamkrelidze 208
Geckeler 278, 289
Geeraerts 234, 235
Gek Lin 232, 236
Gibbs 11, 67, 71, 82, 85, 86, 144, 276
Giegerich 33
Gilquin xxv, 45, 198, 259, 265
Giora 67, 149
Gläser 28, 32, 35, 37, 192–194, 230, 231
Glucksberg 67, 100, 105, 131, 132, 238
Goatly 131, 136, 139
Goldberg 12, 14, 15
Goldman 382
Gonzáles Rey 82
Goossens 144
Gordon 177
Gosselin 91
Gouws 333, 341, 345
Gréciano 192, 198, 199, 207, 223
Grady 132
Gramley 34
Granger xix, xxi, xxii, 22, 27, 198, 272, 289, 334, 340, 397
Grice 131
Gries xxi, 3, 5, 7–9, 19–21, 33, 34, 45
Grondelaers 234, 235
Gross 27, 32, 33, 82–84, 86, 87, 89, 340, 341
Grossmann 351

H
Haarmann 245
Halliday 391, 407
Hamilton 185
Hanks 89, 320, 351, 382
Hanse 264
Hansen 201
Hartley 299
Hasada 277, 289
Haspelmath 246
Hausmann 56, 314, 351
Healey 363
Heid xxi, xxvi, 21, 33, 333, 337, 338, 341, 345, 346, 351, 352, 379, 380, 397
Hendrickson 70
Hengst 244
Herbst 33, 314
Hessky 192, 220, 221
Hjelmslev 192
Hoey xxiv, 16, 33, 105, 162, 181, 313
House 232
Howarth 4
Huddleston 112
Hudson 82, 83, 89, 107, 315, 323
Hunston 4–6, 17, 18, 20, 33, 316

I
Iñesta Mena 217
Ikehara 21
Ishida xxv, 275–278
Ishiwata 278, 290
Israel 15
Itō 276–278, 290
Ivanov 208

J
Jackendoff 11, 12, 260, 305, 313, 346, 348, 363
Jackson 295
Jarrett 208
Jayez 264
Jazyki 245
Jeans 193
Jespersen 299
Johansson 198
Johnson M. 52, 68, 71, 131, 144, 145, 152, 195, 260
Johnson S. 319, 320, 328
Jones 21, 392
Justeson 20

K
Kövecses 71, 131, 145, 154, 217, 277, 290
Kaalep 341
Kahneman 51
Katz 20, 67, 340
Kay 12, 14
Keil 341, 346
Kempcke 195
Kennedy 131, 132, 161
Kermes 397
Keysar 100, 105, 235, 238
Kilgarriff 41, 351, 382

Kindaichi 280, 281, 290
Kirkness 244
Kiss 266
Kita 9, 21
Kleiber 214
Klein xix
Klotz 323
Kluckhohn 209
Knappe 222
Koller 182, 200
Korhonen 192, 194
Kraif 392
Krenn 20, 341, 345, 346, 351, 352, 386, 397
Krishnamurthy xvi, 40
Kroeber 209
Krug 111
Kuiper 208, 215, 232, 236, 346

L
Löfberg xxv, 293, 294
Lakoff 22, 68, 71, 131, 144, 145, 152, 195, 217, 260
Landau 323
Langacker 12, 13, 15, 17, 72, 82, 85, 89, 90, 144, 145, 151, 195, 219
Lapshinova 346, 354
Laviosa 201
Le Pesant 82
Lea 332
Lee 140
Leech 131, 147, 259
Leuschner 246
Levin xxiii, 143, 153, 156
Lewis D. xxiii, 111
Lewis M. xvii, 35, 261
Liberman 124
Lin 353, 382
Lindquist xxiii, xxvii, 143, 153, 156
Lindstromberg 85, 259–261
Liu 236
Locke 259, 263, 266
Lodge 131
Louw 31, 153, 317, 409
Luque Durán 216
Lyons 32, 82, 118

M
Maalej 218
Machida 280, 290
Mackin 11, 365
Maclagan xxiv, 175, 179, 180, 185
Makkai 276, 290
Maks 61
Manjón Pozas 216
Manning 39, 392
Marín-Arrese 193, 195
Mari 245, 257, 264
Martin R. 82, 84, 90
Martin W. xxii, 51, 53, 58, 61
Mason 9, 21, 176, 179, 185
Matlock 67
Matsuki 277, 278
Matsumoto 364
Matsumura 284, 289
Matta 244
McCarthy 353
McEnery 41, 361
McVeigh 277, 290
Mejri 32, 82
Mel'čuk xxvi, 29, 31, 32, 34, 36–38, 45, 55, 56, 61, 90, 299, 306, 314, 344, 347, 377, 380
Melchers 214
Mena Martinez 193
Meynard 377
Michaelis 15
Mieder 214
Miller 127, 131, 294
Minsky 51–53
Mittmann 323, 341
Miyaji 276, 290
Mokienko 222, 244, 253
Montoro del Arco 32, 33
Moon xxi, xxvi, xxvii, 29, 33–35, 82–84, 95–101, 103, 128, 135, 144, 148, 159, 176, 182, 197, 276, 277, 290, 313, 319, 321, 327, 339, 363–365, 391
Moore 180
Mori 21
Mudraya xxv, 293, 294, 296, 363, 366
Muischnek 341
Mukherjee 22
Munske 244
Muraki 278, 290

N
Nagao 21
Nagashima 283, 290
Nattinger 35
Nerlich 152
Nesselhauf 29, 261, 271
Newman 5, 193
Nilsson 371
Nivre 371
North 305
Nowattny 131

O
Oakes 392
Oakley 69
Odijk 345, 346, 350
Oflazer 341
Oksefjell 198
Olsen 342
Omazić xxii, 67, 74, 76
Ortony 131
Osselton 319
Ostapovič 251
Overstreet 176

P
Pätzold 34
Palmer 28, 144, 289
Pamies xxvi, 217, 391, 392
Pamies Bertrán xxvi, 217, 391
Papagno 182
Paquot xxi, xxii, 27
Partington 31, 96, 146, 153
Paszenda 193
Pawley xix, 17, 112, 208
Pazos xxvi, 391, 392
Pecman 45
Pedersen 397
Pedrazzini 345
Perebeinos xvi
Perez-Carballo 341
Perkins 28, 35
Petit 82
Petruck 52
Philip xxiii, 95, 100, 102–105
Phillips 409
Piñango 299
Piñel López 192, 195
Pianta 344
Piao xxv, 97, 178, 293, 294
Piirainen xxi, xxiv, xxv, 71, 137, 192–197, 207, 209–212, 216–218, 229, 231, 233, 237, 243, 247, 248
Poirier 200
Polguère 314, 347

Portis-Winner 210
Potter 148
Poulsen 56, 57
Pullum 112
Pulman 346
Puurtinen 201

Q
Quirk 112, 130, 259, 295

R
Röhrich 222
Rajchštejn 220, 244
Rakusan 221
Rayson xxv, 178, 179, 293–295, 306
Reiter E. 342
Reiter N. 246
Renouf 20, 39, 97, 408
Rice 5
Ritz 345, 351, 352, 354
Roberts 200
Rojo 200
Rosch 217
Rothkegel 192, 338
Runcie 332
Ruppenhofer 342, 348

S
Sánchez 21
Sabat 185
Sabban xxiv, xxv, 193, 200, 207, 209, 210, 215, 229–231, 239, 251
Sag 6, 35, 290, 339, 343, 346, 361, 363, 371, 372, 374, 375
Salkie 303
Schönefeld 22
Schütze 39, 392
Schmid 18, 40, 42
Schmidt-Wiegand 222, 232
Schmied 259, 261, 266, 270, 271
Schmitt 61, 208
Schowe 222
Scott B. 384
Scott J. 160
Scott M. xxv, 39, 162, 178, 266, 409
Searle 86, 131
Seelbach 341
Segura García 221
Selva 314
Shannon 208, 315
Sharoff xxv, 293, 294, 301, 303, 306
Shore 209
Siepmann 45, 323, 329, 333
Simon-Vandenbergen 145
Sinclair xv, xvi, xviii, xx, xxi, xxvii, xxviii, 17, 20, 21, 29, 31, 39, 96–98, 104, 144, 147, 153, 159, 197, 198, 214, 260, 278, 290, 293, 294, 313, 340, 392, 407, 408
Skandera 208, 213
Skelton 182
Skog-Södersved 219
Smadja 351, 382, 397
Somers 299
Spalding 222
Spohr 345, 354
Sproat 124
Stedje 219
Steen 132
Stefanowitsch 8, 9, 20–22, 33, 34
Stein 191, 328
Steinvall 83
Stevenson 305, 341, 353
Stolz B. 208
Stolz T. 245
Stubbs 4, 20, 31, 34, 38, 39, 144, 145, 147, 148, 153, 260, 392
Sun 319
Suszczyńska 232
Svartvik 259
Svensén 329
Svensson xxii, 31, 34, 81, 82, 85, 89, 92
Sverrisdóttir 192
Sweetser 145
Syder xix, 17, 112
Szabó 145

T
Takada 278, 290
Tan D. 215
Tan Y. 302, 305
Tanaka 341
Tarp 343
Taylor 195, 290
Tecedor 232
Telija 207, 219, 220
ten Hacken 345
Tiedemann 353
Tirkkonen-Condit 201
Tognini Bonelli 172
Tolstoj 222
Tomasello 15
Traugott 112
Treis 194
Tribble 39
Tschichold xxvi, 35, 345, 361, 375
Tsujimura 281, 290
Tugwell 351, 382
Turner 67–69
Tutin 314, 344, 351
Tversky 51
Tyler 259
Tylor 209

U
Ueda 192, 277, 290
Utsuro 364

V
Váradi 266, 341
Valentin 192
Van Hoof 262
Van Roey 260
Veisbergs 193
Venkatsubramanyan 341
Verlinde 314, 347
Viberg 261
Viereck 246
Villada Moirón 341, 353
Villavicencio 342, 362, 367, 375
Vinay 262, 263
Vinogradov 28, 192, 299, 300
Vossen 294
Vreck 263, 266

W
Wales 127, 136
Wanner 342, 344, 389
Wanzeck 222
Warren 82, 90, 111, 285
Wasow 6, 290
Weight 171
Wermuth 52
Westphal 194
White 212

Widdows 341, 352
Wierzbicka 207, 209, 220
Wikberg xxiii, 127, 129, 214
Williams xxiii, xxiv, 159, 160, 162, 210, 223
Woolard 33
Woolls 266
Wotjak 215
Wray 28, 29, 35, 144, 176, 178, 182, 208, 214, 313, 364
Wulff 6, 22

Y
Yangüela 216
Yatskovich 297
Ying 211
Yu 211

Z
Zé Amvela 295
Zaharia 192
Zelinski-Wibbelt 259, 261, 262
Zholkovsky 61
Zinsmeister 397
Zouogbo 191
Zuluaga 32

Subject index

A
abstract 13, 57, 59, 83, 128, 140, 150, 217, 338, 384, 408
abstract meaning 37, 89, 122, 145, 149, 261, 264
abstract concept 145, 195
abstract domain 71
acquisition xx, 3, 15, 69, 156, 208, 339, 362–366
adjective xvi, xxiii, 17, 30, 33, 43, 44, 84, 91, 98, 101, 111–125, 134, 135, 153, 155, 165, 199, 223, 262, 267, 270, 303, 314–316, 324–326, 330, 332, 333, 340, 341, 344, 348, 351, 372, 374, 379, 382, 391
Alzheimer's disease xxiv, 175–179, 182, 184, 185
analysability xxii, 81–83, 85, 88, 90, 92
areal 243, 249
association measure 41, 350–352, 387
attributive semantics 100, 101, 105, 106

B
binomial 222, 340, 352
British National Corpus (BNC) xxiii, xxiv, xxvii, 6, 7, 9, 11, 41, 111–123, 127–130, 133, 134, 136, 139–141, 143, 146–149, 153, 156, 159, 161, 162, 165, 167–172, 294, 332, 349, 382

C
canonical form xxiii, 72, 74, 78, 95–97, 99–107, 143, 366, 368–370
canonical phrase 100, 103, 104
canonical phraseology 325
chi-square 6, 41
Chinese 211, 215, 218, 236, 251, 253, 258, 294
chunk 96, 104, 111, 129, 313, 341, 350, 351
cluster 16, 21, 39, 53, 97, 99, 267, 271, 340, 341
cognitive grammar 12–15, 17, 19, 22, 144
cognitive linguistics xxi–xxiii, 3, 10, 12, 18–20, 22, 67, 68, 70, 143, 145, 146, 191, 195, 234, 261, 289
cognitive semantics xxiv, 195, 200
collective memory xxiv, 172, 235
colligation 4, 16, 18–21, 29, 31, 33, 105, 139, 159, 201
collocability 35, 36
collocate 16, 21, 31, 33, 39–41, 43, 95, 99, 162–165, 167, 169, 214, 313–316, 322–324, 326, 327, 329, 330–332, 375, 379–381, 388, 392
collocation xvi, xvii, xxii, xxiv, xxvi, 8, 16, 18, 19, 21, 28, 33, 35–43, 51, 54–57, 59–64, 96–100, 122, 144, 159, 160, 162, 164, 167, 168, 170–173, 196, 197, 214, 260, 261, 271, 276, 290, 299, 300, 313–315, 319, 321–326, 328–333, 340–342, 344–347, 350–352, 364–366, 369, 372, 377–389, 391, 392, 395, 396, 400, 409
contextual collocation 159, 160, 165, 168, 171, 173
grammatical collocation 33, 42, 43, 57, 60, 61, 97
lexical collocation 33, 42, 43, 51, 56, 57, 60–64, 387
restricted collocation 36, 159, 160, 162, 165, 167, 168, 214
see also dictionary
collocation extraction 346, 350–352, 378, 382, 383, 388
collocational framework xxiii, 20, 29, 39, 95, 97, 98, 408
collocational network xxiv, 160, 162–164
collocational pattern 51–65, 401
collocational resonance xxiv, 159–162, 172, 173
collocator 43, 51, 54–60, 63
collostructional analysis 20, 34
complex conjunction 33, 42, 44
complex preposition 32, 33, 42, 44, 270, 370, 371
composite 36, 37, 82, 85, 89
compositional xxvii, 7, 8, 29, 31, 56, 82, 86, 87, 90, 91, 299, 339, 341, 342, 345, 348, 351, 353, 361
compositionality xx, xxiii, 31, 35, 44, 81–83, 85, 88–92, 124, 353, 354
compound 32, 33, 42, 43, 57, 90, 114, 117, 119, 122, 123, 162, 276, 315, 317, 319, 320, 328, 329, 340, 341, 343, 348, 362, 363, 370, 379
computational linguistics 52, 293, 294, 305, 338, 350, 361, 362, 364, 367
computational phraseology xxvi, 337–339, 341–343, 345, 350, 354

computing 178, 337
Conceptual Integration Theory xxii, 67, 68, 74, 77, 78
concordance 16, 20, 96, 101, 102, 112, 133, 161, 165, 178, 179, 266
stepped concordance 101
concrete 57, 71, 83, 89, 90, 149, 150, 192, 221, 277
concrete meaning 89, 90, 149
construction 5–8, 10, 11, 13–15, 17, 21, 22, 34, 57, 74, 76, 118, 120, 122, 138, 144, 148, 153, 156, 182, 197–200, 272, 280, 284, 293, 294, 299, 301, 304–306, 340, 341, 348, 349
construction grammar xxi, 4, 8, 10, 12, 14, 15, 17, 19, 20, 22
contextual theory of meaning 31, 408
contrastive analysis 199, 219, 275, 278–280, 288
contrastive phraseology xxiv, 191, 192, 194, 195, 198, 199, 220
co-occurrence xxiv, 4–11, 16, 18, 20, 21, 29–33, 38–42, 98, 99, 162, 196–198, 214, 260, 275, 276, 278–280, 283, 285, 286, 340, 349–351, 364, 388, 392, 393
co-occurrence analysis 38, 39, 41
co-occurrence restriction 30, 46, 278
co-occurrence test 275, 278–280
co-occurrent 40, 45, 160, 315, 342
corpus xvi, xvii, xix–xxvii, 3, 4, 6, 9, 10, 12, 15–21, 34, 40, 41, 54, 89, 95–97, 100–107, 111, 112, 114, 122, 124, 127, 128, 141, 143, 144–146, 148, 151–156, 159–162, 165, 167–172, 175–179, 181, 191, 195, 197–202, 237, 246, 259–261, 264–269, 271, 275, 278, 291, 294, 296, 298, 299, 301–305, 313–316, 318, 323, 326, 332–334, 337–339, 342–354, 361–366, 377, 378, 382–384, 386, 387, 389, 391–402, 408–410
comparable corpus 265, 266, 268, 308
corpus-based xxi–xxiii, xxvi, 18, 19, 22, 27, 34, 40–42, 45, 64, 127, 129, 133, 140, 144, 178, 197, 198, 277, 289, 290, 293, 294, 298, 301, 303, 304, 314, 341, 344, 346, 367
corpus-driven xxvii, 29, 145, 159, 161, 172, 324
parallel corpus 259, 265, 266, 268, 298, 353, 354
reference corpus 95, 96, 101, 161, 168
corpus linguistics xix–xxi, xxiii, xxiv, 3, 4, 10, 12, 15–18, 20, 21, 143, 167, 172, 191, 195, 197, 199, 200, 260, 261, 363, 364, 409
coselection 407, 410
cross-linguistic xvii, xxiv–xxvi, 191, 192, 194, 196, 198, 202, 207, 208, 213, 217, 219, 221, 223, 243, 244, 251, 261, 277, 299
cultural foundation 209, 210, 212, 218, 248
cultural heritage xxv, 248, 252–254
cultural knowledge xxv, 207, 209–213, 216–218, 223, 233–235
cultural model 209, 212, 213
cultural symbol 210–214, 216, 248
culture xvii, xxiv, xxi, xxv, 71, 74, 161, 171, 172, 191, 193–196, 200, 207–216, 218–223, 229–233, 235, 236, 239, 246, 248, 253
culture specific 209
culture-boundness 229, 231, 232, 239
material culture xxiv, 210, 212–214, 248
social culture 248

D
dictionary xvi, xxvi, 39, 54, 56, 69, 95, 111, 112, 114, 116–119, 124, 127, 130, 132, 137, 145, 159, 162, 172, 195, 220, 222, 233, 238, 246, 249, 250, 254, 259, 263, 266, 271, 275, 293, 294, 296, 298–300, 304, 313–315, 318–334, 337, 342–347, 349, 363–368, 377–379, 384, 407, 408
bilingual dictionary xxvi, 69, 263, 266, 275, 299, 323, 328, 377
collocation(s) dictionary 332, 333, 347
learners' dictionary 322, 330, 335, 347
monolingual dictionary xxvi, 112, 114, 117, 119, 313, 318–321, 330
discourse xxii, xxv, 27–29, 32, 34, 35, 37, 38, 42, 70, 72, 118, 124, 128, 133, 139, 140, 161, 176, 182, 185, 199, 232, 236, 275, 277, 278, 379, 387
distributional approach xxvii, 27, 31, 32, 34, 35, 38, 391
Dutch xx, 52, 58–63, 192, 193, 199, 255, 346

E
etymology xxiv, 70–72, 208, 210, 222, 223, 234
Eurolinguistics 245, 246
Europeanism xxv, 244, 245, 247
extended unit of meaning 31, 97, 98, 104, 106
extender xxiv, 175, 176, 178, 182–185
extraction xix, xx, xxiii, xxv–xxvii, 38, 39, 41, 42, 96, 98, 101, 146, 162, 201, 293, 295, 306, 307, 338, 341, 346, 349–353, 377, 378, 382–386, 388, 389, 391, 392, 394, 397

F
figurative expression 61, 67, 68, 78, 128, 139, 141, 238

figurative idiom 28, 36
figurative interpretation 85–87
figurative language xxiv, 67, 71, 78, 105, 191, 193, 195–197, 199, 207, 210, 211, 216, 221
figurative meaning xxii, xxv, 36, 81, 82, 86–89, 91, 92, 145, 154, 196, 220, 244, 247, 250, 277, 348
figurative phrase xxii, 101, 103
figurative phraseme *see* phraseme
figurative phraseology 100, 207, 208, 210, 212, 213
figurative usage 131, 136, 141
figurative use 83, 132, 300, 317, 321, 323, 324, 329, 331
fixed phrase xxiii, 95, 96, 105, 107, 133, 144, 180–182, 318, 320, 325
fixedness xvii, xix, xxii, xxiii, 27, 32, 34, 45, 81, 85, 95, 128, 193, 214, 324, 352, 353, 391
formulae xxiv, 34, 36–38, 42, 44, 61, 102, 159, 163, 165–171, 195–198, 208, 210, 212, 215, 231, 236, 319, 340, 341, 391, 392 *see* routine formulae
formulaic language xvii, 170, 208, 222, 313
frame xvi, xxii, xxiii, 7, 11, 20, 29, 39, 43, 51–64, 74, 76, 77, 97–99, 129, 134, 141, 207, 212, 216, 222, 348
  frame semantics xxii, 348
free combination xxvi, 28, 30, 36, 56, 59, 60, 63, 269, 377, 379, 383, 389, 396
French xvii, xix, xx, xxv, 27, 33, 84, 87, 90, 91, 104, 192, 193, 198, 199, 211, 214, 217, 223, 244, 250, 252, 256, 257, 259, 260, 262–265, 267–269, 272, 314, 318, 320, 328–331, 340, 341, 343, 346, 347, 372, 401
frequency-based approach 27, 29
frequency of occurrence xxi, 5–7, 11, 13, 15, 16, 111, 274, 288, 295, 388
frequency threshold 13, 18, 41, 301

G
German xx, 54, 104, 191–193, 198–200, 209, 211, 215, 217–223, 232, 237, 243, 244, 255, 261, 266, 277, 278, 289, 314, 344, 346, 347, 351, 352, 383, 397, 401
grammatical collocation *see* collocation
grammatical pattern 5, 13, 16
grammatical word 33, 43, 382

I
idiom xvii, xxii, 3, 4, 6, 7, 10, 11, 13, 17, 21, 22, 27–29, 31, 33–38, 42, 43, 56, 60, 61, 68–78, 81–83, 85, 86, 90, 95–97, 99–101, 103–106, 135, 144, 145, 148, 159, 173, 191, 195–197, 199, 200, 207, 208, 210–213, 215–220, 230, 230–239, 244, 243–258, 274–291, 293, 300, 313, 314, 316, 318–321, 325–330, 340–350, 361, 363–368, 374, 391, 392
  idiom modification 67, 68, 72–74, 77
  partial idiom 36–38
  pure idiom 28, 36, 69
  widespread idiom xxv, 244–258
idiom principle 17, 29, 97, 105, 106, 144, 159, 197, 313
idiomatic expression 3, 74, 216, 222, 235, 236, 294, 297, 340, 344, 346–349, 351, 353, 364, 379
internationalism 244, 245
intertextuality xxiv, 159, 160, 172
Italian 101, 256

J
Japanese xvii, xx, xxv, 192, 193, 212, 215, 253, 257, 275–277, 279–287, 289–291

L
language teaching xvii, xix, xx, xxv, xxvi, 45, 271, 272, 307
lexical affinity 30, 387
lexical bundle 35, 39, 41, 46, 176, 179
lexical collocation *see* collocation
lexical database 361, 362, 364, 365, 367, 368, 370–372, 374, 375
lexical field 162, 275, 278, 279, 289
lexical function xxvi, 37, 61, 306, 314, 347, 377–389
lexical item xvii, xxii, 5, 6, 9, 31, 33, 34, 39, 53, 57, 77, 98, 131, 140, 145, 160, 182, 261, 277, 318, 324, 345, 347, 349, 354, 391, 392, 404, 408
lexical network 160
lexical priming xxiv, 33, 162, 313
lexical word 33, 61, 162
lexicalization xxiii, 55, 58, 111, 112, 118, 122–124, 127, 129, 135, 349, 365, 366
lexico-grammar 34, 260
lexico-grammatical xxiii, 97–99, 172, 263, 276, 323, 334
  lexicogrammatical frame 98, 99
lexicographic(al) xxi, xxvi, 22, 29, 35, 41, 99–101, 103, 107, 301, 313, 314, 323, 333, 334, 338, 351–353, 360, 362, 363, 365, 366, 368, 371
lexicography xvii, xxi, xxv, xxvi, 35, 36, 271, 288, 313–315, 332–334, 337, 347, 361, 362, 364–367
  computational lexicography 337, 361, 362, 365, 366
lexis xv, xxvii, 16, 33, 74, 145, 198, 328, 407, 408
light verb xxv, 60, 293–309, 340, 353
linguistic relativity 219
linking adverbial 42, 44
literal comparison xxiii, 128–131, 133, 139

literal meaning 30, 86, 88–91, 139, 149, 150, 152–154, 171, 196, 214, 277, 353
literal sense 153, 214, 299, 300, 303, 321, 325, 326, 329, 331, 367
literal interpretation 36, 87, 150
log-likelihood 41, 377, 378, 382, 384, 388, 389, 392–396, 398–400, 402

M
meaning-text theory xxvi, 36, 377, 380, 383
metaphor xxii, xxiii, 44, 67–72, 76, 78, 84, 96, 105, 128–132, 136, 138, 139, 141, 143–145, 148, 151–153, 156, 182, 191, 193, 195, 196, 200, 216, 217, 235, 236, 238, 253, 260, 277, 278, 348
conceptual metaphor 68, 71, 72, 76, 131, 132, 138, 152, 217, 235, 277, 278
Metaphor Theory xxii, 67, 78, 132
metaphoric model 235–237
metaphorization 243, 253
metonymy xxiii, 44, 67–69, 71, 72, 76, 128, 143–145, 148, 151, 156, 175, 176, 179, 183–185, 277
morphology xxii, 16, 27–29, 32, 33, 338, 349, 366, 367
motivatable 83, 84, 87–89, 91, 216
motivation xxii, xxv, 22, 64, 81–83, 88, 90–92, 195, 216, 223, 231, 233–235
multi-word expression (MWE) xvii, xxvi, 95, 97, 176, 178–181, 268, 276, 293–297, 299, 338–341, 343, 345, 347–354, 361, 362, 365, 366, 368, 370–372, 374, 375
multi-word unit xix, xx, xxiii, xxiv, xxvii, 19–21, 27–29, 31, 32, 35, 45, 164, 208, 223, 229, 295, 409
mutual information (MI) xxvii, 41, 55, 162, 316, 377, 378, 382, 384, 388

N
n-gram xxiii, xxvii, 16, 20, 21, 38, 39, 41, 144–146, 148, 153, 154, 201, 301
n-gram analysis 38, 39, 41
natural language processing (NLP) xvii, xix–xxii, xxvi, 3, 5, 16, 22, 35, 97, 106, 191, 201, 294, 337–339, 341–345, 348, 349, 354, 361, 362, 364–368, 371
non-compositional 6, 7, 10, 16, 31, 35, 36, 82, 86, 89, 90, 122, 197, 299, 300, 339, 353, 354
non-compositionality xix, xxi, xxii, 4, 8, 10, 12, 13, 15, 17, 31, 43, 45, 81, 82, 84, 85, 89, 92, 128, 276

O
opacity xxii, 69, 70, 81–84, 88, 89, 92, 195, 213, 380
open choice principle 17, 103, 105, 144, 293
orthographic word 32, 43, 367

P
paremiology 214, 246
parser 371, 374, 382–385
parsing 338, 351, 367, 368
phrasal pattern xxiii, 143–156, 370
phrasal verb 5, 7, 8, 11, 42, 43, 196, 276, 293–298, 304–306, 319, 320, 327
phrase-frame 39
PhraseManager 345, 368, 370, 375
phraseme xxiv, xxv, 31, 34, 35, 37, 42, 43, 56, 197–199, 207–9, 211–215, 219–223, 228–231, 234, 239, 243, 339
communicative phraseme 42, 44, 215, 231, 232
figurative phraseme 207, 208, 210, 213, 216, 223
full phraseme 37
interactional phraseme 34
pragmatic phraseme 37, 38
quasi-phraseme 37, 90
referential phraseme 42, 231, 232
semantic phraseme 36, 37
semi-phraseme 37, 56
structural phraseme 42
textual phraseme 42, 44
phraseological approach xvii, 27, 29, 261
phraseological continuum 36
phraseological density 393, 394, 396–400
phraseological framework 97
phraseological item 29, 104, 327, 333, 408
phraseological pattern(ing) 100, 103, 145, 147, 313, 317–319, 321, 322, 326–329, 333, 342
phraseological skeleton xxii, xxiii, 97, 98, 105
phraseological unit (PU) xx–xxiv, xxvii, 18, 27–29, 31–33, 35, 37, 38, 42, 67, 76, 97, 102, 127, 141, 162, 293, 294, 299, 300, 337–339, 342, 343, 351, 352, 354, 391–393, 397, 400, 408
communicative phraseological unit 38
referential phraseological unit 37, 38
structural phraseological unit 38
phraseologism xxi, 3–22, 61, 339, 342, 361, 367
phraseology xv–xvii, xix–xxvii, 3, 4, 10, 14, 15, 17, 18, 21, 22, 27–35, 38, 41, 45, 46, 81, 82, 95, 96, 100, 106, 107, 111, 127–129, 143, 144, 151, 156, 178, 191–202, 207–214, 216–223, 229–232, 238, 243, 244, 247, 251, 254, 259, 268, 271, 313–315, 318–328, 332–334, 337–343, 345, 348–350, 354, 361–364
polylexicality 32, 193

polysemy xxv, 67, 122, 152, 195, 259, 260, 267, 268, 271
POS tag 178, 350, 351, 404
POS tagging 397
pragmateme 37
prefab 82, 83, 92, 111
preposition xxv, 17, 32, 33, 35, 43, 57, 61, 98, 128, 129, 130, 138, 148, 151, 198, 259–272, 296, 306, 315, 318, 324, 330, 340, 341, 343, 344, 365, 370, 371, 382, 392
  bound preposition 259, 260, 263, 267, 268
  complex preposition 32, 33, 42, 44, 270, 370, 371
  free preposition 260
  multi-word preposition 341, 343, 365, 366, 370
prototype 28, 59, 60, 64, 136, 264, 267, 289
prototypicality xxii, 58, 59, 62, 81, 82, 84, 89, 92, 211, 214, 250, 261, 264, 267, 271, 276, 277, 299, 377–380, 386, 388
proverb 27, 29, 38, 42, 44, 61, 76, 81, 82, 96, 101, 148, 159, 193, 207, 209–215, 219, 220, 233, 246, 276, 316, 319, 328, 340, 391, 392, 396

R
recurrence 32, 269, 350, 410
recurrent phrase 25, 29, 49, 157
recurrent sequence 39, 47
recurrent word combination 35, 39
resonance xxiv, xxvii, 159–165, 167, 168, 170–173, 223
restricted collocation *see* collocation
rhetoric 34, 130
routine formulae 34, 36, 38, 44, 195–198, 212, 215, 231, 340, 341, 391
Russian xx, xxv, 28, 37, 192, 193, 209, 217, 219, 220, 248, 256, 278, 289, 293–306, 332

S
salience xxii, 67, 81, 82, 89, 92, 122
semantic class 9, 34, 101, 105, 344
semantic correspondence xxv, 275, 277–279, 287, 288
semantic feature 209, 236, 275, 278, 279, 281, 283–285, 287–289, 296
semantic field xxiv, 32, 85, 145, 216–219, 223, 294
semantic frame xxii, 51–64
semantic preference 31, 32, 98, 105
semantic prosody 31, 32, 105, 153, 159, 317
semantics xv, xxii, xxiv, 6, 7, 11, 13, 16, 27–31, 72, 74, 85, 91, 96, 101, 105, 106, 144, 193–195, 198, 200, 202, 207, 223, 236, 260, 289, 322, 332, 338, 342, 343, 347, 348, 353, 379
  see attributive semantics
sentence stem 42, 44
set phrase 191–202, 379
simile xxiii, 42, 43, 84, 88, 96, 127–141, 213, 214, 221, 391, 392
Sketch Engine 40, 41, 351
slogan 42, 44
source concept xxiv, 208, 212, 213, 216–218, 223
source domain 105, 144, 213, 215, 217, 231, 232, 236
source frame 212, 222, 348
Spanish xx, xxiv, 33, 192, 193, 216, 232, 248, 256, 328, 347, 389, 399, 401
statistical measure xvi, xxiv, xxvi, 39, 41, 377, 382–384, 387–389, 392–394, 398, 400
statistical method xxvi, xxvii, 24, 362, 378, 382, 398
statistics 20, 22, 54, 112, 301, 302, 316, 345, 382, 392, 395, 397
stress xix, xxv, 32, 98, 111, 112, 122–124, 138, 191, 200, 276, 304
syntagmatic xv, xvi, 29, 33, 43, 54, 61, 260, 315, 333, 380, 391
syntax xxii, 16, 27–29, 33, 34, 43, 51, 72, 77, 191, 193, 194, 198, 200, 202, 207, 299, 338, 345, 347, 348, 370, 372

T
t-test 41
target concept 208, 213, 216–218, 236, 250
target domain 67, 105, 231
terminology xvi, xxi, xxii, 14, 17, 27, 28, 36, 39, 41, 45, 81, 144, 194, 201, 229, 268, 281, 289, 299, 361–363, 377, 379, 380
textual sentence stem 42, 44
token-bound xxii, 51, 56–62
translatability xxv, 266, 268, 271
  mutual translatability xxv, 266, 268, 271
translation xvii, xxiv–xxvi, 52, 84, 87, 100, 161, 191, 192, 197, 199–202, 221, 259, 261, 262, 266, 268, 269, 271, 275, 287–289, 293–296, 298, 299, 301–306, 329–331, 337, 338, 341, 342, 362, 368, 378, 383, 384, 398
  translation equivalent xxv, 261, 288, 293–308
translationese 201, 268
transparency xxii, xxiii, 81–84, 88–92, 112, 124, 143, 379
transparent 3, 28, 36, 74, 84, 87–91, 136, 156, 238, 315, 367, 369
trinomial 42, 43
type-bound xxii, 51, 55–63
typology xx, xxi, xxiv, 37, 38, 40, 59–61, 207, 210, 244, 246

U
unanalysability xxii, 81, 82, 85, 88, 89, 92
unit of meaning 31, 97, 98, 104, 106, 128, 407, 409
  extended unit of meaning 31, 97, 98, 104, 106

V
variability xx, xxvi, 106, 185, 306, 339, 344, 349, 361, 363–367, 369, 372, 375
variant xxiii, 41, 95–107, 199, 222, 230, 250, 277, 325, 344, 345
variation xvii, xxiii, xxvi, 4, 7, 34, 39, 95, 96, 98, 100, 101, 103–107, 129, 178, 185, 197, 243, 316, 324, 327, 344–346, 349, 351, 352, 362, 365–367, 370–373, 408

W
word combination xix, xxi–xxiii, 27–29, 31, 32, 34–36, 38, 39, 43, 51, 63, 214, 294, 299, 300, 332, 349, 351–353, 364, 365, 391
word grammar 33
word-play 95, 96, 105, 143
WordManager 345, 368
WordSmith Tools 162, 178, 266

Z
z-score xxiv, 162, 393, 394, 396, 400